# INTRODUCTION TO MACHINE LEARNING

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**NPTEL**

**NPTEL ONLINE CERTIFICATION COURSE**

**Introduction to Machine Learning**

**Lecture 5**

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**Probability Basics - 1**

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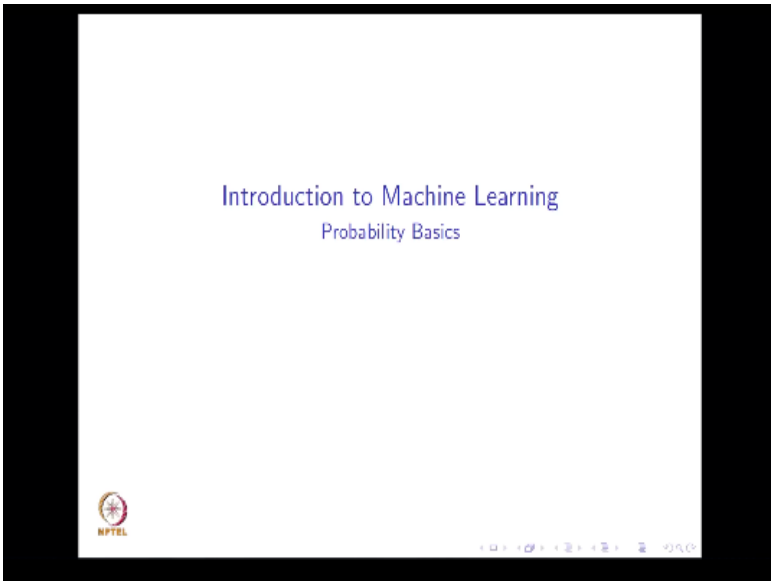


Hello, and welcome to the first tutorial in the introduction to machine learning course. My name is Priyatosh, and I am one of the teaching assistants for this course. In this tutorial, we will be looking at some of the basics of probability theory. Before we start, let us discuss the objectives of this tutorial. The aim here is not to teach the concepts of probability theory in any great detail instead, we will just be providing a high-level overview of the concepts that will be encountered later on in the course. The idea here is that for those of you who have done a course in probability theory or are otherwise familiar with the content, this tutorial should act as a

refresher. For others who may find some of the concepts unfamiliar, we recommend that you go back and prepare those concepts from say an introductory textbook or any other resource so that when you encounter those concepts later on in the course, you should be comfortable with them.

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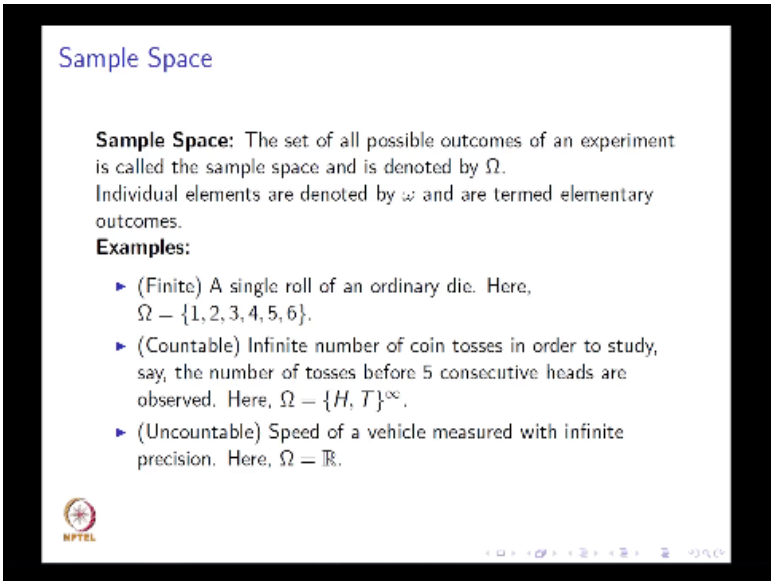


Okay, to start this tutorial, we look at the definitions of some of the fundamental concepts. The first one to consider is that of the *sample space*. The set of all possible outcomes of an experiment is called the *sample space* and is denoted by capital Ω. Individual elements are denoted by small Ω and are termed *elementary outcomes*.

Let us consider some examples. In the first example, the experiment consists of rolling an ordinary die. The sample space here is the set of numbers between one and six. Each element here represents one of the six possible outcomes of rolling a die. Note that in this example, the *sample space* is finite.

In the second example, the experiment consists of tossing a coin repeatedly until the specified condition is observed. Here we are looking to find five consecutive heads before terminating the experiment. The *sample space* here is countable infinite. The individual elements are represented using a sequence of the Hs and Ts where H and T stand for heads and tails, respectively.

In the final example, the experiment consists of measuring the speed of a vehicle with infinite precision. Assuming that the vehicle speeds can be negative, the sample space is clearly the set of real numbers. Here we observe that the *sample space* can be uncountable.

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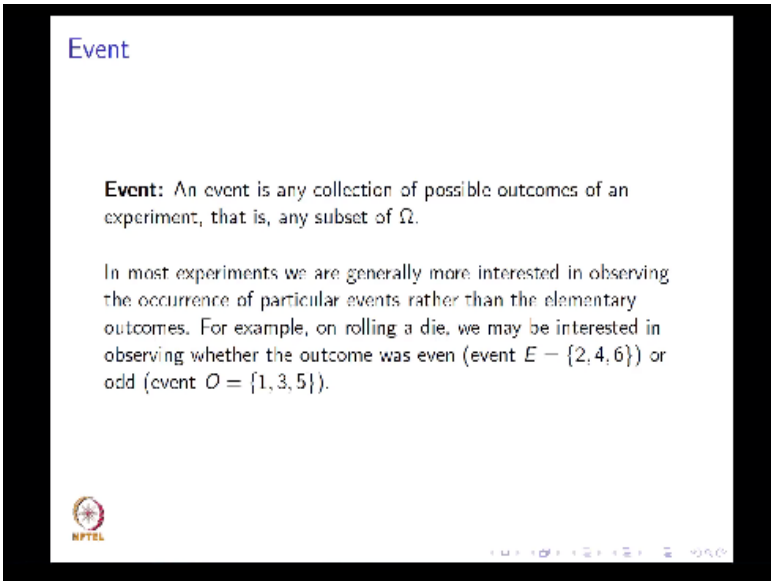


The next concept that we look at is that of an *event*. An *event* is any collection of possible outcomes of an experiment that is any subset of the sample space. The reason why events are important to us is that in general, when we conduct an experiment, we are not really that interested in the elementary outcomes rather, we are more interested in some subsets of the elementary outcomes, for example, on rolling a die we might be interested in observing whether the outcome was even or odd.

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Set Theory Notations

$$A \subset B \Leftrightarrow x \in A \Rightarrow x \in B$$

$$A = B \Leftrightarrow A \subset B \text{ and } B \subset A$$

$$A \cup B = \{x : x \in A \text{ or } x \in B\}$$

$$A \cap B = \{x : x \in A \text{ and } x \in B\}$$

$$A^c = \{x : x \notin A\}$$

On a specific role of a die, let us say we observe that the outcome was odd. In this scenario whether the outcome was a one or a three or a five is not as important to us as the fact that it was odd. Since we are considering sets in terms of *sample spaces and events*, we will quickly go to the necessary set theory notations. As usual, capital letters indicate sets, and small letters indicate set elements.

We first look at the *subset relation*.

$\forall X$ , if $X \in A \Rightarrow X \in B$ then we say that $A \subset B$ or A is contained in B. Two sets A and B are said to be equal if both $A \subset B$ & $B \subset A$ holds.

The *union* of two sets A and B ( $A \cup B$ ) gives rise to a new set which contains elements of both A and B.

Similarly the *intersection* of two sets A and B ( $A \cap B$ ) gives rise to a new set which contains of only those elements which are common to both A and B.

Finally the *complement* of a set A ( $A^c$ ) is essentially the set which contains all elements in the universal set except for the elements present in A. In our case the universal set is essentially the sample space.

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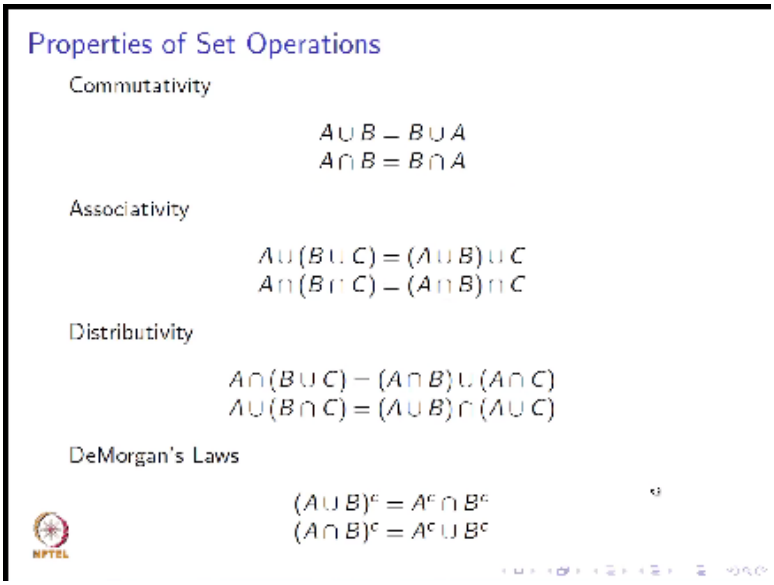


This slide lists out the different properties of set operations such as commutativity, associativity and distributivity, which you should all be familiar with it. It also lists out the DeMorgan's laws, which can be very useful. For instance, DeMorgan's laws states that:

$(A \cup B)^c = A^c \cap B^c$ and $(A \cap B)^c = A^c \cup B^c$

The DeMorgan's laws presented here are for two sets. They can easily be extended for more than two sets.

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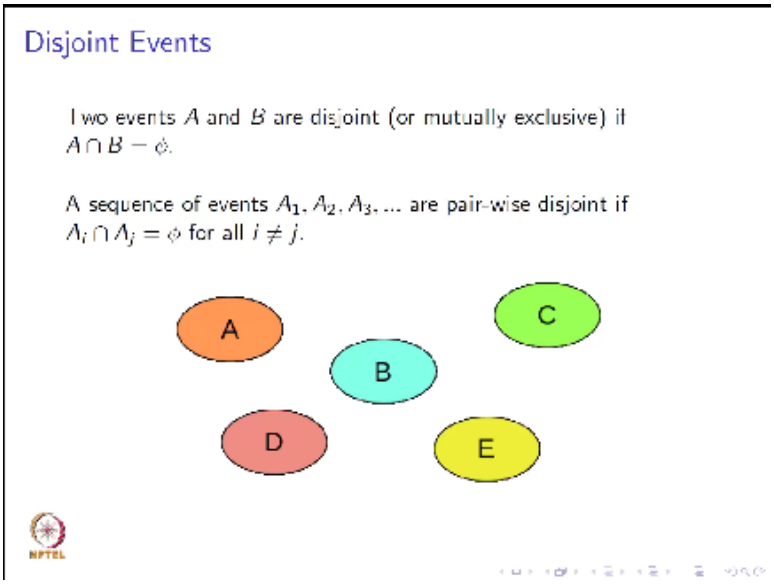


Coming back to events two events A and B are said to be disjoint or mutually exclusive if $A \cap B = \phi$. Extending this concept to multiple sets we say that a sequence of events $A_1, A_2, A_3 \ldots$ are pairwise disjoint if $A_i \cap A_j = \phi \quad \forall i \neq j$.

In the diagram above, if each of the letters represents an event, then the sequence of events A through E are pairwise disjoint since the intersection of any pair is empty.

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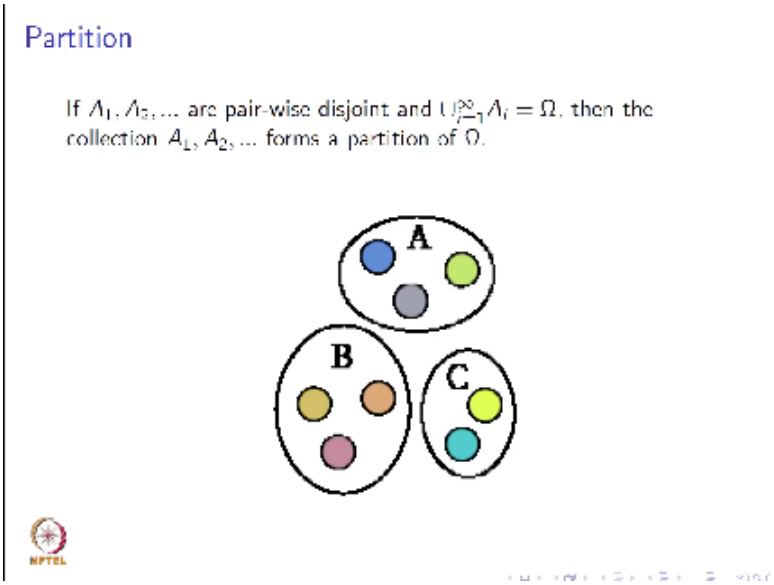

If events $A_1, A_2, A_3 \cdots$ are pairwise disjoint $\bigcup_{i=1}^{\infty} A_i = \Omega$, then the collection $A_1, A_2, A_3 \cdots$ is set to form a partition of the sample space Ω. This is illustrated in the figure above.

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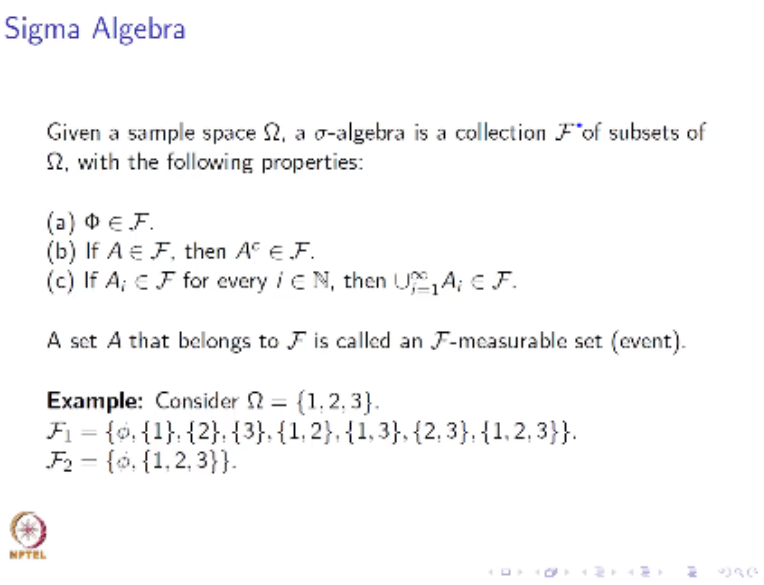


Next, we come to the concept of $\sigma$-algebra. Given a sample space Ω, a $\sigma$-algebra is a collection $F$ of subsets of the sample space, with the following properties.

(a) $\phi \in F$

(b) If $A \in F$ then $A^c \in F$.

(c) If $A_i \in F, \forall i \in N, then \bigcup_{i=1}^{\infty} A_i \in F$

(d) A set $A$ that belongs to F is called an $F$-measurable set. This is what we naturally understand as an *event*.

So going back to the third property, what this essentially says is that if there are a number of events that belong in the $\sigma$ -algebra then the countable union of these events also belongs to the $\sigma$ -algebra.

Let us consider an example. Consider $\Omega = \{1,2,3\}$ , this is our sample space. With this sample space, we can construct several different $\sigma$ -algebras. The first $\sigma$ -algebra $F_1$ is essentially the power set of the sample space. All possible events are present in the first $\sigma$ -algebra.

However, if we look at $F_2$ , in this case, there are only two events the null set ( $\phi$ ) or the sample space itself $\{1,2,3\}$ . You should verify that for both $F_1$ and $F_2$ all three properties listed above are satisfied. N

Now that we know what a $\sigma$ -algebra is let us try and understand how this concept is useful. First of all, for any Ω (countable or uncountable) the power set is always a $\sigma$ -algebra. For example, for the sample space comprising of two elements {H,T}, a feasible $\sigma$ -algebra is the power set. However, this is not the only feasible $\sigma$ -algebra as we have seen in the previous example but always the power set will give you a feasible $\sigma$ -algebra.

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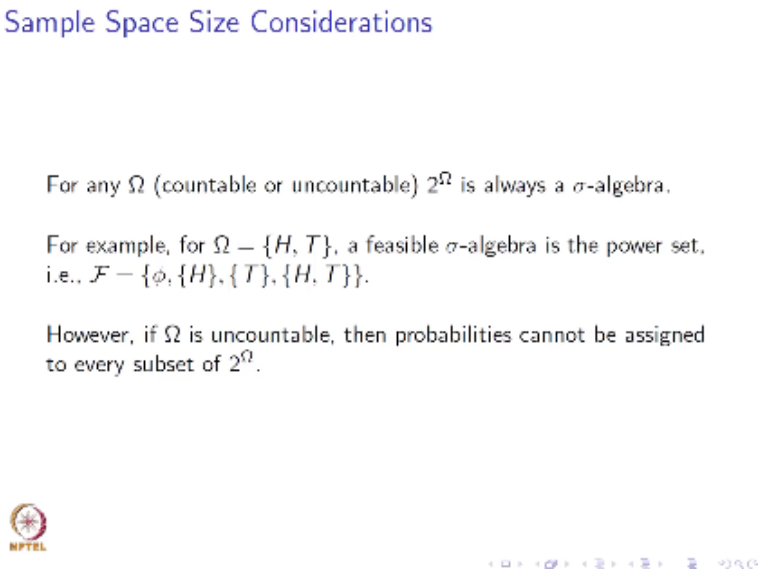

However, if Ω is uncountable, then probabilities cannot be assigned to every subset of the power set ( $2^{\Omega}$ ). This is the crucial point which is why we need the concept of $\sigma$ -algebras. So just to recap, if the sample space is finite or countable then we can kind of ignore the concept of $\sigma$ -algebra, because in such a scenario we can consider all possible events that is the power set of the sample space and meaningfully apply probabilities to each of these events however this cannot be done when the sample space is uncountable that is if Ω is uncountable then probabilities cannot be assigned to every subset of $2^{\Omega}$ . This is where the concept of $\sigma$ -algebra shows its use. When we have an experiment in which the sample space is uncountable for example, let us say the sample space is the set of real numbers. In such a scenario we have to identify the events which are of importance to us and use this along with the three properties listed in the previous slide to construct a $\sigma$ -algebra and probabilities will then be assigned to the collection of sets in the $\sigma$ -algebra.

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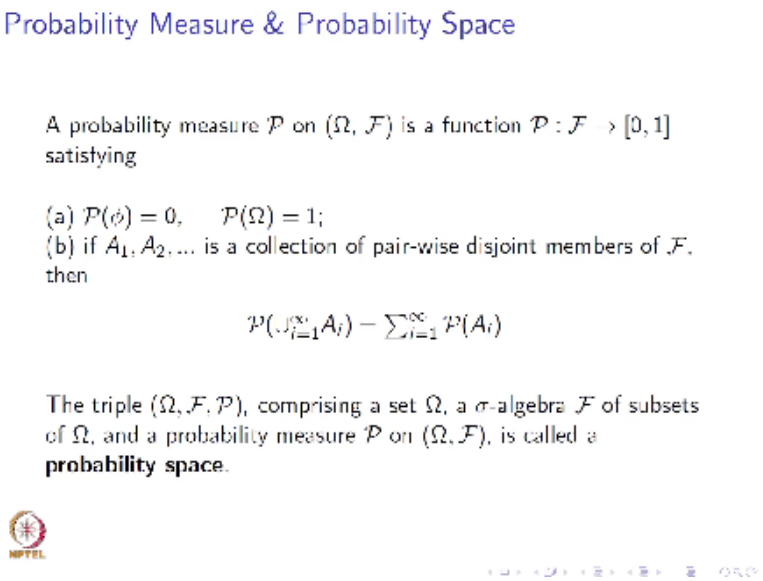


Next, we look at the important concepts of probability measure and probability space. A *probability measure P* on $(\Omega, F)$ is a function $P: F \to [0,1]$ satisfying the following properties:

(a) $P(\phi) = 0$ , $P(\Omega) = 1$

(b) if $A_1, A_2, A_3 \cdots$ is a collection of pairwise disjoint members of $F$, then

$$P(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$$

Or, the probability of the union of all such members is equal to the sum of their individual probabilities. Note that this holds because the sequence $A_1, A_2, A_3 \cdots$ is pairwise disjoint.

The triplet *(Ω, F, P)* comprising of sample space Ω, a $\sigma$ -algebra F which are subsets of Ω and a probability measure P defined on (Ω, F) this is called a probability space.

For every probability problem that we come across, there exists a probability space comprising of the triplet (Ω, F, P) even though we may not always explicitly take into consideration.

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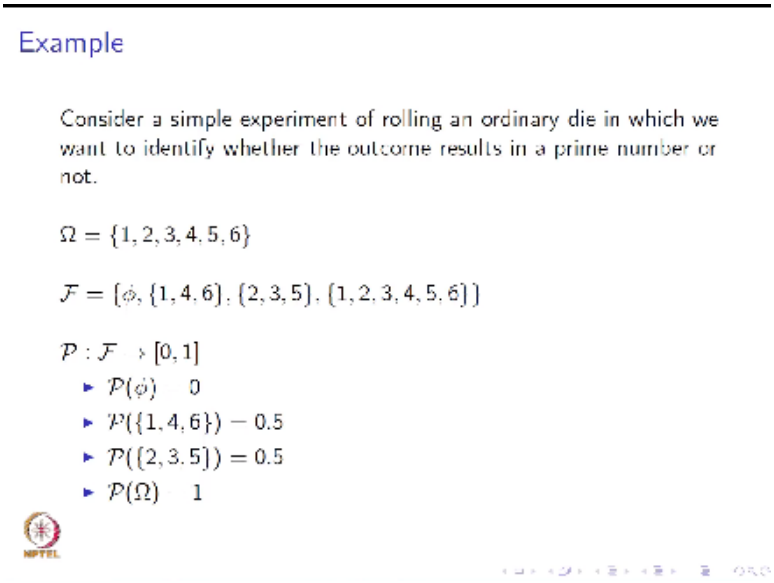


This concept of a probability space, should always remain in the back of our heads when we solve a problem. Let us now look at an example where we do consider the probability space involved in the following problem.

Consider a simple experiment of rolling an ordinary die in which we want to identify whether the outcomes ends in a prime number or not.

The first thing to consider is the sample space. As there are only six possible outcomes in our experiment, the sample space here is $\{1,2,3,4,5,6\}$.

We look at the $\sigma$ -algebra. Note that since the sample space is finite, you might as well consider all possible events. That is the power set of the sample space. However note that the problem dictates that we are only interested in two possible events that is whether the outcome is prime or not thus restricting ourselves to these two events we can construct a simpler $\sigma$ -algebra. Here we have two events which correspond to the events we are interested in and the remaining two events follow from the properties which is $\sigma$ -algebra has to follow.

Please verify that the $\sigma$ -algebra listed here does actually satisfy the three properties that we had discussed above. The final component is the probability measure. The probability measure assigns a value between zero and one that is the probability value to each of the components of the $\sigma$ -algebra. Here the values listed assumes that the die is fair in the sense that the probability of each phase is equal to $1/6$.

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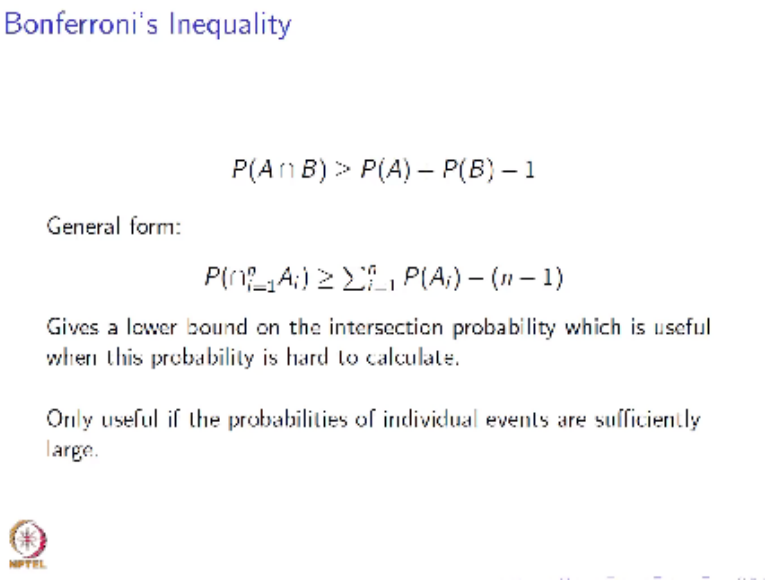

Having covered some of the very basics of probability, in the next few slides we look at some rules which allow us to estimate probability values.

The first thing we look at is known as the Bonferroni's inequality. According to this inequality,

$P(A \cap B) \geq P(A) + P(B) - 1$

The general form of this inequality is also listed. What this inequality allows us to do is to give a lower bound on the intersection probability. This is useful when the intersection probability is hard to calculate. However if you notice the right hand side of the inequality you should observe that this result is only meaningful when the individual probabilities are sufficiently large. For example if the $P(A), P(B)$ both values are very small, then this -1 term dominates and the result doesn't make much sense.

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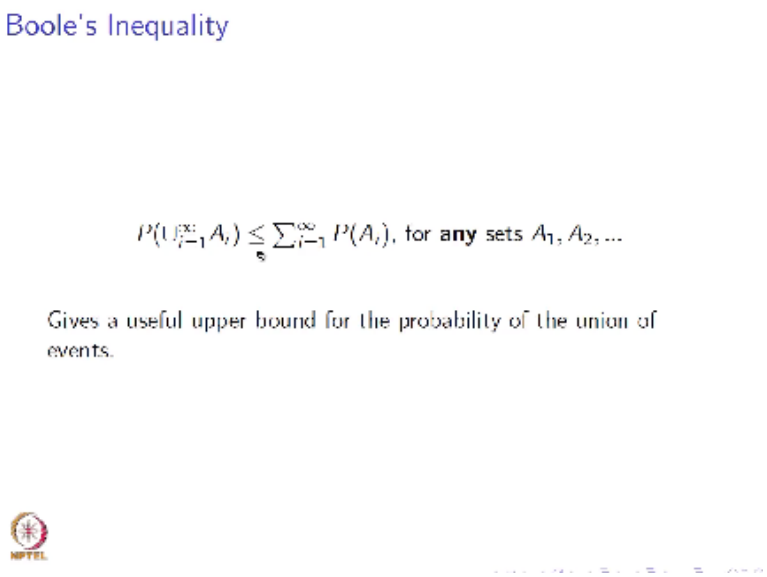


According to the Boole's Inequality,

$P(\bigcup_{i=1}^{\infty} A_i) \leq \sum_{i=1}^{\infty} P(A_i)$, for any sets $A_1, A_2, A_3 \ldots$

Clearly this gives us a useful upper bound for the probability of the union of events. Notice that this equality will only hold when these sets are pairwise disjoint.

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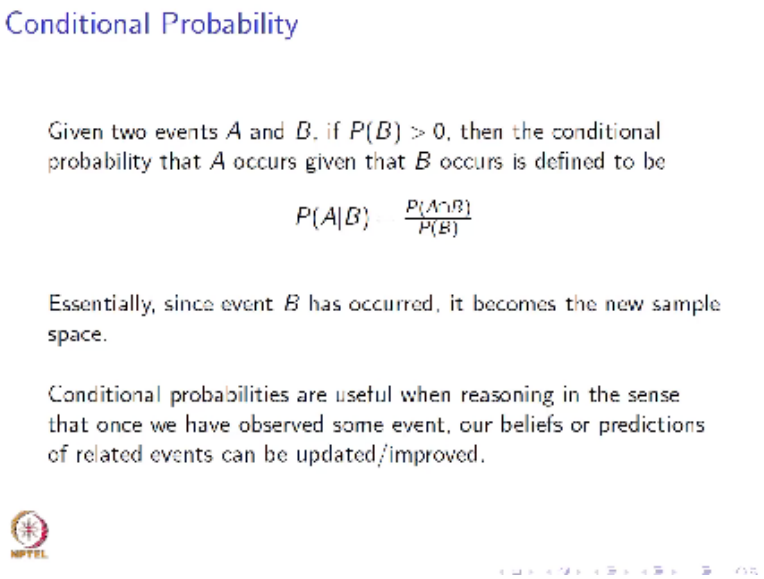


Next we look at conditional property given two events A and B. If $P(B) > 0$, then the conditional probability that A occurs given that B occurs is defined to be

$$P(A \mid B) = P(A \cap B) / P(B)$$

Essentially since event B has occurred, it becomes the new sample space. And now the P(A) is accordingly modified. Conditional probabilities are very useful when reasoning in the sense that once we have observed some event our beliefs or predictions of related events can be updated on or improved.

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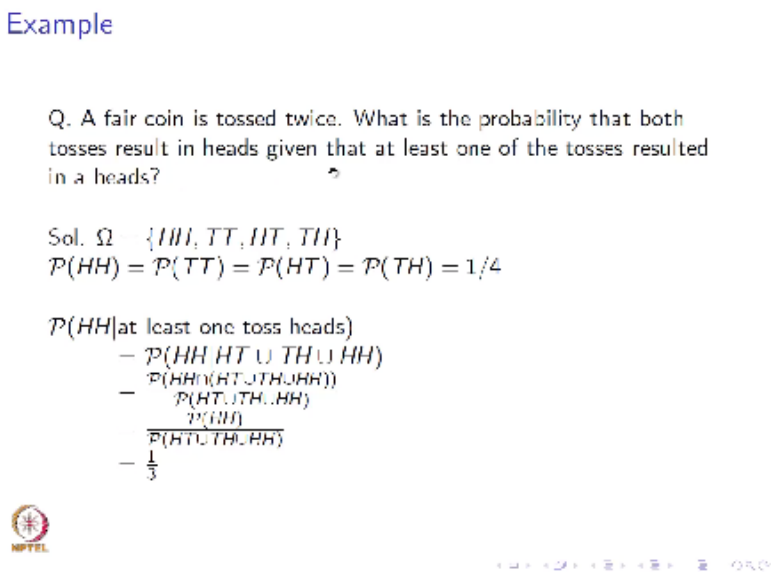


Let us try working out a problem in which conditional probabilities are used.

**Problem:** A fair coin is tossed twice. What is the probability that both process results in heads given that at least one of the tosses resulted in a heads? (Go ahead and pause the video here and try working out the problem yourself)

**Solution:** From the question it is clear that there are four elementary outcomes: $\Omega = \{HH, HT, TH, TT\}$

Since we are assuming that the coin is fair, $P(HH) = P(HT) = P(TH) = P(TT) = 1/4$

Now we are interested in the

$P(HH)|$ (at least one toss heads)

= $P(HH \mid HT \cup TH \cup HH)$

(using the conditional probability formula)

= $P(HH \cap (HT \cup TH \cup HH)) / P(HT \cup TH \cup HH)$

= $P(HH) / P(HT \cup TH \cup HH)$ = $1/3$

Note that in the denominator, each of the events is mutually exclusive. So, $P(HH \cup HT \cup TH) = P(HH) + P(HT) + P(TH)$.

As an exercise, repeat the above problem with the only modification that instead of finding the "probability that both process results in heads given at least one of the tosses resulted in heads" solve "probability that both process results in heads given only the first toss resulted in heads". Does this change the problem?

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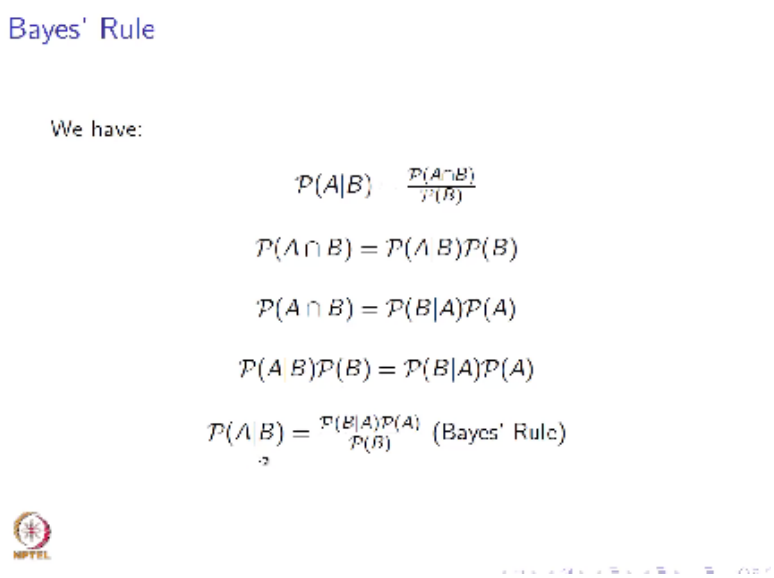


Next we come to a very important theorem called the Bayes' theorem or the Bayes' rule. We start with the equation for the conditional probability.

$P(A \mid B) = P(A \cap B) / P(B)$ Rearranging we have, $P(A \cap B) = P(A \mid B)P(B)$

Similarly, $P(B \mid A) = P(A \cap B) / P(A)$ Rearranging we have, $P(A \cap B) = P(B \mid A)P(A)$.

Equating the right hand sides, we get, $P(B \mid A)P(A) = P(A \mid B)P(B)$

So, $P(A \mid B) = P(B \mid A)P(A) / P(B)$ [Bayes' Rule]

This is what is known as the Bayes' rule. Note that what it essentially says is if I want to find the $P(A \mid B)$, I can use the information of $P(B \mid A)$ along with the knowledge of $P(A)$ and $P(B)$ to find this value.

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## Bayes' Rule

Let $A_1, A_2, \ldots$ be a partition of the sample space, and let $B$ be any subset of the sample space. Then, for each i = 1, 2, ...,

$$\mathcal{P}(A_i \mid B) \quad \frac{\mathcal{P}(B \mid A_i)\mathcal{P}(A_i)}{\sum_{j=1}^{\infty} \mathcal{P}(B \mid A_j)\mathcal{P}(A_j)}$$

Bayes' rule is important in that it allows us to compute the conditional probability $\mathcal{P}(A \mid B)$ from the "inverse" conditional probability $\mathcal{P}(B \mid A)$.

As you can see this is a very important formula. Here we again present the Bayes' rule in an expanded form, where $A_1, A_2, A_3 \cdots$ form a partition of the sample space. As mentioned Bayes' rule is important in that it allows us to compute $P(A \mid B)$ from the inverse conditional probability $P(B \mid A)$.

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## Example

Q. To answer a multiple choice question, a student may either know the answer or may guess it. Assume that with probability $p$ the student knows the answer to a question, and with probability $q$, the student guesses the right answer to a question she does not know. What is the probability that for a question the student answers correctly, she actually knew the answer to the question?

Sol. Let $K$ be the event that the student knows the question, and $C$ be the event that the student answers the question correctly. We have $\mathcal{P}(K) = p$, $\mathcal{P}(\neg K) = 1 - p$, $\mathcal{P}(C \mid K) = 1$, $\mathcal{P}(C \mid \neg K) = q$.

$$\mathcal{P}(K \mid C) = \frac{\mathcal{P}(C \mid K)\mathcal{P}(K)}{\mathcal{P}(C)} = \frac{\mathcal{P}(C \mid K)\mathcal{P}(K)}{\mathcal{P}(K)\mathcal{P}(C \mid K) + \mathcal{P}(\neg K)\mathcal{P}(C \mid \neg K)}$$

Let us look at a problem in which the Bayes' rule is applicable.

**Problem:** To answer a multiple choice question, a student may either know the answer or may guess it. Assume that with probability p the student knows the answer to a question, and with probability q, the student guesses the right answer to a question she does not know. What is the probability that for a question the student answers correctly, she actually knew the answer to the question? (Again pause the video here and try solving the problem yourself )

**Solution:** Let us first assume that K is the event that the student knows the question and let C be the event of the student answers the question correctly. Now from the question we can gather the following information.

We have, $P(K) = p, P(\neg K) = 1 - p, P(C \mid K) = 1, P(C \mid \neg K) = q$ .

We are interested in finding $P(K \mid C)$ .

Applying Bayes' rule we have,

$$\frac{P(C \mid K)P(K)}{P(C)} = \frac{P(C \mid K)P(K)}{P(K)P(C \mid K) + P(\neg K)P(C \mid \neg K)} = \frac{p}{p + q(1-p)}$$

The probability of answering the question correctly can be expanded in the denominator to consider the two situations, probability to answer the question correctly given that the student knows the question and probability of answering the question correctly given the student does not know the question. Now using the values which we have gathered from the question we can arrive at the given answer above. Note here that the Bayes' rule is essential to solve this problem because while from the question itself we have a handle on the value $P(C \mid K)$ there is no direct way to arrive at the value of $P(K \mid C)$ .

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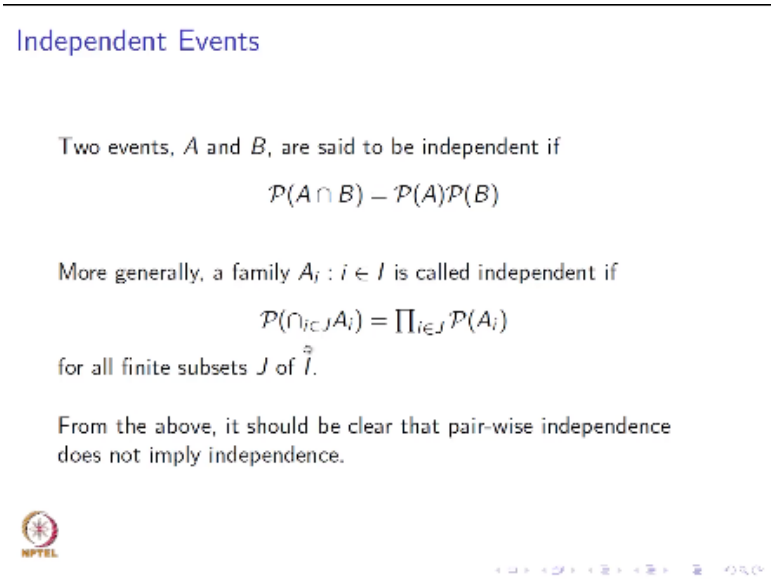


The two events A and B are set to be independent if $P(A \cap B) = P(A)P(B)$ .

More generally a family of events $A_i : i \in I$ , is called independent if $P(\bigcap_{i \in j} A_i) = \prod_{i \in j} P(A_i)$ For all finite subsets J of I.

Essentially what we are trying to say here is that if you have a family of events $A_i$ , then the independence condition holds only if for any subset of those events this condition holds.

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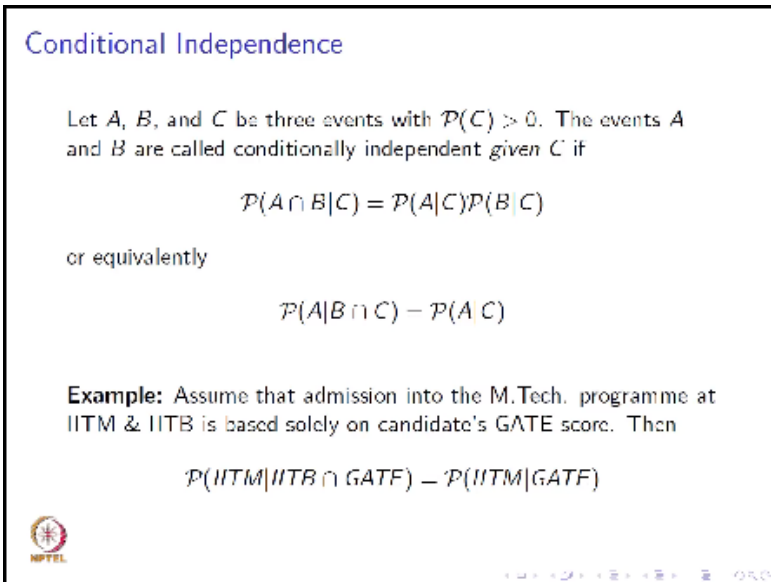


From this should be clear that pairwise independence does not imply independence that is pairwise independence is a weaker condition.

Explaining the notion of independence of events we can also consider conditional independence. Let A ,B and C be three events with $P(C) > 0$. The events A and B are called conditionally independent given C if $P(A \cap B \mid C) = P(A \mid C)P(B \mid C)$ This condition is very similar in form to the previous condition for independence of events.

Equivalently, $P(A \mid B \cap C) = P(A \mid C)$

This latter condition is quite informative. What it says is that the probability of A calculated after knowing the occurrence of event C is same as the probability of A calculated after having knowledge of occurrence of both events B and C. Thus observing the occurrence or non-occurrence of B does not provide any extra information. And thus we can conclude that the events A and B are condition independent given C.

Let us consider an example. Assume that admission into the M.Tech program at IIT Madras in IIT Bombay is based solely on candidate's gate score. Then,

$P(IITM \mid IITB \cap GATE) = P(IITM \mid GATE)$

The probability of admission into IIT Madras given knowledge of the candidate's admission status in IIT Bombay as well as the candidate gate score is equivalent to the probability

calculated simply knowing the candidates gate score. Just knowing the status of the candidate's admission into IIT Bombay does not provide an extra information. Hence since the condition is satisfied, we can say that admission into the program at IIT Madras and admission into the program at IIT Bombay are independent events given knowledge of the candidate's gate score.

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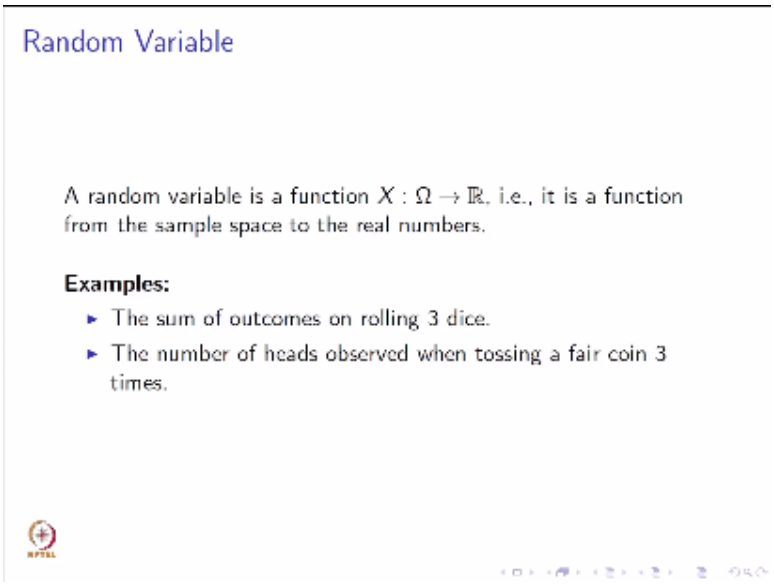


One of the important concepts in probability theory is that of the random variable.

A random variable is a variable whose value is subjected to variations. That is, a random variable can take on a set of possible different values each with an associated probability. Mathematically a random variable is a function from the sample space to the real numbers $X : \Omega \to R$.

Let us consider some examples. Suppose we conduct an experiment in which we roll three dice and are interested in the sum of the outcomes, the sum of fives, say. It can be observed if two of the dice show up two each and the other die shows up as one. Alternatively the sum of five can also be observed if one die shows up as three and the other two dice show up one each. Since we are interested in only the sum and not the individual results of the dice rolls we can define a random variable which maps the elementary outcomes that is the outcomes of each die roll to the sum of the three rolls.

Similarly in the next example we can define a random variable which counts the number of heads observed when tossing a fair coin three times. Note that in this example the random

variable can take values between 0 and 3 whereas in the previous example the range of the random variable is between 3 and 18 corresponding to all dice showing up one and all dice showing up six.

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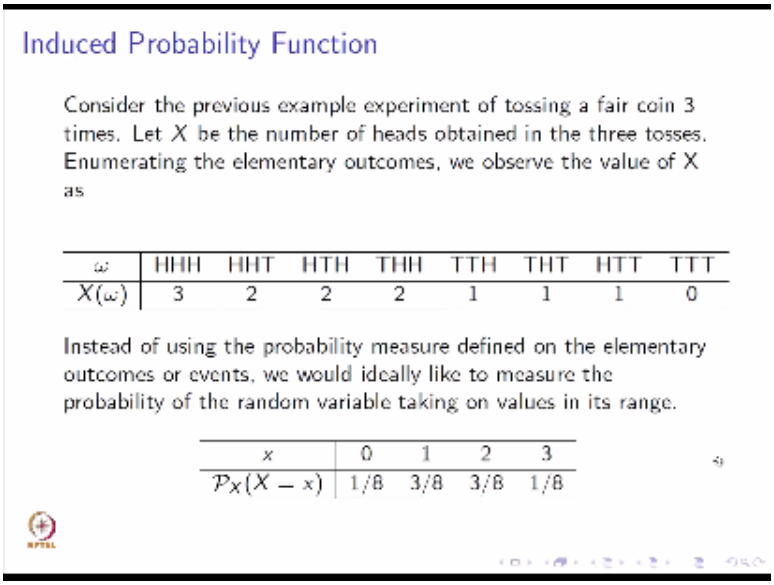


Consider the previous example experiment of tossing a fair coin 3 times. Let X be the number of heads obtained in the three tosses, that is X is a random variable which maps each elementary outcome to a real number representing the number of heads observed in that outcome as it is shown in the first table. The first row lists out each elementary outcome and the second row lists out the corresponding real number value to which that elementary outcome is mapped that is the number of heads observed in that outcome.

Now instead of using the probability measure defined on the elementary outcomes or events we would ideally like to measure the probability of the random variable taking on values in its range. What we are trying to say here is that when we define probability measure we were associating each event that is the subset of the sample space with a probability measure. When we consider random variables the events correspond to different subsets of the sample space which mapped to different values of the random variable.

This is illustrated in the second table, the first row lists out the different values that the random variable X can take and the second row lists out the corresponding probability values assuming that the coin toss is a fair coin. This table describes the notion of the induced probability function which maps each possible value of the random variable to its associated probability value. For example, in the table the probability of the random variable taking on the value of 1 is given as 3/8. Since there are three elementary outcomes in which only one head is observed, each of these elementary outcomes has a probability of 1/8.

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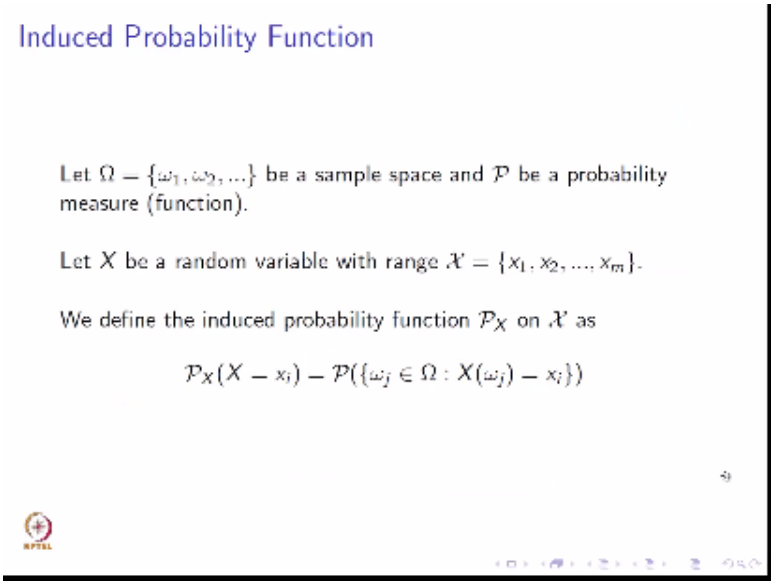


From the previous example we can define the concept of the induced probability function. Let $\Omega$ be a sample space and P be a probability measure. Let X be a random variable which takes values with range X= $\{x_1, x_2, ...x_m\}$ . The induced probability function $P_x$ on X is defined as $P_x(X = x_i) = P(\{\omega_j \in \Omega : X(\omega_j) = x_i\})$ or it equals to the probability of the event comprising of the elementary outcomes $\omega_j$ such that the random variable X map $\omega_j$ to the value $x_i$ .

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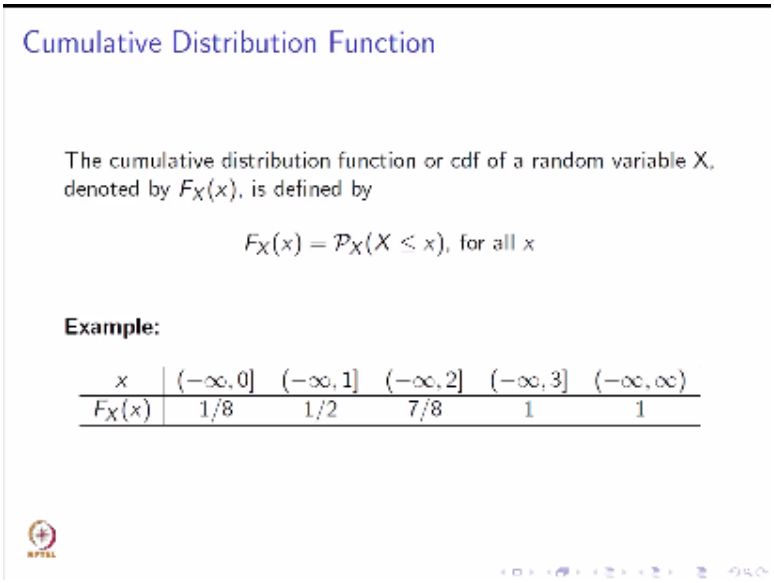


The cumulative distribution function or CDF of a random variable X denoted by $F_x(x)$ is defined as $F_x(x) = P_x(X \leq x) \forall x$

For example, going back to the previous random variable which counts the number of heads observed in three tosses of a fair coin. The following table shows the intervals corresponding to the different values of the random variable X along with the corresponding values of the cumulative distribution function.

For example, $F_x(x) = F_x(1) = 1/2$, because the probability that the random variable X has a value of 1 is 3/8, and the probability that the random variable X has a value of 0 is 1/8. And therefore, the probability that the random variable X takes on a value with less than or equal to 1 is 1/8+3/8=4/8 or 1/2.

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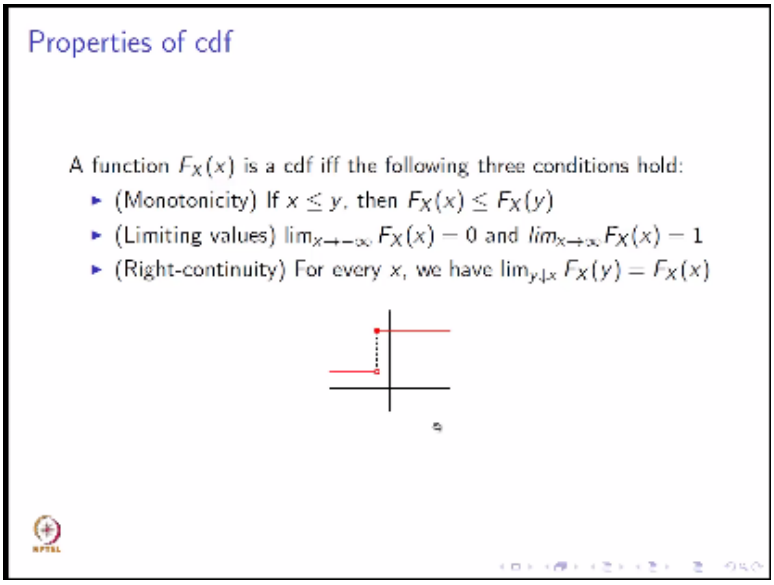


A function is a valid cumulative distribution function only if it satisfies the following properties.

1. The first property simply states that the cumulative distribution function is a non decreasing function.
2. The second property specifies the limiting values, $\lim_{x \to -\infty} F_x(x) = 0$ and $\lim_{x \to \infty} F_x(x) = 1$
3. The third property specifies right continuity that is no jump occurs when the limit point is approached from the right and is also shown in the plot.

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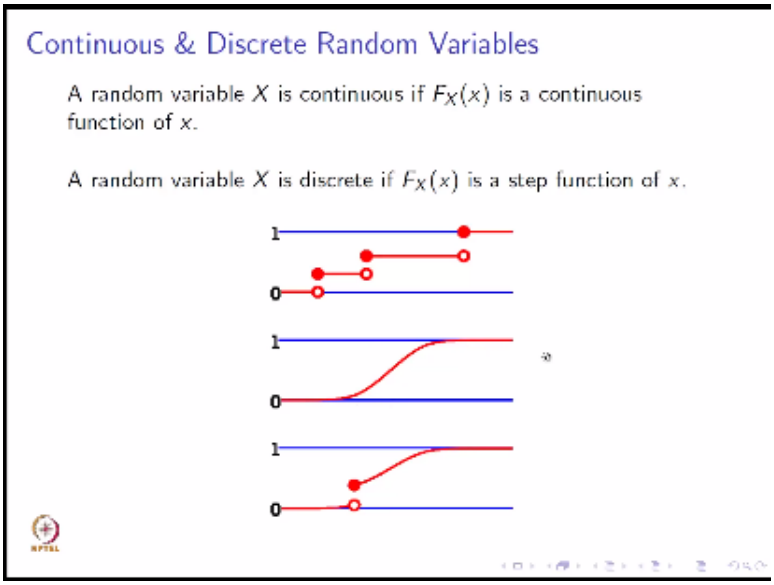

A random variable X is continuous if its corresponding cumulative distribution function is a continuous function of X. This is shown in the second part of the diagram above.

A random variable X is discrete if its CDF is a step function of X this is shown in the first part of the diagram.

The third part of the diagram shows the cumulative distribution function for a random variable which has both continuous and discrete parts.

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## Probability Mass Function

The probability mass function or pmf of a discrete random variable $X$ is given by

$$f_X(x) = \mathcal{P}(X = x), \text{ for all x}$$

**Example:** For a geometric random variable $X$ with parameter $p$,

$$f_X(x) = \begin{cases} (1-p)^{x-1}p & \text{for } x = 1, 2, \ldots \\ 0 & \text{otherwise} \end{cases} \qquad (1)$$

The probability mass function or PMF of a discrete random variable X is given by

$f_x(x) = P(X = x), \forall x$

Thus for a discrete random variable the probability mass function of that variable gives the probability that the random variable is equal to some value.

For example, for a geometric random variable X with parameter p the PMF is given as

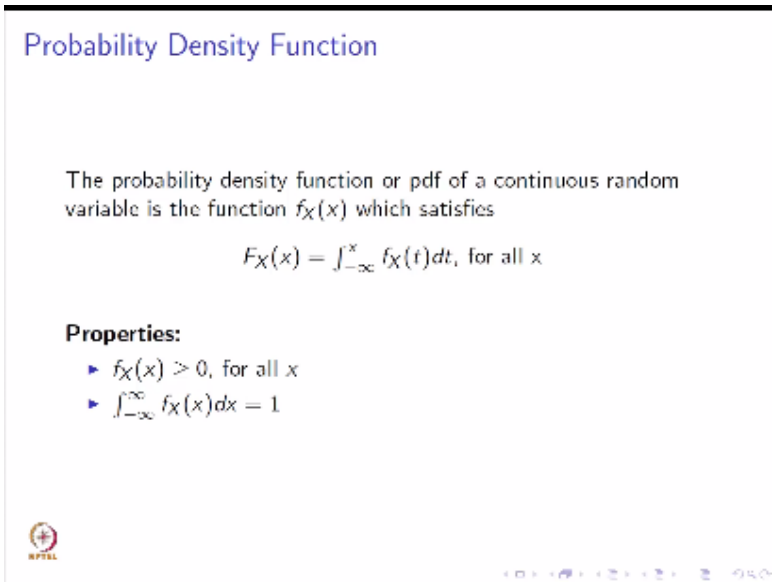
$f_x(x) = (1-p)^{x-1} p$ for x = 1,2,...

And for other values of x the PMF = 0.

A function is a valid probability mass function if it satisfies the following two properties.

1. First of all the function must be non-negative.

2. Secondly $\sum_{x} f_x(x) = 1$




For continuous random variables we consider the probability density function. The probability density function or PDF over a continuous random variable is the function $F_x(x)$ which satisfies the following.

$$F_x(x) = \int_{-\infty}^{x} f_x(t)dt, \forall x$$

Similar to the PMF, the probability density function should also satisfy the following properties.

1. First of all the probability density function should be non-negative for all values of x or $f_x(x) \geq 0, \forall x$.
2. $\int_{-\infty}^{\infty} f_x(x)dx = 1$

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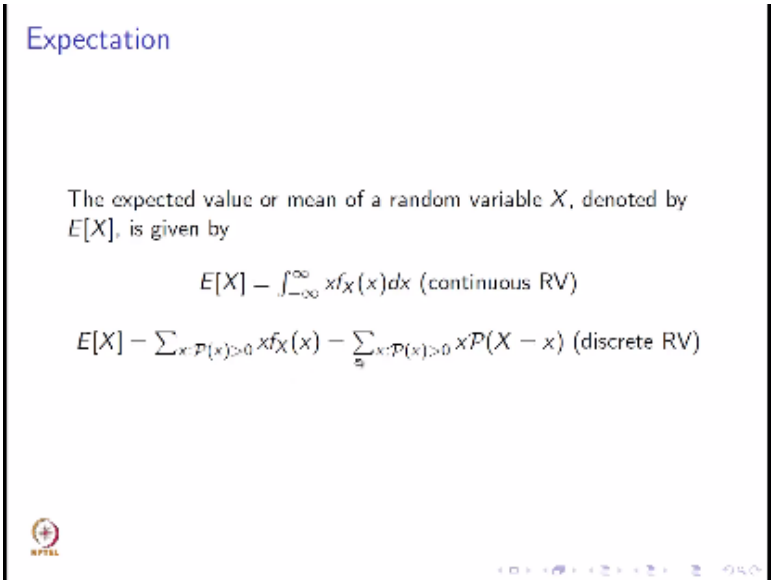


Let us now look at expectations of random variables. The expected value or mean of a random variable X denoted by E[X] is given by integral $E[X] = \int_{-\infty}^{\infty} x f_x(x) dx$ (continuous RV).

Note that $f_x(x)$ here is the probability density function associated with random variable X. This definition holds when X is a continuous random variable. In case that X is a discrete random variable we use the following definition. $E[X] = \sum_{x:P(x)>0} x f_x(x) = \sum_{x:P(x)>0} xP(X = x)$ (discrete RV).

Here $f_x(x)$ is the probability mass function of the random variable X which essentially gives the associated probability for a particular value of the random variable thus leading to this definition.

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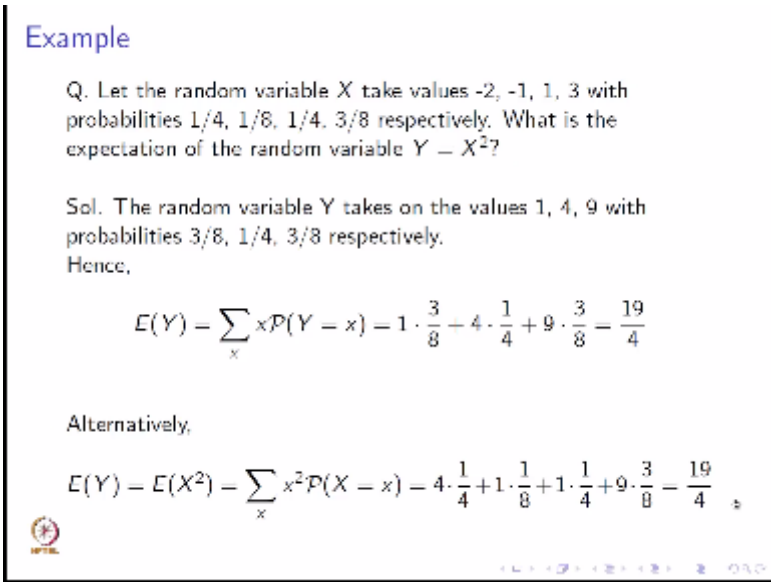


Let us now look at an example in which we calculate expectations.

**Problem:** Let the random variable X take values -2, -1, 1 and 3 with probabilities 1/4, 1/8, 1/4, 3/8 respectively. What is the expectation of the random variable $Y = X^2$ ?

**Solution:** So what we can do is we can calculate the values that the random variable Y takes along with associated probabilities, since we are aware of the relation between Y and X. Thus we have Y taking on the values 1, 4 and 9 with probabilities 3/8, 1/4, and 3/8 respectively. Given this information we can simply apply the formula for expectation and calculate the expectation on the random variable Y giving a result of 19 /4.

Another way to approach this problem is to directly use the relation $Y= X^2$ in calculating expectation.

$$E(Y) = E(X^2) = \sum_x x^2 P(X = x) = 4.\tfrac{1}{4} + 1.\tfrac{1}{8} + 1.\tfrac{1}{4} + 9.\tfrac{3}{8} = \tfrac{19}{4}$$

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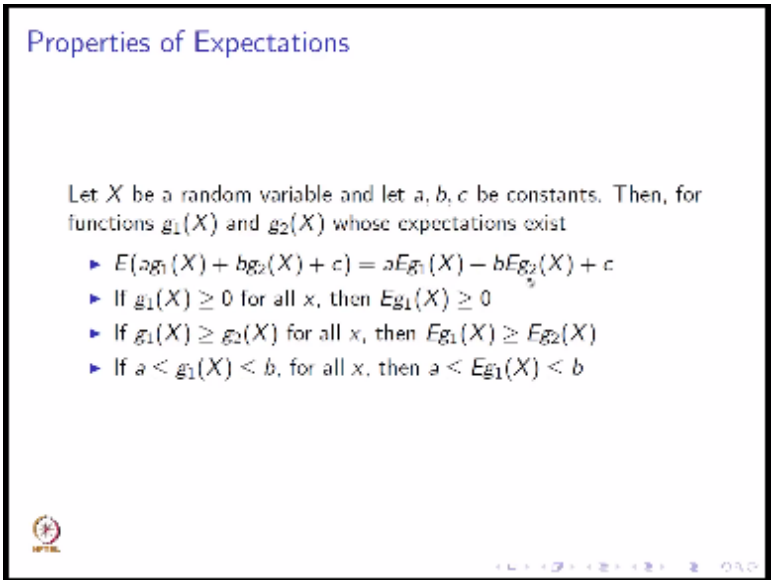


Let us now look at the properties of expectations.

Let X be a random variable and let a, b, c be constants. Then for functions $g_1(X)$ and $g_2(X)$ are functions of random variable X, such that their expectations exist that is they have finite expectations.

1. $E(ag_1(X)+bg_2(X)+c) = aEg_1(X)+bEg_2(X)+c$ .This is called the linearity of expectations. There are actually a few things to note here first of all expectation of a constant is equal to the constant itself expectation of a constant times the random variable is equal to the constant into the expectation of the random variable and the expectation of the sum of two random variables can also be represented as the sum of the expectations of the two random variables. Note that here the two random variables need not be statistically independent.
2. According to the next property if a random variable is $\geq 0$ at all points then the expectation of that random variable is also $\geq 0$
3. Similarly if one random variable is > another random variable at all points then the expectation of those random variables also follow the same constraint.
4. Finally if a random variable has values which lie between two constants then the expectation of that random variable will also lie between those two constants.

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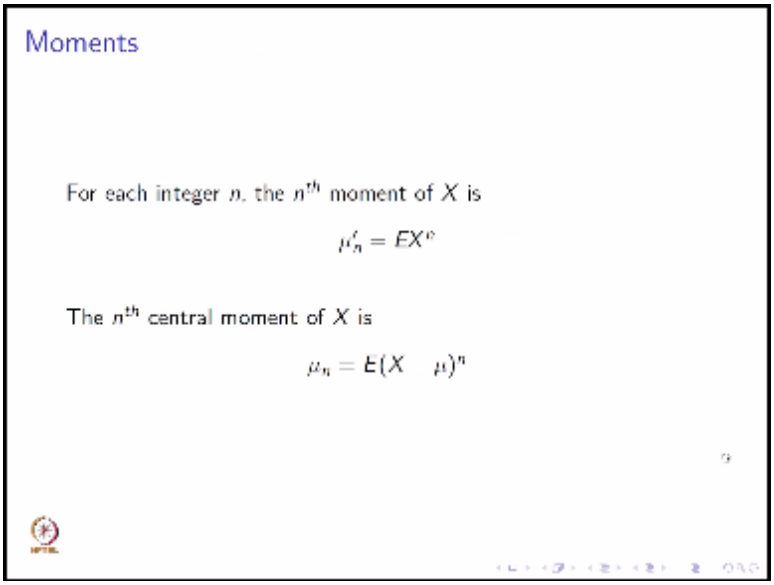


Let us now define moments. For each integer n the $n^{th}$ moment of X is $\mu_n^{'} = EX^n$ . Also the $n^{th}$ central moment of X is $\mu_n = E(X-\mu)^n$ . So the difference between moment and central moment is that in central moment we subtract the random variable by the mean of the random variable or expected value. The two moments that find most common use are the first moment which is nothing but $\mu_1$ = E[X] that is the mean of the random variable X and the second central moment which is $\mu_2 = E(X-\mu)^2$ which is the variance of the random variable X.

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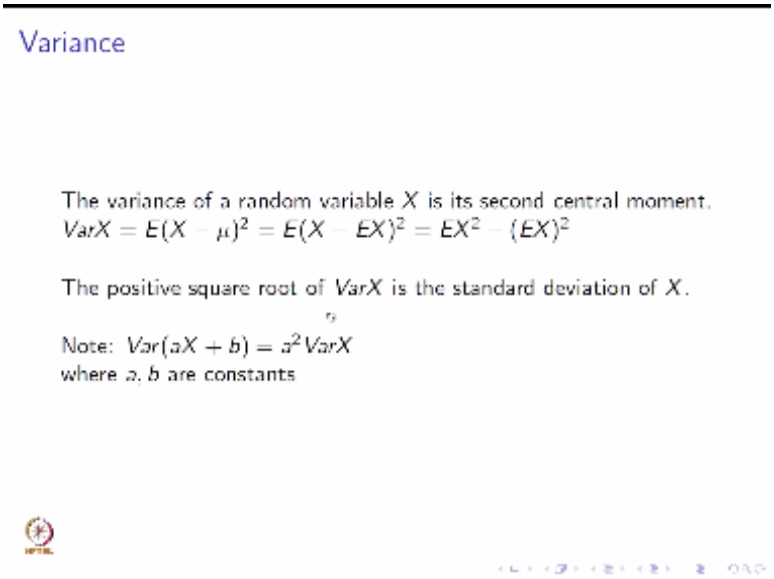


Thus the variance of a random variable x is a second central moment. Variance of X = $E(X-\mu)^2$. Note that $\mu$ is just the first moment which can be replaced by E[X]. Thus we have $VarX = E(X-\mu)^2 = E(X-EX)^2 = EX^2 - (EX)^2$. Another very useful relation to remember is $Var(aX+b) = a^2 VarX$ where a and b are constants.

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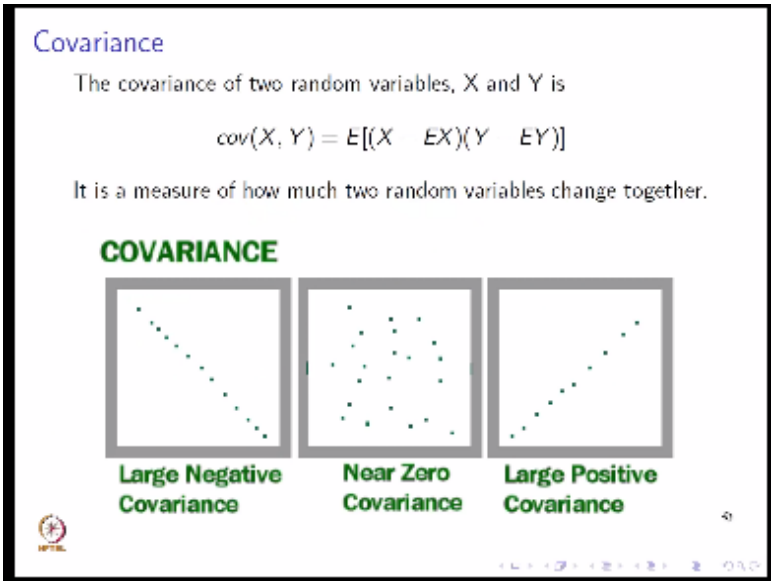

The covariance of two random variables X and Y is $\text{cov}(X,Y) = E[(X - EX)(Y - EY)]$. Remember that the variance of our random variable X is nothing but the second central moment thus the variance of a random variable measures the amount of separation in the values of the random variable when compared to the mean of the random variable. For covariance the calculation is done on a pair of random variables and it measures how much two random variables change together. Consider the diagram above. In the first part assume that the random variable X is on the x-axis and the random variable Y is on the y-axis. We note that as the value of X increases the value of Y seems to be decreasing thus for this relationship we will observe a large negative covariance. Similarly in the third part of the diagram we can see that as the variable value of variable X increases, so does the value of the variable y. Thus we see a large positive covariance. However in the middle diagram we cannot make any such statement because as X increases there is no clear relationship as to how Y changes thus this kind of a relationship will give zero covariance.

Now from the diagram it should immediately be clear that covariance is a very important term in machine learning because we are often interested in predicting the value of one variable by looking at the value of another variable we will come to that in further classes.

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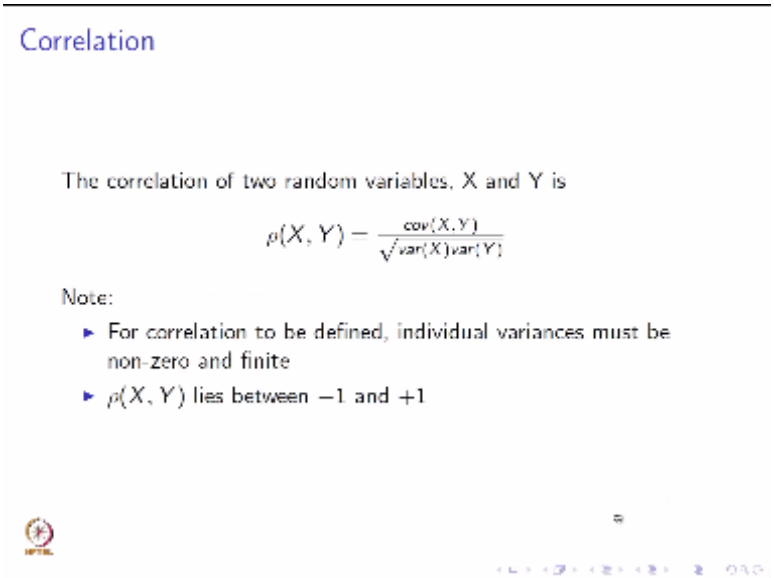

Closely related to the concept of covariance is the concept of correlation. The correlation of two random variables X and Y is nothing but the covariance of the two random variables X and Y divided by the square root of the of the product of their individual variances.

$$\rho(X,Y) = \frac{\text{cov}(X,Y)}{\sqrt{\text{var}(X)\,\text{var}(Y)}}$$

Basically correlation is a normalized version of covariance. So the correlation will always be between - 1 and 1 also since we used the variance of the individual random variables in the denominator for correlation to be defined, individual variances must be nonzero and finite.

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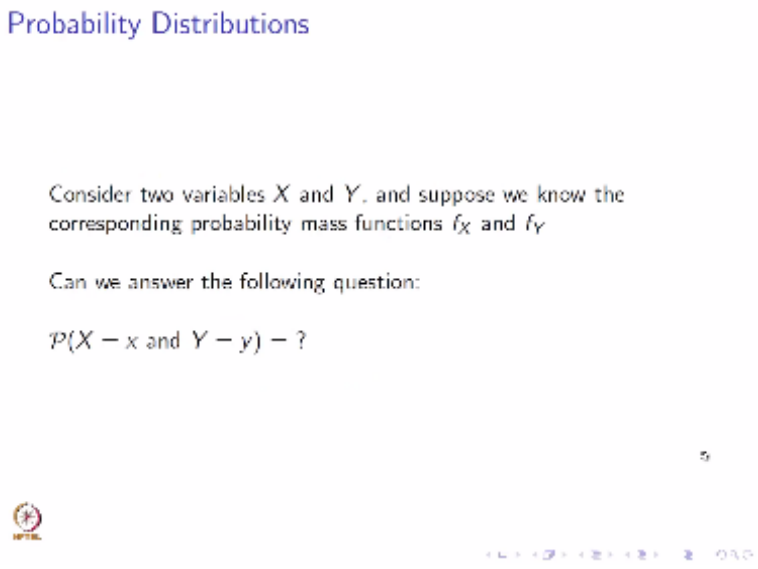


In the final part of this tutorial on probability theory we will talk about probability distributions and list out some of the more common distribution that you are going to encounter in the course. Before we proceed let us consider this question. Consider two variables X and Y and suppose we know the corresponding probability mass function $f_x$ and $f_y$. Corresponding to the variables X and Y, can we answer the following question: P(X=x and Y=y)=?

What is the probability that X takes a certain value small x and Y takes a certain value small y? Think about this question. If you answered no then you're correct let us see why.

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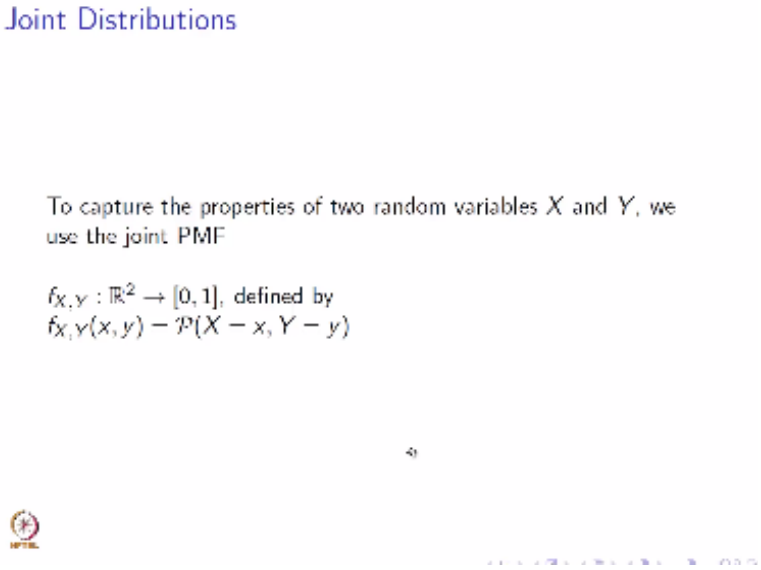


Essentially what we were looking for in the previous question was the Joint Distribution which captures the properties of both the random variables the individual PMFs or PDFs. In case the random variables are continuous they capture the properties of the individual random variables only but miss out on how the two variables are related thus we define the joint PMF or PDF as $f_{X,Y}$.

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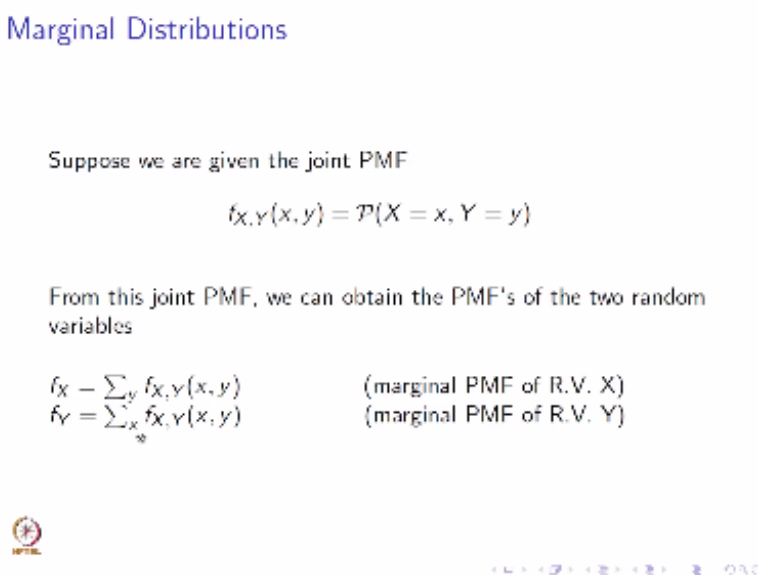


Suppose we are given the joint probability mass function of the random variables X and Y. What if we are interested in only the individual mass functions of either of the random variables. This can be obtained from the joint probability mass function by a process called marginalization. The individual probability mass function thus obtained is also referred to as a marginal probability mass function. Thus if you are interested in the marginal probability mass function of random variable X we can obtain this by summing the joint probability mass function overall values of Y. Similarly the marginal probability mass function of random variable Y can be obtained by summing the joint probability mass function over all values of X. Note that in case the random variables considered here are continuous we substitute summation by integration and PMF by PDF.

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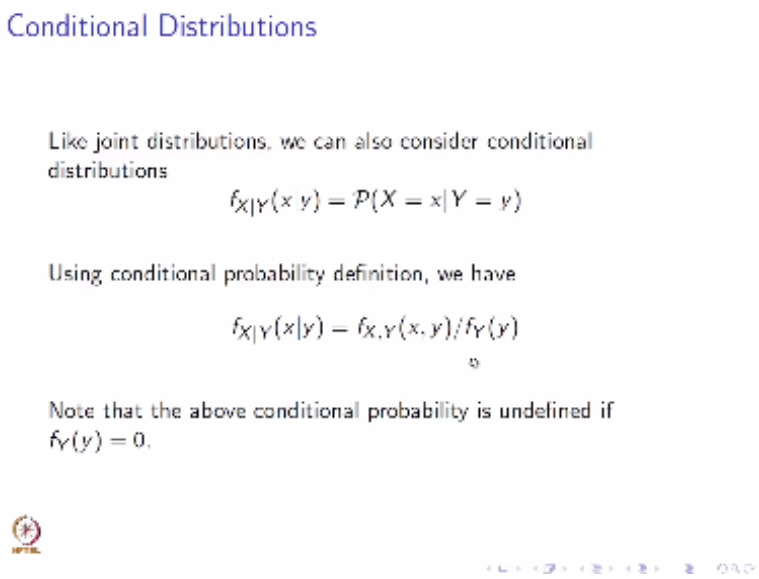


Like joint distributions we can also consider conditional distributions. For example here we have the conditional distribution $f_{X|Y}(x, y)$ which is the probability that the random variable X will take on some value small x given that the random variable Y has been observed to take on a specific value small y. The relation between conditional distributions, joint Distribution and marginal distributions are shown here. This relation should be familiar from the definition of conditional probability that was seen earlier. Note that the marginal distribution $f_y(y)$ is in the denominator and hence it must not be equal to 0.

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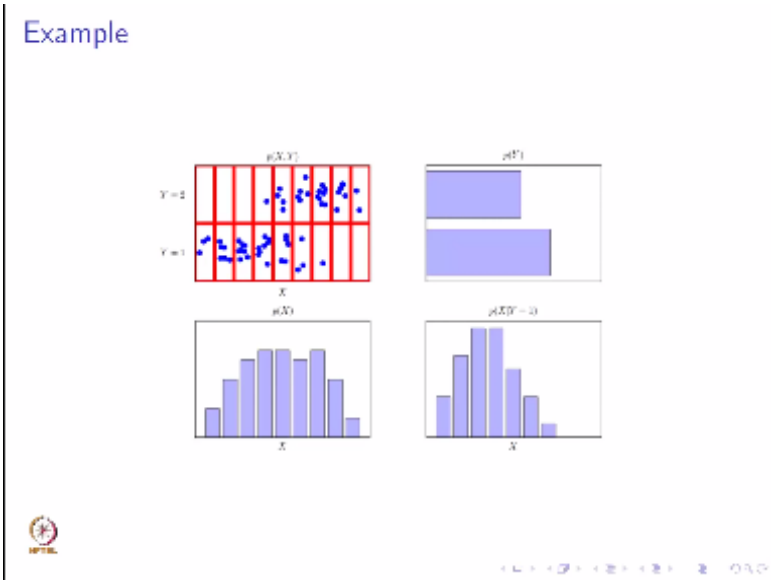


The overall idea of joint marginal and conditional distributions is summarized in this figure, The top left figure shows the joint Distribution and describes how the random variable X which takes on 9 different values is related to the random variable Y which takes on two different values. The bottom left figure shows the marginal distribution of random variable X. As can be observed in this figure we ignore the information related to the random variable Y. Similarly the top-right figure shows the marginal distribution of random variable Y. Finally the bottom-right figure shows the conditional distribution of X given that the random variable Y takes on a value of 1. Looking at this figure and comparing it with a joint distribution we observe that in the bottom-right figure is simply ignore all the values of X for which y equals to 2 that is the top half of the joint distribution.

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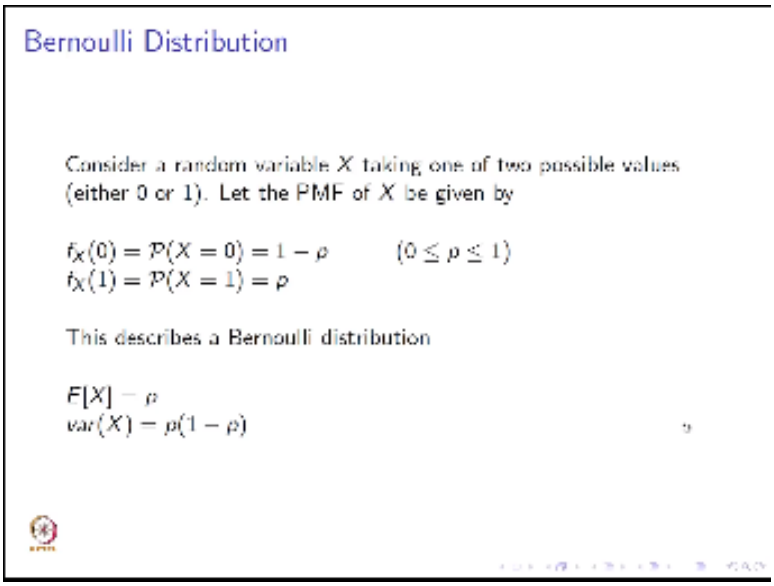


In the next few slides we will present some specific distributions that you will be encountering in the machine learning course. We will present the definition and list out some important properties for each distribution. It would be a good exercise for you to work out the expressions for the PMFs or PDFs and the expectation and variances of these distributions on your own. We start with the Bernoulli distribution. Consider a random variable X taking one of two possible values either 0 or 1.

$$f_x(0) = P(X = 0) = 1 - p \quad (0 \leq p \leq 1)$$

$$f_x(1) = P(X = 1) = p$$

Here p is the parameter associated with the Bernoulli distribution. It generally refers to the probability of success so in our definition we are assuming that X = 1 indicates a successful trial and X equals to 0 indicates of failure. The expectation of a random variable following the Bernoulli distribution is p and the variance is $p(1-p)$. The Bernoulli distribution is very useful to characterize experiments which have a binary outcome such as in tossing a coin we observe either heads or tails or say in writing an exam either pass or fail.

Such experiments can be modeled using the Bernoulli distribution next we look at the binomial distribution.

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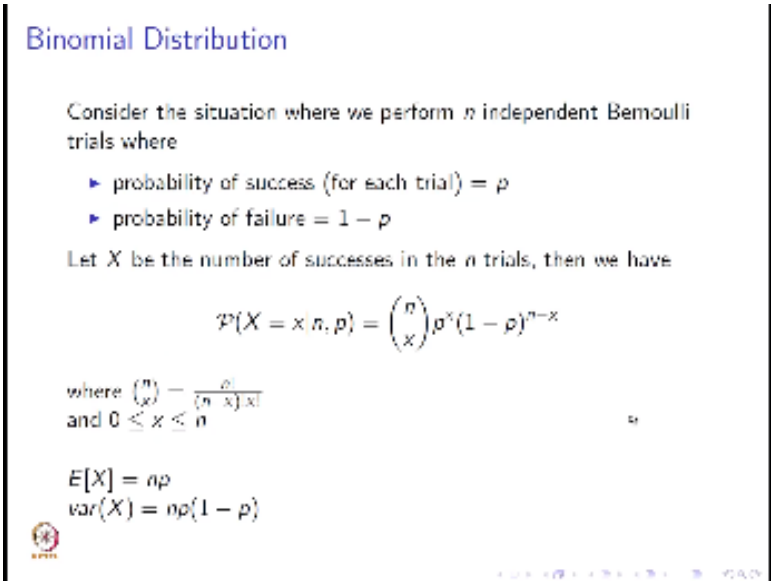


Consider the situation where we perform n independent Bernoulli trials where the probability of success for each trial equals to p and the probability of failure for each trial equals to 1 – p.

Let X be the random variable which represents the number of successes in the end trials then we have probability that the random variable X will take on a specific value of small x given the

$$P(X = x \mid n, p) = \binom{n}{x} p^x (1-p)^{n-x}$$

parameters n and p is

Note that here X is going to be a number between 0 and n. The expectation of a random variable following the binomial distribution equals to np and the variance equals to $np(1-p)$. The binomial distribution is useful in any scenario where we are conducting multiple Bernoulli trials that is experiments in which the outcome is binary. For example suppose we have a coin and we toss the coin 10 times. We want to know the probability of observing three heads given the probability of observing a head in an individual trial. We can apply the binomial distribution to find out the required probability.

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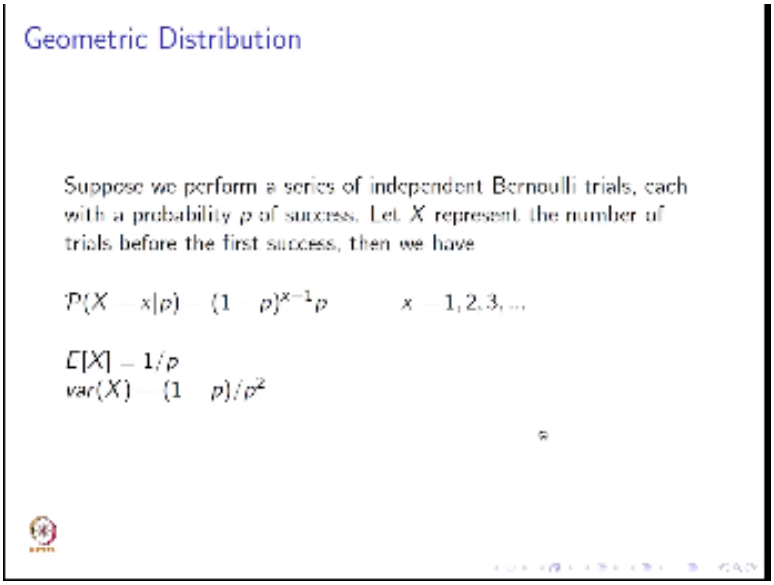


Suppose we perform a series of independent Bernoulli trials each with the probability p of success. Let X represent the number of trials before the first success. Then we have probability that the random variable X will take a value small x given the parameter p is $P(X = x \mid p) = (1-p)^{x-1} p$. Essentially we are trying to calculate the probability that it takes us small X number of trials before observing the first success. This can happen if the first $x-1$ trials failed that is with probability $1-p$ and the last succeeded with probability p.

A random variable which has this probability mass function follows the geometric distribution. For the geometric distribution the expectation of the random variable equals to 1 / p and the variance equals to $\frac{1-p}{p^2}$.

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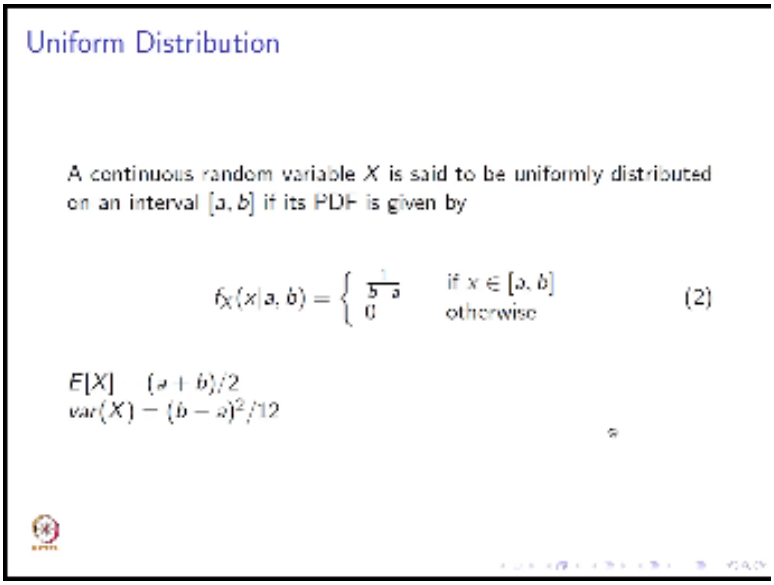


Many situations we initially do not know the probability distribution of the random variable under consideration but can perform experiments which will gradually reveal the nature of the distribution. In such a scenario we can use the uniform distribution to assign uniform probabilities to all values of the random variable which are then later updated. In the discrete case, say the random variable can take n different values then we simply assign a probability of $1/n$ to each of the n values. In the continuous case if the random variable X takes values in the closed interval (a,b) then it is PDF is given by

$$f_x(x \mid a,b) = \frac{1}{b-a} \quad if x \in [a,b]$$

For a random variable following the uniform distribution the expectation of the random variable X is $(a+b)/2$ and the variance equal to $(b-a)^2/12$.

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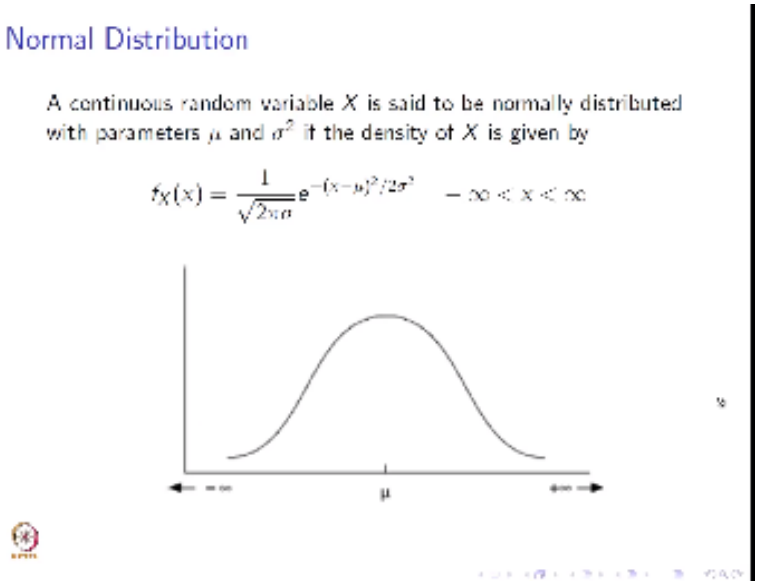


A continuous random variable X is said to be normally distributed with parameters $\mu$ and $\sigma^2$, if the PDF of the random variable X is given by the expression above. The normal distribution is also known as the Gaussian distribution and is one of the most important distributions that we will be using. The diagram represents the famous bell-shaped curve associated with the normal distribution.

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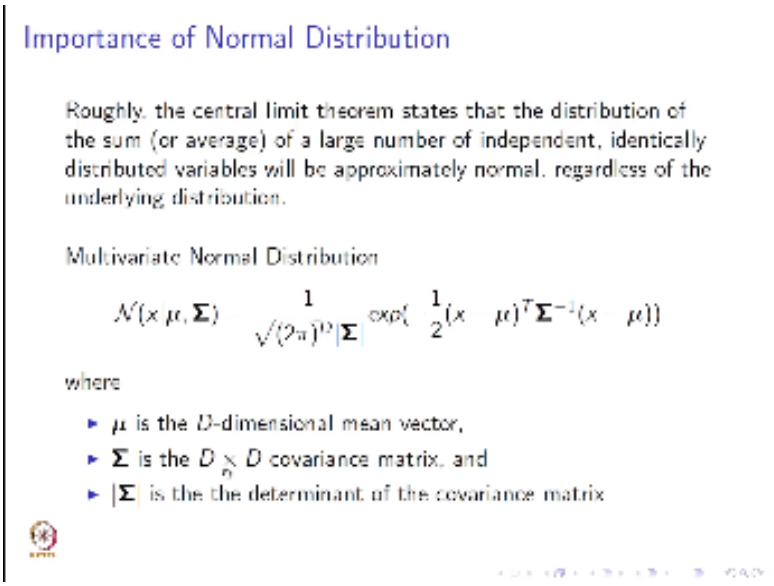

The importance of the normal distribution is due to the central limit theorem. Without going into the details the central limit theorem roughly states that the distribution of the sum of a large number of independent identically distributed variables will be approximately normal regardless of the underlying distribution. Due to this theorem many physical quantities that are the sum of many independent processes often have distributions that can be modeled using the normal distribution.

Also in the machine learning course we will be often using the normal distribution in its multivariate form here we have presented the expression of the multivariate normal distribution where μ is the D dimensional mean vector and Σ is the $D \times D$ covariance matrix.

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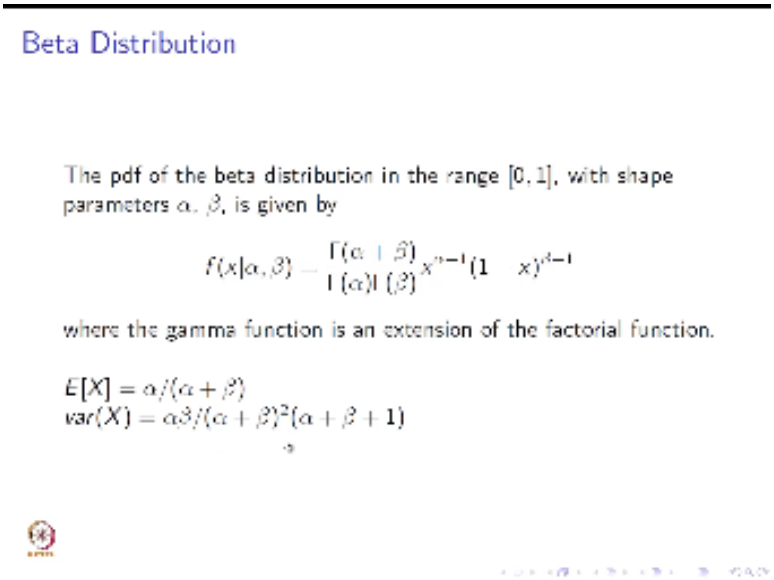


The PDF of the β distribution in the range 0 to 1 with shape parameters α and β is given by the following expression, where the λ function is an extension of the factorial function. The expectation of a random variable following the β distribution is given by α / (α + β) and the variance is given by $\alpha\beta / (\alpha+\beta)^2(\alpha+\beta+1)$.

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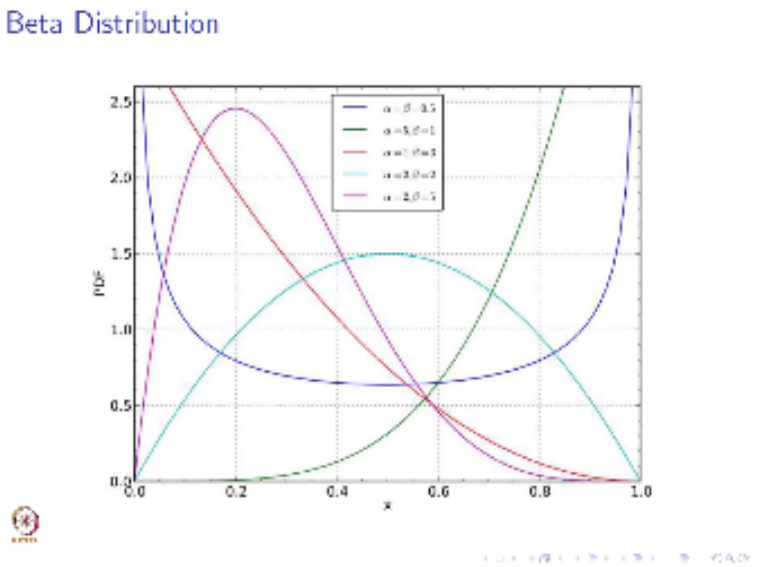


This diagram illustrates the β distribution similar to the normal distribution in which the shape and position of the bell curve is controlled by the parameters μ and $\sigma^2$ . In the β distribution the shape of the distribution is controlled by the parameters α and β and in the diagram we can see a few instances of the β distribution for different values of the shape parameters. Note that unlike the normal distribution a random variable following the beta distribution takes values only in a fixed interval.

Thus in this example the probability that the random variable takes a value less than 0 or greater than 1 is 0.

This ends the first tutorial on the basics of probability theory. If you have any doubts or seek clarifications regarding the material covered in this tutorial please make use of the forum to ask questions. As mentioned in the beginning if you are not comfortable with any of the concepts presented here do go back and read up on it there will be some questions from probability theory in the first assignment. So hopefully going through this tutorial will help you in answering those questions and note that there we will be having another tutorial next week on linear algebra.

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# Linear Algebra Tutorial
# CS5011 – Machine Learning

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**Department of Computer Science**
**IIT Madras**

**January 17, 2016**

Hi everyone welcome to the second tutorial of the introduction to machine learning course. So in this tutorial we shall be taking a tour of the aspects of linear algebra which you would need for the course. We will cover a variety of concepts such as subspaces, basis, span, decomposition, eigenvalues and eigenvectors over the course of the tutorial.

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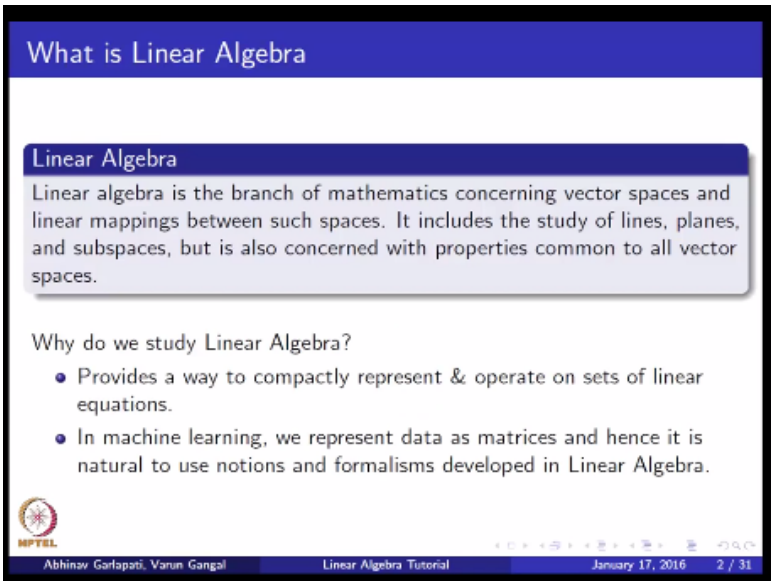

So the first question one would ask is why we need linear algebra at all and what is linear algebra? So you may have come across this in school or your +1 or +2 level but just to recap linear algebra is the branch of mathematics which deals with vectors and vector spaces and linear mappings between these spaces. So, why do we study linear algebra here? Especially in the context of machine learning, firstly it gives us a way to manipulate, represent and operate sets of linear equations. And why do these linear equations pop up in machine learning in the first place? So the reason for that is in machine learning we represent our data as a $n \times p$ matrix where n is the number of data points and p is the number of features. So it is natural we have to use notions and formalisms developed in linear algebra. The data or the parameter is you use are represented as vectors so as a result linear algebra has an important role to play in machine learning.

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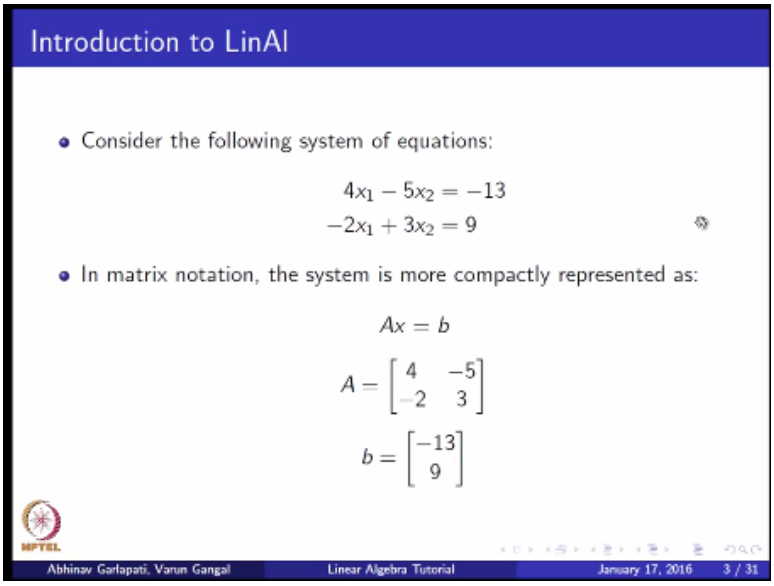


So you can see here a system of linear equations with two equations and two variables

$4x_1 - 5x_2 = -13$

$-2x_1 + 3x_2 = 9$

So we can right away see here the advantages of a matrix notation. If you see below you can represent the same system of two equations directly as one equation in the form of $Ax = b$ where A is the set of coefficients and x is the $2 \times 1$ matrix or you may also call it a 2-dimensional vector $(x_1, x_2)$.

So we can see that when you multiply the matrix A with $(x_1, x_2)$ the 2 x 1 matrix you will get back the same L.H.S or the set of two L.H.S and $b$. The matrix $b$ represents the RHS so it is very easy to verify that if you represent this in matrices you can get back the original representation. So to solving this in general, without matrices would require first solving for one variable and then substitute to get the other. But using matrices you can even solve this directly, so just multiply both the sides by a inverse.

So you would get $x = A^{-1}b$. Of course you would have to care about the fact that all matrices do not have an inverse. But in most cases they do, so in that case you can directly get the solution of x in the form of $x = A^{-1}b$. As said earlier, linear algebra gives us this freedom to manipulate several equations at once and multiple variables.

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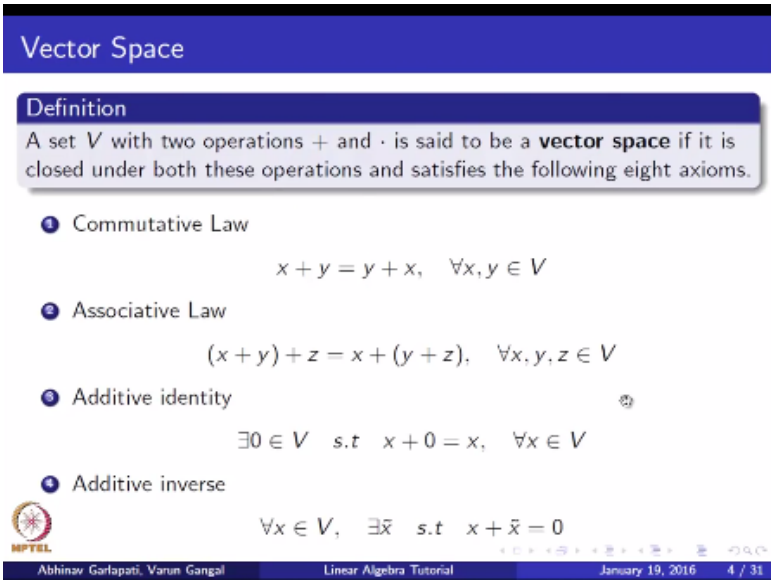


A fundamental definition in linear algebra is that of a vector space. A set of vectors $V$ is said to be a vector space if it is closed under the operations of vector addition and scalar multiplication and in addition satisfies the axioms we have listed here. Like if we take two elements from this set $X \& Y$ then $X+Y$ will also lie in the set $V$. In addition to this, if we take a scalar $\alpha$ (a real number) and multiply a vector from this set $y$ with it then $\alpha y$ also belongs to $V$. If both these properties are satisfied then the set of vectors is said to be closed with respect to vector addition and scalar multiplication. Now let us have a look at the axioms.

1. The first one is a commutative law, the commutative law states that if you pick any two elements from the set $V$, $x \& y$ then $x+y=y+x$.
2. The associative law says that if you pick any point from this set $x, y, z$, $(x+y)+z=x+(y+z)$
3. The additive identity loss states that there exists an additive identity or a 0 such that if you pick any element from the set and add this 0 to it you get back the same element.
4. The additive inverse law states that there exists for every element $x$ a corresponding $\bar{x}$ such that $x+\bar{x}=0$.

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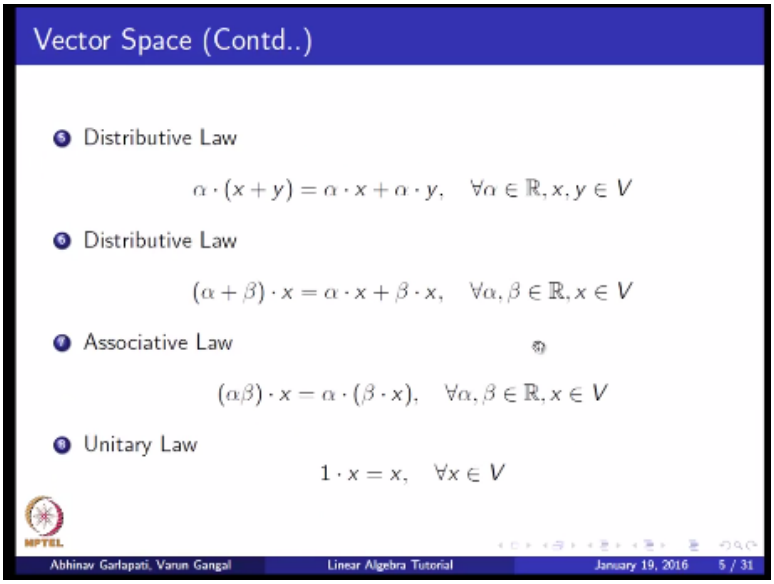


5. The fifth law is the distributive law. This law says that, if you have a real scalar $\alpha$ with which you multiply the sum of two vectors $x+y$, then that should be equal to $\alpha x+\alpha y$. The second distributive law says that if you have two scalars $\alpha, \beta$ and vector $x$, then $(\alpha+\beta)x = \alpha x+\beta x$
6. The associative law says that if you will first multiply two scalars $\alpha, \beta$ and then multiplying the vector $x$ with them that should be equal to multiplying the vector $x$ first with the second scalar $\beta$ and then with $\alpha$. Or, $(\alpha\beta).x = \alpha.(\beta.x)$
7. The unitary law says that on multiplication by the scalar real number 1, you get back the same vector. This is important because you would not want multiplication to force any unexpected scaling like if you multiplied a vector $x$ by the scalar $k$ then you need to be sure that it will be exactly $k$ times the initial value and should not be say $\sqrt{k}$.

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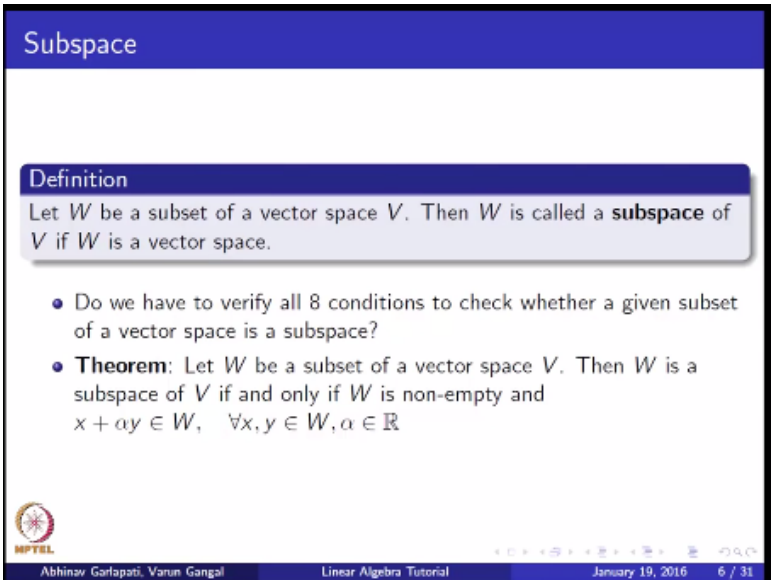


A second related definition is that of a subspace. A subset W of a vector space V is said to be a subspace, if W is a vector space. Now this means that W should be closed under vector addition and scalar multiplication. It should also satisfy the eight axioms we stated earlier. Now the question that arises is do we need to verify all these eight conditions given that we know that W is already a subset of a vector space. No, it is enough to check just for the following two conditions firstly that W is non empty that in other words it has at least a single element. And secondly that if I pick any two elements $x \& y$ from this set and any real number $\alpha$ then $\alpha x + \alpha y \in W$.

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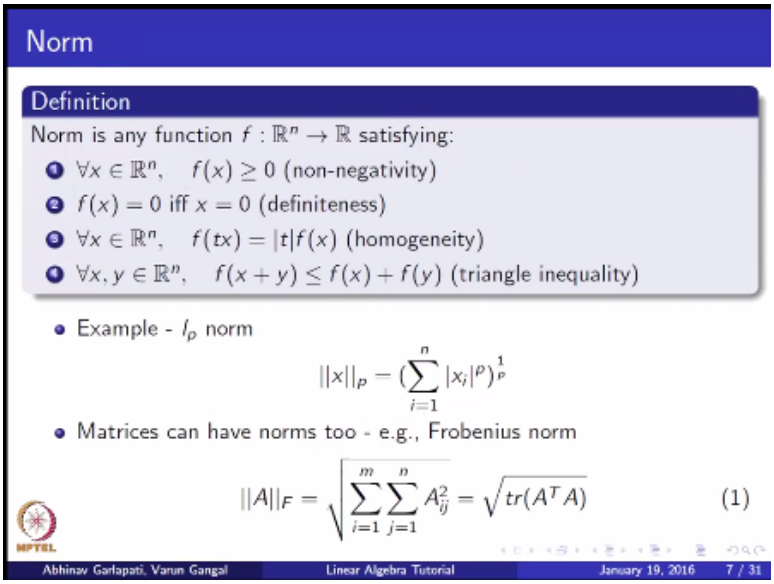


Now let us have a look at the definition of a norm. So intuitively a norm is a measure of the length of a vector or its magnitude. It is a function from a vector space which mostly happens to be $\mathbf{R}^n$ where n is a dimension of a vector to the space of real numbers $\mathbf{R}$. So for a function to be a norm it should satisfy the four conditions which we have given here. Firstly it should be always be non-negative. Secondly, it should be zero if and only if the vector is zero. Thirdly for every vector if you multiplied by a scalar its norm should get multiplied by the modulus of the scalar. By modulus here we mean the absolute value. The fourth being that if we take any pair of vectors in our vector space which is $\mathbf{R}^n$, the norm of the sum of these two vectors should be less than the sum of their norms, which this is also known as the triangle inequality. So this is namely related to the fact that the third side of a triangle should always be less than sum of the other two sides.

Now an example of a norm is the $l_p$ norm there you sum up the absolute values along each dimension raised to p and then take the $1/p$ th root of this. So when p = 2 you get the L2 norm which is the magnitude of a vector as we have learnt in our earlier studies $\sqrt{x^2 + y^2}$ if you are looking at just the space $\mathbf{R}^2$. There are other norms for instance there are norms defined for

matrices as well. Here we had defined all norm for vectors, so the Frobenius norm is a matrix norm. So what it does is it essentially sums up the squares of all the elements and then takes the root of that. So this also happens to be equal to the trace of $A^T A$. Now the trace of a matrix is simply the some of its diagonal elements.

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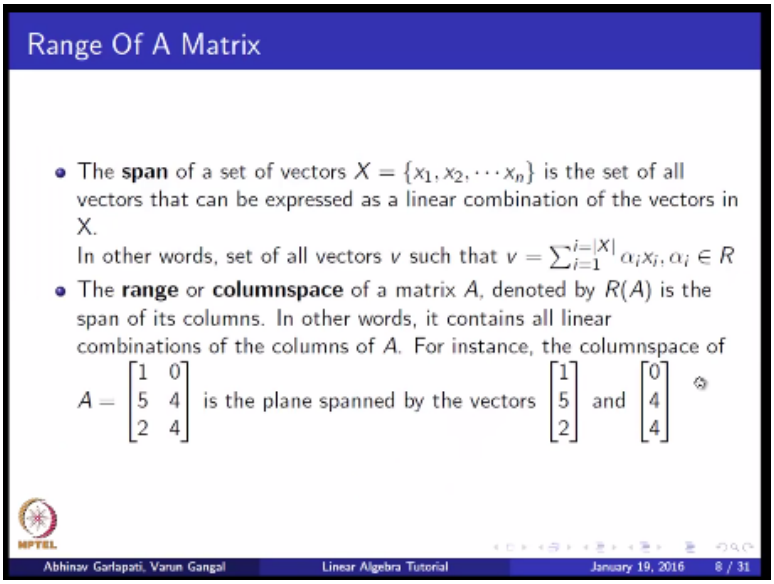


The span of a set of vectors is the set of all vectors which can be composed using these vectors by using the operations of vector addition and scalar multiplication. So the name span comes from the fact that this set of vectors spans a potentially larger set of vectors which is then called a span. So to define this more formally it is the set of all vectors V such that $v = \sum_{i=1}^{|x|} \alpha_i x_i$, $\alpha_i \in \mathbf{R}$. Now a related definition is that of range or column space. So if we think of a matrix each of its columns is a vector so the set of all columns of a matrix is like a set of vectors. Now the span of this set of vectors is called the range or column space of that matrix. So if you consider the matrix given here the columns of this matrix A are (1, 5, 2) and (0, 4, 4), so what would be this column space of this matrix, it would be this the span of the vectors (1, 5, 2) and (0, 4, 4) which is essentially the plane which is spanned by these two vectors.

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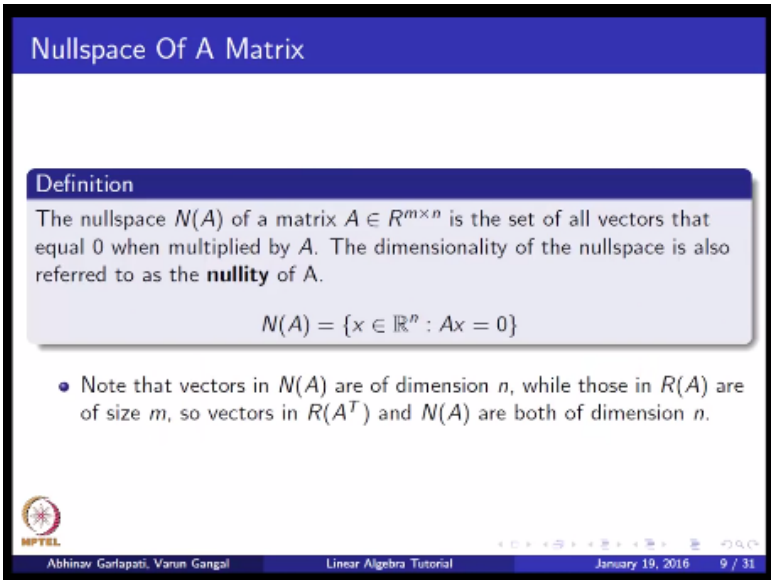


If we have matrix A of dimensions $m \times n$ then the null space is the set of $n \times 1$ vectors which gives $m \times 1$ zero vector on being multiplied by A. In other $N(A) = \{x \in \mathbf{R}^n : Ax = 0\}$. Nullity is the rank or dimensionality of the null space. We will revisit the definition of nullity later once you defined rank more clearly. Another interesting fact about null spaces is that the null space of A is of dimension n while the range of A or the column space as we defined it earlier is of dimension m or m x 1. This means that vectors in $R(A^T)$ so note that $A^T$ is $n \times m$. So vector in the range of $A^T$ will be of dimension $n$ similar to the vectors in the null space of A. So this means that the vectors in range of $A^T$ and null space of $A$ would both be of the dimension $n$ *x 1*.

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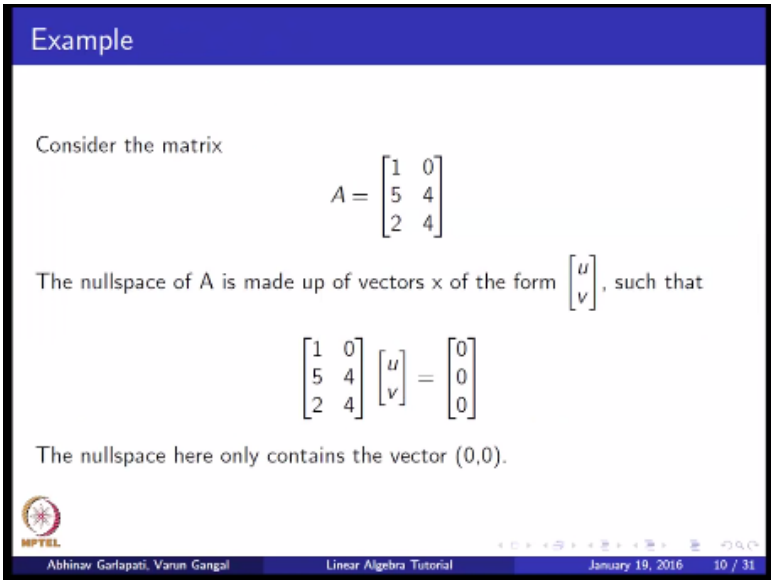


Let us consider an example to illustrate the concept of a null space. Consider the matrix A given here, A is a *3 x 2* matrix hence the null space of A will be made up of vectors of dimension 2 or 2 *x 1*. Now we see that the on solving being we get $u = 0, v = 0$. This means that the null space only contains the 0 vector the two dimensional 0 vector (0, 0).

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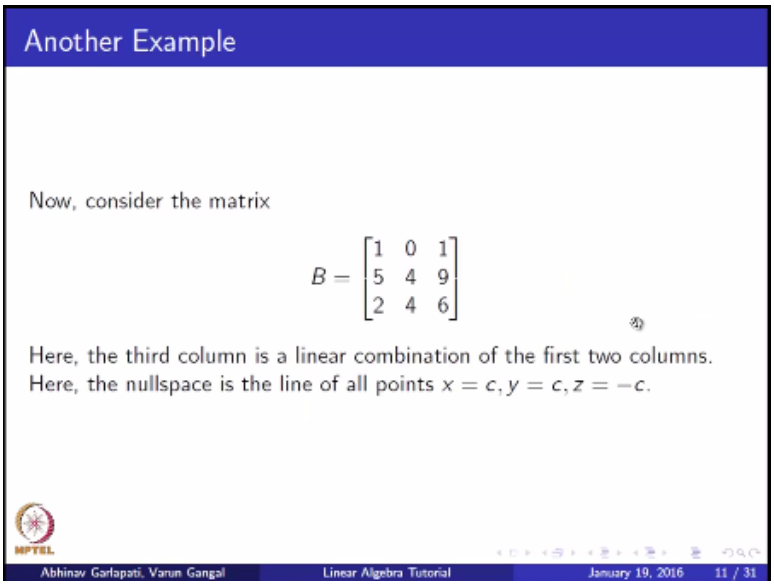


Let us consider another example to illustrate null spaces better. Take the matrix B which is a 3 x 3 matrix. The null space would consist of 3 x 1 vectors. We leave the finding of the null space to the audiences as an exercise. However when on solving we get the null space to be the set of all vectors of the form $x = c$, $y = c$ *and* $z = -c$, where $c$ is any real number and $x, y, z$ referred to the first second and third dimension respectively.

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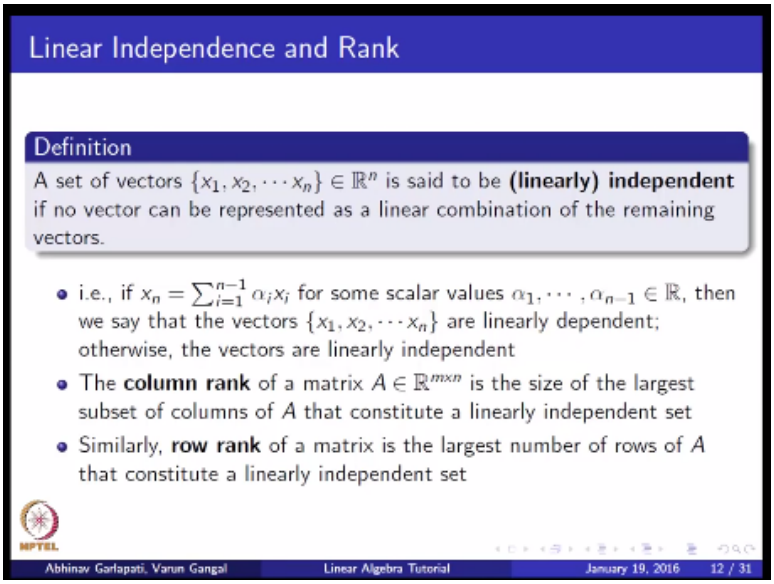


Before we dive into defining linear independence, recollect how we define the linear combination. A set of vectors is linearly independent if no vector in the set can be produced using the linear combination of the other vectors in the set. Now let us have a look at the related concept of rank. So the column rank of *m x n* matrix *A* is the size of the largest linearly independent subset of columns. Note that our columns here are *m x 1* vectors. The row rank is defined in a similar way for rows.

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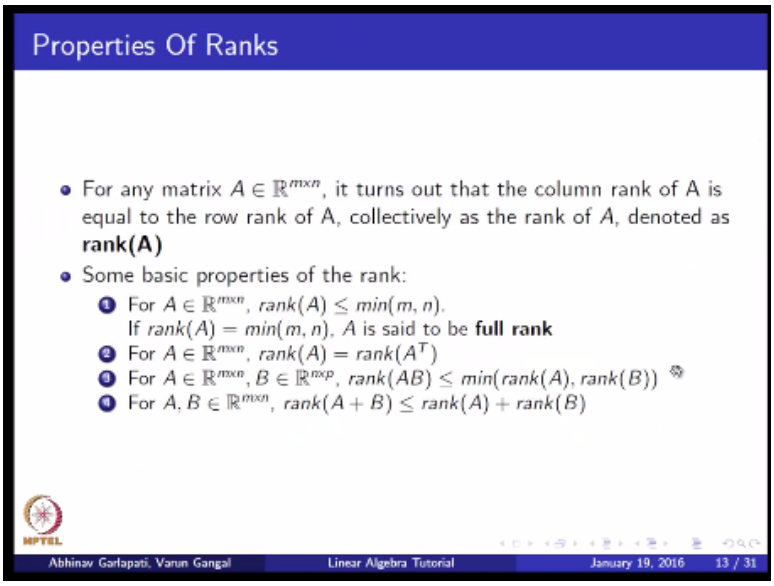


Let us walk through some interesting properties of ranks. For any m x n matrix of real numbers the column rank is equal to the row rank. We refer to this quantity as the rank of the matrix. Earlier we had looked at the quantity called nullity. Nullity is the rank of the null space of A. Some other interesting properties of ranks are also listed here. For instance the rank of a matrix is at most the minimum of its two dimensions, the row dimension and the column dimension. Secondly the rank of a matrix is the same as the rank of its transpose. Thirdly if you multiply two matrices A and B the rank of the resultant matrix is at most the minimum of the ranks of A and B. If you add up two matrices the rank of the resultant matrix is at most the sum of the ranks of A and B.

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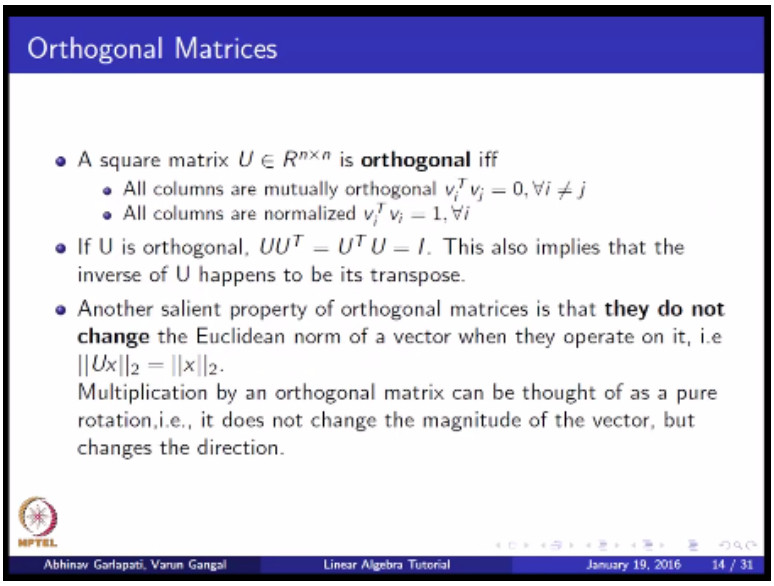


A square matrix U of dimension n x n is defined to the orthogonal if and only if the following two conditions hold. Firstly all pairs of distinct columns should be orthogonal. By columns being orthogonal we mean that the dot product of any pair of distinct column vectors is zero. In other words the $v_i^T v_j = 0, \forall i \neq j$ . The second condition is that the dot product of any column with itself or $v_i^T v_i = 1, \forall i$ .

In other words all the column vectors should be normalized. An interesting implication of a matrix being orthogonal is that $UU^T$ and $U^T U$ both end up being equal to the n x n identity matrix I. This also means that $U^T = U^{-1}$ . Or the transpose of such an orthogonal matrix is also means inverse. An additional interesting property is seen when we multiply a *m x 1* vector *x* by m x m orthogonal matrix U. The Euclidean or L2 norm of such a vector *x* remains the same on multiplication by U. Intuitively we can understand this as orthogonal matrices *U* performing only

pure rotation on multiplying the vector $x$. In other words they only change the direction of a vector but do not change its magnitude.

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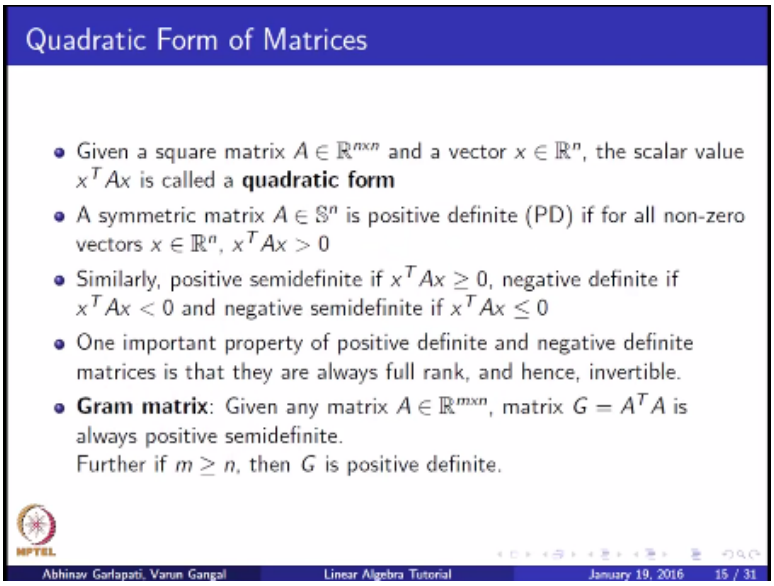


We often encounter the quadratic form which is the vector equivalent of a quadratic function. The quadratic form with respect to the matrix A of a vector $x$ where the matrix A is m x n and the vector $x$ is n x 1 is given by the real number $x^T Ax$. Based on the quadratic forms of matrices we can classify them as positive definite, negative definite, positive semi definite and negative semi definite.

A matrix A is said to be positive definite if its quadratic form is greater than zero for any vector $x$. Similarly we can define it to be negative definite. A matrix A is positive semi definite if the quadratic form is greater than equal to 0 for any vector $x$. Note that equality with 0 may also hold here. One important property of positive and negative definite matrices is that they are always full rank. An implication of this is that $A^{-1}$ always exists. For a matrix A, which is of dimension m x n, one can define a special matrix called a gram matrix. The gram matrix is given by $A^T A$. One of the property of the gram matrix is that it is always positive semi definite Moreover if the

number of rows exceeds the number of columns in other words if $m >= n$ this means that G is positive definite.

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**Introduction to Machine Learning**

**Lecture 8**

**Prof. Balaraman Ravibdran**
**Computer Science and Engineering**
**Indian institute of technology**
**Linear Algebra (2)**

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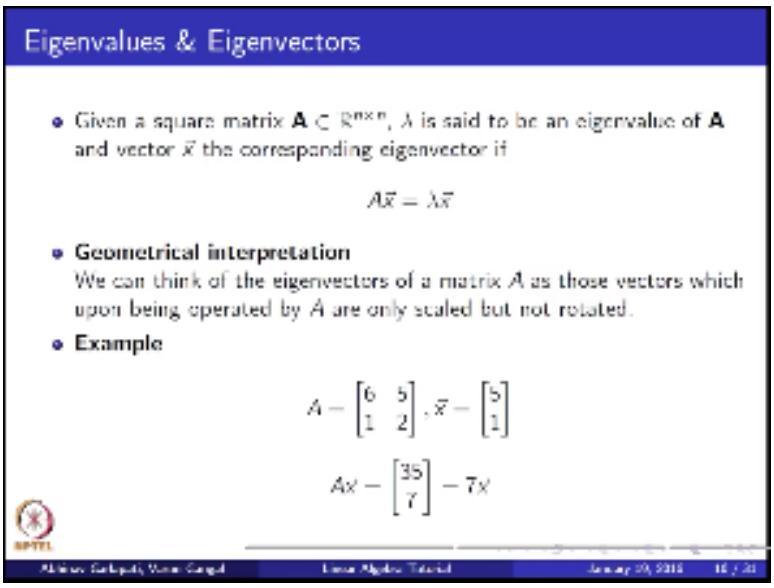


Eigenvectors and eigenvalues of A tied together, which means that every eigenvector has an associated eigenvalue. We often characterize square matrices in terms of their eigenvectors. One way of looking at eigenvectors is as follows $\vec{x}$ can be thought of as a vector and in $R^n$ and the square matrix acts A acts like an operator which transforms $\vec{x}$ into another n-dimensional vector

$A\vec{x}$. Now the Eigen vectors of A are those vectors which on being transformed by A or operated upon by A are only scaled by $\lambda$ but not rotated in other words their direction does not change.

We can have a look at this example here the $2\times2$ matrix A. On multiplying the vector X = $(5,1)$ gives back the vector X multiplied by the real value 7. So here X is an eigenvector of A and 7 is an eigenvalue of A.

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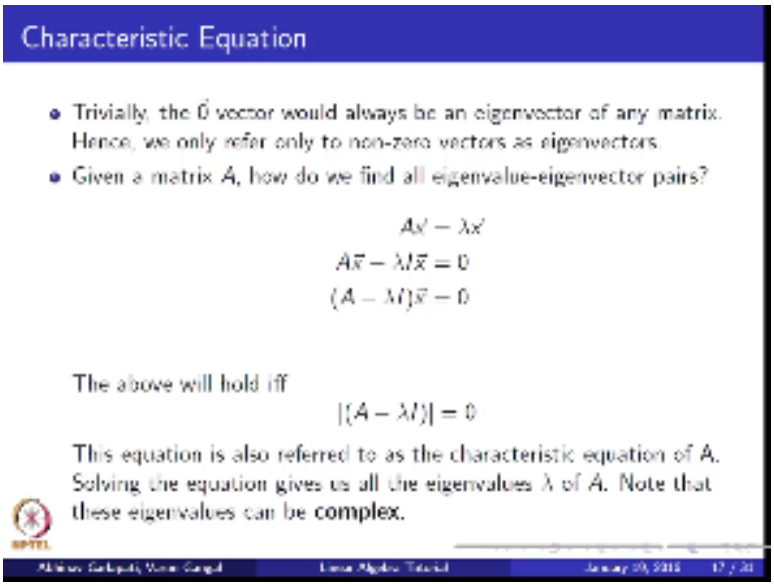


We can see that zero would always be an eigenvector of any matrix if we simply go by the $A\vec{x}$ $=\lambda\vec{x}$ definition. Hence we only refer to nonzero vectors as eigenvectors. So the question is given a matrix A how does one find all the eigenvalue eigenvector pairs. By simplifying $A\vec{x}=\lambda\vec{x}$ we get (A-λ I) $\vec{x}$ =0. Now since we are only looking at nonzero vectors $\det(X)$ cannot be 0 and $X$

can't be a zero vector which means that $\det(A-\lambda I)$ should be zero. So the equation $\det(A-\lambda I)=0$ is called a characteristic equation of A and solving this equation gives us all the eigenvalues of A. One thing you note that even though all the values of A are real and A is a real matrix the eigenvalues can be complex.

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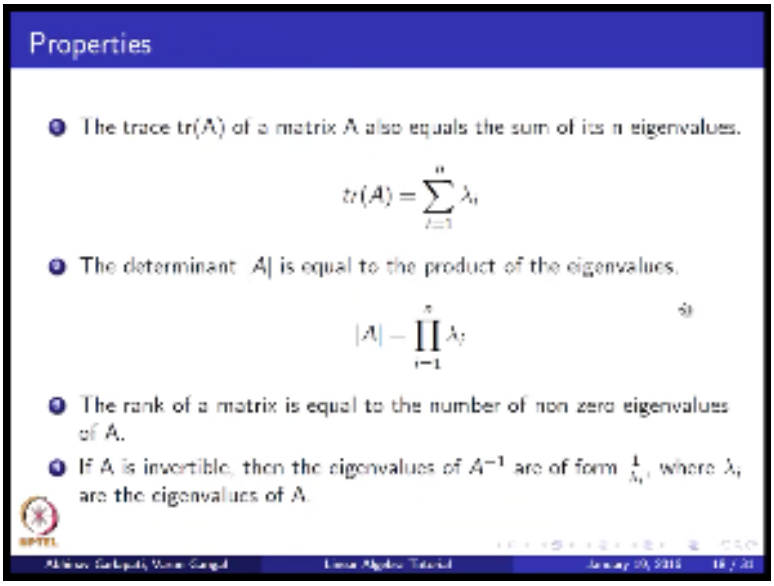


There are interesting relations between some properties of a matrix and its eigenvalues. For instance,

1. The trace of a matrix is equal to the sum of its eigenvalues $tr(A)=\sum_{i=1}^{n}\lambda_i$
2. The determinant of a matrix is equal to the product of its eigenvalues $\det(A)=\prod_{i=1}^{n}\lambda_i$

The rank of a matrix is equal to the number of nonzero eigenvalues. Note that if an eigenvalue has multiplicity greater than 1, for instance, if two distinct eigenvectors $x_1, x_2$

both have eigenvalue λ we would count λ twice. Also we can describe the eigenvalues of $A^{-1}$ in terms of the eigenvalues of A provided of course A is invertible. The eigenvalues of $A^{-1}$ will be of the form $\frac{1}{\lambda_i}$ where $\lambda_i$ is an eigenvalue of A.

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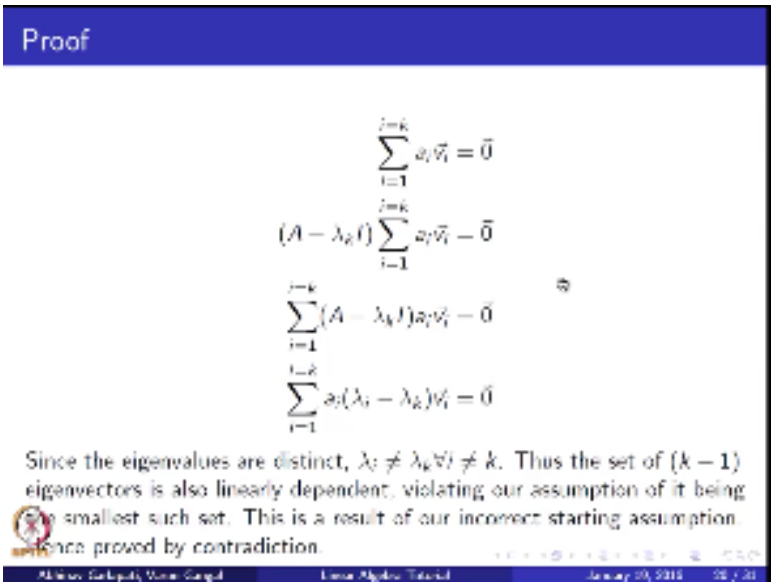


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Now let us have a look at an interesting theorem about eigenvalues and eigenvectors. The theorem goes as follows:

*If a matrix has all its eigenvalues distinct then its eigenvectors are linearly independent.*

We shall prove this by what is called a proof by contradiction. If this theorem does not hold that means there is a set of $k$ eigenvectors such that it is linearly dependent. Let the i$^{th}$ vector in the set be $v_i$ and the corresponding eigenvalue be $\lambda_i$. Note that we are considering the smallest such

set. Since the set is linearly dependent this means there exists real constants $a_i$ such that $\sum_{i=1}^{i=k} a_i v_i = \vec{0}$. Now let us multiply both sides of the equation by $A - \lambda_k I$. Since $v_k$ is an eigenvector of A, $(A - \lambda_k I)v_k = 0$. We can understand this from the characteristic equation hence the term corresponding to $v_k$ disappears from the equation since it goes to zero.

Now for the remaining eigenvalues since we know they are distinct, the term $\lambda_i - \lambda_k \neq 0$. Note that $(A - \lambda_k I)v_i$ simplifies to $(\lambda_i - \lambda_k)v_i$ since $Av_i = \lambda_i v_i$. However now we can think of $a_i(\lambda_i - \lambda_k)$ as a new constant $b_i$. This means now that we have $\sum_{i=1}^{k-1} b_i v_i = 0$. However, we had assumed that the set of size $k$ was the smallest set of linearly dependent eigenvectors however now we have an even smaller set. This contradicts our starting assumption. Hence such a set of $k$ linearly dependent eigenvectors cannot exist for any $k \geq 2$. Hence all our eigenvectors are linearly independent hence our theorem stands true.

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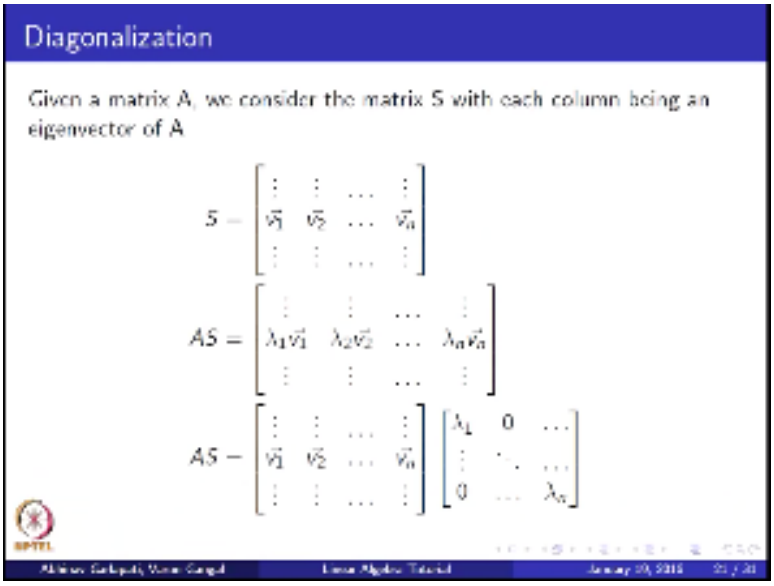


Diagonalization gives us a way of representing a matrix in terms of its eigenvalues and eigenvectors. Let us consider a $n \times n$ square matrix A. We denote the matrix where every column is an eigenvector of A by S. On multiplying S by A each column would get multiplied by $\lambda_i$ since the column itself is an eigenvector of A.

This right hand side can then be simplified as the product of two matrices the first one being S itself while the second one being the diagonal matrix where the i$^{th}$ diagonal element is the eigenvalue $\lambda_i$. Remember that the LHS is AS. Now we have the equation $AS = S\Lambda$ where $\Lambda$ is the diagonal matrix of eigenvalues. On simplifying this we get $A = S\Lambda S^{-1}$. This is a diagonalization of A. Note that $S^{-1}AS$ is a diagonal matrix since $S^{-1}AS$ is nothing but $\Lambda$, the diagonal matrix of Eigen values.

This result is dependent on S being invertible. It will hold if the eigenvalues of a matrix are distinct since the eigenvectors would then be linearly independent. This would mean the columns of S would be linearly independent and hence S would be full ranked and as a consequence invertible.

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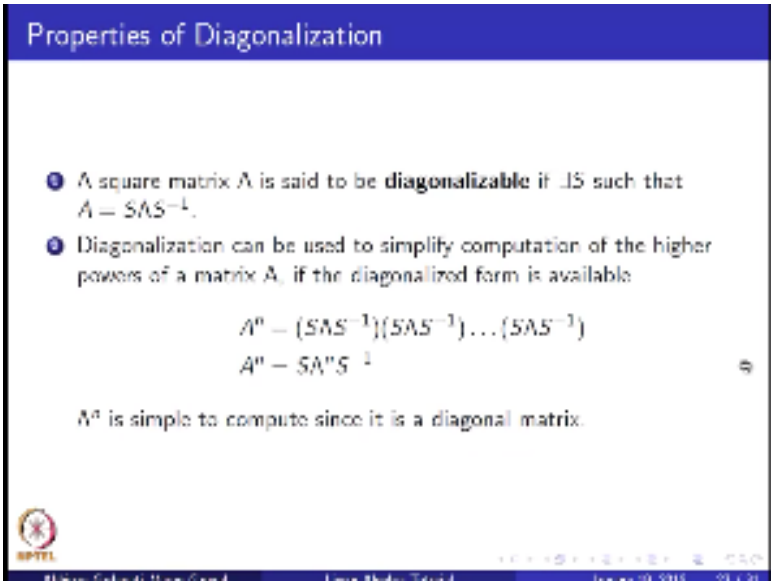


Then do we say that the square matrix is diagonalizable? Well when such a diagonalization exists we saw that we needed S to be invertible for the diagonalization to exist.

Another advantage of diagonalization is that it simplifies the process of computing $A^n$. We first represent every A in diagonalized form now we can see that the $S^{-1}$ of the first term and the S of the second term would multiply to give us I. Similarly for the second third fourth and so on in this way by regrouping the terms we get $A^n = S\Lambda^n S^{-1}$. Note that it is very easy to compute the $n^{th}$ power of a diagonal matrix since you just have to raise every diagonal element to the power

of n. In this way the diagonalization has helped us simplify the process of computing $A^n$. Without this simplification we would have needed to multiply non-diagonal matrix n times.

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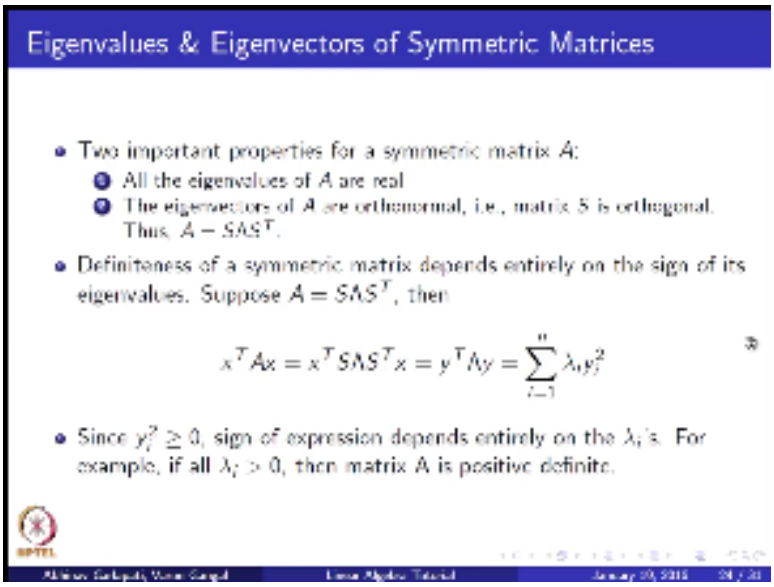


If a matrix is symmetric then all its eigenvalues are real numbers. Also its eigenvectors are orthonormal, that is they are mutually orthogonal and normalized. This means that the matrix of eigenvectors S is also orthogonal. We have seen that for orthogonal matrices the inverse and the transpose are the same, hence, we can write $A = S\Lambda S^T$ as per the diagonalization we defined earlier. For symmetric matrices their definiteness can be inferred from the signs of their eigenvalues. Suppose that $A = S\Lambda S^T$, now taking the quadratic form with respect to A for vector $x$, $x^T Ax$ simplifies to $y^T \Lambda y$ ,where $y$ is $S^T x$. This further simplifies to $\sum_{i=1}^{n} \lambda_i y_i^2$. Now for a

matrix to be positive definite this term must always be positive. Since $y_i^2 \geq 0$, anyway the sign of this term depends on the eigenvalues.

If all the eigenvalues are positive the matrix is positive definite. If we know that the matrix is positive semi definite or PSD then what can we say about its eigenvalues. Since the quadratic form of a PSD matrix is non-negative for any vector $x$, this should hold for the eigenvectors too. Now since $Ax = \lambda x$, $x^T Ax$ simplifies to $\lambda \|x\|^2 \geq 0$.
Since eigenvectors are nonzero by definition the square of the norm is always positive. This means that every eigenvalue of A is non-negative.

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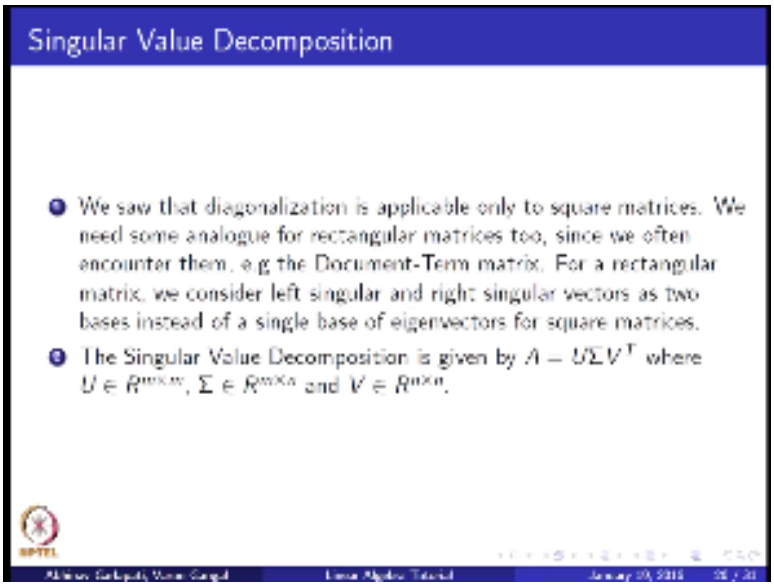


We learnt about diagonalization which took in a square matrix of size $n \times n$ and represented it in terms of its eigenvectors. However we cannot directly apply the same diagonalization for rectangular matrices since the notion of eigenvector is defined only for a square matrix. We need

a diagonalization for rectangular matrices since we come to them often. For instance the matrix of n data points or features or the matrix of n documents and r terms. For the rectangular matrix A of size $m \times n$ they can be represented in terms of the eigenvectors of $AA^T$ and $A^T A$ both of which are square matrices. This is known as the singular value decomposition. A is represented as $A = U\Sigma V^T$ , where $U \in R^{m \times m}$ , $\Sigma \in R^{m \times n}$ and $V \in R^{n \times n}$

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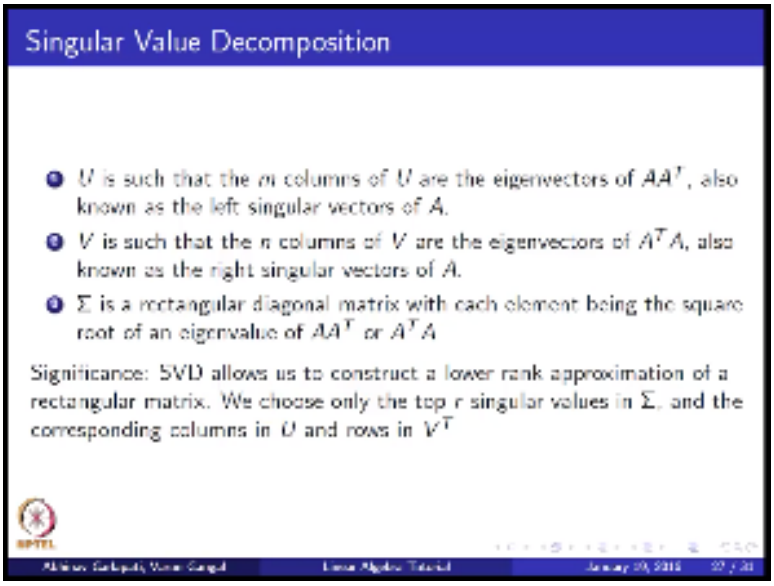


The three elements U, $\Sigma$ and V are as follows. In U every column represents an eigenvector of $AA^T$ . In V every column represents an eigenvector of $A^T A$ and $\Sigma$ is a rectangular diagonal matrix with each element being the square root of an eigenvalue of $AA^T$ or $A^T A$ . Now look at $AA^T$ or $A^T A$ have different eigenvectors but the set of eigenvalues is the same. This is because suppose $A^T Ax = \lambda x$ for some eigenvector and eigenvalue $\lambda$ . Now multiplying both sides by A we get AA transpose times $AA^T Ax = \lambda Ax$ . Hence $Ax$ is an eigenvector of $AA^T$ and $\lambda$ is also an eigenvalue of $AA^T$ .

This is why $AA^T$ and $A^TA$ have the same set of eigenvalues. The significance of this decomposition is that if we ordered U, V and $\Sigma$ such that the eigenvalues whose magnitude is large will come first both in U and V in the column order also along the diagonal in $\Sigma$ . Then we can drop everything greater than index r to get a r dimensional low rank approximation of the original matrix A. This approximate form of A will be represented as U which is an $m \times r$ matrix. $\Sigma$ which is a $r \times r$ matrix and V which is a $n \times r$ matrix.

Consider a function f which takes in matrix of dimension $m \times n$ and outputs real numbers. The gradient is the matrix of partial derivatives then $(\nabla_A f(A))_{ij} = \frac{\partial f(A)}{\partial A_{ij}}$ . Consider a different type of function which takes in a n dimensional vector and returns a real number. The Hessian for this function is defined as follows the $(\nabla^2{}_x f(x))_{ij} = \frac{\partial^2 f(x)}{\partial x_i \partial x_j}$ . You can see that hessian would be an $n \times n$ matrix.

Now let us study how we can find the gradient for some simple vector functions consider the function $f(x) = b^T x$ , where x is an n dimensional vector and b is also an n-dimensional vector. Then $f(x) = \sum_{i=1}^{i=n} b_i x_i$ . On differentiating this with respect to the kth component of the vector X we get $\frac{\partial f(x)}{x_k} = b_k$ . Hence the gradient of $f(x)$ is given by the vector b. We can see how this intuitively relates to the first derivative of the scalar function $f(x) = Ax$ which is equal to A.

We had earlier looked at a type of function called the quadratic form defined for a $n \times n$ matrix A. The quadratic form with respect to matrix A is a function $f(x) = x^T Ax$ which takes in an n-dimensional vector x. Now let's have a look at how one can find the gradient and hessian on the quadratic form of a known symmetric matrix A. They can write down $f(x) = \sum_{i=1}^{n}\sum_{j=1}^{n} A_{ij}x_i x_j$. We can split-up this summation into four terms based on whether i and j are equal or not equal to k.

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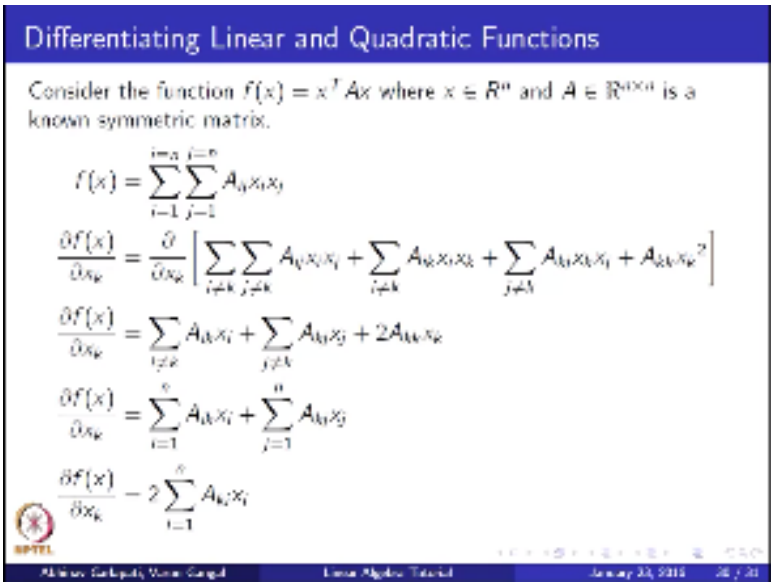


Finally, $\frac{\partial f(x)}{\partial x_k} = 2\sum_{i=1}^{n} A_{ki}x_i$. Note that the simplification from the second last step to the last step can only be done if A is symmetric thus we get the gradient of $x^T Ax = 2Ax$. Similarly on further differentiating every element of the gradient by $x_k$ we can derive the hessian of the function. The hessian of this function comes out to be 2A.

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**Introduction to Machine Learning**

**Lecture 25**

**Prof: Balaraman Ravindran**
**Computer Science and Engineering**
**Indian Institute of Technology Madras**
**Optimization**

Hello everyone. I am Abhinav and in this unit we will be covering the basic concept of optimization, which should be useful in this course.

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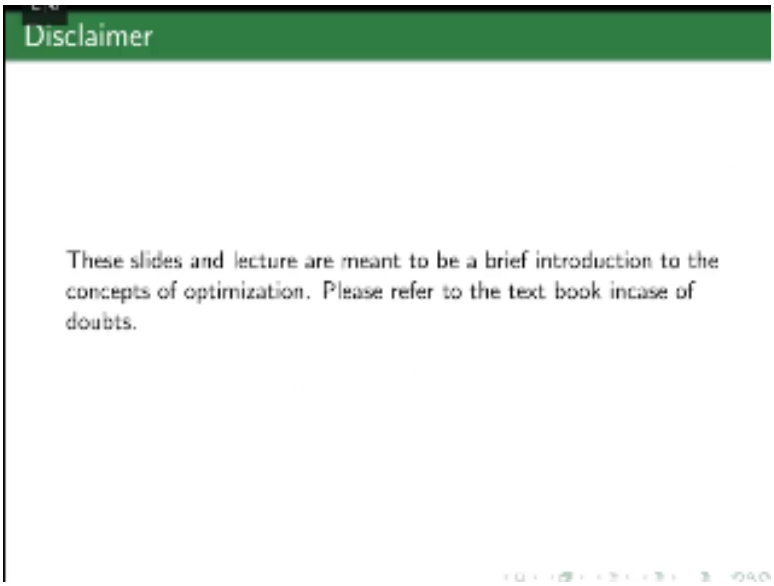


So before going in to details a small disclaimer this tutorial is meant to be a small introduction for a complete understanding of these concepts please refer to any standard text book.

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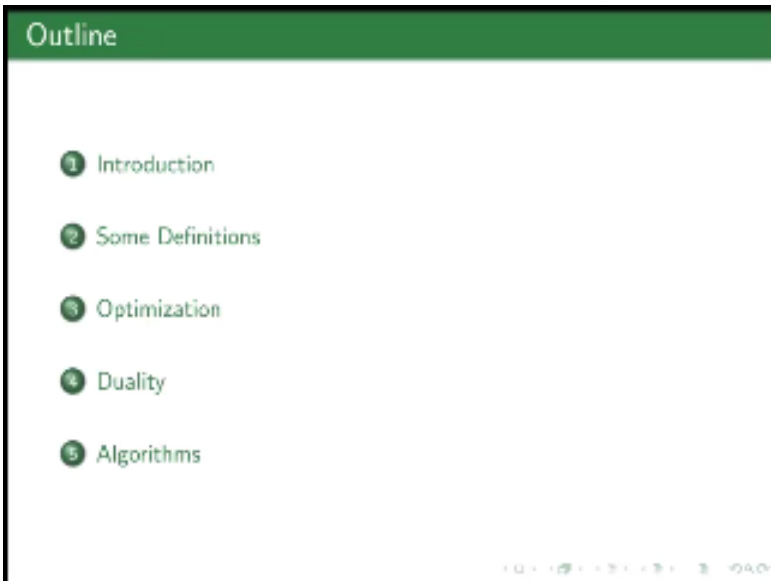


This tutorial is broken in to five chunks first let us start off with the introduction.

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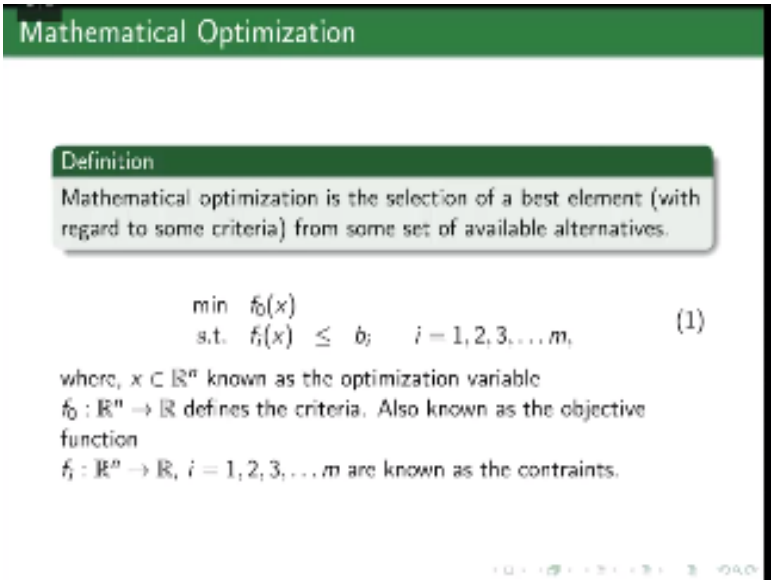


What is mathematical optimization? Mathematical optimization according to Wikipedia is a selection of a best element with regard to some criteria from some set of available alternatives. Now let us look at the mathematical formulation for the same. Here we are trying to minimize $f_0$ (x) subject to m constraints of the form $f_i(x) <= b_i$. $f_0$ is also known as the objective function $f_i$ are the constraints and x is known as the optimization variable.

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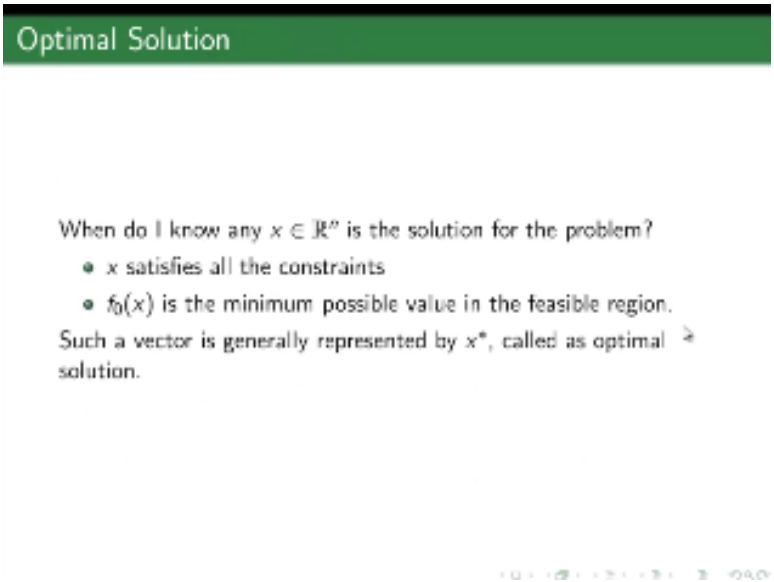

x is known as the solution for the problem if it is satisfies all the constraints and it minimizes $f_0$ (x). Such a solution is known as the optimal solution and it is represented by $x^*$ so through all this tutorial whenever you see x* it represents the optimal solution for the optimization problem.

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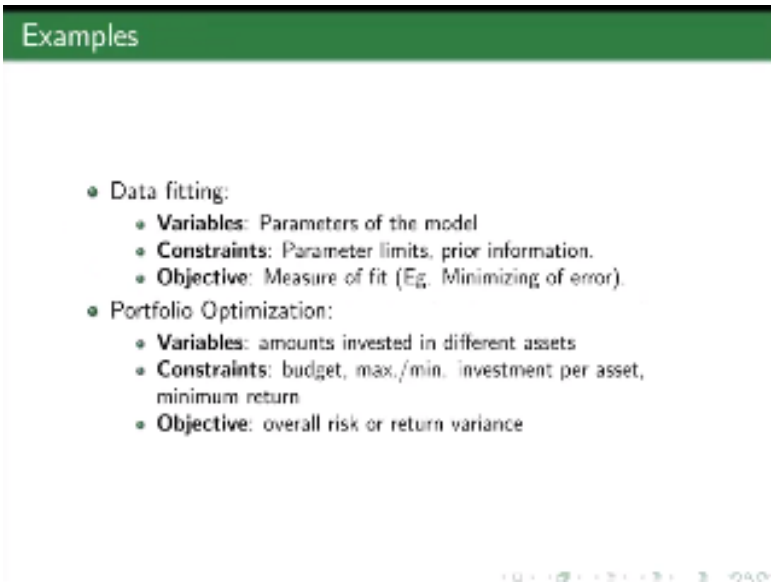


Now let us look at some examples where optimization is used. First data fitting, data fitting is a very common problem in the field of machine learning. What I mean by data fitting? Data fitting is fitting of a parametric model given some data. So one such example both linear regression in linear regression we are trying to fit a linear model whose parameters are $\beta_i$'s, so those translate to the optimization variables here and constraints, constraints in general encode something like parameter limits or prior information which you want to encode. So what is the specific example of linear regression? We do not have any constraints. And what would be the objective? You would try to fit get the best fit for the model. So one way of doing this would be minimizing this error and in linear regression we have seen how see to minimize this squared error.

So that forms the objective of the optimization problem. Another example of application of optimization is portfolio optimization. So by portfolio optimization we mean to optimize the amount of money I can invest in various assets. So these assets could be something like shares

from different companies or any other investment options. So the variables would be the amount I invest in all the options available. The constraints would bead it overall budget the maximum or the minimum investment per asset. And the minimum return I expect from each asset, objective would be to minimize overall risk or minimize the return variants. You have seen what optimization problems are and you seen some examples. Now the next big question is how do we solve them.

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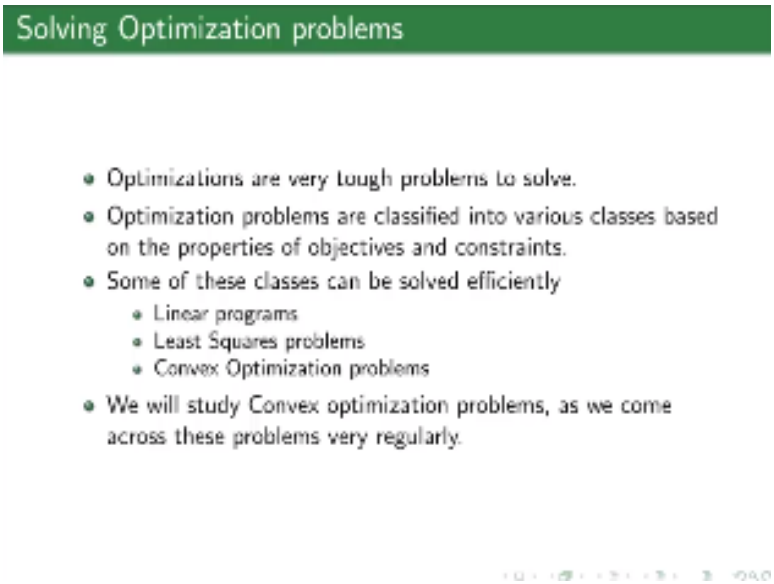


Optimization problems very difficult problems to solve in general optimization problems are classified in to different types based on the properties of objectives and constraints. Some of the examples are linear programs, least square programs and convex optimization problems. These problems are well studied and can we solved efficiently. Not all class of problems can we solve very efficiently. In this tutorial we will be covering convex optimization problems in detail.

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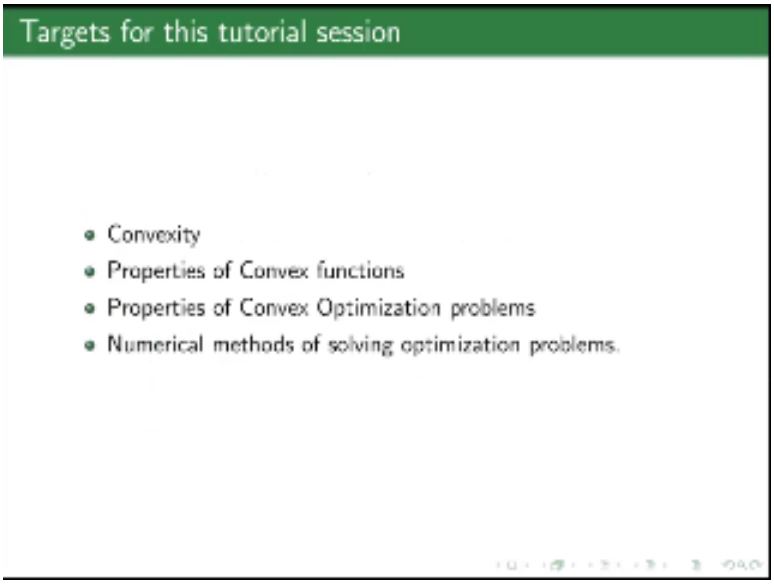


In this tutorial first we will be looking at convexity. What convexity means and how do we define it? Then we will look at properties of convex functions and then we will look at properties of convex optimization problems. And at the end we have briefly cover some numerical methods for solving optimization problems.

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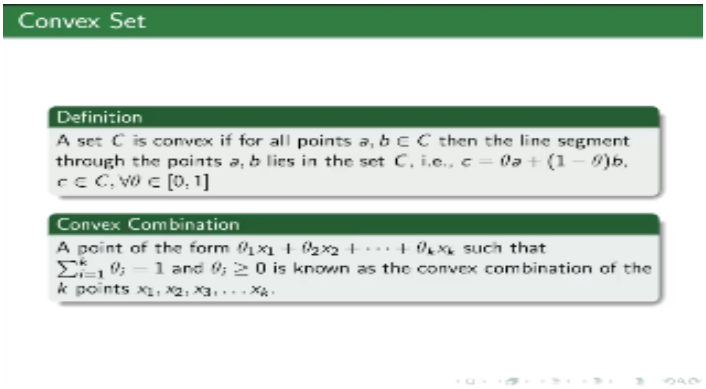


A set C is set to be convex if for all point a,b belong to the set the lines segment passing through this points should also lie inside the set. So mathematically we can see the sets all the points of the forms θa + (1-θ) b, when θ lies in [0,1] should also belong to the set C. Next let us look at the definition of convex combination. A point of the form $\theta_1$ $x_1$ + $\theta_2$ $x_2$ so what it $\theta_k x_k$ such that the coefficient sum up to 1 and the coefficients are non negative is known as the convex combination of this k point.

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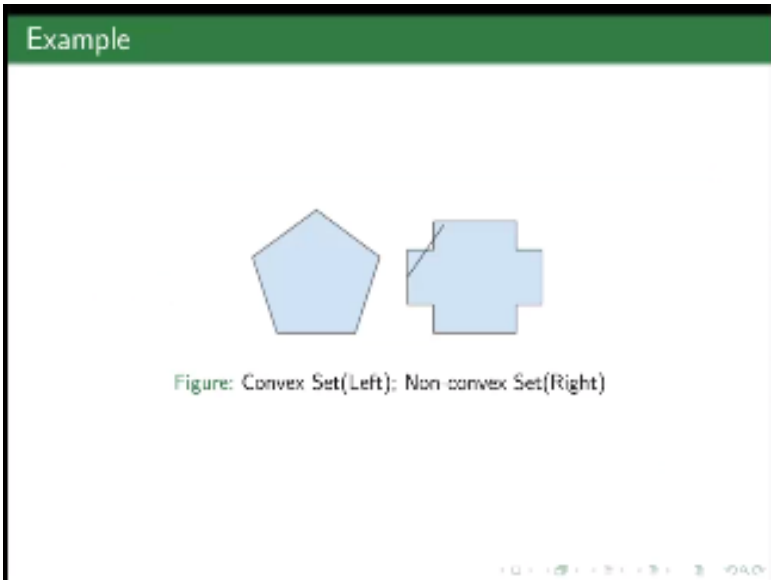


Now let us look at examples of convex set. This pentagon is a convex set because any line joining two points inside the set lies inside the set. Whereas this set is a non convex set because this lines are going with joints 2 points here passes outside the set. Thus theses points do not lie inside the set hence this does not satisfied the definition of convex set right it is not a convex set.

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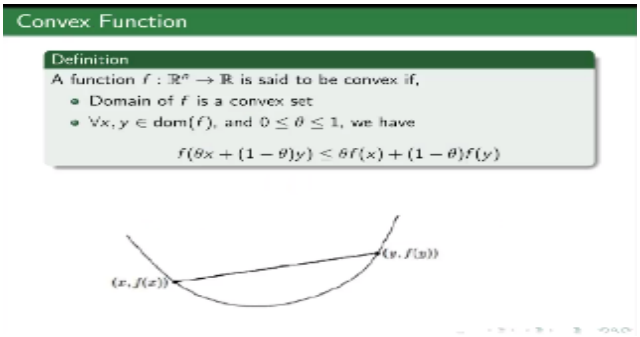


Let us look at the definition of convex function. A function f is set to be convex if the domain is a convex set and if for all x, y which belong to the domain f the convex the value of the convex combination of these two points is less than or equal to the convex combination of the values at these individual points. So what I mean is $f(\theta x+(1-\theta)y) \le \theta f(x)+(1-\theta)f(y)$. So

geometrically you can see that the line joining (x, f(x)) and (y, f (y)) should lie above the curve. So if this happens we can see that the value $f(\theta x+(1-\theta)y)$ are the points along the curve $\theta f(x)+(1-\theta)f(y)$ are points along the line segment joining (x, f(x)) and (y, f (y)). So by ensuring that this always above the function we ensure that the inequality holds this making it a convex function.

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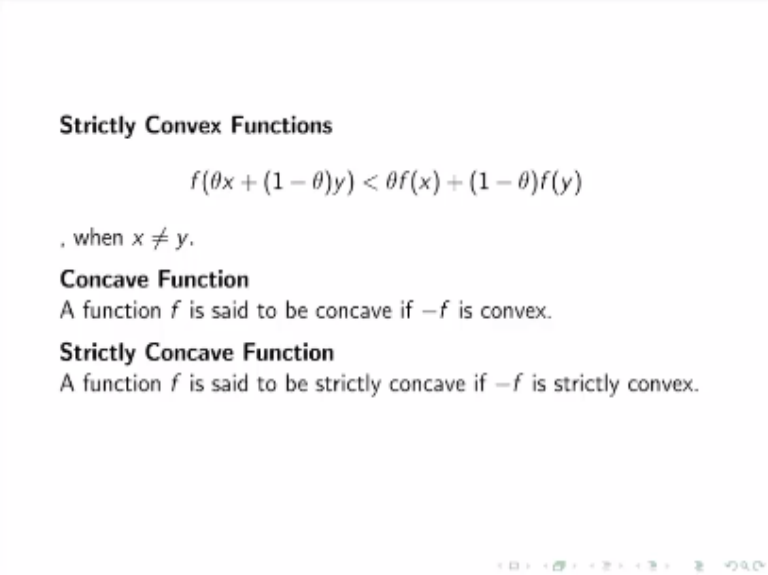


Now let us define what a strictly convex functions is. In strictly convex functions the inequality becomes a strong in equality that is $f(\theta x+(1-\theta)y)<\theta f(x)+(1-\theta)f(y)$ . And now let us define what is concave function is a function f is said to be concave if, –f is convex and then similarly we define a strictly concave function a function f is said to be strictly concave function if, –f is strictly convex.

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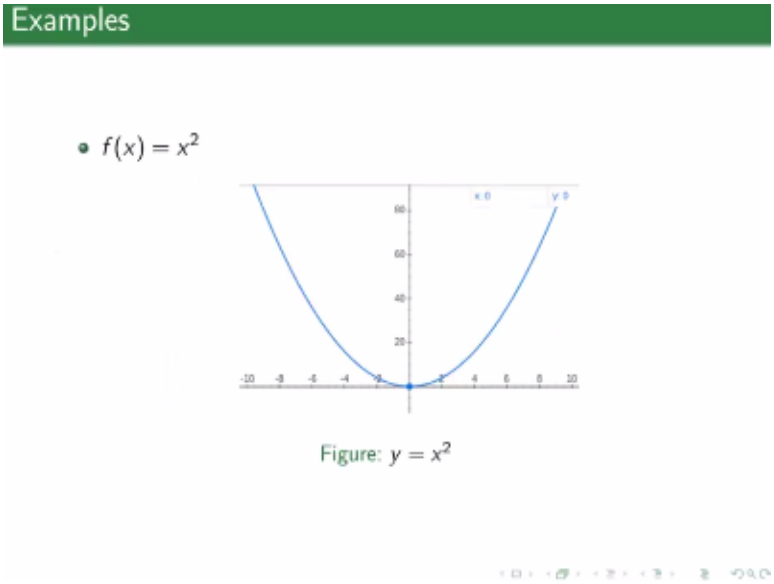


Now let us look some examples. First f(x) =$x^2$ is a convex function from the graph it is clearly evident that any line joining two points this will lie above the curve between these two points. This claim can also be verified by using the definition.

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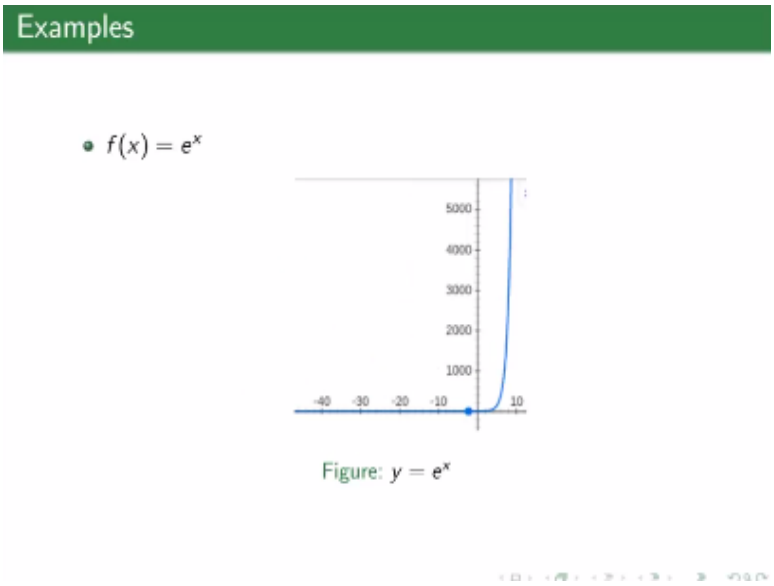


The next example that we see is graph of f (x)= $e^x$ . Again graphically you can clearly see that this is the convex function. If you try to prove this according to the definition you can see that this is not trivial. So we would like to see if any other ways to check the convexity of function.

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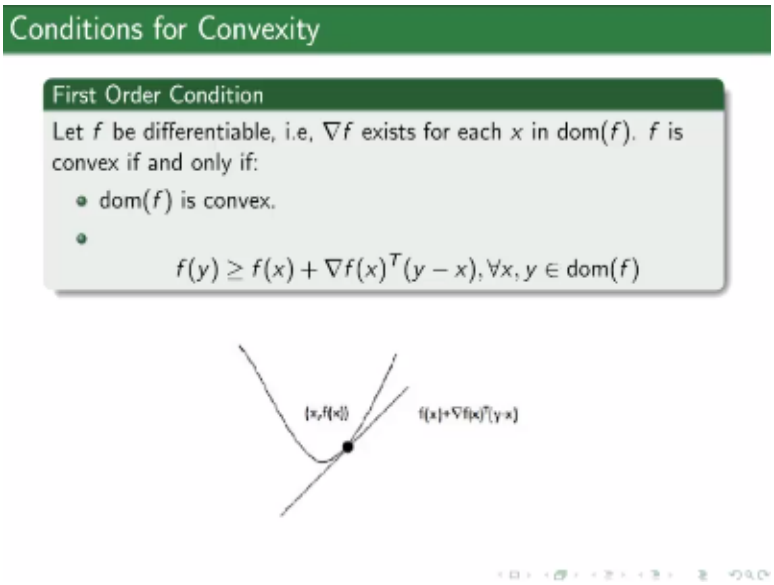


Let us look at the first order condition for the convexity. Let f be a differentiable function that is $\nabla f$ exists for all x in a domain of f. So a function f is convex if and only if the domain of f is convex and this inequalites satisfy. This inequality states that function should always lie above all it is tangents. If you look at the right hand side carefully it is nothing but the equation of the tangent at (x,f(x)) and we expect this value to be less than f(y) this is nothing but the condition saying that the curve should be above the tangent.

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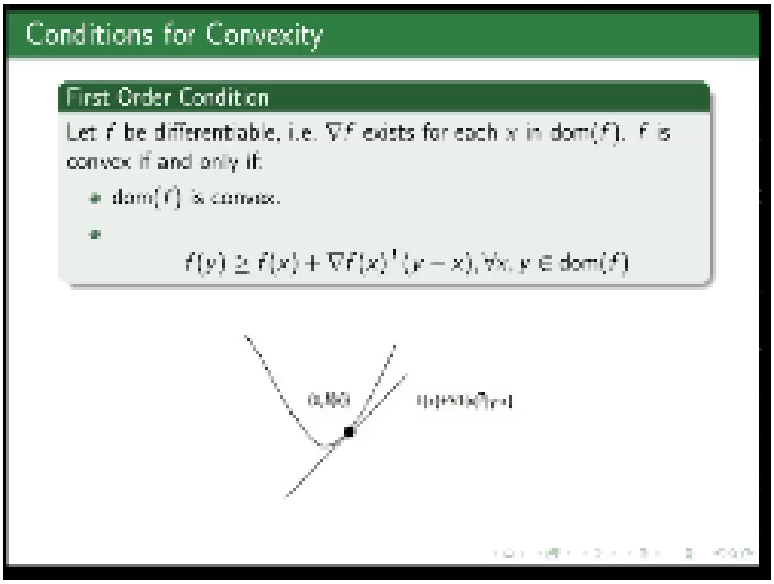


Now let us look at the second order condition for convexity. Let f be twice differentiable function and the f will be convex if and only if domain of the f is convex and its hessian is positive semidefinite. So if you look at the second example of the $e^x$ the second derivative is always positive hence it can be proved that it is a convex function

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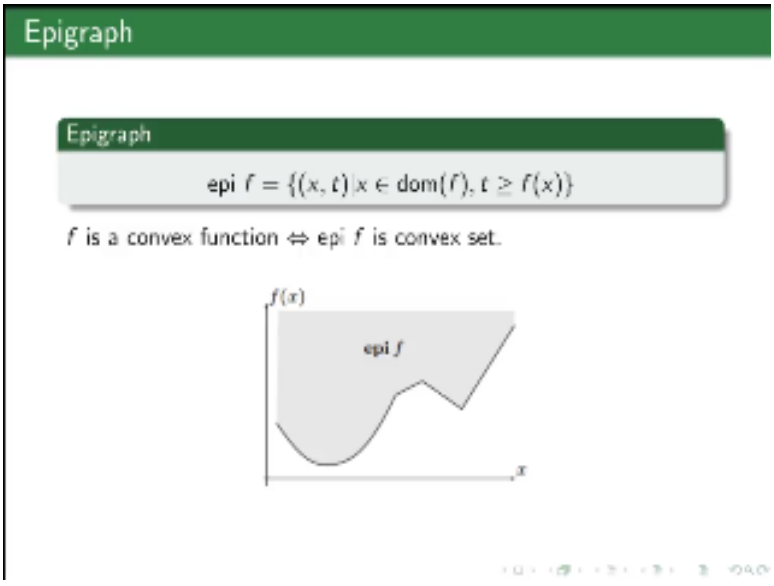

Now we defined a epigraph of function. For a given function f, the epigraph of f, is defined as the set of all pairs (x,t) such that x belong to the domain of f and t is greater than or equal to f(x). So if you look at the graph you can see that the area above the curve is belongs to the epigraph of the function. One important property to know is for a convex function, the epigraph is always a convex set and the converse also holds, that is if for a function the epigraph is a convex set then the function is convex.

So we till now we have seen three ways of checking for convexity of a function. First you can do the first order test or the second order set or you can check for convexity of the epigraph of the function.

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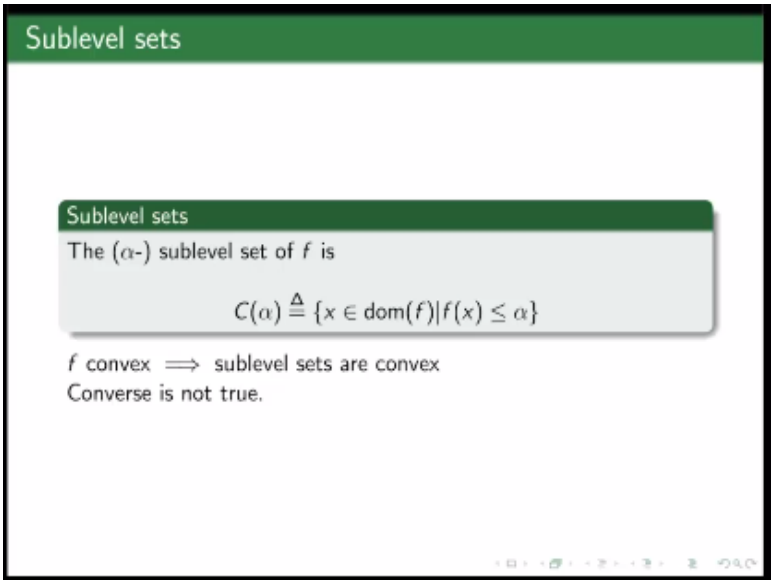


Now let us look at what is sublevel sets of function are. An alpha sub level set of a function f is set of all pints x which belong to the domain of f such that the value of the function at these points is <= alpha. There is one important property that if the function is convex, the sublevel set sets of the function are also convex it is important to note the converse is not true.

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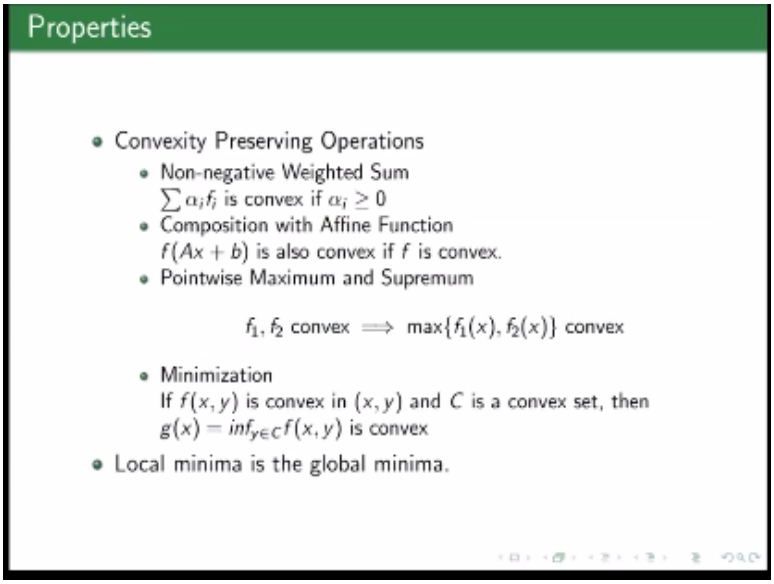


Now let us look at some other properties of convex functions. First we will look at the operations which preserve the convexity of the function. First non-negative weighted sum. That is a non negative weighted some of various convex functions which still remain a convex function. Consider $f_i$ is the series of convex functions $\sum \alpha_i f_i$ where $\alpha_i$'s are greater than 0, will also remain a convex function. Next, composition with affline function. Affline function is a linear transformation of x so ax + b is an affline function. If f is convex then f(ax + b) is also convex. Point wise maximum and supermum of two convex functions will also remain convex. Minimization if you look at as two variable function f(x,y) which is convex, then if you try to minimize the function allowing any one variable in a convex set, the resultant function is also convex function. The most important property of convex functions is that the local minima is also the global minima. It is a very powerful result which can proved easily. This result gurantees that the minima option while searching for the minimum of a convex function is the optimal solution. Another important property of convex function is that they satisfy the Jensen's inequality. This is a generalization of the inequality which we have seen in the definition of the convex function. Here instead of two points, we have n points. So the value of the convex combinations of n points is less than or equal to the convex combination of the values each of the

function at each of the individual points. A colloquial way of saying this is that the value of the average is less than the average of the values. Here by the average I mean a weighted average.

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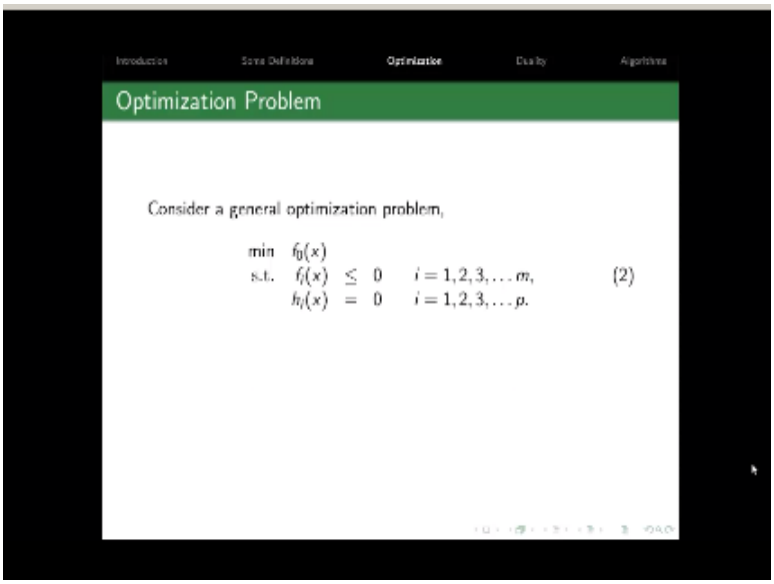


Now let us look at a general optimization problem. Any optimization problem in generally can be reduced to this form. Of minimize an objective function subject to few inequality constraints and few equality constraints. So the optimal value p* can also be written as

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The optimal value $p^*$ is given by

$$\inf\{f_0(x) | f_i(x) \leq 0, i = 1,2,\ldots,m \text{ and } h_i(x) = 0, i = 1,2,3,\ldots,p\}$$

Now the next question is why did we use infimum instead of minimum? In some function the minimum might not be attainable; it might just tend to a minimum value but not actually attain it. Hence we write infimum instead of minimum.

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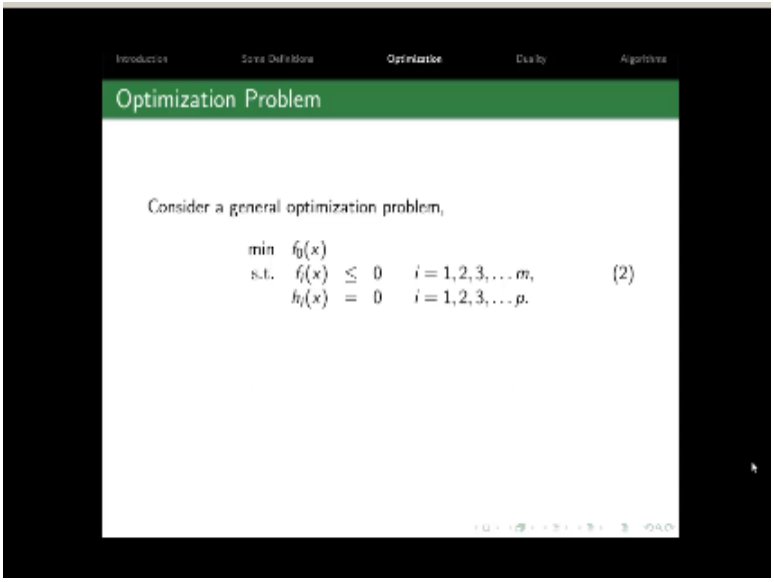


An optimization problem with satisfies the given three condition is not as a convex optimization problem. So first $f_0$ the objective function should be convex then the inequality constraints $f_i$ should also be convex. And the equality constraints should be affline. When I say affline it should be at the form $a_i^T x = b_i$. So one can observe that the domain has become a convex set right now. So these inequality constraints represent a sublevel set of a convex function so it is a convex set. And a convex set intersection with affline function is a convex set. So why a convex problems are so interesting. So convex representation problems are interesting because with the properties of a convex functions and the convex set. So first the most important property which is useful for us is that the if there is local minima anywhere, it is guarantee that is the global minimum for the function. So it makes are like very simple and we do not have search a lot for global minimum.

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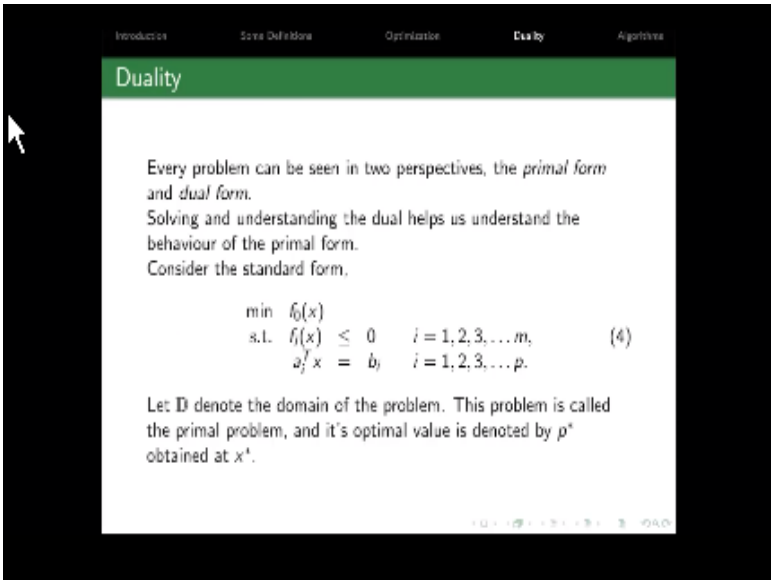


Every optimization problem can be seen in two perspectives. One the primal form and the dual form, so whatever we seen till now is generally known as the primal form. And we will now develop the dual form. So why do we need another view of the problem? So sometimes the primal form might be very difficult to solve it. So the dual form might be easier to solve and also cases some understanding on how the solution of the primal form may be. So before going ahead let me just recap the notation which we going to use. So this is the standard optimal convex optimization problem and when I said p* it denotes that the optimal value of this problem and the value of p* is attained at x* which is the solution of a solution. Now let us consider the alternative relaxed problem. Instead of minimizing the $f_0$ will may the weighted some of the objective functions and the constraints.

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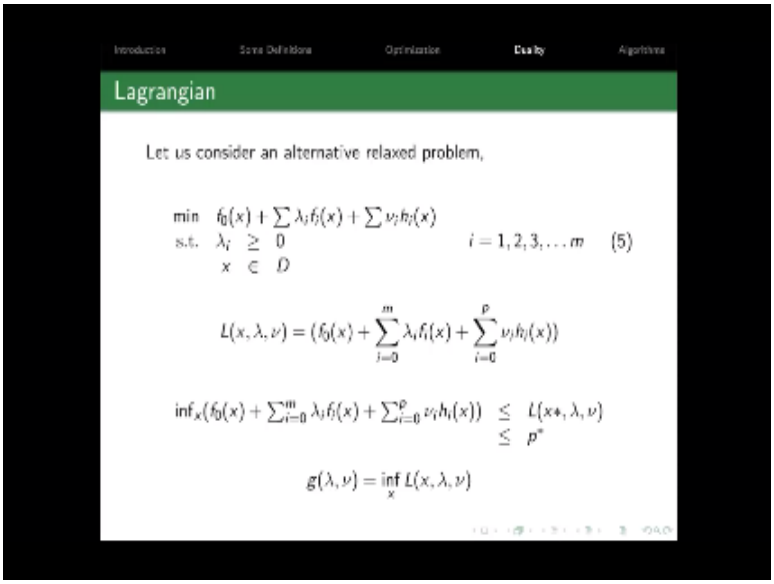


So we will be minimizing

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$$\begin{aligned} \min \quad & f_0(x) + \sum \lambda_i f_i(x) + \sum \nu_i h_i(x) \\ \text{s.t.} \quad & \lambda_i \geq 0 \qquad i = 1, 2, 3, \ldots m \qquad (5) \\ & x \in D \end{aligned}$$

Here we also have an addition constraint that λ should be greater than or equal to 0. And as usual x should belong to the domain we call the object of this optimization of this optimization problem as the Lagrangian. So $L(x, \lambda, \upsilon)$ is defined

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$$L(x, \lambda, \nu) = (f_0(x) + \sum_{i=0}^{m} \lambda_i f_i(x) + \sum_{i=0}^{p} \nu_i h_i(x))$$

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$$\inf_x(f_0(x) + \sum_{i=0}^{m} \lambda_i f_i(x) + \sum_{i=0}^{p} \nu_i h_i(x)) \leq L(x*, \lambda, \nu) \leq p^*$$

$$g(\lambda, \nu) = \inf_x L(x, \lambda, \nu)$$

Infimum of the Lagrangian over x is less than equal to p* this can be seen very easily. But thing to be noted as in equality is valid only when x is feasible. So now we define g as a function of λ and $\upsilon$ as the infimum of the Lagrangian over x.

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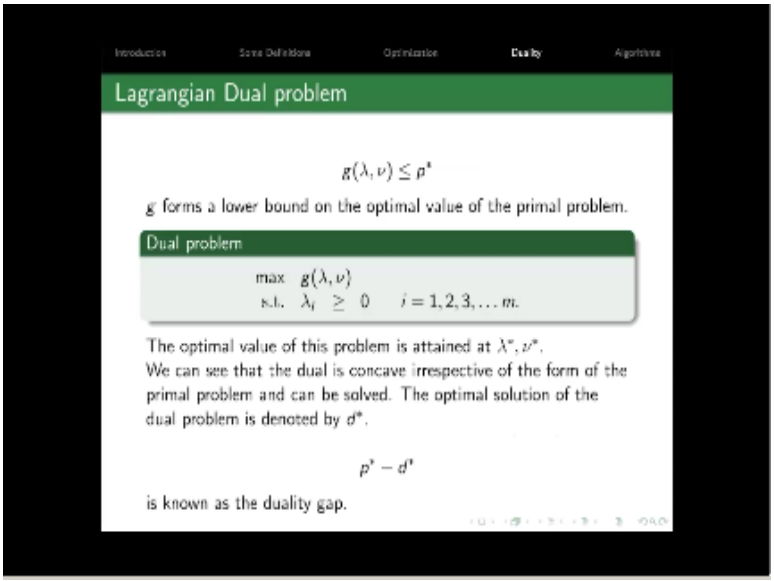


So we have seen of a function g which cases lower bound of the optimal value of the primal problem. So if you try to maximize the function g we will achieve a very good lower bound of the optimal value. So this is what is may known as the dual problem. So maximizing g(λ , $\upsilon$ ) such that $\lambda_i$>= 0.

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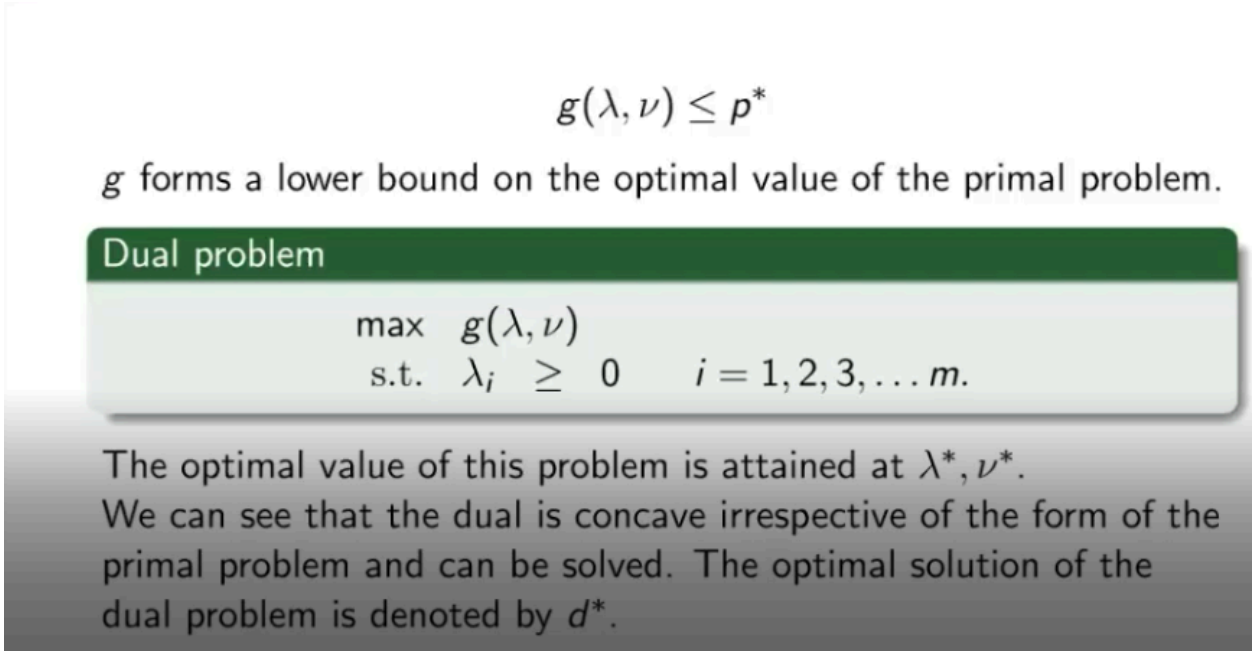


The optimal value of the problem is λ* and $\upsilon$ *, we can see that this function g is concave irrerespective of the form of the primal problem. So if you go back and see we started with the general form of primal problem and we achieve when reached with g which is concave. So g can always be solved the optimum value of the dual problem is denoted by d* so now we would like to see how far is this d* from the actual value p*. So p* - d* is known duality gap

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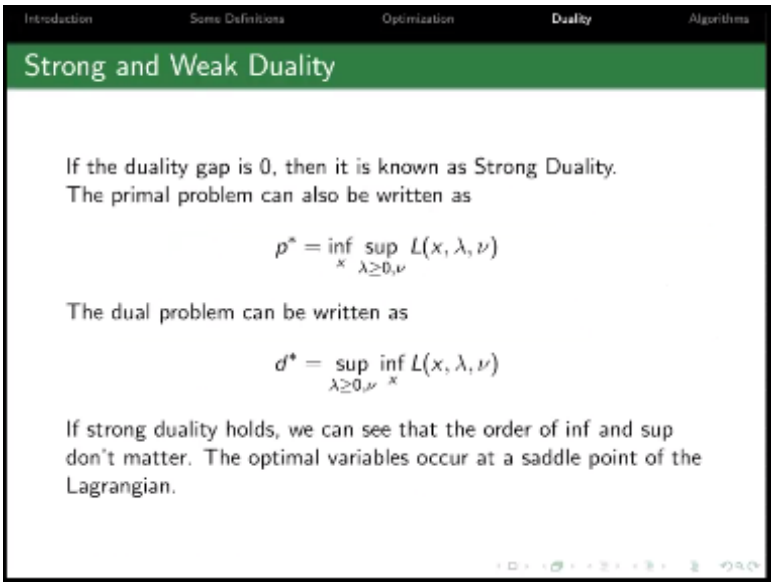


The next obvious question is to find out when this p* -d* will be 0 and when it is not so whenever it is 0 it is known as the strong duality and when it is not it is known as the weak duality. So next we will try to further characterize when what can occur. So first decide we can see that p* can be written as

The primal problem can also be written as

$$p^* = \inf_x \sup_{\lambda \geq 0, \nu} L(x, \lambda, \nu)$$

d* can be written as

The dual problem can be written as

$$d^* = \sup_{\lambda \geq 0, \nu} \inf_x L(x, \lambda, \nu)$$

So when this strong duality holds we know that p* = d*. So you can see that the order of the infimum supremum can be interchanged and it is equivalent. So this means that at the same point we have maxima in one direction and the minima in another direction. So it is a saddle point, so we have one good result here, that is whenever that is strong duality optimal variables occur at the saddle point of the Lagrangian.

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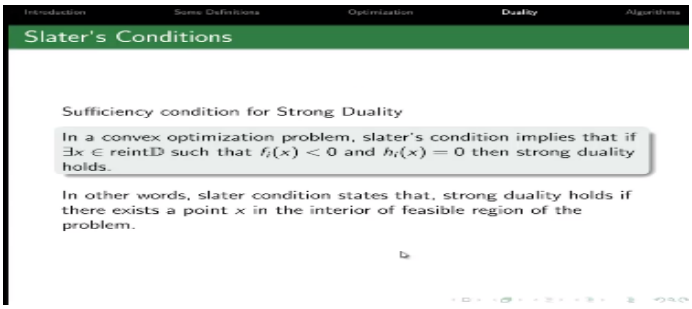


Now let us look at sufficiency conditions for strong duality. So we look at slater's conditions which gives us conditions for a convex optimization problem to be strongly dual. So Slater's conditions states that for a convex optimization problem, there exists an x, such that it belongs to the relative interior of the domain such that $f_i(x) < 0$ and $h_i(x) = 0$, then strong duality holds. So here we require the inequality constrains to be strongly strictly unequal and the function should be the point should belong to the relative interior and not the boundary.

So Slater's conditions state that for any convex optimization problem if that exits a point inside the feasible region then strong duality surely holds. So note that this is only for convex optimization problems and not a general result.

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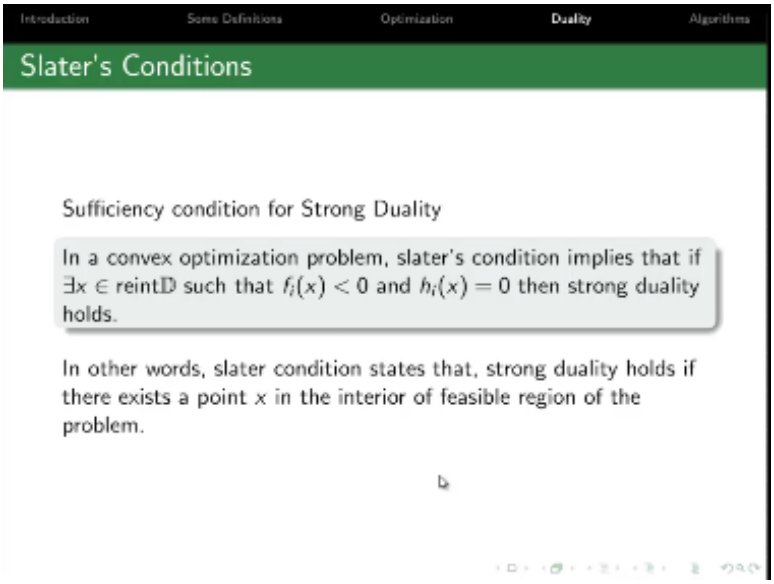


So now we look at complementary slackness. Assume strong duality holds and x* is the primal variable and λ* also dual variables. So when I say strong duality holds, we know that

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$$
\begin{aligned}
f_0(x^*) = g(\lambda^*, \nu^*) &= \inf_x \left(f_0(x) + \sum_{i=1}^{i=m} \lambda_i^* f_i(x) + \sum_{i=1}^{i=p} \nu_i^* h_i(x)\right) \\
&\leq f_0(x^*) + \sum_{i=1}^{i=m} \lambda_i^* f_i(x^*) + \sum_{i=1}^{i=p} \nu_i^* h_i(x^*) \\
&\leq f_0(x^*)
\end{aligned}
$$

So by expanding g by this definition and looking at some simple inequities, we can reach to a conclusion that for all i, $\lambda_i^* f_i(x^*)$ should be 0. Okay so basically we know that $f_i(x^*)$ is $\leq 0$ because x* is a feasible value. So whenever $f_i(x^*)$ is not equal to 0 we know that $\lambda_i = 0$. So this is known as complementary slackness. That is either $\lambda x^* = 0$ or $f_i(x^*) = 0$ then strong duality holds.

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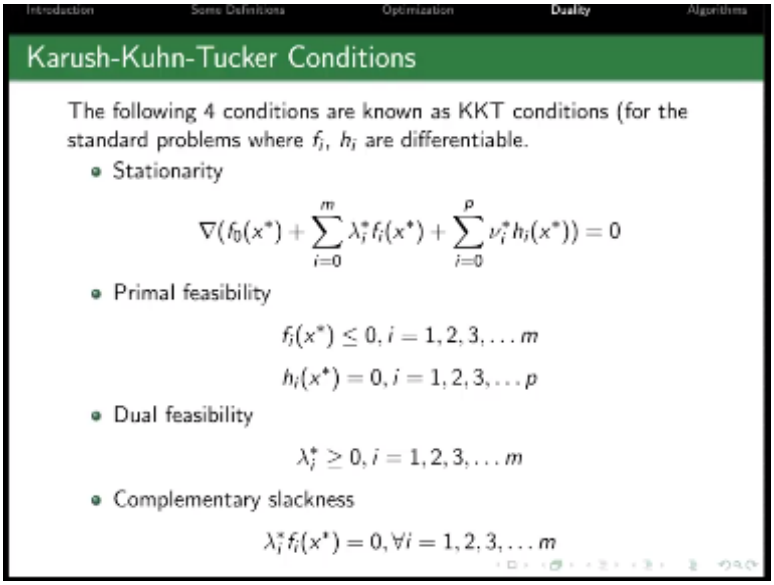


Now we will look at Karush Kuhn Tucker conditions, also known as KKT conditions. So these provide us the necessary conditions for a point x*λ* $\upsilon^*$ to be optimal. So consider any point x* λ* $\upsilon^*$ if it has to be a optimization these things have to be satisfied. So first stationarity so since you already seen that at the optimal point l has a saddle point so the gradient at that point should be 0 so that is trivial to c and then primal feasibility and dual feasibility should hold and then you have seen complementary clackness as you seen previously we also be valid at this point.

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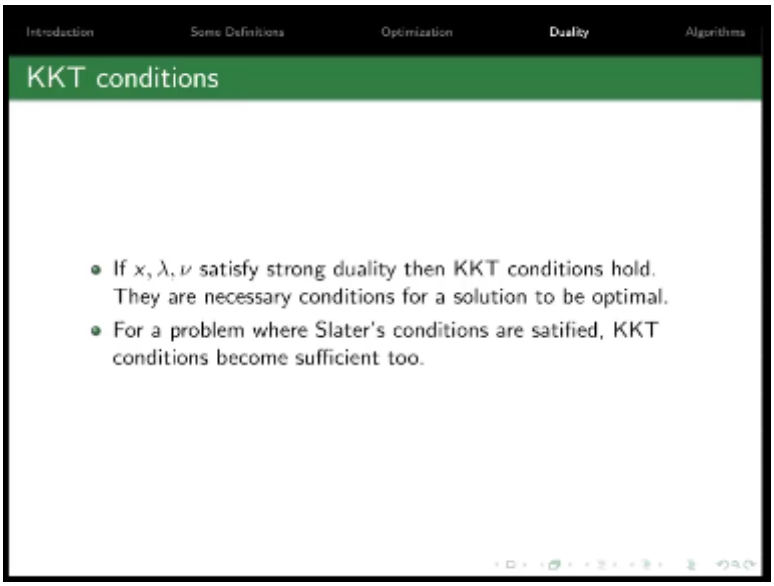


So just to reiterate what you already seen if x, λ, ν satisfy strong duality then KKT conditions hold. So these are just necessary conditions and sufficient but for our optimization problems when Slater's conditions are satisfied then KKT became sufficient also.

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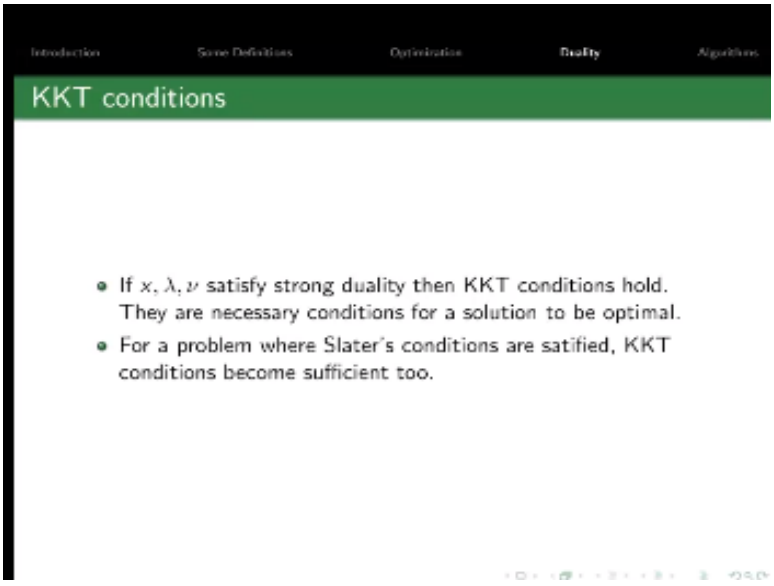

Now we look at some examples first is the most popular example of least squares, so we are trying to minimize a least square function so we are trying to minimize a least square function so we are trying to minimize this $\min \| Ax-b \|_2^2$ with no constraints so we can clearly say that this a convex function and there is no constraints and we solved in this thing in while solving linear regression to give x* as $(A^T A)^{-1} A^T b$. So this is a very trivial convex optimization problem which you are able to solve but just by differentiating.

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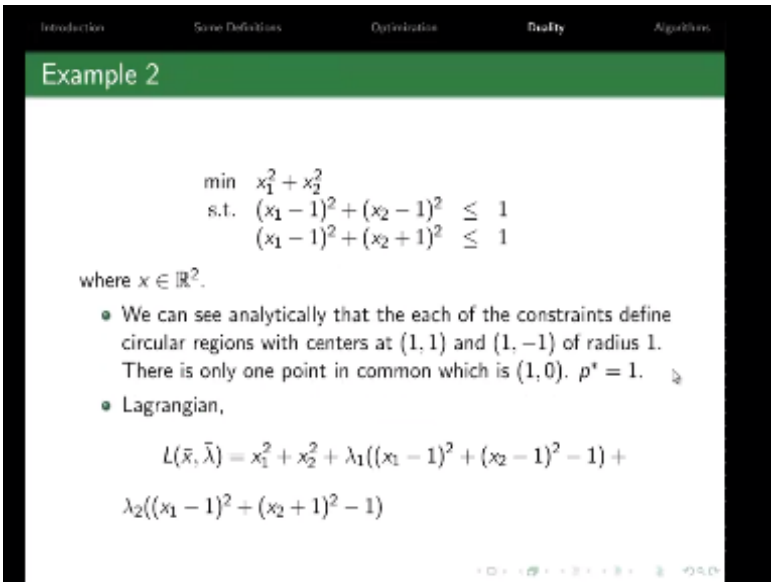


Now let us look at another example, so here we are trying to minimize $x_1^2 + x_2^2$ subject to these two linear quadratic constraints. So you look at these constraints carefully both of them are circular regions one centered at (1, 1) and the other centered at (1, -1). Each of which is radius at one. So if you just plot them and see that you can see that there is only one feasible point that is (1, 0) so trivially the optimal value will become one. But now let us do analysis which we have learnt and how to do and then try to arrive at the same answer, so first when you have a convex of machine value or for that matter optimization problem, the first thing you do is write that like Lagrangian so here the lagrngian will be

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- Lagrangian,

$$L(\bar{x}, \bar{\lambda}) = x_1^2 + x_2^2 + \lambda_1((x_1 - 1)^2 + (x_2 - 1)^2 - 1) + \lambda_2((x_1 - 1)^2 + (x_2 + 1)^2 - 1)$$

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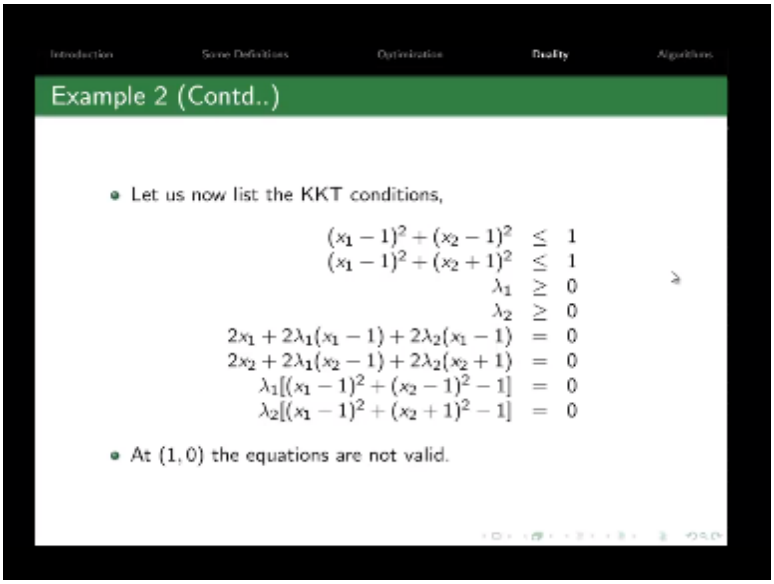


So now that you have seen the lagrangian let us try to list out the KKT conditions. So here the first two are the primary feasibility conditions second two are the dual feasibility conditions and the next two are obtained by differentiating the lagrangian with $x_1$ and $x_2$ respectively and the next two are obtained by writing the complimentary slackness equations. And we have seen that there is only one feasible point (1, 0) and at that point these conditions are not valid. You get contradictory answers for $\lambda_1$ and $\lambda_2$ and you try to solve. See that is KKT conditions are not

valid. But this is tricky so we have already seen that we have an optimal value but KKT conditions are not satisfied. We will try to see why this is happening here. Now let us try to investigate what exactly is happening so we will try to solve that your problem now.

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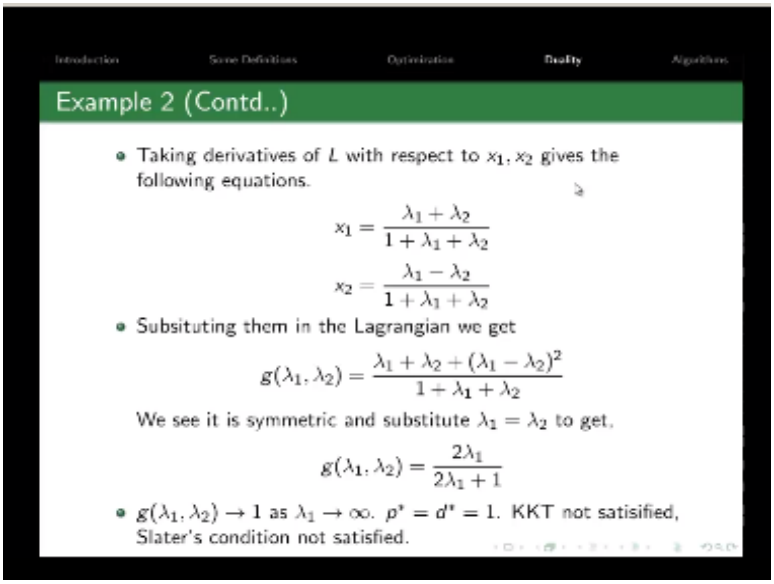


So for solving the dual problem we have find the maxima for g. So first let us find out what the function g is. g is in few form of the lagrangian over x. So we will substitute we will try to take the derivative of l with respect to x solve it and then if we arrive at this g function which is the function of λ1 and λ2 now you can see that this is a concave function which is symmetric λ1 and λ2 so we can substitute this λ1 =λ2= λ1 and the go ahead. So when we do that we get this $2\lambda_1 / (2\lambda_1 + 1)$ as that g function. So if you see that under the limit λ1 tending into ∞, g tends to 1 but otherwise there is no maxima achieved. So under a simple conditions p* = d* = 1 and because this is these points are not been attained at point KKT conditions and Slater's conditions are not satisfied. So this example is just to show you that just solving KKT conditions or checking first latest condition is not sufficient we might have to solve sometimes the dual problem and see what exactly is happening.

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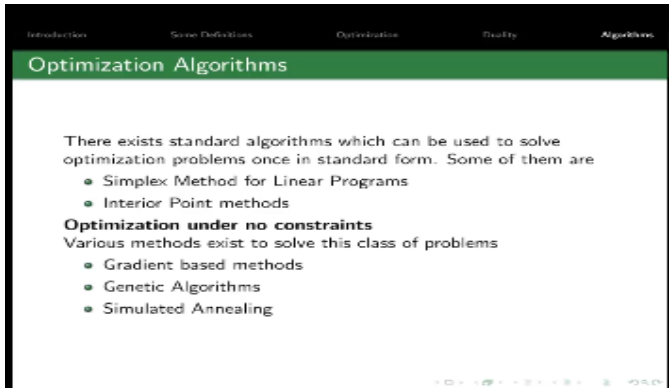


So we have seen the mathematical characterization for optimization problems, now we will try to see how to solve them. So there exists very many standard algorithms to solve optimization problems once you taken them to standard form. So for linear programs there is this feel known simplex method and the most popular methods for solving general optimization problems right now are interior point methods. We will not be covering in these methods in detail at all will be looking at simpler class problems that is, optimization under no constraints. So that is given an objective function under no constraints, how can we solve this? We will look at algorithms, so to do this there exist a lot of algorithms; gradient based methods, genetic algorithms and simulated annealing. First we will look at gradient based methods which are very popular used in machine learning.

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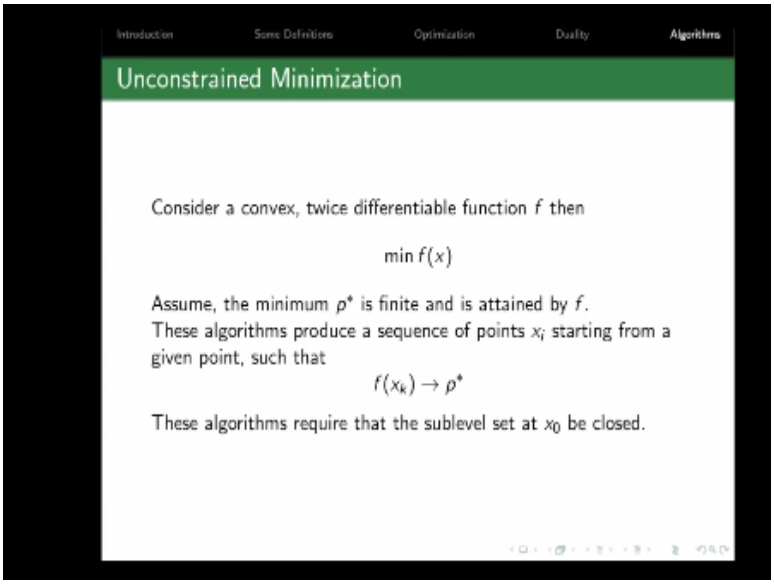

So first let us look at proper mathematical definition of unconstrained minimization. Consider a convex which is twice differentiable function is and we want to find minimum of this function. So assume p* is minimum and it is finite and it is attained by f, so we want algorithms, start from some point and give a series of $x_i$'s, such that value of $f(x_k)$ tends to this optimal minimum. So these algorithms required one condition that is the sublevel set should be closed.

So what exactly this condition means is, so when I start from $x_0$ and I go to some other point is which is < then so basically each time I am trying to reduce the value of f. So x1,x0 to x1 where $f(x_1)$ < then $f(x_0)$. So that is, this belongs to the subset of $f(x_0)$ and this point $x_1$ should be inside the set. So we just need this condition, so that we get a chain of points, which are in the domain of the function.

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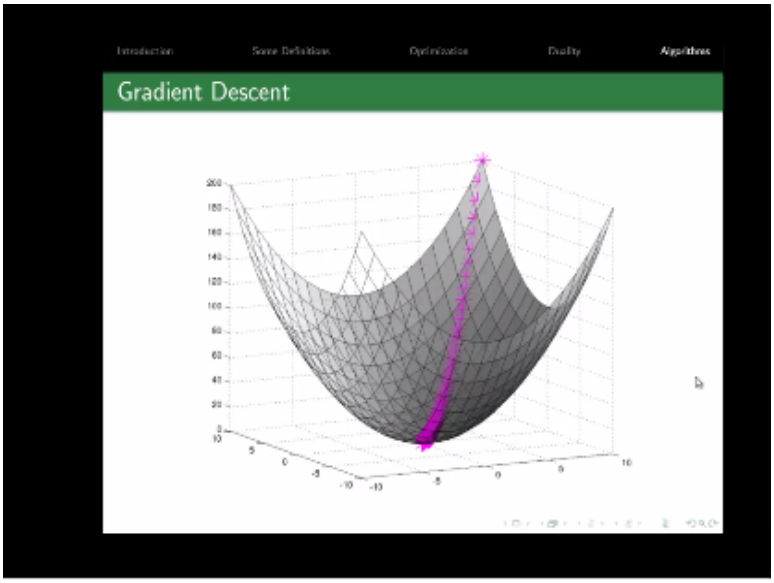


So now we will look at what are the most popular algorithms gradient descent. So this works in convex problems where there exist in minimum and you start from one point from the top and go down according to the gradient, so if you see this visualization gives you the 3-dimensional surface, which is basically $f(x_1,x_2)$ say. So if we start of f(x1, x2) at the top point and we take the gradient there and move along the negative direction slowly.

As we keep going down we reach the bottom of this, so and the bottom is where the minima exist, at the last point the gradient become 0. So this is the motivation for gradient descent algorithms that is by going along the negative direction of the gradient, we reach the minima in convex functions.

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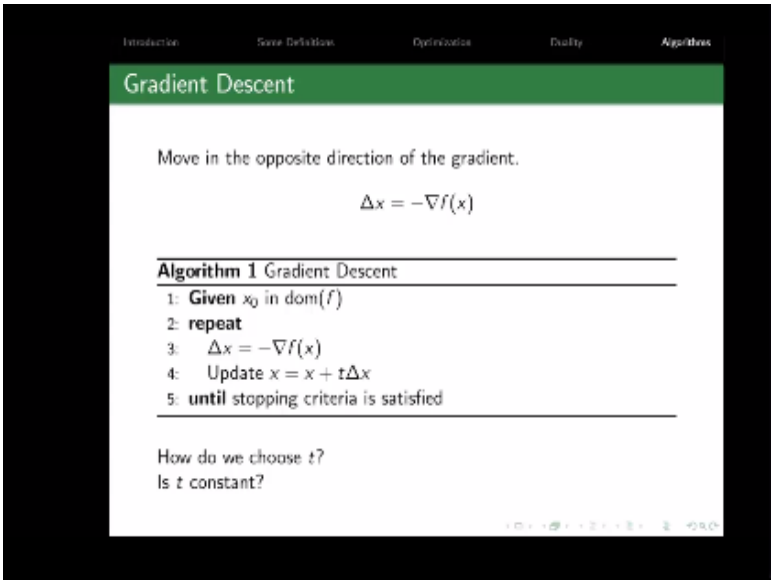


So let us formally look at the gradient descent, we intuitively seen that if you move in the direction opposite to the gradient we will reach the minima, so will state that as an algorithm,. So if we start $x_0$ in the domain of f, you can update in every iteration x as x = x+ tΔx. so essentially what we are doing is, we are moving along the negative direction of the function in some step size of t, basically this t is the multiplicative factor, which will magnify or minimize step size that you are taking in the direction.

So the next question is how do we choose t? Should t be constant? So in the ideal case t should be depended on the curvature of the functions? So if you look at the graph in the previous slide carefully, so where ever there is low curvature you could afford to take larger steps, where ever there is high curvature at the bottom especially where there are minima, you should take small steps, so you do not jump over the minima value.

Methods which choose t according this are out of the scope of this tutorial, so but we will just answer this question, is t constant is enough for us. In most conditions a small t if you take a small enough step size it is fine and you will very reasonably very close to the minima. So in practice the constant t works. So we will end this tutorial session with this. So the main take home message from this tutorial session should be, what optimization problems are? What is the generic form? What are convex optimization problems? What is duality? What is strong duality? What is KKT and Slater's conditions?

So knowing these will be enough for you to navigate, whatever the optimization that come across this course but ideally we can look up other resource online if you are not clear with these basics still.

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**Introduction to Machine Learning**

**Lecture 1**

**Prof. Balaraman Ravindran**
**Computer Scince and Engineering**
**Indian Institute of Technology Madras**

**Introduction to Machine Learning**

Hello everyone and welcome to this NPTEL course on an introduction to machine learning. In this course, we will have a quick introduction to machine learning and this will not be very deep in a mathematical sense but it will have some amount of mathematical rigor. And what we will be doing in this course is covering different paradigms of machine learning with special emphasis on classification and regression tasks. Also we will introduce you to various other machine learning paradigms.

In this introductory lecture set of lectures, I will give a very quick overview of the different kinds of machine learning paradigms and therefore I call these lectures machine learning, a brief introduction with emphasis on "brief".

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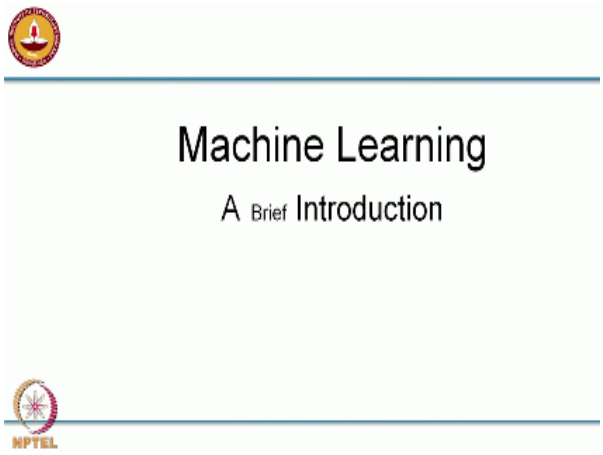

So the rest of the course would be a more elongated introduction to machine learning.

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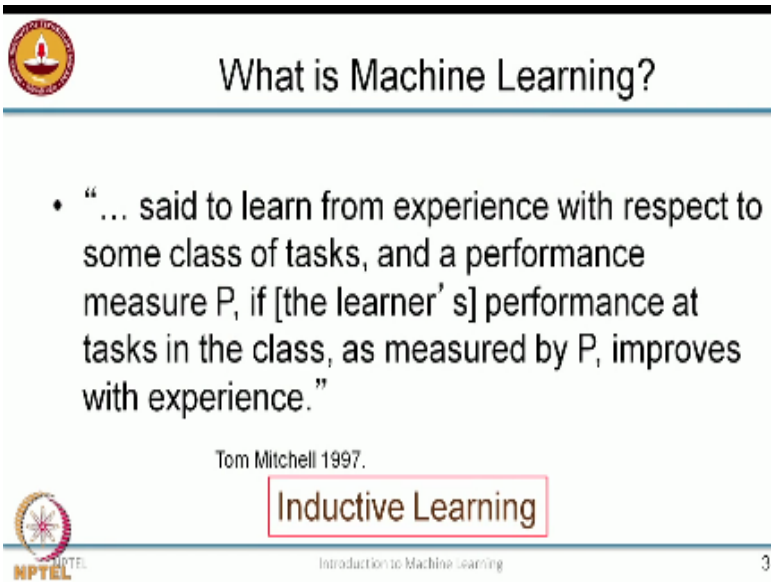


So what is machine learning? So I will start off with a canonical definition put out by Tom Mitchell in '97. So a machine or an agent (I deliberately leave the beginning undefined because you could also apply this to non-machines like biological agents). So an agent is said to learn from experience with respect to some class of tasks and the performance measure P if the learner's performance tasks in the class as measured by P improves with experience.

So what we get from this first thing is we have to define learning with respect to a specific class of tasks. It could be answering exams in a particular subject or it could be diagnosing patients of a specific illness. But we have to be very careful about defining the set of tasks on which we are going to define this learning, and the second thing we need is of a performance measure P. So in the absence of a performance measure P you would start to make vague statement like "Oh I think something is happening or there seems to be a change in something learned; There is some learning going on and stuff like that".

So if you want to be clearer about measuring whether learning is happening or not you first need to define some kind of performance criteria. So for example if you talk about answering questions in an exam your performance criterion could very well be the number of marks that you get or if you talk about diagnosing illness then your performance measure would be the number of patients that you say are the number of patients who did not have adverse reaction to the drugs you gave them. There could be variety of ways of defining performance measures depending on what you are looking for and the third important component here is experience.

So with experience the performance has to improve and so what we mean by experience here in the case of writing exams is that it could be writing more exams. So the more the number of exams you write the better you write it, the better you get at test taking or it could be, in the case of diagnosing illnesses like the more patients that you look at the better you become at diagnosing illness.

So these are the three components. So you need a class of tasks, you need a performance measure and you need some well-defined experience. So this kind of learning where you are learning to improve your performance based on experience is known as inductive learning.
The basis of inductive learning goes back several centuries. People have been debating about inductive learning for hundreds of years now and are only more recently we have started to have more quantified mechanisms of learning. But one thing I always point out to people is that you should take this definition with a pinch of salt. For example you could think about the task as fitting your foot comfortably. So you could talk about whether a slipper fits your foot comfortably or let me put it this way: the slipper is supposed to give protection to your foot and a performance measure for the slipper would be whether it is fitting the leg comfortably or not or whether it is you know as people say that it is biting your leg or is it chafing your feet. And with experience you know as the slipper knows more and more about your foot (as you keep wearing the slippers for longer periods of time) it becomes better at the task of fitting your foot. This is measured by whether it is chafing your foot or whether it is biting your foot or not.

So would you say that the slipper has learned to fit to your foot well by this definition? Yes. So we have to take this with a pinch of salt and so not every system that confirms to this definition of learning can be said to learn usually.

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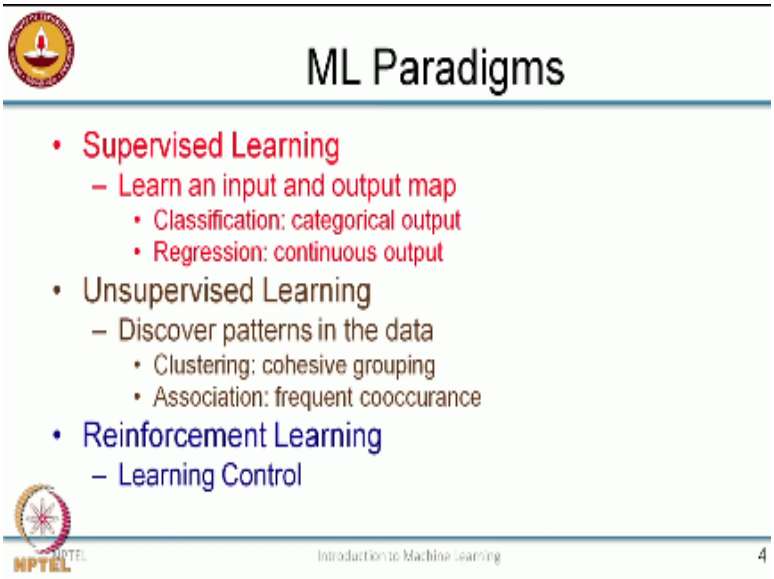


So going on so there are different machine learning paradigms that we will talk about and the first one is supervised learning where you learn an input to output map. So you are given some kind of an input. It could be a description of the patient who comes to comes to the clinic and the output that it has to produce is whether the patient has a certain disease or not. So we have to learn this kind of an input to output map or the input could be some kind of equation and then output would be the answer to the question or it could be a true or false question I give you a description of the question you have to give me true or false as the output.

And in supervised learning what you essentially do is a mapping from this input to the required output. If the output that you are looking for happens to be a categorical output like whether he has a disease or does not have a disease or whether the answer is true or false then the supervised learning problem is called the classification problem. And if the output happens to be a

continuous value like, so how long will this product last before it fails or what is the expected rainfall tomorrow, so these kinds of problems they would be called as regression problems.

These are supervised learning problems where the output is a continuous value and these are called as regression problems. So we will look at in more detail classification and regression as we go on. So the second class of problems are known as unsupervised learning problems, where the goal is not really to produce an output in response to an input but given a set of in data we have to discover patterns in the data. So that is called unsupervised learning – there is no real desired output that we are looking for, we are more interested in finding patterns in the data.

So clustering is one unsupervised learning task where you are interested in finding cohesive groups among the input pattern, For example I might be looking at customers who come to my shop and I want to figure out if there are categories of customers like so maybe college students could be one category and IT professionals could be another category and so on so forth and when I'm looking at these kinds of grouping in my data, so I would call that a clustering task.

So the other popular unsupervised learning paradigm is known as the Association rule mining or frequent pattern mining where you are interested in finding a frequent co-occurrence of items in the data that is given to you. So whenever A comes to my shop B also comes to my shop. So those kinds of co-occurrence, I can always say that if I see A then there is likely very likely that B is also in my shop somewhere. So I can learn these kinds of associations between data .

And again we look at this later in more detail. There are many different variants on supervised and unsupervised learning but these are the main ones that we look at so the third form of learning which is called reinforcement learning. It is neither supervised nor unsupervised in nature and typically these are problems where you are learning to control the behavior of a system and I will give you more intuition into reinforcement learning now in one of the later modules.

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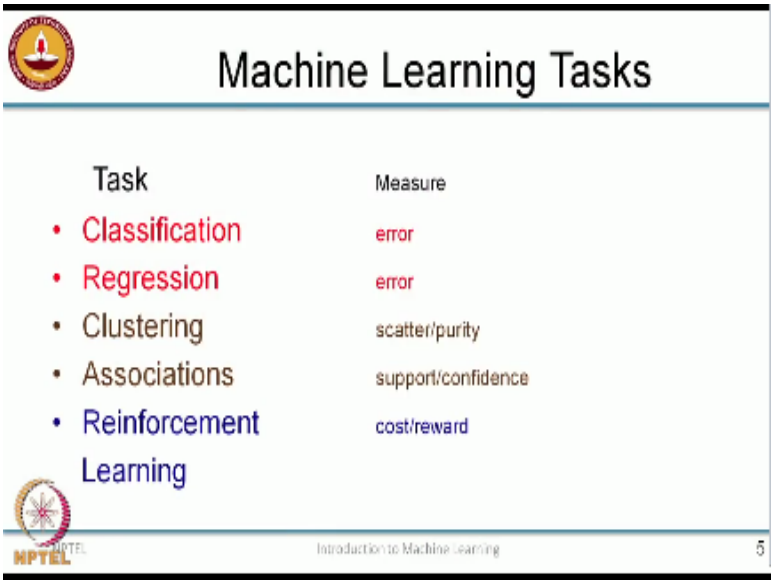


So like I said earlier, for every task, so you need to have some kind of a performance measure so if you are looking at classification the performance measure is going to be classification error so typically. So we will talk about many, many different performance measures in the duration of this course but the typical performance measure you would want to use this classification error – it's how many of the items or how many of the patients did I get incorrect so how many of them who are not having the disease did I predict they had the disease and how many of them that had the disease that I missed.

So that would be one of the measures that I would use and that would be *the* measure that we want to use but we will see later that often that is not is not possible to actually learn directly with respect to this measure. So likewise for regression again we have the prediction error. Suppose, I say it is going to rain like 23 millimeters and then it ends up raining like 49 centimeters. So that is a huge prediction error. And in terms of clustering, this is a little trickier to define performance measures. We don't know what a good clustering algorithm is because we do not know what how to measure the quality of clusters.

So people come up with all different kinds of measures and so one of the more popular ones is a scatter or spread of the cluster. That essentially tells you how spread out the points is that belong to a single group. If you remember we are supposed to find cohesive groups, so if the group is

not that cohesive, then not all of them are together and then you would say the clustering is of a poorer quality. If you have other ways of measuring things like I was telling you, so if you know that people are college students and then you can figure out that how many what fraction of your cluster or college students.

So you can do this kind of external evaluations. One measure that people use is known as purity. And in the Association rule mining we use variety of measures called support and confidence and it takes a little bit of work to explain support and confidence, so I will defer it till I talk about Association rules in detail. And in the reinforcement learning tasks, so if you remember I told you it is learning to control and you are going to have a cost for controlling the system and also the measure here is cost and you would like to minimize the cost that you are going to accrue while controlling the system. So these are the basic machine learning tasks.

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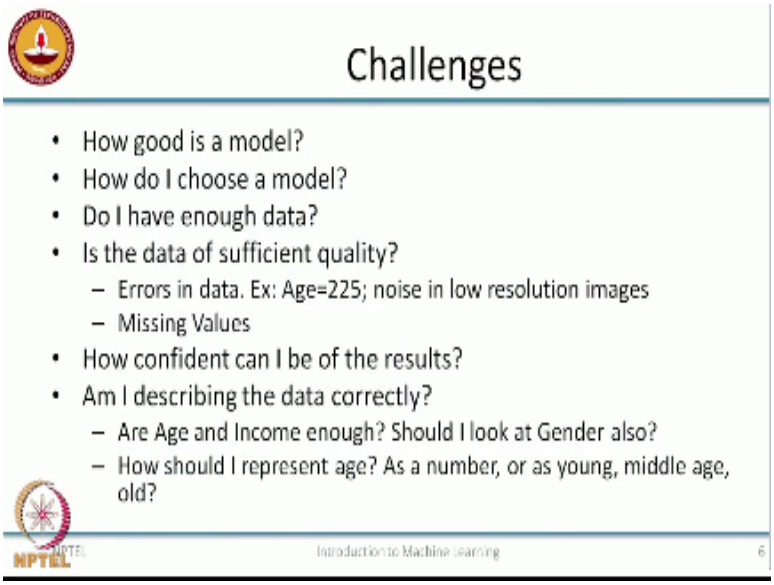


So there are several challenges when you are trying to build a build a machine learning solution and a few of these I have listed on this slide. The first one is you have to think about how good a model you have learned. So I talked about a few measures on the previous slide but often those are not sufficient there are other practical considerations that come into play and we will look at some of these towards the a middle of the course somewhere and the bulk of the time would be spent on answering the second question which is how do I choose a model .

So given some kind of data and experience, how I would choose a model that somehow learns what I want to do. So how that improves itself with experience and so on so how do I choose this model and how do I actually find the parameters of the model that gives me the answer.

So this is what we will spend much of our time on in this course and then there are a whole bunch of other things that you really have to answer to be able to build a useful machine learning or useful data analytics or data mining solutions. Questions like do I have enough data? Do I have enough experience to say that my model is good? Is the data of sufficient quality? There could be errors in the data. Suppose I have medical data and age is recorded as 225, so what does that mean it could be 225 days in which case it is a reasonable number it could be 22.5 years again is a reasonable number or 22.5 months is reasonable.

But if it is 225 years it's not a reasonable number. So there is something wrong in the data and how do you handle these things or noise in images or missing values (so I will talk briefly about handling missing values later in the course) but this is as I mentioned in the beginning is a machine learning course and this is primarily concerned about the algorithms of machine learning and the and the math and the intuition behind those and not necessarily about the questions of building a practical systems based on this.

So I will be talking about many of these issues during the course but it's just that I want to reiterate what the focus of the course is? And so the next challenge I have listed here is how confident can I be of the results and about that I certainly we will talk a little bit because the whole premise of reporting machine learning results depends on how confident you can be of the results and the last question is whether am I describing the data correctly.

So that is a very, very domain dependent question and the question that you can answer only with your experience as a machine learning or a data sciences professional or with time, but these are typical questions that you would like to ask that are on the slides and from the next module, we will look at the different learning paradigms in slightly more detail.

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**Introduction to Machine Learning**

**Lecture 2**

**Prof. Balaraman Ravindran**
**Computer Science and Engineering**
**Indian Institute of Technology Madras**

**Supervised Learning**

So in this module we will look at supervised learning .

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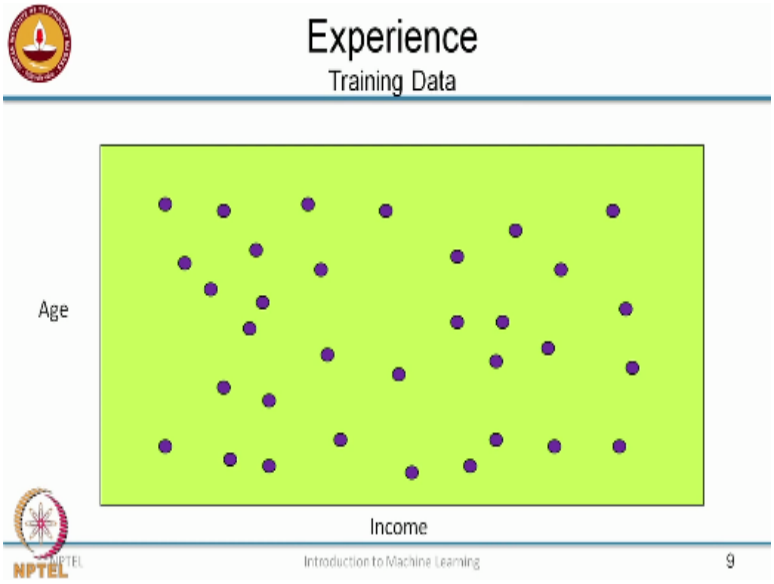


If you remember in supervised learning we talked about experience where you have some kind of a description of the data. So in this case let us assume that I have a customer database and I am describing that by two attributes here, age and income. So I have each customer that comes to my shop I know the age of the customer and the income level of the customers.

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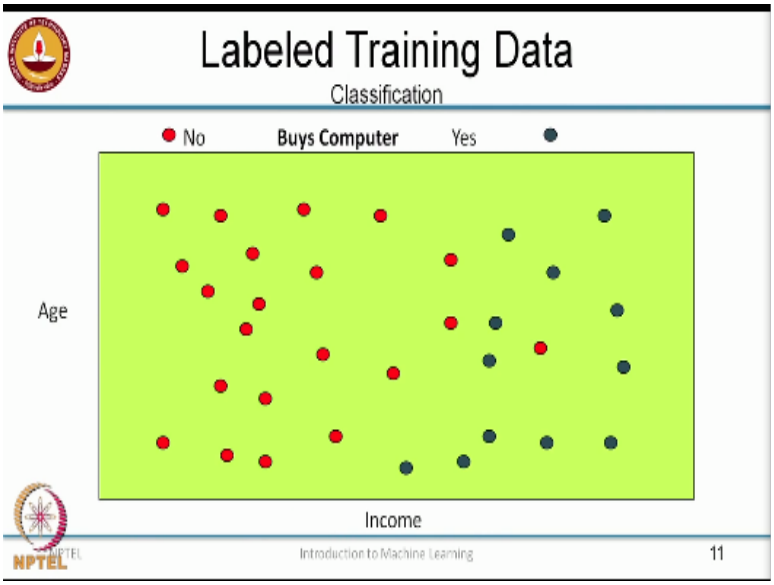


And my goal is to predict whether the customer will buy a computer or not buy a computer. So I have this kind of labeled data that is given to me for building a classifier, remember we talked about classification where the output is a discrete value in this case it is yes or no, "yes" means this person will buy a computer, "no" means the person will not buy a computer. And the way I describe the input is through a set of attributes in this case we are looking at age and income as the attributes that describe the customer.

And so now the goal is to come up with a function, come up with a mapping that will take the age and income as the input and it will give you an output that says the person will buy the computer or not buy the computer. So there are many different ways in which you can create this function.

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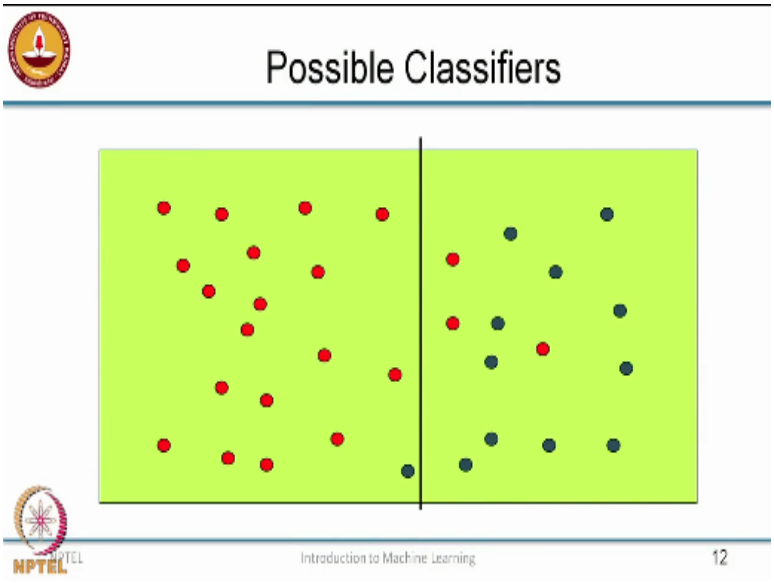


And given that we are actually looking at a geometric interpretation of the data, I am looking at data as points in space, the one of the most natural ways of thinking about defining this function is by drawing lines or curves on the input space. So here is one possible example, so here I have drawn a line and everything to the left of the line in red, would be classified as will not buy a computer, everything to the right of the line where the predominantly the data points are blue will be classified as will buy a computer.

So how would the function look like? It will look like something like this. If the income of a person (remember that the x-axis is income and the y-axis is age). So in this case it basically says that if the income of the person is less than some value (less than some X) then the person will not buy a computer. If the income is greater than X the person will buy your computer. So that is the kind of a simple function that we will define.

Just notice that way we completely ignore one of the variables here which is the age. So we are just going by income, if the income is less than some X then the person will not buy a computer, if the income is greater than X the person will buy a computer. So is this a good rule more or less I mean we get most of the points correct except a few. So it looks like we can survive with this rule. So this is not too bad, but then you can do slightly better.

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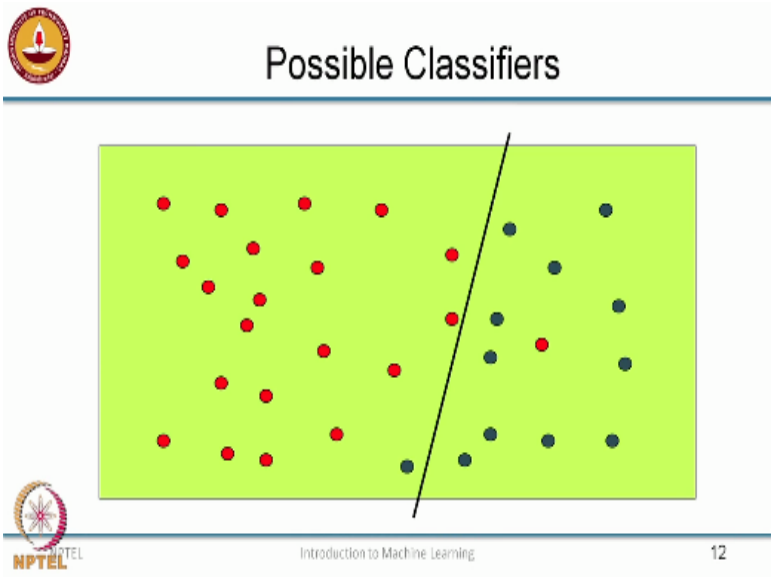


So now we got those two red points that those just keep that points are on the wrong side of the line earlier now seem to be on the right side . So everything to the left of this line will not buy a computer, everything to the right will buy a computer. So if you think about what has happened here, so we have improved our performance measure.

So the cost of something, so what is the cost here? So earlier we are only paying attention to the income, but now we have to pay attention to the age as well. So the older you are, the income threshold at which we will buy a computer is higher. So the younger you are, younger means lower on the y axis, so the younger you are the income threshold at which you will buy a computer is lower.

So is that clear, so the older you are, the income threshold is shifted to the right here. So the older you are, you need to have a higher income before you buy a computer and the younger you are your income threshold is lower, so you do not mind buying a computer even if your income is slightly lesser. So now we have to start paying attention to the age, but then the advantage is you get much better performance .

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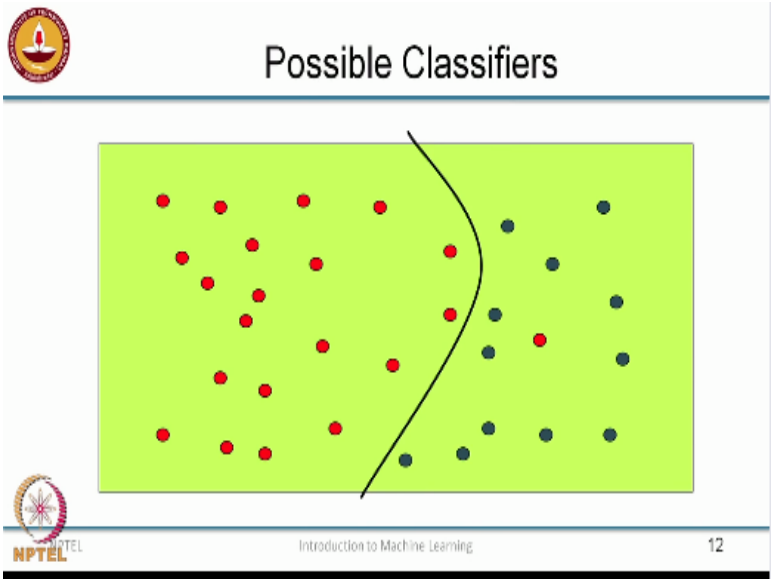


Can you do better than this yes? Now almost everything is correct except that one pesky red point, but everything else is correct. And so what has happened here we get much better performance, but at the cost of having a more complex classifier.

So earlier if you thought about it in geometric terms, so first you had a line that was parallel to the y-axis therefore, I just needed to define a intercept on the x-axis. So if X is less than some value then it was one class was greater than some value was another class. Then the second function it was actually a slanting line like that, so I needed to define both the intercept and the slope. And now here it is now a quadratic so I have to define three parameters. So I have to define something like $ax^2+ bx+c$. So I have defined the a, b and c – the three parameters in order to find the quadratic, and I am getting better performance.

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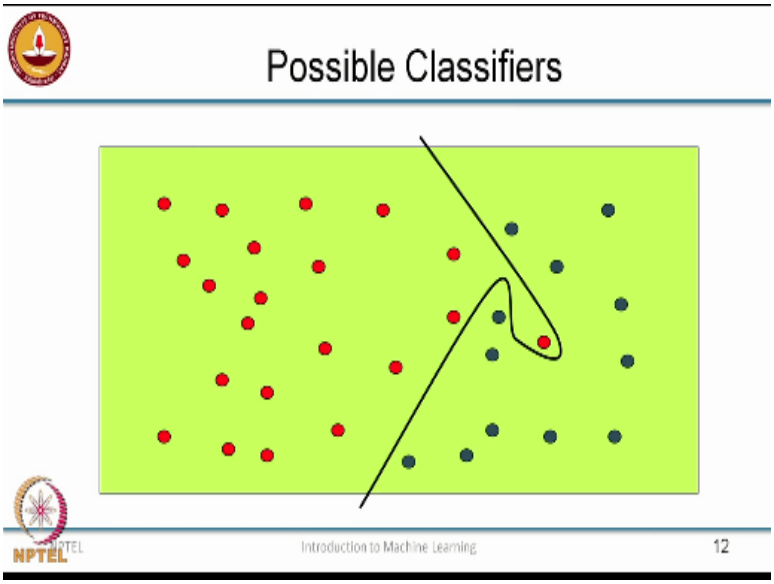


So can you do better than this? That somehow does not seem correct. It seems to be too complex a function just to be getting this one point there. And I am not even sure how many parameters you need for drawing that because Microsoft PowerPoint use some kind of spline interpolation to draw this curve I am pretty sure that it is got lot more parameters than it is worth. Another thing to note here is that that particular red point that you see is actually surrounded by a sea of blue.

So it is quite likely that there was some glitch there either the person actually bought a computer and we have not recorded as "having bought a computer". Or there are some extremist reasons such as the person comes into the shop pretty sure that he is going to buy a computer but then gets a phone call saying that some emergency please come out immediately and therefore he left without buying a computer. There could be variety of reasons for why that noise occurred and this will probably be the more appropriate classifier.

So these are the kinds of issues I would like to think about what is the complexity of the classifier that I would like to have and versus the accuracy of the classifier. So how good is the classifier in actually recovering the input-output map and or are there noise data in the input. In the experience that I am getting is it clean or is there noise on it and if so how do I handle that noise these are the kinds of issues that we have to look at.

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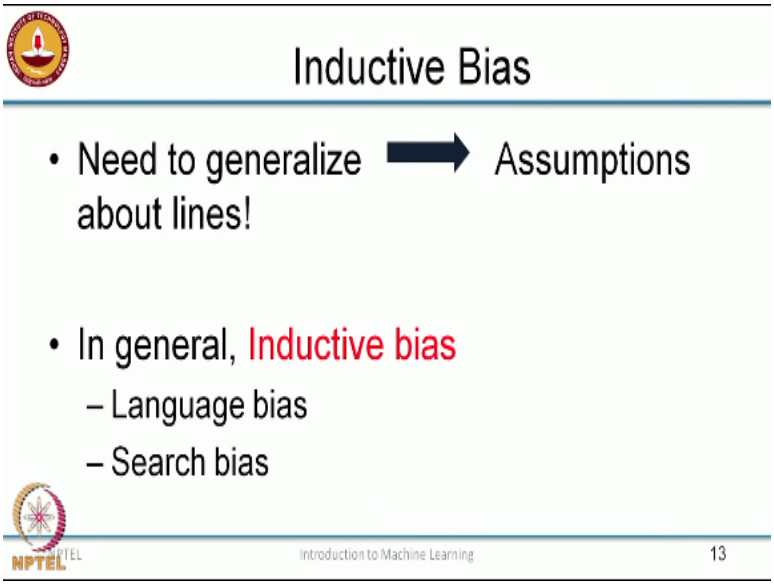


So these kinds of lines that we drew kind of hiding one assumption that we are making. So the thing is the data that comes to me comes as discrete points in the space and from these discrete points in the space I need to generalize and be able to say something about the entire state space so I do not care where the data point is on the x and y-axis I should be able to give a label to that.

If I do not have some kind of assumption about these lines, the only thing I can do is if the same customer comes again or somebody who has exact same age and income comes again, I can tell you whether the person is going to buy a computer or not buy a computer but I will not be able to tell you about anything else outside of the experience.

So the assumption we made is everything to the left of a line is going to do one thing or the other; so everything to the left of the line will not buy the computer everything to the right or everyone to the right will buy a computer. This is an assumption I made about the lines to enable me to segregate people who buy from who do not buy. So that is a kind of an assumption I made about the distribution of the input data and the class labels.

So this kind of assumptions that we make about these lines are known as inductive biases. In general inductive bias has like two different categories one is called language bias which is essentially the type of lines that I am going to draw. Am I going to draw straight lines or am I going to draw curves and what order polynomials am I going to look at and so on so forth these for my language bias. And search bias is the other form of inductive bias that tells me how or in what order am I going to examine all these possible lines? So that gives me a search bias. So putting these things together we are able to generalize from a few training points to the entire space of inputs. I will make this more formal as we go on in the next set of modules.

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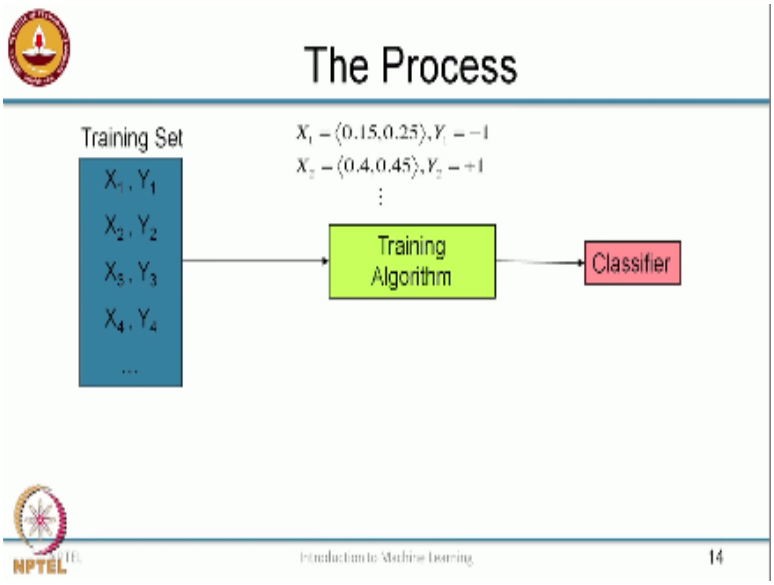


And so here is one way of looking at the whole process. I am going to give you a set of data which we will call the training set. So the training set will consist of an input which we'll call as X and an output which we call as Y. So I am going to have a set of inputs $X_1$, $X_2$, $X_3$, $X_4$ likewise I will have $Y_1$, $Y_2$, $Y_3$, $Y_4$ and this data is fed into a training algorithm. So the data is going to look like this in our case.

So remember our X's are the input variables. In this case that should have the income and the age, so $x_1$ is like 30,000 and 25 and $x_2$ is like 80,000 and 45 and so on so forth and the y's or the labels they correspond to the colors in the previous picture. So $y_1$ does not buy a computer $y_2$ buys a computer and so on so forth. This essentially gives me the color coding so $y_1$ is essentially red and $y_2$ is blue and I really if I am going to use something numeric this is what we will be doing later on I really cannot be using these values. First of all the X's vary too much .

So the first coordinate in the X is like 30,000 and 80,000 and so on so forth and the second coordinate is like 25 and 45 so that is a lot a lot smaller in magnitude so this will lead to some kind of numerical instabilities. So what will typically end up doing is normalizing these so that they form approximately in the same range so you can see that I have try to normalize these X values between 0 and 1 .

So I have chosen an income level of say 2 lakhs, it is the maximum and age of 100 and you can see the normalized values and likewise for buys and not buy I have taken not by as - 1 and by as computer is + 1. These are arbitrary choices, now but later on you will see that there are specific reasons for wanting to choose this encoding in this way. And then the training algorithm chugs over this data and it will produce a classifier so now this classifier I do not know whether it is good or bad. So we had a straight line in the first case. An axis parallel line if we did not know the good or bad and we needed to have some mechanism by which we evaluate this.

So how do we do the evaluation? Typically you have what is called a test set or a validation set so this is another set of x and y pairs like we had in the training set, so again in the test set we know what the labels are it is just that we are not showing it to the training algorithm. We know what the labels are because we need to use the correct labels to evaluate whether your training algorithm is doing good or bad. So this process by which this evaluation happens is called validation. Then at the end of the validation, if you are happy with the quality of the classifier we can keep it. If you are not happy then go back to the training algorithm and say hey I am not happy with what you produced give me something different, so we have to either iterate over the algorithm again we will go over the data again and try to refine the parameter estimation or we could even think of changing some parameter values and then trying to redo the training

algorithm all over again. But this is the general process and we will see that many of the different algorithms that we will look at during the course of these lectures actually follow this kind of a process.

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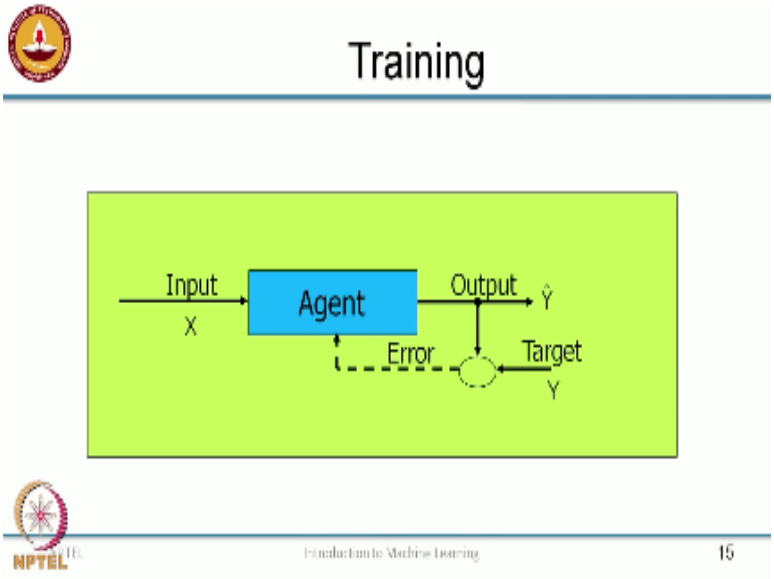


So what happens inside that green box? So inside the training algorithm is that there will be this learning agent which will take an input and it will produce an output $\hat{Y}$, which it thinks is the correct output, but it will compare it against the actual target Y it was given for the in the training. So in the training you actually have a target Y, so it will compare it against a target and then figure out what the error is and use the error to change the agent. So then it can produce the right output next time around. This is essentially an iterative process so you see that input produce an output $\hat{Y}$ and then you take the target Y, you can compare it to the $\hat{Y}$, figure out what is the error and use the error to change the agent again . And this is by and large the way most of the learning algorithms will operate; most of the classification algorithms or even regression algorithms will operate and we will see how each of this works as, we go on .

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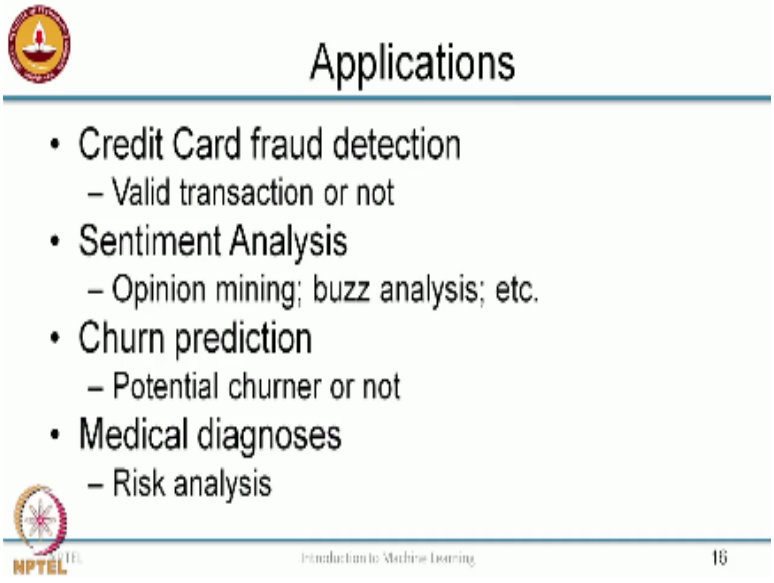


There are many, many applications. I mean this is too numerous to list. Here are a few examples you could look at say a fraud detection, so we have some data where the input is a set of transactions made by a user and then you can flag each transaction as a valid transaction or not. You could look at sentiment analysis you know variedly called as opinion mining or buzz analysis etc., where I give you a piece of text or a review written about a product or a movie and then you tell me whether the movies whether the review is positive or whether is negative and what are the negative points that people are mentioning about and so on so forth and this again a classification task. Or you could use it for doing churn prediction where you are going to say whether a customer who is in the system is likely to leave your system or is going to continue using your product or using your service for a longer period of time, so this is essentially churn. So when a person leaves your services you call the person churner and you can label what the person is churner or not. And I have been giving you examples form medical diagnosis all through apart from actually diagnosing whether a person has a disease or not you could also use it for risk analysis in the slightly indirect way I will talk about that when we when we do the algorithms for classification.

So we talked about how we are interested in learning different lines or curves that can separate different classes in supervised learning and, so this curves can be represented using different structures and throughout the course we will be looking at different kinds of learning mechanisms like artificial neural networks support vector machines decision trees nearest neighbors and Bayesian networks and these are some of the popular ones and we look at these in more detail as the course progresses.

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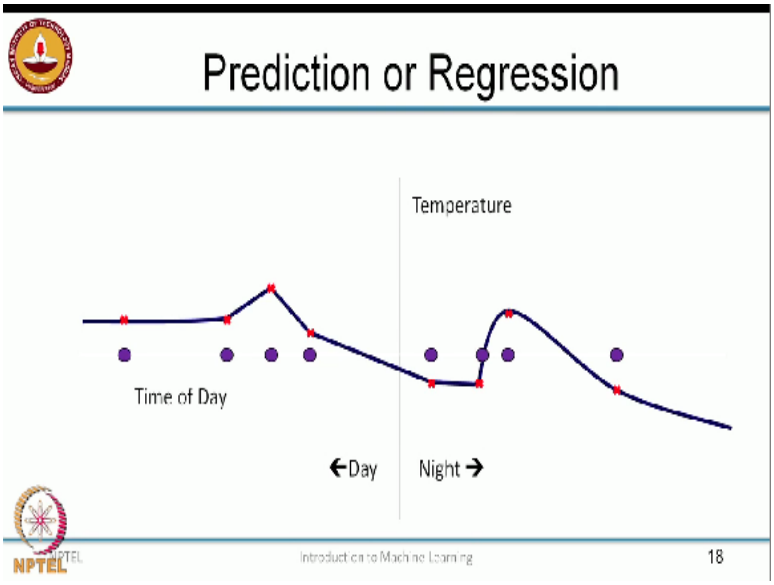


So another supervised learning problem is the one of prediction or regression where the output that you are going to predict is no longer a discrete value it is not like – will buy a computer or does not buy a computer – it is more of a continuous value so here is an example, where at different times of day you have recorded the temperature so the input to the system is going to be the time of day and the output from the system is going to be the temperature that was measured at a particular point at the time . So you are going to get your experience or your training data is going to take this form so the blue points would be your input and the red points would be the outputs that you are expected to predict.

So note here that the outputs are continuous or real value and so you could think of this in this toy example as points to the left being day and the points to the right being night . And just as in the previous case of classification, so we could try to do these as simple as possible fit in this case which would be to draw a straight line that is as close as possible to these points. Now you do see that like in the classification case when you choose a simple solution there are certain points at which we are making large errors, so we could try to fix that.

And try to do something fancier. But you could see that while the day time temperatures are more or less fine but with the night times we seem to be doing something really off. Because we are going off too much to the right-hand side. How if you could do something more complex just like in the classification case where we wanted to get that one point? So we could try and fit all these temperatures that were given to us by looking at a sufficiently complex curve.

And again this as we discussed earlier is probably not the right answer and you are probably in this case surprisingly better off fitting the straight line. So these kinds of solutions where we try to fit the noise in the data we are trying to make or the solution predict the noise in the training data correctly are known as over fitting. Over fit solutions are one of the things that we look to avoid in machine learning. So we will talk about this again in due course.

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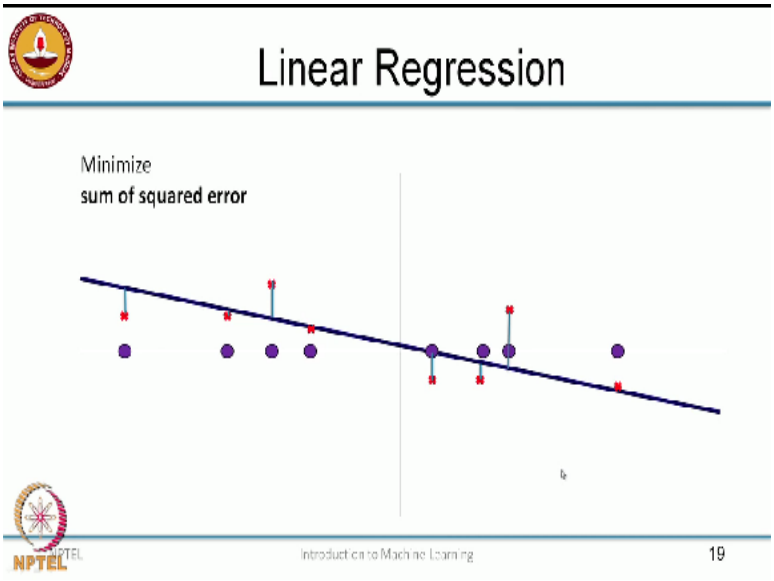


So now we would look at linear regression. Some of you might have come across this under different circumstances and the typical aim in linear regression is to say take the error that your line is making so if you take an example point, let us say I take any I take an example point somewhere here. So this is the actual training data that is given to you and this is the prediction that your line is making at this point so this quantity is essentially the prediction error that this line is making. And so what you do is you try to find that line that has the least prediction error so you take the square of the errors that your prediction is making and then you try to minimize the sum of the squared of the errors. Why do we take the squares? Because errors could be both positive or negative and we want to make sure that you are minimizing that regardless of the sign of the error.

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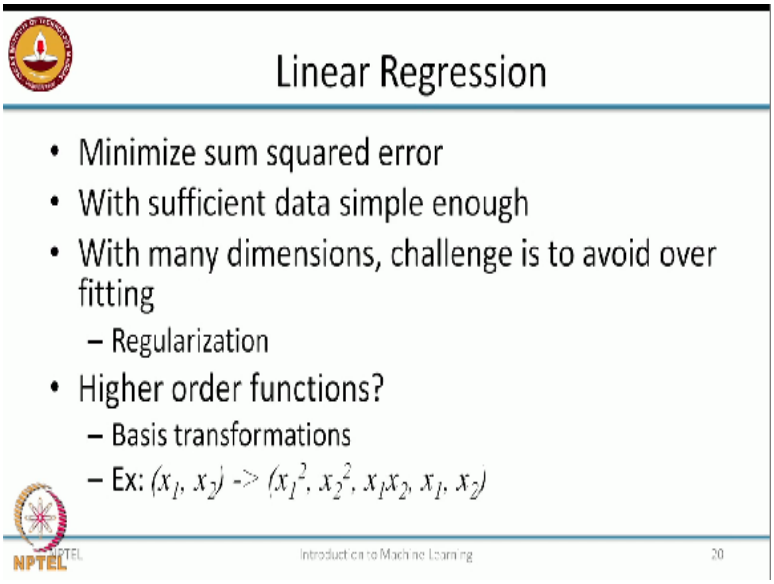


So with sufficient data a linear regression is simple enough you could just solve it using matrix inversions as we will see later but with many dimensions the challenge is to avoid over fitting like we talked about earlier and then there are many ways of avoiding this.

And so I will again talk about this in detail when we look at linear regression. So one point that I want to make is that linear regression is not as simple as it sounds. So here is an example so I have two input variables $x_1$ and $x_2$ and if I try to fit a straight line with $x_1$ and $x_2$ I will probably end up with something like

$$a_1 x_1 + a_2 x_2$$

And that looks like a plane in two dimensions.

But then if I just take these two dimensions and then transform the input so instead of saying just the $x_1$ and $x_2$ if I say my input is going to look like $x_1^2$, $x_2^2$ , $x_1x_2$ instead of the $x_1$ and $x_2$ as it was in the beginning. So instead of looking at a two-dimensional input, now I am going to look at a 5 dimensional input , and now I am going to fit a line or a linear plane in this 5 dimensional input so that will be like

$$a_1 x_1^2 + a_2 x_2^2 + a_3 x_1 x_2 + a_4 x_1 + a_5 x_2$$

Now that is no longer the equation of a line in two dimensions so that is the equation of a second-order polynomial in two dimensions but I can still think of this as doing linear regression because I am only fitting a function that is going to be linear in the input variables .

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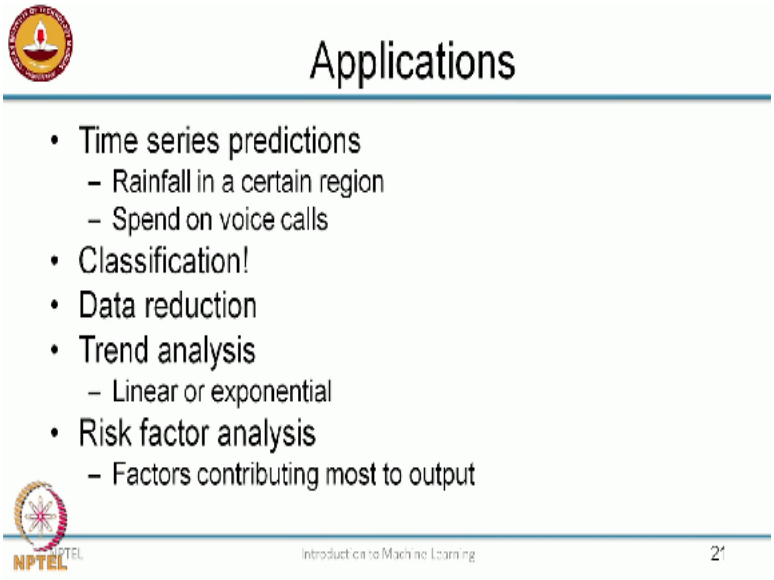


So by choosing an appropriate transformation of the inputs, I can fit any higher-order function so I could solve very complex problems using linear regression. So it is not really a weak method as you would think at first glance. Again, we will look at this in slightly more detail in the later lectures and our linear regression prediction can be applied in a variety of places – one popular place is in time series prediction you could think about predicting rainfall in a certain region or how much you are going to spend on your telephone calls you could think of doing even classification using this: if you remember our encoding of +1 and -1 for the class labels. So you could think of +1 and -1 as the outputs and then you can fit a regression line regression curve to that and if the output > 0 you would say this class is +1 its output < 0 you see the class is -1. So it could use the regression ideas to solve the classification problem and you could also do data reduction. So I really do not want to give you all the millions of data points that I have in my

data set but what I would do is essentially fit the curve to that and then give you just the coefficients of the curve .

And more often than not that is sufficient for us to get a sense of the data and that brings us to the next application I have listed there which is trend analysis. So I am not really interested in the actual values of the data but more in the, the trends. So for example I have a solution that I am trying to measure the running times of and I am not really interested in the actual running time because 37 seconds to 38 seconds is not going to tell me much.

But I would really like to know if the running time scales linearly or exponentially with the size of the input. So those kinds of analysis again can be done using regression. And in the last one here is again risk factor analysis like we had in classification and you can look at which are the factors that contribute most to the output so that brings us to the end of this module on supervised learning.

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**Introduction to Machine Learning**

**Lecture 3**

**Prof: Balaraman Ravindran**
**Computer Science and Engineering**
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**Unsupervised Learning**

Hello and welcome to this module on introduction to unsupervised learning, right. So in supervised learning we looked at how you will handle training data that had labels on it.

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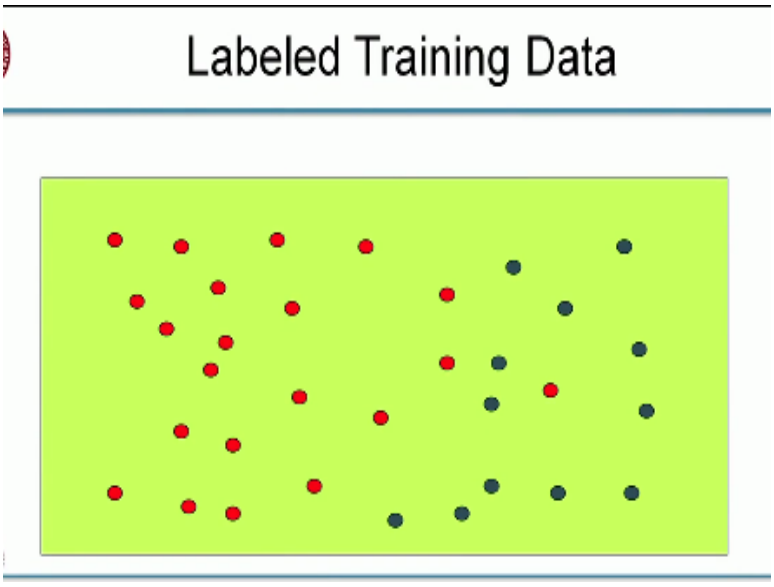


So this is this particular place this is a classification data set where red denotes one class and blue denotes the other class right.

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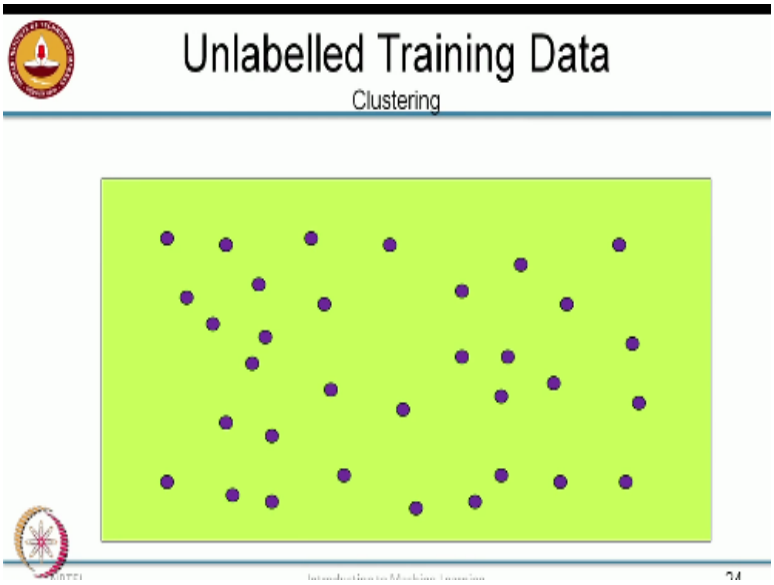


And in unsupervised learning right so you basically have a lot of data that is given to you but they do not have any labels attached to them right. So we look at first at the problem of clustering where your goal is to find groups of coherent or cohesive data points in this input space right so here is an example of possible clusters.

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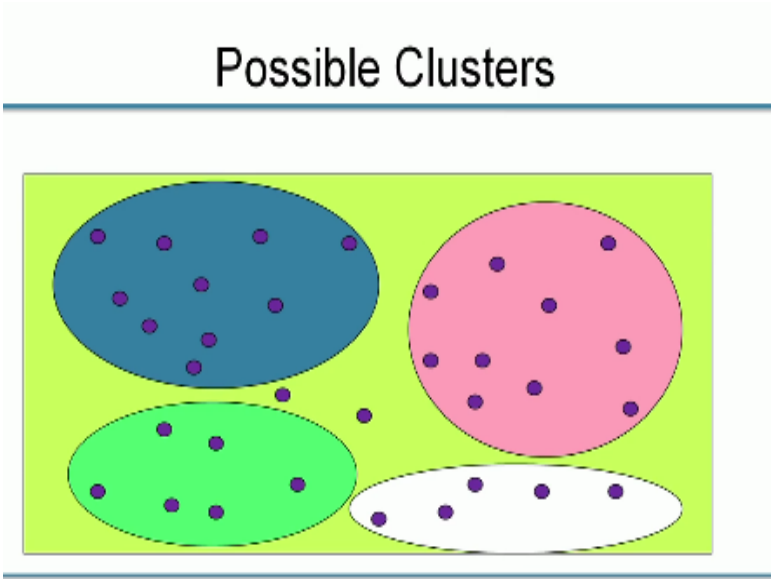

So those set of data points could form a cluster right and again now those set of data points could form a cluster and again those and those. So there are like four clusters that we have identified in this in this setup. So one thing to note here is that even in something like clustering so I need to have some form of a bias right so in this case the bias that I am having is in the shape of the cluster so I am assuming that the clusters are all ellipsoids right and therefore you know I have been drawing a specific shape curves for representing the clusters.

And also note that not all data points need to fall into clusters and there are a couple of points there that do not fall into any of the clusters this is primarily a artifact of me assuming that they are ellipsoids but still there are points in the center is actually faraway from all the other points in the in the data set to be considered as what are known as outliers. So when you do clustering so there are two things so one is you are interested in finding cohesive groups of points and the second is you are also interested in finding data points that do not conform to the patterns in the input and these are known as outliers all right.

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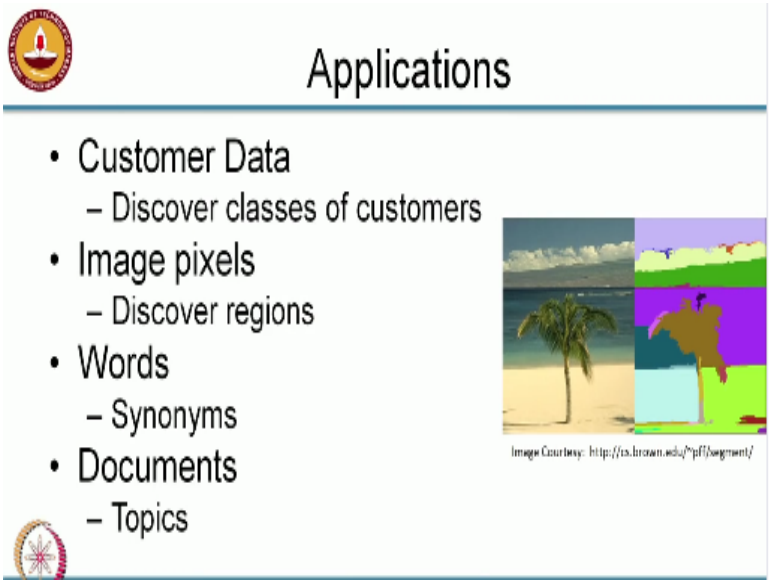

And there are many many different ways in which you can accomplish clustering and we will look at a few in the course. And the applications are numerous right so here are a few representative ones. So one thing is to look at customer data right and try to discover classes of customers. So earlier we looked at in the supervised learning case we looked at is that a customer will buy a computer or will not buy a computer. As opposed to that we could just take all the customer data that you have and try to just group them into different kinds of customers who come to your shop and then you could do some kind of targeted promotions and different classes of customers right.

And this need not necessarily come with labels you know I am not going to tell you that okay this customer is class 1 that customer is class 2 you are just going to find out which of the customers are more similar with each other all right. And as the second application which we have illustrated here is that I could do clustering on image pixels so that you could discover different regions in the image and then you could do some segmentation based on that different region so for example here it have a picture of a beach scene and then you are able to figure out the clouds and the sand and the sea and the tree from the image. So that allows you to make more sense out of the image right.

Or you could do clustering on word usages right and you could discover synonyms and you could also do clustering on documents right and depending on which kind of documents are similar to each other; if I give you a collection of say 100,000 documents I might be able to figure out what are the different topics that are discussed in this collection of documents and many many ways in which you can use clustering.

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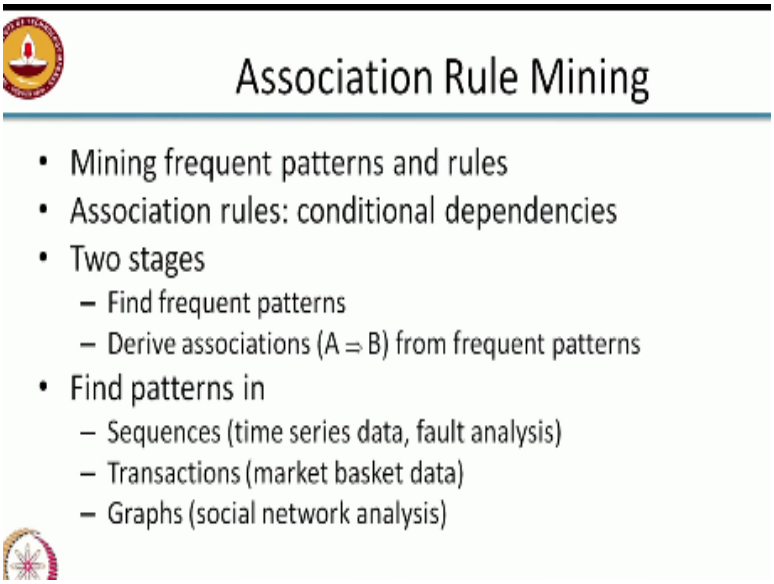


Rule mining: I should give you a side about the usage of the word mining here so many of you might have heard of the term data mining and more often than not the purported data mining tasks are essentially machine learning problems right so it could be classification regression and so on so forth. And the first problem that was essentially introduced as a mining problem and not as a learning problem was the one of mining frequent patterns and associations. And that is one of the reasons I call this Association rule mining as opposed to Association rule learning just to keep the historic connection intact right. So in Association rule mining we are interested in finding frequent patterns that occur in the input data and then we are looking at conditional dependencies among these patterns right.

So for example if A and B occur together often right then I could say something like if A happens then B will happen let us suppose that so you have customers that are coming to your shop and whenever customer A visits your shop custom B also tags along with him right, so the next time you find customary A somewhere in the shop so you can know that customer B is already there in the shop along with A.

Or with very high confidence you could say that B is also in the shop at some somewhere else may be not with A. But somewhere else in the shop all right, so these are the kinds of rules that we are looking at Association rules which are conditional dependencies – if A has come then B

is also there right and so the Association rule mining process usually goes in two stages so the first thing is we find all frequent patterns.

So A happens often so A is a customer that comes to my store often. And then I find that A and B are pairs of customers that come to my store often. So if I once I have that right A comes to my store often and A and B comes to my store often then I can derive associations from these kinds of frequent patterns. And also you could do this in the variety of different settings you could find sequences in time series data right and where you could look at triggers for certain events.

Or you could look at fault analysis right by looking at a sequence of events that happened and you can figure out which event occurs more often with a fault right or you could look at transactions data which is the most popular example given here is what is called Market Basket data. So you go to a shop and you buy a bunch of things together and you put them in your basket; so what is there in your basket right so this forms the transaction so you buy say eggs, milk and bread and so all of this go together in your basket.

And then you can find out what are the frequently occurring patterns in this purchase data and then you can make rules out of those or you could look at finding patterns and graphs that is typically used in social network analysis so which kind of interactions among entities happen often right so that is another question that is what we looking at right.

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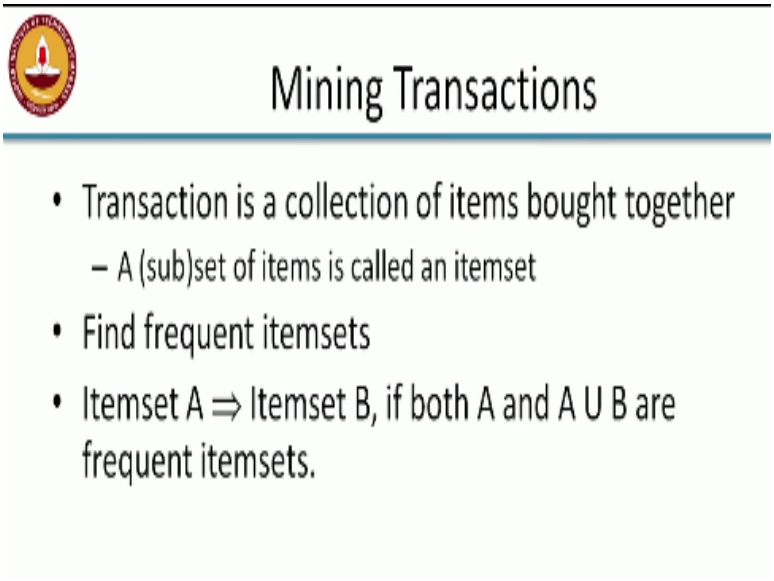


So the most popular thing here is mining transactions so the most popular application here is mining transactions. And as I mentioned earlier transaction is a collection of items that are bought together right and so here is a little bit of terminology. A set or a subset of items is often called an item set in the Association rule mining community and so the first step that you have to do is find frequent item sets right.

And you can conclude that item set A, if it is frequent implies item set B if both A and AUB or frequent item sets right so A and B are subset so AUB is another subset so if both A and AUB are frequent item sets then you can say that item set A implies item set B right. Like I mentioned earlier so there are many applications here so you could think of predicting co-occurrence of events.

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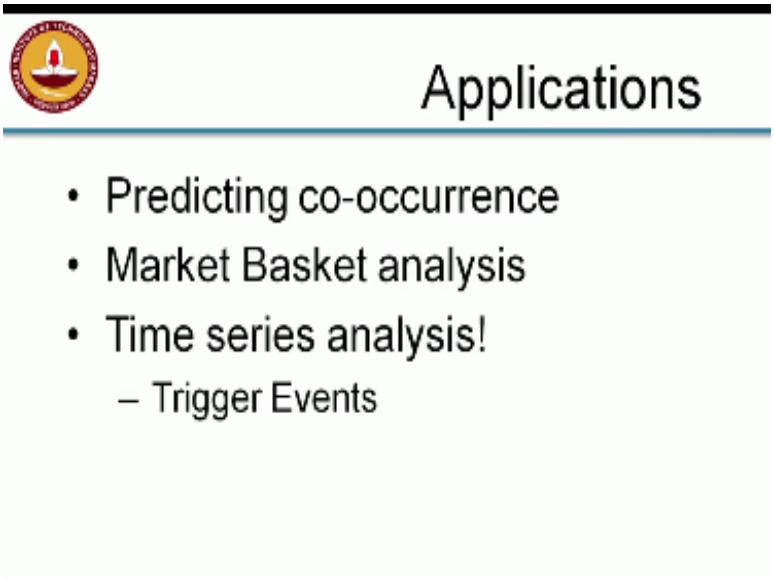


And Market Basket analysis and Time Series analysis like I mentioned earlier you could think of trigger events or causes of Faults and so on so forth right so this brings us to the end of this module introducing unsupervised learning.

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**Introduction to Machine Learning**

**Lecture 4**

**Prof: Balaraman Ravindran**
**Computer Science and Engineering**
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**Reinforcement Learning**

Hi, and welcome to this module that introduces reinforcement learning. So far, we have been looking at popular models of machine learning, such as supervised and unsupervised learning. In supervised learning, we looked at the classification and the regression problem, and in unsupervised learning, we looked at clustering and frequent pattern mining.

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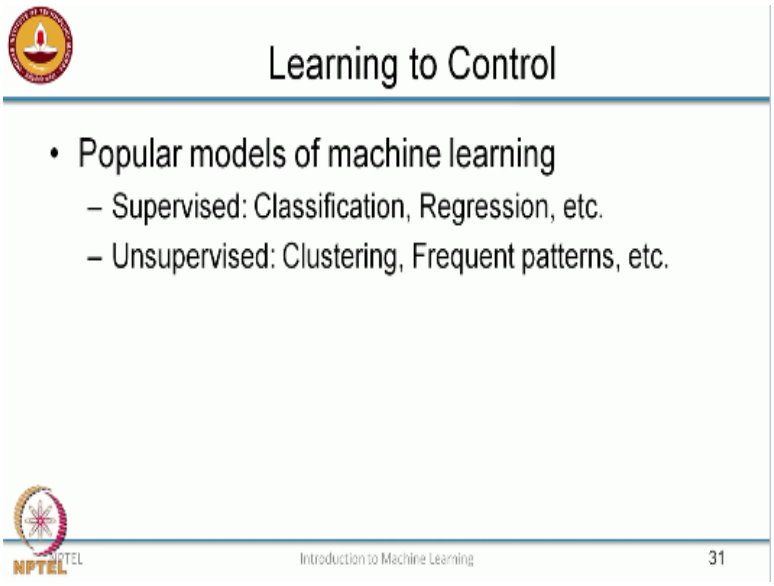


I have a question for you. So how did you learn to cycle? Was it supervised learning or unsupervised learning? There is no one telling you how you should cycle.

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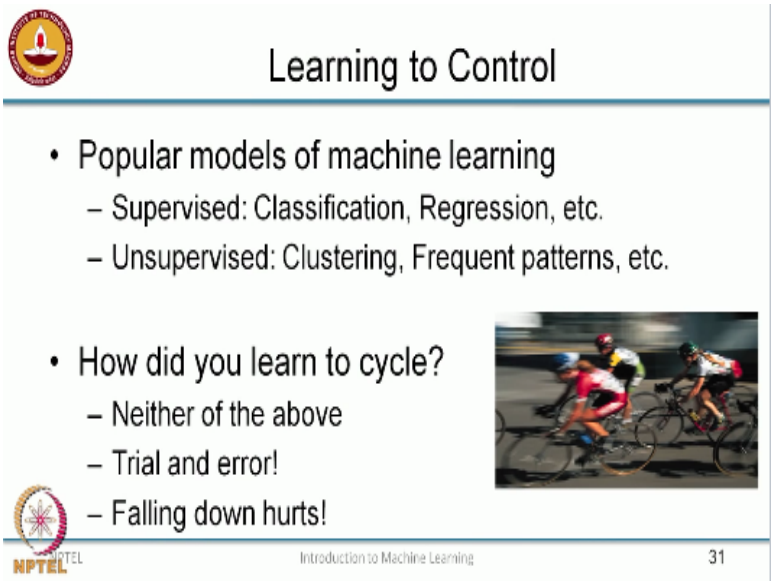


It is never explicitly stated how many pounds of pressure you should put with your left foot and what angle should be leaning and so on so forth. In other words, this is not a supervised learning problem. It is not wholly unsupervised either because it is not like you watch people cycling and then figure out what the pattern that you should move to cycle, and then you just magically got on a cycle and started cycling.

So the crucial thing here is that there is a significant trial and error component. You had to get on the cycle and try things out yourself before you could learn how to cycle acceptably. So you have some feedback, and it is not entirely unsupervised. For instance, when you are learning to cycle as a kid, there will be somebody standing there and clapping and motivating you to continue. That is the feedback. Another feedback that you have is the fact that if you fall down, you might get hurt.

So there is some amount of trial and error component and a feedback that you are getting from the environment. This kind of learning where you are learning to control the system through the trial and error and the minimal feedback is essentially what reinforcement learning is. The mathematical formalization that captures this kind of learning is referred to as reinforcement learning. So in the RL framework, you typically think of a learning agent.

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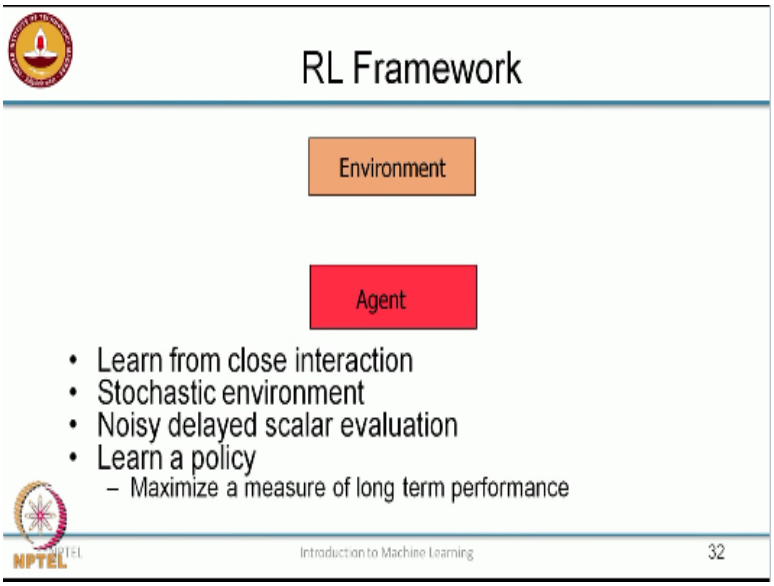


We have already looked at learning agents. It could be the supervised learner, or it could be some supervised agent. In this case, you have a reinforcement learning agent that learns from close interaction with an environment that is outside the control of the agent. The RL agent learns from close interaction with an environment. What I mean by close interaction here is that the agent senses the state in which the environment is and it takes an action which it then applies to the environment. This causes the state of the environment to change, thereby completing the interaction cycle.

The agent senses what the state of the environment is, so if it is a cycle it is going to sense what angle is the cycle tilting in at what speed I am moving forward and what speed I am falling etc. All this constitutes the state of this environment. Based on that, the agent is going to take an appropriate action such as lean to the right or push down with your right leg etc. Then this action is then applied to the environment, and that in turn changes the state of the environment.

So the agent learns from such close interaction with the environment, and we typically assume that the environment is stochastic, meaning every time you take an action you are not going to get the same response from the environment so things could be slightly different. There might be a small stone in the road that you did not have the last time you went over this place and an

otherwise smooth ride suddenly turned bumpy etc. There is some noise associated with cycling and you need to learn to react to that noise. So apart from this interaction, the mathematical formulation also assumes that there is some kind of an evaluation signal that is available from the environment that gives you a measure of how well you are performing in this particular task. If you remember we needed to have an evaluation measure for every task. We are assuming that this comes in the form of some scalar evaluation from the environment. It could be someone clapping and saying that here you are doing well or it could be falling and getting hurt. So all of this would be translated to some numeric scale.

And that's the mathematical abstraction that we make. So the goal of the agent is to learn a policy, which is a kind of mapping from the states that you sense to the actions that you apply so as to maximize a measure of long-term performance. I'm not just interested in staying up for the next two seconds, but I am interested in getting from point A to point B. So I have to make sure that I stay balanced throughout the entire duration of the ride. This is the basic idea behind the reinforcement learning problem. For each reinforcement algorithm the goal is to learn a policy that maximizes some measure of long-term performance.

So there have been many successful applications of reinforcement learning. So some of the marquee applications come from the domain of game playing like with many classical approaches. Backgammon is a board game based on die rolls.

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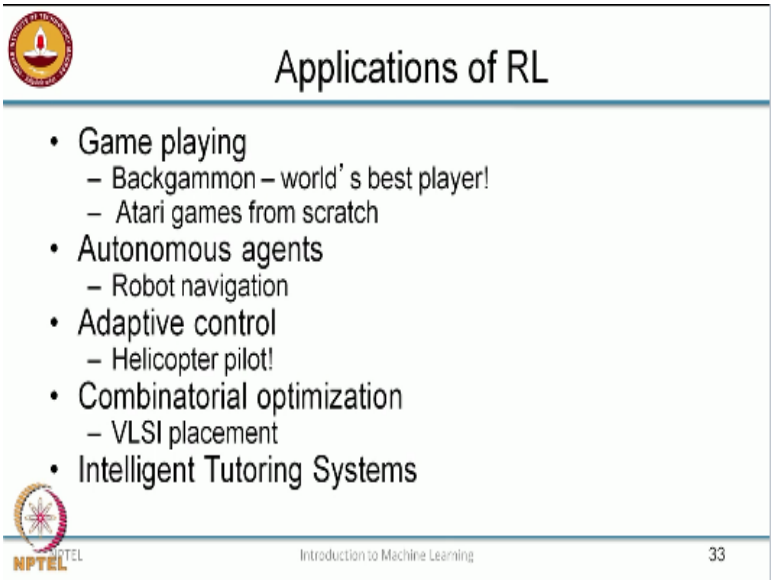


If people are not familiar that backgammon, it is similar to the game Ludo but it's also got a rich history. People have been playing it for several centuries. There are even world championships in backgammon, and the world's best player of backgammon is a reinforcement learning engine. So notice that I did not qualify it saying the world's best computer player or anything. It was the world's best player and that managed to beat the world champion in backgammon over a tournament.

So people have also trained reinforcement learning agents to play video games such as atari video games from scratch. The input to such a system were like pixels from the screens and the output from the system were joystick controls. They managed to play this game from scratch. In autonomous agents like in robots, reinforcement learning is almost always the learning algorithm of choice. Also in adaptive control, again a very prominent success stories of reinforcement learning is this helicopter pilot that was initially trained by Andrew Ng at Berkeley and later at Stanford where he trained a reinforcement learning algorithm to fly a helicopter and at near human level competence. There are other applications where people have looked at applying reinforcement learning algorithms to solve combinatorial optimization problems and also in personalization, adaptive systems like intelligent tutoring systems.

And so to wrap up this set of introductory modules, I wanted to recap the different machine learning paradigms that will be covering in the course. So the first one we will be looking at is supervised learning where we will be looking at learning an input-output map.

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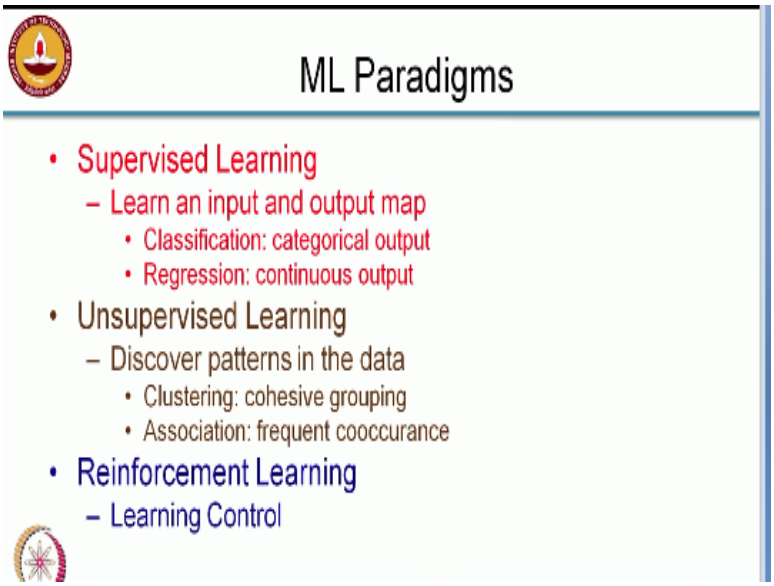


The tasks that we look there are classification where the outputs that we are looking to predict or categorical outputs like yes or no or blue or red or buy a computer or not by a computer and the second supervised learning problem we look at is a regression where the output is continuous. The second class of problems we look at is unsupervised learning problems where we are interested in discovering patterns in the data. Here, the goal is not necessarily in predicting a specific output. The canonical task that we look at here is clustering where we are interested in finding cohesive groups in the data and also association rules where we are interested in finding frequently occurring patterns. And the third paradigm which we will spend very little time on is reinforcement learning where you are interested in learning control or learning to control a system based on minimal feedback. So from the next module onwards, we will start looking at each of them separately, taking a little bit more mathematically rigorous look at machine learning.

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**Introduction of Machine Learning**

**Lecture 9**

**Prof. Balaraman Ravindran**
**Computer Science and Engineering**
**Indian Institute of Technology Madras**

**Statistical Decision Theory –**
**Regression**

Hello and welcome to this module on statistical decision theory. So the goal here is try to give you a framework that we will keep using for the rest of the course or at least for the majority of the rest of the course and introduce you to some of the basic notations and also to talk about some kind of a unifying idea behind what we will look at in different classification algorithms and regression algorithms.

To set the tone let's consider the inputs which will denote by $X$ as being gone drawn from some $p$ dimensional space so which we will call $\mathbf{R}^p$. So if you think about what we did in the previous modules, we talked about input that had age and income as the attributes, so that would mean that $p$ was two dimensions. So one of the dimensions represented age and the other dimension represented income.

So what we are doing here now is trying to move to a more general setting where I am talking about any kind of a $p$ dimensional space and what $p$ could be much larger than two and the output that we are going to be looking at least in the initial regression case that we will see I will assume that the output is drawn from the real numbers again. So this will be like the temperature that we saw in this second example in the previous modules.

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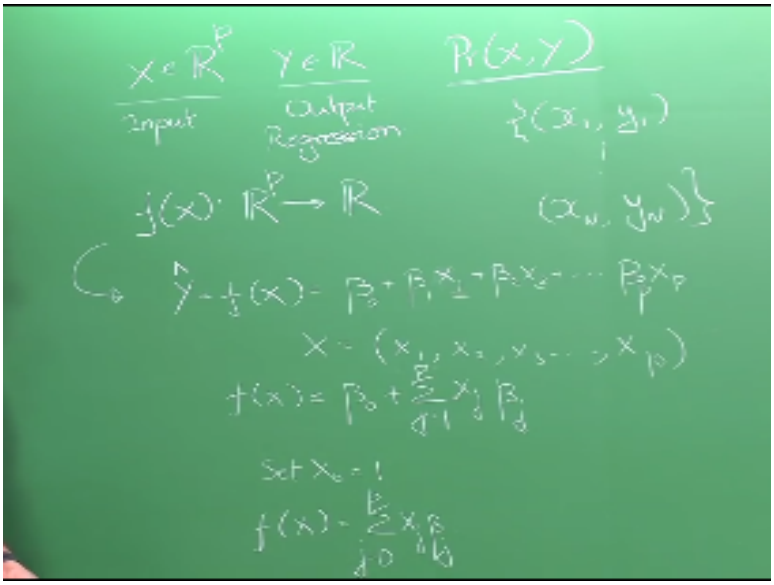


So the input $X$ is drawn from a $p$ dimensional real space and the output $Y$ is drawn again from the real numbers and in the case of regression. So the case of classification will see the little bit later the output will come from a discrete space and we will also make an assumption that the data comes to you from some kind of a problem joint probability distribution $\Pr(X, y)$.

So you do not know this joint distribution apriori and so nobody tells you what is the distribution from which the data is coming. But the assumption that we are going to make is that there is an underlying data distribution like a joint distribution over the inputs and the outputs and that it is fixed. You are going to be given samples drawn from this probability distribution $\Pr(X, y)$. So this will be your training data which you will use for both training and possibly for validation if required.

So you are going to get an $x_1$ with a corresponding $y_1$, $x_2$ with the corresponding $y_2$ and so on so forth. So the goal is given such a set of training data we have to learn a function $f(x)$ that goes from a $p$-dimensional space to the real line, where the p dimensional space essentially corresponds to a point in the input space and the real line corresponds to the output space. So the

function *f* is going to take any input that is given to it and produce a number. So the *f* could take different forms for instance we looked at *f* being a straight line in the example that we saw earlier.

So one example of *f* would be saying that I am going to predict $y$ where $y = f(x) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + ... + \beta_p x_p$ . So one thing which I want you to note here is that this ( $x_1$ , $x_2$ … $x_p$ ) are essentially the coordinates of *X*. So when I say *X* here, *X* essentially comprises of age, income etc. One of these corresponds to a different attribute that describes the data.

So I can look at this and then I can write $y = f(x) = \beta_0 + \sum_{j=1}^{p} x_j \beta_j$ . An alternate way of writing this is to set $x_0 = 1$, and then I can just remove this special treatment of $\beta_0$ and I can just write it as $y = f(x) = \sum_{j=0}^{p} x_j \beta_j$ . So this is essentially what do you do in when you are doing linear regression. So another example of doing this is a very popular classifier which we will call the nearest neighbor classifier.

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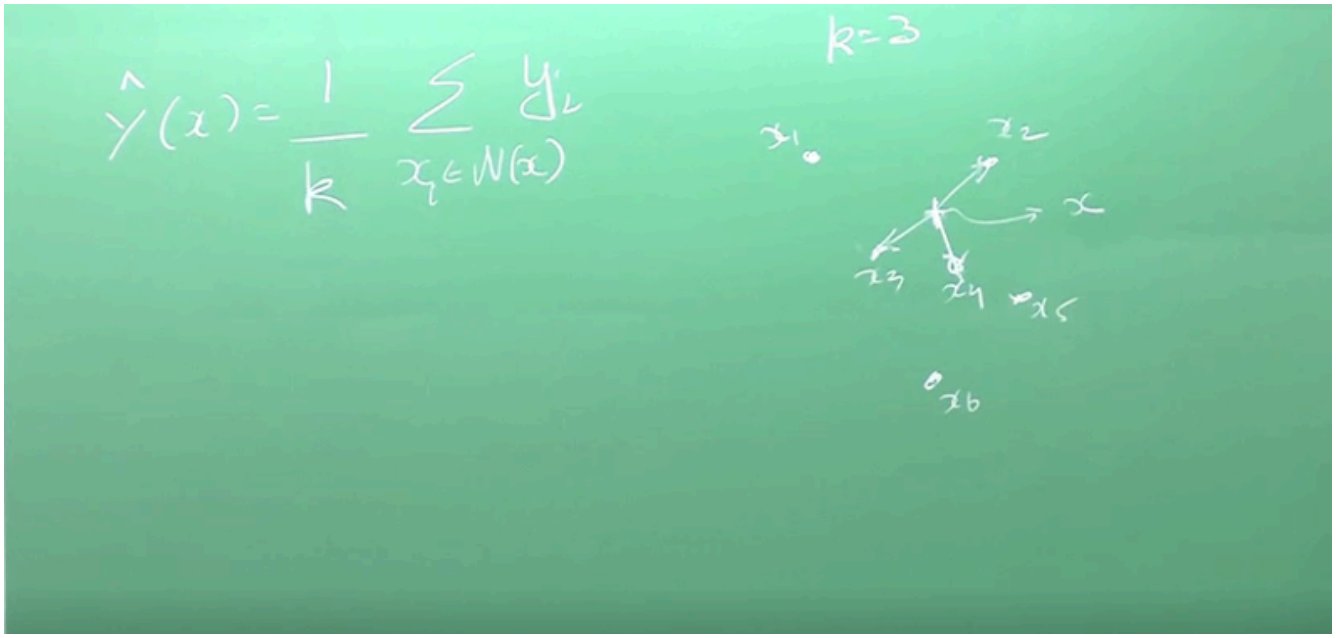


Here, $y = \frac{1}{k} \sum_{x_i \in N(x)} y_i$. So let us assume that my training data looks something like this. Take note of the different points labeled as $x_i$'s. Let us say my $k$ is 3, so if I get a query point (middle x) say somewhere here I get a query so this is my $x$ and my training data the $x_1$ *to* $x_6$ are my training data and $x$ is the point for which I want to predict the output. So these are the places which I have already measured it. This is a new point and I want to produce the output here so in this case what do I do? I pick the three nearest neighbors because $k$ is 3, I pick the 3 points that are closest to this data point right find the corresponding $y$'s. So in this case I will pick $y_2$ $y_3$ *and* $y_4$ and I will take the average of these three points and I will report the value of the function at $x$. That will be the average of this three point. So this is called the $k$- nearest neighbor regressor.

So I will just take the average of the outputs of $y_2$ $y_3$ *and* $y_4$ and report that as the value of the $x$. So depending on where $x$ is, I will be picking different three neighbors and reporting their values. This is the $k$-nearest neighbor algorithm. So there are different ways in which you can define this function $f$ but remember that we had this discussion in the last set of modules that unless you make an assumption about the form of $f$ you really cannot do any generalization. We needed to talk about lines in the previous class but now we are talking about different assumptions for the

function *f* need not necessarily be lines. In the previous case it is a straight line but in this case it is an average. It is a local averager and that gives me the function that I want to learn.

So how do you choose this function? So there could be many different ways in which you can define the $\beta$'s. Given that I have chosen that this is the way to model the function how do I pick the $\beta$? So how do I even choose this form for my predictor and how do I know that this is a valid form to choose? So we have to look at some performance measure which we will consider in this case as the loss function. This will compare the true output *y* right with the predicted output *f (x)*. I have the true output *y* and I have a particular *f (x)*, so I will have some loss function that compares *f(x)* with *y*. My goal is to find an *f(x)* such that this loss function is minimized. One of the most popular loss functions that people use in the literature is known as the squared error. So basically I am interested in is the expected prediction error of the function *f* . That is equal to the expected value of $y - f(x)^2$ . In the case of squared error so the expected prediction error is $E\{(y - f(x)^2\}$ .

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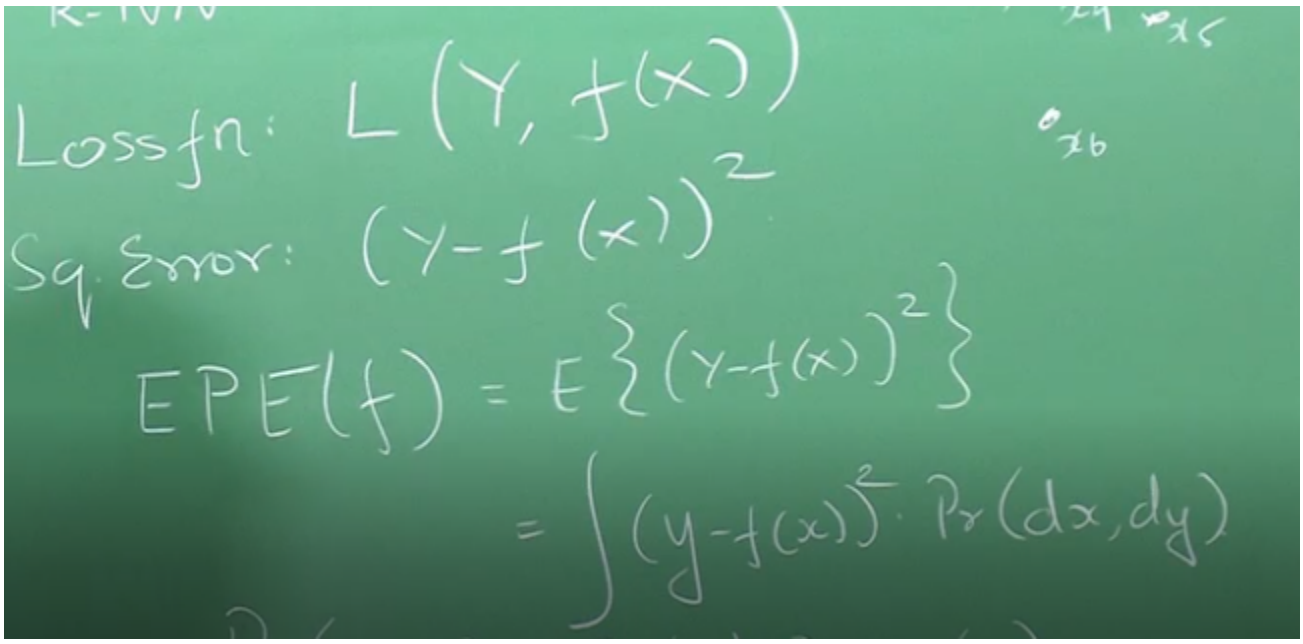


So what is the distribution with respect to which you are taking this expectation? Whenever we talk about the expectation of a random variable we want to talk about the underlying distribution . So what is the distribution with respect to which you are taking this expectation? The answer is

exactly the joint distribution of X and Y or $\Pr(X,Y)$. So I can do a little bit more sleight of hand here right and talk about the conditional distribution. If you remember $\Pr(X,Y) = \Pr(Y \mid X)\Pr(X)$. This is just the product rule in probability. So what does this tell me that okay there is some chance with which I can choose a data point X and having chosen a data point X so what is the probability of seeing a particular output value.

So why are we looking at probabilities here again? So this helps us to kind of you know model a variety of different scenarios. The first one is that if there is noise in the measurement then we should we do?

I am talking about $\Pr(Y \mid X)$, suppose I am telling you that I am measuring the temperature at 3 o'clock every day so there will be some kind of a natural variation in the temperature is measured at 3 in the morning right. So that is modeled by this probability and there will be some set of temperatures that are very probable and some set of temperatures that are not. So for example if I am measuring temperature at 3 a.m, $40^0$ C is not a probable value. So those will have lower probabilities and then say something in the 20s will have a higher probability. So, I am talking about Chennai if people are wondering how you are getting $20^0$ early in the morning.

The second factor is that this allows us to look at is our ignorance about the whole system. So I might have just chosen the time of day maybe there are other factors I should have taken into consideration while I am forming my data, so these factors about which I do not know anything will appear as noise. So it is not important whether I take the temperature at 3:00 a.m, maybe it is important where in the building I do the measurements. Maybe I am measuring it next to the kitchen where things will be warmer or maybe I am measuring it next to an air-conditioner where things would be actually warmer. If I am measuring it on the external of the building or it could be measuring it on the inside of an air-conditioned room the temperatures could be lower so even though I say I measure it at 3 a.m. there could be many such factors for natural variations which I have not modeled. So this is beyond the natural variations in the system. One way of arguing about it could be to say that hey the natural variations are due to factors that you do not know anything about. So that is a valid argument so it could very well be that. So one might argue that really there is nothing like a natural variation and there is no real noise so all the uncertainty in the data arises from my lack of knowledge but that is a philosophical question.

So there are things that are measurable which we do not measure right and that I would call as lack of knowledge and things which are immeasurable which I would call as noise. There could be both of these sources which introduced the probability into system. So it is not just a mathematical whimsy that we model this as a joint distribution but there is an actual practical reason for talking about probability distributions. So now I can go back and write my expected prediction error as $EPE = E_X E_{Y|X}([y - f(x)]^2 \mid X)$ .

It is the same quantity earlier the only difference is now I am conditioning it on the value of *X*. So what this expression says is a I will tell you what *X* is, then you tell me what the error will be? So the uncertainty here is over the value of *Y*. I will give you *X* , you tell me what Y is? I am going to look at the error just after conditioning on *X* this will only look at the variation on Y and the outer expectation gives me the variation of *X*.

Now I can try to find the minimum of this prediction error by conditioning on a specific value for X. So I will not look at this expectation, I am not making any assumptions about *f* and I am just assuming that *f* can be anything like any function in the world. What I want to do now is I want to look at each and every value of *X* and I want to say that I will pick an *f* such that for every value of *x* it makes the best possible prediction. So what does that mean? It will produce the prediction so *f(x)* for a specific value of *x*, *f(x)* will give the output such that this inner expectation is minimized.

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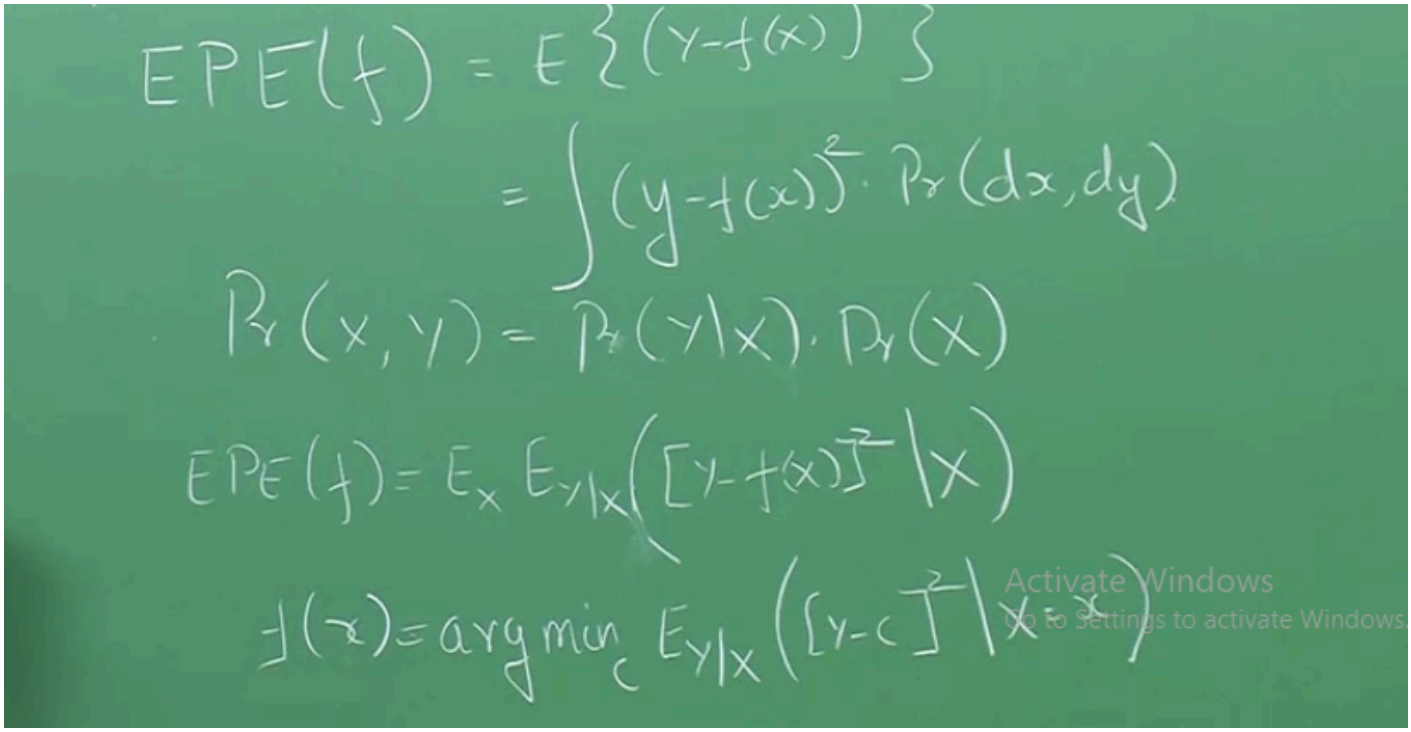


So I am going to write it down like this. For a specific value of *x* this is *f (x)* for a specific *x.* I was writing capital X which is a random variable but here I am using a specific *x*. So given an *x*, *f(x)* has to be a specific number. Let us say somebody with an age of 25 an income of 15,000 rupees walks into my shop I can only give one output. Whether that person is going to buy a computer or does not buy a computer or I am going to say I am measuring the temperature at 3 a.m what will be the value? And I can give you only one number since I have already fixed the input description, so I can give you one output corresponding to that input description. Let me call that output as *c*. So it should be such that the error which is $(y-c)^2$ is as small as possible. I am minimizing over the different possible values of *c* that I could assign for *f(x)*. I am trying to pick that *c* which gives me the smallest error. So argmin means first minimize with respect to *c* and take that argument which achieve this minimum. If there are multiple values that gives you the minimum, I can pick any one right. So this is essentially called conditioning on a point. Instead of conditioning on the random variable *X*, where we conditioned on a specific point where *X*=*x*, I can find this. So now what happens for every possible input *X* that I could have small *x,* I will find the corresponding *c* and I will say *f(x)* equal to that *c*. The thing to note here is I have not made any functional assumptions about what this what should $f$ look like.

So *f* could be something really, really jagged I do not care. This is a recipe for disaster as we saw earlier that you might end up over fitting the data, but just work with me here because we are just trying to build some general principles. We can go little further right now that we have decided to say that the minimizer is the one that that you have assigned to *f(x)* so what is the value of *c* that will minimize this expression? So I have to look at $(y-c)^2$ and I have to assign a single value for *c*. Suppose I give the input as *x*, I make a measurement let us call it say $y_1$ and I give it same input *x* again I make another measurement say $y_2$. I give the same input *x* again I make another measurement $y_3$. So I have three measurements $y_1, y_2, y_3$ for the same input *x*. Right now I am asking you to give me a prediction for what will be the output given *x*.

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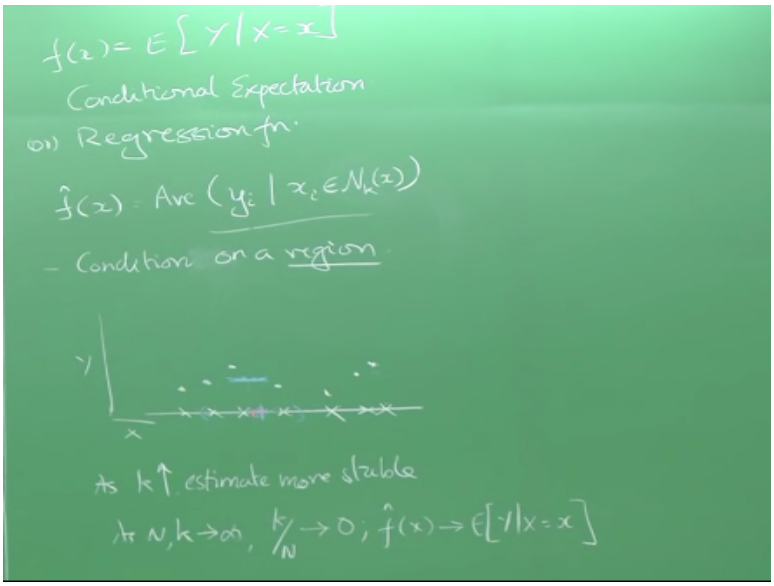


So what should your prediction be? It will be the average of these three or $(y_1+y_2+y_3)/3$. Why is that the case? Because we are talking about squared error, the quantity that will minimize the squared error is essentially the average. I will end up writing that $f(x)=E[Y \mid X=x]$. So this is essentially what my prediction would be. This is the known as the conditional expectation or also sometimes called the regression function.

What are some of the problems with this equation?

a. I do not know the distribution with respect to which I am taking the expectation. Or I do not know $\Pr(Y \mid X)$. If I know that, then my life would be a lot simpler. I actually have to estimate it from the data. What is the data that is given to me? I have this pairs of $\{(x_1, y_1), (x_2, y_2)...(x_n, y_n)\}$. That is the data that has been given to me and I have to do this estimate of this expectation from that data. So how will you do that? You know that you can always estimate the expectation by taking averages so what you would do is from your data you pick all the training data points that have this value of $x$ right find the corresponding $y$ take an average. So one simple way of thinking about it is to say that okay I cannot find the true $f$ so I am going to find an estimate of that which is called as $\hat{f}$ which is equal to average of all the $y_i$'s such that $x_i$ equal to $x$ *or* $\hat{f}(x) = Ave(y_i \mid x_i = x)$.

There is a problem here. First, how many samples do you think you are going to get of the same input $x$. Second, you are not going to be able to make a prediction for any data point which is not there in the input. We are trying to make an estimate of the expectation by using the averages but if you don't have enough measurements than your average is going to be bad and second thing is you are making an average at that point and if the point does not exist in your training data you are not going to be able to return an estimate for it. S

So we need to address this somehow. What we will do here is we will relax the conditioning. Instead of conditioning on a point we will be conditioning on a region. So I am taking the average here of all those data points for which $x_i$ equal to $x$. Now that is not going to work because there are too few data points. What I am going to do is I am going to take this as the average of all the data points which belong to some region around $x$ which is essentially the neighborhood that we are talking about. That circle there would correspond to the neighborhood around $x$ and I am going to be conditioning on this region which is given by this neighborhood around $x$. So we are not going to condition on the point we are conditioning around the region so the one assumption that we are making is an implicit one. Why are we conditioning around a

region so that instead of taking an average of one data point, I have at least $k$ data points of which I will be taking the average. That gives me a better estimate of the expectation. That is the reason we are doing this conditioning over a region but more importantly we are also making an implicit assumption. If you remember our inductive bias said that we needed to make assumptions. The one we are making here is that the output of the function over this region is going to be a constant. Let us let us try and do a little example so that becomes a thing clear to people. Let me go back to my one-dimensional example so it makes it easier for me to draw things. I have let us say I have multiple data points like this. I have a query point and then the corresponding outputs. So these are the $y$'s this is $x$ and that is $y$ so these are my $x_i$'s and $y_i$'s . This is the training data I have and now given a query point let us say I am given a query point here. I want to know what is the output value for this $x$. Let us say I pick my three nearest neighbors which would be these three data points right and then I will try to take the average of this, this and this which will be somewhere here.

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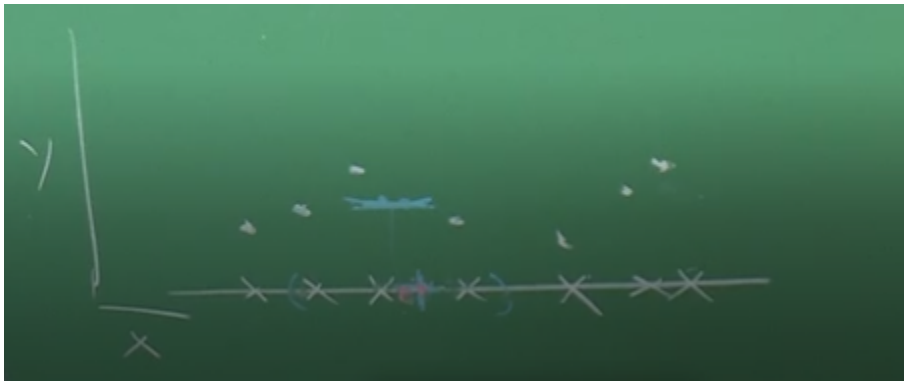

I am going to make cases for variety of data points but one thing which I want to point out here is that I assume that my data point lies here so what if I had assumed that my data point was here if my query point was here so what would have been the output. So my neighbors remain the same. The three neighbors do not change and these are my three nearest neighbors whether the query was here whether the query was here or the query was here my nearest neighbors do not change. So what will be the output for this input point? I will be taking the average of these three points so the output will be the same. In fact for a certain region around here where these three are the nearest points the output will be a constant. I said output will always be a constant so this is what I mean by saying that we make the assumption that the output is constant in a region. For all those data points for which these three are the nearest neighbors, the output is going to be

here. So this is essentially the assumption we are making that the output is going to be consistent over a region. I can write an expectation over the region as my substitute. If you think about it so what have we come up with here this is essentially your nearest neighbor classifier. You take the generic idea of minimizing the expected prediction error and then add certain conditions to it. You are going to take averages and you cannot do an average on the training data and therefore you are going to do an average over a region assuming that the output is constant over a region. Relaxing the conditioning on a data point principle gives us conditioning on a region. This is also known as the nearest neighbor classifier. So in some sense you can argue that one way of minimizing the expected prediction error yields a nearest neighbor classifier.

In fact it is a very powerful classifier and you can show that as *k* increases so the estimate becomes more and more stable. So for small changes in the input data the classifier does not change tremendously and so as and as *n* and *k* tends to infinity or they become large , your ratio at $\frac{k}{n} \to 0$ in such a case your $\hat{f}(x) \to E[Y \mid X = x]$. As K increases the estimate becomes more stable in particular as *k* and *n* becomes large and my number of data points is very, very large, the number of points I can look in the neighborhood also becomes larger and larger such that the data points have to grow at a faster rate than the size of the neighborhood. That is what $\frac{k}{n}$ means. In which case I can show that my actual prediction I make using this average actually approaches the true prediction that I am interested in.

There are a few caveats here that I need to point out. I assuming I am saying that *n* goes to infinity. That is a pretty blank statement to make because *n* rarely goes to infinity. In fact coming up with large data sets are hard except for very rare cases, therefore you cannot really have a classifier that gives you the right output.

And the other problem is as *p* becomes larger as the dimensionality becomes larger. Generally the data tends to becomes sparse, so if I am looking at *k* neighbors like a thousand dimensional space, the area or the volume covered by these *k* neighbors would be very large because they are very sparse space and it is usually not a good assumption to make that the output is a constant over this large area.

One thing is if $p$ is large then if the dimension of the input is large if you have like 10,000 dimension vector as your input then using $k$ nearest neighbors is not really a good idea. Alternatively you should also remember that in some cases having a little bit of a bias is actually not a bad thing and therefore we have to look at an appropriate way of representing the function $f$. Remember we did not make any assumptions about the function $f$ so the function $f$ could change as drastically as we as we want. That means that we are trying to keep the bias as minimal as possible.

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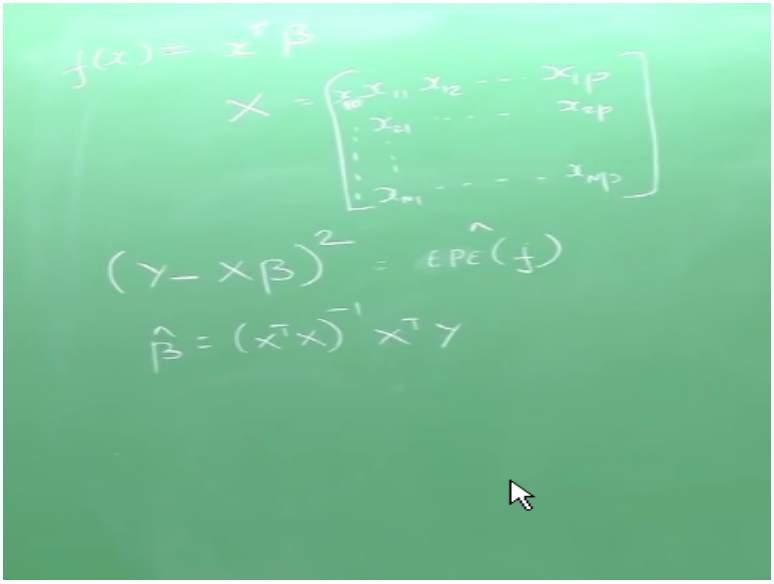

We would like to remove that assumption. Moving on let us look at the linear regression case where we actually made a significant assumption about the form of the function *f*. Specifically we assume that *f* is going to be linear in the input parameter. So *f* can be written as $y = f(x) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + ... + \beta_p x_p$ . Essentially $f(x) = x^T \beta$ . And so if you look at it from the training data point of view I can think of having a vector notation for this. I can think of a matrix in which each row corresponds to an input $x_1$ so *X* will be something like

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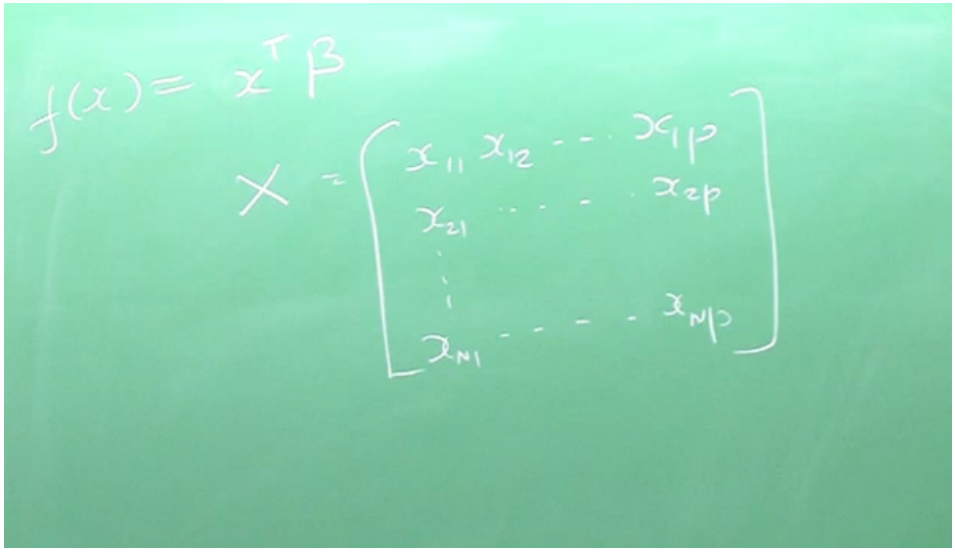


$\beta$ would be a vector of the coefficients so $\{\beta_0 ... \beta_p\}$ and the zeroth dimension is going to be 1. So the final equation would be $(Y - X\beta)^2 = EPE(\hat{f})$ . I can minimize this by taking the derivative and then I can equate it to zero. I can do some minimization to get the value of $\beta$ and that is essentially taking the differential of this equate it to zero simplify for β. I am going to get $\hat{\beta} = (X^T X)^{-1} X^T Y$ . Remember that the X that we put in here is essentially a matrix where the the rows are the data points and the columns are the features.

So this will be the like the age of every customer that comes and this will be the income of every customer and so on so forth and each row is a complete data point. So what we have done here is make the assumption that my function is globally linear and then I have tried to solve for it to

give you the parameters $\hat{\beta}$. In the nearest neighbor case we made the assumption that my function is locally constant.

So we start off with the same formulation we wanted to minimize the expected prediction error and we make different assumptions. One assumption leads us to linear regression the assumption that we made was the data is going to be globally linear and another assumption that we made where the data is going to be locally constant gives us *k* nearest neighbor.

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**Introduction to Machine Learning**

**Lecture 10**

**Prof. Balaraman Ravindran**
**Computer Science and Engineering**
**Indian Institute of Technology Madras**

**Statistical Decision Theory –Classification**

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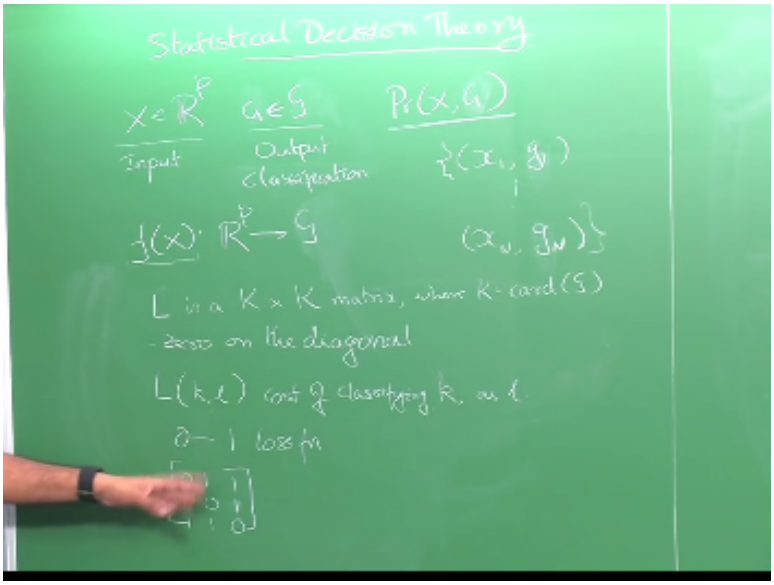


In this module we are going to look at the case where the output variable is drawn from a discrete space or in other words we are going to look at the classification problem. As before the input is coming from a $p$ dimensional space $R^p$ and the output which I am denoting by $g$ here, I am going to assume is coming from some space $G$ which is a discrete value. It could be "Buys a computer" or "does not by a computer" so $g$ could just consists of buys a computer does not by computer or it should consist of like 5 different outcomes "has the disease" , "a mild form of the disease", "a severe form of the disease", "does not have the disease" and so on so forth right.

It could be a variety of outcomes but it is a small discrete set. So that space is denoted *G* which is the random variable corresponding to the output, then like before we are going to have a joint distribution on the input on the output. The training data is going to consist of $\{(x_1, g_1), (x_2, g_2), ...(x_n, g_n)\}$ and the goal here is to learn a function *f(x)* that is going to take you from a *p*-dimensional input space $R^p$ to the discrete space *G*.

And so the thing that we have to look at now is what is an appropriate loss function in this case. What is an appropriate loss function in this case since we are talking about the discrete output? So I really cannot talk about squared error as a loss function even though in cases where the discrete values have been encoded as numeric outputs people do use squared error and we will see that later. So people do use squared error is an appropriate measure as long as your space *G* has been encoded numerically.

So but in general so we are going to define the loss in as a $k \times k$ matrix where *k* is the cardinality of the discrete space *G* that we are looking at. Suppose there are 5 classes then my last matrix is going to be a *5 by 5* matrix. so it will have zeros on the diagonal and so the $kl^{th}$ entry *L(k,l)* in the loss matrix essentially is the cost that you incur of classifying the output *k* as l. So the true output is *k* but you output you say *l* so that is essentially the cost of classifying *k* as *l* . That is denoted by the $kl^{th}$ entry of the loss matrix.

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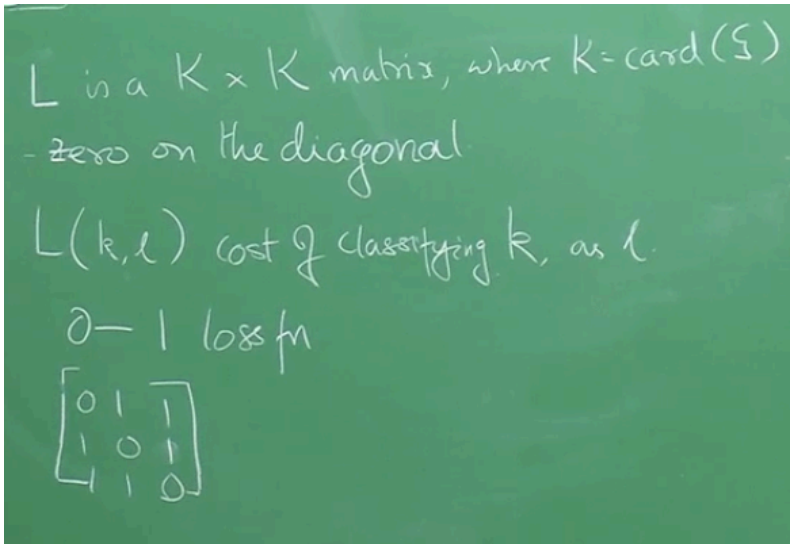


So frequently the most popular loss function that you use is known as the *0 - 1* loss function. The *0-1* loss function essentially says that suppose I have three classes, so my loss function would look like this, $\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$.So if I if I classified to the right class I get a penalty of zero but if I classify to the wrong class right I get a penalty of one regardless of which wrong class I classify to. So this entry says that okay the data point actually belongs to class one I have classified it as class two what is the penalty so 1 data point belongs to class 1 I classify it as class 3 what is the penalty one and so on so forth. This is called the 0-1 loss function because all the entries in the loss matrix are either 0 or 1.

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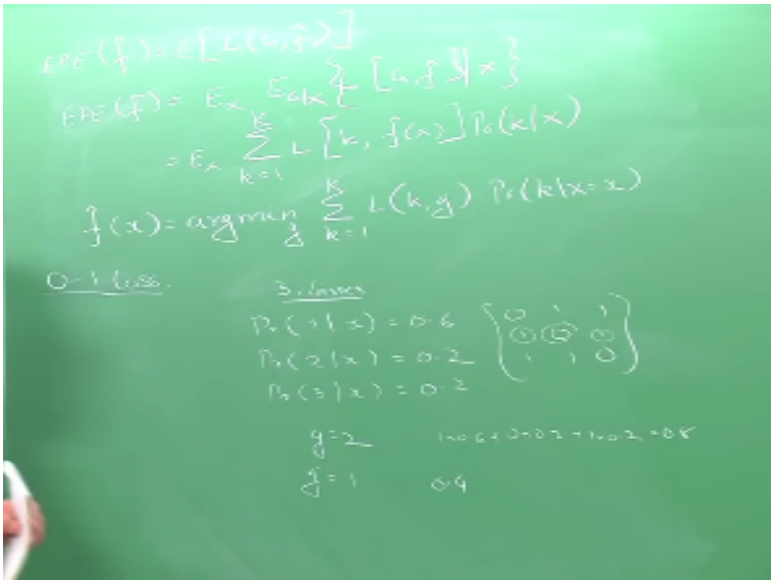

So what we are again going to look at is the expected prediction error of $\hat{f}$, or, $EPE(\hat{f}) = E[L(G, \hat{f})]$.

We can do the same thing that we did earlier so I can start conditioning it on *x* and the expected prediction error and then the expectation of *g given x* which essentially becomes $EPE(\hat{f}) = E_X E_{G|X}\{L[G, \hat{f}] \mid x\}$.

So the loss of g, $\hat{f}$ given that the input is x but if you think about it this is not a continuous distribution this is actually a discrete distribution because *G* can take only finitely many values. So instead of writing it out as this expectation I can actually simplify that and write it as $E_X = \sum_{k=1}^{K} L[k, \hat{f}(x)] \Pr(k \mid x)$. So this is a loss that I will incur if *k* was the true class and my prediction was $\hat{f}$ *(x)* times the probability that *k* is the true class given the input *x*. So here I am essentially writing out the expectation because it is a discrete distribution. I am able to write it out in a compact form and again I can do point wise minimization of this like we talked about earlier. So point wise would mean that I make a specific assumption about what is the value of *x*.

So I am going to look at $\hat{f}(x) = \arg\min_g \sum_{k=1}^{K} L(k, g) \Pr(k \mid X = x)$

We are essentially following the same treatment that we did with the regression case except that we are using a discrete output space since of a continuous output space. So this essentially says that I am going to pick the prediction *g* that gives me the smallest expected error. Suppose, I have the 0-1 loss function, so what does this mean? I should essentially set my *g* to be that *k* which has the highest probability. So if we think about it this probability term $\Pr(k \mid X = x)$, contributes to every element in the summation. So what I can do is among all these probability terms I can pick one term and set it to 0 by my choice of *g*. Suppose I choose *g* to be 1 then my $l(1,1)$ will become 0 and but my *l(2,1),l(3,1)* so on so forth will all be 1. What will happen then is that $\Pr(2 \mid x), \Pr(3 \mid x)$ all of this will actually appear in this summation.

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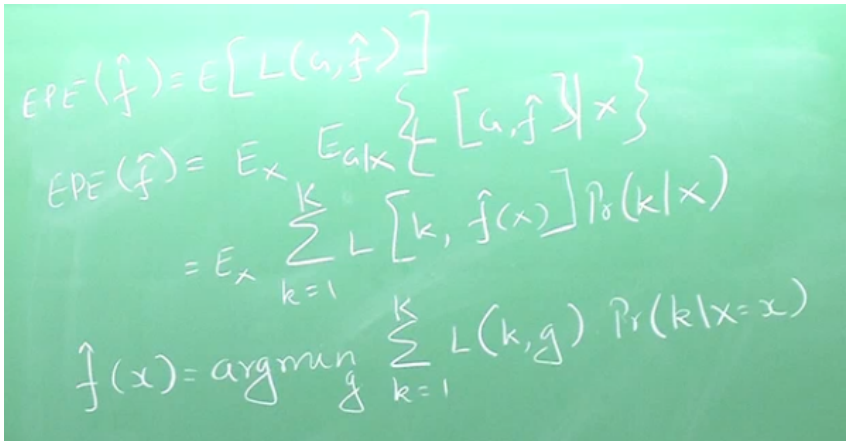


So if I set my *g* to that value of *k* which has the highest probability then that will yield the best possible solution here. If you are not able to see that let us assume that there are 3 classes. I assume that there are 3 classes so and my true distribution is says that the $\Pr(1 \mid x) = 0.6, \Pr(2 \mid x) = 0.2, \Pr(3 \mid x) = 0.2$ and of course my loss function is going to be such that and $\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$.

So if I guess that my class label is going to be 2 let us say so I said $g = 2$ , what is going to happen? If the class label is 1 right so I am going to look at the loss corresponding to 1, 2 which is the loss matrix entry (2,1), so I will get 1 times 0.6 then if the class label is 2 so I will be looking at loss matrix entry (2,2) I will get 0 times 0.2 + if the class label is 3 I look at loss matrix entry (2,3) I will get 1 times 0.2, so I will get a score of 0.8.

So as you can see depending on which value I choose, say, if I choose g = 2 then I will be zeroing out the second entry if I choose g = 1 I will be zeroing out the 1st entry and by choosing g equal to 1 I will basically get a score of 0.4. So what I have to do in order to get the minimum here is to pick that g for which this probability is the highest.

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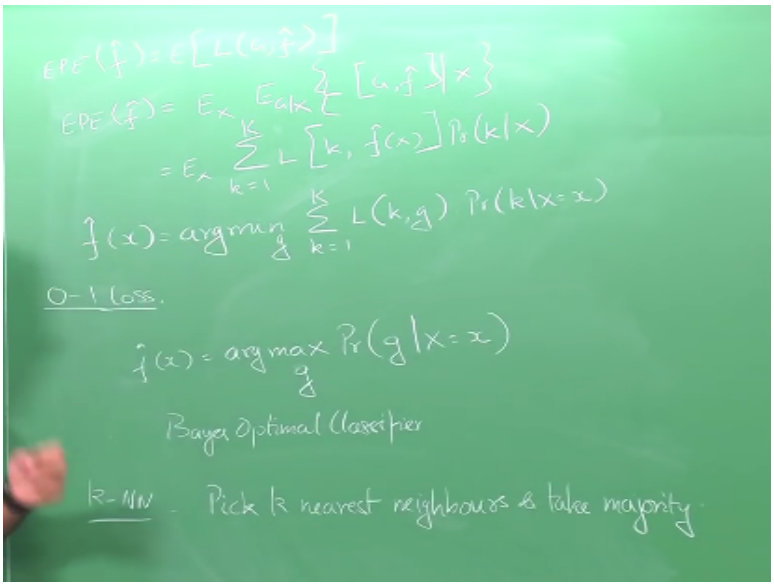


So now can you realize why the min here became the max? This is based on the argument that we just did, so this is essentially saying that from your training data classify it to the most probable class and if I knew this probability, what will I do? I can set it to the most probable output. So this is this kind of a classifier which is known as the Bayes optimal classifier. It says that I can look at the conditional distribution given $x$ look at the probability of $g$, take the $g$ that has the highest probability and assign it as the output so this is essentially what the Bayes optimal classifier would say.

But then you do not know *g*. So what we have to do is you have estimate this probability. How would you estimate this probability? Do we know of any method for estimating this probability? Of course we do. We know how to do nearest neighbor, so what you would do in this case is that instead of taking the average over the neighbors like we did in the regression case you would do estimate the probabilities in the neighborhood. What you would do is that you will take a data point look at the *k* nearest neighbors of the data point find out what their class labels are and then for each label count the number of occurrences of that label in the *k* neighbors and divide by *k*. This will give you the probability of the class label in the neighborhood but we really do not have to do this much work because we are not interested in the actual probability. All we need is the one that has the maximum probability since the denominator is going to be *k* for all the probabilities we can ignore the denominator we can just look at the numerator.

So what we can do is we can count the occurrences of the class label in the neighborhood and whichever occurs more often we can assign that as the class label. Think about it for a minute. What we are essentially doing when we take the majority is actually estimating this probability and taking the max probability so take the majority label in the neighborhood and use that as your prediction. So this essentially gives you the *k*- nearest neighbor classifier.

Like earlier, all the caveats that we talked about for the k nearest neighbor regressor apply to the k nearest neighbor classifier as well. You have to be careful about using it in very high dimensions and you really need large values of *k* and large values of *n* before you can get stable estimates. But having said all that I should say that it turns out to be a really powerful classifier in practice and we will come back to that a little later as to why it is such a powerful classifier.

Now can we use linear regression or the linearity assumption here? It turns out that you could use linear regression in almost directly for solving this problem. So the way you do it is the following. You take this data set $\{(x_1, g_1), (x_2, g_2), ...(x_n, g_n)\}$ and convert it into a data set suitable for doing regression, so how do I do that? I take that ($x_1$, $g_1$)

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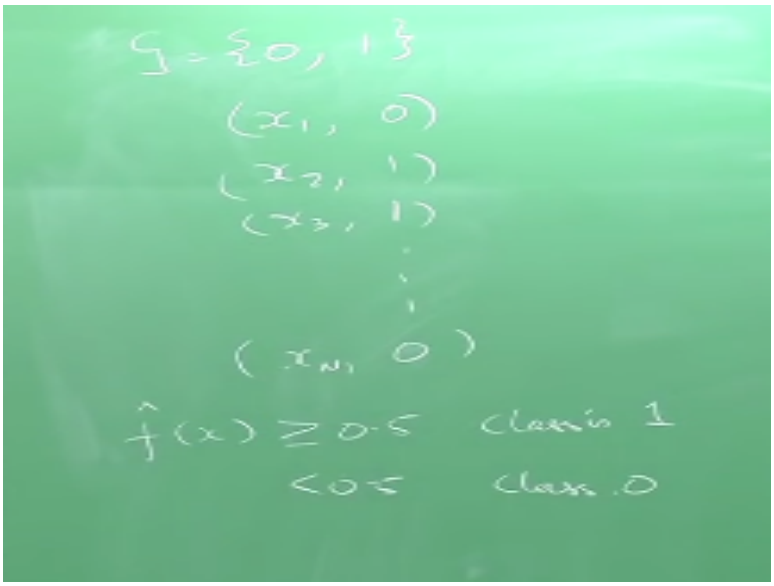


Let us say that I have only two classes for simplicity sake let us say I have only two classes. I have g1 and g2. I will say I will say that 0 or 1 right so instead of having some arbitrary classes I am going to say it is 0 or 1. so what I am going to now do is my thing will become something like this right so instead of having some arbitrary symbols g's ($g_1$ $g_{2)}$, I am going to have $(x_1,0),(x_2,1),(x_3,1),...(x_n,0)$. What I can do is I can solve this as a regression problem I can just solve this as a regression problem and whatever output I get I can treat that as an estimate of the $\Pr(g=0 \mid x) or \Pr(g=1 \mid x)$ .So how to find the probability of g = 1 given x. For the same value of x, say, if there are multiple ones. Suppose I the same value x occurs say 5 times in my training data 3 of the times it was 1 and 2 of the times it was 0 , when I am trying to do a prediction I would expect to end up at the average of this prediction right, so just be like 3/5 and it also turns out to be the probability with which the output is 1 given an x. So if I do regression with this as my training data, what I will be learning is the probability that g = 1 given x . Roughly there are

lot of caveats in this which we will look at when we do regression later obviously you cannot treat this directly as probabilities because the regression curve can become negative.

You cannot really treat it as probabilities but it is a useful intuition to have. So the output that you learn here is $\hat{f}(x)$. In this case if it is $\geq 0.5$ then you say in the class is 1 if it is $<$ than 0.5 you say the class is 0. Let us say can use a linear regression to solve this as well so what we have done in this couple of modules is to look at a unifying formulation for classification and regression problem and looked at a couple of different classifiers that arise out of making certain assumptions about classifiers and regressors that arise out of making certain assumptions about the function that we are trying to learn.

In the subsequent classes we will start looking at each of these in more detail starting off with linear regression. We'll look at this different classifiers in greater detail thank you.

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**Introduction to Machine Learning**

**Lecture 11**

**Prof. Balaraman Ravindran**
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**Bias-Variance**

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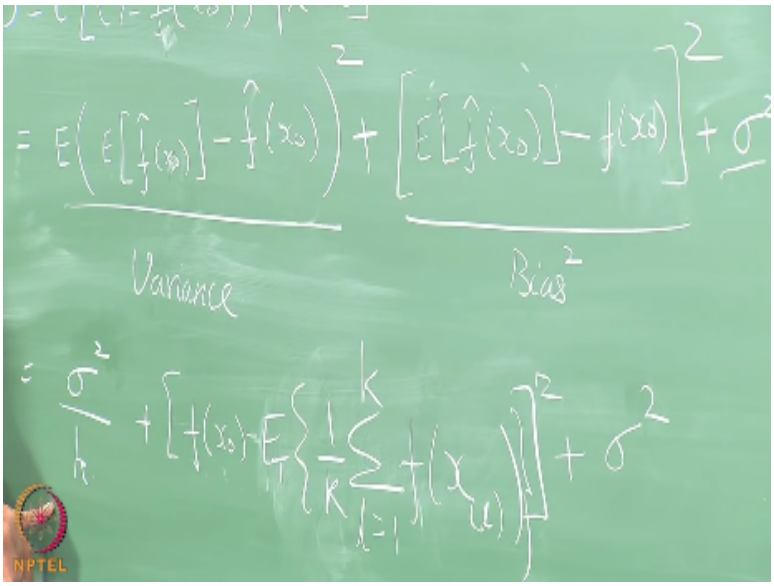


I will give a very preliminary introduction on bias variance in this class and later on, as we progress we will come back to this. So let us start off with the assumption that and in many cases I will be looking at regression because it is easier to write but you can extend similar concepts for classification also. I am going to assume that your actual data is being generated by a system of this form $y = f(x) + \varepsilon$ . So there is a function $f$ which is what you are trying to learn about but the data that is given to you or the y's that are given to you are actually corrupted by some kind of noise.

If you remember from the last class or at least the last class I was teaching, we were talking about a joint distribution over *y and x*. You do not know what the joint distribution is, you are only given samples drawn from that distribution. Here we are making a specific assumption about the form of the joint distribution. I am assuming that there is some kind of an underlying deterministic function *f* here which is operating on my input *x*.

But then it is corrupted by some stochastic noise which we will call epsilon. And that gives me the joint distribution over X and Y and we are going to assume that $E[\varepsilon] = 0, Var[\varepsilon] = \sigma^2$.

So the expected prediction error at some point $x_0$ right is $EPE[x_0] = E[(y - \hat{f}(x_0))^2 \mid X = x_0]$ .So it turns out that I can rewrite this expectation as a sum of three terms. So what are the three terms? The first term is essentially the error that I am going to see by looking at the estimate that I will get from a specific data instance or from the estimate that I will get as an expectation over the sample from which the training data is being drawn. So if I build a classifier multiple times so this ( $E[\hat{f}(x_0)]$ ) is the expected output that I am going to get for $x_0$ and this ( $\hat{f}(x_0)$ ) is the output I am getting for this specific instance of data that I have. So that is one component of the error the other component is, look at the expected prediction I will make for $x_0$ taken over multiple training instances ( $E[\hat{f}(x_0)]$ ) minus the expected output I am going to get or the true output *(E[y])* I am going to get. What is expected true output I will get? That is *E[y]* and what will be the *E[y]* in this case? That is $f(x_0)$. Then there is an underlying error $\sigma^2$ , that just comes from the fact that I have a variance of $\sigma^2$ . I am going to make any single prediction even if I am going to give you the output as *f(x),* there will be an expected error or $\sigma^2$ because my y has that noise in it. So this term $E[\hat{f}(x_0)] - f(x_0)]^2$ is typically called the bias. And this term $E(E[\hat{f}(x_0)] - \hat{f}(x_0))^2$ is typically called the variance of the estimator.

Or $EPE[x_0] = E[(y - \hat{f}(x_0))^2 \mid X = x_0]$ = $E(E[\hat{f}(x_0)] - \hat{f}(x_0))^2 + [E[\hat{f}(x_0)] - f(x_0)]^2 + \sigma^2$

So, one way to think about it is the following. So *f* is my true function, and regardless of how much ever data I am getting, or regardless of whatever data I get I expect to make at least this much error from the true function *f*. So that is the bias and the variance. So it is essentially given

a specific instance of the training data, what is the expected error I am going to make. You cannot do anything about the noise, I mean regardless of how powerful your classifier is you cannot get rid of that $\sigma^2$ because that is inherent noise in the data. So now by choosing your classifier appropriately you can trade-off between the bias and the variance. I will just for simplicity sake take the example of our *k*-nearest neighbor classifier. All of you know about K-nns right. It is very easy to talk about bias and variance in KNNs. So what do you think this variance term will be for the KNN case? Since I am looking at a prediction I am making over many, many instances right and the specific prediction I make for one training set, what would that be? If you think about it, the prediction I am making is essentially just the mean of *k* numbers. What will be the variance of that prediction from many many different samples drawn? It will be the base variance divided by the number of samples. You should have seen that in probability theory course if you have not okay, later I will be doing a session on statistics, and a little bit on hypothesis testing and so on. So further at that point we will go back and look at it.

So I have some distribution and I take samples from it. I draw samples from that distribution *p* and I try to estimate the mean and variance of that distribution through those samples. So now the variance of the estimates of the mean made from this samples is essentially given by the variance of the underlying distribution divided by the number of samples which you are drawing every time. This assumes that you have drawn the *k* samples many times and they have made an estimate of what the mean will be right.

And this is the specific estimate of the mean this essentially the variance of that estimate right, so the variance is $\sigma^2 / k$. Now what about the bias$^2$? $[f(x_0) - E\{\frac{1}{k}\sum_{l=1}^{k} f(x_{(l)})\}]^2$

So this is essentially my expected prediction this there should be an expectation here over the training data so this is the expected prediction I am going to make. So I am going to take the all the *K* nearest neighbors of a data point then take the average of that and that will be the prediction I am making so this is the prediction that I am going to make.

Now let us try and look at this what happens when I change my k. If I increase my k, what will happen to the variance? It will decrease. What will happen to the bias? This is an interesting

question. If increase my $k$, essentially what is going to happen is that I am going to start pulling in data points that are further and further from $x_0$ therefore my estimate of $f(x_0)$ is going to be an average of a lot of dissimilar data points, so the error is going to be higher. So for a fixed dimension increasing $k$, is going to essentially pull in data points that are more and more dissimilar than the query point so I am going to go further out and therefore this will essentially become larger. So as K becomes larger my bias increases in KNN and my variance decreases. Variance decreases just because I am taking an average of more data points.

There is nothing to tell you that the average is correct. Its just that I am telling you the average will look the same even if I change the training data. Because I am averaging so many data points and this part of course we cannot do anything about this is the irreducible bias ( $\sigma^2$ ). So last class we had this discussion about increasing $k$. What did we say when K becomes larger I did not say anything about it becoming more correct, I said that it will look more stable.

Why does it look more stable because my variance goes down. So when I say that classifier is a more stable estimator because the variance has gone down. Also, if you look at the classification surface that you will get, the separation surface that you will get will be a lot smoother if $k$ is very large. But I told you when $k$ is 1, you are going to get lot of isolated islands of different classes and so on so forth and for small values of $k$ , you will find that the classification surface is very complex like it is not like a linear thing or you can predict very complex functions also.

Easy to think of the complexity in terms of the classification surface but function wise also you can think of very complex functions if $k = 1$. If $k$ is larger and larger, the function has to be smoother and smoother. It cannot have rapid variations in the function. That essentially means that when $k$ becomes larger your function class becomes simpler. This kind of looks counterintuitive. I am giving you a lot of $k$ but then your function class typically becomes simpler because it has to have all the smoothness constraints on it and as $k$ becomes smaller, then your function class can be larger. So your regressor or your classifier is more complex. If $k$ is smaller, it is less complex. If $k$ is larger and in general that is the case that if your classifier is more complex your variance will be higher and bias will be lower. If your classifier is less complex the bias will be higher and their variance will be lower. So this is usually the case and

this also lets us understand why k-means does not perform that well in high dimensions. Why is that? So if you take a very high dimensional data, then you can, with a little bit of analysis show that the very high probability, the nearest neighbors will be far away from any query point. If you take any query point and draw a ball around it the ball is more likely to be empty then filled. So the radius of the ball depends on the dimension. It becomes larger and larger as the $p$ becomes larger and so essentially it means that even for small values the bias will be high even for small $k$ the bias will be high because the expected distance to the nearest point will be larger in the high dimensional space. So not only will the variance be high because you have small $k$, the bias will be high now. Basically increasing $k$ is not helping you in that case.

This is just pretty rudimentary discussion on bias and variance tradeoff. But I just wanted to give you a feel of that and you have to keep this in mind later on as we are looking at every classifier that we will see and specifically now we are going to go into linear regression. So, what about bias and variance in linear regression? Does linear regression have any bias? It must be right as it seems to be a very simple classifier. I will talk about that later but the point is yes so any classifier that you are going to be building or thinking about in the future, you will have to start thinking about what the bias and variance okay is. Is it appropriate to use this classifier in this setting and things like that.

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**Introduction to Machine Learning**

**Lecture 8**

**Prof. Balaraman Ravindran**
**Computer Science and Engineering**
**Indian Institute of Technology Madras**

**Linear Regression**

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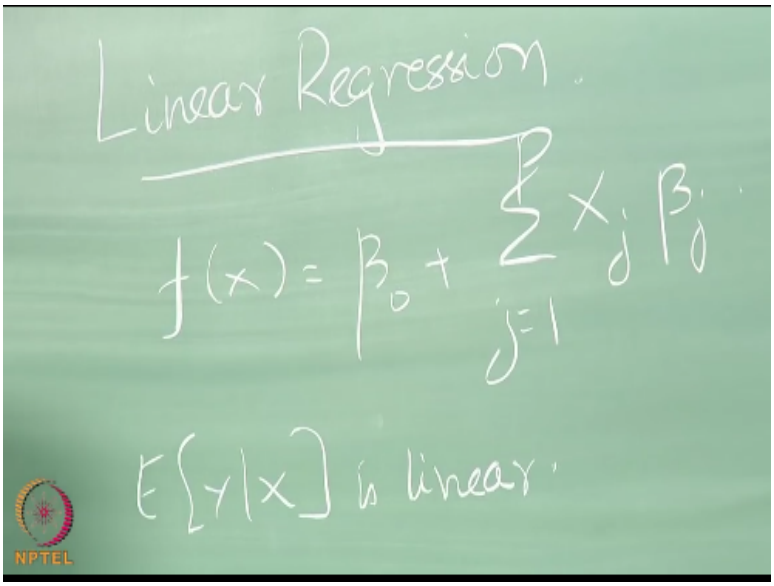


So there is a basic assumption that we had earlier. $f(x) = \beta_0 + \sum_{j=1}^{p} x_j \beta_j$ So we are going to assume that $E[Y \mid X]$ is linear. So what does this mean? *f(x)* is the expected value of y. So there is some kind of a noise corrupted training data that is given to you and $E[Y \mid X]$. So we had this discussion earlier, the nice thing is linear regression is not as weak as you think.

So X can be a variety of different things, X need not just be real valued inputs. They are assumed to be drawn from $R^p$ or X are assumed to be drawn from some *p* dimensional real valued space.,

but they did not just be real valid inputs. They could be any kind of encoding. We talked about basis expansions, which essentially is blowing up your input space by some kind of transformation of the input variables. So if my original data is $x_1 \ldots x_p$ , basis expansion will be: $x_1^2, x_2^2, \ldots x_p^2$ . I could also think of interaction terms such as $x_1 x_2, x_1 x_3 \ldots$

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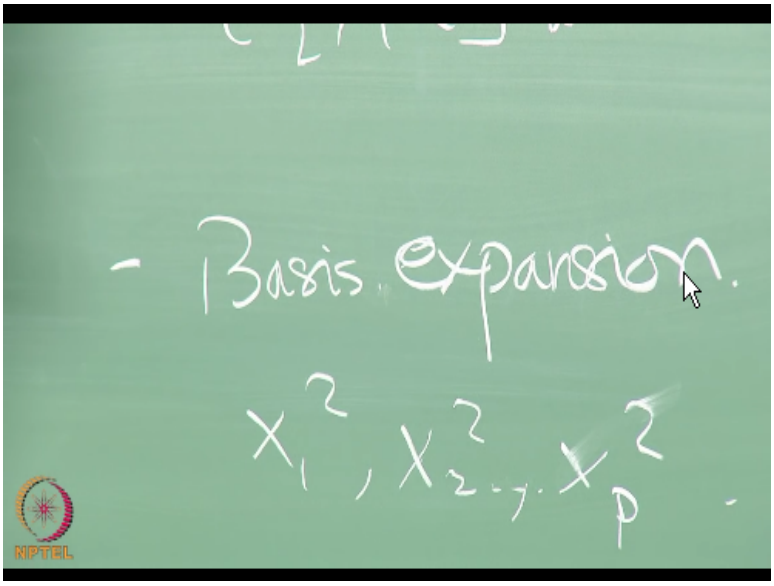


I could think of more complex transformations $sin(x_1)$ etc. X could be qualitative inputs as well. What I mean by that: It can be (hot, cold), (tall, short, medium height) etc. How I would handle that? Weights has levels in the input or it could be just (red, blue, green) meaning it does not really correspond to any level. I mean young and old we can think of saying 'young' is 1 and 'old' is 2 and middle age is '1.5', say, but what about (red, blue and green)? We have to encode each color. But how do you do the encoding? I could do some kind of binary encoding, so I can think of saying that okay I have four colors, so I will have two bits to encode the four colors .Four gets translated into two bits. It turns out that that is usually too much of a compression in the encoding right and if you have four possible values this thing can take it is better to sometimes use four bits. So it is sometimes called 'one of n' encoding. So, only one of those four bits will be one for any input. 'Red' means the first bit will be one blue means the second bit

will be one and so on and so forth. Or sometimes it is called 'one hot encoding' because one of the inputs will be hot the others will all be cold. So sometimes it is called 'one of n' or 'one hot encoding'. So you could take care of qualitative or categorical inputs like that. And whatever you do or however you have expanded your basis or doing your encoding, finally the model you fit will be linear. Its just that if your original dimension was 1 in this case because I had a one color input, it could take four values now my input dimensions become 4. Similarly I had *p* input earlier now input dimensions has become the case above. It really depends on whether I am feeding in $x_1 \ldots x_p$ also. If I only feed in the second order terms it is still *p* but the class of functions I can model is restricted. And if I feed in $x_1 \ldots x_p$ as well as the squares then it is *2p* and the class of functions I can model will become larger.

So that is basically the underlying set up. The model is still linear. Why is four-bit encoding better than 2-bit encodings? The point is so when I have two bits encoding so there will be the same input variable that gets activated for two different colors. Suppose I am using red this 01 okay and blue is 11 so that 11 will get activated for both red then blue. And likewise when there is 10 and 11 right. So so the same bit gets activated for multiple inputs. And that gives you some amount of interference in the training. We can still train it with two bits you probably need a lot more training data to take care of the interference from one to the other. When you have these kinds of 4 bits essentially you have independent weights modeling the influence of each of the levels. So for red there will be one weight and by weight I mean one $\beta$ here. So for red there will be one β that will be modeling the effect of red and for blue there will be another $\beta$ modeling the effect of blue. That way there will not be much interference between the variables. So technically you can model it with two bits and get away with it is that you will probably need more data for the estimation. That is why I say in practice four bits is better.

Let us continue looking at this. So my training data is : $(x_1, y_1), \ldots (x_n, y_n) \quad x_i = (x_{i1} \ldots x_{ip})$ .I am going to assume this of the form and that each $x_i$. I am going to assume that I have *n* data points each of the form $(x_1, y_1), \ldots (x_n, y_n)$ .

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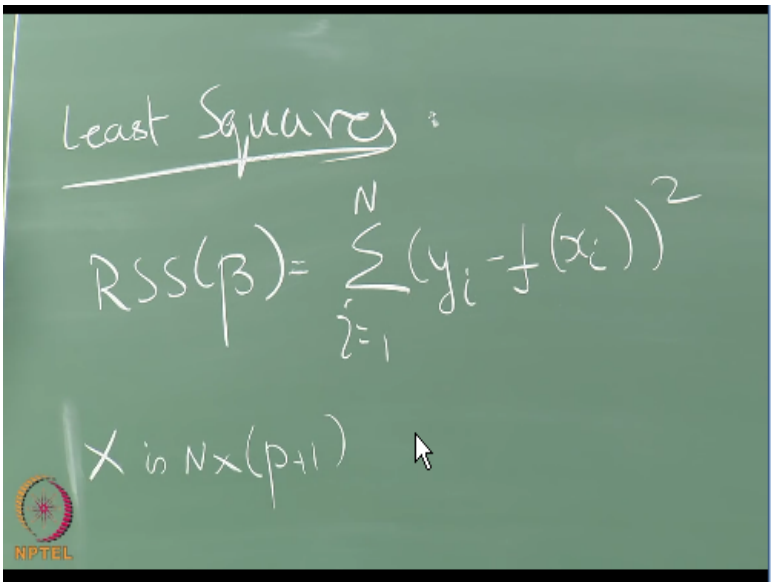


And the way we are going to fix this is using least squares. So we are going to translate this into matrix notation to show you some things and in matrix notation when I write *X* at least for today it is an N x (P +1) matrix, where the first column is all ones. So we have seen this already so it is N x P matrix where the first column is all ones. So I can write it like this in matrix form as below.

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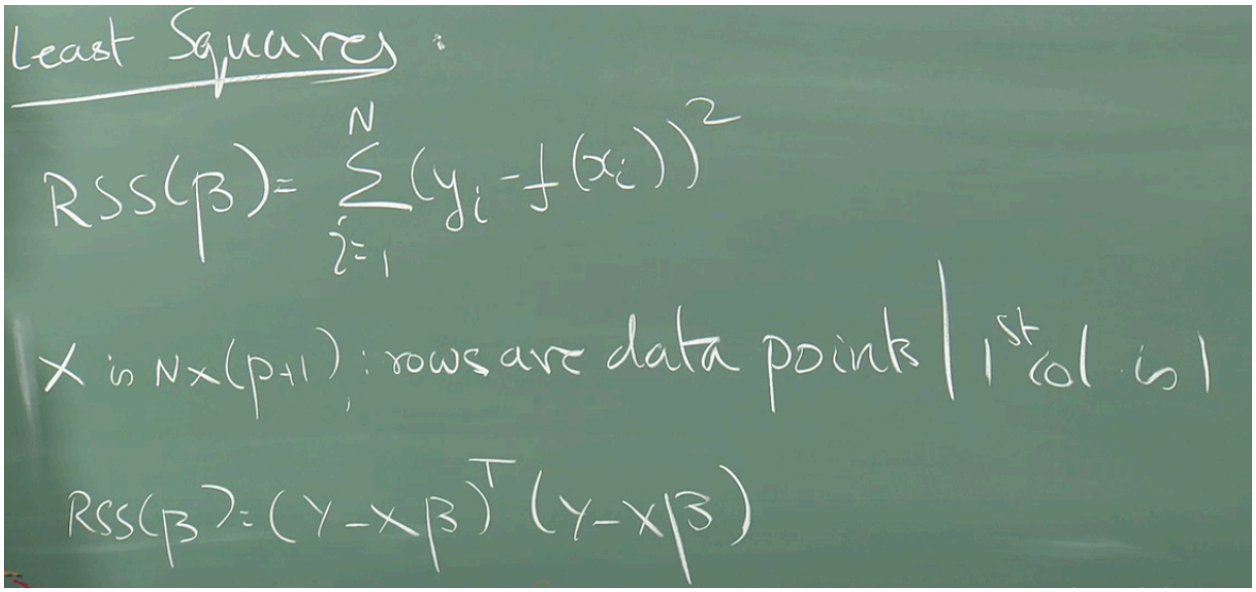


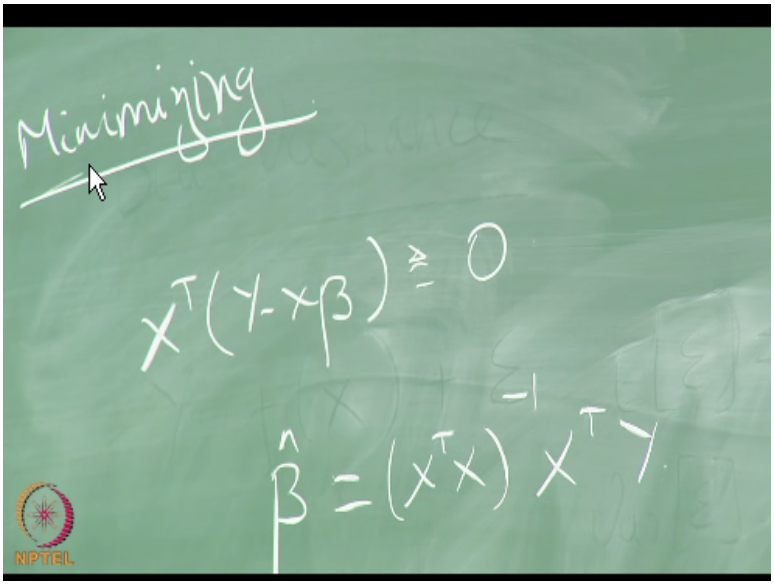

The square in the first equation becomes that way in the last equation. Because *f(x)* now becomes just *Xβ*. What would that be? It will be $\frac{\partial RSS(\beta)}{\partial \beta} = -2X^T(Y - X\beta)$

So I am going to let you think about this if you cannot see that immediately. I am going to let you work it out yourself. We should get really familiar with doing this kind of derivatives of matrices. Because we will be using this quite often whenever we write this kinds of error functions in terms of matrices be ready to use this.

Intuitively you can see it but you just need to work out the math here. $\frac{\partial^2 RSS(\beta)}{\partial \beta^2} = -2X^T X$ .

And at least this seems easy enough I am taking a derivative of this with respect to β and the only term where β appears is $X^T X$. So if X has full column rank, $X^T X$ will be positive definite. So it will certainly be invertible. So it will be and no it is not just invertible it will be positive definite and therefore we can assume that it has a maxima or minima.

Now anyways so if I equate this to 0, I will get an extremism point, I will get either a maximum or a minimum. So what would it be? Anyway think about it I am going to anyway minimize the error that that should give you a clue right. So essentially I have to set this to 0 if I want to find the minima of the error and this is going to give me the following result $\hat{Y} = X\hat{\beta} = X(X^T X)^{-1} X^T Y$ . (Refer Slide Time: 13:26)

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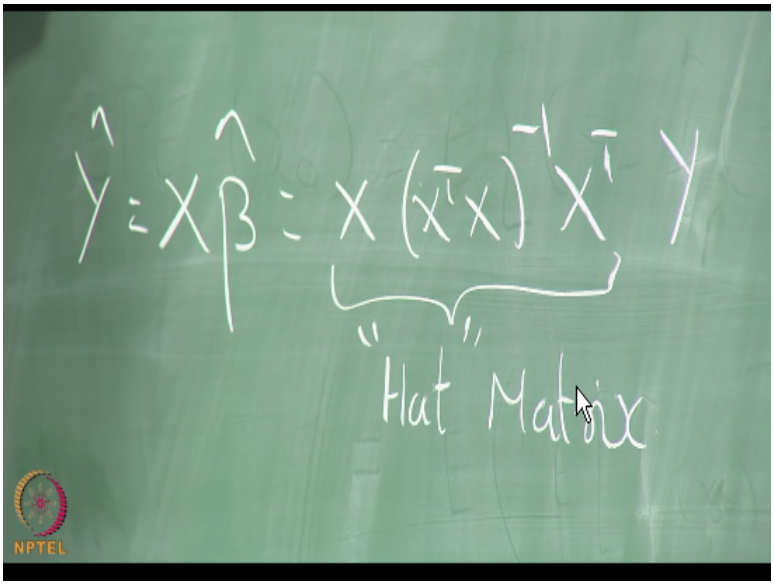


This is all standard if you already know what the solution of linear regression is we saw that in the last class and you should have revised things by now.

They tell you that if you read in the previous whatever we have covered till the previous class and come the next class will be easier. So we already seen the solution and so if I put this

together I basically get $\hat{Y} = X\hat{\beta} = X(X^T X)^{-1} X^T Y$ . As shown above, this expression is sometimes called the "hat matrix". You know why? Because it takes Y and puts a hat on it. So it is called the "hat matrix" so hat essentially is the short hand for estimates. Hats are short hands for estimates and they denote that it is not the true quantity so Y is the true random variable and $\hat{Y}$ is an estimate of the value of Y. So this is essentially the estimator matrix. So in that sense you can think of it as a hat matrix, so another way of thinking about it is the following. So what can we say about Y? The vector Y and not the output random variable Y. I am talking about the vector Y, I should say that. So X is (n x p +1) and Y is (n x 1). So Y is actually a point in (n x 1), so you can take the (p + 1) columns of X and X is going to have (p + 1) columns. You can take the (p +1) columns of X as the set of basis vectors. So what is the dimensionality of each column? It is n. So each column is a vector in $R^n$ and I have (p + 1) such vector in $R^n$.

Now I can think of these vectors as a set of basis function or basis vectors. So ideally I would like them to span a p +1 dimensional subspace of $R^n$. It is where all the linear algebra tutorial supposed to help. So you have a p + 1 subspace of $R^n$, and your $X\beta$ will be a point in that p + 1 dimensional subspace. Because X are my basis vectors and I am combining the basis vectors by some set of scalars β like (β1 β2), all those scalars just am getting just getting a linear combination of my basis vectors, so it is going to give me some point in the p + 1 dimensional space. In fact if I am doing linear regression all I can do is express a point in that p +1 dimensional space. If I take the columns of my X matrix any output that I can learn will be a point in that p +1 dimensional space. This makes sense. So what is the best possible point in that p +1 dimensional space that I can predict? So let us say I have two vectors $X_1, X_2$ . These are not the data points. These are the column vectors. Let us suppose that I have a vector Y which is in the n dimensional space. What is the best prediction I can make?

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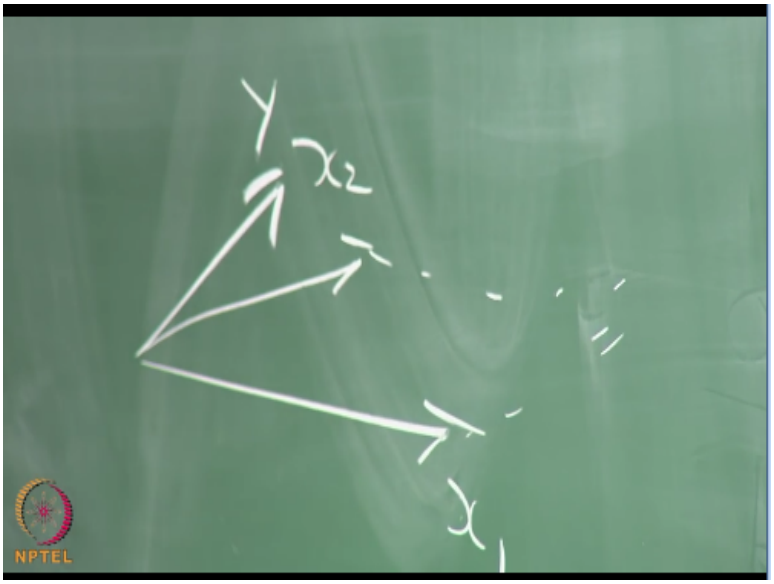


So Y is in $R^3$ and if you can buy into my drawing skills , so X1 and X2 span that two dimensional subspace of $R^3$ and Y is a point in $R^3$. So that is what Y is. What is the best prediction that I can make that fits into the X1 X2 space? It will be the projection of Y. Am I making the prediction? Yes, because if you look at the error, or $Y - \hat{Y}$ is essentially orthogonal to the space spanned by X. So that is what our minimization condition is telling us. $X^T Y - X\beta = 0$ . So essentially it is telling us that ,this vector this vector is orthogonal to the plane spanned by X. That is essentially what the minimizing condition is telling. So this is the best possible estimate that you can make for y given that you are restricted to the plane spanned by the columns of X.

That make sense so this is a geometric interpretation of what linear regression is doing. Lets us think about some other things. So what happens if X is not full rank. That would mean that some of the columns are dependent on each other or linearly dependent on each other. That essentially means that it is not really spanning a p +1 dimensional space its spanning a smaller subspace. It is spanning a smaller dimensional subspace therefore your approximation is going to be worse. That is one part of it. Then anything else the formula would not be valid. So we have to think of different ways of doing it. So that is the next thing but still regardless of that the best fit that you can get will still be the projection of your Y onto the space spanned by the Xs. You have to have to come up with different ways finding it. But it will still be the projection so that is the thing. So

one of the easiest ways of doing it is what? Now we know exactly, that it is in the space that is spanned by these vectors and we are supposed to find the projection onto the space. And if there are redundant vectors that will not help us define the space. We can throw them out but even though I have all this p+1 dimension. Whatever is redundant that is not helping me define the subspace I can throw them out.

So there are some very simple checks that you can do. In fact if you use some standard tools like R and you are trying to do linear regression unless you explicitly tell it not to it will automatically do the check for you it will automatically do the check and throw out the independent columns. It will pick some subset of independent bases and then use that to figure out what the projection should be. So what about the case with the number of dimensions is much larger than the number of data points? Do you think that will happen? Yes? No? Possible? How many of you here work with images or have done any work with image data? So more often than not that is the case right. So because image data is very high dimensional and unless you are able to generate huge volumes of such data and more often than not p will be greater than n.

So you have to think of some kind of regularizing the fit so that you get actually a valid answer. If p is larger essentially what it means that you have a much larger space and Y actually exists in a smaller space than what is given to you. So we have to figure out a way of regularizing the problems so that adding additional constraints on what kind of projections you are looking for. Because otherwise it does not make sense to talk about the projection of Y on this P plus 1 dimensional space okay.

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**Introduction to Machine Learning**

**Lecture 13**

**Prof. Balaraman Ravibdran**
**Computer Science and Engineering**
**Indian institute of technology**

**Multivariate Regression**

So far I have really not worried about the fact that we have multiple dimensions in the input space. That we just had this way of handling it. But then if you actually look at how statisticians typically present linear regression, they will start off with a univariate regression or they start off with one input variable and one output variable or one independent variable and one dependent variable, so the independent variable is the input variable.

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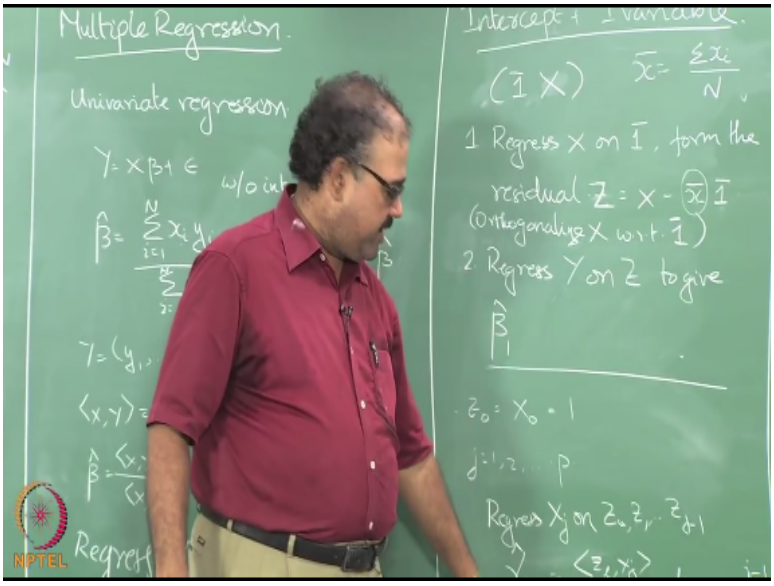


So whatever we have looked at so far is usually called multiple regression. We will still typically start off with univariate regression as people usually start off with univariate regression because

it is easier to analyze you can derive a lot of intuition into what exactly is happening with the regression. In fact if you think about it, this picture I drew for you is a univariate regression with an intercept so that there is a column of ones and then there is one other variable that is all.
This is essentially univariate in the regression with the bias term. So you can very easily develop all kinds of intuitions and also the analysis will be very clean and more importantly you can understand multivariate regression by a series of univariate regressions. So let us look at it very quickly and then we will see what happens. So this is the basic model that we have $Y = X\beta + \varepsilon$ . But here we are going to assume that X is a single number so it is a single vector now. So my data will be of the form some (x,y). This is not a that is not a vector is just a simple x.

So why is it called the intercept? So the constant value you add is what the value of where it will cut the y-axis. That is why it is intercept. So I have no intercept that means there is no $\beta_0$ here.

So now this one β is given by $\beta = \frac{\sum_{i=1}^{N} x_i y_i}{\sum_{i=1}^{N} x_i^2}$ . Essentially our original case or $\hat{\beta} = (X^T X)^{-1} X^T Y$ for a univariate case. I am going to denote by $r_i$ the residual error that I am making, or $r_i = y_i - x_i\hat{\beta}$ . So I made the prediction $x_i\hat{\beta}$ and $y_i$ is the actual output that I saw in the training data, so $r_i = y_i - x_i\hat{\beta}$ is the residual error.
I hope people are familiar with the inner product notation of this form essentially. Now can you tell me what $\hat{\beta}$ should be? This is fairly simple, so one thing I just point out in passing here is that I am not going to cover it. If you want you can go through Hastie Tshibrani Friedman later chapters.

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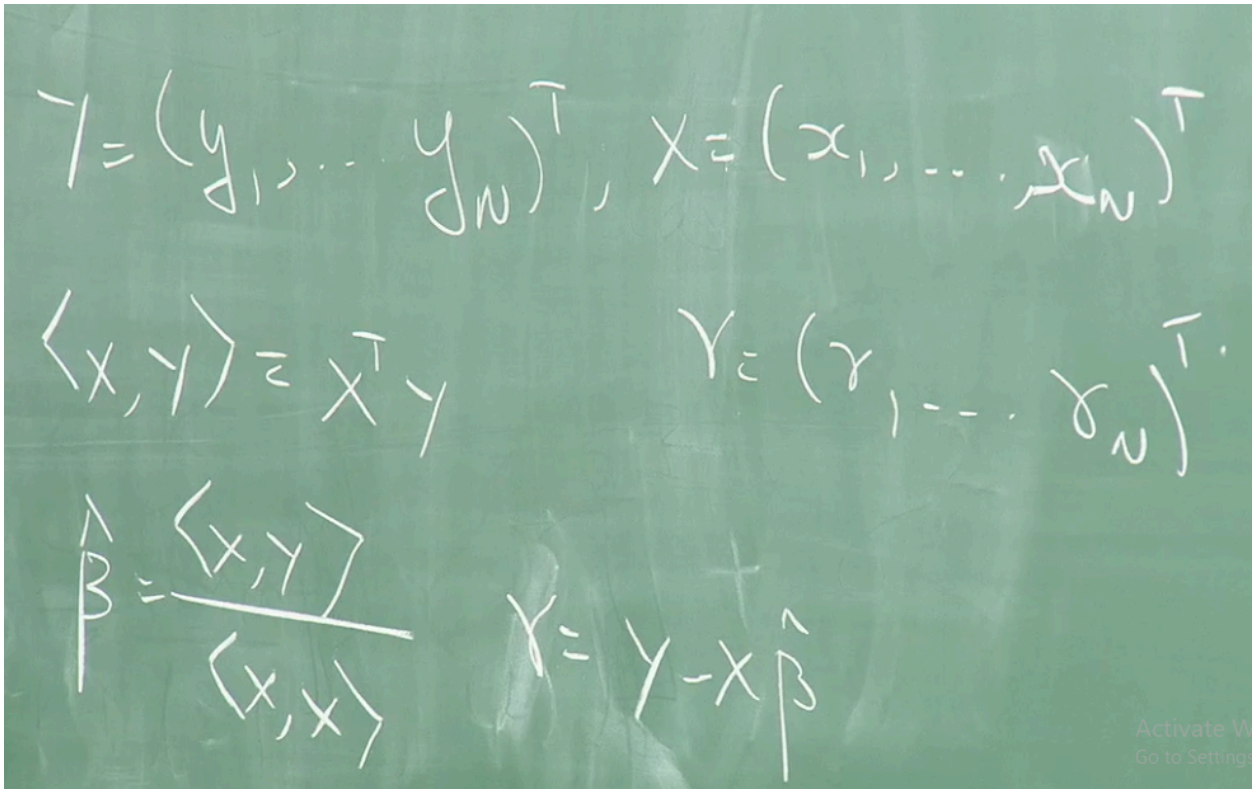


But the fact that I am using inner products here should tell you that I can apply the ideas of linear regression on any inner product space and not just in real number space. So I will leave it at that gives you a good generalization. So what we are doing here I will call this as, "Regressing Y on X" and we get the coefficient $\hat{\beta}$ . So what we are talking about so far is a univariate regression no intercept nothing.

Suppose that your columns are all orthogonal. Not only are they independent they are all orthogonal with a little bit of thought you can convince yourself that each β, so β1 β2 β3 and so on so forth are just given by regressing Y on X1, X2, X3 and so on so forth. So β1 this regression of Y on X1, β2 is regression of Y on X2 . Why is that the case? So now my X1 and X2 are orthogonal they are actually the orthogonal basis an orthogonal basis for the p dimensional space the p + 1 dimensional space I am talking about and each coefficient that I am going to get essentially would mean will be the intercept on each of the individual dimensions.

The projection on each of the individual dimensions because they are orthogonal in the lower space. So that is easy to convince yourself. What is interesting is what happens if the Xs are not orthogonal, they are independent or they are still spanning a p + 1 dimensional space but they are not orthogonal. So what do the coefficients represent? In that case so that is essentially what we are going to look at. So we will start off by taking one step at a time.

Look at the intercept plus one variable. So far I said that is one variable without intercept and now I am adding the intercept. So what will be what does it essentially mean for us? My X becomes (1,x). So what I am going to consider is a column of ones and my original vector x. This is my new vector that I am going to consider. So what I am going to do is the first step, I am going to do is tell you about that, so this upper case X is the actual column vector Xs of consists of x this is the actual input I am going to look at. So let me define $\bar{x} = \frac{\sum x_i}{N}$ as the average of the all the inputs I have seen all the inputs I have received is my training data.

So I regress X on 1 and form the residual. So what will the residual be? In this case what would it be. Because I am regressing on one all ones, so if all ones is the only input variable I have what should be the best possible prediction I can give? It will be $\bar{x}$ . So $\bar{x}$ is the only output I can give that will be the one that minimizes the prediction error. Because I am looking at squared error the output should be $\bar{x}$ . So my $\hat{\beta}$ will be $\bar{x}$ in this case.

So the residual which I will denote by is by $z = x - \bar{x}\bar{1}$ . This one bar is just to indicate that it is a vector of ones. So this x is a vector so this $\bar{x}$ is a scalar value which is the average of all the inputs and $\bar{1}$ is the vector of ones so that this gives you the residual. Does it make sense? This is the vector of residuals, so I usually put the bar on the x and the z then the middle to differentiate it from two. But sometimes it looks like lower case z, so is it fine let us adopt the convention that even if I put the bar there it is still uppercase z? Because either way I will have to be very careful about distinguishing 2 and Z.

So the second thing is shown below. 2 separate univariate regrssions (1) and (2)

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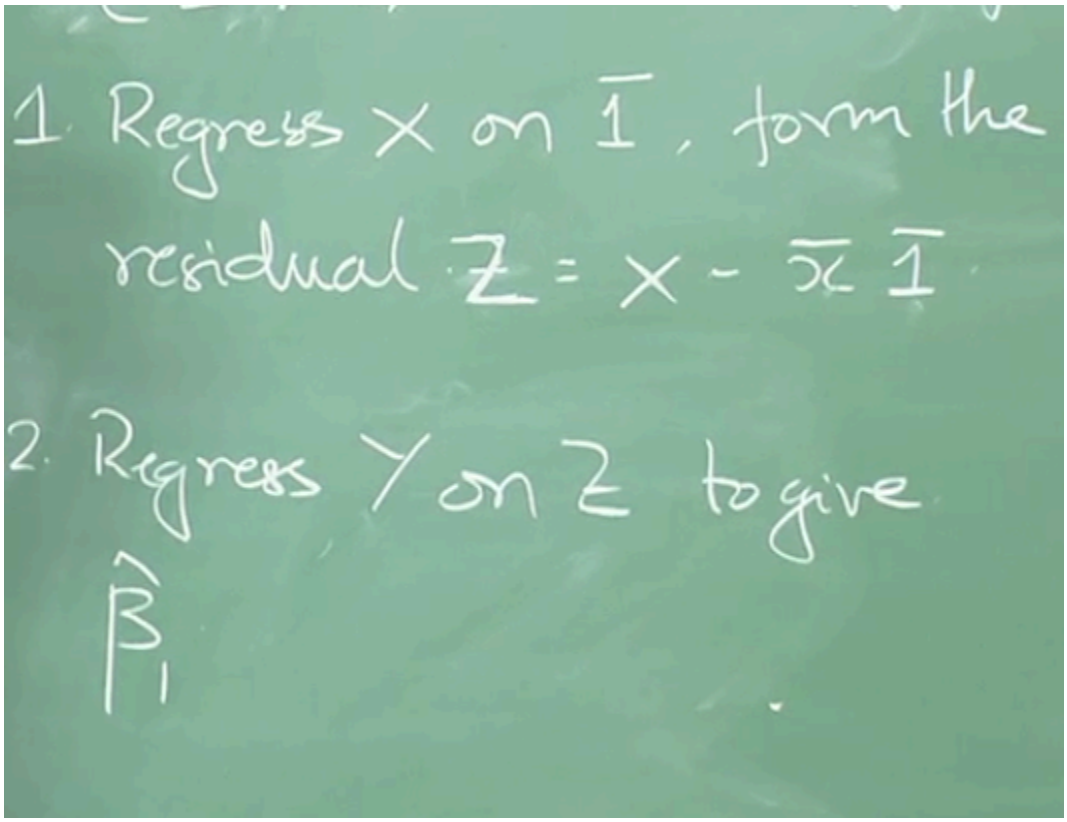


This tells me that I have taken the average value out of the input variables because the average value can be used to predict the average output if I want I have taken out the average value so whatever is left is the individual variations on the data point and use that to predict my output value y.

So this essentially means that so given that there are two dimensions 1 and x. So the $\hat{\beta}_1$ tells me what is the contribution of x after I have used one to explain the output. So given I have taken care of one already what is x? So if you think about what I have done here this is essentially making it orthogonal to the 1 vector, the z vector is essentially the x vector the component of x that is orthogonal to the 1 vector going back to how we did the univariate regression. That is what we have done here so does this remind you people of anything? We have already looked at Gram Schmidt orthogonalization. This is essentially something very similar to that.

So I start off with one bar and the x as the 2 vectors that span some space now I am orthogonalize I am essentially giving you an orthogonal basis now one is one vector and z this the other vector but together they span the same space that was spanned originally by one and x. Except that they are orthogonal and people agreed with me earlier when I said that if the columns of x are orthogonal then they can independently do regression on each of those columns that is essentially what we are doing here. So I have done a regression on this to get me $\beta 1$.

So going back to our picture here, so essentially I had some $x_1$, I had some $x_2$. So what I did was I first regress $x_2$ on this and so essentially I am getting so that is my z . I am getting a orthogonal component to that. So now I have $x_1$ I have z and they are spanning the same well yeah they are spanning the same space that z is a projection of $x_2$. Imagine the plane is going into the board so it does not look to you but the plane is going into the board.

So z is actually perpendicular to $x_1$ and it is formed by projecting, by regressing $x_2$ on $x_1$ that is direction z . And my original y which was going out of the plane, now I essentially project it on z to get the coefficient. It does not matter see this is still going to project here so earlier when I wanted the coefficient for $x_1$ and $x_2$. I would have looked at this these points here now I will basically look at these points that is essentially what I have done there is no change in the actual space. So the same $\hat{y}$ is what I will get but the coefficients I will be using for representing the $\hat{y}$ will be different. We can generalize this to p dimensions.

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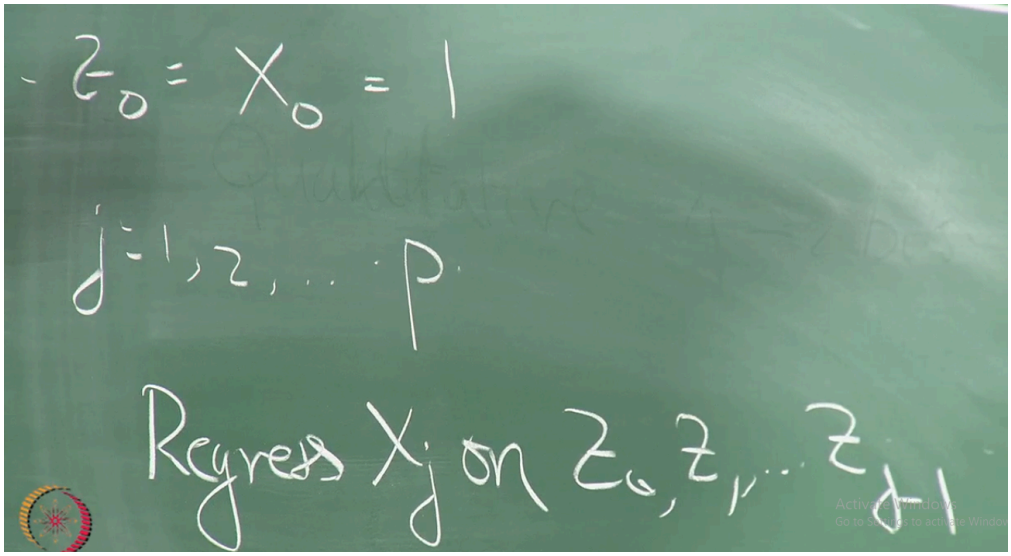
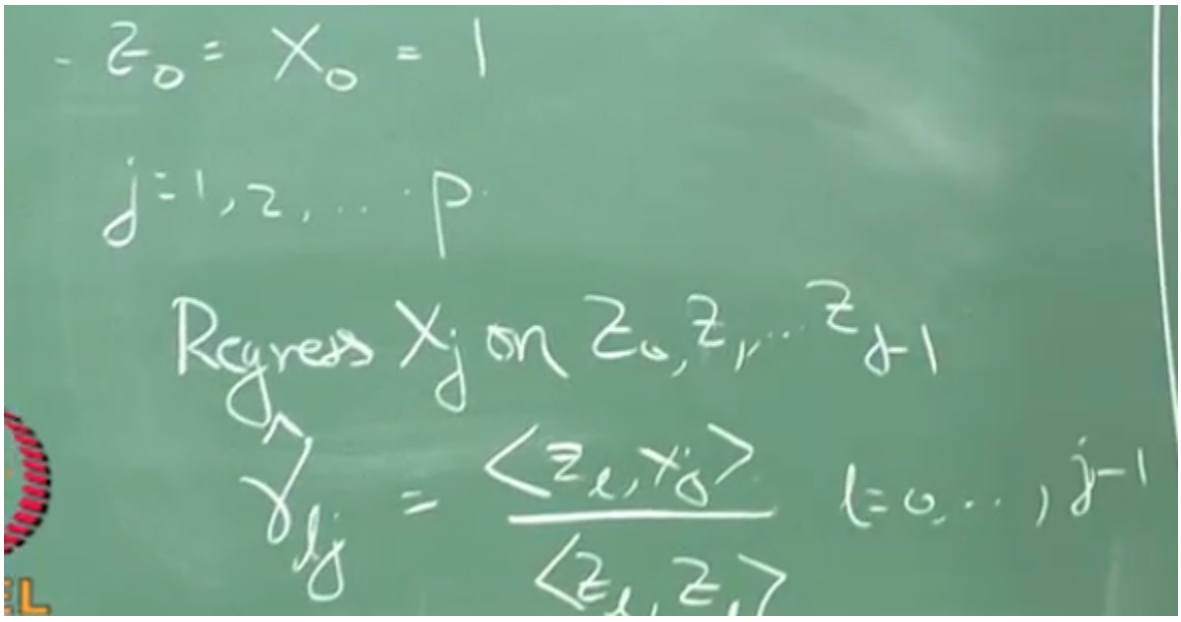



So what will you get so j runs from 1 to p so I will regress $x_j$ on all the previous z directions that I have determined. So how would I determine $z_0$ . I start off with $z_1$ would have been obtained by regressing $x_1$ on $z_0$ and then finding the residual. So that gives me so that is what we do here , so I take $\bar{x}$ I basically I regressed x on $\overline{1}$ and then take the residual and make that as z. So likewise I will regress $x_1$ on $z_0$ take the residual and use that as $z_1$. Then I will regress $x_2$ on $z_0$ and $z_1$ and then take the residual. So that is the $\gamma$ coefficient which will so in this case it was $\bar{x}$. (Refer to slide time 19.48)



So if you think about it. That will be $\bar{x}$, so the inner product with $z_l$'s when z is all ones is n and the $z_l$ inner product with $x_1$ when z it is all ones is just $x_1$. So this essentially will be the summation of $x_1$ this will be just $\bar{x}$. The first case that is this is our gamma. Does it make sense or was it too quick? Yes or No. So I am saying this $\bar{x}$ was derived by just using the same formula. This happens when I regress on the first variable so start off with $z_0$ is 1 I am regressing $x_1$ on $z_0$ when I regress $x_1$ on $z_0$ what do I get is $z_0$ inner product $z_0$ is n. Well just ones and they sum up all the ones. So that is where dimension is n that will be n and $z_0$ inner product $x_1$ will be summation $x_1$. So summation $x_1$ / n is essentially the average. So that is essentially what we had here. So that is the same formula. Now I am generalizing to other dimensions so I am still continuing the loop here , so that loop that runs for J = 1 to P.

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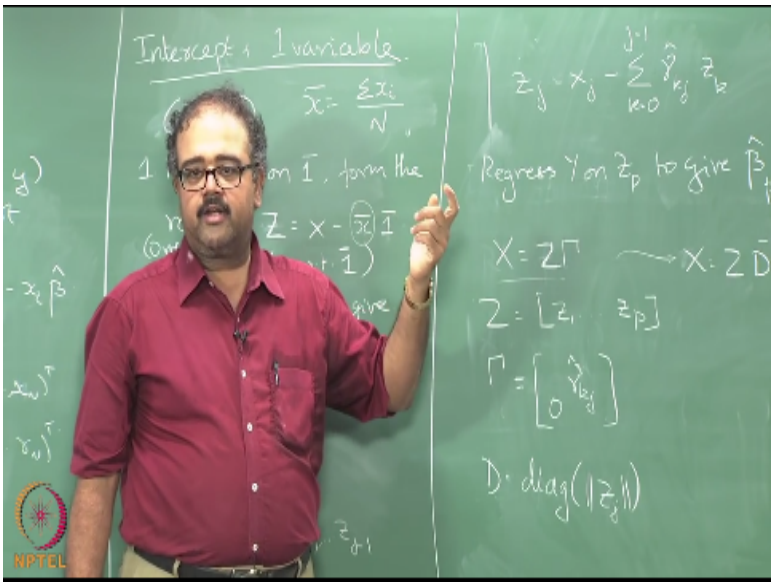


So for every j and that is how I derive my $z_j$. So I take the current coordinate under consideration $x_j$ subtract all the previous dimensions I have basically looked at. So what I am left with is the orthogonal component of $x_j$. Orthogonal with respect to the dimensions I have already looked at, so in some order I am considering, so once I have done this for in some more I am considering it in some order when they come to the pth dimension, so what do I get, so what is $\hat{\beta}_p$ ? It is the

actual coefficient that I will find for the pth variable if I had done the regression as we talked about earlier. If I had done that if you estimated my β like this .

This is essentially what I will end up with but because of the process we have generated it we can interpret it in a slightly different way which is essentially β hat P tells you how much the pth variable contributes to the output given that we have adjusted for all the other input variables, how much does the pth variable contribute to the output, now we can go back and think about non independent vectors, if any of the variables is not independent , so what will happen in this case the residue will be 0 and it essentially will be trying to find, how much would 0 contribute to the output that is not going to be a lot .

But it becomes even more interesting if my vectors are nearly dependent but not exactly so. What will happen is if I subtract out everything else from that vector, so think of it like this: this is $x_1$ that is $x_2$ this is the 2D plane it is not like this is the plane , so $x_1$ and $x_2$ are very close to each other there, so if I subtract out $x_1$ from $x_2$ I am going to get a small component that is orthogonal to this , I am going to get something like this. Now if I take the inner product of that, so that will be a small number here, so this can become very large.

So if my vectors are nearly dependent but not exactly so that the residual is not zero exactly but close to zero then the whole thing can become very unstable. The whole estimation process can become unstable, so that is essentially what will happen even if you eliminate perfectly dependent columns, there could still be possibility of getting numerical instability so to avoid all of these things people have come up with various techniques, that of course one of them is to essentially get rid of all the correlated or the nearly correlated columns, but there are there are other ways of actually trying to get this to be stable .

So just an aside, so let Z be the matrix that we create by taking $z_1$ to $z_p$ columns. So I have done this $Z_1$ to $Z_p$ in this elimination process in some order. So I will take this $z_1$ to $z_p$ columns and γ is the matrix where I store all my γ hat $k_j$ there is an upper triangular matrix, so for every combination $k_j$, I will have 1 γ hat value I will just put it in the upper triangular part and the

lower triangular side I will just keep it as zero. So an upper triangular matrix there and you can think about it you can write the x as z times gamma x can be written as z times $\gamma$.

So essentially the I am just stacking all of these things you have done together and we are writing it as is z times $\gamma$ and so D is a diagonal matrix where the diagonal entries are the norm of the inner product of $z_j$ with itself , so the j entry or the $Z_j$ entry in the D matrix would be the inner product of $z_j$ with itself that is the norm of $z_j$, so I can write it like this, so this is called the QR decomposition of x , so the thing about Q is it is orthogonal . Q is orthogonal and R is upper triangular .

So this kind of a representation for the data matrix or a QR representation of the data matrix essentially gives you some kind of a orthonormal basis but Q is not just orthogonal it is orthonormal also. Why because I am dividing by the norm here? So the product will be ones or zeros because they are orthogonal to begin with and anyway I made them orthonormal so Q gives me an orthonormal basis and R is said upper triangular matrix that lets me reconstruct the inputs x and this kind of a decomposition is very convenient and it is used widely in other kinds of representation or transformation of the data and so on so forth.

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**Introduction to Machine Learning**

**Lecture 14**

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**Subset Selections 1**

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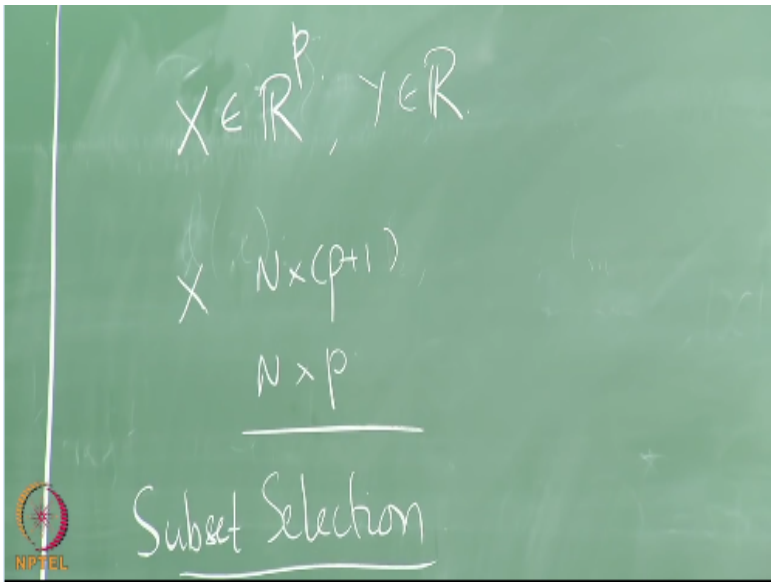


So we were looking at linear regression. We are assuming that the X is coming from $R^P$ but I told you that it's not necessary that it has to come from the set of real numbers and we talked about various ways in which you can encode the data and so on so forth. And then if we assume that the Y comes from R and I told you depending on the circumstances, for example, so we will talk about the input matrix X okay which might be of $N \times p+1$ or $N \times p$. So it will be $N \times p+1$ when we actually have an explicit intercept or a $\beta_0$ term. For the $\beta_0$ term we will have a

column of ones added to the data. The input instead of thinking of it as a $p$-dimensional vector we will think of it as $p + 1$ dimensional vector and when I do not have the intercept, it just becomes a $p$-dimensional input. So that is that is the basic setup that we have and we are essentially looking at minimizing some squared error. So we looked at the simple linear regression we looked at the case where there were multiple inputs and we looked at how you can interpret that in terms of single variable regression univariate regression is essentially what we looked at in the last class. And so this class we will go on to look at a little more complex things that you can do with linear regression. So linear regression is great because it is so easy to solve and it is very efficient, runs very quickly and all, that but there are a couple of drawbacks to linear regression. So the first one is that if you remember I was mentioning in last class also that by doing this least squares fit you are actually getting a fit that has very low variance but it also has how much bias? What I said is that if you do least squares fit and if linear happens to be the right choice then least squares which gives you the 0 bias solution. So the least squares fit gives you 0 bias solution but the problem is the variance can be relatively high and it turns out that by not just doing the straightforward least squares fit, by doing more tricks with their data with the models that we have we can trade off a little bit bias. I am going to get a biased estimator for the fit for the line. I am still fitting straight lines and I have not done anything different I am still fitting straight lines but the fit I am getting will no longer be a bias-free fit but then I can reduce the variance a lot more by adding certain additional constraints to the problem. So essentially what I would really like to do is reduce the number of variables which I am trying to fit, Instead of trying to fit $p + 1$ variables, if you can somehow reduce the number of variables what does is equal to? It is equivalent to setting some of the $\beta$ to 0. So if I can somehow set some of the β to 0 then essentially what it means is that I have lot fewer parameters that I am estimating . So the fewer the parameter set I am estimating the lower the variance still but because I am now restricting the class of models that I am going to be looking at, since, I have to set some numbers to 0 so my bias will go up slightly. This is assuming that I am fitting straight lines you know the straight lines are the right thing to do but still my bias will increase a little bit in this case. Of course there is always this residual bias because the language of straight lines is not powerful enough. So that bias is still there. I am saying even assuming straight lines are the right thing to do if I am going to force certain coefficients to be 0, I am increasing the bias in the estimator. But the variance will go down because I have a lot fewer parameters that I am going to estimate. So

that is essentially what we are going to look at and so what I mean by subset selection here is that I am going to select a subset of the input variables to use for fitting the line. So one thing is we can reduce the variance significantly and that is where we can improve the prediction accuracy of the model. That is one of the reasons we would like to do subset selection. Are there any other reason you can think of for wanting to work with the few smaller subset? Less computation but it is a question of where you have to do more computation to figure out what the subset is and so on so forth. But yeah less computation is one answer but there is another one. So related to this, there are variables which could potentially have high noise and so it will end up with a small coefficient. But then if I tell you okay here is this model M and it has like 135 coefficients and it becomes hard for you to even figure out what this means. Instead of that, if it is okay here is this model you gave me a 135 variables but these are the 4 important variables that I need for doing a linear fit.

Now it is easy much easier for you to interpret what is going on. So interpretation is a very big component of any kind of data analytics that you want to do. Ultimately what you are doing with machine learning is trying to understand the data, so one of the things you would like to have this interpretability.

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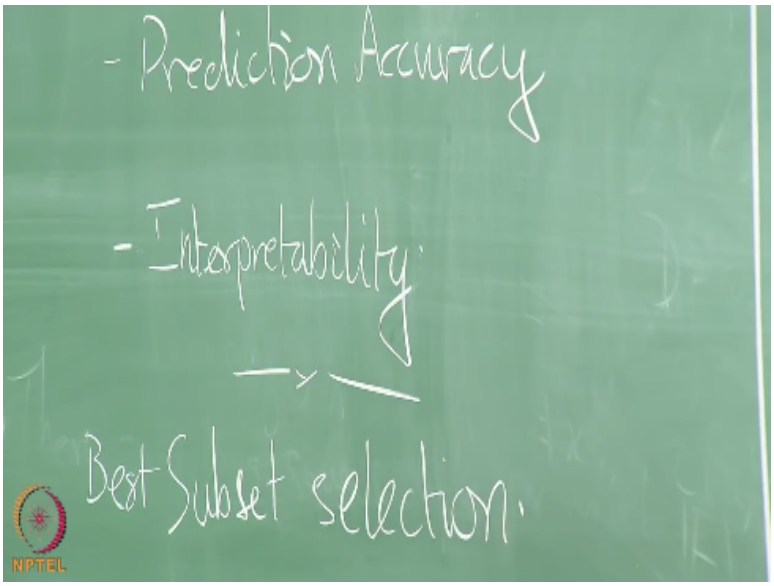

So there are many ways in doing this the simplest kind of approach is essentially to take "subset selection" literally and try to select from subsets of features. So why is that a simplistic approach? It is combinatorial.

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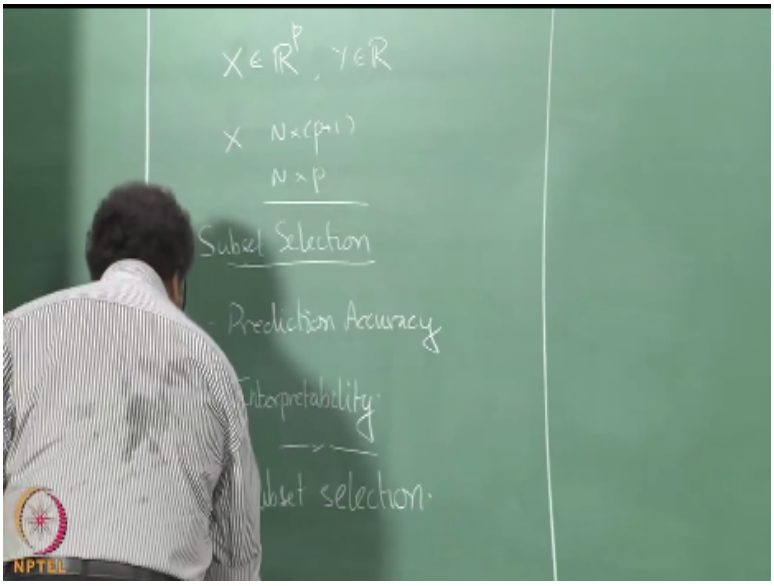


So this will just do subsets in a best subset selection. Essentially I would say that okay here first pick subsets of size one, subsets of size two and subsets of size 3 and so on so forth right, and it turns out that you can see yourself if you start playing around with some linear regression tools that what is the variables that go into the best subset of size one right basically the one best variable need not figure in the best subset of size two so it does not have a nice inclusion property.

So for it to have a nice inclusion property you need to have certain very nice conditions on the data set. So in general it does not have this inclusion property and you have to just redo the whole thing again. So it is not enough if you just do one, say okay you cannot be greedy. Basically I cannot say that I will do the first subset and then I just add the one best variable to it and then I will find the next one, so essentially you have to do a combinatorial selection so people have come up with a very clever ways of organizing this, say, using QR decomposition to

do things more rapidly. I am not going to go into details of that but then up to like 30 or 40 variables you can scale well. But if you go for much larger variables, like many problems like in text or image domains then there is no hope to do an exhaustive search but that is a baseline which you can do people actually come up with algorithms which do this kind of a subset selection. So there is one very interestingly named algorithm called “leaps and bounds” which does pretty efficient subset selection but this is just a more of informational thing for you right.

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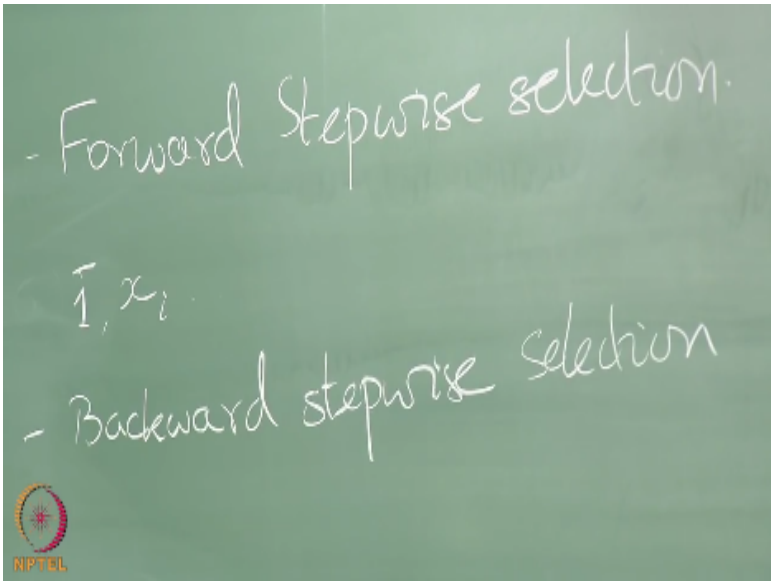


Next is forward step-wise selection. So any guesses what that is? It is a greedy approach. It is just trying to do best subset selection by being greedy. So you start off what is the feature you for sure want to have? All ones or you need the intercept. Otherwise your line has to pass through the origin. So you need the intercepts you start with. So what should be the coefficient for the intercept? We already looked at it before now the mean of the y's right.

So that will be the coefficient? So we already have fitted that now what you do this you start off with that variable, then you add the next one let some $x_i$, add that as the next variable such that it gives you the best fit modulo the set they have already taken. So you are not going to disturb all the stages the variables you have taken to. In step $k$ now, we will add a new variable such that

among all the variables I could add at this $k+1$ stage, this one gives me the maximum of improvement in the performance.

So how will I measure performance? Some kind of residual error on the test data. So that is how I measure performance. I keep doing this until a point where the error does not change much. Is there any other thing I can say? So the residual is orthogonal to any of the other directions I could add right there. So we know that at the end of the right fit you get the residual will be orthogonal to the space spanned by the $x$'s. That means individually if I take any of the $x$'s, the residual will be orthogonal to that individual direction as well. So when you find that none of the directions that are left have any kind of component along the direction of the residual then I can stop.

Or I mean that may take a long time to do because that might happen only when I have the full least squares fit. So I can stop and the residual drops below a certain threshold that you can say okay I am happy with the prediction accuracy I am getting and I will stop here. So there are many ways of doing. So the other way is to do what will you do in this case?
I will start with the fit that has all the variables okay and then I will keep dropping one by one right. So one thing to note is that you can do this if the number of data points is greater than the number of dimensions. So then you can actually find the fit. If $p > n$, as people pointed out last time, the formula we are using is no longer valid. So because if the matrix will not be invertible so we will have to think of other tricks for doing the fit and so on so forth. But forward stepwise selection you can do even if p is greater than n because I am anyway fitting one direction at a time right so it is fine I am adding one direction at a time I can keep going until I reach $n$ directions at which point I should have a full least squares fit. So when people actually even come up with all kinds of variants on this. So one thing which you could do is think of some kind of hybrid approach where I keep adding and deleting directions right. So if you remember we talked about how greedy is not always the best way to grow things. So you can grow up to a certain level again maybe then dropping one of the earlier dimensions will actually give you a slightly better performance could be right so that is one possibility and right they could do some kind of a hybrid version of this. So one thing I should point out here, so you might think that forward stepwise selection because it is greedy is going to be much worse than best subset

selection. So it turns out that in many real cases or many real data sets that you work with right the greedy selection is actually not a bad thing to do. In fact, so much so many statistics packages like R will allow you to do this. So you would not find this in many of the machine learning tools like Weka, since they would not have this kind of a forward feature selection and things like that they have other ways of doing feature selection which we will talk about later right but then statistical packages actually have this stage wise edition of features because they seem to work well on a variety of data sets okay.

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**Introduction to Machine Learning**

**Lecture 15**

**Prof. Balaraman Ravindran**
**Computer Science and Engineering**
**Indian Institute of Technology Madras**

**Subset Selection 2**

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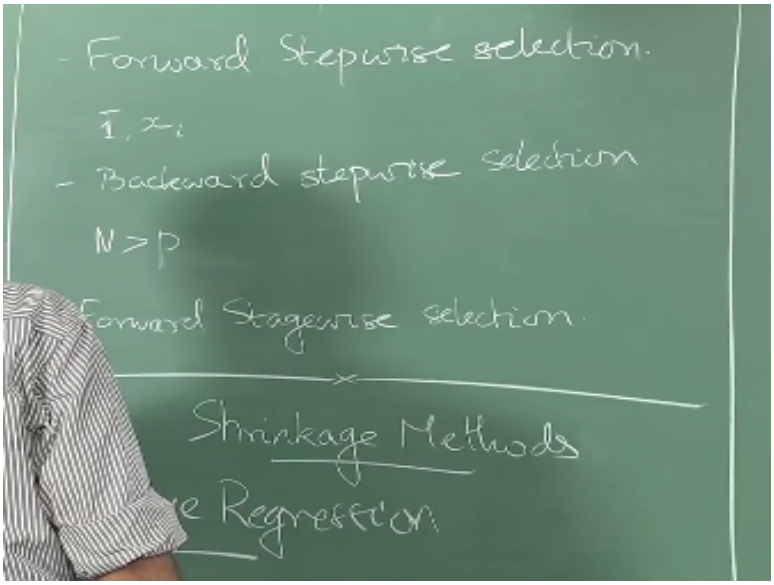


So now we will talk about forward stage wise selection, where at each stage you do the following . Let me rephrase it, on the first stage you do the following. So you pick the variable that is most correlated with the output and then you regress the output on that variable and find the residual. Now what you do is pick the variable that is most correlated with the residual, regress the residual on that variable. Now add it to your predictor. So what is your predictor? You already had one variable then you had a coefficient for that variable which you got by the first regression. Now you have a second variable and they have a coefficient for that variable which you got my regressing the residual on this variable. Essentially what you are trying to do is using

the first variable make some prediction. The second variable is going to try to predict what the error is, so essentially now I will be adding the error to the prediction of the first variable. Did that make sense? So the first variable, let us say, that is the true output that I want. So the first variable will make a prediction saying that okay this is the actual fitted value, and this is the residual. What I am trying to do with the second variable is actually to predict this gap.

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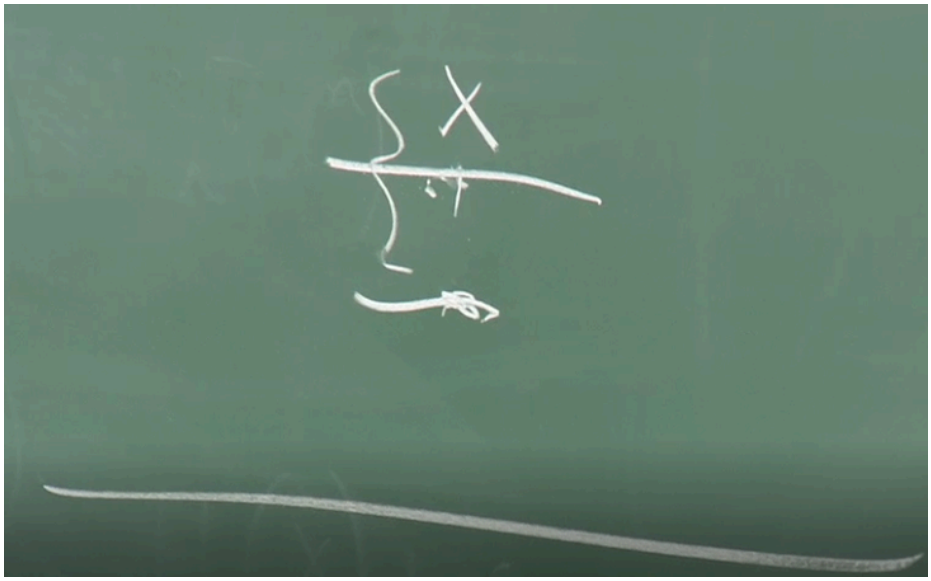

So when I add the second variable with this coefficient to the first variable, what I am essentially doing is okay the first variable will give this as the output, the second variable make some other prediction, let us say that much so I will add the two, so the new output will be that right. Now I still have a residual left, so then I will pick a third variable which is maximally correlated with this residual. And now I add the output of all the three. And then I get my new predictor. Does it make sense? So at every stage I find the residual whatever has not been predicted correctly by the previous 'k' stages, whatever is the residual error after the previous 'k' stages and try to predict that using the new variable. Essentially I find the direction which is most correlated with this prediction and then I try to give you that. This is called forward stage wise selection.

So what is the advantage of stage wise selection? Can you think of any advantage of this?
**Student:** I am not picking variables randomly in forward stage wise selection.

I wasn't randomly picking a variable in the previous methods, I was picking greedily that was not random. Even in the previous case I only pick variables that gave me better fits right. In fact I will tell you that it will probably converge faster in forward step wise selection rather than forward stage wise.

But there is another significant advantage here. If you just thought about the process of fitting the coefficients, at every stage I do a univariate regression. At every stage I am just regressing the residual on one variable. In forward stepwise selection, so every stage I will add a new variable, but then I have to do a multivariate regression, I have to do the regression all over again, I am not able to reuse the coefficients from the previous step right.

So when I add a new variable I basically now I have $k+1$ variables and then do a new regression with $k+1$ variables. But in this case what is happening at every stage is that I just have to do a linear regression. I keep all the work that I have done so far intact. So in fact since we are doing this only one at a time, even though I might have $k$ variables in the system.

But the coefficients I have for the $k$ variables might not be the same $k$ coefficients I would have gotten, if I started with this $k$ variables and did a linear regression on it. So the coefficients could be different, if I take those $k$ variables and do linear regression I will get a better fit rather than doing this stage wise fit. But we prefer to the stage wise, because it saves us a lot of computation.

Eventually everything will catch up and we will get the same kind of prediction at the end of it, but you might end up adding a little bit more variables in this approach, but that is fine.

So the next class of methods we will look at are called shrinkage methods. The idea is to shrink some of the parameters to zero. So in the subset selection here essentially if you think about what we are doing is that all the variables that we did not select you are setting the coefficients to zero. But instead of doing an arbitrary greedy search or stage wise selection and so on so forth, in shrinkage methods what we do is we come up with a proper optimization formulation which allows us to shrink the unnecessary coordinates.

Ideally you would like to shrink them all the way to zero, but there are problems in doing that, but we will try to keep them as small as possible you can do some post-processing and then get rid of really small coordinates and things like that. But we really like to shrink these coordinates. So this is fine from the prediction accuracy point of view. From the interpretability point of view it still leaves a little bit to be desired, because you might have a lot of coefficients with I mean a lot of variables with very small coefficients back in the system.

So it is still a little bit of a thing, but mathematically this is a much sounder method than any of these things we have been talking about. And of course this is the soundest, but also impossible right. So the first thing we look at it is called ridge regression. The whole idea behind all of this shrinkage methods is that you are going to have your usual objective function which is the sum squared error that you are going to try and minimize. In addition you are going to impose a penalty on the size of the coefficients. So you want to reduce the error, but not at the cost of making some coefficient very large. You do not try and shrink the coefficients as much as possible, so what will happen is your optimization procedure will try to find solutions which have as smaller coefficient as possible and give you the similar kind of minimization in this squared error objective.

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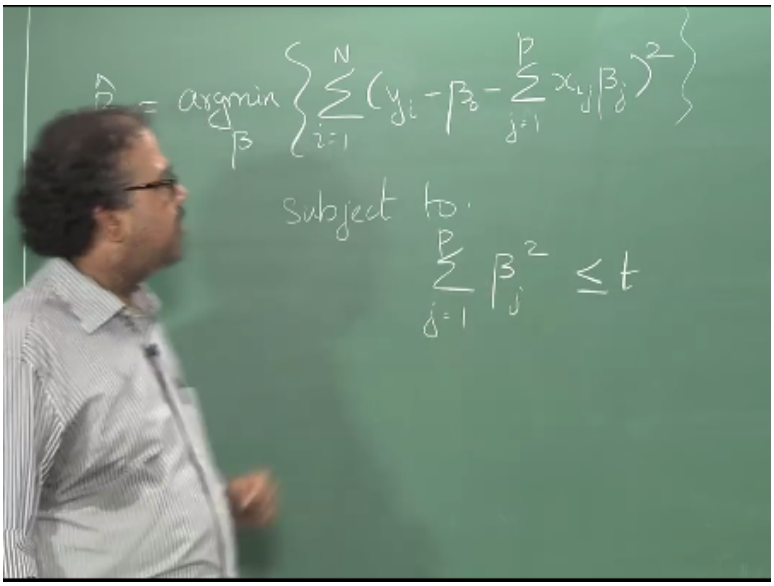

So what is your normal objective function? $\hat{\beta} = \arg\min_{\beta}\{\sum_{i=1}^{N}(y_i - \beta_0 - \sum_{j=1}^{p} x_{ij}\beta_j)^2\}$. So that is a normal objective function for finding their β, and so your $\hat{\beta}$ is essentially this. So now what I am saying is, let us not do this, but let us do this with the constraint right. So what is a constraint? $\hat{\beta} = \arg\min_{\beta}\{\sum_{i=1}^{N}(y_i - \beta_0 - \sum_{j=1}^{p} x_{ij}\beta_j)^2\}$ subject to $\sum_{i=1}^{p}\beta_i^2 \leq t$

Fairly straight forward I have added a squared norm constraint.

So this is essentially the L2 norm. I am taking the root I am just leaving it as a square and it does not matter. So it is like an L2 norm constraint for my data.

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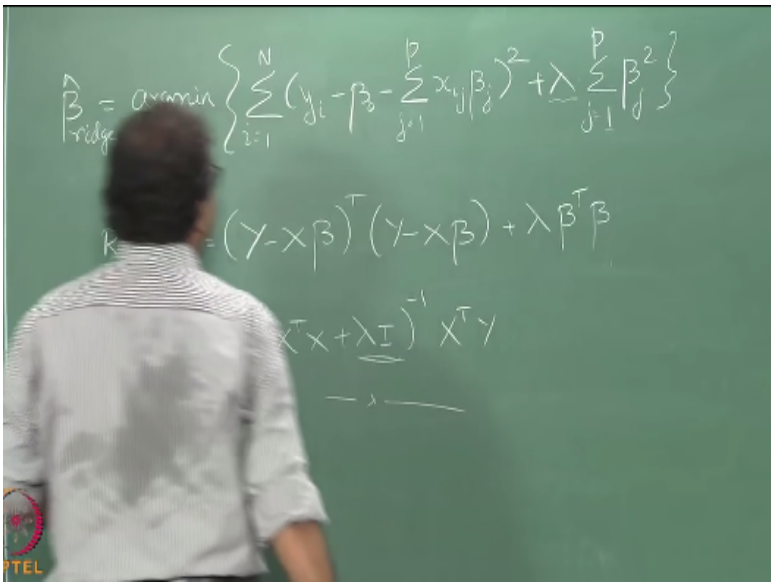


So I can make this into an unconstrained problem right. Because λ has to be greater than zero. Why do I want the βs to be small okay good question actually. So what we wanted to do was to make sure that you are reducing the variance of your model right, so that is essentially what we are trying to do now, all the subsets selection was we set the coefficient to zero, we said you have lot fewer parameters to estimate right.

So now what I am doing is that I am by imposing the size constraint on the parameters. The size constraints on the variables I am actually reducing the range over which these variables can actually move around. So if you think about it if I have moderately correlated input variables or correlated or anti-correlated input variables, so let us say I have two variables which x1 and x2 which are correlated and now I can have a large β1 and a large negative β2, that essentially will cancel out each other in terms of the predictions I am making, because x1 and x2 are themselves correlated. So I can actually make my β1 very large and my β2 is largely negative right, so that it will just cancel out the actual effects of the two variables. So it essentially becomes a difference of β1, β2 that actually matters right, not necessarily the difference in magnitude of β1 β2 that matters not actually not the actual values. So I can basically have a large class of models which will give me the same exact output. This makes my problem much harder to control and then that increases the difficulty of the estimation problem. But now we are saying that no I cannot allow these things to become very large and then I am restricting the class of models I am going to be looking at.

So that is the reason why the decreasing size of β helps. I did not explain this completely last time so thanks for asking the question. So we just have to make sure that our λ are positive we know that little so lagrange multipliers have to be positive and so on so forth.

$$\hat{\beta} = \arg\min_{\beta} \{ \sum_{i=1}^{N} (y_i - \beta_0 - \sum_{j=1}^{p} x_{ij} \beta_j)^2 + \lambda \sum_{j=1}^{p} \beta_j{}^2 \}$$

So now I can go ahead and minimize this right.

So a couple of things which I want to point out now, so one thing is if you notice the penalty here, so what do you notice about this? I am not including $\beta_0$, see the sum runs from 1 to p it is not running from 0 to p also note that I actually explicitly wrote out $\beta_0$ here I did not squish it into the p+1 thing, because I am going to be treating $\beta_0$ specially here mainly, because if I penalize $\beta_0$ then what will happen is if I move my data up right, so let us say this is my X and Y axis and I have this is the data that I had right.

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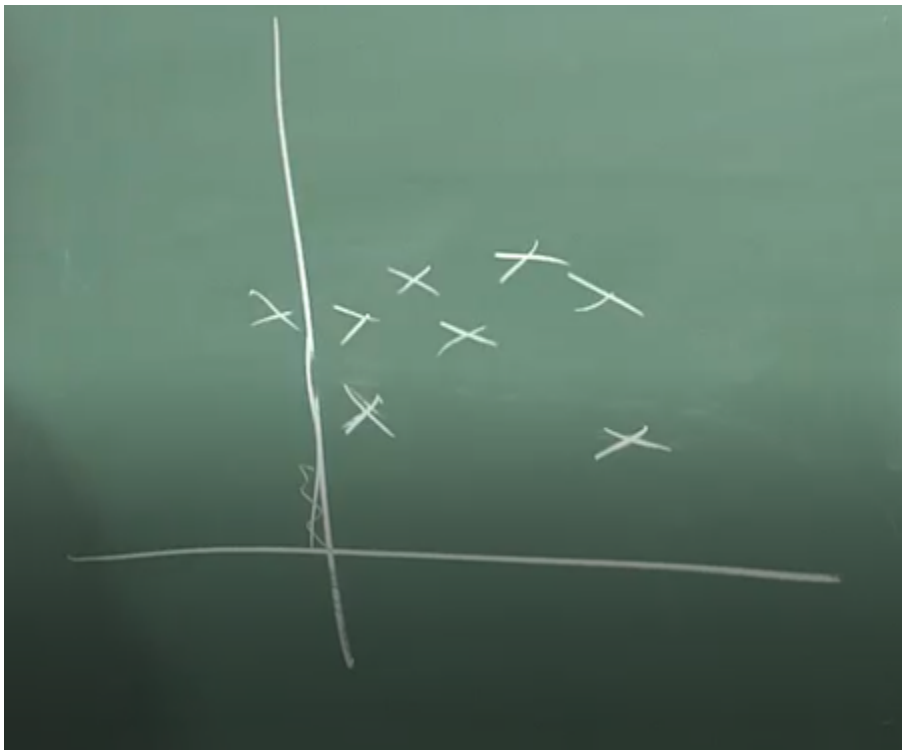

So now I have to fit that line through this right, it is a univariate regression problem. Y is my response and X is my input I have to fit a line. But now the same data points, if I shift them up, so shifting up the data points is hard, so I will just shift the origin. If I shift the origin what will happen if I penalize $\beta_0$? So if you penalize $\beta_0$ it will try to keep this intercept small, penalizing $\beta_0$ will try to keep the intercept small.

So earlier when I had that, if you look at the fit it will pass very close to the origin the intercept will be close to 0. Now when I shifted this, it is going to try and make the intercept small. There is this and line just shifting the slope of the line will change. It is the same data it has just shifted up a little bit, so the slope of the line will change, so it will try to go through somewhere here.

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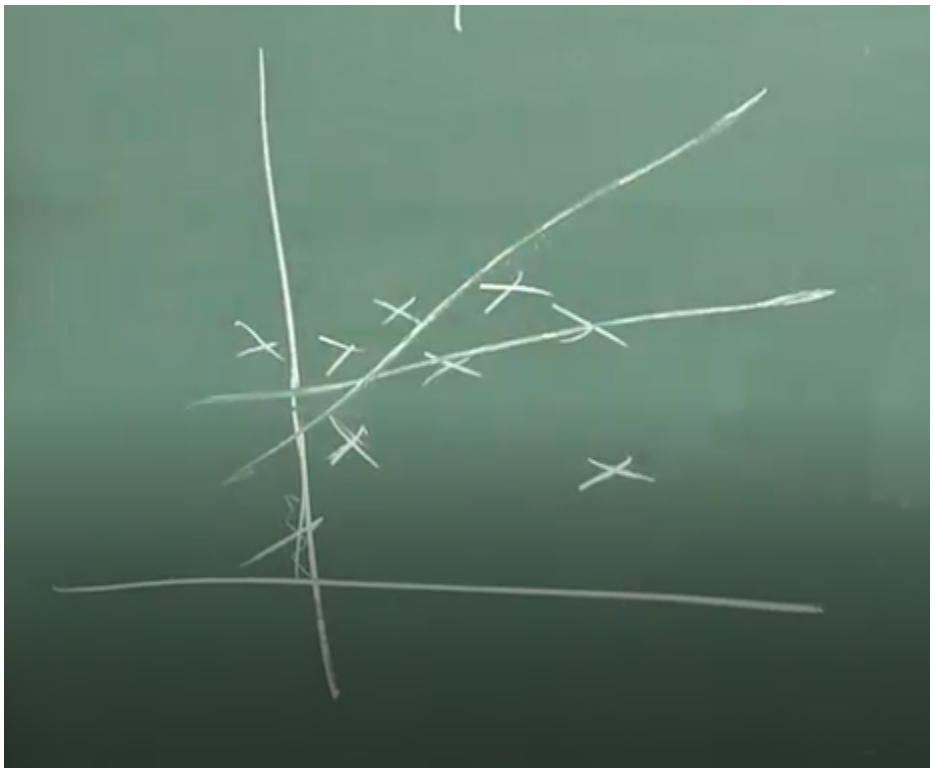

So essentially earlier when the line now the line will become like that because I am penalizing $\beta_0$. So we do not want that to happen so just simple shifts in the data should not change the fit. So we do not penalize $\beta_0$. Does it make sense? And anyway we know what $\beta_0$ should do, you know what $\beta_0$ should be the average of the outputs.

So one way which we can actually get rid of $\beta_0$ from this optimization problem is to say that we will center the inputs. So we will subtract the average from the $y_i$'s and likewise we will subtract the averages from all the x's. So we will center the input and we will make all the X variables centered on zero. So we will subtract the mean from all the X's, we will subtract the mean from the Y's. So this will give me a centered input and then I will just do regression on this centered input well there will be no $\beta_0$. So from now on when I write X it is a *n x p* matrix where the input has been centered. So essentially what I have done here is I have taken my data from there and shifted it so that the fit whatever is the fit I am going to get will pass through the origin.

So that is essentially what I have done. I have taken the original data translated it so that whatever fit will pass through the origin. And I will go back and add the $\beta_0$ later to get my original fit. Does that make sense? So in matrix form I write it like this,

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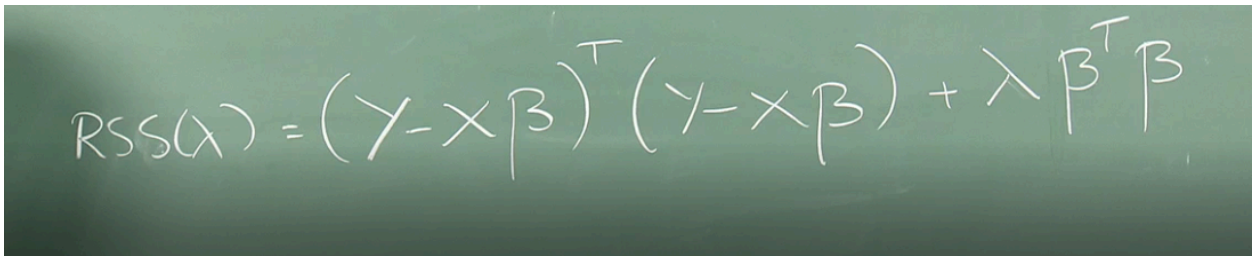


So you can minimize this take the derivative and set it to zero solve for it you will get this. So here, so both my x and y are centered. So I subtracted the mean from Y, I subtracted the mean from the columns of x so they are all centered here. So just remember that once I get this

centered values I can solve for it, this gives me $\hat{\beta}_{ridge}$ for 1 to P in the β0 I estimate as $\bar{y}$ and that gives me the full solution.

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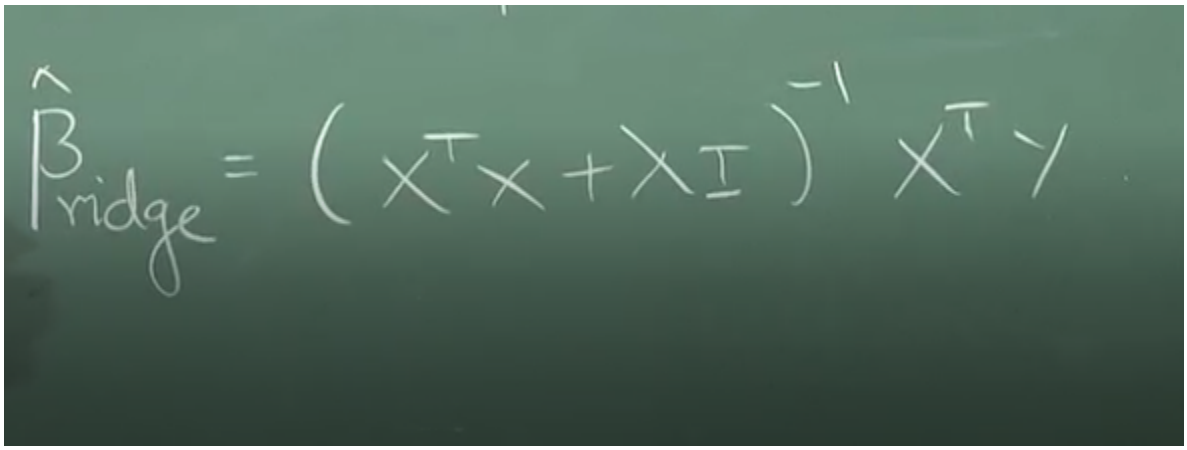
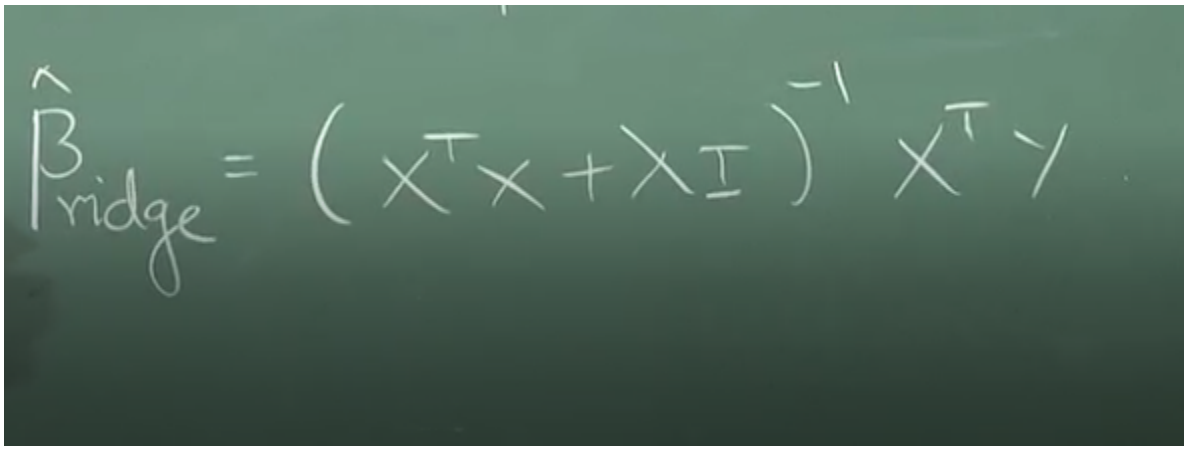


So one thing which I forgot to point out earlier you remember I had this variable 't' here which was upper bound. I said subject to the constraint that it should not be larger than 't', the 't' has vanished but you can show that this λ and the 't' are related.

So it does not matter, so for every choice of 't', you have a choice of λ but typically what happens is you choose your appropriate lambda and then you work with it you do not worry about the 't' formulation.

So this tells you why this is called "ridge regression", because what they have done here is you essentially added a ridge to your data matrix you take the $X^TX$ and then you add a diagonal λ which is like adding a ridge of size λ to the diagonal elements of $X^TX$. So that is why it is called ridge regression. So why are you doing this and can you see one advantage of doing this?




This whole thing becomes invertible right. So as soon as I add the λi, I am sure that this is non-singular. And even if $X^TX$ was originally singular and adding λi makes it nonsingular and it is invertible.

In fact this was the original motivation for ridge regression right, back in the I forget, in the 50s when people came up with ridge regression the original motivation was $X^TX$ could be badly conditioned, even if it is non singular we talked about this in the last class. It could be that some variables are so highly correlated. So even if the matrix is invertible numerically you will get into problems I told you that the residual might be so small right.

So when you try to fit the coefficients you will get into problems. So numerically the inversion might be a problem right, even if the matrix is non-singular, but by adding this λi term to it you make sure that it is invertible and by controlling the size of the λ you can make sure that numerically also the problem is well behaved. So that is the idea behind with original motivation for ridge regression was essentially to make the problem first of all solvable right.

But then people went back and understood rigid regression in terms of shrinkage or variance reduction. And since it makes it convenient for us to talk about a whole class of problems, shrinkage problems, we motivate ridge regression from the view point of shrinkage as opposed to this inversion problem. Any questions? So I am going to encourage you to read the discussion that follows rigid regression in the book right. It requires you to work out some things along with the book you just cannot just sit there and passively read it okay, but then it draws a lot more connections from ridge regression to a variety of other statistical properties about the data which will be useful to know and I am going to make you read, I mean so go read it and I will ask you questions on it later. So go and read the discussion.

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**Introduction of Machine Learning**

**Lecture 16**

**Prof. Balaraman Ravindran**
**Computer Science and Engineering**
**Indian Institute of Technology Madras**

**Shrinkage Methods**

So what are some of the other shrinkage methods you think we can come up with? Each of β is closed right so we imposed a $L_2$ norm on the β. You can impose any other norm constraint on the β, say, I can I can impose an L4 norm or more commonly I can impose a L1 now it is called Lasso.

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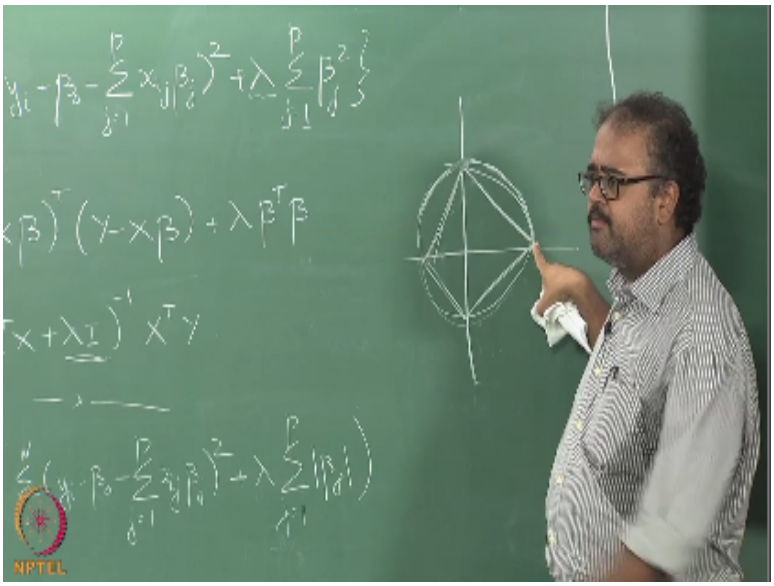

So lasso essentially says that let us just ignore the absolute value of the β. You sum up those and you want to keep them so you can write the same. We can write the constraint formulation where

I can say sum of β has to be less than some 't' or sum of |β| has to be less than some 't', then I could just do this kind of a formulation right.

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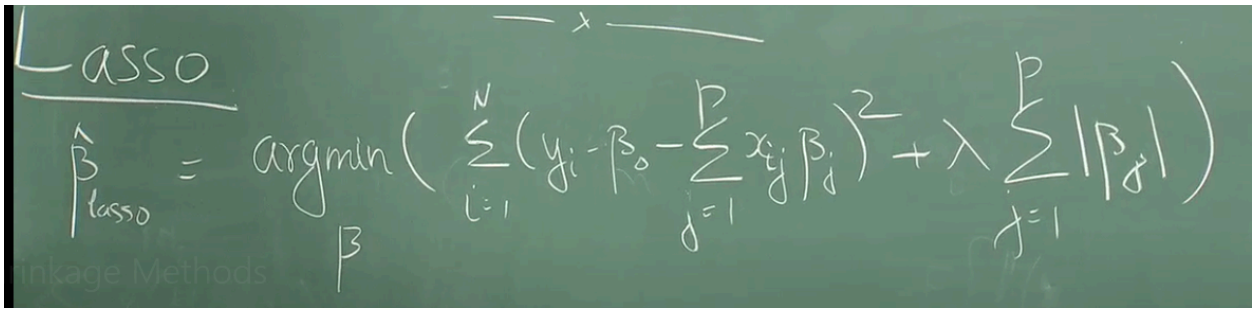


And so to impose a constraint on each individual βs would require you to know something about the variables themselves beforehand, otherwise if you constrain a very important variable to have a small coefficient then it becomes a problem. So you need to know something about the variables and you can say I know that these variables are very important make sure that the other variables I want to be more than 0.5 times the coefficient of these variables, so something like that. You can think of all kinds of complex constraints once you have knowledge about the system but typically you do not write in such cases you will have to have some kind of uniform constraints like this and so this is a very popular constraint. It actually makes life harder for us as it does not have such a nice closed form solution any more. I mean this is no longer differentiable. So I can't write your nice closed form solution like this in fact I have to work very hard to solve this. Although there are packages in R or WEKA where you can always run lasso on it will give you the nice fit. So what is the nice thing about lasso I will try to give you an intuition about it. So think of it this way so suppose I have a non-important coefficient, if I can reduce it from say at 1000 to 0.3 okay. Let us not even look at it that way. So I can reduce some coefficient from say 1000 to 999 and there is another coefficient which you can reduce from let us say 1 to 0. So there are many variables in my fit there is one variable whose coefficient I can reduce from 1000 to 999 there is another variable whose coefficient is one I can reduce it from 1 to 0. And both of them cost the same change in my squared error both of them contribute equally to the squared error and making this change will make the same change to the squared error.

So which one would ridge regression prefer? To reduce 1000 to 999 because that causes a much larger reduction in the squared penalty. And which one would lasso prefer to reduce? Doesn't matter. Either one but then I can make this thing slightly more contrived right now, say from 1.1 to 0. Which one would lasso prefer? So even though in absolute value terms this is a larger reduction, ridge would prefer still preferred 1000 to 999 right.

Because the fall is $1.1^2$ to $0^2$ vs $1000^2$ quite to $999^2$ still that is a larger reduction in error. So what is a take-home message here? LASSO is more likely to drive coefficients to zero than ridge. So ridge would happily leave the coefficient at 1.1 or even more dramatically it will happily leave coefficients at 0.3, 0.2, 0.8. So it will leave it at small values it will not drive it all the way to zero. Well lasso given an opportunity right we will drive the coefficients to zero we need not drive it to zero at the cost of minimizing the error. It will still try to minimize the error but given the chance it will more likely to drive coefficients to zero.

So, sometimes Lasso is also called sparse regression. Because this L1 norm constraint is also called a sparsity constraint because it makes your β vector to have more zeros. So if you know what a sparse matrix is so you have a matrix with a lot of zero entries in it and only few nonzero entries. So you really don't want to have an array representing your sparse matrix because most of the entries are 0 so typically what you do in a sparse matrix representation is you store the index of the nonzero entry and the non-zero entry itself. So that actually takes a lot less memory than actually having a large M by N array with lots of 0 so that's why its called sparse. So here the L1 regression has a tendency to make the β sparse or to have a lot of 0. So it is sometimes called the sparsity constraint.

Yeah no see if I take 0.01 and square it okay and the difference between that squared and $0^2$ is lesser than 0.1 and 0. So but the drop in the value will be bigger in the Lasso than in Ridge. Now it depends on what other competing elements that you have. So lasso typically drives the coefficients to zero. Well ridge does not do this is that I was giving you an intuition as to why that is the case right. It is not mathematically a sound argument but you can give a mathematically sound argument also that LASSO is more likely to find sparse fix than ridge regression okay.

So I am being very careful here. So I mean I can also think of a geometric intuition for it so if you think about the the lasso constraints it will be something like this. Let me think about ridge constraints that suppose to be a circle. So the ridge constraint will be something like this so here if you know where the sum has to be a constant. So the sum has to be a constant in the sum of squares has to be a constant, so one will be a circle other one will not be a circle.

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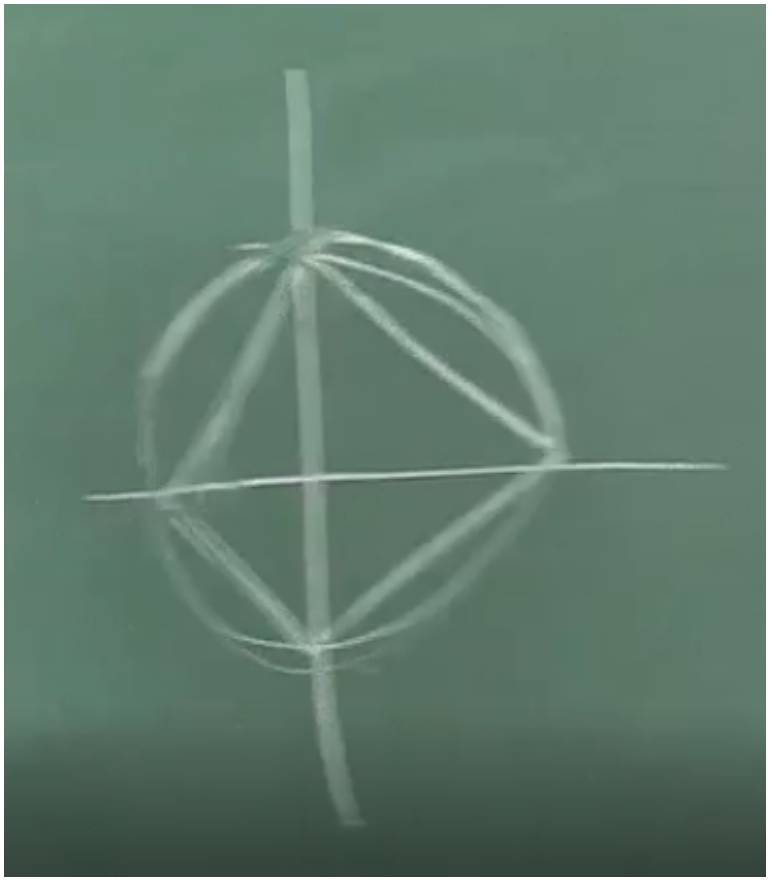

So when you're looking at the error surface corresponding to this, essentially you will have to find solutions that lie on this or lie within this for lasso and lie within this for ridge. And it turns out that you are more likely to hit a corner off you can show that more formally that you are more likely to hit a corner in the latter case than in the ridge case. So the probability of hitting something because this is the whole thing is convex the probability of hitting that side it is higher

.

This is just the very rough geometric intuition. I do not want to get into showing things formally but you can show that the probability that lasso will give you these kinds of corners in the fit corner obviously you can see that has one of the coefficients a 0, so that you will get a corner as a fit is much higher then you will get one of these axis points in the ridge. So in fact you can think of having higher-order penalties also. Like I said you can think of an L4 norm penalty. So far we looked at two methods for variance reduction. One was subset selection and the other one was shrinkage based methods. Now there is a third class of methods which people use for getting better fits with possibly fewer variables or fewer parameters.

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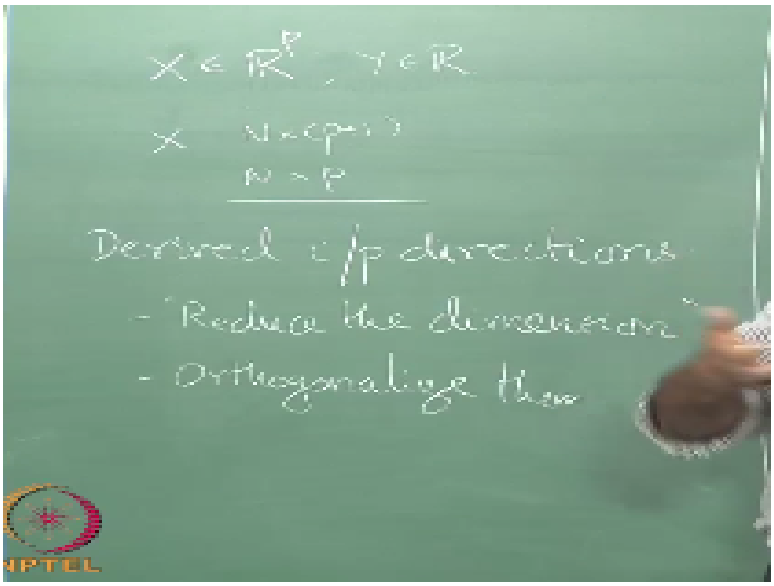


This is based on the derived input directions. So we talked about reducing the number of variables so far at least in the subset selection part we retain some of the variables and then we ignore some of the other variables. Likewise whether we are doing implicit subset selection by doing lasso or ridge regression we are reducing the coefficients of some variables and retaining some other variables. But at all points we were operating with the original set of basis vectors that were given to you. So what were the basis vectors we are talking about here? The columns of the X matrix are the basis vectors.

So we are working with the original basis vectors we are working with the same columns that were given to us. In one case we picked some columns and threw out some of the other columns and in the other case we tried to continuously adjust the weights of the column so that some of them were given more weight and some were given less weight. So when we talk about derived input directions now I am not going to stick with the original columns. I am going to find a new set of columns and I am going to find a new set of features or new set of directions which I will then use for doing my regression okay.

So we actually talked a little bit about it in the when we looked at the orthogonalization. I told you that in orthogonalization essentially what you are doing is you are finding an orthogonal basis for the input space and then you are trying to find the coefficients there. So likewise what we will do here is we will reduce the dimensions. So what is the advantage of orthogonalizing the dimension well we could do univariate regression on each dimension separately and then that will give us the coefficients.

So we do not have to actually do a multivariate regression we can do univariate regression on each dimension because once I orthogonalize the directions they do not interfere with one another. So I can do univariate regression. So typically when I try to do these derived input directions I try to orthogonalize the directions and I also try to find a reduced set of dimensions that will give me the original fit or as close to the original fit as possible.

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**Introduction to Machine Learning**

**Lecture 17**

**Prof. Balaraman Ravindran**
**Computer Science and Engineering**
**Indian Institute of Technology Madras**

**Principal Components**
**Regression**

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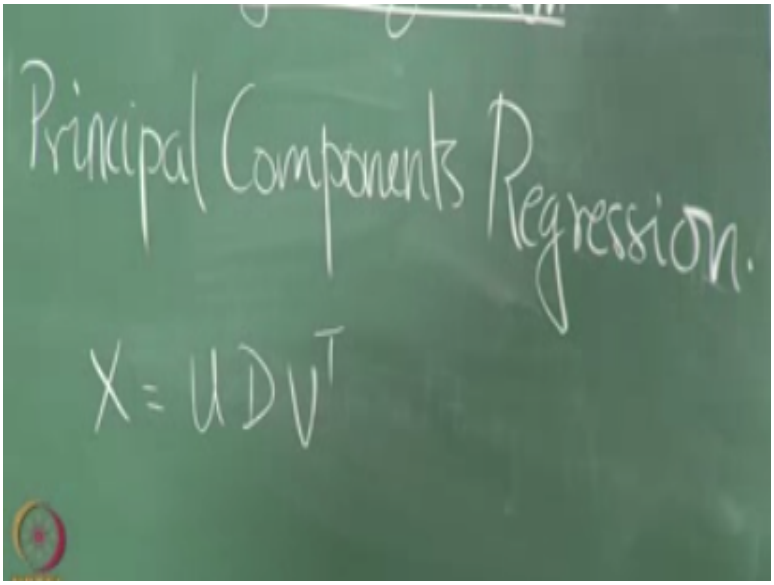


So D is a diagonal matrix where the diagonal entries are your eigenvalues (ideally) or otherwise known as singular values. V is a P x P matrix which has your eigenvectors and U the N x P matrix which typically spans your column space as X . So this is essentially your singular value decomposition that we talked about.

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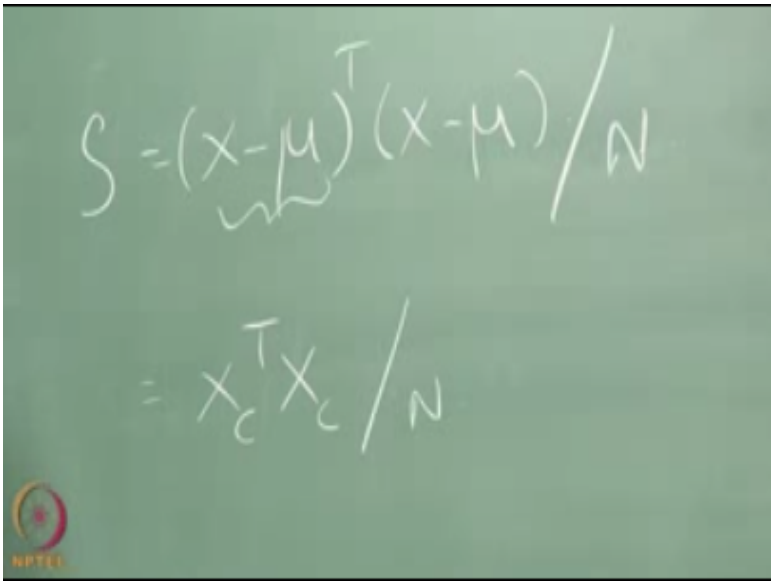


So if you look at singular value decomposition or what is called the principal component analysis literature, you will find the following. You will find that they will talk about the covariance matrix S. What is the covariance matrix? $S = (X - \mu)^T (X - \mu) / N = X_C^T X_C / N$

It is essentially if you think of whatever we have been doing so far what would be this centered data. So I take the centered data and I find the eigendecomposition of that. I find the Eigen decomposition of the covariance matrix.

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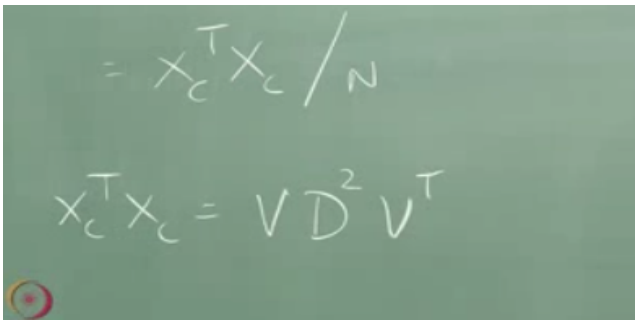


So I can essentially write this as $X_C^T X_C = VD^2V^T$ . So the same V and D . So it is the same V and D that I wrote here. If I take $X_C$ so basically I am going to get the same thing. So it is essentially like doing singular value decomposition and retrieving the V matrix. I am essentially taking the

$X^TX$ which is the covariance matrix of the centered data and I am finding the Eigenvalue decomposition of that. So $D^2$ would be the Eigen vectors of $X^TX$ so this is standard stuff you should know.

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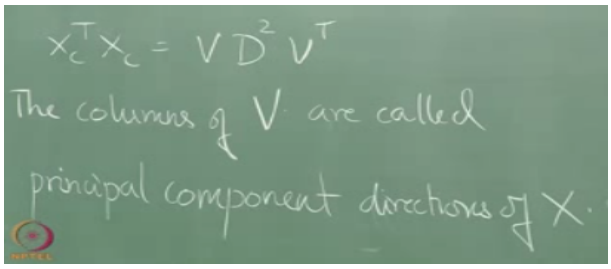


So the columns of V are called the principal component directions of X. So there are a couple of nice things about the principal component directions. We will talk about just one. So I will actually come back to PCA slightly later when I talk more about generally about feature selection not just in the context of regression but when I talk with generally about feature selection I will come back to PCA and tell you or at least show you why PCA is good. Right now I will just tell you why PCA is good I will come back later and then I will show you why PCA is good.

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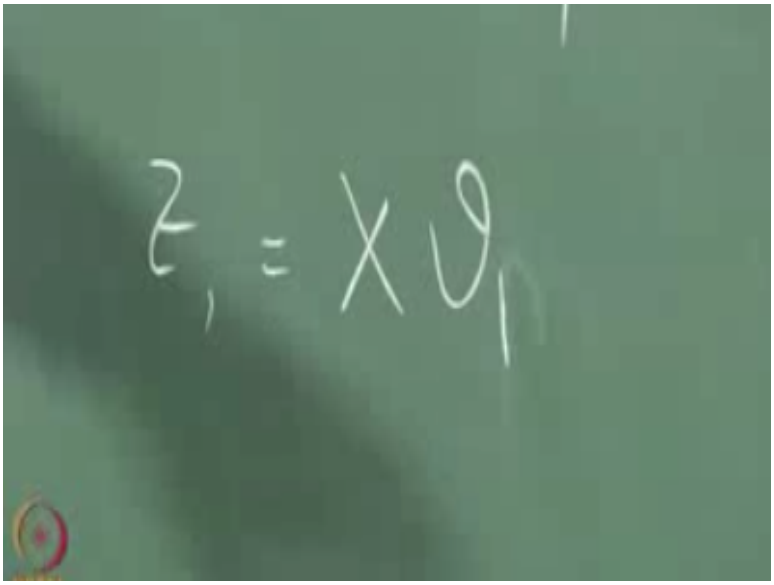

So suppose I take $z_1 = Xv_1$ where $v_1$ is the eigenvector corresponding to the first eigenvalue. So essentially what this means is that I am projecting my data X on the first eigenvector direction so the resulting vector $z_1$ will have the highest variance among all possible directions in which I can project X.

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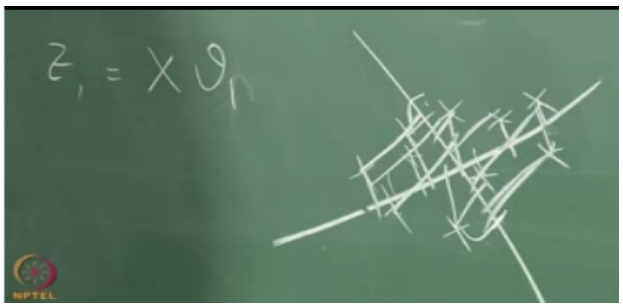


So what does that mean? Suppose this is X okay this is not x and y, so it is a two-dimensional X this is X now I am claiming that $v_1$ will be such that when I project X onto $v_1$ I will have the maximum variance. So in this case it will be some direction like this and projecting X onto this essentially means that so you can see that the data is pretty spread out it goes from here to here. On the other hand if I had taken a direction let us say that looks like that right so if I look at projection of the data right, so you can look at the spread it is a lot lesser in that direction than in the original direction I did the projection I know it looks pretty confusing to look at but the people can get my point. It is in the original direction that way the data was a lot more spread out as opposed to this direction where the data is lot more compact when I project it on to that direction.

So that is essentially what I am saying. So $z_1$ is essentially the projected data onto that direction onto X like $z_1$ actually has a highest variance among all the directions in which I can project the data and consequently you can also show things like if I am looking to reconstruct the original data and I say that you can only give me one coordinate, so you have to summarize the data in a single coordinate and now I am going to measure the data measure the error in reconstruction. If

you looked at it so the error in reconstruction would have been these bars that I did the projection over.

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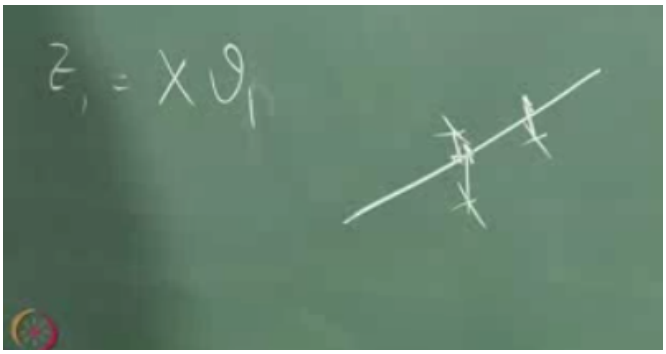

That would be the error in reconstruction. So I have the original data so that is the data and now I will give you these coordinates. Now I have to reconstruct the data so essentially this will be the errors so the first principal component direction. The first principal component direction is the one that has the smallest reconstruction error. First principal comment direction will be the one that has the smallest reconstruction so we can show a lot of nice properties about this. So I will actually come back and do this later, when we talk about the general feature selection. But here the first thing you can see what each one $v_1$ to $v_P$ will be orthogonal. So I have gotten my orthogonal directions and the thing to notice is a lot of the variation in the data is explained by $v_1$ or $v_1$ has the maximum variance likewise you take out $v_1$ and so now what you have your data lies in some kind of a p - 1 dimensional space and the direction in that the space which has the highest variance is $p_2$ it turns out that so $v_1$ has the highest variance over the data. So in this space orthogonal to $v_1$ , $v_2$ has the highest variance right in the space orthogonal to $v_1$ and $v_2$, $v_3$ will have the highest variance and so on so forth. So essentially now what you can do is hey I am going to take all this directions one at a time and I will do my regression because each is orthogonal I can independently do the regression I can add the outputs and I can keep adding the dimensions until my residual becomes small enough that make sense. So I will just keep adding this orthogonal dimensions until my residual becomes small enough at that point I stop.

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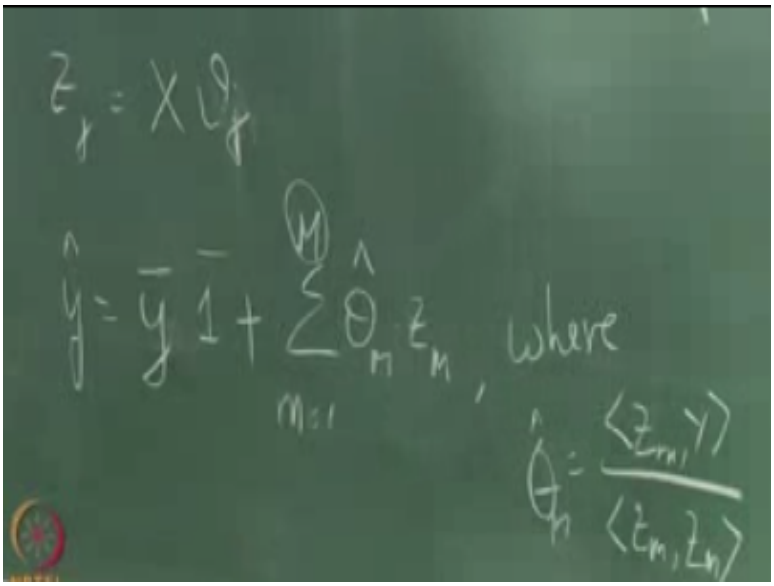


So this is essentially the idea behind PCR.

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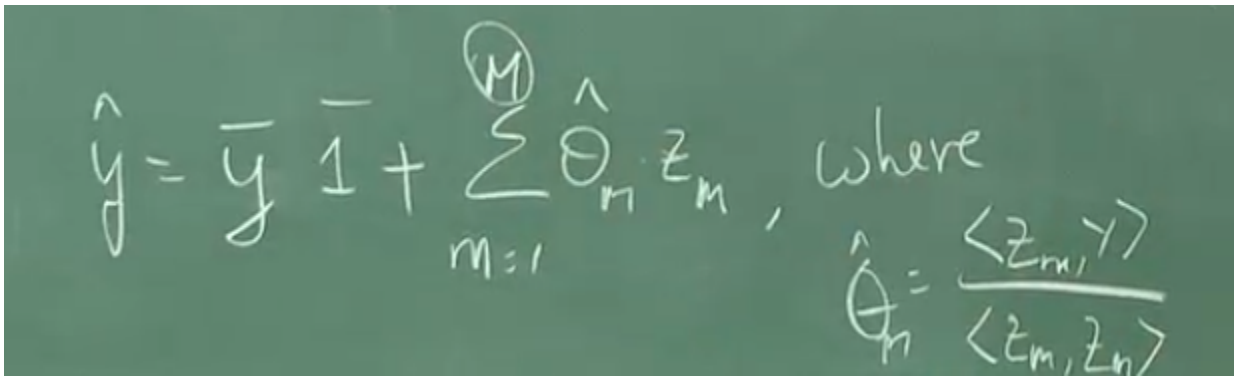


So remember we are working with the centered data a and you automatically add in your intercept which is $\bar{y}$. The coefficient is $\bar{y}$ and then if you choose to take the first 'm' principal

components your thing will be $\theta_m$ $Z_m$ where $Z_m$ is given by $z_j = Xv_j$ and $\theta_m$ is essentially regressing y on $Z_m$. So that is a univariate regression expression we know that well now. So this gives you the principal component regression fit. So one of the drawbacks of doing principal component regression is that I am only looking at the input right I am not looking at the output so it could very well be that once I consider what the output is right I might want to change the directions a little bit right, so I can give you an example is easier for me to draw if I think of classification.

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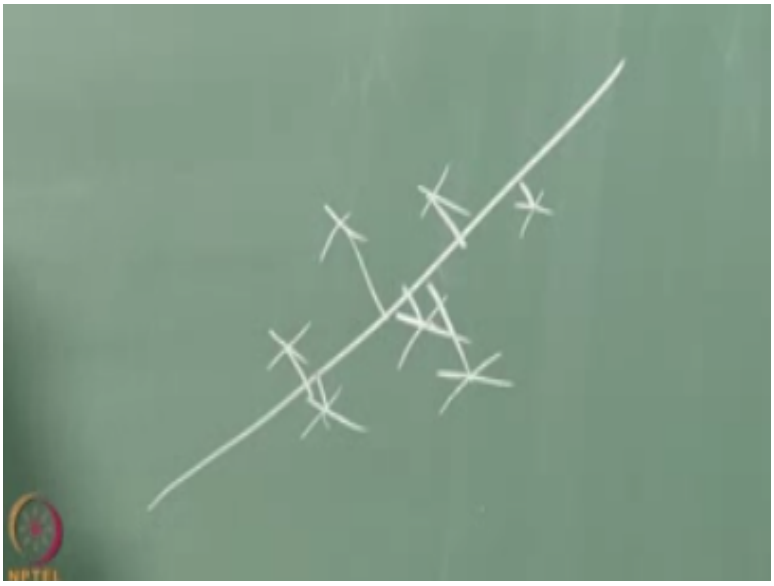

Let us say this is the data and what would be the principal component direction you want to choose something like this. So that would be the ideal direction that you would want to choose okay so now what will happen the data will get we get projected like this right but suppose I tell you that.

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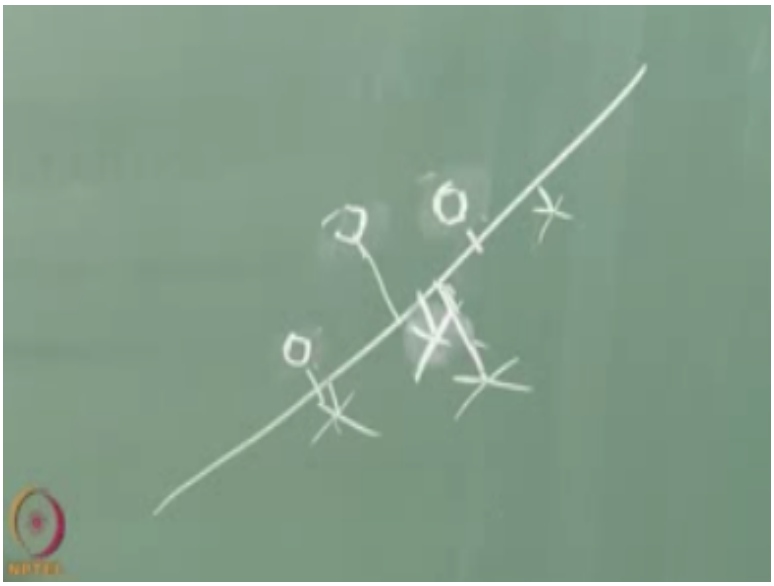

Suppose I tell you that that is fine that these three were in a different class and if you want to think of it in terms of regression let us assume that these three have an output of -1 and these four have an output of +1. Now if you think of this direction so the +1 and -1 are hopelessly mixed up and I cannot I cannot draw or give a smooth prediction of which will be +1 which will be -1. On the other hand, if you project onto a direction like this right the variance is small. I agree the variance is much smaller. But if you think about it, all the -1 go to one side right all the +1 go to one side, so now if I want to do a prediction on this so it will be like okay this is this side is -1 and that side is +1. I can essentially do a fit like this which will give me a lot lesser error than the other case. So in cases where you are having an output that is specified for you already it might be beneficial to look at the output also when trying to derive directions as opposed to just looking at the input data. So in classification this will be say class 1 this will be class 2 and having this direction allows you to have a separating surface somewhere here right we talked about classification in the first class.

So just having a separating surface here will be great but in this case if I am projecting on this direction coming up with a linear separating surface is going to be hard as everything gets completely mixed up.

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**Introduction to Machine Learning**

**Lecture 18**

**Prof. Balaraman Ravindran**
**Computer Science and Engineering**
**Indian Institute of Technology Madras**

**Partial Least Squares**

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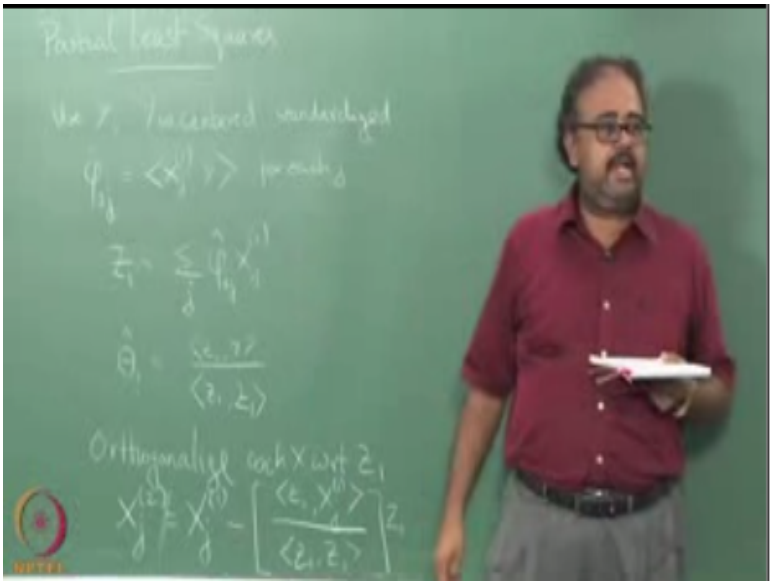


Okay so we will continue from where we left off. As I promised, so we are looking at linear regression and we looked at subset selection and then we looked at the shrinkage methods and then, finally we came to derived directions. I said there are three classes of methods so we are looking at a couple of examples of each of those classes of methods the first one we looked at was subset selection so we looked at forward, backward selection and stage wise selection in step wise selection and all that and then we looked at shrinkage methods where we looked at ridge

regression and lasso and then we started looking at derived directions where we looked at principal component regression. I said the next one we look at is partial least squares and gave the motivation for looking at partial least squares. It is mainly because principal component regression only looks at the input data, does not pay attention to the output and therefore you might sometimes come up with really counterintuitive directions. Like an example I showed you with the +1 and -1 outputs, so the basic idea here is that we are going to use the Y also.

Just like the usual case I am going to assume that Y is centered. And I am also going to assume that the inputs are standardized. This is something which you have to do for both PCA and partial least squares as they essentially assume that the each column is going to have 0 mean and unit variance on the data that is given to you make it 0 mean unit variance. So that you are not having any magnitude related effects on the output. So what I am going to do is the following. If you remember how we did orthogonalization earlier something very similar so I am going to look at the projection of Y on $X_j$, then I am going to create a derived direction which essentially sums up all of these projections. I have computed basically the projection of Y on Xj. So this is essentially the direction is a vectorized version of it then I am going to sum all of this up. So essentially what I am doing here is I am looking at each variable in turn. I take each $X_j$ in turn I am seeing what is the effect on Y, so how much of Y, I am able to explain just by taking Xj alone and I am using all of that I am combining that and making that as my single direction. So individually taking each one of this all by itself how much of Y can I explain. And that becomes my first derived direction that is my $z_1$. So that is the coefficient for $z_1$ in my regression fit. For that one you can see what it is like so I have taken Y and regressed it on $z_1$ and that essentially gives me what the coefficient for $z_1$. So I am looking at how much of Y is along each direction Xj, so in some sense you can think of it as if I have one variable Xj, how much of Y can be explained with that one variable Xj. I am looking at that and then my first direction $z_1$ is essentially summing that univariate contributions over all my input directions I suppose, I have two input directions. Unfortunately I have to do this in 3d suppose I have two input directions so what I am going to do is I am going to take my Y and project it on x1 alone first and on x2 alone.

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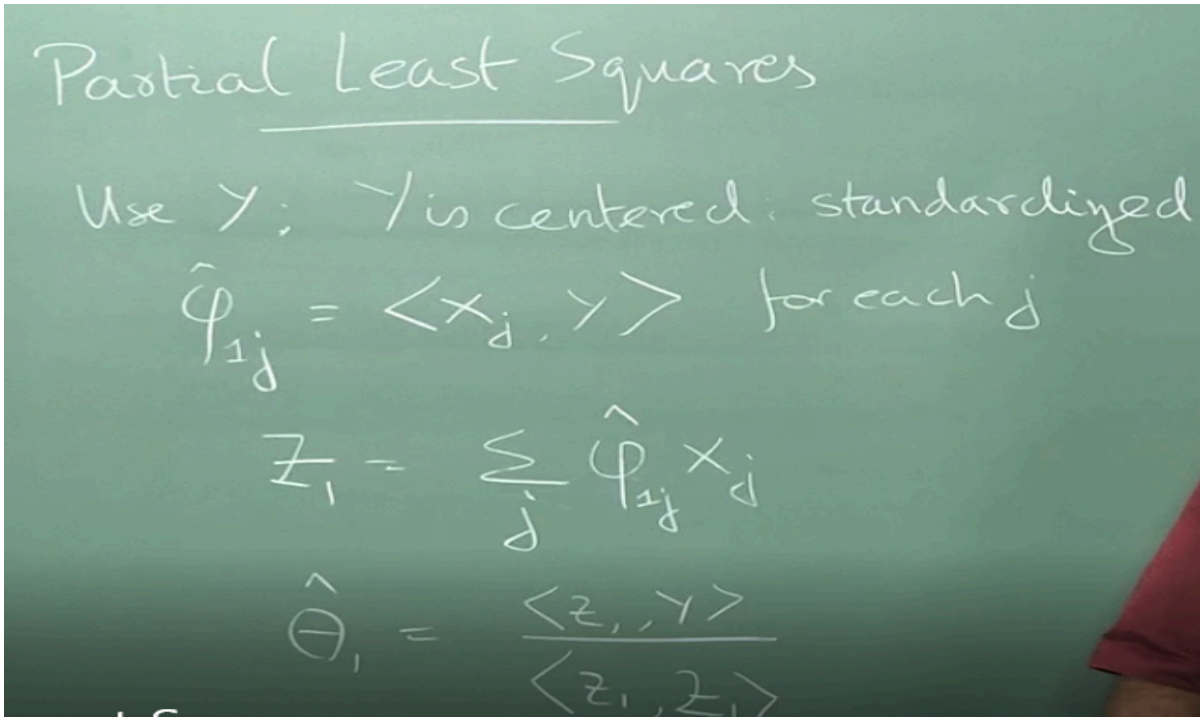


But the basic idea is I take Y and I find the projection of Y along x1, then I find the projection of Y along x2. Now I am going to take the sum of these two and whatever is the resulting direction and I am going to use that as my first direction.

In PCR what we did was we first found directions X which had the highest variance here we are not finding directions in X with the highest variance but we are finding directions in X. In some sense, components of X which are more in the direction of the output variable Y, so eventually you can show that which you are not going to do but you can show that the directions you pick that $z_1$, $z_2$, $z_3$ that you pick or those which have high variance in the input space. But also have a high correlation with Y. It is actually an objective function which tries to balance correlation with Y and variance in the input space Y. PCR worries only about variance in the input space and does not worry about the correlation, but partial least squares you can show that it actually worries about the correlation as well . We find the first coordinate now what do you do you orthoganalize, so what should I do now I should regress xj on z1.

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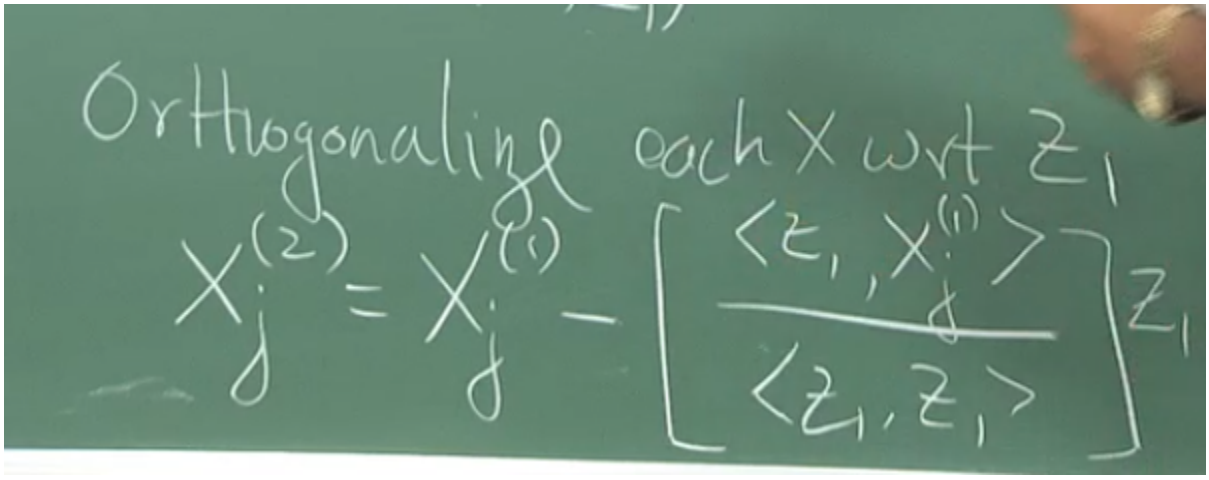


This is how we did the orthogonalization earlier. So you find one direction then you regress everything else on that direction then subtract from it that gives you the orthogonal direction. So essentially that is what you are doing here. The expressions look big but then if you have been following the material from the previous classes then it is essentially just reusing the univariate regression construction we had earlier.

So now I have a new set of directions which I call $X_j^{(2)}$. I have a new set of directions which we will call $X_j^{(2)}$ and then I can keep repeating the whole process, I can take Y projected along $X_j^{(2)}$ and then combine that and get $Z_2$ and then regress Y on $Z_2$ to get $\theta_2$. So I can keep doing this until I get as many directions as I want. So what is the nice thing about $Z_1$, $Z_2$ other things. They themselves will be orthogonal because they are being constructed by individual vectors which are orthogonal with respect to their all the previous Zs that we have. Each one will be orthogonal and therefore I can essentially do univariate regression. So I do not have to worry about accommodating the previous variable, so when it when I want to fit the $Z_k$ I can just do a univariate regression of Y on $Z_k$ and I will get the coordinates $\theta_k$. So once I get this $\theta_1$ to $\theta_k$ how do I use it for prediction? Can I just do like $X\beta$ or can I do $X\theta$? I know what should I do well so I can do $\theta Z$ and predict it but then I do not really want to construct this Z directions for every vector that I am going to get. So I do not want to project it along those Z directions, so instead of that what I can do if you think about it each of those Zs is actually composed of the original variables X . So I can compute the $\theta$ and then I can just go back and derive coefficients for the

Xs directly because all of these are linear computations and all I need to do is essentially figure out how I am going to stack all the thetas so that I can derive the coefficients for the Xs okay think about it you can do it as a short exercise.

So where can I derive these coefficients $\hat{\beta}$ from these θ's? So I will derive $\theta_1$, $\theta_2$, $\theta_3$ so on so forth. I can just go back and do this computation. So you will have to think about it. It's very easy you can work it out and figure out what is the number should be. And what is the 'm' doing? That number of their directions, I actually derive the number of directions I derive from the PLS. So here the first direction I can keep going suppose I derive p directions what can you tell me about the fit for the data if I get 'p' PLS directions? It essentially means that I will get as good a fit as the original least squares fit. So I essentially get the same fit as least squares fit and anything lesser than that is going to give me something different from the least squares fit. Here is a thought question if my X were actually orthogonal to begin with what will happen with PLS? Z will be the same as X's and what will happen to $Z_2$ ? Can I do the $Z_2$? No. PLS will stop after one step because there will be no residuals after that. So I will essentially get my least squares fit in the first attempt itself okay so that is essentially what will happen. So we will stop with regression methods.

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**Introduction to Machine Learning**

**Lecture 19**

**Prof. Balaraman Ravibdran**
**Computer Science and Engineering**
**Indian institute of technology**

**Linear Classification**

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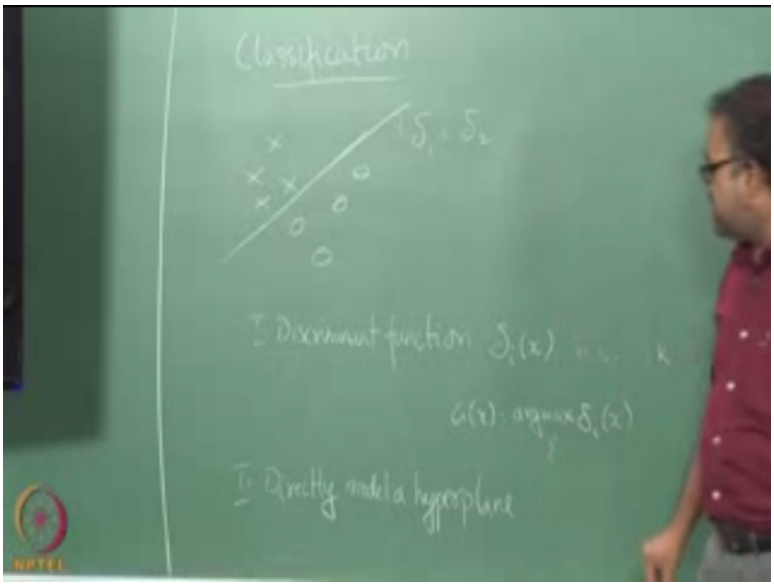


So we move on from linear methods for regression to linear methods for classification. So far we have been looking at linear methods for regression but I did tell you that you could do "nonlinear regression" also by doing appropriate basis transformations. So what do I mean by linear methods for classification? For linear regression you can understand that the response is going to be a linear function of the inputs, so what do I mean by linear classification? So when I am going to separate two classes the boundary of separation between the two classes will be linear.

So that is what I mean by linear classification. So this boundary that I draw between two classes will be linear, so you can think of when we did look at an example in the first class where we had drawn quadratics and planes and things like that. So instead of that we will assume that the separating surface will be a hyper plane. So there are two classes of approaches that we will look at for linear classification and the first one is essentially on modeling a discriminant function.

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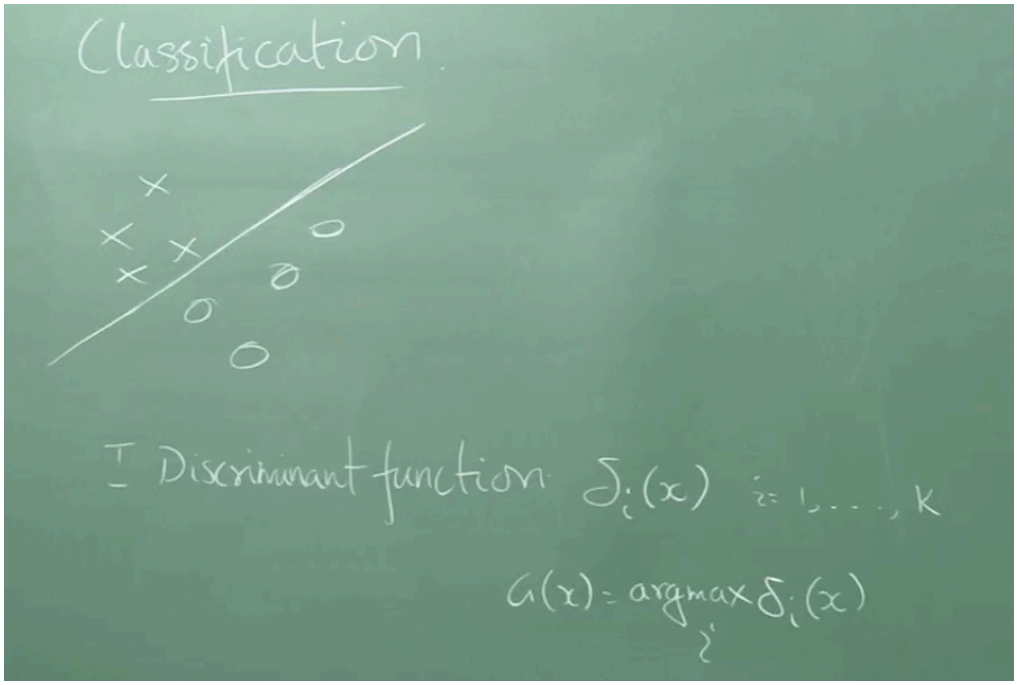


So one rough way of thinking about it is to say that I am going to have a function for each class and if the functional output for "class I" is higher than for all the other classes I will classify the data point as belonging to "class I". So I am going to have some function for each class and depending on whichever is the highest I will output it to that, so this is essentially the idea behind discriminant functions. I am going to have to figure out a way to learn this $\delta_i$'s. Suppose let us just keep it simple think of a two class problem .

Here is a question. Think of a two class problem and I have δ1 and δ2 so where will be by separating hyper plane? Wherever δ1 > δ2 it is class 1, when δ2 > δ1 it is class 2 and wherever they are equal it will be the boundary. So if I need this to be a linear surface what conditions

should δ1, δ2 satisfy? Should they be linear? Not necessarily but this is sufficient condition if you are linear the surface will be linear.

So what else can they be? They can be non linear as long as I have some kind of a monotone transformation of them which will become linear. So we will see examples of this. We will actually look at discriminant functions or we look at the assumptions which will appear to be. We are doing something heavily nonlinear but at the end of the day you will find that the surface will be linear or the separating surface will be linear, so we look at that as we go along. So the few approaches that we look in this class are essentially linear regression. You could do a linear regression and try to treat that as your discriminate function it for each class you could do we talked about this in the very first class or the 2nd class, where you could do a linear regression on an indicator variable, so that will give you a discriminant function or you could do logistic regression or it could do linear discriminant analysis which is like principal component regression but taking into account the class labels you will think of deriving directions and which will be doing the classification. You can look at any 3 of those.

The second class of methods which will come to this is directly model the hyper plane. So it is related to this in some sense. If I give you the discriminant functions I can always recover the hyper plane but here instead of trying to do a class wise discriminate function will directly try to model a hyper plane. This is second class of problem we look at one classic approach for doing that which is the perceptron. We will also talk about some more recent well founded ways of doing that which is essentially looking at the question of what an optimal hyper plane is and trying to solve for it directly. So these are the two approaches we look at. So this is basically just setting things up. Do people remember the basic set up for classification?

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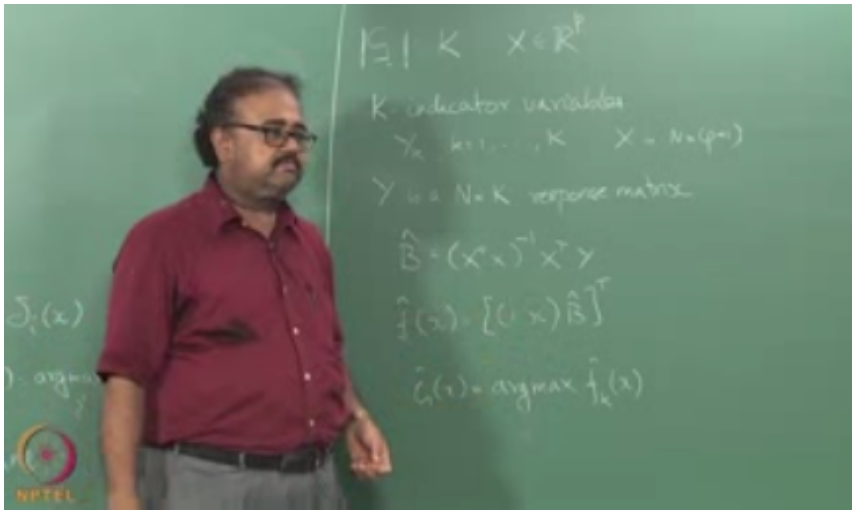

So I am going to assume that I have some space *G* which has *K* classes. I will first conveniently index them as 1 to *K*. *X* is going to come from $R^p$ as before and the output is going to come from this space *G*. So that is our setup and so if there are *K* classes I am going to have when have K indicator variables.

Remember when we talked about "one of *K*" encoding, one of these *K* indicator variables will be one for any input, depending on what class that data point belongs to. I have augmented ones so my $\hat{\beta}$ is equal to $(X^T X)^{-1} X^T Y$ . That is linear regression for you. I can just do linear regression on my response matrix. $\hat{\mathrm{B}}$ is capitalized here because it is also a matrix. So each class I have a set of β so I can produce a vector of outputs f given an input X by essentially taking the product with the β that gives me a vector of outputs f and finally class label that is sent to the data point this argmax of they have f. $\hat{f}(x) = [(1x)\hat{\mathrm{B}}]^T$ and $\hat{G}(x) = \arg\max \hat{f}_k(x)$ .

So I am going to get a vector f(x) one for each class and the one that I assign finally is the one that gives me the maximum output. I am not doing that any complex math here at all, so only bit of math here we already saw in the very first linear regression fitting.

So X is the input data point I add a 1 to the front of it to for the bias. So what does it mean? What does this $f_k(x)$ mean? So if you think about it whenever the input take as of some classes let us pick a particular class let us call it j let us have a class or even make it more confident and instead of 'j' let's consider class 3. So the input belongs to class 3 whenever, the input belongs to class 3, $y_3=1$ from the training data.

So if you think about it, and look at the expected output that you should get for a particular x the expected output you should get for particular x is: what is the average number of times it is going to be one? So I am going to see the x again and again and whenever that the x belongs to class 3 the output will be 1 and the x does not belong to class 3 the output will be 0. So what is the output I expect? It is the average of the outputs. The prediction should be the average of the outputs. Does it make sense? So I have many x,x,x there are different x they are the same x many times I am getting x again and again so sometimes it is class 3 sometimes it is not sometimes it is not sometimes it is class three .

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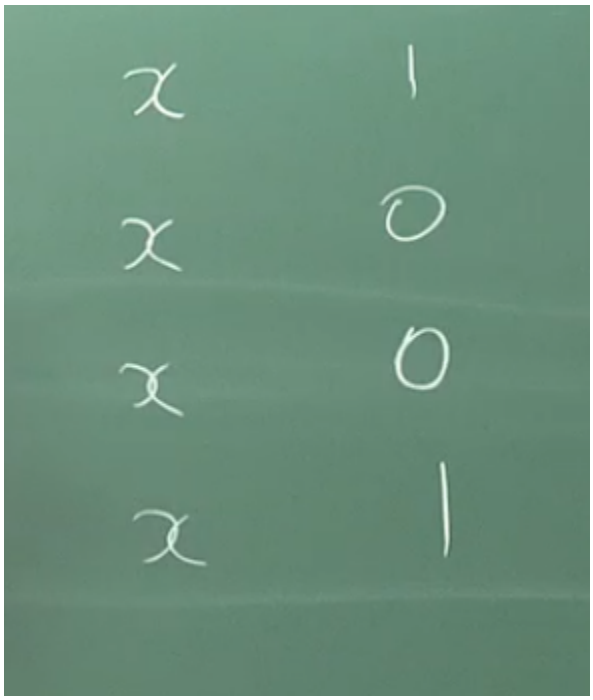


$E[Y_k \mid X = x] = \Pr(G = k \mid X = x)$ . So if you take the average of all of these outputs, what am I getting? Probability that x is class 3. We know that when I am trying to do the linear regression what I am trying to predict is the expected value. Ideally I should be trying to predict the expected value of this but since its linear you will not be able to get there but we are trying to do is probability of the class. There is a problem of using linear regression and that is what I am coming to it.

So you cannot really interpret these as probabilities because linear regression is not constrained. So we will come to that in a minute but this is one I am working upto that it is telling you the interpretation of what you want to do is that it is a probability? So I really would like to interpret this so the expected value of $y_k$ given x you would ideally like it to be probability that the output is K given the input is X. This is ideally what you want and the linear regression gives you hope of getting there and people sometimes still use linear regression because it is easy to use.

We would have to think of other ways of getting to it. I will come to that in a minute, so before that I just want to point out one other pitfall of using linear regression for classification. But is it clear? Any questions? This is the same indicator variable thing, so it is either 0 or 1.
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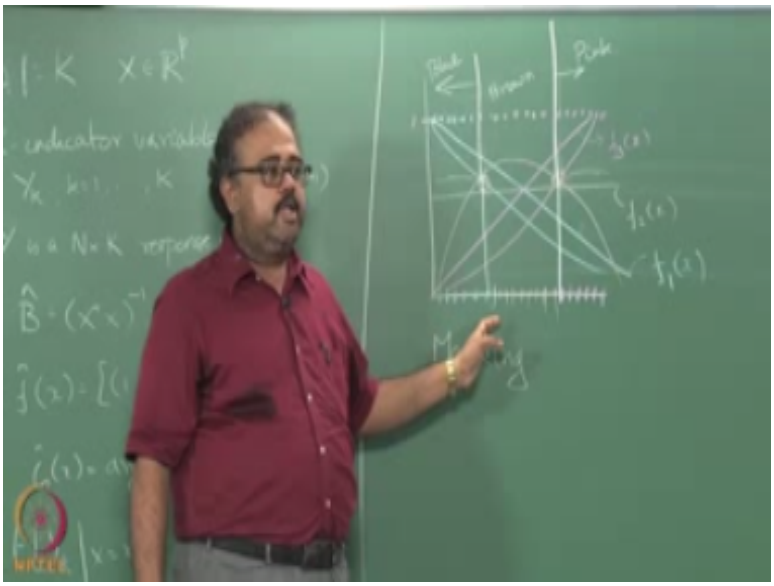
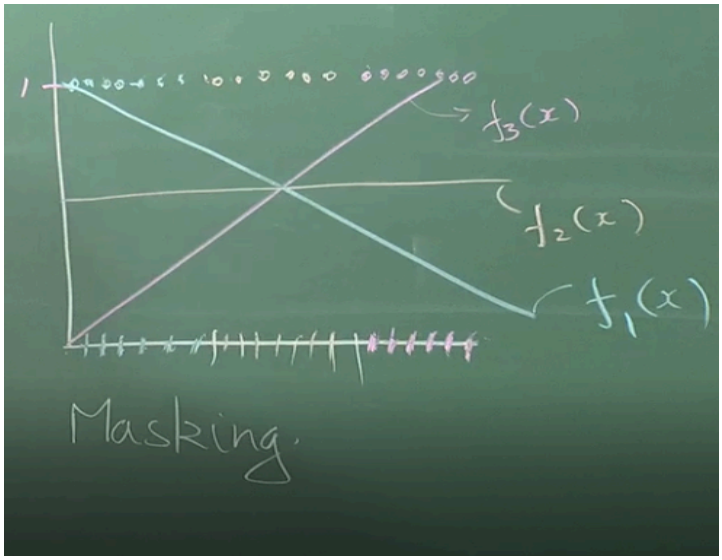


So I am going to assume there is a single input direction. So let us say that there are data points here that belong to one class and data points here that belong to another class r. So if you think about it let us say this is this is encoded by pink so the training data will look like this and 0 elsewhere for pink training data for blue will look like this and 0 elsewhere.

(Refer to slide time 21.03) 


So now if I try to fit a straight line to this so what do you think will happen? I will get a line that goes like that and I will probably get a line that goes like that. So this is essentially what your outputs will look like. Directly trying to interpret this as probabilities is not a good idea obviously but you can see that wherever this is greater than this that should probably belong to class blue, where ever it is pink is greater than blue it should belong to class pink. So at least this much you can conclude from the output of the linear regression.

So that is essentially how you would interpret the output? So whenever one output is greater than the other or greater than all the others you will assume that it is the correct class. Directly interpreting that as probability it is a problem, so this is what you would like to do. But you do not want to do this. Having said this let us see how visible this color is, suppose I have a 3 class they are sitting in the middle like this, so the outputs were this will be somewhere there . Now if I try to fit a straight line for this what is going happen?

Remember the rest of the points are all sitting here they are a bunch of 0 here a bunch of 0 here and a bunch of ones there, so I try to do linear regression on this so I am going to get that line like that I know. What is the problem with that? Blue and pink completely dominated, there is no part in the input space, where brown no part of the input space where brown actually dominates. The output of brown never dominates anywhere. So this is essentially what your $f_1$ $f_2$ $f_3$ will be so it turns out that for class two will never output any input point as class two , so this problem is called this problem is called masking. So this is one thing which you have to be aware of well

you are doing linear regression for making your predictions. Is there any way to get to over masking? So instead of looking at pairs you just look, at higher order basis transformations. Instead of regressing on x, I could regress on $x^2$.

So if I am going to do that essentially I am going to get curves that look like that, with interesting curve is this guy how is this brown curve going to look like. So these are the crossover points it is anything, that this side will be blue anything to that side will be pink and anything in between will be brown , you can see . So but remember the input space is just on this line , so here this is the output whatever is going up is the output, so the input is only on this line just a single dimensional input. So it is no region but say it is only a line segment here so in this part of the input space it will be blue this part it will be brown in this part it will be pink thank the almost ideal except there is a small here.

That is the just drawing error so you can you can choose appropriate data points such that you can actually, with the quadratic transformation, if you regress on $x^2$, recover the actual boundaries. So the rule of thumb is that if you have K classes in your input data you need at least K - 1 basis transformations. So in fact with a lot of work you can show that even with $x^2$ regression you will have masking if you have four classes so in four classes you have to regress on the cubic transformation, so that you can still get away with it.

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**Introduction to Machine Learning**

**Lecture 20**

**Prof. Balaraman Ravindran**
**Computer Science and Engineering**
**Indian Institute of Technology Madras**

**Logistic Regression**

So let us go back to whatever has been bothering all of us. So what we are essentially choosing when we did regression here was we are going to make sure that the output is either 1 or 0. And then we are trying to regress on that. So what do you want our function to do? Basically you want your f(x) to give you P(K|X) but then trying to do that is a little harder so what we are going to do is you are going to look at some kind of a transformation of the probability.

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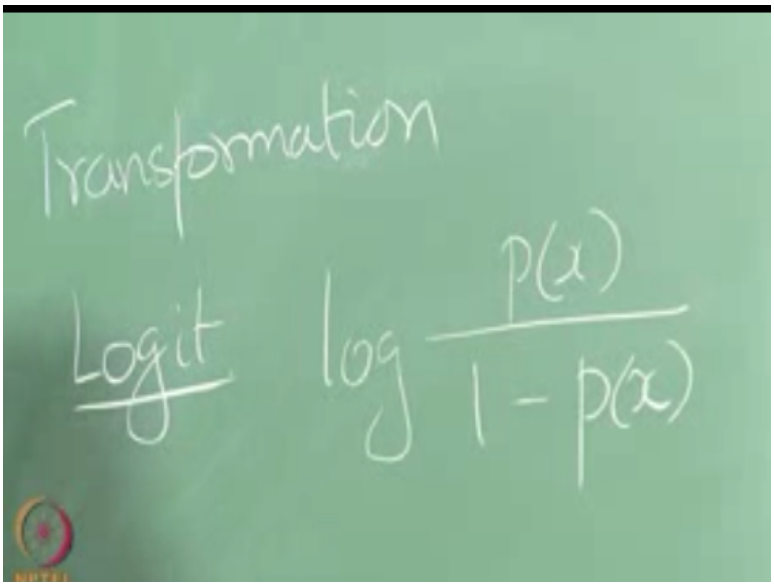

And we are going to try and fit that. Let me look at the logit transformation. This is essentially $\log \frac{p(x)}{1-p(x)}$.

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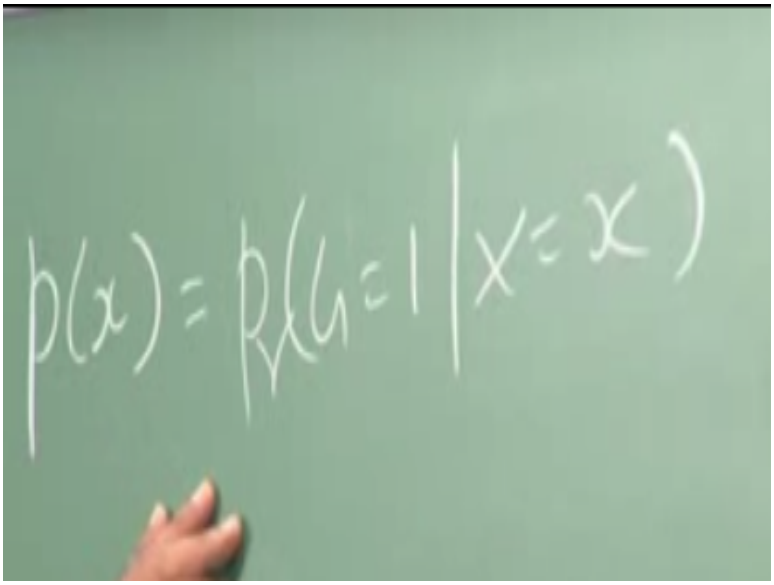


To make my life easier for the next few minutes I am going to assume we are dealing with binary classification. So the class label is either 0 or 1 and p(x) is essentially probability that the output is 1 given the input is X. So this makes my life a little easier when I write the next part.

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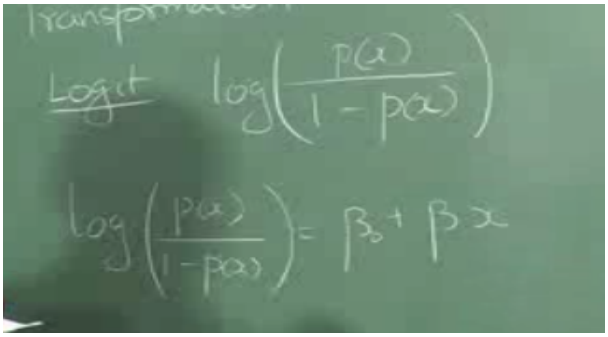

So given that $p(x) = \Pr(G = 1 \mid X = x)$, what is 1 - p(X)? It is probability 0. We are talking about binary classes so this is sometimes called the probability of success divided by the probability of failure or "odds". So this is sometimes called the "log odds" function or the "logit" function. So this is essentially the transformation that we want to look at so what I am going to do is I am going to try fit a linear model to the log odds. So what is p(x) in this case?

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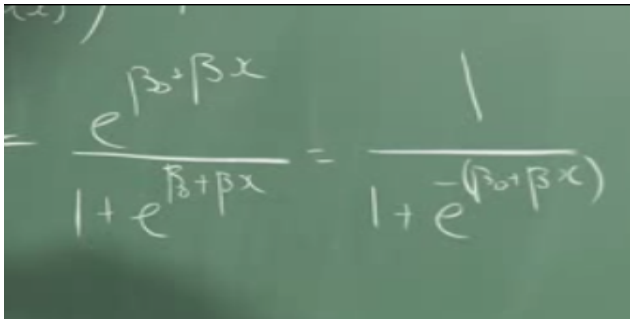


So what is this function going to look like? That is a sigmoid. So essentially we are saying that my p(x) is going to given by probability that x is 1 will be given by a sigmoid.

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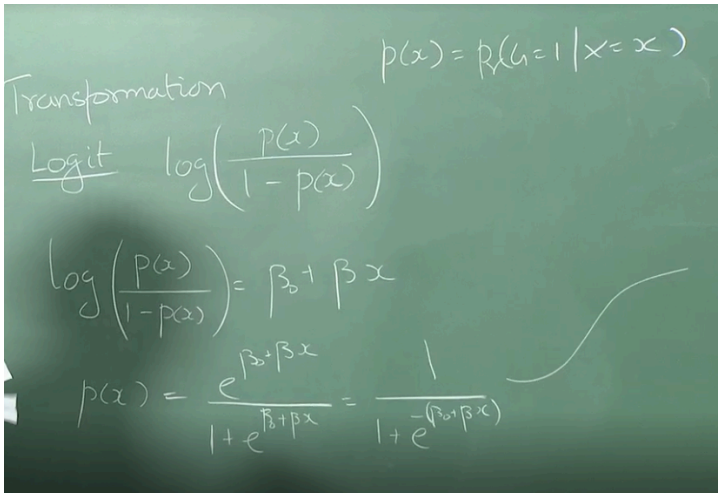


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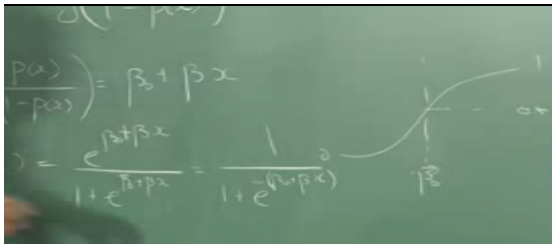
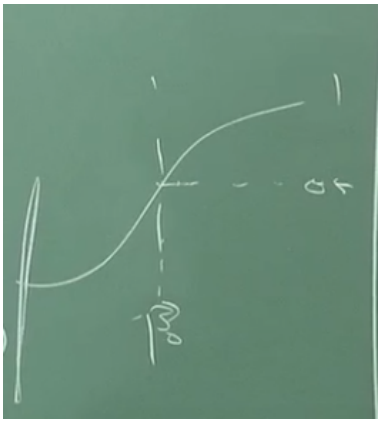


That is the term in the power except that here it is a minus before that. So what do we have here? What do we do if p(x) is greater than 0.5?

(Refer to slide time 7.46) 

We will output this one and if p(x) is less than 0.5 will output it as 0. So it is because even though I am doing linear regression and linear regression is unbounded I am going to plug it into this expression and therefore this will make sure that my probability is between 0 & 1. What is that point? This 0.5 depends on what I had put for my $\beta_0$. So what about the classifier that I am learning here? What the separating surface or the decision boundary is between class one and class two? Think you have p(x) = 0.5 but what does it mean? Look at the expression that we have here? So in p(x) equal to 0.5 this be 1 we have log of that will be 0.

So essentially the thing is $\beta_0 + \beta x = 0$ and that is the straight line I mean assuming X is uni-dimensional that is a straight line. So even though I did something complex, I used an exponential to define my probability, the decision surface turns out to be still hyper plane. So plug in p(x) equal to 0.5 here and I'm going to get 0 on the left hand side. I am essentially solving $\beta_0 + \beta x = 0$. That is just a straight line assuming that it is a hyper plane. Maybe I should do the whole class in one dimension it makes it easier for people to visualize things. So one thing I

should point out is that logistic regression looks simple but it yields a very powerful classifier it works very well in practice.

And it is used for not just for building classification surfaces but it is also used a lot in what people sometimes call "sensitivity analysis". So they look at how each factor contributes to the output and so how each how much is each factor important in predicting the class label so for doing that they do logistic regression and then they look at the $\beta$ vector and figure out how much each variable is going to be contributing to the output. So people use that a lot I mean of course you can use anything that we have seen for doing this kind of sensitivity analysis I am just telling you what people use in practice .
So logistic regression is something that is used very widely in practice both by machine learning folks and by statisticians in fact when I work with few doctors it was almost impossible to get them to accept anything else other than logistic regression as a valid classifier because they were so sold on logistic regression and with good reason because it does work very well in very well in practice.

So that is for two classes, so what do you do for multiple classes? So multiple classes I'm essentially going to take recourse to this form. I am going to say keep the probability that the output is class 1 given X is given by an expression like this and the probability that the output is class 2 given the input is X is given by another expression like this where which will have a different set of $\beta_0$ and $\beta$s .

Likewise for every "class I" am going to say is given by a different set of $\beta_0$ and $\beta$. So do we have to do that for all the k classes? I have to do it only for k-1 classes because the kth class probability will be automatically determined. I will have to have k-1 sets of $\beta$. If I have k classes I have to figure out how to estimate those.

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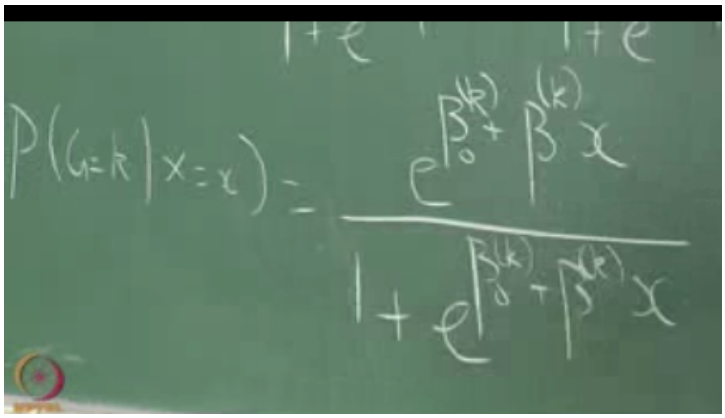

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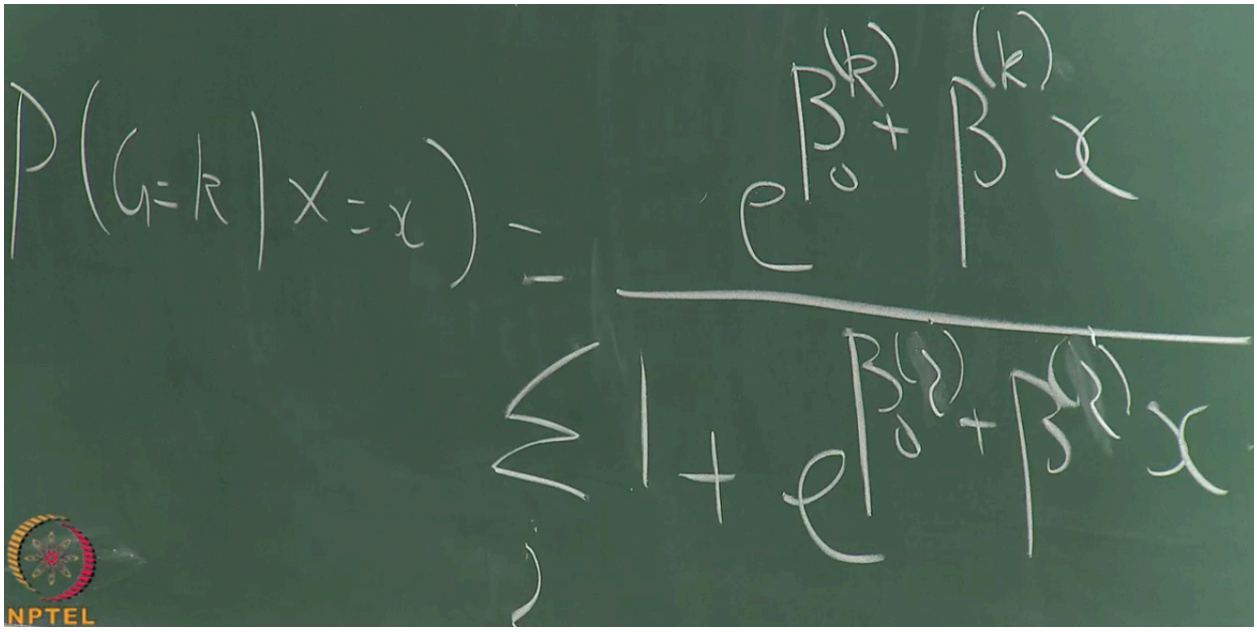

So we are going to write like this for that K minus 1 class by convention either the first class or the last class (whichever arbitrary numbering you choose). For either the first class or the last class the coefficient is set to zero. Setting the coefficient to zero will essentially give you the answer that you want. So now you agree with me setting it to zero is fine so how do we estimate the parameters for logistic regression? So it's a little tricky since we are anyway trying to model directly the probabilities. What we are going to try and do is maximize the likelihood of the data? So far we have always looked at some kind of error function and we have been trying to optimize the error function within those linear regression we looked at squared error and then we did the

optimization and soon so forth but here we are going to look at a slightly different criterion we are going to optimize the likelihood of the data.

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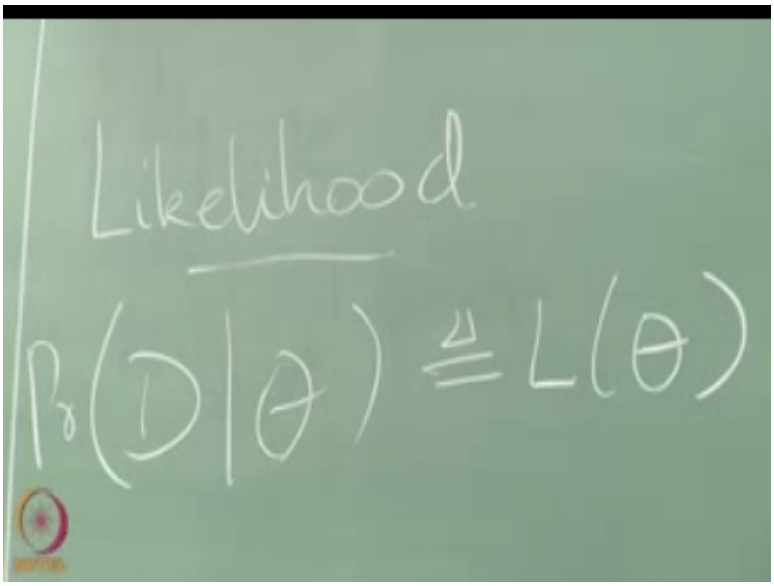


So just to keep it together I am going to do this today but I have a whole session planned on maximum likelihood and other ways of estimating parameters so when we come to that I will do maximum likelihood in more detail in a generic form. So right now I will just look at logistic regression and maximum likelihood. So what is likelihood? Suppose I have some training data D the training data has been given to me the probability of D given parameters θ is known as the likelihood of θ.

So D is fixed. Think about it. I am given a training data D, D is fixed what is it that I am actually looking to find? It is θ. So this I will write as likelihood of θ. So we are always used to thinking of something that comes after the slash as the conditioning variable and the one that comes before the slash is the actual argument in this case it turns out that theta is the argument and the probability of D given θ is the likelihood of θ. D is fixed I am really trying to find what θ is correct. So finding the likelihood of theta so the scoring function should be on θ and I am usually interested in the log of θ.

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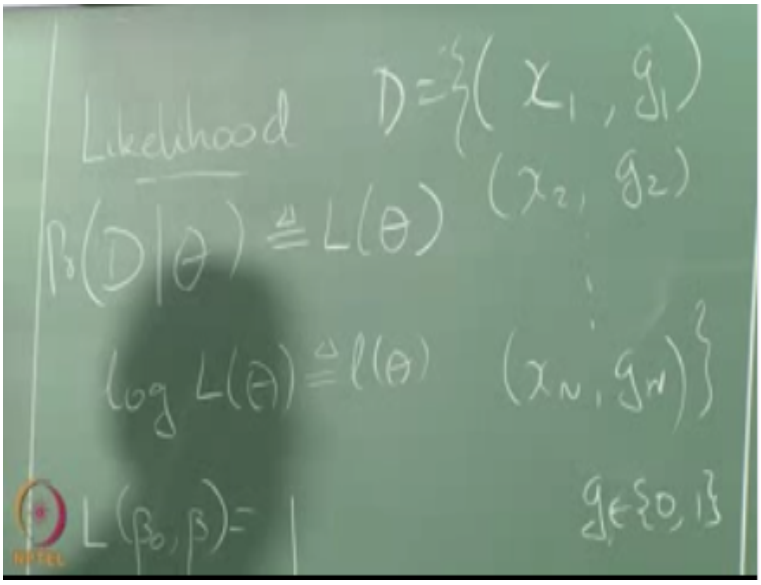


Because it allows me to simplify a lot of the distributions that I will be considering and we will denote this by "l" mostly. So what is the likelihood so in our case? θ is our βs so my input my D is going to consist of $\{(x_1 , g_1), \ldots (x_n , g_n)\}$. It is going to consist of pairs of data points like this. So X is the input G is the output we are talking about classification so G is the output X is the input. What is the likelihood? So I wanted to stay in the two class domain for y so G is either G belongs to 0 or 1 so 0 means is class 0, 1 means is class 1.

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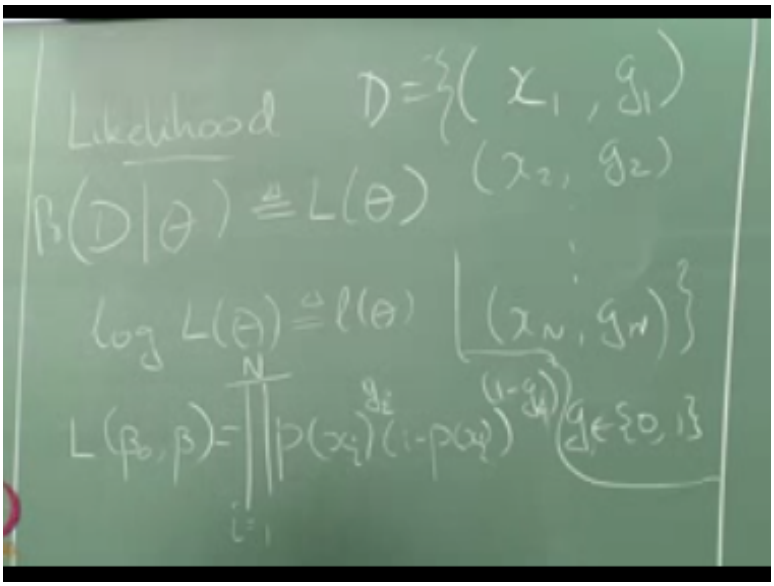
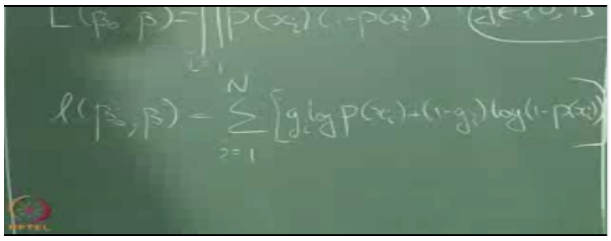


This is a funky expression that is written and we will come back to this. We will see this again so this is the probability of one pair "xg" occurring? So what is this is the probability that the X has a label 1? This is the probability that X as the labels 0 and what is this? This is the actual label of X. If the actual label of X is 1 then the term $\log(p(x_i))$ will appear in the equation if the actual level of X is 0 then the term $\log(1-p(x_i))$ will appear in the equation this will become 1 so if the actual level of X is 1 what should be the probability.

Probability that X equal to 1 and that is what I should be looking at. The actual label is 0 then I should be looking at the probability that X is 0 that is what this term is. So you can see that this gives me the probability of one "XG" pair. I do this for all of them assuming that they are all sampled independently because I am assuming they are independent I can take the product. So now we know why we love logarithms. (Refer Slide Time: 18:26)



So that is the expression and is simple enough so now comes the interesting part we want to do when we want to maximize likelihood. So we need to take the derivative of this and equate it to zero. It's fine because log is a monotone transformation we can take the derivative of the log equate it to zero and then solve for $\beta$ . Unfortunately life is not so simple. Let us try and do the simplification which I am multiplying this out and gathering the terms. Multiplying this out I gather the terms here and we know what that is what is that yet we already know that so that we can insert that and simplify that there and what about this guy one minus this I can again write it in a simpler form write log of one minus z will give me .

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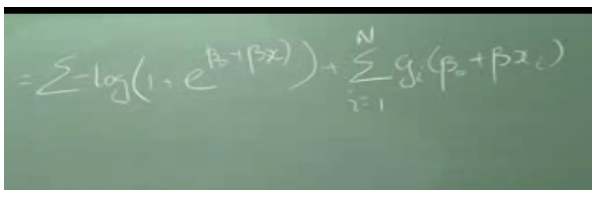
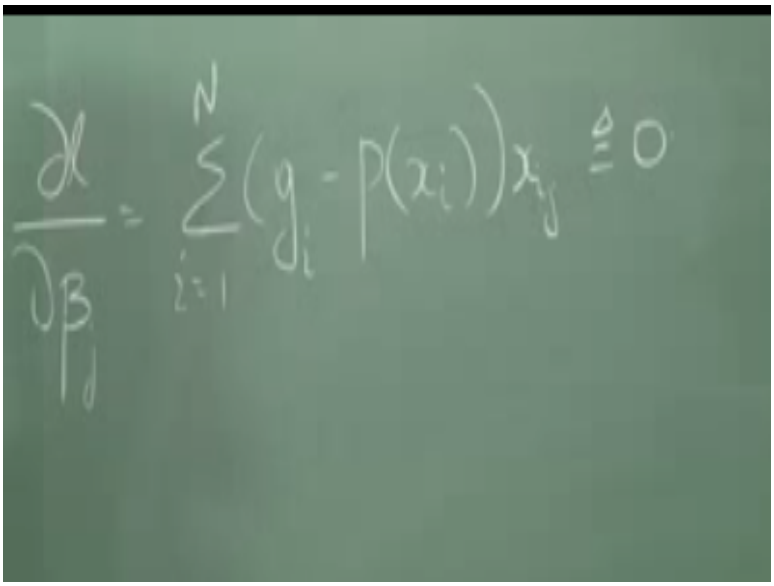


So now I can take a derivative of that with respect to β and equate it to zero. Then what do I get? (Refer Slide Time: 22:11)




So take the derivative of this term you are going to end up with minus P. So this will go down and I will get $e^{\beta_0 + \beta x}$ as the numerator. I will get minus P times X since I am doing it with respect to a specific $\beta_j$. I will get $X_{ij}$. Does it make sense? So this first term I will get minus P times $X_{ij}$ and this one if I take the derivative of that I will get $g_i$ times $X_{ij}$ . So that is essentially what I have done here and this looks like a nice and easy expression to solve. But unfortunately it is not so because this will it is an exponential function here so it is not really easy to solve this you have to look at some other iterative method for solving this and the most popular method used is Newton-Raphson. I am not going to go into the depths of Newton-Raphson. People are encouraged to look it up.

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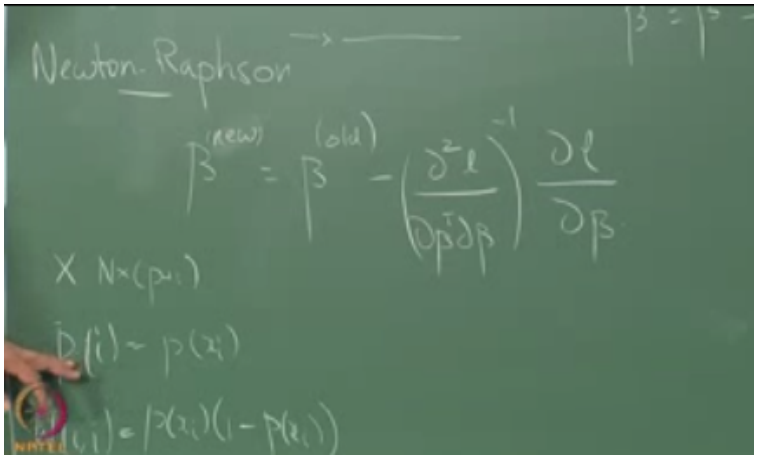


But the basic idea is that so people were more comfortable looking at it this way so take the old estimate of your values or your take your old solution and look at the first derivative of the function that you are maximizing. l' divided by l'' so you adjust this by that so that is essentially the basic idea behind Newton-Raphson. I am just defining some terms here so X is going to be my (n x p + 1) matrix as usual and my P is going to be a vector where each entry is going to be the probability of $x_i$. So what will be the dimensionality of P? It will be n. So it is a n vector that tells me what is the probability of each $x_i$ being one. So P and W is going to be a diagonal matrix where each diagonal entry is P into 1 minus P for that particular at a point $x_i$ so this makes it convenient to rewrite things and I am going to assume that g is the vector of outputs like zeros and ones depending on what class it is.

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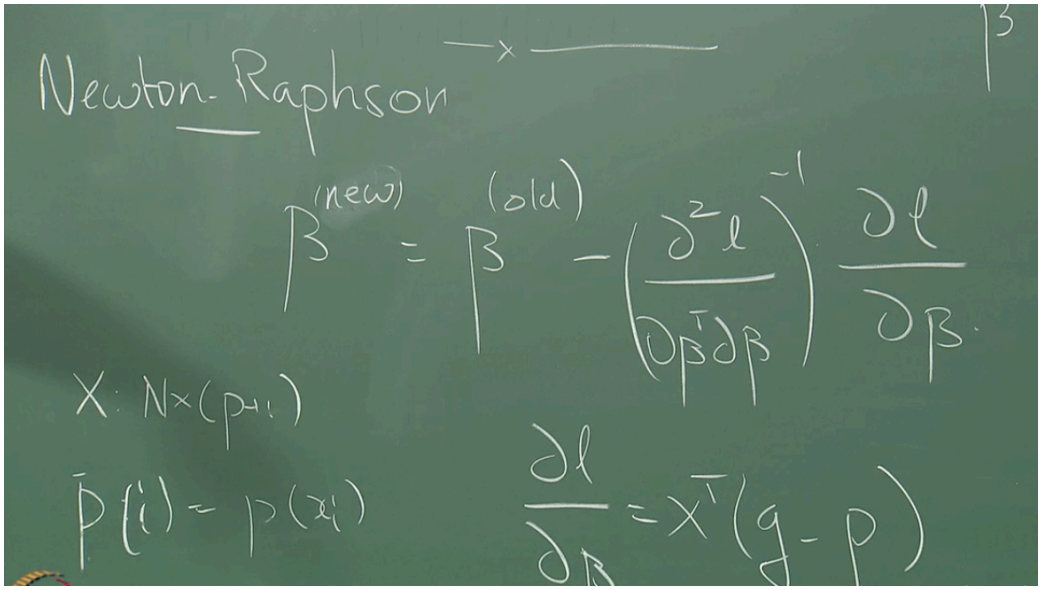


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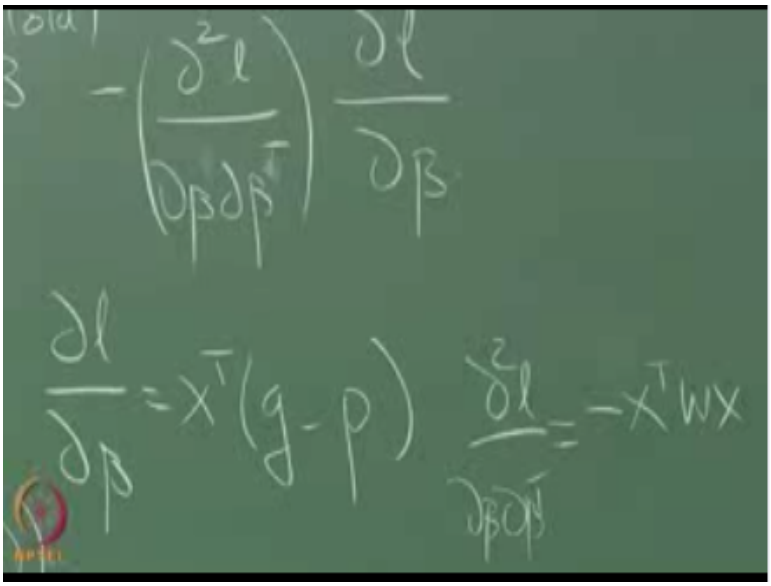


So I can write my $\partial l / \partial \beta$ is. It is $X^T(g - p)$. In terms of the matrices it is P and the vector Y is the vectors of zeros and ones corresponding to the class label and X is my input. So I am just basically written this in vector notation so you already found the derivative I have just rewritten it in vector notation that it make sense. So what about the second derivative? I am not going to work it out. But it is $X^{-T}$ WX and so W is essentially the diagonal matrix with this entries. So that is my second order I made my second derivative so what do I get now putting this together.

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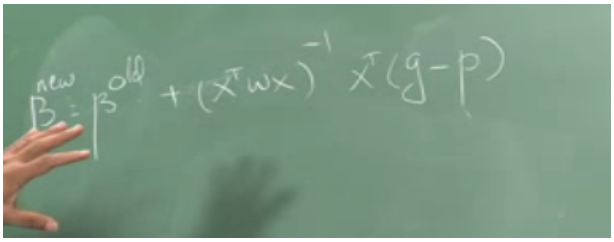


The beginning looks something like regression. Here you are getting your $(X^{-T} X)^{-1} X^{-T}$ and so we just have to do a little bit more work little bit of algebra to make it look more like regression so that's what we will do now. I just substituted the derivatives here, nothing fancy. So you want to solve this problem when this becomes 0 so you can see the β in here. The β is in the P. I have erased the p off now but so β is in their P is the P is $e^{\beta 0+Bx}$/ 1 + $e^{\beta 0\beta}$ is here I really want to solve for this. I want to find the 0 of this function.

But it is not easy to do because of the fact that we have it exponential in there. So what we have to do is look at some kind of iterative method for solving this problem and so the way we do this is by adopting an iterative approach. You start off with a guess called $\beta_{old}$. And then you do some computation you get a new guess call βnew. So one very popular way of doing this kind of iterative thing is to do "gradient following". Have you have you looked at that? I mean you must have might have come across that this side so suppose I have a function like this, I am here this is my current solution. This is what I will call as $X_{old}$. So now I will compute the gradient here and I will move in the opposite direction of the gradient to find the minimum. So instead of going all the way I can go a small step that gives me the Xnew.

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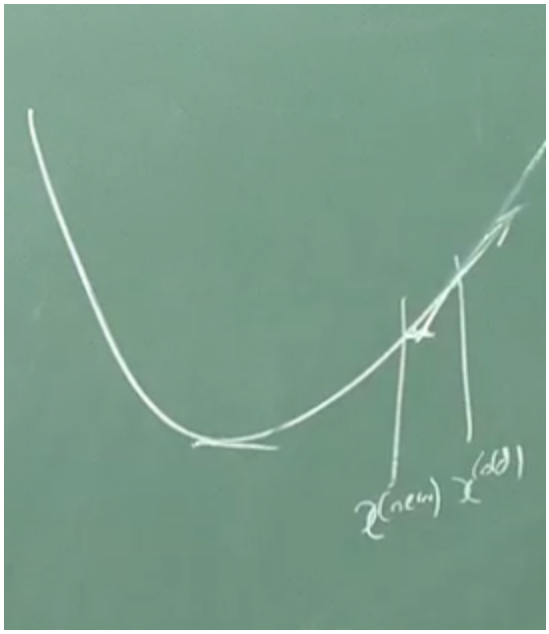
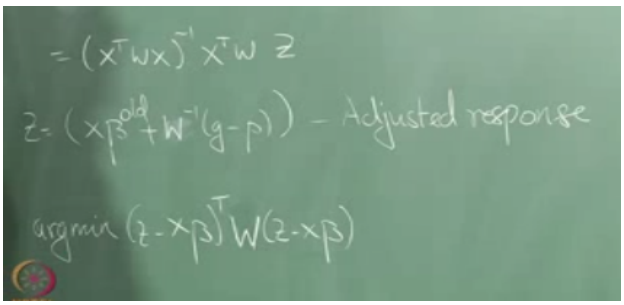


Normally what you will do is you will find the gradient try to equate it to 0 and get it but you can do this in iterative fashion also. You can take small steps in the direction of the gradient so likewise what we are going to do is we will start off with βold with a guess. I can start off by saying all my β at 0 and then try to find a βnew. So what I will essentially be doing is I will find so β0 will put me somewhere here on the L function, I will find out what is the first order and the second order derivatives at this point with respect to β and then use that for changing my β values so people agree with me so this is the $X^T W X^{-1}$. (Refer Slide Time: 32:08)




If you take the product here this will be $X^T W X$ so that is just the identity and I take the product here I will get $X^T W^{-1}$ back this $W^{-1}$ and will get cancelled out. So I have just done some algebra to get it this way. Think about it. What is $X\beta_{old}$? Since it is like linear regression, it is like the original response I will get if βold is my variables and I am actually making a linear prediction based on X. So the $X\beta_{old}$ is this and I am essentially adjusting it. So this is the prediction I make

with my old parameters this is some kind of adjustment I am making to the prediction so this is called the "adjusted response" and this turns out to be the solution of something known as weighted linear regression it is in weighted linear regression essentially what you do.

So in linear regression that is what you are trying to minimize that is a squared error. Here you are trying to minimize weighted linear regression and you essentially have a waiting term in your error function. Since I am just saying I am going to minimize the squared error for every term in the squared error I am going to assign a different weight, so for some data points I want to be more aggressive in minimizing and for some I want to be less aggressive.

So data points in which I have to be more aggressive I will have a higher weight for data points for which I want to be less aggressive I will have a lower weight so that will allow me to trade-off the importance of data points. This is idea behind weighted linear regression. So this is essentially weighted linear regression. So to minimize this, you can do the usual now. You can take the derivative set it to zero and solve it this is easy enough to solve this is actual linear regression.

So the solution is $(X^TWX)^{-1} X^T WZ$. So essentially we are solving a weighted linear regression or weighted least squares problem this adjusted response. So this is called iterative reweighted least squares. But all it does this essentially does Newton-Raphson essentially is doing Newton-Raphson but the way iteratively reweighted least squares is described to you is start off with a guess for $\beta$ form the adjusted response and as soon as I guess or I have a value for $\beta$, I can find out what my P is so G is given to me already in the data and my W can be constructed once I know P. I make a guess for $\beta$, I construct my P. I construct my W.

Form this adjusted response solve this weighted least squares problem and get a new $\beta$. Keep repeating this until my predictions are accurate enough. So that is basically the most popular way of solving logistic regression but there are many other ways people have come up with more efficient ways of solving logistic regression actually and but if you pickup any popular package like R or something so IRLS is the base logistic regression solver that would be implemented . So this just to give you a flavor about how hard it can be to optimize things.

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**Introduction to Machine Learning**

**Lecture 21**

**Prof. Balaraman Ravindran**
**Computer Science and Engineering**
**Indian institute of technology**

**Linear Discriminant Analysis I**

So we started looking at linear classification and in the last class we looked at logistic regression. So you remember the assumptions that we made for logistic regression. We assume that the log odds can be modeled as a linear function. So each of the individual the probabilities were given by sigmoid functions but the log odds was assumed to be linear. That is where we started off with and that gave us a linear decision boundary.

So the separating surface between two classes ended up being linear. If people remember what log odds was, it was probability of class one divided by probability of class zero and the log of that we assumed that was linear. So that's the assumption we made in logistic regression and today we look at another one of those discriminant based approaches. So we already looked at two, one was linear regression on an indicator variable and the second one was logistic regression. Now we look at the third popular classifier called Linear Discriminant Analysis also known as LDA.

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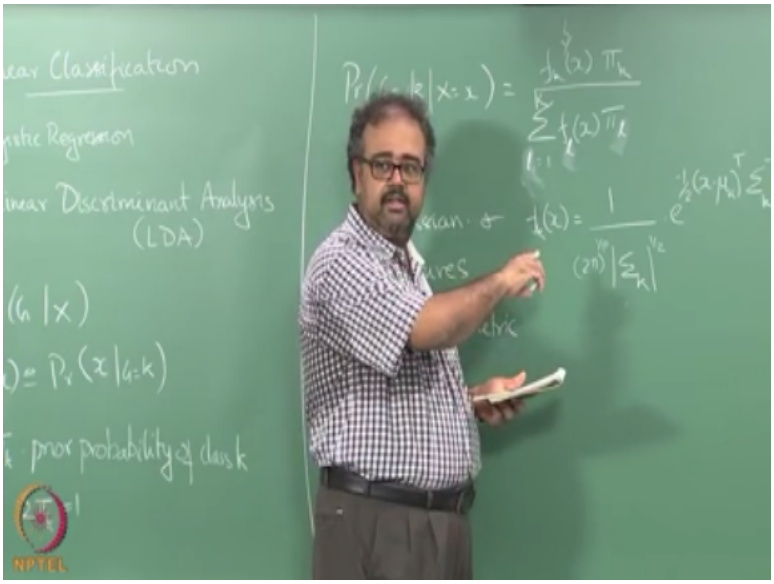


Unfortunately in machine learning there are two very popular algorithms both of which are abbreviated as LDA so this was the older one: linear discriminant analysis there's also something called Latent Dirichlet Allocation which we will not get into, which talks about a completely different approach to modeling distributions. It has nothing to do with classification. It is more on modeling distribution that's also sometimes abbreviated as LDA. So be context-sensitive when you use LDA. If you remember so we are really interested in the probability of a class given the data point. We are really interested in the probability of the class given the data point and you can get this using Bayes' rule. If you have the probability of the data point given a class and probability of the class. So probability of the data point given class times probability of the class divided by probability of the data point.

So what we will do is we will start by making assumptions on probability of the data point given the class is k.

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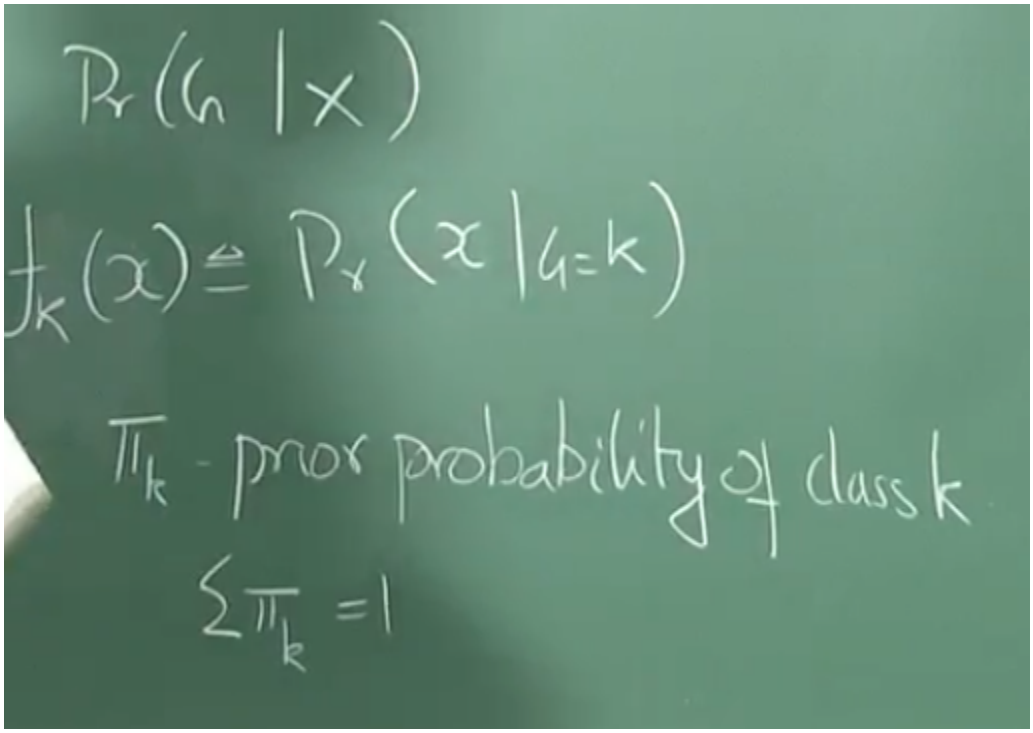


So these are also known as class conditioned densities. The class conditioned density of the data point which I am going to denote by $f_k(x)$ probability of x given that the class was k. This is the class conditional density and I am going to assume that $\pi_k$ is the prior probability of class k.

So we assumed that all data points belong to some class or the other. So that's going to be one. So now I can write

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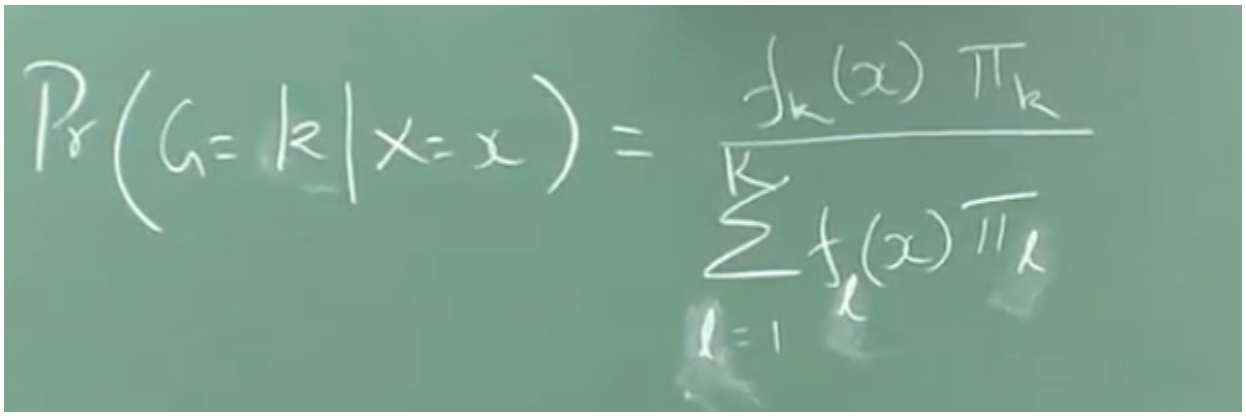

So we have been using ‘l’ throughout so that makes sense. That is why I told you that we do not need the probability of the data. I can always fake that by saying that since the data has to belong to some class so I can just sum over all the classes I will get the probability of the data.

So this is essentially marginalizing over class and I get the probability of the data. And now depending on the kind of assumptions we make for our $f_k$ the form of $f_k$ , we will get different classifiers. So some of the most popular assumptions about $f_k$ are that $f_k$ is Gaussian for both LDA and a related method called QDA. Any guesses what QDA is? Quadratic discriminant analysis. Both of them assume that the class condition density $f_k$ is given by a single multivariate gaussian. You could also assume that the class conditional densities come from mixtures so instead of a single Gaussian, you assume that there are multiple gaussians which jointly generate the data for you. So people are familiar with the concept of a mixture distribution? Very simple it is like let us do a little segue you here suppose I want to model this following distribution over univariate data . So this single dimension and the axis actually tell me the probability of seeing something but what I want to do something like that can you think of a parametric form that will give me this kind of a distribution. Looks a little daunting, doesn’t it? Can you come up with like a closed form expression for this? But if you think about it I can look at another Gaussian like that and I can suitably weigh the two of them and I can combine their distributions. So the combined distribution will look like it has two peaks and so this is essentially the idea behind mixture distributions. So if the form of the distribution I want seems rather complex and I want to have a simpler functional form for the distribution I can think of writing it as a combination of several simpler distributions.

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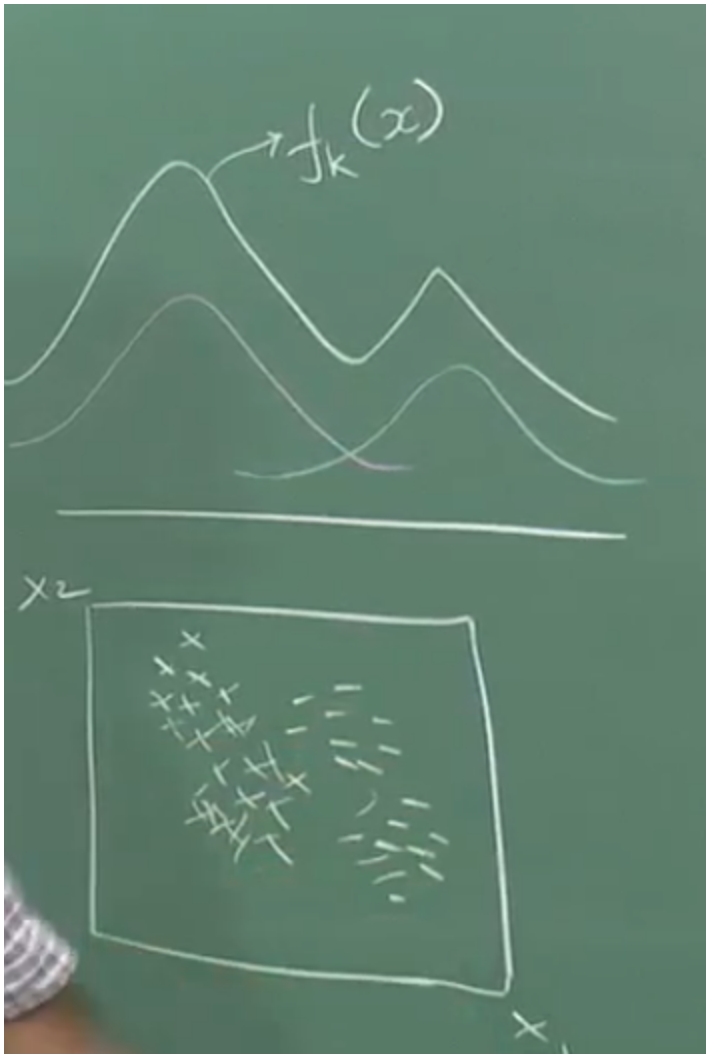


So likewise suppose my positive class look like this. So how will it look like in a 2D setting? So let us think of it. So my data looks like this. This is the positive class there's another class that comes to here and so if you think about it this there is more data points here and then there are more data points here and there is a slight region of lesser density in between the two.

If I try to model this as a single Gaussian and if I use any kind of maximum likelihood estimate where will the peak go? It peak probability will be somewhere here which is obviously incorrect. So where's the peak probability? It will be somewhere here for the negative class which is obviously incorrect. I suppose to that if I say that they're both the positive class and the negative class are created by two Gaussians each, then the mixture of this so I can have one Gaussian which has a peak somewhere here other Gaussian which has a peak somewhere here likewise

one for this and one for this and then I can combine them using some kind of weighting mechanism.

So this is what we mean by mixture distributions giving you class densities. So you can think about this I mean I can have more arbitrarily complex kind of distributions here and then instead of having two Gaussians I can say, I am going to have ten and also in they need not be Gaussian say could be other functional forms but the more complex the forms I take the harder it is going to become solving this problem. So $f_k(x)$ if you remember it's the probability of x given G given that class is k. So likewise if I have mixture of two Gaussians here that'll be the probability of the data point given that the class is X and here they are given that the class is whatever. So that is what we are mapping here. So mixtures are fine if we still want to stay in a parametric space. Well then yeah so that is a hard problem. So usually you take some guess from whatever knowledge you have about the domain or you can do some preliminary experiments you can try to run some kind of rough clustering by varying the number of clusters and trying to see whether you can decide on the number of mixture components alternatively. You could do nonparametric methods they are more complex but in the last 5, 6 years so lots of tools have been developed to be able to handle these kinds of nonparametric tests. So nonparametric is actually slightly misleading and it does not mean that it does not have any parameters. It only means that it has an unbounded number of parameters. So it does not mean it doesn't have any parameters it is just that I do not fix it a prior like we are fixing the mixture component where I'm fixing the Gaussians per class. When you fix these things we call them parametric and nonparametric methods typically can add parameters if the data needs it. You can start off with just one Gaussian and then figure out oh no I need more then I can add another Gaussian here I can add another Gaussian and so on so forth. So that is essentially what nonparametric methods. The ability to grow the number of parameters needed if it is supported by the data and if the data warrants it. So obviously we'll have to be very careful about doing things like over fitting the data but there are other ways of adjusting for it so I like I said in the last five years a lot of powerful techniques have come up for nonparametric reasoning but I'm not going to cover any of that in the class like I keep reminding you people is "intro to ml" course. If at all we do an advanced topics in machine learning course then we will probably cover some of that.

So I do not fix the bound a priori. I do not say that you can use only three Gaussians per class. So you can keep adding more Gaussians if the data warrants it that's what I meant by unbounded. So I don't put the bound a priori and obviously I mean that is always a physical bound but in the modeling sense I don't bound it apriori and I don't say that oh you have to use ten Gaussians.

So the most popular of assumptions that people typically make on $f_k$ is sometimes called the Naive Bayes assumption. We will deal with this separately and just putting it out here to just to tell you that all of this come under the same class. So the Naïve Bayes assumption is essentially to factor my class condition density along each dimension assuming given the class one dimension does not influence the other dimension. So if I have two dimensions here x1 and x2. So I will say that I can write the probability of x given K as probability of x1 given K times probability of x2 given K and that is a very strong assumption if you think about it. And I'm saying given the class x1 is independent of x2 - if I don't know what the class looks like, there is some dependence between x1 and x2 but given k, I'll ask you x1 is independent of x2. So this is essentially called the Naive Bayes assumption because looks like a very simplistic assumption or looks like a very naive assumption to make about the data so it's called Naive bayes and it turns out to be powerful in many settings we will come back to Naive bayes separately in one of the later classes and so now I am going to go back to Gaussian. So I am going to assume that. Does it look familiar? That's the Gaussian distribution.

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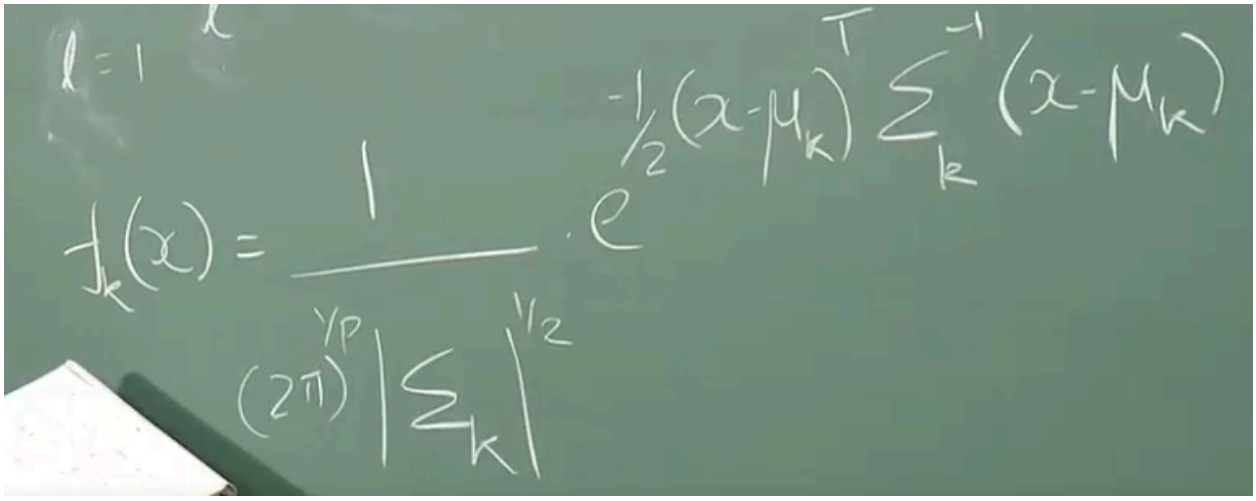


So that is $\Sigma$. So if I'm looking at a univariate Gaussian I will write the variance here looking at multivariate Gaussian this is the covariance. This is the covariance matrix this is multivariate Gaussian and here again this is what $(X-\mu)^2/\sigma$. So people must be familiar by now when I say $(X-\mu)^T(X-\mu)$, that is actually the above in the vector sense and then the sigma becomes sigma inverse and this is the covariance matrix.

So this is the called the multivariate Gaussian. So the multivariate Gaussian will capture these kinds of scenarios where the input rate of dimensions itself is 2 and now I have to have a Gaussian that is actually jutting out of the board.

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**Introduction to Machine Learning**

**Lecture 22**

**Prof. Balaraman Ravibdran**
**Computer Science and Engineering**
**Indian Institute of Technology Madras**

**Linear Disciminant Analysis II**

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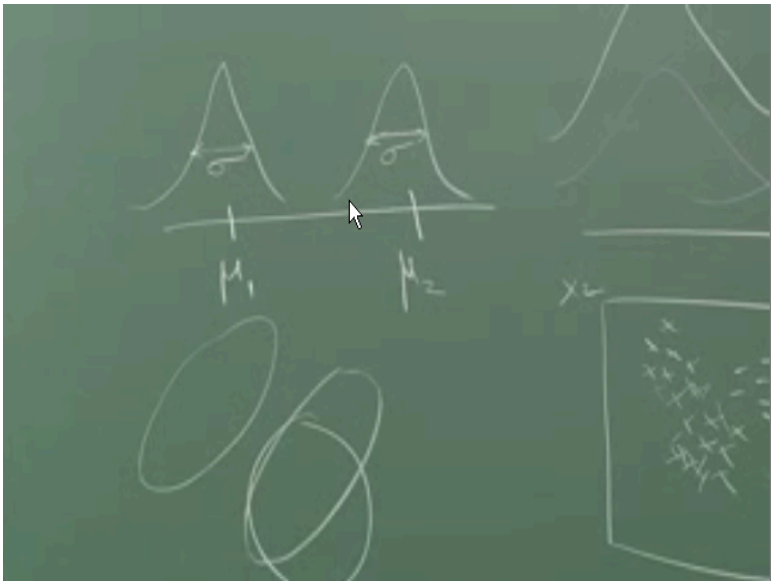


In LDA we make an assumption that the class conditional density is Gaussian. So what else can you assume? So I am going to assume $\Sigma_k = \Sigma \forall k$ . It is the same for all the class k. So what does that mean? It means that I will do that in 1D, say, class one could have the mean of my Gaussian here, class two could have the mean of the Gaussian somewhere else but when I look at this, so that is the same.

And all I can do is shift the Gaussian around but I cannot change the shape of the Gaussian. Is it clear? When I say $\sum_K$ it is the same for all the classes? It essentially means that I can just shift the Gaussian around, I cannot change it. So in 2D it will be like, let us assume that that is the equivalent of one $\sum$ contour. If that is the case for one class and for another class also it has to be similar .

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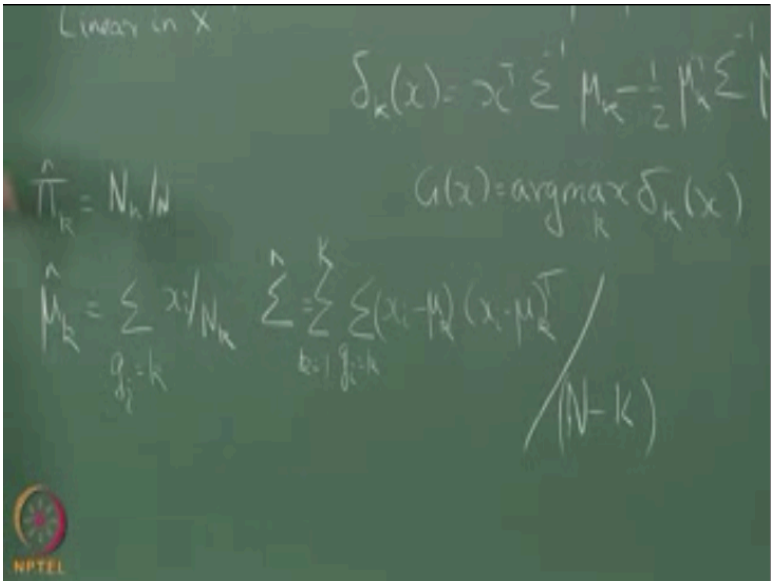


So I cannot have one class looking like this and the other class looking like that. Does it make sense? People are able to visualize what I mean? So it has to be that both the classes look similar in terms of the variance that is essentially the assumption we make here. So in looking at logistic regression we saw that we could look at the log odds, likewise I am going to look at $\log \frac{\Pr(G = k \mid X = x)}{\Pr(G = l \mid X = x)}$. So when I am at the class boundary what will this ratio is? One and log of that is going to be zero.

So I can actually solve for this ratio equate to zero and I can get my boundary, So now we know what is the form of $\Pr(G = k \mid X = x)$. Now I can put that in here and I can solve for what? What

are the parameters I should be solving for? μ and ∑ and anything else we are talking about this we had it is all for $\pi$ as well. Solving for $\pi$ is rather straightforward.

I mean you just count the number of data points that belong to a class divided by the total number of data points you have that gives you $\pi$ . So it is not complex but you still have to solve for it as it is not that it is given to a priori. So you have to estimate from data so all these three parameters your estimate. So this gives you the boundary so when the probability of it belonging to k is higher than the probability of it belonging to l then you will put it in k .

So you will have to do this for every pair of classes to make sure that you have the right class so assuming that only two classes just you have to make one comparison but of course there are really k classes then will have to make k-1 comparisons to figuring out which class it belongs to. So this essentially will give you the boundary. So when the probabilities are equal then you know that well it could go either way so this is going to be 0. In the problem it actually belongs to class k the numerator will be higher when it belongs to class l the denominator will be higher so based on that you can decide which side it is going to go. What about now solving for this? This is essentially log of

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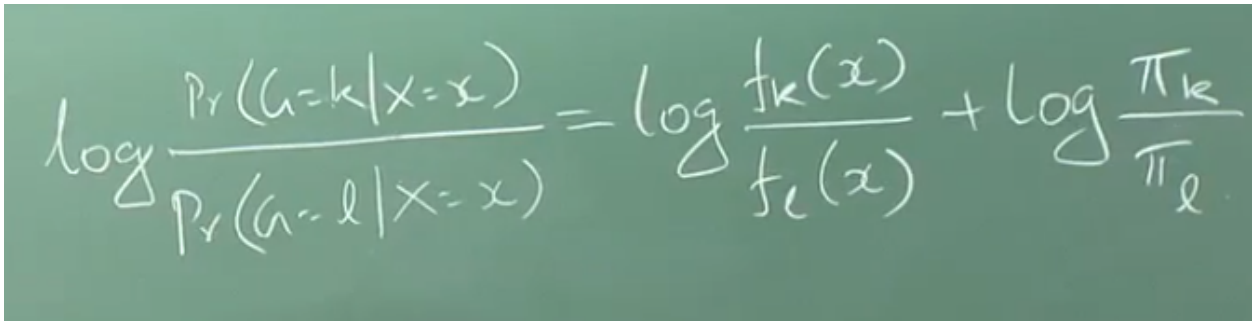


So the denominator will get canceled out like you only worry about the numerators and the fact that we assume that the variances are the same is also going to allow us to cancel out a whole bunch of other terms.

So what other terms can cancel out? So this is $\sum_k$ it will become $\sum$ so when I take the ratio of the two things this thing can go so I do not have to worry about that and it is log and there say e so all of that will go away so.

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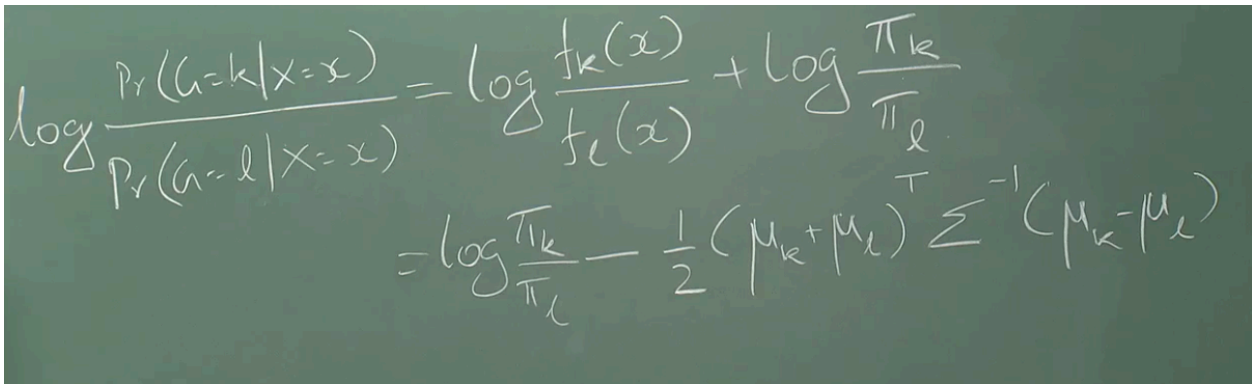


Roughly the way to think of it is if I had taken out the product here. If you think of the term by term product here, so I will have some terms that will have an $x^2$ sometimes that is going to have $X\mu$ and some terms that will have $\mu^2$ .

So I am taking the ratio so I am going to get $\mu K^2 - \mu l^2$. So that is essentially what I am writing out here. The first term here corresponds to $\mu K^2 - \mu L^2$ so you have to get familiar with doing this in the vector notation it makes life a lot easier. I am just giving you the intuition here if you can write it out and see that this is the way to simplify it. So if you think of this essentially you are going to have a $x^2$ term $\mu x$ term under $\mu^2$ squared term. And we take the ratio so you will have e power this divided by e power something else so that is going to become minus $\mu$ in the numerator so you are going to have $\mu k - \mu L^2$. So that is essentially what I am writing out here. What about $x^2$ terms? $x^2$ will get cancelled out because $\sum$ are the same so $\sum_K$ and $\sum_L$ is the same. So $x^2$ will get cancelled out I only have the X terms left. So I will have $X\mu K - X\mu L$ so I am going to get that term as well. So turns out that this separating hyper plane that we have this is essentially the solving this for zero gives us the separating hyper plane. The separating hyper plane turns out to be linear in X. So to get the linearity we needed to make this assumption if you do not make this assumption so what will happen? The $x^2$ term will stay there and what we get this is QDA. I told you about QDA. So if I do not make this assumption I will get you QDA. I

said in discriminating function case we are we always have some function like this $\delta_k(x)$ and if $\delta_k(x)$ is greater than any other $\delta_l(x)$ then we will classify the x into k .

This is what we said was the idea we had disciminant functions in the very beginning so what would be the disciminant function version of LDA? Please note that for most of the time I will make sure I write limits whenever I write this summation ∑. So this is essentially the discriminant function. So you can just compare this and this is whichever has the highest discriminant value will become the class so $\hat{\pi}$ like I said earlier you count the number of data points in the training data there belongs to class k divided by the total number of data points you get $\hat{\pi}$ and $\hat{\mu}_k$ . You pick out all those data points for which the class was k from the training data find the center of them that gives me $\hat{\mu}_k$ . Does that make sense?

So what about ∑? What is that μ? Like well presumably so because these are all data points that belong to one class, so when the training data comes, I am assuming that I have sufficient number of data points of each class otherwise I will not be able to learn anything. It will work to the extent possible with the small data set. Suppose I give you only ten data points for training then you are anyway in a soup like most of your parameter estimation algorithms will not work if you have very little data there are some class of algorithms is work with very little data .

So one such thing which we will look at depending on time today or tomorrow it is support vector machine. So it works with very little data but most of the other parameter estimation methods require you to have some amount of data. So what we do with the variance? So if you remember that variance is not limited to the class it is across all the classes we really want the same variance so essentially we do what is called a pooled estimate.

So we essentially use all the data points for estimating the variance not just the data points belonging to one class. But then you know what is variance. So variance you add that up and divide by minus one. So remember that you always do a minus one for variance estimate. Given a sample mean and sample variance, sample mean is add up the data points divide by N and

sample variance is data point minus $\mu^2$ divided by N -1 usually to adjust for the fact that mean is a dependent variable on all the data points.

So the N-1 essentially gives you an unbiased estimate of the variance. But then what is the mean? So the mean of mean is a bad idea. Thanks for bringing it up because that is a natural confusion when they say you are going to estimate the variance across the entire population so the natural mean you are going to plug in is the mean of the mean of the whole data. But that is not correct. Why? Because I am only worried about the variance within the class so I should plug in the mean of the class of $x_i$. Remember all of this is from the training data so I know what class $x_i$ actually belongs to. So I take the class of $x_i$ so I ask you I will take all those data points belonging to class k and then I will use $\mu k$ here in computing this quantity. Then I will do this over all classes.

And then I will divide by k well divide by n-k. So this is a slightly different way of doing the variance as opposed to computing the variance of each class and taking some kind of a mean this gives you a slightly more robust estimate of the variance so this is called a "pooled estimate". So if you think about let us say I have three classes that look like that so this will become what the separating surfaces that I learn if this had been completely spherical if this had been like this then the separating hyper plane would have been perpendicular to the line joining the means.

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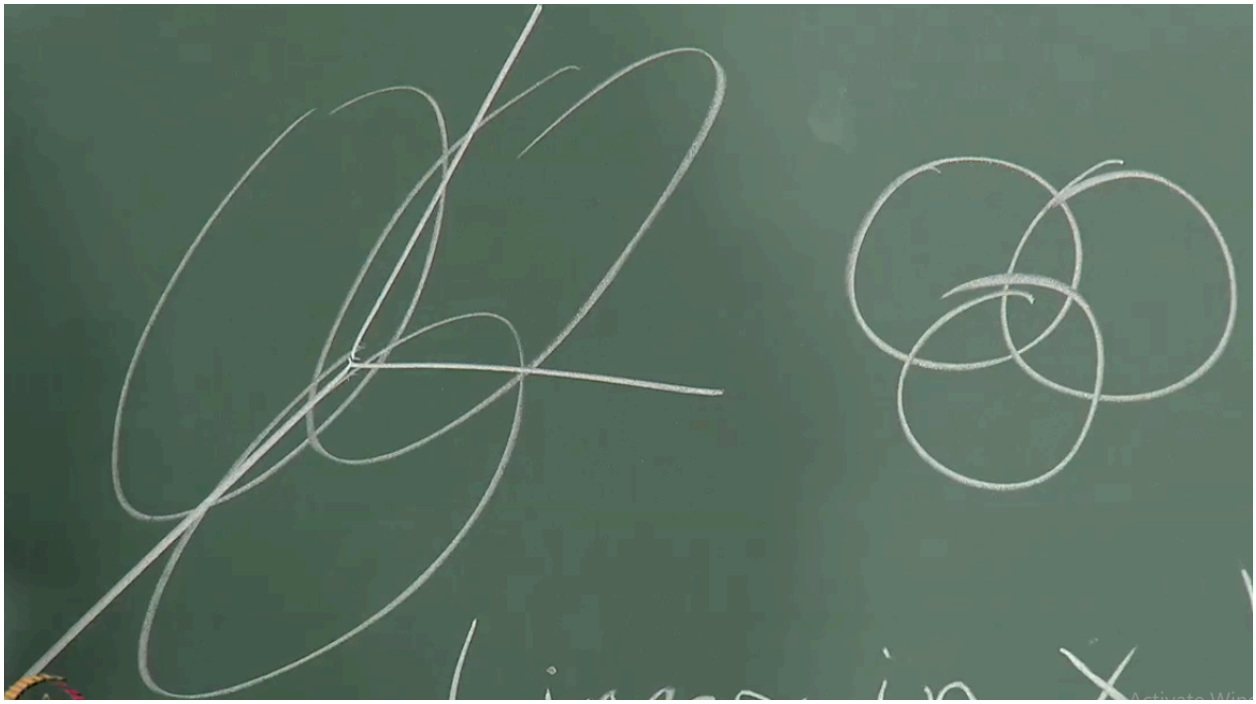

This is something which I just want you to note in fact I can ask you to show that it is fairly straightforward on univariate Gaussians. But if it had been spherical it would have been perpendicular to the line joining the means because I have now this is slanting so the separating hyper plane will also be at an angle to the line joining the means. If you look at many pattern recognition text books they will talk about LDA as a feature selection mechanism .

So you remember we looked at PLS when we did regression. So we did the principal component regression where we said we are looking at the directions in the input only taking into consideration the input so that we are looking at the direction that maximizes the variance in the input and then when we did PLS we took into account the class labels as well.

So the equivalent of that in classification is LDA. So you can think of LDA as actually finding directions along which the variance between the classes is maximized at the same time minimizing the variance within the classes. So PCA just maximizes the variance of the data LDA maximizes the variance between the classes how does it achieve that it tries to find a direction say set the means or mean of the data of each class is a spread apart as possible. So in

this case, suppose this is the mean and I am choosing this where I am using some direction where the mean such as spread apart as possible. So this is essentially the idea behind LDA and we will just take that as our assumption.

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**Introduction of Machine Learning**

**Lecture 23**

**Prof. Balaraman Ravindran**
**Computer Science and Engineering**
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**Linear Discriminant Analysis III**
**- Another view of LDA**

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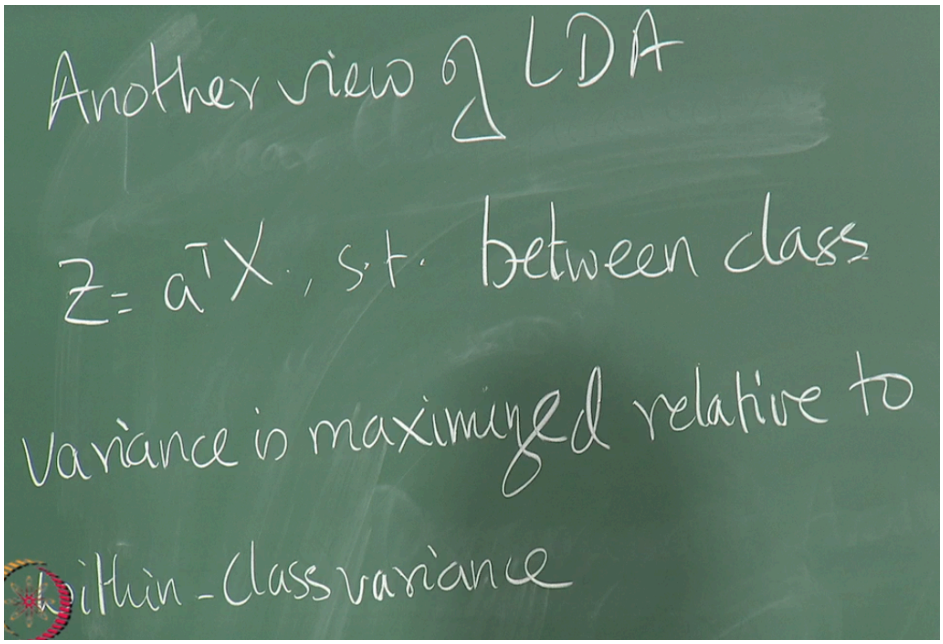


Okay so when I say between class variance, I say it is the variance of the class means. So I will take the classes look at the means of those classes and look at the projected means of those classes and compute the variance among the projected means. Suppose I have "k" classes I can compute the variance among those. If I have two classes, what will this amount to? Maximizing the distance between the projected means and if it is "k" classes, it will be maximizing the variance among the k centers relative to the within class variance and what would be the within class variance? For each class the variance with respect to the class mean and that is what we

already computed there but for each class. So within class variance that essentially what I'm looking at here. So let us just treat the first condition alone. So for simplicity's sake start off with a two class case and then we can think of the generalization to multiple classes. So I am going to have a surface defined by $w^T x$ so $y = w^T x$ if it's greater than some $w_0$ , I am going to classify it as class one just less than some $w_0$ or less than or equal to, I will classify it as class two.

I am going to say $\bar{m}_1$ and $\bar{m}_2$ are the means of $c_1$ and $c_2$ and well we know how to compute $\bar{m}_1$ just like you do a $\hat{\mu}$ there and I am going to assume that when I write the $m_k$ without the bar, this the projected one.

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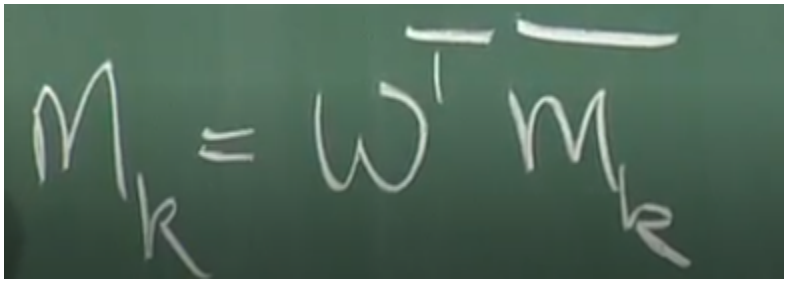


So $\bar{w}\bar{m}_k$ I should see the projection of the mean in the direction $w^T$. So that is essentially what this is. So the reason I am using this funny notation is in the textbook if this is bold it is $m_1$ if it is unbolded it is a projection but I cannot write bold every time on the board.

So I am just using the bar right then when you read the book you can translate back and for this you read this part alone from PRML (Pattern Recognition and Machine Learning) by Bishop the textbook reference is there. The rest you can get from so till that part it is from Hastie Tibshirani Friedman the ESL. So what is my goal when I say I want to maximize between class variance. It is essentially to maximize the quantity $w^T(\bar{m}_2 - \bar{m}_1)$ . $w^T m_2$ is the projection of $m_2$ on w, $w^T m_1$ is a projection of $m_1$ on w. I'm trying to maximize this quantity so that is essentially my first criterion right.

The direction w that maximizes this right so there should be some alarm bells ringing for you what is the problem? If I do not have any bounds on w, I can just arbitrarily scale my w and get larger and larger values. So I will have to have some constraints assuming summation over.

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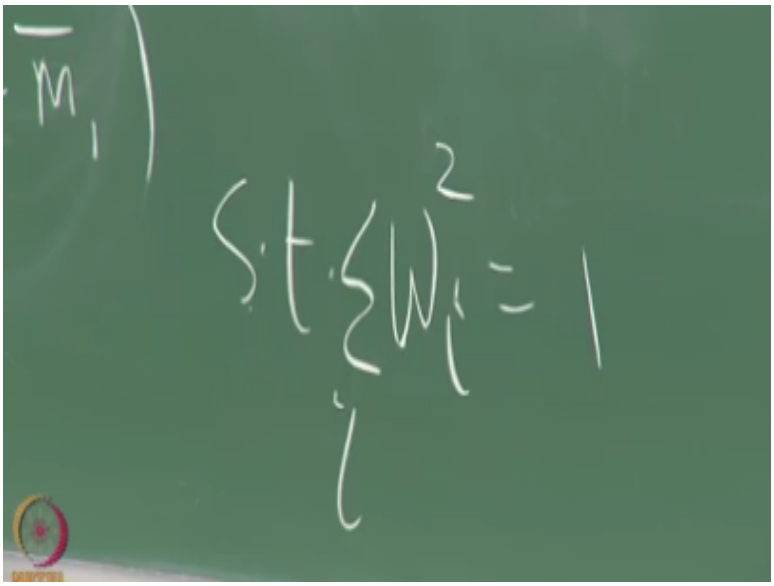


So essentially the norm w is one. That is an assumption we will make frequently to make sure that we do not get unbounded solutions. So this is numerically unbounded.

**Student question:** Why can't we impose an inequality here?

Yeah good question so you could impose a inequality constraint saying that summation w square is less than 1 but what do you think will happen? You are maximizing the value and you can just scale it so essentially what will happen is you will scale it such that $w_i$ hits 1 anyway. So even if you have the lesser than or equal to constraint because you are maximizing over w you will hit it you will essentially scale w till you hit 1. So you might as well leave it as equal to 1 right.

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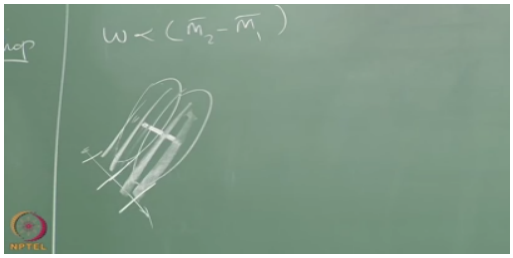

So you can solve this right but the take-home message is that your w is going to be $w \propto (\bar{m}_2 - \bar{m}_1)$ . So w will be proportional and you add that here right you take the derivative 'w' will go and that will become w, so there will be some constants here but essentially you are going to get w will be in the direction of $m_2 - m_1$. So what does this mean? Take the means and if again you can go back and show that if it is spherical then the constant will be half so it will be the midpoint of the line dividing that two means.

So let us do it again so I have two classes and I take the means. So this will be the direction of the projection. I predict everything on to this right this way this will become class one that will become class two. Does it make sense? So in this line and this line are actually parallel to each other I know you really did not want me to repeat the drawing but I think that will help. So I have class one I have class two, so I mean if you look at the data point that comes to me so the people understand when I say class one class two like this do you know the direction what I mean.

So this is the Gaussian corresponding to class one I am drawing the 1σ contour of that this is likewise the 1σ contour of the second Gaussian. So the data point that comes to me could be something like this. This could be the training data that I am getting it will be a mixture up of + and – in this region. That could be minuses here also. Already I drew one that could be minuses here that could be pluses here because the Gaussian still does extend beyond the contour I have drawn, the contour is only the most probable region for the data points to lie does not mean that outside this contour the probability is 0 okay. So this is essentially what it means. So I am going to get data like this and I am going to model it I am modeling the Gaussian by these contours.

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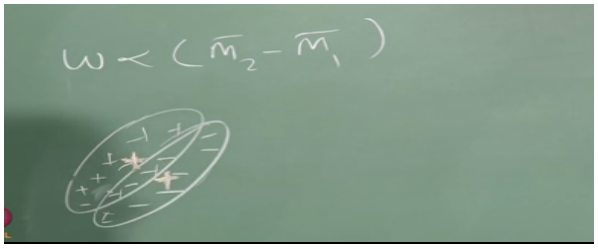


Roughly that these points are the centroids of the data that I get. So what this tells us is that can you join this by a straight line okay and essentially you take direction that is all right like this and project all the data points to that so you will get all the data points lying here now fix up threshold that what that is what I wrote here as $w_0$ pick a threshold such that above that it is class 1 below then it is class 2 right.

In this case in fact if this had been spherical you can show that the threshold would lie in the midpoint now we cannot because well you can I would guess I mean depending under special circumstances but now the point will be somewhere here and all the data points is projected above this I will say it is plus all the data points are projected below this I will say it is minus that makes sense right.

But then this is not what we are looking for. We are missing something important. What is that? The within-class variance. So this is the inter class variance within class variance is what we are missing. So what we will do now start looking at that right.

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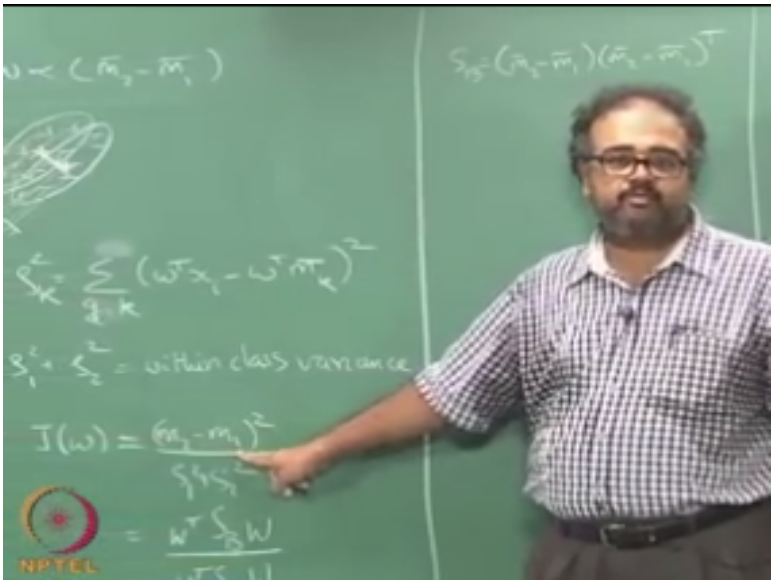


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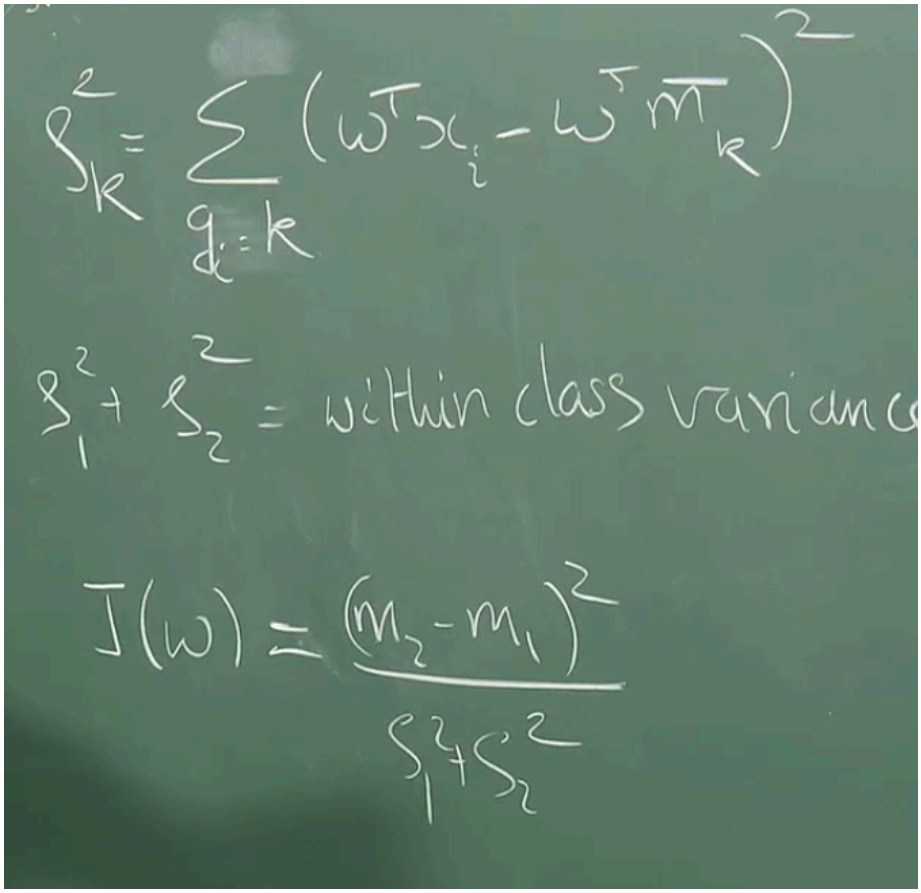

So that is a projected mean these are the projected data points belonging to class one. Keeping in with the terminology are using there so I'm picking on all the data points training data points which had class k and looking at the projected distance from the projected mean. This gives me the total within class variance. Why is there no n in the denominator? Where I'm going to maximize everything at the end. So I am just ignoring the things that do not affect the maximization of that. The squared term $S_k^2$ is essentially the projected data and that is a projected mean and just taking the variance of that is exactly what we did that except that I have not divided by the number of data points okay right.

So this criterion is called the official criterion it is called the "Fisher criterion" after Fisher who was a very famous statistician who came up with LDA several decades ago. So here I am going to do something confusing so I am going to rewrite it so this is the between class covariance matrix. So if you think about it so what I wanted was $m_2$ -$m_1$ what is $m_2$ the projected right so the projected one, so $m_2$ will actually be $w^T \bar{m}_2$ right so essentially I have $w^T \bar{m}_2$ - $w^T \bar{m}_1$ so I can take out the $w^T$ and just have the square of the $\bar{m}_2 - \bar{m}_1$ and I am adding the $w^2$ back in okay by doing $w^T S_B w$ okay. Now what about $S_w$?

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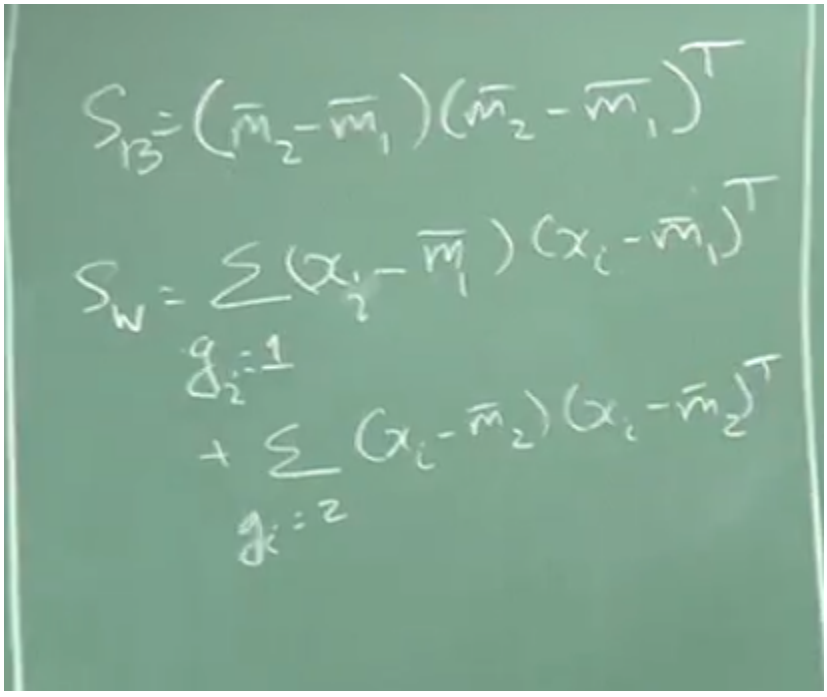

So likewise so I have this as my $S_k^2$ so an $S_1^2 + S_2^2$ is essentially this is this $S_1$ right I had took take out the W from there and this is $S_2$ I take out the W from there so that gives me the $w^T S_w w$. So now what we want to do we want to maximize this. We want to maximize the between class variance relative to the within class variance that is what we said between class variance is maximized relative to the within class variance so that is between class variance is within class variance I have to take the ratio now I am maximizing this ratio.

So differentiate with respect to w and set it equal to zero. So this is what you use u/v method for differentiation. So people want to tell me what the differentiation will be? I will write it but you should recall all of this childhood memories and you should not forget whatever you studied to get in here. Like so the denominator in the thing will become zero because I equated to zero already so when you take the derivative of this you're going to get some term in the denominator.

So that will go to zero so I will just have to equate the two half's in the numerator and I will get this. So just refresh your derivatives the only thing that I am pretty sure putting everybody off is the fact that we are doing all of this in the matrix notation. Just practice it makes life a lot easier do it a couple of times. The best way to do it is try and write it out in matrix form in gory detail okay do the term by term the derivative of it and then look at how it simplifies after you do the derivative right then you will see the pattern and then you will know exactly what we are writing it it's a very simple things like there are quadratics so you should know how to differentiate quadratics that is the only thing that is throwing you off right $w^T w$ is actually a quadratic in w. So that is the only thing so it becomes a linear in w so that is all nothing more to it actually if you think about it $S_B W$, will always be in the direction of $\bar{m}_2 - \bar{m}_1$. You already saw that here when we had only the constraint on $S_B$. So here that the constraint was only on $S_B$ only on the between class variance and when he had the constraint only on the between class variance we ended up finding out that the solution is going to be the direction of $m_2 - m_1$.

And a little bit of work you can show that always that $S_B w$ will be in the direction of $m_2 - m_1$. So I can actually drop that and replace that with a vector proportional to $m_2 - m_1$. So now it makes our life a lot easier right I only have one w left so what about these guys.

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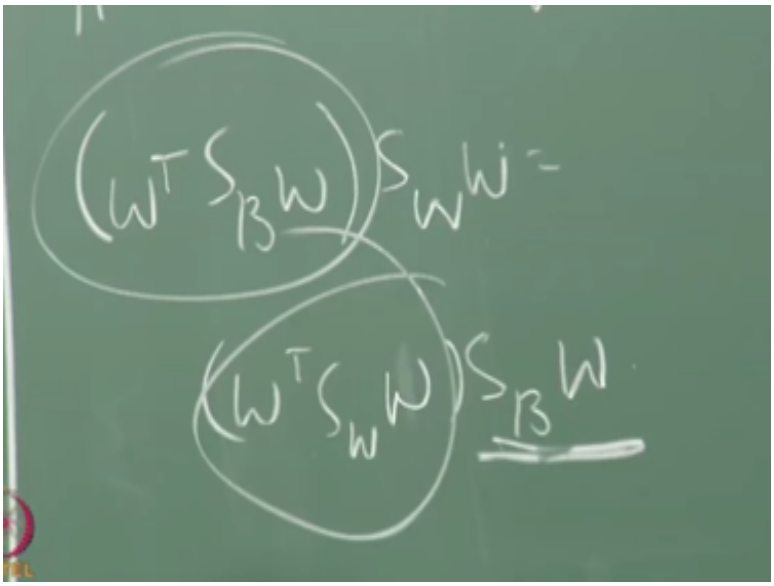


They are all simplified to some kind of scalar quantities right so finally what I will get this w is not equal to but proportional to so that is essentially what I will get so if I did not have the $S_w$ constraint what I got was "w" as proportional time to $m_2$-$m_1$ right. But now if I am taking into account the within class variance also then I will have to pay attention to the within class covariance matrix.

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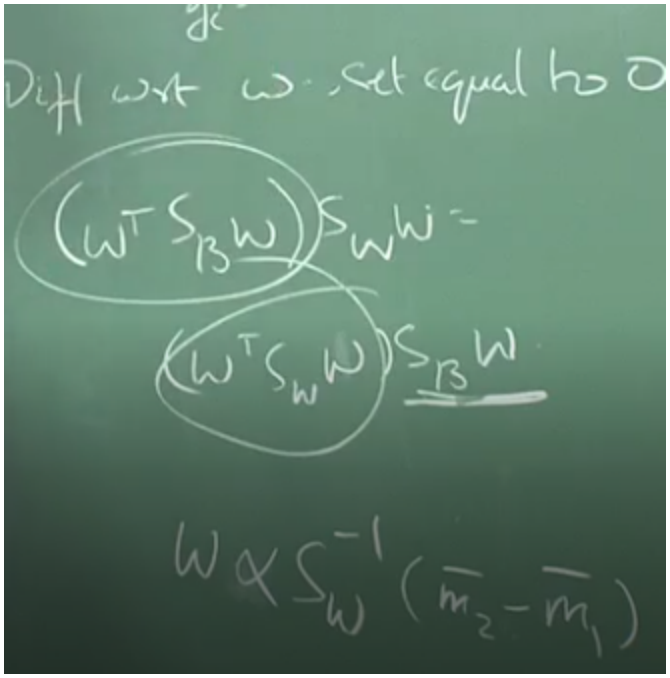


So I will have to pay attention to the within class covariance so that is basically all there is to it. But how does this relate to this? I see any relation between this and that think about it that is basically what we are doing there right. So $\Sigma^{-1}$ is $S_w^{-1}$ just using different notation here. So $S_w^{-1}$ is just taking the variance between the in the data right in the within class variance so $\Sigma$ if you remember is the within class variance matrix right.

So that gives me $\Sigma^{-1}$ here and this how I got $\Sigma^{-1}$ here and then I have $m_2$ - $m_1$ and I have $\mu_k$ - $\mu_l$ here. So essentially for modulo all of these other non X related terms so we are essentially finding the same direction right so whether you do it this way starting with that is your objective function right between class variance and within class variance or you start off by saying that your class condition density is Gaussian and then you are trying to find out the separating hyper plane.

So in both cases you end up with the same direction modulo some scaling factors right, so you can use either motivation for deriving it but what is the nice thing about this motivation we did? It does not make any assumption about the class conditional distribution the Gaussian

assumption is missing here and we worked only with sample means and sample variance and so on so forth.

So it just tells you that LDA does not work only when the distributions are Gaussian. They are fine even when the underlying distribution is not Gaussian that is actually well-defined semantics to doing LDA. People are with me on that so far. So any questions let us let them move on to the next thing what does J(w) represent? So I want to look at the between class variance relative to the within class variance right so the numerator is the between class variance and the denominator is the within class variance so I'm trying to maximize the relative score.

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**Introduction to Machine Learning**

**Lecture 24**

**Prof: Balaraman Ravindran**
**Computer Science and Engineering**
**Indian Institute of Technology Madras**
**Weka Tutorial**

Hello and welcome to this tutorial on Weka. Weka is an open source and freely available software package containing a collection of machine learning algorithms. The algorithms present in Weka are all coded in java and they can be used by calling them from your own java code. However, the algorithm also provides a graphical user interface from which the algorithms can directly be applied to data sets. For the programming assignments in the introduction to machine learning course, we will mostly be using Weka in its GUI form.

This will allow us to spend more time on understanding how the algorithms which we come across in the lectures actually work and how to use them in analyzing data. You can download different versions of Weka for different operating systems from the website of the University of Waikato. This tutorial is mainly aimed at people who have never used Weka before we will look at some of the basic features and options provided by the software and also do some linear regression experiments to help you in solving the questions in the third assignment.

Before we start with Weka let us create a synthetic data set on which we can then apply linear regression. Since this is just for illustration purposes, we will create a simple one dimensional data set. We will create this dataset using a few lines of Python code. So here we have imported the numpy package. The statement creates the input data which ranges from -25 to 100 and consists of 100 data points. Now we will get the output data which will have a linear relation with the input.

Since we are creating a data set here, we know how the input and the output are related. However if this data was given to us, then our objective would be to try to run this relation that is the parameters $\beta_0 = 12$ and $\beta_1=3$. As we will see when this input that is xy pair is provided to the linear regression algorithm. It will be able to learn a perfect model; this is because there is no noise in our data. So to make things a bit more challenging we will add some noise to the output.

The variable z is essentially the noise character output where we have used Gaussian noise with parameters 0 and 3. We will now save this data and apply linear regression on it using Weka. We have saved our data in a text file.

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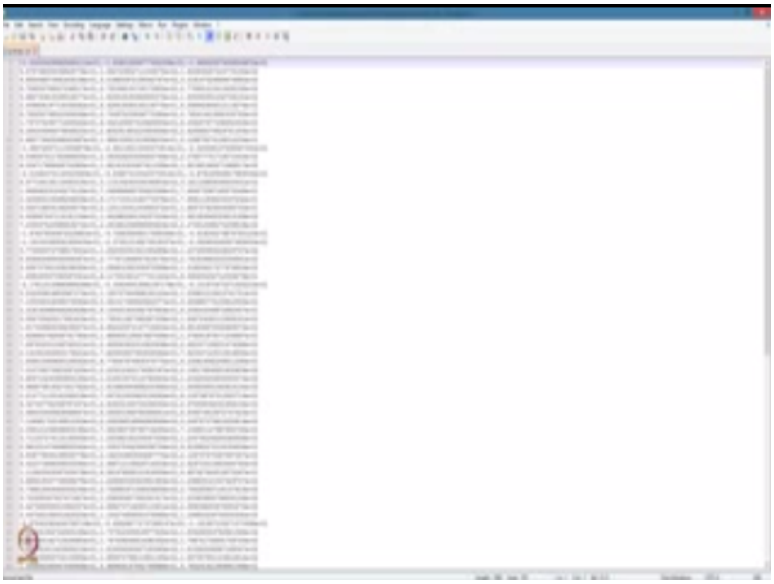

So let us have a look now Weka uses a specific format for its input it this format is known as a ARFF and we have to add a few bits of information to what is essentially a CSV file to make it suitable for use Weka. We essentially have to provide three pieces of information this first one just gives a name to the data. Eventually specifying what relation this data is showing, so since

we have cooked up this data we are just given it the name synthetic. Next we provide the attribute information.

We have used x, y and z as the names of the three columns and specified the data type as numeric. There are other data types which will be seen a little later such as nominal and string data type. The final piece that has to be provided is the data which we have already listed @data specify the start of the data. Now we should save this in the ARFF format that is with the extension .arff. Hopefully this gives you an idea of how to represent data in the ARFF format suitable for use with Weka.

Just to recap you have to provide the relation and attribute information and the data is listed row wise that is each row specifies one data point with the values being separated by commas. We will now open this data in Weka and apply linear regression on it.

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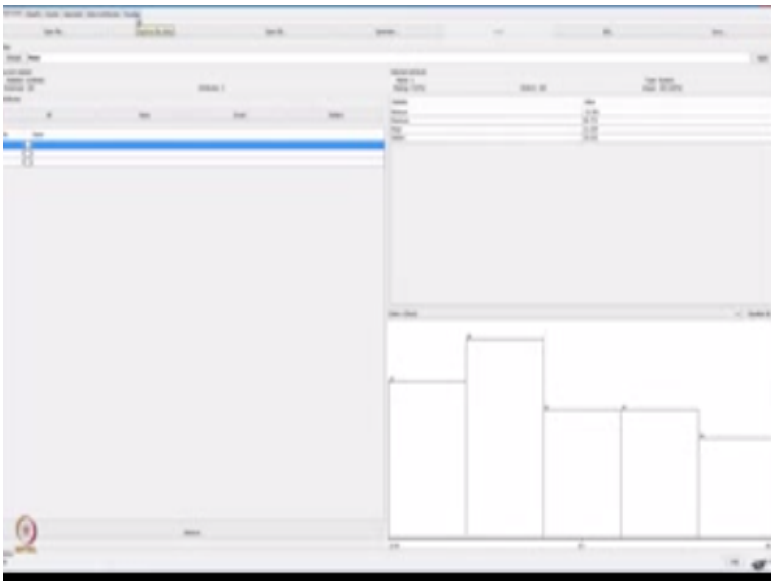

So this is the opening screen for the Weka application. We will be using explorer. So this the start screen for the explorer application. As you can notice most of the options are grayed out this is

because we do not have any data selected yet. So let us do that this is the synthetic data that we had just created, so let us open that. Right a lot of things to notice here. First of all at the top we have these tabs which allow us to specify different actions. So the first tab is preprocess where we can do different preprocessing activities such as normalizing the data, filling in missing values, in case the input as missing values and so on.

Under the classify tab are listed all the supervised learning algorithms, that is both classification and regression algorithms which will be looking at soon. Clustering algorithm was listed on the cluster tab. Association rule mining algorithms under the associate tab in. Under the select attributes tab we can perform attribute selection activities such as is subset selection PCA and soon and finally visualize allows us to visualize the data.

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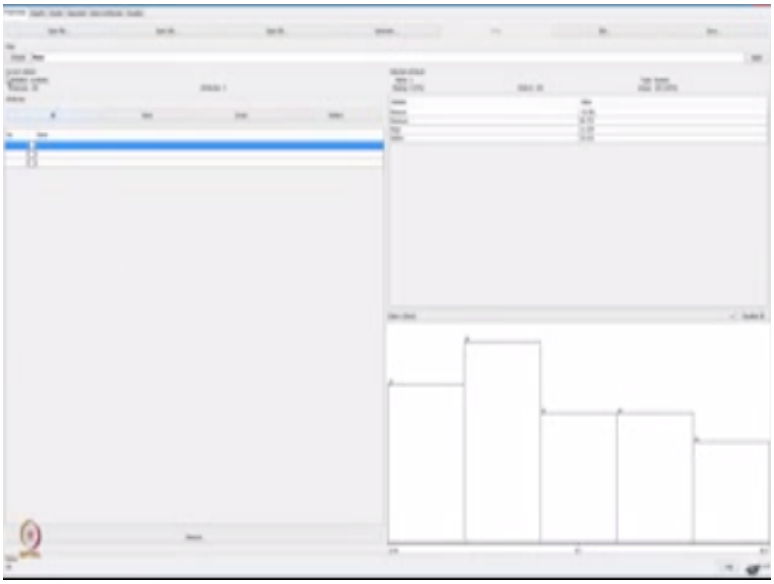

Let us have a look at that here we have the scatter plots between each pair of attributes in the data; this allows us to visualize the distribution of the data. For example, if we look at the scatter plot between y and x, we can observe the perfect linear relation between the two variables. Since that is how we created the data, however if we look at the scatter plot between z and x.

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We can see the effect of adding noise to the output. Getting back to the preprocess tab here, we have the relation information, the name of the relation is synthetic and there are 100 instances and three attributes. The attribute window gives the list out the attributes. We have three attributes x, y and z, on the right-hand side for the selected attribute, we can use some information. So right now the attribute x is selected, so it has 0% missing data there are 100 distinct values with each value being unique. Some basic statistics which has mean maximum mean and standard deviation and at the bottom we have a histogram. Now let us go ahead and apply linear regression on the data. For our first attempt we will use the first two columns that is x and y and remove the noise character output set. So first we come to the classify tab.

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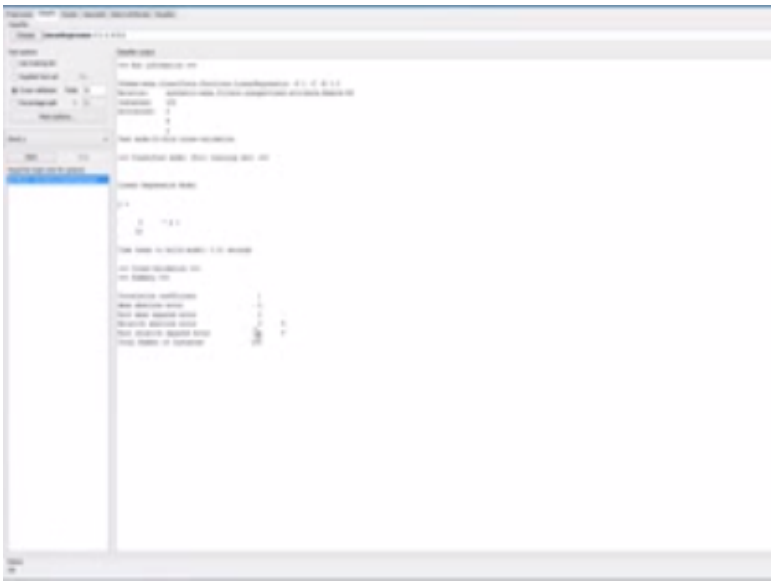

Here we have to choose the algorithm, so which we will choose functions and linear regression. Note that there is a simple linear regression function which is actually suitable for this specific task because we have only one dimensional input but when the dimensionality of the input is more, we need the linear regression function. So let us just use this function these are the default parameters, for the linear regression function. We can change them by clicking here; the first parameter is the attribute selection method that is the method used to eliminate attributes which not contribute to the learning of the model. Since we would like to handle this ourselves, we will select no attribute selection we will not be using the debug mode the third attribute the third parameter is eliminate collinear attributes which essentially allows Weka to identify and remove attributes which have a high correlation. We will set this too false for now. The final parameter is the value of the regularization parameter. Note that we are using Ridge regularization here.

So initially we will take this to zero, that is we will not be using any regularization. Having set the parameters of the linear regression algorithm, we now look at the different evaluation options. The first option is to use the training data that is we use the training data to build a model and then use the same data to evaluate the model. In case we have a separate data set for testing that is a portion of the data which has not been used in training the model. We can supply that here.

More common you will be using cross-validation. In cross-validation, we will iteratively partition the training data into testing and training slits. In each iteration we will train on the training split and evaluate on the testing split. The partition will be done in such a way that each data point will appear in the train and the testing split at least once. The purpose of doing this is to get a robust estimation of the performance of the model. We will be discussing the concept of cross-validation in more detail in upcoming lectures. For now we can go ahead and use this evaluation method. Note that the number of folds simply indicates the number of your iterations and the sizes of each set. We have ten folds which mean we will partition, the data and build models ten times with each partition being a 90:10 split between training and testing. Only the

percentage slit option allows us to split the training data and keep a portion for testing. Next this drop-down box allows us to select the output attribute. That is the attribute which we are trying to predict. With all settings in place we can execute.

The main window displays the results of the execution of the algorithm. Going through the output, we see the linear regression function used with the following parameters. The relation synthetic on which the filter applied is removing of the column 3. There are 100 instances two attributes namely x and y we use 10-fold cross-validation for evaluation and this is the linear regression model that we obtained. Note that we are able to recover the exact parameters values used in constructing the data.

The correlation coefficient is 1. The correlation coefficient specifies the correlation between the predicted output and the actual output. Here it is perfectly correlated and each error term is 0. Now for more meaningful evaluation we will use the noise corrupted output. So we can recover the column which we deleted by selecting the undo button and now we will remove the y column. Going back to classify we will have all the parameters will leave it same and just execute the algorithm. Note that Weka uses the z attribute for prediction. In general, Weka will use the last attribute listed for prediction or classification. Here we observe the effect of noise in the output variable. The parameters estimated parameters differ from the actual parameters, due to the noise. We also observe that the noise the error is no longer zero. Hopefully you will now be a little comfortable with the Weka interface and the steps involved in performing linear regression on the data set. Up to now we have worked with a very simple synthetic data, so next we will apply regression on the more realistic data set.

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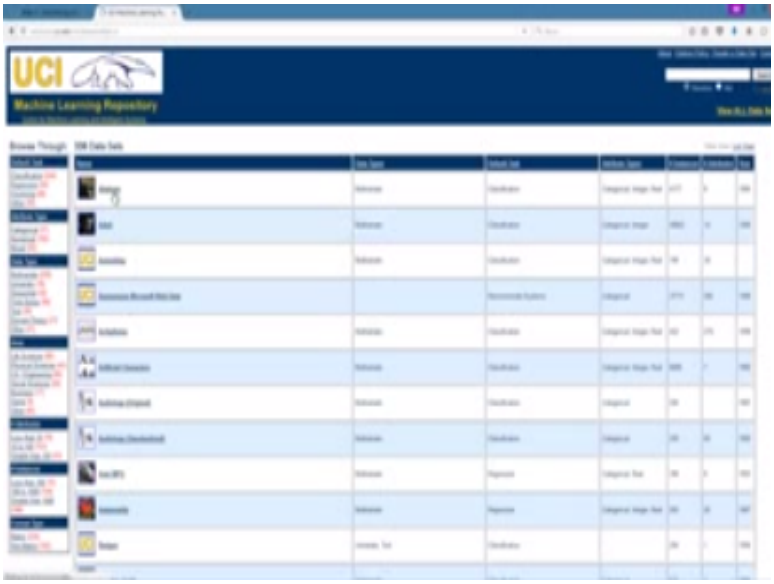


The UCI machine learning repository is home to a number of real-world datasets. Here you can see the different data sets characterized by the primary tasks they support the type of data they contain the area from which the data has been generated and so on. We will use the abalone data set for our next experiment.

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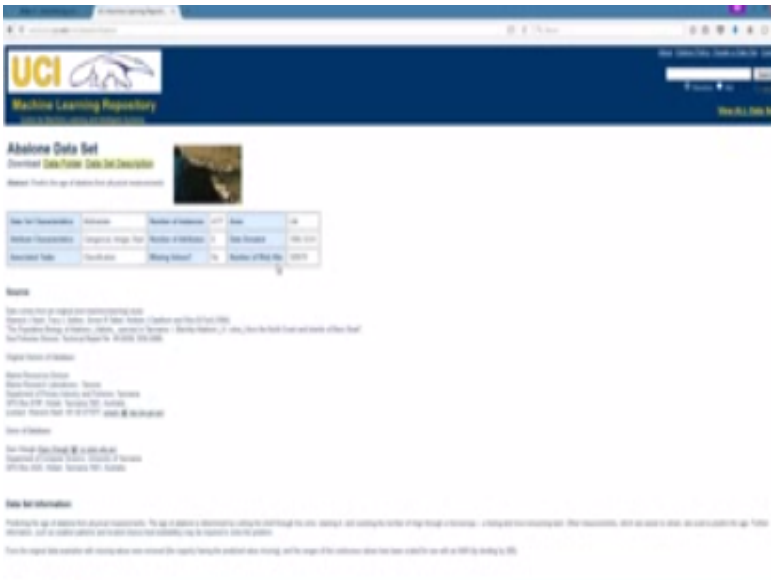

Here you can see the basic information about the data set such as the associated tasks, the number of instances, the number of attributes whether the dataset contains any missing values and so on. The data description page contains more detailed information.

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Most importantly lists out the attributes and their data types. This information will be needed for creating the ARFF input format.

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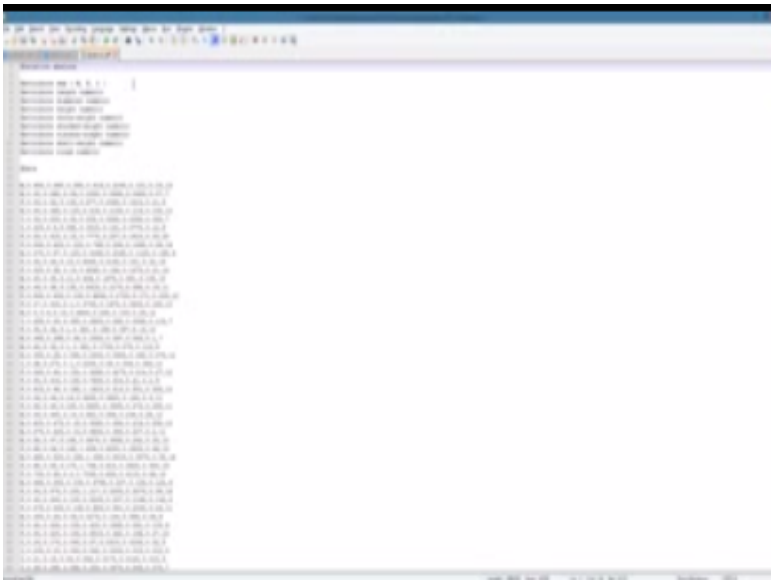

This is the raw data and here we have added the information necessary for the ARFF representation. Note, that this data set contains categorical attributes. For such attributes we specify their data type by listing all the possible values they can take. In this case the first attribute can take one of three possible values (M,F or I). Note also that the last attribute is of integer type ranging from 1 to 29. Here we have specified it as a numeric attribute but in case, we wanted to consider this as the class level for a classification task we would have specified it as a categorical attribute. Let us get back to Weka and apply linear regression on this data set.

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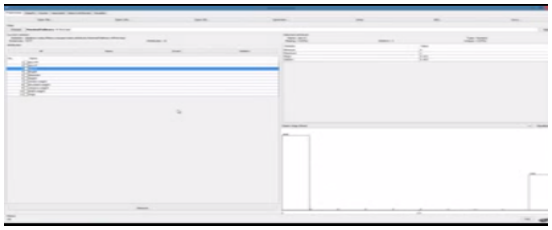

Here we see the nine attributes along with the associated information on the right. The first thing to do is to handle the categorical attribute. Recall from the lectures we learned about one hot encoding this can be done in the preprocess tab using an appropriate filter. First of all we select the attribute then choose the appropriate filter. This filter comes under the unsupervised attribute folder and it is the nominal to binary filter. On applying this filter we observe that from one

attribute we have now created three attributes. One corresponding to each of the possible values that the original attribute would have taken. Also from the histograms, we can see that each of the attribute is now 0 or 1 or that is binary variable, we can now apply linear regression on this data set.

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We will start by using the same parameter values as used previously. We will stick with cross validation for evaluation and try and predict the Rings attribute. Here we observe the results. These are the estimated β parameters and below we see the error measures. The question here is can, we improve this result. One idea is to apply regularization, however before applying regularization, we should normalize the data since ridge regression is not invariant to scale. This can be done in the preprocess tab by selecting the normalization filter. Note that except for the nominal attributes and the output attribute each of the other attributes has been normalized to a range within 0 & 1. Now we can try ridge regression. Let us start with a value of 0.5. However to compare this result we should apply linear regression on the normalized data. So we will run linear regression again with ridge parameter set to zero. Here we have the result of running linear regression with regularization parameter set to zero on the normalized data. Looking at the β parameters as well as the error values we notice some slight changes. Especially if we look at the

β parameters for some of the attributes we see that in the result for rigid regression the parameter values have shrunk this is understandable, since in ridge regression we are adding a penalty term ways to are on the magnitude of the parameters. We can also observe a slight change in the erro terms. By trying out different values of the regularization parameter, we can attempt to improve on the performance. However trying out parameters manually is not feasible for this. We will use a meta learning algorithm called CV parameter selection. Essentially CV parameter selection will take as input a learning algorithm, a parameter of that algorithm and a range of values to try out for that parameter. Let us specify this, so first we select the algorithm which is linear regression. We set its parameters. Notice that the regularization parameter is specified with the letter R. So this will allow us to specify the regularization parameter along with the range. So let us say we want to vary the regulation parameter between 0 and let us say 5 in 50 steps. We stick with the same cross-validation evaluation and execute. The result of the execution of the meta learning algorithm shows the optimal value of the regularization parameter as follows subject to the range constraints provided by us.

The β parameters and the corresponding error measures are shown here. Comparing this result with the results of the previous two executions, where the regularization parameter was set to 0 and 0.5, we see that there are small changes but not nothing drastic. This seems to suggest that regulation does not seem to have much effect on this model or perhaps that we have not found the right range of parameters. In case of the latter we can run the meta learning algorithm again and specify a larger range.

One very useful technique when searching for the optimal parameters for any learning algorithm is to start with a large range and the large step size. This initial step performs a coarse-grained search over the range of parameter values. Next we perform a fine-grained search in the vicinity of the value which gave the best results in the previous step. That is we restrict the range but decrease the step size. We can leave it to you to apply these two-stage parameters search on this data set this concludes the tutorial on Weka. We hope that people encountering Weka for the first time will feel a really comfortable with the basic features of the software. In this tutorial we covered most of the concepts which will be required for the first set of programming assignment

questions. In future assignments we will simply mention the algorithms that need to be used and expect you to apply them on the datasets supplied using Welka. We also encourage you to explore the algorithms provided in the package as and when we cover them and related one in class. For this the UCI machine learning repository is a very good source for data sets that can be used for all kinds of learning experiments.

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**Introduction to Machine Learning**

**Lecture 26**

**Prof. Balaraman Ravindran**
**Computer Science and Engineering**
**Indian Institute of Technology Madras**

**Separating Hyperplane Approaches**
**Perceptron Learning**

So I said there are two ways in which we can build linear classifiers. So one was the discriminant function approach, where by comparing discriminant functions, we decide what the separating hyperplane is. What is other approach? Modeling the separating hyperplane directly.

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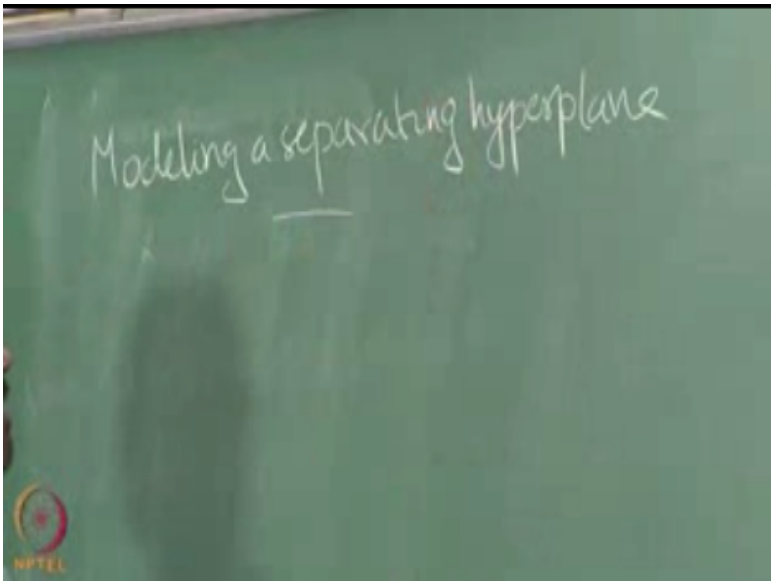


So there are many ways in which this can be done and we look at two in particular one because it leads us into neural networks later and the other because it is the most popular way of building classifiers nowadays.

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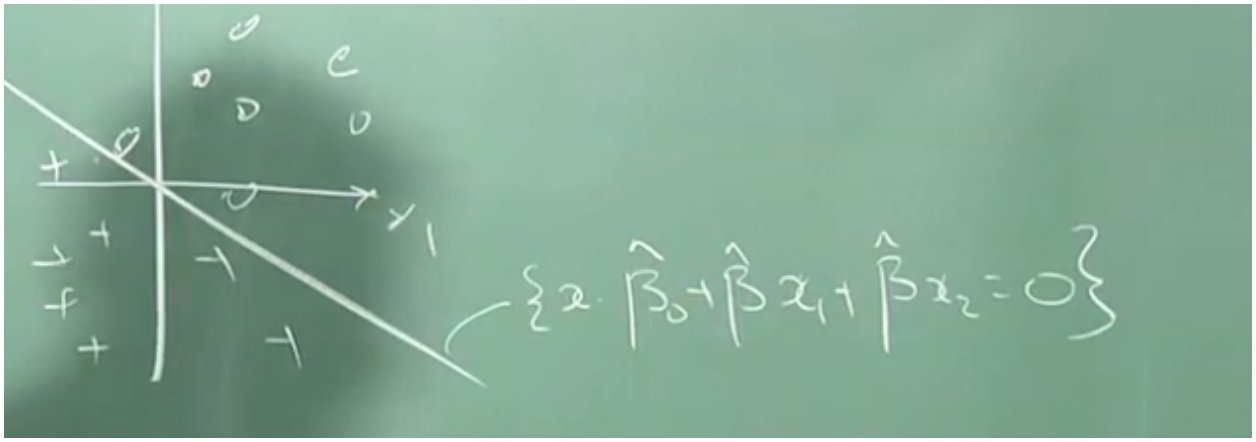


Before we go on let us just have a little fundamentals again. Recall the definition of a separating hyperplane. So if we equate f(x) to 0 that defines our hyperplane and if f(x) is whatever if it is greater than 0 it implies in this case g(x) is +.

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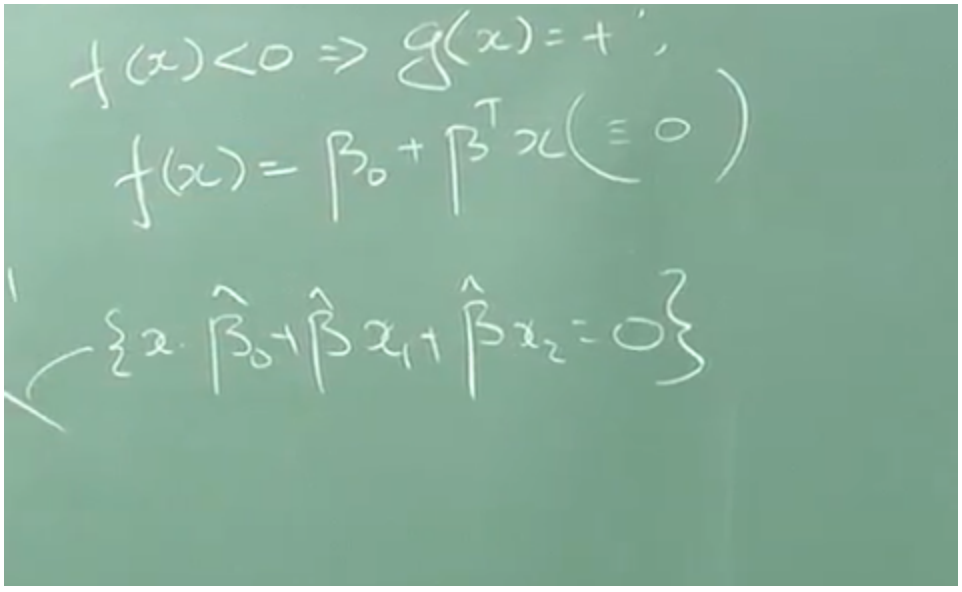


So some properties are listed here. If $x_1$ and $x_2$ both belong to the line and I am going to call this L here if x1 and x2 both belong to L then we know that $\beta^T(x1 - x2)$ will be 0. So what does it mean? The β is actually a normal to L but actually let me rephrase this: β* is normal to L, β is perpendicular to L. β* is a normal is unit direction.

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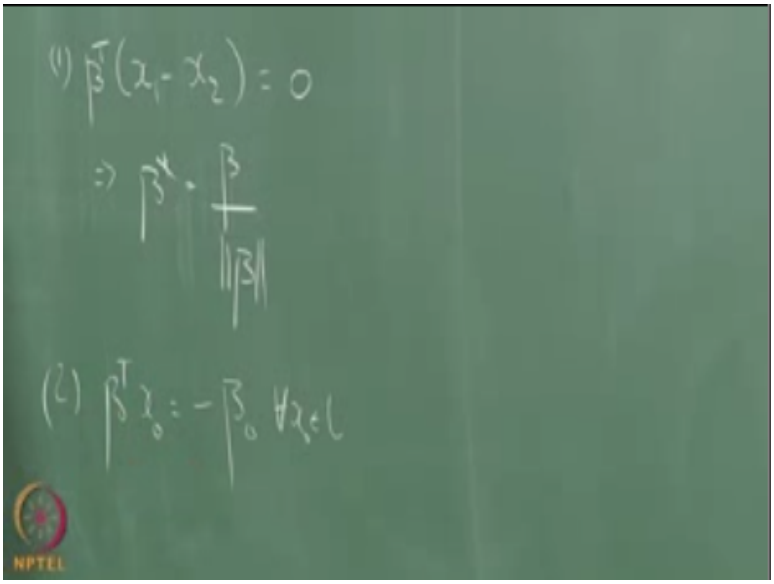


This also we know so $\beta^T x_0$ equal to $-\beta_0$, so I mean that essentially gives you this $1 - \beta_0$ as $+\beta_0$ so it will be 0. This is $\beta$ remember the $\beta_0$ is left out here so $\beta^T x_1$ will be $-\beta_0$ and $\beta^T x_2$ will be $-\beta_0$ again so this expression will become 0. That is why $x_1$ and $x_2$ belong to L. So $\beta^{*T}(x - x_0)$ where $x_0$ belongs to L, x does not belong to L. $\beta_0^*$ is a direction that is perpendicular to L, so this is essentially the distance of x from L. This is actually the signed distance of x from L depending on which side of L it is this is going to be different. But because I am multiplying with $\beta^*$ it gives me the projection on the perpendicular, so this gives me the distance to some x and this will be the direction of $\beta^*$. So this is essentially projecting that on to this direction so this will give me a $\beta^*$. We know is beta by norm $\beta$, so I replace that so I get $\beta^T x + \beta_0$ because $x_0$ times $\beta$ is $-\beta_0$ so I get $+\beta_0$ so and this expression is actually f(x). So this is f(x) divided by norm $\beta$ and $\beta$ is what? $f'(x)$ if we take the derivative of f(x) with respect to x that gives me $\beta$ .

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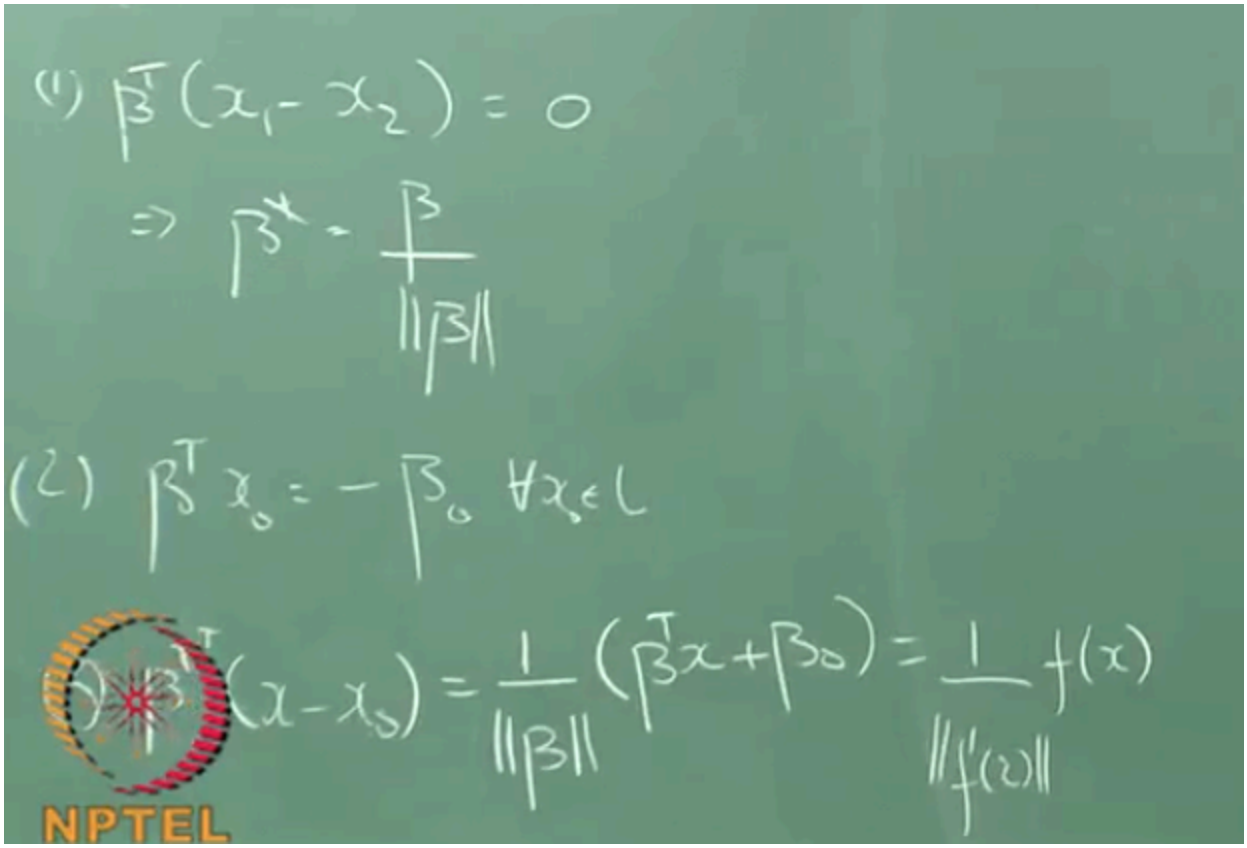


So this is f(x) divided by the norm of f'(x). So what did we say this expression was? The signed distance to the hyperplane. So f(x) gives me a quantity that is proportional to the signed distance to the hyperplane. And if I find a hyperplane such that I normalize my $\beta$ to be one f(x) gives me the signed distance to the hyperplane. It is not proportional it actually gives me the signed distance to the hyperplane. Because whenever I say I am going to minimize f(x) it essentially means I am going to try and minimize the distance of the data point to the hyperplane. All I am going to say were maximizing f(x) I am going to maximize the distance of the data point to the hyperplane, so just to keep that clear I am just introducing these notations beginning.

So the first thing we look at it is a perceptron learning algorithm so it is got a hoary tradition. People are familiar with have heard the term neural networks artificial neural networks. So you know that we are going through the third boom of artificial neural networks, so in the back in the 50s and 60s there was this initial boom of artificial neural networks everybody was so taken up

with artificial neural network they said oh! here is something that can solve the human learning problem.

And we have talked about this already, so that was started by the perceptron learning algorithm and what about the second boom? People know what the second boom was? Now we will come to that later. So the second boom was started by the back propagation algorithm which took care of some of the problems with perceptrons and then that also died away and the third boom has been shattered by what people call deep learning now and the hype is scary but apparently not as scary as it was I mean I recently read some newspaper clippings from the 60s it is really scary.

So the idea behind the perceptron learning algorithm is very simple. So I have this decision boundary and I want to make sure that any point that I actually misclassify is as close as possible to the decision boundary. In fact if I put it on the other side I am happy. I am doing the signed distances, so if I put it on the side I am happy as I have put it on the wrong side and I try to keep it as close to the hyperplane as possible so that early there is some kind of a satisfaction that we are not getting something very egregiously wrong. So I am going to minimize, so we will assume that for data points which is such that $x_i\beta + \beta_0$ is greater than 0 and output as class plus 1 , whenever it is less than 0, I will output the class as – 1.

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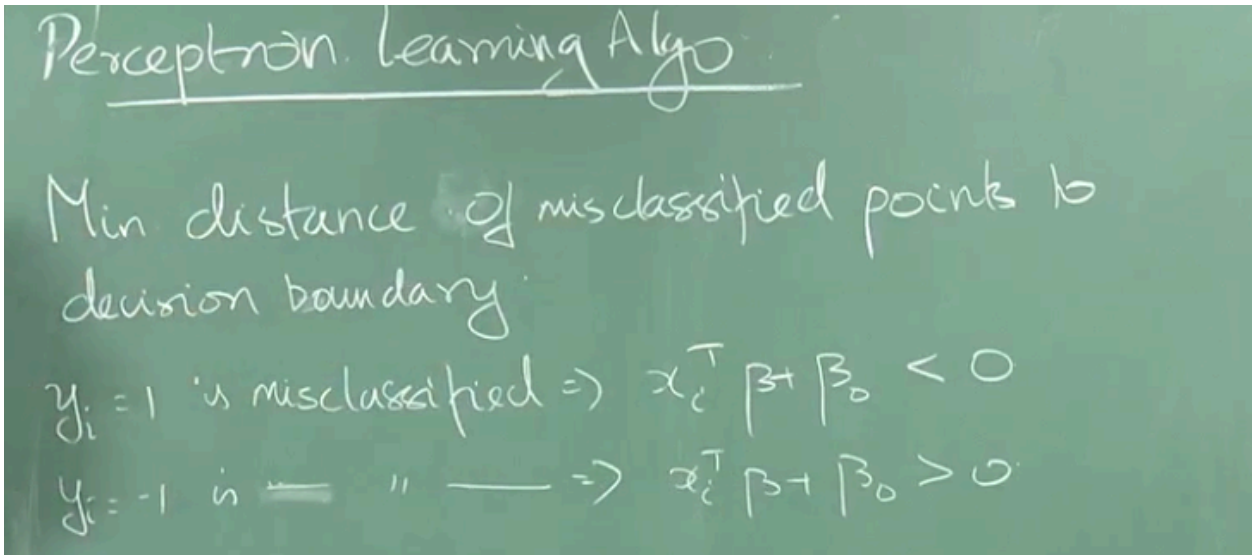

So if the true class level is + 1 but if the f(x) is < 0, I mean if f(x) < 0 that means I will be outputting minus 1 but the true class is + 1 so that means this has been misclassified . Does that make sense? People have understand this point now. So if the true class was +1 but f(x) is < 0 then x will get misclassified the true class was - 1 and f (x) > 0 then x will get misclassified, so these are the two conditions under which you will have a point being misclassified. So if I take this quantity so what it will be for misclassified points it will be negative for misclassified points, it will be positive for correctly classified points. So now take that and I minimize it. So take this for all the misclassified points and I try to minimize this, so D is sometimes called the perceptron criterion of the perceptron objective function. Does it make sense?

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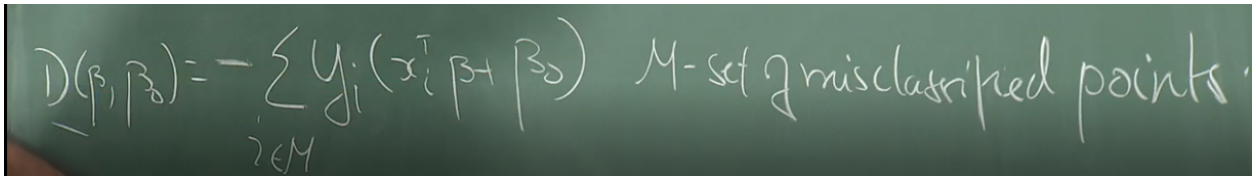


Point here is that if you take the misclassified data points so this is going to be a negative quantity and I want this to be as close to 0 as possible. In fact I want M to be empty. I want M to be empty, so that is the goal. This is the way to minimize to make sure that M is empty as long as M is non-empty I still not achieved the optimum. When can M become empty? So if my data is actually linearly separable. If you remember I drew some data points sampling from Gaussians there and I drew a minus somewhere here and then pluses there and so on so forth. That kind of a data will never be linearly separable. I can never draw a straight line separating those data points so here things are nicely linearly separable. All the x's are one side all the zeros are on the other side. Here the data was nicely separable so the perceptron objective function works well if the data is linearly separable if it is not linearly separable,

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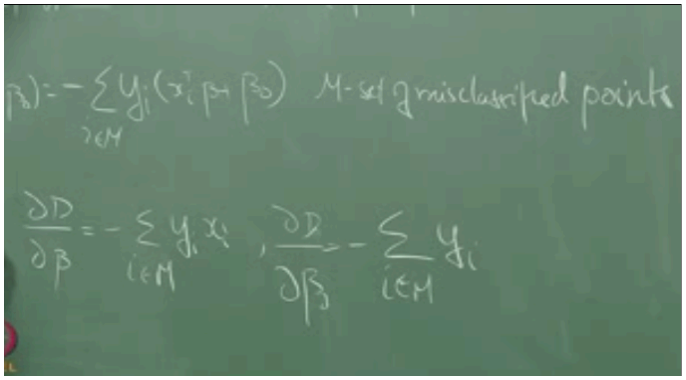


we will run into problems. So now what we do is just use gradient descent. So I will differentiate D wrt $\beta$. What will that be? So this is the derivatives and then so what I do now is essentially the technique that is very popular nowadays but it was not really called by that name in the olden days it is called stochastic gradient descent.

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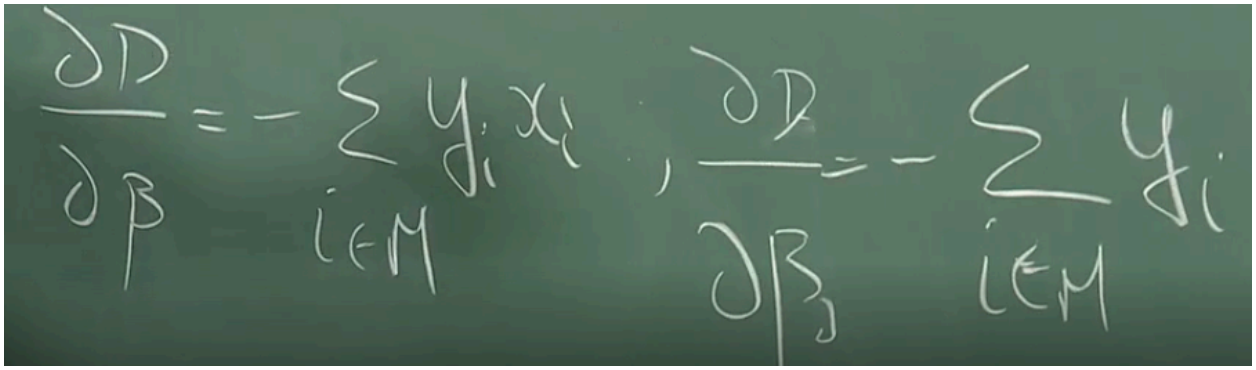

So people know what gradient descent is. What is gradient descent? Usually find the direction of steepest ascent and go in the opposite direction. How far do you go in the opposite direction? Proportional to the gradient. Gradient is very large in what you do? So it depends , so if you know the gradient properly I can compute the gradient set it equal to zero solve for it. And then just jump there. I do not have to go in small steps if I actually know where the gradient becomes zero I can just set my solution to that point.

So the problem comes when things are little iffy why as soon as I move my $\beta$ in some direction what will happen in the perceptron case? If I move my β say in the direction of the

gradient what will happen? Set M changes. So immediately my definition of gradient changes. So even if I change β slightly I have to recompute the gradient. So the gradient I am computing is valid for wherever I am sitting in the β space when I move from there then I have to re compute the gradient. So I cannot just move in the direction all the way. I can't really move a long distance pointed to by that particular point in space or a particular gradient. So I have to iteratively re compute the gradient so in such cases what we do is usually take a step in the direction indicated with a gradient and hope that you are moving in the generally expected direction. So if you take the expectation of the gradient you are moving in the right direction. So the stochastic gradient descent has a lot of nuances and other things to it so we will not get into that whenever we used to stochastic gradient descent I will point out to you what are the things to be concerned about but will not get into the details of that. Possibly anyone doing the optimization course and look yeah you might I am not sure whether they are going to get to stochastic gradient descent but they might covered it in the course.

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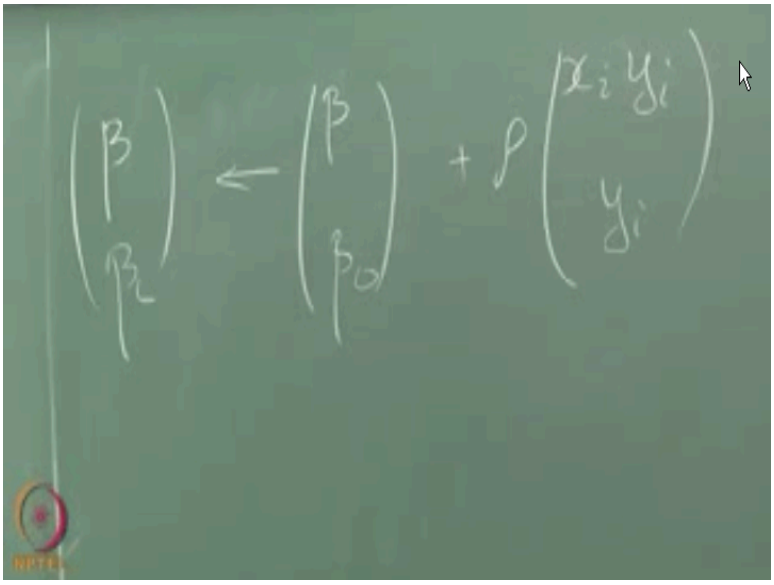

So what we are going to do is essentially take β so the gradient is $x_i y_i$ but since I am anyway doing stochastic gradient, as soon as I find one misclassified data point I am going to find the estimate of the gradient and move in the direction. I am not going to wait till I find all the misclassified data points under that particular β for the current β, I find one misclassified data

point and then I will change my β in the direction of the gradient. So what does it look like? So I just take the misclassified data point multiplied by the decided output and add it to the weight vector. I have written this in a really funny spacing that is because what do you think should go here now? I can convert this into matrix notation. So every time I encounter a misclassified data point I change my $\beta$ . So we are doing it in this particular form because this is the perceptron learning algorithm. So if you want to find all the misclassified data points and then compute the cumulative gradient and then change your β within that direction you are welcome to do that and that is a perfectly valid way of doing it. The reason we are doing it once to one data point at a time is saying this is exactly how the perceptron learning algorithm was derived. So the $\rho$ has to be really small if you are operating in a very, very stochastic environment. Because so in effect what you are doing when you are doing stochastic gradient descent is that you are making many, many different estimates of the gradient in a local region and you are trying to move in the expected direction of the average direction of the gradient. So if the $\rho$ is very large it turns out that you just make one estimate of the gradient and then move out of the region. So you will take a large step so we will go somewhere else the gradient will be completely different or this could have been the wrong estimate for the gradient direction. So to make sure that you are following a reasonable expected gradient $\rho$ has to be really small. Having said that in the perceptron learning algorithm setting $\rho$ to one actually works and that is how the original perceptron algorithm was also stated $\rho$ was one.

And normally for stochastic gradient descent one is very very large. You typically think of the order of $10^{-3}$ to $10^{-4}$ and things like that for step sizes but it turns out in this case it is fine $\rho$ equal to 1 is fine and it will work it will converge. You can show convergence with ρ equal to 1 but if in typically in stochastic gradient descent you want ρ to be small because the idea is what I told you just like we talked about taking a replacing expectation with an average over a region and into nearest neighbors. So something like that at a very, very gross level so you want to be able to take an average of the gradient in the local region but if you move very fast then you might be in a completely wrong direction. So therefore you go slowly.

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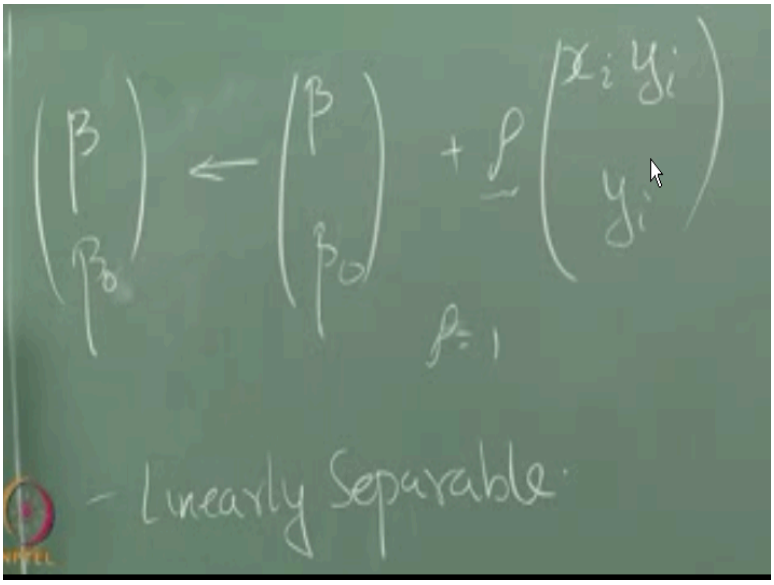


So what you mean by linearly separable. That there exists a linear separating hyper plane that will make no mistakes. So if the data is linearly separable then the perceptron algorithm will converge to some solution. So what do I mean by some solution? So let us say these are the data points that are given to you.

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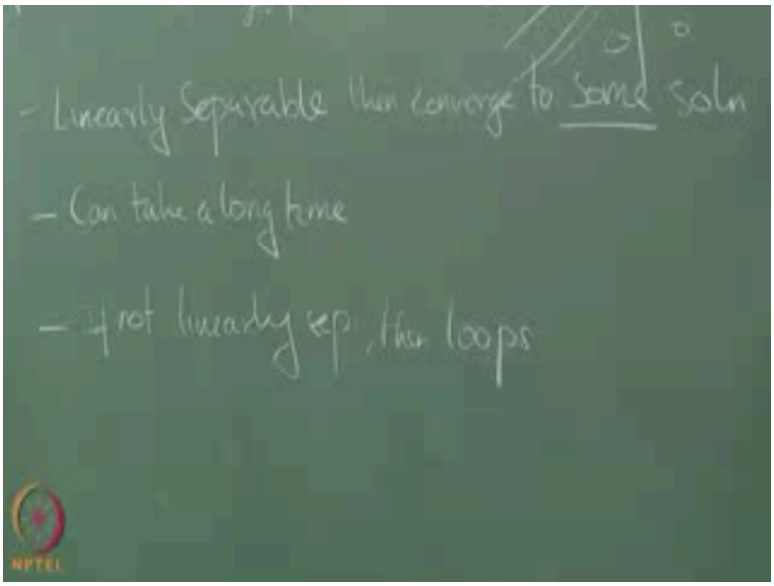


There are many, many solutions that work. I mean assuming all of those are straight lines so you have many, many different straight lines that you can draw that separates these two classes. Infinite number of them actually. So the perceptron learning algorithm will converge to one of these three or one of these many infinitely many number of separating hyper planes which one would it converge to depends on the starting point. While it is hard to say a priori, but given a starting point can you tell me what it is going to converge to? You can just have to run the perceptron algorithm what is the problem. We will come to that. That is a way of defining one of these infinite number of hyper planes as the most desired hyper plane. So if you think there is a way to pick one of these yes we will come to that the second problem is it can take a long time and take a longtime especially if the gap between the two classes is very small and if the gap between the two classes is very small then it can take a long time. So partly it is a function of $\rho$ to 1 but then it is hard I mean so see setting $\rho$ to 1 essentially makes your thing oscillate a little bit. So you will have something that makes a mistake here then you will go back you will make a mistake there and then you have to keep going back and forth multiple times before you converge to the answer. But then setting $\rho$ to something small will also make you move only small steps at a time so that also may take a long time so there is always a trade-off between how large you make $\rho$ and how small you make $\rho$. And so one way of fixing this is what $\rho$ is one

thing but can increase the gap between the data points using transformations. Use basis expansions that we talked about earlier so instead of doing it in the original space say do it in the $x^2$ space or three x space or whatever I mean you can think of some way of scaling the data or transforming the data. So that the gap between the classes widen and therefore it converges quicker.

But the third problem is the harder one. The data is not linearly separable what will happen to the perceptron algorithm? There will be no hyper plane where M is empty so as long as M is not empty I am going to keep adding something to the β again and again as long as M is not empty I will keep adding it up was every iteration I will be changing the β it is not linearly separable. Then it enters then it loops basically then it loops but the problem is sometimes the loops can be very, very large and if you have very large data sets it might actually be setting up a very large loop so it may not be easy to detect also. So normally it takes a long time to converge that in that if it is looping so you have no way you do not know if that the algorithm is taking a long time to converge or whether it is simply looping so how could you indirect loops.

So do you think that every time it has to be monotonically decreasing. If it is going to converge then yes. So if there is no guarantee that cardinality of M will monotonically decrease also, so it is not very easy its cardinality of M is not the only thing that tell you when it will converge. So it is yeah has no efficient way to get around that.

So people discovered a lot of drawbacks to perceptrons and the biggest drawback was they discovered that some very, very simple problems that you want to solve are not linearly separable. So XOR is not linearly separable. I simply cannot even solve simple problems like XOR on what a useless thing is I do not care how well you make the baby speak you cannot solve XOR. So do people remember when I talked about the example how they trained a perceptron to reproduce speech and as the training progress it just sounded like a baby learning to speak and people just went gaga over it. Any way so we'll stop here it looks like I am actually out of time so the next class we will try to explore a way of fixing this. So what is the problem

there? Not that it is linearly separable but the fact that it could converge to any solution that is linearly separable we start by trying to define a single optimal solution.

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**Introduction to Machine Learning**

**Lecture 27**

**Prof. Balaraman Ravindran**
**Computer Science and Engineering**
**Indian Institute of Technology Madras**

**Support vector machine 1**
**Formulation**

We are looking at linear classification and I will quickly remind you of the properties of hyper planes that we wrote down in the last class.

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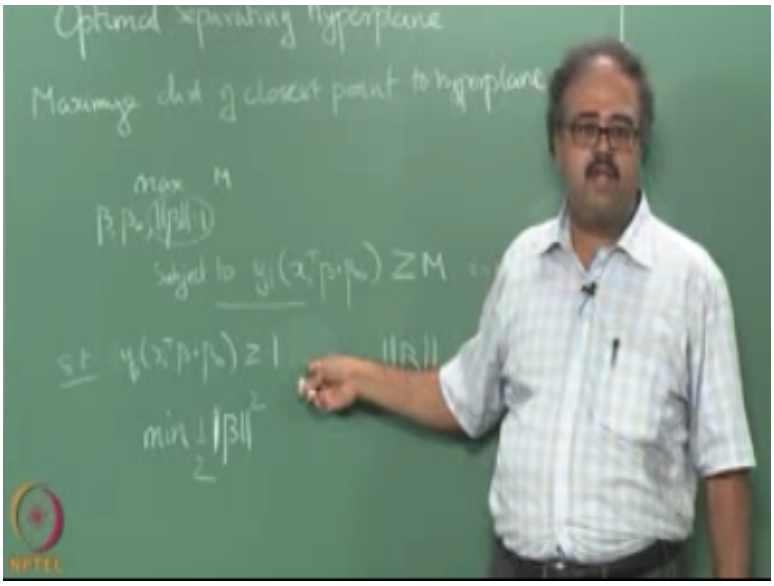
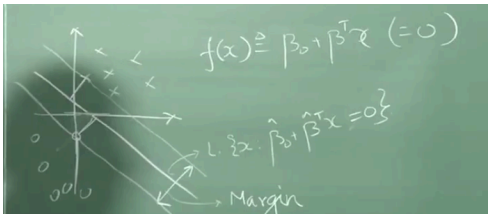



Ok we will denote that by f(x). So the hyper plane is essentially given by solving the f(x)=0 and what I really want is, so I do not care about the other properties what I really want is for you to recall that the signed distance to the hyper plane is given by f(x). So then we looked at the

perception learning algorithm and do you recall what the problem with the perception learning algorithm was? Convergence. So if it is linearly separable it will converge but it might converge slowly if it is not linearly separable it will cycle. But if it is linearly separable what can you say about the convergence apart from the fact it is slow? It depends on the starting point and it is not definite as to where it will actually converge. There is no particular solution to which it will converge so now what were going to try and do is try to characterize a specific optimal solution. We will first start by considering the case of linearly separable data. So just like whatever I have drawn here so the data point is given to you is actually perfectly separable by a hyperplane. So we will start with this case now I am going to try and characterize what I mean by an optimal separating hyper plane. Give me some options for what could be optimal. The modulo sum of distances is maximum between the points and separating hyperplane . Sum of the distance of all the data points to separating hyperplane or maybe this point is that is closest. Exactly so that makes a lot of sense so that is exactly what we are going to use so we are going to maximize distance of closest point to the hyper plane. So essentially so if you think of this data so that is close but this is closer so if I want to maximize the distance of the closest point what should I do? Move it like that or like that whatever some are moved further away from this so how much further away can I move it? Until I reach the other side. Also I say the closest point from both sides should be at the same distance. For both classes the closest point should be at the same distance from the hyper plane and I have to choose an appropriate orientation for the hyper plane so that this distance is maximized. So that is essentially what we are going to try and do so instead of erasing the hyper plane and redrawing it again I just move the data point so that this is closer now. I'm essentially going to have a slab in some sense around the separating hyper plane which will have no points.

So on the thickness of the slab will be the same on either side of the hyper plane. So that is essentially what I am looking for and this is called the margin. Whatever you cleaned up around the hyper plane is called the margin so that is why these kinds of optimal hyper plane classifiers or sometimes known as max margin classifiers, because they are trying to maximize the margin is the distance of the closest point to the hyper plane is the margin.

(Refer to slide time 07.06) 


So in fact so the margin would be that. So we know what $x_i^T(\beta + \beta_0)$ is the signed distance from the hyper plane so I multiply by $y_i$ so that I always get it positive. So essentially what I am saying is I look at the quantity this is essentially the distance a data point is away from the hyper plane and I am saying that every data point has to be at least M away from the hyper plane that is my constraint. So go through all my training data points 1 to n and I am saying that every data point should be at least a distance M away from the hyper plane and under that condition maximize M. So I cannot make M arbitrarily large because I might not be able to find a $\beta$ which will satisfy for every data point. So this is how I will write down the optimization function what we wanted was maximize the distance of the closest point to the hyper plane.

So now what I am going to is that say every point should be at least M away. Now maximize M. This will automatically maximize the distance of the closest point. So what do you think will be the distance of the closest point? Whatever M you end up with so whatever is optimal M for this will be the distance of the closest point and that will be the margin. So that is essentially what you are doing is that you are directly maximizing the margin.

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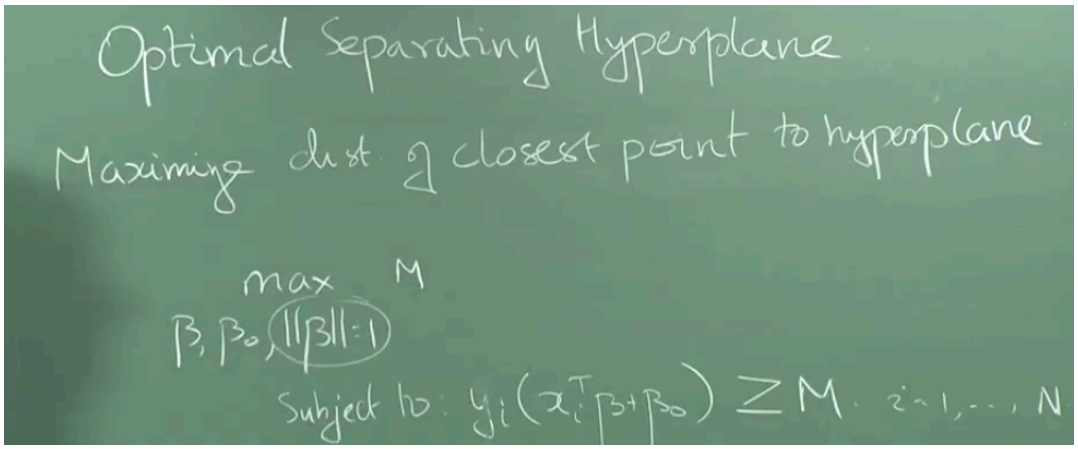
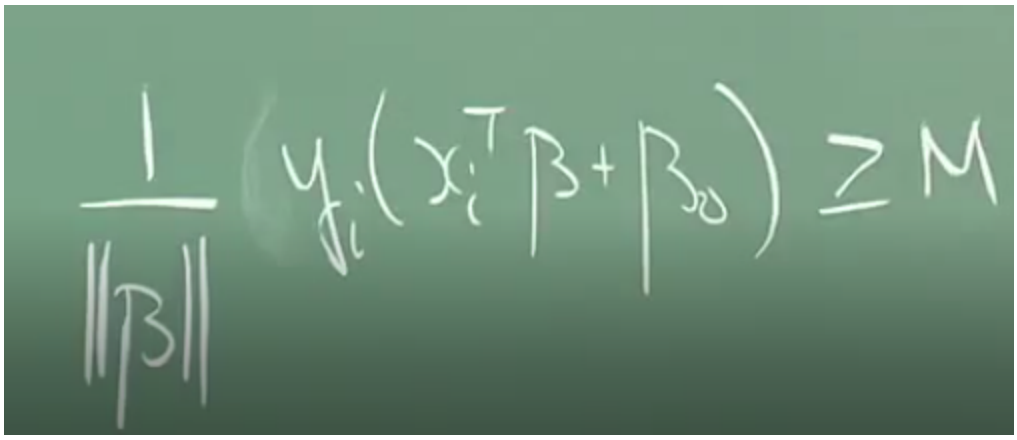
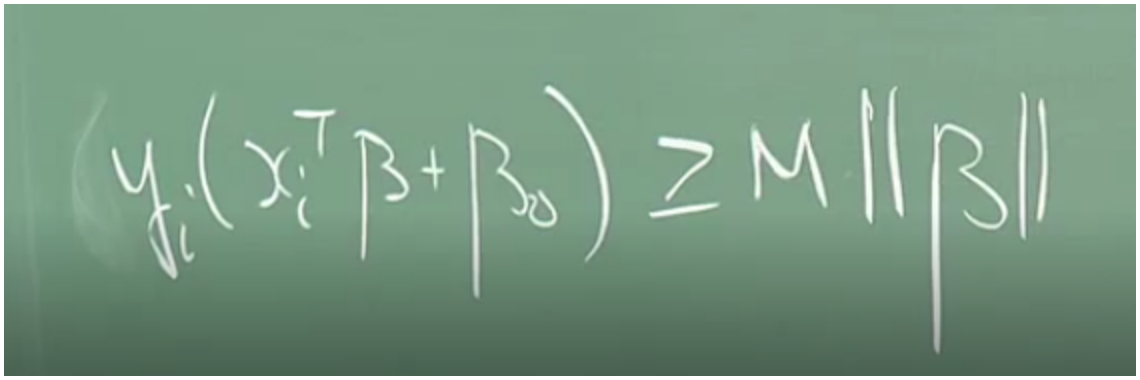


So we have this constraint that ||β||=1 because we do not want the solution to blow up arbitrarily. So instead of that we will get rid of that by that. If I didn't have the constraint of ||β||=1, I can arbitrarily make β large here and make things larger than M. So now I am normalizing by β so that I do not have to worry about that. I can remove this constraint here that instead of that I put the ||β|| here in the denominator in the constraint. It make sense but then I can do something more interesting now can I do that great now let us step back and think about it for a minute so if something satisfies this constraint .




I can arbitrarily scale it. So it will still satisfy the constraint if you think about it. Because I have ||β|| on this side so if β already satisfies this constraint, I can just scale the β and then that will satisfy the constraint as well. So I can arbitrarily set and I still have to find out the direction of orientation of the β. I'm just saying that whatever orientation of the β you pick you normalize it so that the ||β|| is 1/M.

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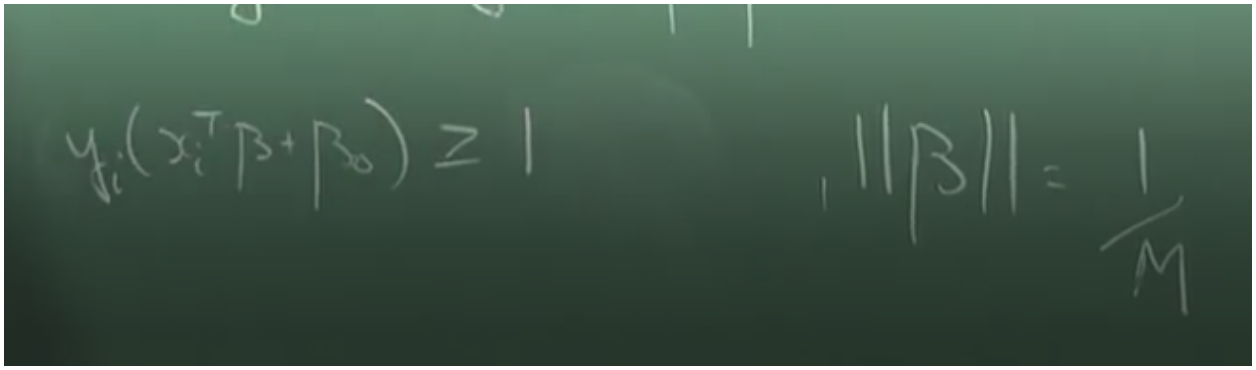


So I started with this constraint started with that optimization problem made the assumption that well ||β|| = 1 / M and I came up with this problem with just a little algebraic manipulation in fact it is geometric as well. I am just not drawing the geometry here. But then the objective function also now can change. It is maximizing M again minimize norm β because now β is 1 by M.
Now we do this instead

$$\min \frac{1}{2} || \beta ||^2$$

So I can do that subject to the constraints here. I am going to do all of that so that it makes it easy for me to take derivatives minimize and things like that so I just made it a squared function so that can manipulate it more easily does not matter minimizing ||β|| is the same as minimizing β square so norms or anyway positive or non-negative.

Now I can go and say that this margin is actually going to be 1/||β||. So another way of thinking about it is what I am trying to find is a minimum norm solution such that all the data points are what are correctly classified. So what does this mean when $yi(x_i^T\beta+ \beta_0) >=1$, that essentially means that $x_i$ is in the side of the hyper plane remember that so $y_i$ is + 1 then this is also positive I mean at least + 1. So not only it should this be positive, you will get a 1 here and similarly if $y_i$ is -1 which is the other side of the hyper plane so this product this term has to be at least -1 so that you will get a +1 it will be greater than or equal to + 1 so you know that all the data points are correctly classified correct and minimizing β essentially is finding smallest possible β. So finding the smallest both β set data points are correctly classified and not only correctly classified, they should be at least certain distance away from the hyper plane.

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**Introduction to Machine Learning**

**Lecture 28**

**Prof. Balaraman Ravindran**
**Computer Science and Engineering**
**Indian Institute of Technology Madras**

**Support Vector Machines II**
**Interpretation & Analysis**

So this is the optimization problem, which is actually a simple optimization problem, since it is a quadratic objective and a set of linear constraints. We already saw how to solve this. You guys had a convex optimization tutorial. So one of the things that we are loing from the convex optimization tutorial is that you will know how to solve this problem. So what we do after this is write the Lagrangian .

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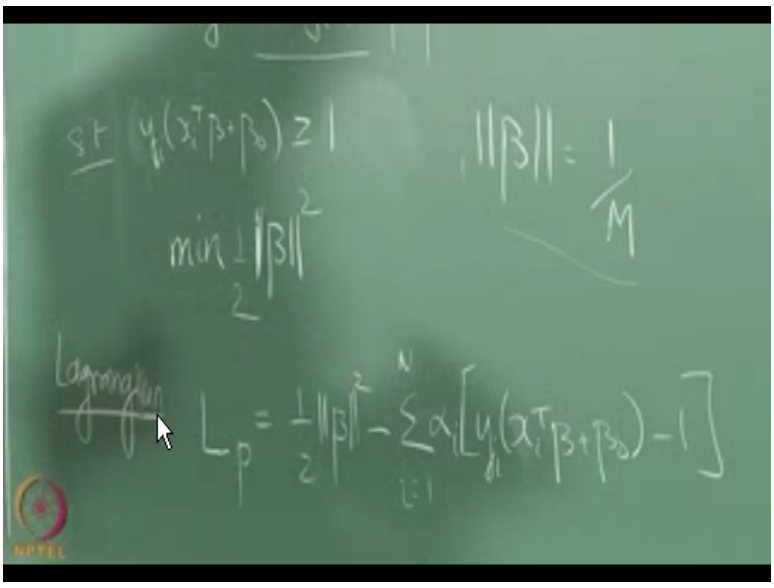

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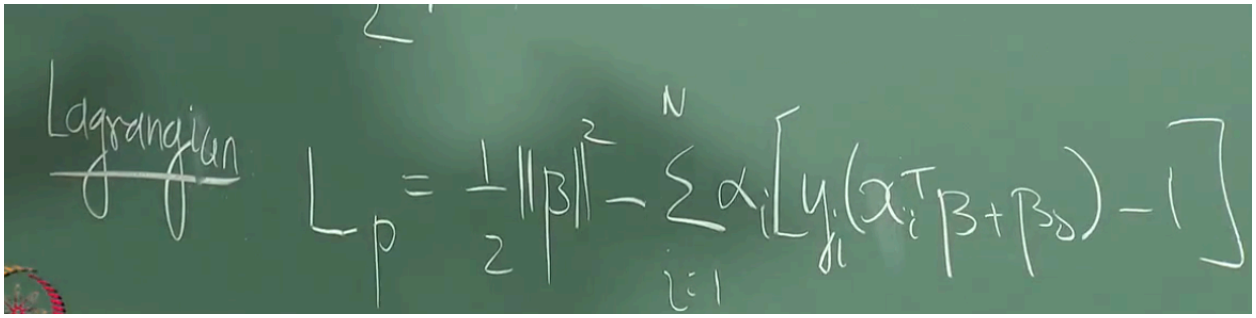


I have to apply this for every data point so i=1 to n . And I put a p there so there is a primal so we will have to form the dual of this and the dual is a lot easier to solve. So we will go ahead and do the dual. So for first I will take the derivatives. So derivative with respect to β and you can do that and solve it we will get that derivative with respect to $\beta_0$. So now I'm going to do some hand waving but you can go through this computation so take that substitute into this and do a lot of simplifications. So remember we have this β squared here therefore I am going to get a $\alpha_i\alpha_j y_i y_j$ kind of terms.

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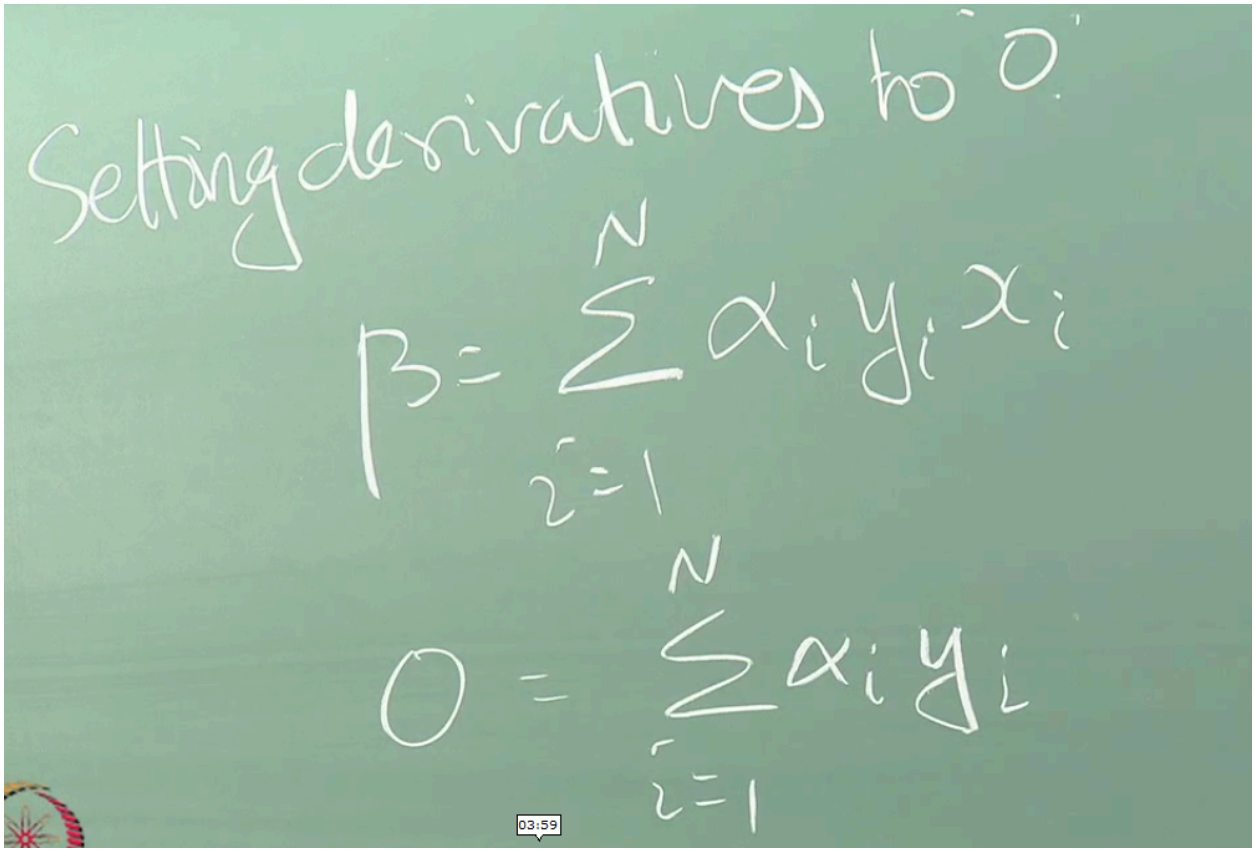


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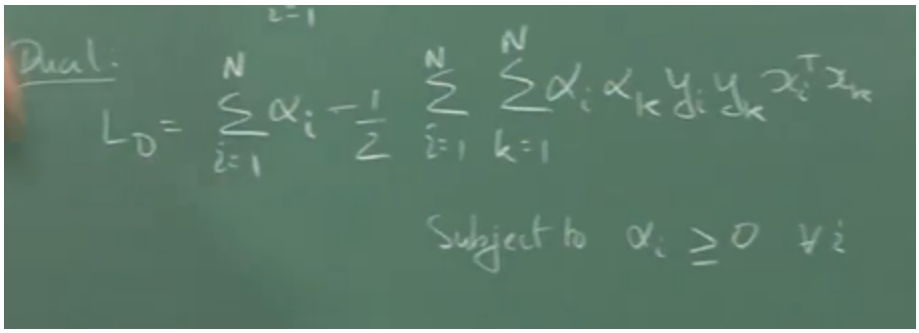


So the dual is going to be a slightly simpler form. Why is it a slightly simpler form? So my constraints have become of lot simpler here. It just going to be $\alpha_i$'s should be non-negative. Also it turns out that there are efficient ways of solving optimization problems of this form. You do not have to worry about it. Here are lots of packages that solve SVMs for you.

But then you just need to know what kind of optimization problem we are solving. I do not want you to use it as a black box. Essentially what you are going to be solving is this. So when you

have a solution, when you have something that is both primal and dual so we can actually show that the duality gap is 0 in this case so it is not going to that. But the point is when I have a solution to the problem and that it has to satisfy certain conditions.

We have already looked at the KKT conditions. If people do not remember it please go back and revise that. So there are a whole bunch of things you need to have the solution to be primal feasible . Primal feasible would mean that well your $\alpha_i$'s have to be greater than zero and it is a solution for the primal and there you have your complimentary slackness. So in this case becomes

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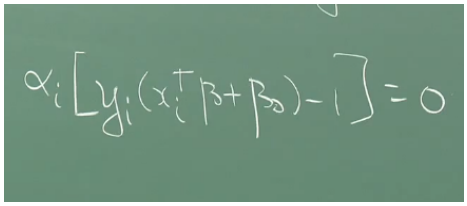


So I know if in the notes I think you saw it as $\lambda_i f_i$ . So this essentially is $\alpha_i f_i$. So all the KKT conditions need to be satisfied . And so what does this tell us?

Tells us a couple of things: one so we know what the form of $\beta$ should be what is the form of $\beta$? It has to be $\alpha_i y_i x_i$ . So it is essentially what you are going to do is your $\beta$ will be taking out certain data points from your training data and adding them up. So suitably multiplying it by the output the desired output, so if $x_i$'s output was positive then this will be +1 if $x_i$'s output was negative this will be -1.

So it is going to take a few of those and they are going to add them up. So this should remind you of perceptrons. So if you remember what we did in perceptions is we to whatever was misclassified we just kept adding it to the weight vector. So in some sense you are doing something very similar to that but instead of having some kind of a heuristic approach to optimizing things. We did do a gradient descent but then we just said that we will arbitrarily pick the set of misclassified points and we will do the gradient descent and so on so forth. But here we

started off by saying that we will minimize the distance to the closest point and from there we derive something very similar to the perceptron update rule. In fact nowadays when people say I am going to train a perceptron they are actually doing this more often than using the perceptrons learning rule right away. So now something else that you can observe is that this condition has to be satisfied. So let us look at it. There are two terms here so when will $\alpha_i[y_i(x_i^T\beta+\beta_0)-1]$ be 0? When either $\alpha_i$ is 0 or $y_i(x_i^T\beta+\beta_0)-1$ is 0. These are some condition when this has to be 0. Yeah! You are right but geometrically can you give me an answer? So $\alpha_i$ has to be zero is when the other term is not 0. So when will the other term be not 0? When it is not the closest point. If $x_i$ is the closest point it will be bang on the margin and for a point here that term will be 0. For a point here that term will be greater than 1. You see that so since the term will be greater than 1 the term in the square brackets will be non 0 so $\alpha_i$'s have to be 0. Correct, so what does this mean? It means that points that are further away from the hyper plane do not contribute to finding β. Because the α's will be zero, points that are far away from the hyper plane are not going to contribute in finding β. In fact the points that will contribute to β are exactly those points that are on the margin.

So in fact for this data set that they drew here, there are only two important points: that one and this one because only two points are on the margin . Such points which lie on the margin are known as "support points" or "support vectors". And your β is going to depend only on the support points. What about $\beta_0$? So we can plug in any data point here, and we can solve for $\beta_0$.

One of these support points you can plug it in here and you can solve for $\beta_0$. Which support point do you pick? Ideally all of them should give you the same answer but usually does not happen because of numerical reasons. So what typically people do is they plug in all the support points. Solve for $\beta_0$ and take the average. So each one in turn, for every support point you are going to get slightly different $\beta_0$ you just take the average.

So that is how you compute the hyper plane at the end of it is basically how here when would α be 0? If your data is on the hyper plane. So that will be the one case when that happens. Essentially you have two points which are on the same point. It is not collinear but repeated things; I give you two data points that are on the same point. So by definition most of the support vectors will lie on the same line so it cannot be collinear. So in such cases that could be the case but, yeah! These are generally degenerate cases yeah! So sure call them support vectors if you want. So one thing to note is my $\hat{f}$ . This is essentially going to look like I can flip these things around anyway that plus $\beta_0$.

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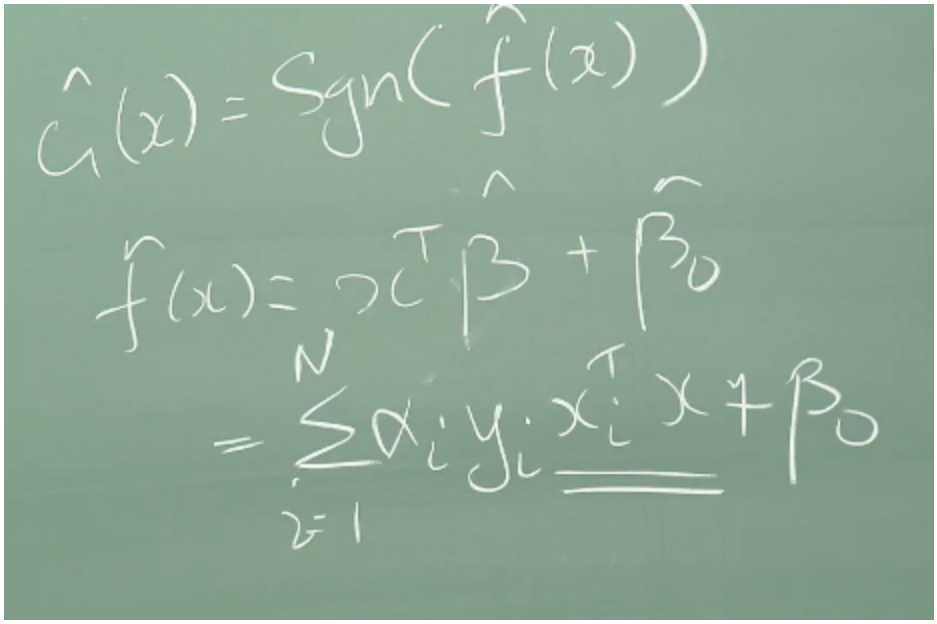


So, so if you think about it I will come back to this point later so if you lo at the dual I only have $X^TX$ and if you look at the final classifier I am going to use I am going to have $X^T X$. So if I have a very efficient way of computing $X^TX$, I can do some tricks with this whole thing we will come back to that .I just want you to remember this. So any questions on this?

So before we move on I just wanted to point out something so if you think about, how LDA works. So LDA tries to do density estimation eventually, if you think about it you make some assumptions about the form of the probability distribution. What assumption will you make it is Gaussian with equal covariance across all the classes'?

Though, that essentially means that every data point in your training set is going to contribute towards the parameters that you are estimating. So the β we estimate there will depend on all the data points that were given to you, whether they are here, close to the hyper plane or whether they are very far away from the hyper plane. All the data points will determine your class boundary so that means that it becomes little susceptible to noise.

And if I have one or two data points that are generated through noise even that will contribute to determining the separating plane hyper plane. On the other hand, with this kind of optimal hyper plane we are only worried about points that are close to the boundary. So I can do whatever I want here. I can change move a few points over here and things like that it does not really matter.

What matters is if any noise enters close to the boundary? So in some sense if my noise is uniform, the LDA will get more affected. Because even if noise insert some points there, LDA classifier will change. Well my optimal hyper plane classifier will not move it will be affected only by that fraction of the noise that changes the actual decision surface. Having said that I should point out that if, if your data is truly Gaussian with equal covariance LDA is actually optimal. It is probably optimal. While this one will depend on the actual data that you get but in general would say this is more preferable because this is more stable. People remember what stability is? Small changes in the data will not cause the classifier to change significantly.
So here small changes in the data will not cause it to change significantly in an expected sense. If I go and take the support vector and move it somewhere else, the class boundary will change. But then I have whole bunch of other vectors which I can move around nothing will happen to the class boundary unless I move it closer to the hyper plane than the existing support vectors. If I take a point from here and move it here of course the class boundary will change. As long as I do not modify which are the support vectors, I will get back the same classification surface again and again. So in that sense SVM or will come to SVM in a little bit, this kind of optimal hyper plane are very stable.

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**Introduction to Machine Learning**

**Lecture 29**

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**Indian Institute of Technology Madras**

## SVMs for Linearly Non Separable Data

Suppose I have some data which is not linearly separable. So that is the problem that we have seen with perceptions? So what happens if the data is not linearly separable? Perceptrons do not converge. So can we tweak our objective function that we have here to make sure that we can handle non-linearly separable data. Is it to say non linearly separable data? It should be linearly inseparable data , so you have to be careful where you put the not the negation there.

So what we do in this case? Somebody had a suggestion. So there are many ways there are many choices you can make let me not play around with it are many choices you could make but there is one particular choice which is seems to yield a very nice optimization formulation. So what is a choice I am going to say that I would really like to maximize the margin and I would like to get as many data points correct as possible .

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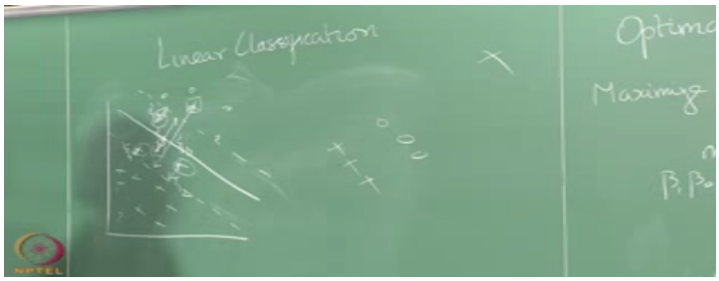

So if you think about it so there are a couple of things. So this is the margin that I want so what are the problems here? Well these data points are within the margin so I have some data points that are within the margin so I would like to minimize such cases there are some data points that are within the margin and erroneous. I would like to minimize such cases as well. If you think of what if I had tried to get this correct, there is a gap here and there seems to be a gap here between the points if I try to get this correct and move my classification surface below then the margin would have been reduced even further. So it is to get this wrong but then what about this case is it within the margin or outside the margin? Within. So the margin for that class is defined on the other side and so the margin for that class is this side. So anything to this side and x is within the margin. This will be $y_i$ times this and so this will actually be negative and it is within the margin. If we want things to be greater than one $y_i$ times f(x) we want it to be greater than one $\geq 1$, this is going to be negative.

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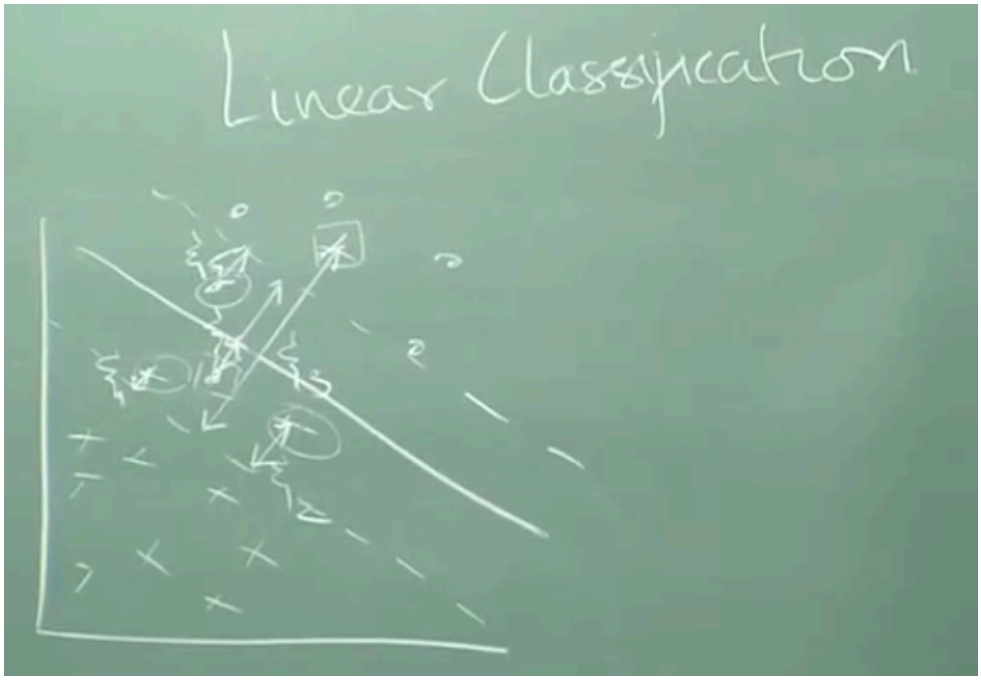


So obviously this is within the margin. So essentially what I want to do is minimize these distances so you can see the distance that I marked here are these distances I would like to minimize. So this is a certain small distance inside the margin, this is a large distance inside the

margin is a very large distance inside the margin, likewise so I can mark each one of these and I want to minimize these, so let us denote them says $\xi$ 1 to $\xi$ 5 and I want to minimize those essentially.

So if I minimize the sum of these deviations I make along with my original objective function. Why do we not minimize the maximum here? Because that would essentially mean that I will try to get as many things correct as possible. So in this case I do not mind getting something wrong as long as the overall deviation does not exceed a certain limit. See that the difference between minimizing the maximum and minimizing a sum is that I might as well give up all of the sum to a single data point it might be something that is very hard to classify.

And I might have one single outlier somewhere here let us let us let us draw so this data might be perfectly separable and I might have an outlier then so now if I say minimize the sum of the things it is fine. But if I say minimize the max, then it is going to actually give me a hyper plane somewhere there but like I said many different formulations are possible this one actually yields a very nice computation that is one of the reasons people use this.

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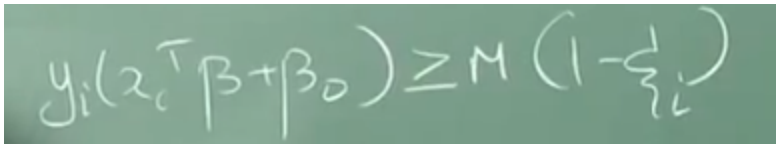


So what I am going to do is write it here. So I am going to say that this has to be that we had already found out and I am going to introduce a slack variable so it does not have to be greater than M. It can be some fraction lesser also M is what I would really like but I allow it to have a slack. Ideally I would want most of these $\xi$i's to be 0 and I force $\xi$i's to be zero I am back here but I really like some lee way .

So I am allowing myself that leeway by introducing $\xi$i here. This is a very standard technique for relaxing constraints in optimization. That is one of the reasons people adopt this is a standard constraint. So another thing which I could have chosen is that in fact this is a little bit more

common constraint but it turns out that in this particular case if I choose M - $\xi i$ instead of M(1-$\xi_i$) I end up getting a non convex optimization problem.

So we do not want that so we end up doing this. So I drew this figure first because I wanted to get an idea of what these slack variables actually mean. So the slack variables essentially tell you by what fraction you are violating the margin. So is $\xi 1$ is essentially what fraction of distance you are coming in here from the margin $\xi 2$ is what fraction of the distance you are coming in from the margin.

So the margin is M so I have moved some fraction of the distance inside. So essentially that is what the $\xi$ tells me. So what are the constraints we have? So the first constraint I have is all $\xi_i$ have to be $\geq 0$. I do not care about points going to that side of the margin. So all $\xi_i \geq 0$ and the second thing is whatever we have been talking about. I do not want the $\xi_i$ is to be very large taken in total so I want to upper bound them by a constant.

So because I am talking about going that side of the margin, if $\xi_i$ are negative so essentially I will be imposing a tighter constraint than what I was hoping for so this will be like it will larger than M. I will be having a thing that is larger than M and it is a relative distance, so essentially this becomes M- $M\xi_{i..}$ So the original should be M, so it is now $M\xi_i$ away from there. So $\xi_i$ is essentially a relative distance and if I make $\xi_i$ is negative so this will become plus so that will essentially mean that not only do I want the data points to be away than M actually asking it to be further away than so it just imposes a tighter constraint. So I do not want that to happen so and here we are essentially giving it a budget that we do not want it to be greater than the budget fine. So we saw such a constraint earlier. Where did we see such a constraint earlier? We had a budget we did not want it to be greater than a budget.

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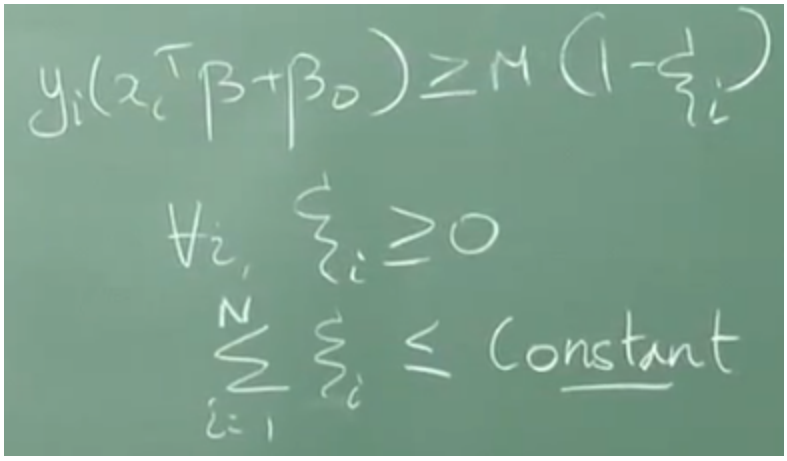


So yeah ridge regression and LASSO and other things we had this thing. So wherever we are looking at this regularized regression, so we had this greater than or lesser than a constant and what did we do in those cases? We push it into the objective function and then add a multiplier there. So then there is a relationship between this constant and the multiplier that we put in the objective function. So likewise will do the same thing here I will do all the other transformations that we need to do to normalize β and things like this.

So essentially I will end up with the same objective function I had there and you want ≥ to because they have gotten rid of the M. Why how do we get rid of M? Because M is 1 / β. So 1 by ||β|| so we got rid of that just anything else we need here. So now that we have this objective function what should be the value of C? If I want a linearly separable case we want to solve the linearly separable problem or I want to ensure that all ξi are 0 what should the value of C? This is the simple question. The answer is C should be infinity.

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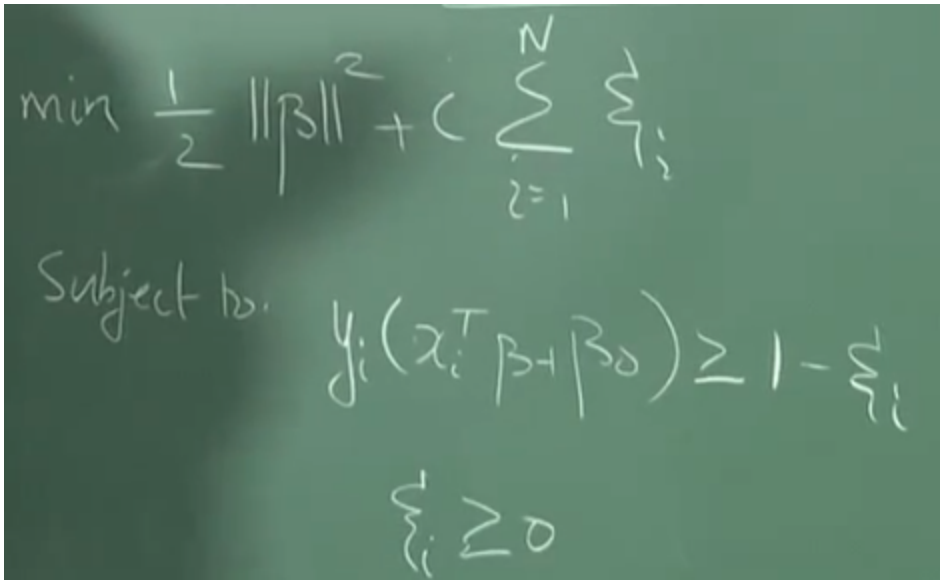


So the larger the value of C the more you are penalizing the violations. So the smaller the ξi should be. So the larger the value of C the smaller the ξi should be so there is a trade-off. So the larger you make C, the smaller will the margin be. But we will be getting more of the training data correct so for large values of C you are allowing a little bit more leeway. So C is very small then you are allowing lot more errors to happen if C is very large then you are forcing the classifier to classify as much of the training data is correct as possible .

The data is truly linearly separable and you make C very large what will happen? You will find the correct linear separator. But if the data is truly linearly separable but you keep C small what might happen? You might trade off errors in the training data for a larger margin even if the data is linearly separable. Is that a desirable thing? When exactly? So if the data is noisy such that there are some data points that are closer to the margin it may be one or two data points that are closer to the margin. So if you are trying to find the perfect linear separation you will pay attention to them as well and therefore you will end up having a low margin, but then if you are willing to ignore a few noisy data points, even if the training data los perfectly separable you might end up making a few errors on it but you will get a more robust classifier. So can people visualize a situation I am going to try and do something here let us see that works it has los perfectly separable. That is noise is it still separable. There you go there it is still separable and if you try to solve it as a perfectly separable problem that is the separator that you are going to get

but if you allow errors so that will probably be the separating hyper plane you get and that is probably a more appropriate hyper plane .

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Apart from being robust it is it is also correct in an expected sense. We will move on to the primal. So I just wanted to leave this on board till I wrote this note so that you can compare it.

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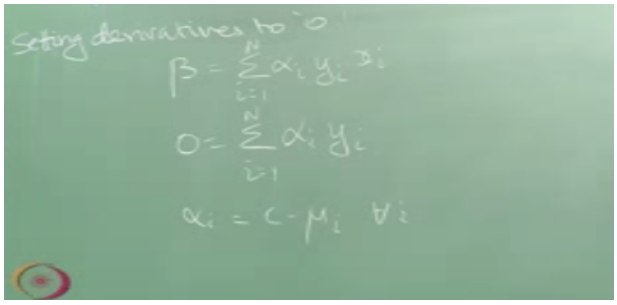


So that is the primal value also having have $\alpha$ and $\xi$, $\alpha$, $\mu$ has to be $\geq 0$.

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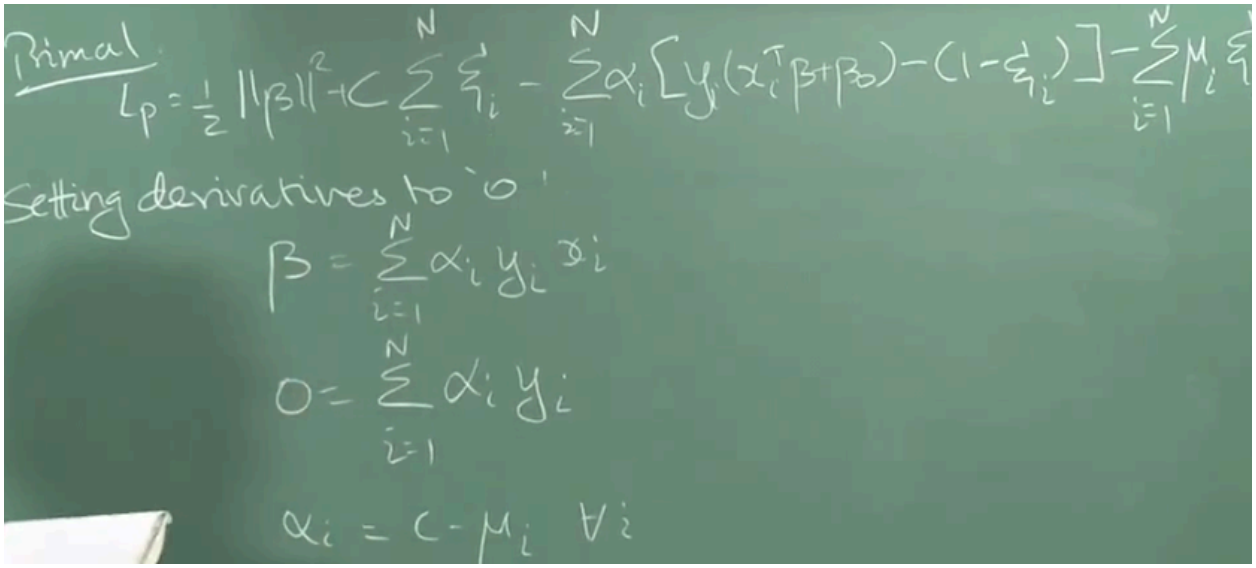


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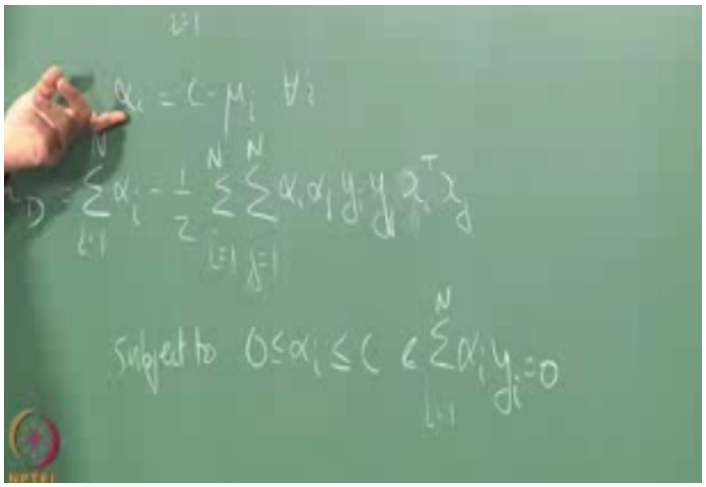

Yeah I do not have to do this. It is not a single condition that is there for each i. Do we need the $\sum_{i=1}^{n} \xi_i \leq const$. No , so that is why we put that into the optimization objective function itself , so by minimizing this, we are ensuring that $\sum_{i=1}^{n} \xi_i$ will be less than some limit. And like I was mentioning in the ridge regression discussion, so you can find a relationship between this constant and this C.

So basically they are equivalent ways of writing the optimization problem except that you have this constant and the C will not be the same there. It will have different values so this constraint is gone this is no longer present here that went into the objective function.

So putting all of this back in and doing some algebra can be surprised at the algebra outcome of this. Anyone has already solved it? Los familiar ? It is essentially the same dual you will get but your constraints are different. This is already there so it is just added for completeness sake but what is important here is earlier while I had only a non negativity constraint on $\alpha$ now I have a upper bound on the value of $\alpha$. So why is that because $\alpha$ is only C- $\mu$ and since $\alpha$ is only C - $\mu$ so there has to be a upper bound on $\alpha$ good so what about the other KKT conditions.

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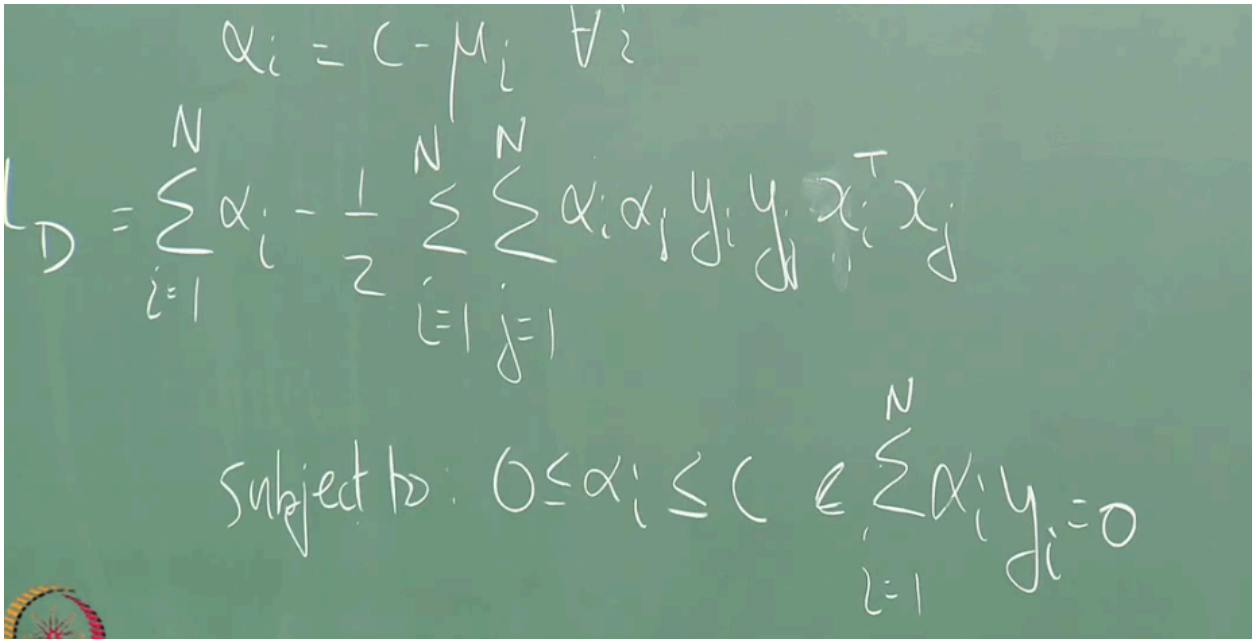


So 1 to 7 are the KKT conditions.

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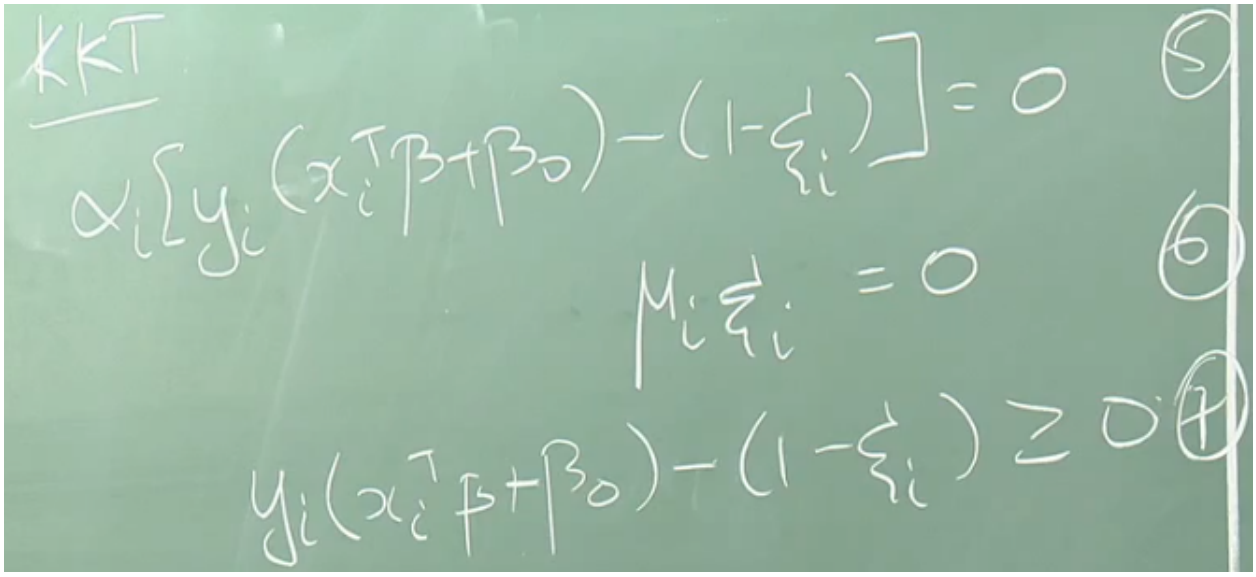


So what do you notice here again? Well you notice again that your β is determined by your $\alpha_i y_i x_i$ just like you hand earlier. β is given by those $x_i$ for which α will be nonzero. So like we had earlier those xi's for which α is non zero or called support points. Our support vectors depending on how we want to lo at it. Now let us go lo at when it will be nonzero. So when will α be 0? When will this whole thing be nonzero? Well it lies at a large enough distance on the side of the margin what about ξi is a will be 0. Then ready somewhere here the ξi will be 0. So in the ξi = 0 so it will be left with this term alone -1 that just takes exactly the same condition that we had earlier. So if this is far enough away from the margin then this will be nonzero so α is have to be 0. So we know for sure the same thing. Things that are on the side of the margin means correct side. Things that are on the correct side of the margin then α will be 0 so they would not contribute anything.

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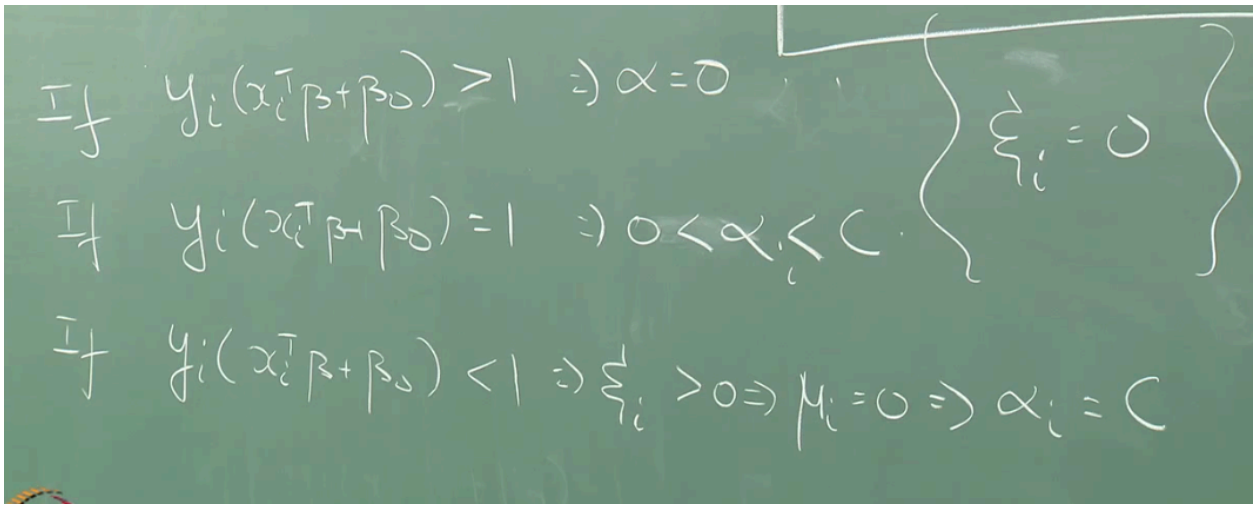


So now what about things that are on the margin? Is that a third case? We have to consider third case now . We have to consider the third case in which case what happened as ξi will start increasing ξi's will become non zero. If ξi is nonzero what does it imply? Because my α is C – $\mu_i$ , so if ξi is nonzero then μi will become 0 that for my α is will become C. So now how will this term go to 0 by suitably makings ξi a large enough.

So I will make ξi a large enough. So that this term will go to 0. Because this will be negative will be less than 1. So I will make this I will just ξi so that this term in the square bracket goes to 0 because my αi will be C what is that in case this is because I do not want to penalize this case. This case also ξi will be 0. So this case ξi is 0 this case also ξi will be 0 because what I really need is that is my condition ≥ 1- ξi so if is equal to 1, so I can set ξi to 0. In both these cases ξi= 0.

So what are all the support vectors? Everything on the margin and everything on the wrong side of the margin as well. Everything for which alpha is nonzero now becomes support vectors so at the end of the day I am going to say that you are just going to use a package to solve all of these things but it is like saying yeah anyway you are going to use Microsoft Windows or I mean Mac OS X or something why do you learn operating system . So you need to know what the internals are it is not the fact that you just use the tools that matters it when you can just use the tools well

yeah we can do a tool course how to use the tools how will you start up limbsvm and it is not trivial. So many people I know actually run experiments with SVM's by just using the default parameter settings that the package will give. The thing is you need to understand what is it that you are tuning. So now I told you about the C parameter , so you have some idea of what a large C means versus what a small C means hey instead of blindly saying that I am going to queue C from some number to some other number, so to have an appreciation of what these things are doing actually helps you even use the tools better so that is the whole idea behind doing all of this. Ideally I am not going to expect you to come and derive a large margin classifier tomorrow.

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**Introduction to Machine Learning**

**Prof: Balaraman Ravindran**
**Computer Science and Engineering**
**Indian Institute of Technology Madras**

Lecture 24 NOC

So if you remember I asked you to note the fact that I am using a inner product there right x i transpose x is just the inner product of 2 vectors and the way I wrote the dual also I had only inner products in there. so in fact if I want to evaluate the dual I need to only know the inner products of the two vectors likewise if I want to finally evaluate the use the classifier that I learn I still need to only find inner products right so if I can come up with a way of efficiently computing this inner products right I can do something interesting so what is that So what do we normally do to make linear classifiers more powerful basis transformations this is somebody said basis transformations right so I can just take my hedge. right replace it with some function h of x that gives me a larger basis right it could be just replace it with the square right I take x and replace it with x squared right and then I will get a larger basis and now it turns out that I can do a lot of math but I can get up get with get a dual that looks like this. so that is the inner product notation right so if I can compute the inner product so I can just solve the same kind of optimization problem right but I can do this in some other transform space okay so and likewise our f of X is going to be So essentially what I need to know is h of x for whatever pair x and x prime that I would like to consider right. So in the training it is the pairs of training points right while I am actually using it, it is one of the support point and the input data that I am looking at right. So at any point I just take this pairs of data points and I need to compute the inner product right.

So I am going to call this as some function which is a kind of a distance function or a similarity measure between h and h . Such similarity measures are also called as kernels. right so kernels are nothing but I mean so you might have heard of kernels in the context of support vector machines we have been trying to use lib SVM or any of the other tools for some projects over the summer you would have heard of kernels right kernels are nothing but similarity functions okay so the nice thing about the kernels that we use right is that they actually operate on X and X prime okay they operate on x and x prime but they are computing the inner product of h of x and h of x prime, okay. Did you see that? They are going to work with x and x prime, right but they will be computing the inner

product of h of x and h of x prime.

So I will give you an example. So the kernel function k should be symmetric, right and and positive semi-definite okay positive definite live semi-definite is fine in some cases positive definite people remember what positive definite is right right and that if it is semi-definite right essentially we want the quadratic forms to be to be positive we do not want to take X transpose AX and suddenly find it is negative so it is in fact you remember I told the X transpose AX is usually the quadratic form that we are trying right and that will actually messes up big time in the computation if the quadratic form becomes negative. Then we will have problems in all the optimization thing going through. So that is the mechanistic exam reason for wanting it to be positive semi-definite. There is a much more fundamental reason for it which I have not developed the math or the intuition for you to understand.

So it has to come at a later course. So hopefully in the kernel methods course if you are taking it you will figure out why that is needed. So there are many choices which you can use for the kernel. so the something called the polynomial kernel which is essentially 1 plus inner product of XX prime the whole raised to the power D. So D is a parameter you can have D of 2, 3, 4 you can even have D of 1 which is essentially your whatever you have solved so far right you could have D 2, 3, 4 whatever and this next one is some called the Gaussian kernel or the RBF kernel right so where the the distance is given by e power minus gamma x minus x prime square norm of x minus x prime square this is essentially the Gaussian without your without the normalizing factor right.

So that is why it is called the RBF kernel so if you want to call it the Gaussian kernel you actually have to make it Gaussian okay it is otherwise called the RBF kernel and then this is called the neural network kernel or the sigmoidal kernel sometimes that is just the hyperbolic tangent right this is kappa 1 and kappa 2 some constants some arbitrary constants k1 and k2 which are your parameters that you tune and this is x, x prime inner product okay. So these are some of the popular kernels which can be used for any generic data but then depending on the kind of data that you are looking at right where the data comes from people do develop specialized kernels. For example for string data people have come up with a lot of kernels when you want to compare strings how do I look at similarity between strings. So the nice thing about whatever we have done so far is that you can apply this not just to data that comes from RP right you have been assuming so far that your X. comes from some p-dimensional real space as long as you can define a proper kernel right you can apply this this max margin classification that we have done to any kind of data does not have to come from a real valued space okay which is not true of many of the other things you have looked at right all those inherently depend on the fact that the data is real value right because of this nice what is called the kernel trick right so

you can you could do all of these nice things so as long as you can define appropriate kernel right you can actually apply this to any kind of data so that is one very powerful idea so just to convince you so let us look at the polynomial kernel of degree 2 right.

operating on vectors of two dimension, so there are two twos here, so the degree is 2, the d is 2 and the p is also 2, but they need not necessarily be the same that I could have had a much larger thing about the CCA for me to write. right I have just squared it. Now, if you think of H which does the following so what is this function h it is essentially the quadratic basis expansion right so I have two two features x1 x2 right so I give so remember that x right x is x1 x2 right they consider of two coordinates x1 and x2 right so this is essentially the quadratic expansion the first thing is 1 the second coordinate is x1 third coordinate is x2 so it keeps it as it is okay then fourth coordinate is x1 squared fifth coordinate is x2 squared the sixth coordinate is x1 x2 it has done all the quadratic basis expansion right now if I make this operate on x and x prime and take the inner product so what will be the terms 1 2 x1 x1 prime 2 x 2 x 2 prime x 1 square x 2 x 1 prime square right x 2 x 2 prime the whole square and 2 x 1 x 1 prime x 2 x 2 prime that is exactly what we have here right. So, what is the nice thing about it is I can essentially compute the inner product of x and x prime first add one and square it so numerically what I will end up with it is the same as what I would have ended up with if I had done the basis expansion right and then taken the inner product that makes sense right if I had just taken whatever is the vectors original vectors let us say I have some 2, 3 and 4, 5 so instead of doing this basis expansion and then computing the inner product right I can just take the inner product right away right this is essentially 22 square this answer so this well for degree 2 it might not seem great what about degree 15 polynomial I am essentially doing similar amounts of computation except that I have to raise something to the power of 15 right so so any questions so far Any questions so far? We are all good? That is what basis expansion is. If you thought something else about basis expansion, please correct it.

This is basis expansion. So, I take the original data I then essentially I said you could have a new components at our sin x cos x I mean does not matter you could think of variety of different ways of expanding the basis in this case I am just doing the quadratic basis expansion. right is it clear to people so whatever we have done so far and so this whole idea of a kernel and other things are rather straightforward. So what I cannot write now for you is what is the basis expansion for the RBF kernel it turns out that the computation is doing is actually in an infinite dimensional vector space okay so here the computation is a six-dimensional space right I took some data point from a two-dimensional space okay I did computation in a six-dimensional space right and I gave you back the answer but all the time doing computation only in a two-dimensional space right I only took the inner product of these two and then added one squared so I am essentially doing computation only in R2. while the actual number I am returning to you

is the result of computation done in R6 that is why it is called the kernel trick right.

So likewise the RBF kernel I will do something in whatever is the original dimensional space you give me but the resulting computation has an interpretation in some infinite dimensional vector space okay. So it is not even easy to write it down okay so that is why the RBF kernels are considered pretty powerful they work on a variety of data right but they are not all powerful though just have to be careful about it. So that is all there is to support vector machines. So we have done this support vector machines as well. So I do not know if people have used lib base VM or one such tool for that.

For most RBF kernels you would have tuned two parameters. One is C which we already saw. That is essentially how much penalty you are giving to the thing other one you will tune is gamma it is essentially this right it is some kind of a width parameter for your Gaussian means how wide your Gaussian is it just this controls that so that is gamma. So those are the two parameters you tune and for polynomial kernels you have a D and you have your C right and for sigmoidal kernels you have constants k1 k2 and you have C and this form of defining a support vector machine is called as CSVM. there are other ways other constraints that you can impose on it not just the penalty on the size you can impose penalty on the number of support vectors you consider right you want to say so suppose I run the data and it comes back and says okay everything is a support vector right so that is not something interesting how can everything be a support vector can all the data points be equal distance from the separating hyperplane not if you are considering linear but when considering RBF kernels right the separating hyperplane can be very very complex right.

So in which case you might end up with a lot of support vectors typically I do not know if you have not thought too much about it and you are setting some very high values for C and trying to run this thing you will end up with like 60% of your data as being support vectors So instead of trying to do that empirically saying no I only want 20 support vector so let me try different C different gamma and so on so forth you can use something called the new SVM new not any W the Greek new okay new SVM which gives you a upper bound on the number of support vectors that you are going to get you can say do the best you can but do not give me more than 30 support vectors or something like that to that effect okay.

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**Introduction to Machine Learning**

**Lecture 31**

**Prof. Balaraman Ravindran**
**Computer Science and Engineering**
**Indian Institute of Technology Madras**

**Hinge Loss Formulation Of the SVM Objective Function**

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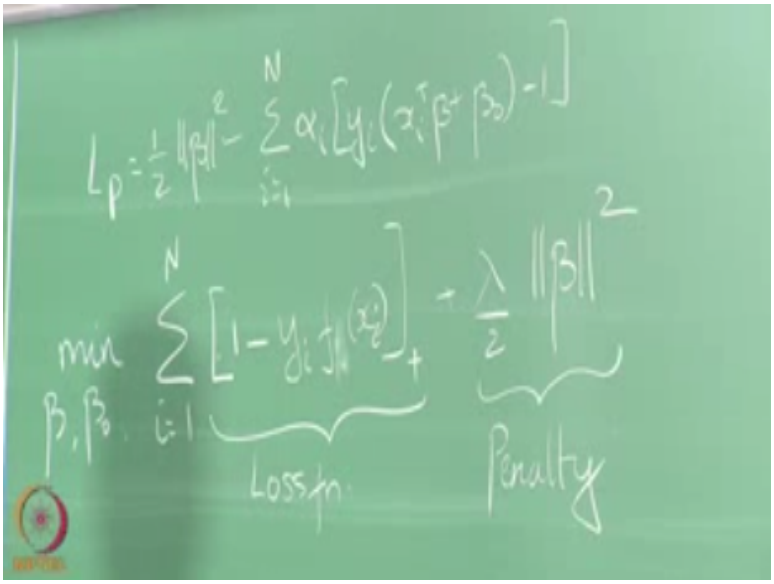


Okay so people remember the primal objective function that we had for SVM's. So this is a primal objective function we had for SVM's. So one way of thinking about it is to say that I am going to write it the following way. Maybe with some jugglery so the α's I have replaced it with a λ here and well you know $xi^T\beta + \beta_0$ is actually $f(x_i)$ and then essentially the same objective function except for this plus thing here.

So what is this plus thing?

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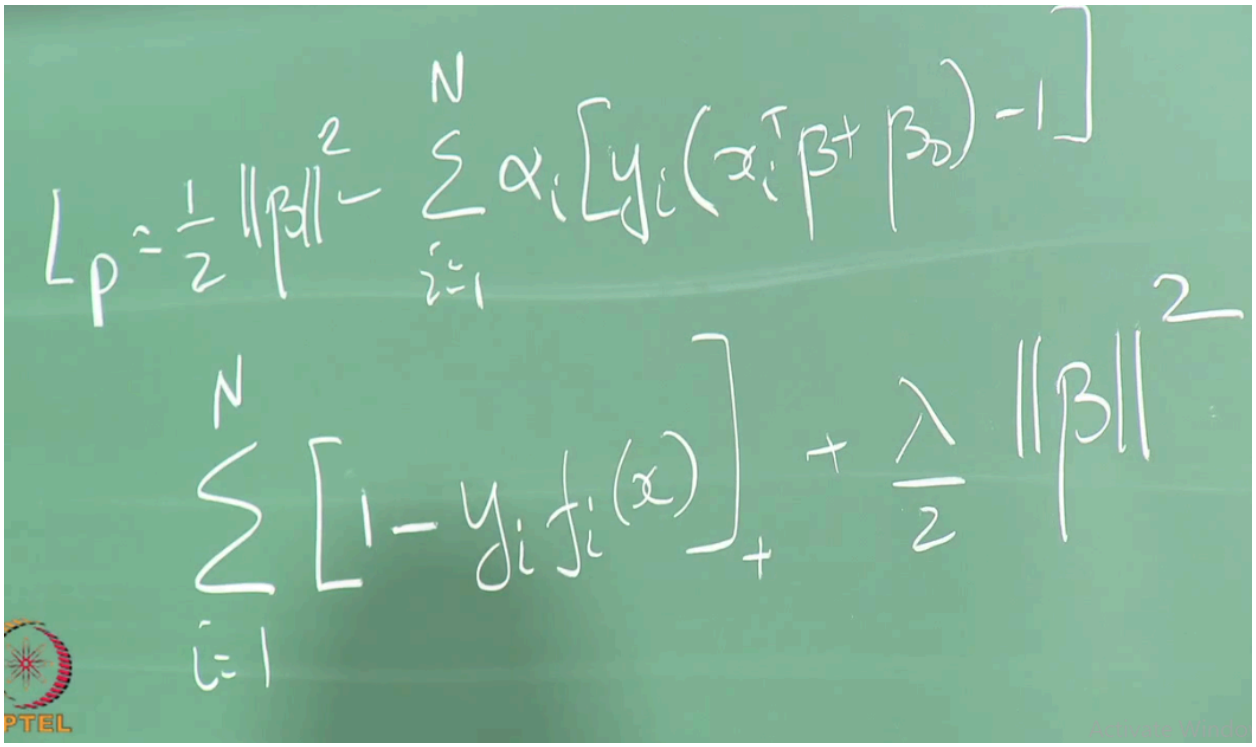


It means that I will call this only whenever this is positive , whenever it is negative I will read it as 0. Does it make sense? I will count this only whenever it is positive, whenever it is negative I will make it I will consider it as 0. So that is what the plus term here indicates that went into λ. I mean I kind of redid this thing, so I divided everything by some factor of α and moved to λ. So if you stop a minute this should look familiar to you. What does it look like? Ridge regression. In ridge regression so you have a loss function and you have a penalty term and doesn't it look like that? So far we have been talking about $\|\beta\|^2$ as being the objective function that you are trying to minimize and the other thing is constraints and then we then wrote the Lagrangian and then we got the constraints into the objective function. So now I am saying you can think of another way of writing the objective function which is to say that there is this loss function which is accounted whenever it is negative, so now your goal is to minimize this.

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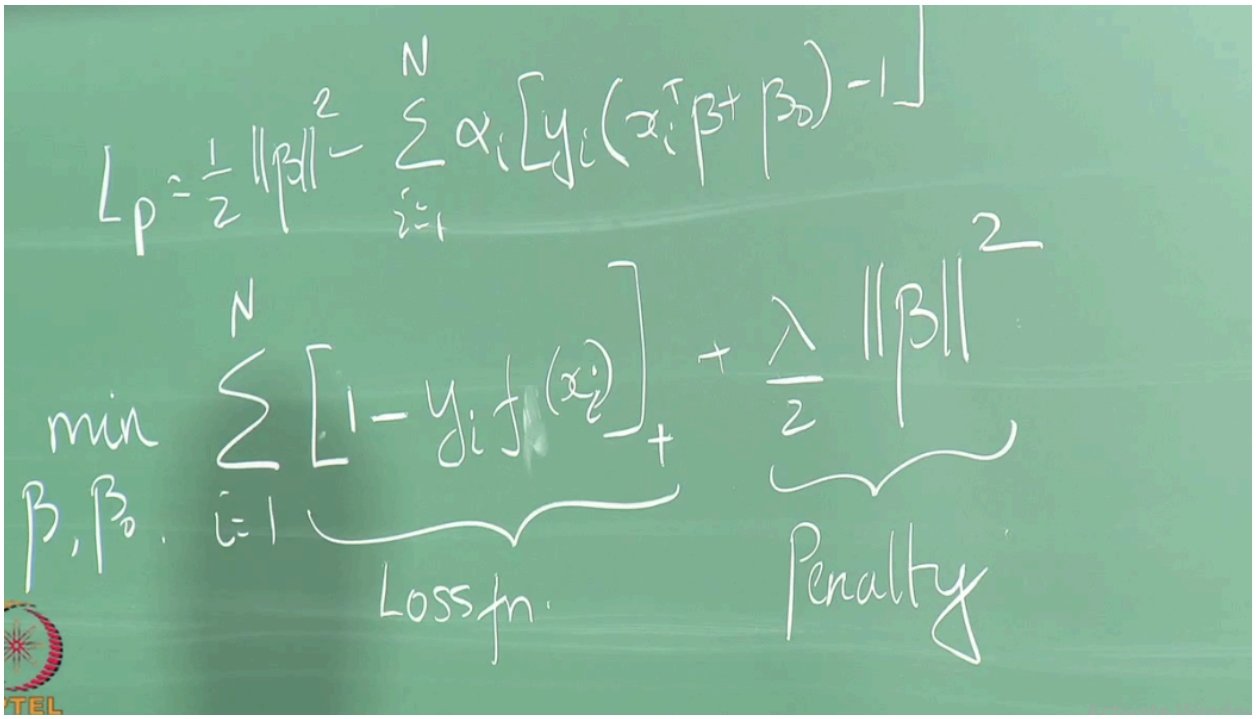


So how will this loss function look like? So when yf(x) is one, after that it will be 0. We talked about loss functions and not about the penalty term but till yf(x) becomes one it is going to be a linear function. You can see that I it is just 1 – yf(x) so it is going to be a linear function of yf(x). This kind of a loss function where this is like a door or a book opening on a hinge if you think about it this is like two flaps of a book or a door and it is opening on the hinge which is here . So it is also called “hinge loss”. So sometimes if you have read about SVM's elsewhere you might have heard that the SVM's minimize hinge loss. So this is exactly what we are doing here. So the hinge loss actually arises from the constraints that we are imposing on the SVM. But if you think about where the constraints come from or why were the constraints imposed? What is the semantics of the constraint? What was that we wanted to make sure? That they are correct and a certain distance away and that is the reason for this.

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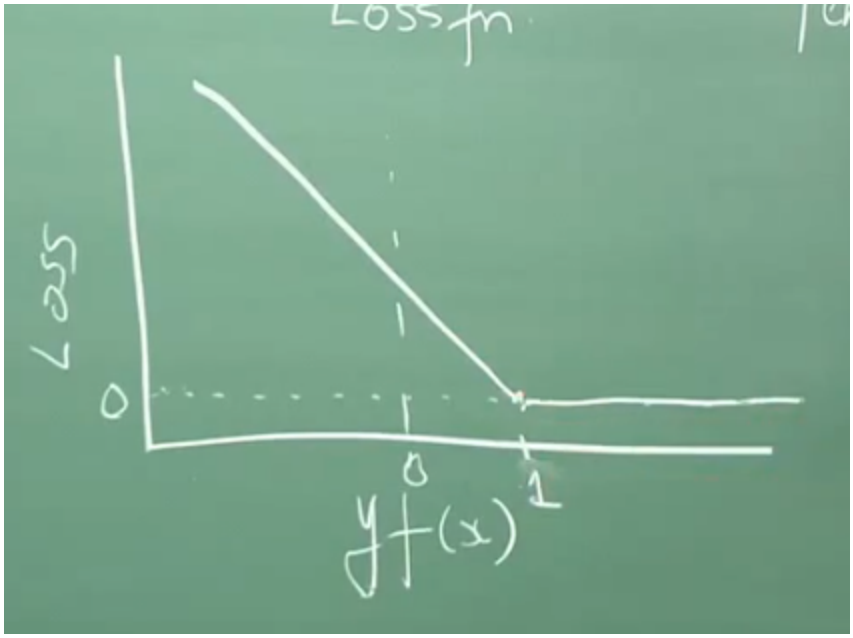


So in effect the constraints are enforcing the correctness of the solution and what the objective function originally was enforcing was essentially the robustness of the solution how far away are from the hyperplane. The constraints were making sure that you are on the right side of the hyperplane and if you think about it so in effect the constraints are an important part of what you are trying to optimize it is just not the distance from the hyperplane that matters but it is also matters that you should be on the right side of the hyperplane .

So the putting it is a hinge loss makes it explicit and I am saying okay this is the loss function I am interested in is the correctness and I want to make sure that all my data points are correctly classified and the penalty tells me okay make sure it is a small norm solution it essentially becomes like Ridge regression. You make sure that the squared loss is as little as possible at the same time make sure that the norm of the solution is also small. So that is what we did. We enforce the L2 norm in the ridge aggression case and we are doing the same thing in the SVM case. Does it make sense now? We can ask interesting questions like okay if I replace this with some other norm penalty what will happen? Can you do L1 regularized SVM's? No that was regression so L1 regularised regression was LASSO. So for SVM's instead of the $\beta^2$ if you put $\beta$ what do you think will happen?

Well you will have a much harder optimization problem on your hand but it is actually a valid thing, so what it will try to do if you remember we talked about this in LASSO I did in a admittedly a little hand wavy fashion but we talked about how it will enforce sparsity. We said it will try to make as many coefficient 0 as possible, so in this case what do you think will happen? If I put norm can attend for sparsity will it reduce the number of support vectors does that enforce sparsity think about it. What is that? Now the squared loss is actually like this if you think about it is little weird, so if you are to this side you are actually correct but, the further away you are from the hyperplane on the -hand side also you still contribute to the loss because of this squared error function whether you are on the right side of the wrong side of the hyperplane you still contribute to the loss. So that is why sometimes the squared error function is not the ideal thing to minimize. So the hinge loss more often than not gives you a much better solution than optimizing squared error. So what will the squared error be?

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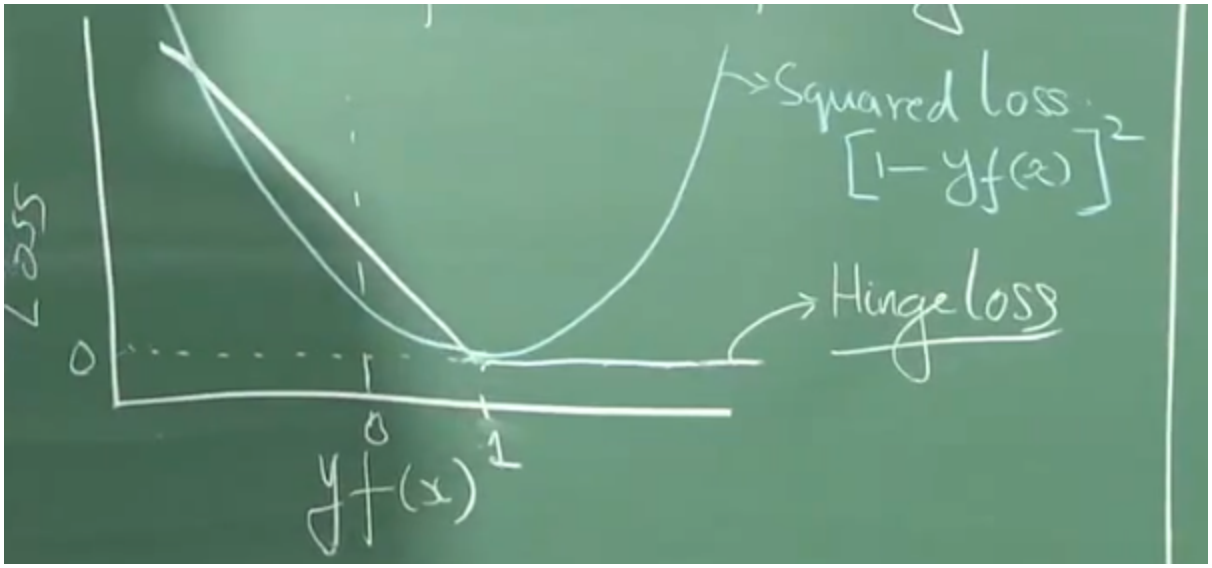


That is what the square loss function is so normally you are used to seeing this as $(y - f(x))^2$ but I have written it as $1 - yf(x)$ that is also fine because if it is correct $yf(x)$ will be one all the time. So what is the actual loss function that you want? What is the loss function called? 0-1 is what you really want it should be 0 if it is correct and it should be one if it is incorrect at 0-1 is what you really want and a lot of this just like a segue I am not really not going to test you or anything

on it just for your interest a lot of work in theory in machine learning goes into showing that if you optimize some other loss function will end up with the same solution as if you optimize the 0-1 loss. So if you take the 0-1 loss I try to find a solution for it, I am trying to find the β that gives me the smallest possible 0-1 loss. It is as small as possible 0-1 loss 0 depends on the data and you say linearly separable. But why because you chose to use a linear classifier. So depending on what family of classifiers you choose and the and the data. The minimum 0-1 loss could be 0 or it could be something higher, so you say minimizes 0-1 loss I mean whatever is the minimum possible achievable given the data distribution and the class of I mean the class of classifiers of the family of classifiers your chosen given that what is the minimum achievable will you be somewhere close to that. If I minimize a different loss function. So that is interesting question to ask  so I can arbitrarily come up with other loss functions I can come up with hinge loss or a squared loss so if you minimize hinge loss or squared loss will I get the same solution as I would have gotten if you had minimized 0-1 loss? So that is something people do think about so we did look at one other loss function which is the I guess it goes something like that so that is what we minimize actually in the logistic regression case.

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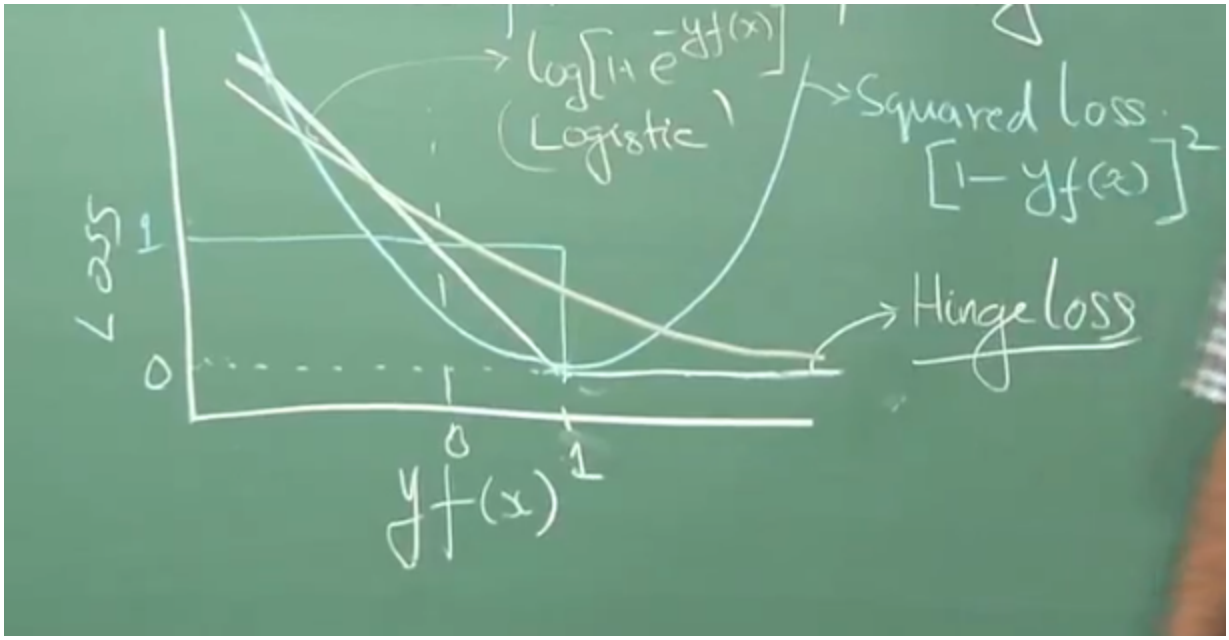

Even though we did not write it out explicitly as a loss function, if you think about it this is what we actually minimize in the logistic regression case. Also what we are trying to do? To estimate parameters it is maximum likelihood, so we made some assumptions about the distribution and then we try to maximize the likelihood and so on so forth. So if you work through that you can write it out as a loss function. It turns out that this is what you are trying to do so you can see that this never goes to 0. But then you can still think of minimizing that so we will just an aside you do not have to worry about the logistic loss function now. We will come back to that later.

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**Introduction to Machine Learning**

**Lecture 32**

**Prof. Balaraman Ravindran**
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**Indian institute of technology**

**Artificial Neural Networks I –**
**Early Models**

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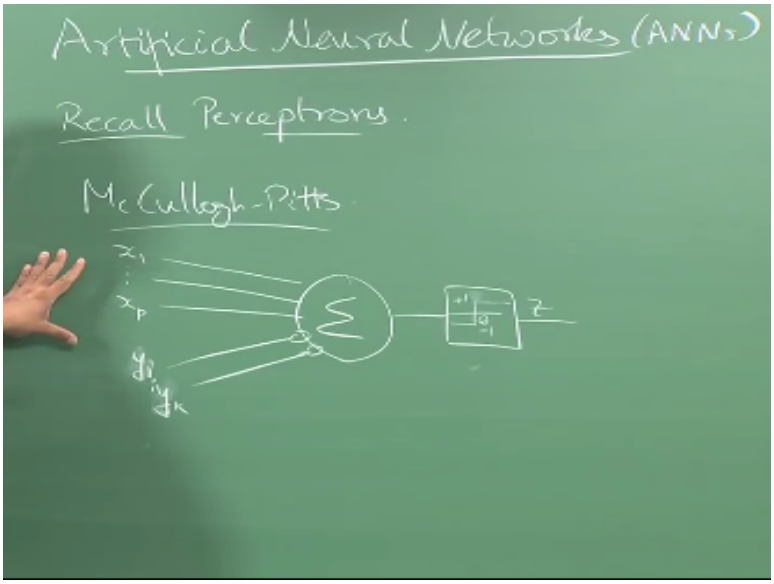


Okay so we have been having discussions about neural networks off and on right, the last time we had a discussion about $ANN_S$ was went and we did perceptions right. So, basically so the whole class of solution methods which are lumped under $ANN_S$ are artificial neural networks were primarily inspired by trying to emulate the brain architecture right yeah, so then after a while the field split into two right one class of researchers who are looking at neural networks as

just computing elements right trying to interpret it in terms of linear algebra right and partial analysis and other mathematical tools and then trying to understand what computing these artificial elements were doing.

And others who are still trying to make the neural models biologically relevant right, so now the communities have become fairly divergent right, so there are this set of people neuroscience people who are trying to build computational models of the brain right and then there are machine learning people who just use neural architectures without worrying anything about biological relevance right, so yeah we will take the latter approach right there will not be looking too much about the biological relevance of the neural networks I will just try to understand it in terms of computing.

So do not two people have ever been seen a neuron synapse did not write all those pictures so you do not have to really tax my drawing skills right so I will not draw anything on the board about the neurons right, so the whole idea behind all these biological neurons is that it is pulling in inputs from a variety of other neurons right and some computation is goes on within the neuron and there is a result of this it might actually fire right in which case it will cause it another input to be activated for yet another neuron right or a set of other neurons depending on who they are connected to.

Then these neurons are all connected in a very complex networks and even though it is a very simple computing element each neuron can be thought of as a very simple computing element the whole the fact that they are connected together in a large network allows them to do all kinds of cool things right, I mean if you really want to know what a neural network can accomplish they stop and think what you can do right, so one of the earliest, all of the earliest models of a neuron was the Mcculloch Pitts model right.

So it is a very simple model so it essentially it has a $\Sigma$ unit right so it has a set of inputs right and then it has not set off inhibitory signals okay right, so it has a set of inputs right and a set of

inhibitory signals if we what we are talking about artificial neural networks and I started explaining the Mcculloch Pitts model so you have a simple unit which has P inputs right and then it also has k inhibitory inputs right, so what does this neuron do is essentially adds up these k inputs right.

And if the if they exceed a certain threshold right some threshold θ then it will output a 1 if it is below a threshold θ it will output - 1 or a 0 depending on how you are encoding it right and the inputs are all considered either to be 1 or – 1 or 0 is depending on how we encode it right what are these inhibitory in inputs for if any one of these inhibitory inputs was one there was no output no I 0 okay yeah if any one of these inhibitory outputs is inputs is one then there will be the output will be - 1 okay or 0 depending on how we were encoding it again right.

So that is the basic Mcculloch Pitts model but then just to give you the historic perception right and then what happened people modified this way to propose the perceptron right.

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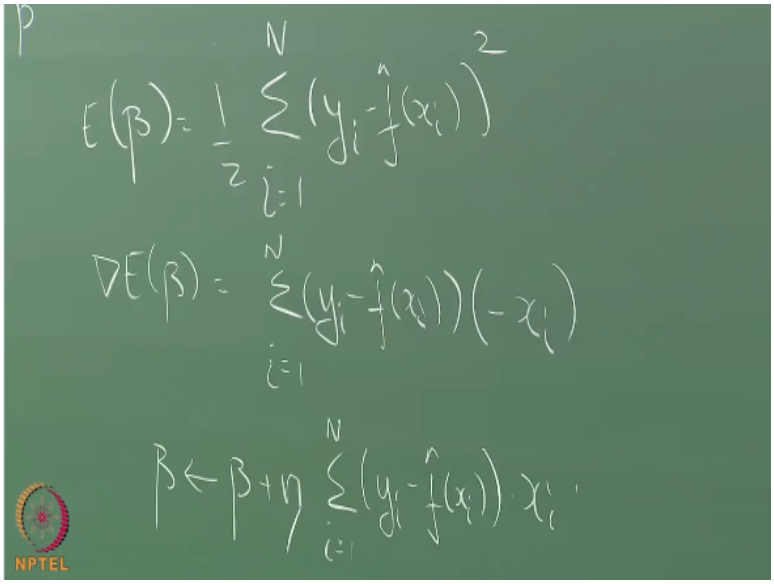

So if you think about what the perceptron does it is essentially saying that right so instead of doing $x_1$ to $x_p$ and adding it up I am going to multiply it by $\beta_1$ to $\beta_P$ so $X_1 \beta_1 + X_2 \beta_2 + X_3 \beta_3$ and so on so forth now if this entire thing is greater than a threshold okay I will output a +1 right the entire thing is lesser than a threshold I will output -1 so what is the threshold meet or not right so another way of doing it is to actually add a add another input label that $\beta_0$ I mean way is that as $\beta_0$ and put a 1 and then and right okay.

So you could think of it that way right so usually it is written without the - sign right that essentially means $\beta_0$ will be a negative quantity okay so this is essentially what our perceptron this you can see it is very closely related to the original Mcculloch Pitts model except that there are no inhibitory inputs and they the actual inputs are weighted okay and we all know how to estimate the parameters of the perceptron right how did we get there we use the gradient descent rule right.

We started with that error function and then we took the derivative and quite look like the perceptron ruled as it does not quite look like the perceptron what it do the perceptron update so wherever there was a missed classification what did you do with this we just added those data points to added those data points times Yi to so we did essentially read $X_i Y_i$ to the input right we did not do anything like this right no right, so what we essentially done here is minimize the squared error all right.

So this is another way of training perceptron is it something wrong here what is wrong here yeah so with the with the perceptron training so we did something very different from whatever I have done so far so it is perceptron training algorithm we had a very different objective function that we were optimizing right, so what were we trying to do people remember that you have a quiz like two days away should have revised all of this by now yeah minimize the mean you might see what minimize the distance to the hyper plane of the misclassified points right.

And the way you can minimize it is we get a distance of zero total distance of zero is when all the points are correctly classified right, so what we had actual objective function we wrote down was we are trying to minimize the distance to the hyper plane of the misclassified points we remember we wrote it only over the misclassified points right so in this case what we are doing you are writing a plane with plain old squared error function right but this is actually the squared error there right.

If you think of it otherwise this is anon linearity right this is a non-linearity the threshold is a non-linearity right I can just take the derivative of z with respect to x it does not make sense I cannot take the derivative of z with respect to x because that is a nonlinear function of x right.

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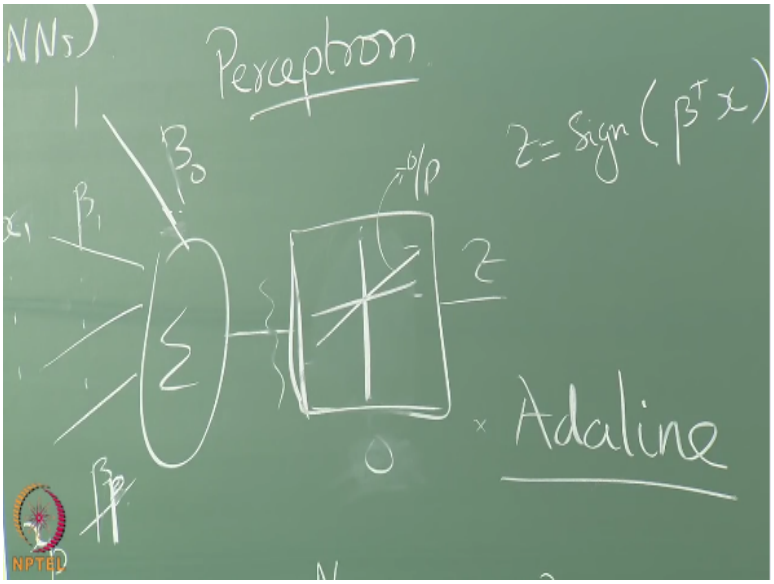


So z is what right and sign is a nonlinear functions non differentiable so I cannot really take the derivative. So in fact what I have written down here is for a slightly modified model of the neuron where the output is taken at this point okay the non-linearity is not there, so one way of thinking about this is that is the output like it is like a straight line right, so whatever comes as

the input it will be produced as the output right, so fool yourself for a minute that this line has a slope of 1 okay.

So whatever comes as the input will go as the output right in that case I can take the derivative of the output with respect to the input does it make sense right, so this is not the perceptron by the way they are sometimes called the but what is either line stand for any guesses adaptive closed linear adaptive linear so it is called adaptive linear units so they are Adaline right so on the way you train Adeline is just you straight forward gradient descent right and you end up with this okay.

So people are familiar with gradient descent right I do not have to explain gradient descent to people everyone familiar with gradient descent right okay, so why am I using η here step size yeah why do I need a step size okay so why would what would produce oscillations okay great yeah.

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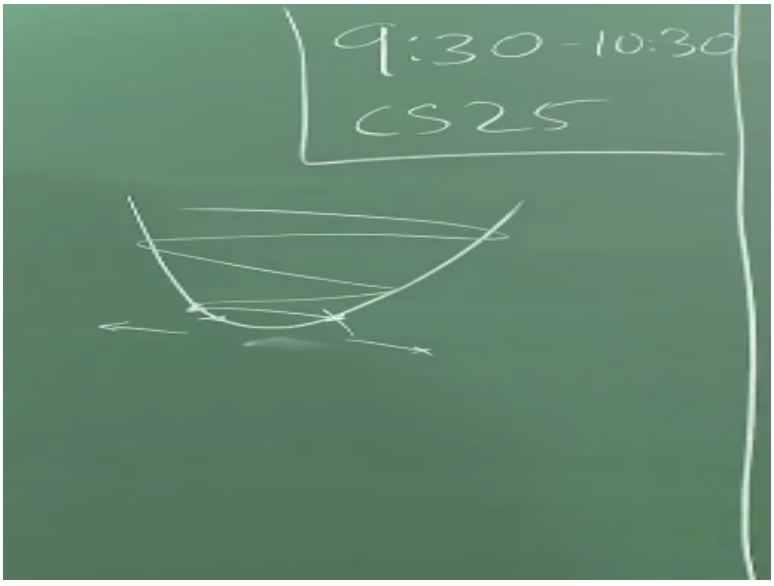

So let us look at it in a 1 dk may suppose I have a function that I want to optimize right let us assume that I cannot measure the gradient properly and I am actually making estimates of the gradient right so I am somewhere here right and I find the gradient what direction is the gradient here that way hey that is the way I have to change x to go up right and what they have to do I have to move in the opposite direction.

So if I take a large step in the opposite direction what will happen is I will actually end up this same look I will end up somewhere here now again I will I the x the gradient is in that direction here again I take a large step I would end up here like I could keep going back and forth then I might not actually converge okay that is it make sense in fact it is even worse I could go back and forth and I might even diverge if I am taking very large steps right so that is why I have to take small steps when you are following the gradient right.

And this these kinds of methods are called stochastic gradient why no it is essentially the so if you think if you just think about it a little bit right where do these excise come from for you not from some underlying distribution you do not know right so what you are essentially doing is there is an error function which is the error function of β on the entire input space right what I am really trying to minimize is the error on the entire input space right but I cannot compute that gradient because I do not have the data distribution I do not have the the entire data set available to me right.

I only have a sample and the sample was chosen in a stochastic fashion, so if I am trying to minimize the error with respect to the entire data so what I am actually finding out here is some sample radiantly it whatever sample data said was given to me I am finding out the gradient with respect to that sample data right that is one part right the second thing is typically I end up doing this one data point at a time instead of doing it n data points right if you I spoke about this in perceptron, so when I am doing it one data point at a time that is essentially not even the correct computation of the gradient.

So even given the data that is available to me I am only doing it one data point at a time that means I am actually doing some incomplete computations of the gradients right and if I take really tiny steps in the direction I am hoping that on an average I will move in the right direction so if I had computed for all the data points and taken a step then well given the data that is the best direction I can move in what if I computed one point at a time then I have to really take tiny steps.

So that I am sure that on an average I am going in the right direction okay, so that is that is idea here, so remember this is useful for the rest of the class so what is the problem with the perceptron people remember let goodness in something as simple as Excel because it was not linearly separable right.

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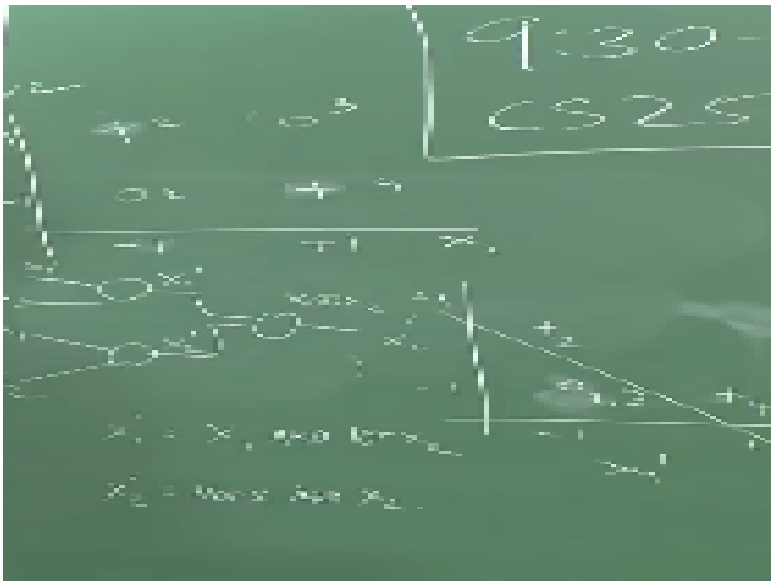

So it is not able to solve this is there some way you can think of solving this let us call this $x_1$ and $x_2$ should be -1 how do I move it to a different space change the basis function right I cannot rotating the axis would not be sufficient for me to still be not like I still not be linearly separable right I think I said that right so what do we do now too many x's on the board yeah so I can

define a new feature but what do what should I define will it one x times okay $x_1$ x $-x_2$ and $x_2$ into $-x_1$ okay.

So now what will happen if I do the projection of this data okay, so we are at this point go -1 -1 -1 is it, so where would 1 go -1 -1 where would 2 go 2 is here, so there will be -1 x -1 will be +1 okay then this will be +1 x +1 will be +1 is it okay does not sound very promising what about 4 is also + 1 what about 3 which is +1 4 so where will 4 go is my question 4 when there 4 is what +1 x +1 okay -1 x -1 +1 +1 that is 4 goes there right now this is -1 that will be -1 is +1 yeah there you go what about that k it is -1- 1 - what are they the same see you okay you negate $x_2$ you first negate $x_1$ and then pick the product okay.

Is essentially the same no see the idea is that I want my $x`_1 = x_1$ and $0x_2$ they are not the same thank okay now tell me so we know that we can actually implement and using a perceptron right you can implement or not using a perceptron okay, so now do this so where will $x`_1$ be for 1 to next one is -1 $0x_2$ so there will be -1 right $x`_2$ will be -1 okay what about 2 -1 that is not the same thing okay does that make sense, so in this transformation where $x`_1$ x is x1and not x2 right and $x`_2$ is not $x_1$ and $x_2$ right where true is +1 false is -1 and if you do that then my projection will be 1 and 3 will get projected to -1 -1 and 2 and 4 will get projected to the same coordinates as they were earlier in this modified space okay.

Now is this linearly separable right, so now I can separate this right, so I can listen sleep construct a perceptron that does that so what have we done here is that xr is hard to solve using a single perceptron but I can hookup several of them and I can solve the problem right, so is that computable using a single perceptron yes or no that is computable using a semi perceptron that is computable using a single perceptron huh first one is not yeah so I am just giving people a chance to change their minds you need to for each of them one for the not do you need to do a not can I feed in the $x_2$ to do that can you do or not okay so if I can do $x_1$ and $x_2$ is it sufficient in invert the weight on $x_2$ to get $x_1$ and not $x_2$ you think an answer anyway so the point is essentially what I have done this I can actually build a perceptron that can compute $x'_1$ I can build another

perceptron that can compute $x'_2$ so take these fill it to another perceptron which will actually compute right.

So essentially we have somebody sink another so essentially we have found they have shown that I can actually put this perceptron in layers and I can solve what was originally thought to be a problem that cannot be solved by perceptron right if you remember I was telling you in the beginning neural networks are very powerful because they are all hooked up in a very complex network right.

Individually they need on saw a very simple computing elements individually the neuron was a simple computing element and therefore you could not find the answer to $x\hat{}$ but then by hooking them up in appropriate Cascades you can find the solution to $x\hat{}$ right does it make sense, so sorry for the initial confusion on the feature transformation but now it should be clear right yes okay great.

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**Introduction to Machine Learning**

**Lecture 33**

**Prof. Balaraman Ravindran**
**Computer Science and Engineering**
**Indian Institute of Technology Madras**

**Artificial Neural Networks II –**
**Backpropagation**

Now we move to artificial neural networks I suppose to looking at neurons.

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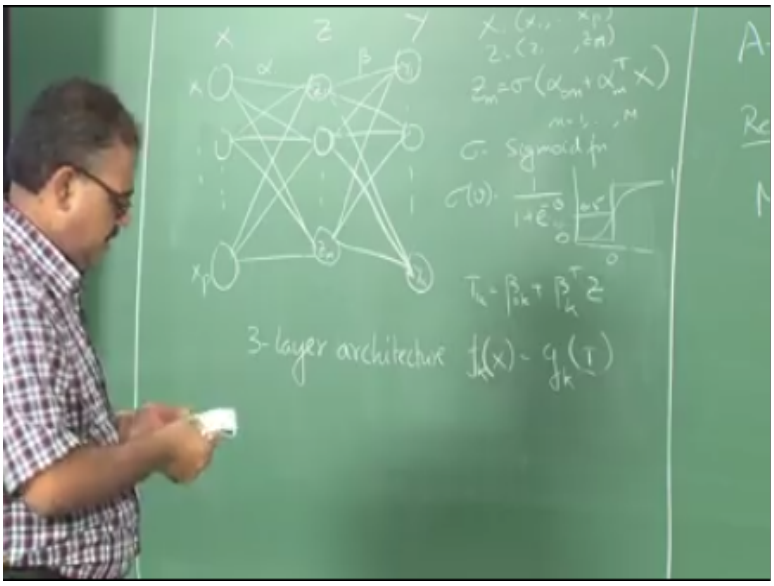


So we are going to start hooking up neurons in multiple layers like this and the way we are going to train this is using gradient descent right, where we are going to Train this neural network is using gradient descent but what is the problem with using gradient descent for perceptrons the

non-linearity right. So we had a threshold function which was not differentiable for the Adaline we got around it by getting rid of the threshold function altogether as I said we will use a linear output right.

So what is the problem in using linear outputs in multi layers multi-layer perceptrons so if you think about it let us call the first layer weights α and the second layer weights I will call them β right, so if you think about it so what you are producing is some $\alpha^T$ x is the output here if it did not have any non-linearity if I just use a linear neuron the output from here will be $\alpha^T$ x right so Z will be $\alpha^T$ x and what will go in here Z so Y will be that which is equal to just like having one layer of weights given by α β.

There is no point in doing all this layering so if you only have linear neurons then I do not get the power of doing all the layering and I might as well have done a single layer of neurons, so I really need to do something nonlinear in the middle I need to have a threshold function for me to get the power of layering right so the threshold is actually needed but whatever we did that if I did kept it as linear neurons run into trouble great, so we need the threshold so how do we get around the fact that it is.

So I am going to say that okay now you get the drill Y1 to Yk. Yeah, so sigma is a. So sigmoid is like a soft threshold right, so I can throw in a slope parameter here which I have not done that will give me different rates at which this sigmoid will assign so the actual threshold will be that way sigmoid will give me a soft way of doing the threshold the nice thing about the sigmoid is it is differentiable. There are many choices that you can have for the non-linearity a differentiable threshold function.

So the sigmoid is one of them so the thing with the sigmoid is that it will be between 0 and 1 so if you want it to be between -1 and + 1multiplied by something, is it? That will go from – 1 and + 1 right so you could have different choices, so for the time being I will stick with the regular the sigmoid that we are familiar with okay so each Z here, will be given will be given by that

expression right so T is the quantity that you will see that goes as the input to the sigmoid in the output neuron Z the output from here.

From the middle are a hidden layer this one not one of this one 0 m $\alpha_0$ m + $\alpha$ $m^T$ x can you zoom into that a little bit. At back, can you see it better now? People at the back who did not complain the fonts are small can you see it better now? So what are you doing on the laptop actually seen the video feed of this or something right so that is the output of the first layer okay and the output of the second layer is given by some other function G acting on this input T okay right.

So like to see if anyone notices something funny? Actually all can be different. Each one of these is a unit like this of that like that right, each one of those it is a one block this evil there are a good point okay yes there are three layers and this is sometimes called the standard three layer architecture okay but the first layer is really a fake layer this layer is really a fake layer it just takes x1 and gives x1 out on all the outputs okay, this one takes x2 in and gives x2 out on all the outputs okay this is this is really not a neuron okay so sometimes I like to call this as a two layer network because for me there are two layers of weights okay so it is a two layer Network right but in the literature for some reason this is called a three layer architecture.

So this is called the input layer. this is called the output layer okay one in the middle is called the hidden layer, so why is it called the hidden layer because I do not see the outputs of that layer directly okay so they are called the hidden layer so this is called the standard three layer architecture but there are other ways of doing it where actually you take outputs from the middle layer as well right and we can do and we can have inputs feeding into somewhere in the middle so you can have all kinds of craziness okay.

So the standard architecture is EC and we will stop with that okay I am not going to going to all the crazy neural network architectures are out there so you might want to take another course on ANNs specifically, if you want to I know more about thee all the crazy architectures outside so there is time permitting I will come back and do something very quickly at the end of the course

not today but this is the standard architecture will stick with okay right so still people have not told me is there is something odd about this yeah.

Lastly it always has to be linear not necessarily but it did not also be sigmoid that is why I written it as decay, so the last layer could be sigmoid it could be linear right when we do you want the last layer to be linear when we want it to be linear is when you want to do regression for sure right when you want it to be a sigmoid is when you want to do classification and still people have not told me what is odd about this I am assuming people are thinking but I am waiting for the answer.

So that I can go that is why I am stalling yeah, so why did I write this directly as $\alpha m + \alpha m^T x$ $\alpha_0 m + \alpha m^T x$ but here I split it up into $T_K$ and $F_K$ sorry good point, but why is it TY is it not $T_K$ so if I am doing regression I might as well do it $T_K$ right, so usually I because my regression my regression variables right if I am doing multiple output regression okay I am not talking about multiple input regressions multiple output regression my output variables are typically taken to be independent right.

So I mean, so what is the what value I predict for one I will usually does not affect the prediction I make for the other right so I do not know if you read the book I did not I did not talk about multiple output regression before this but if you read the book they would have actually told you that you can do that regression independently right but if it is classification really they are not independent.

If I am going to be outputting class probabilities right they cannot be independent right if I am outputting class one probability is higher than class two probability necessarily has to be low okay so I had to say do some way of normalizing the outputs to produce probabilities right, that is why I am saying that this will operate on the entire T, I can produce a output probability vector so in case of classification you need to operate on the entirely so we will come back to that in a

second right for classification. We will do a soft max like we did for logistic regression. Do you remember that so E power.

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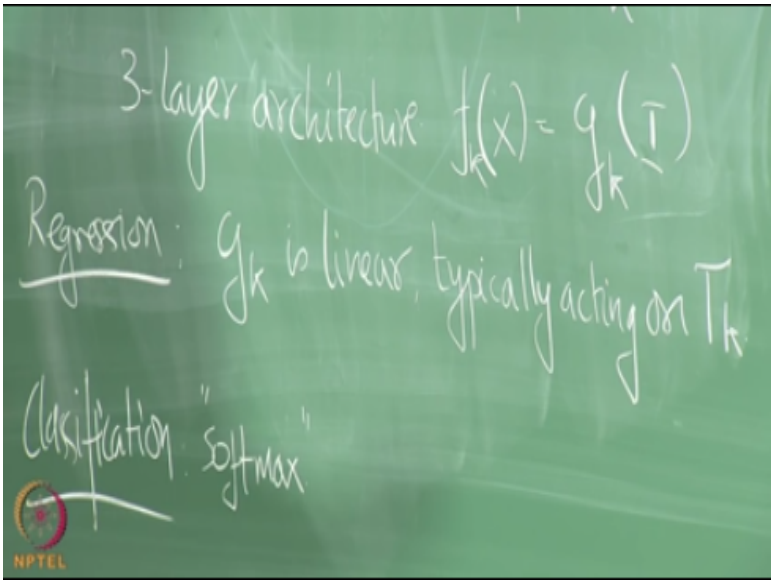


Yeah so I need the ∑ over all the outputs right.

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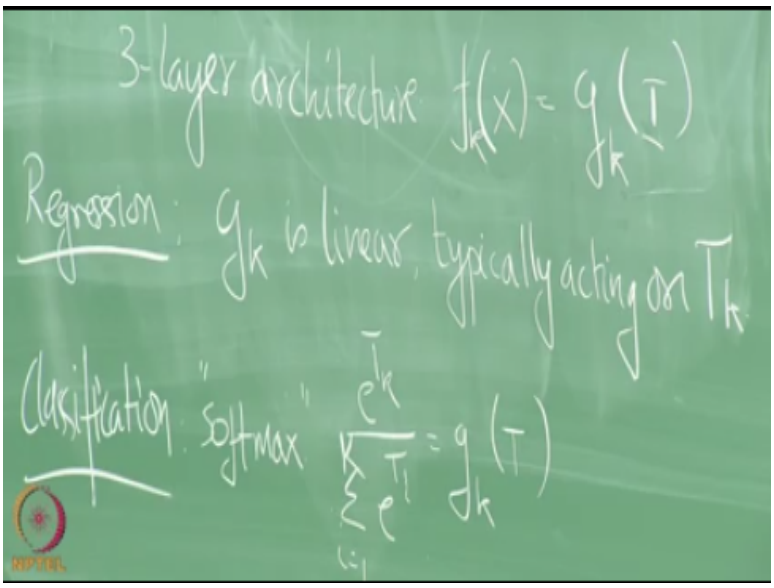


I need a $\sum$ over all the classes in the denominator and that is why my gk operates on T so this will be the soft max thing, so $e^{Tk}$ divided by $\sum^{eTi}$ .At this will be the gk of T so what will happen if it is a single class I mean 2 class problem, it will reduce to a sigmoid it reduces sigmoid I can pick one class and have that output as the sigmoid and I can just say that okay, if this is greater than 0.5 then it is that class if it is lesser than 0.5 this other class correct so this is a 2 class problem that gk will reduce to a sigmoid.

So if I am doing classification with only two classes I can straightaway keep a sigmoid as my output, output neuron and then I can solve it one how many neurons I need one right suppose in

solving a three class problem how many neurons I need as output 3 and you can always have the third one as dummy and then you can say I am going to do 1 - the sum of the rest right but if you are going to do this right, so I can always say that okay I am going to have three outputs which will give me the probability of each of those three classes okay.

That is typically how it is done for two you just have one right but then for more than two classes you typically tend to have as many at the output is there any problem for doing that with doing that think about it I am not going to give you the answer right okay, so how do we fit the neural network parameters now and I have two layers the first layer m into P + 1 parameters am for right, so m for each of the hidden neurons right P for the $\alpha_m$s and + 1 for the $\alpha_{0s}$ and in the second layer okay likewise, so I have that many parameters that they have to fit so I have to find all the $\alpha$ and all the $\beta$ so why do not you do more layers than this where did I stop with only three layers yes so empirically people observed that it is harder and harder to train why does it become harder and harder to train.

I will tell you in a minute okay but there is another reason for stopping with two layers right, so if you think of these as some kind of Boolean gates right if you think of the neural networks are some kind of Boolean gates it turns out that I can implement any Boolean function just using two layers of neurons except that the branching will become very large right but I can still implement any so as long as they do not give you any kind of gate with right I can have as many inputs coming in to a neuron.

As I want and I can implement any Boolean function in just two layers of neurons so why is that all of you know that right you can write midterm expansions right, so all of those things you know that and so you can essentially implement it in two layers of neurons and people thought oh, two layers is sufficient is a universal function approximator I can represent any function I want so let me not even think of what higher layers so that is one school of thought so people stop there but then there are others who are interested in going into more complex neural networks.

Because they did observe that when they got it to work okay adding more layers worked well so people kept at it and they made it work more robustly and so there like I was telling you that third wave of neural networks is all about having deep networks where you have more than two layers.

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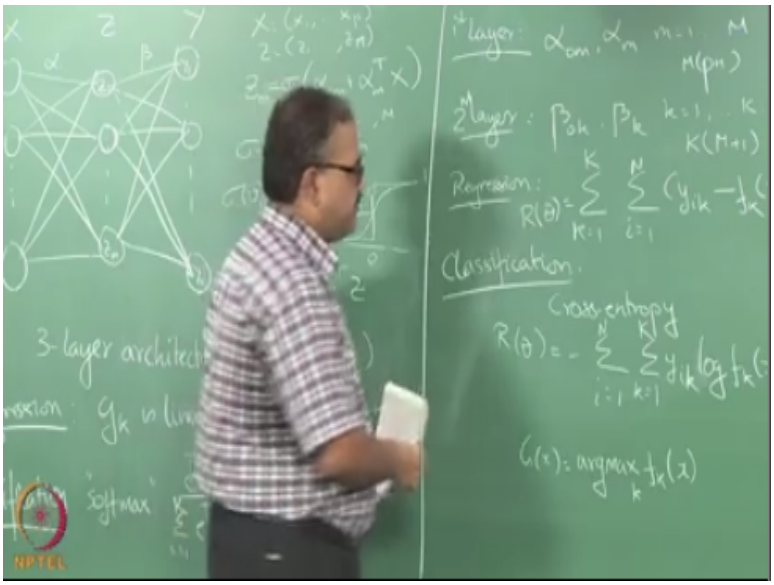


So regression what will be a loss function so my I am going to say define my regression loss for the parameters θ what our θ here all the α and β okay, so I wanted a single notation for the parameters instead of saying α β everyday so I will say θ is given by essentially this quiet loss that should look really familiar to you guys by now because they have been writing squared loss almost once every class if not often so what about classification what can we use for classification.

But 0/1 is incredibly hard loss function to optimize, right? So you could use squared error itself right so what is the rationale for using squared error same rationale that we use to linear regression right, so $Y_{ik}$ is an indicator variable that this one that gives you the probability of this particular data point being class K right and you are trying to fit the probability anyway that is what you are trying to fit and therefore for every data point you can take its probability of being class case 1 or 0.

And then you can try to do some squared error and try to make a prediction okay that is one way of thinking about it the other way is to use what is called the cross-entropy error or the deviance which is related to whatever we did in logistic regression, so here I will define my so we will all put the problem the actual class table as the one that has the highest $F_K$ of x like we did in the discriminant based classifiers except that $F_K$ of x is no longer a simple discriminant function right.

And so this is essentially a error term okay that stands in for likelihood that we maximized earlier, so and to optimize this error term we will essentially train the neural network using maximum likelihood right so I am not going to go there so we will look at a more popular more mechanism for training neural networks which essentially looking at the gradient the squared error and do gradient descent okay.

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**Introduction to Machine Learning**

**Lecture 34**

**Prof. Balaraman Ravindran**
**Computer Science and Engineering**
**Indian Institute of Technology Madras**

**Artificial Neural Networks III-**
**Backpropagation Continued**

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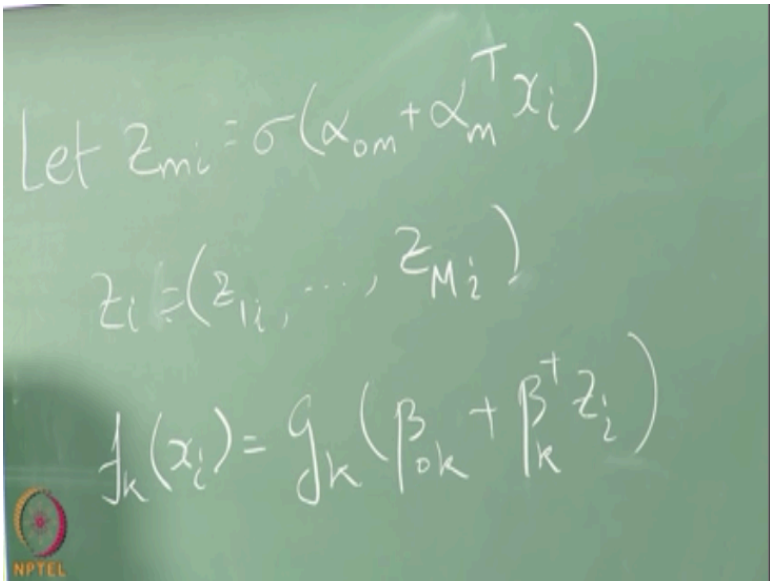


So let $Z_{mi}$ correspond to the output of the $M_{th}$ unit in the hidden layer corresponding to the $i_{th}$ input this is not the $i_{th}$ component of the input corresponds to the vector $x_i$ right so it is the $i_{th}$ input in my training data of n elements okay and I am going to say that and $Z_i$ corresponds to the, the entire activation of the hidden layer for the $i_{th}$ input. it will find so far right.

So now we got rid of over what did we get rid of here? the T right so this is what I was saying in regression Gk is linear and typically d is acting on TK right and so this is acting only on TK so this whole thing is so because we are only talking about the regression, I got rid of that this will make our life a little simpler when we write the write the gradient so I am going to take the basic I am going to use gradient descent right so I have squared error I am going to use gradient descent.

So I am going to take the derivative of the error with respect to the single output layer weight okay this is a weight that runs from some neuron $M_{zm}$ right to some output K right so that is $\beta$ okay just this one, one weight I am taking here right I am taking the derivative of R with respect to that one weight okay is the setting clear right so I am taking the derivative of R with respect to a single weight here.

Let us just designate that as $\beta_{KM}$ so what will this be equal to yeah okay let us do it in a slightly simpler fashion so I am going to assume that each term inside is denoted by RA then I just do the summation over all i okay so that way I do not have to write the summation over all a everywhere so I am going to say this is del RA by del $\beta$ right if you remember the earlier that what we had the thing that I erased here there was just this right $Y_I$-f of X into X was what we had earlier right.

But the input in this case is actually ZM right if you think about what is there on the other end of this weight right so the input that comes from here is actually ZM right so mathematically if you think about it just let them $\beta_m$ okay right so that is what is happening so essentially that is what you are going to get so the i indicates that you are considering it only for the $i_{th}$ input right this is clear so far we just then just taken a derivative right but exactly the same computation that we did earlier the only new thing here is they are the derivative of GK earlier.

We did not have that because we are assuming that GK was linear so GK is linear this will again vanish now comes the interesting part they will just disagreed some the single input layer wait we will consider that so I am calling it alpha $m_l$ how will I take the derivative of the error with

respect to $\alpha_{ml}$ you look at the error $\alpha$ does not appear directly at all it appears indirectly so what is the best way to do this name this using the chain rule.

So $\alpha$ is going to affect the output of the hidden layer right and the output of the hidden layer is obviously going to affect the error right so I am going to take the output of the hidden layer right so I am going to chain it through the output of the hidden layers are going to take del $Z_{mi}$ by del $\alpha_{ml}$ and del $R_i$ by del ZM right so one thing to note is that $\alpha_{ml}$ is going to affect the output only of ZM right it is going to affect only ZM.

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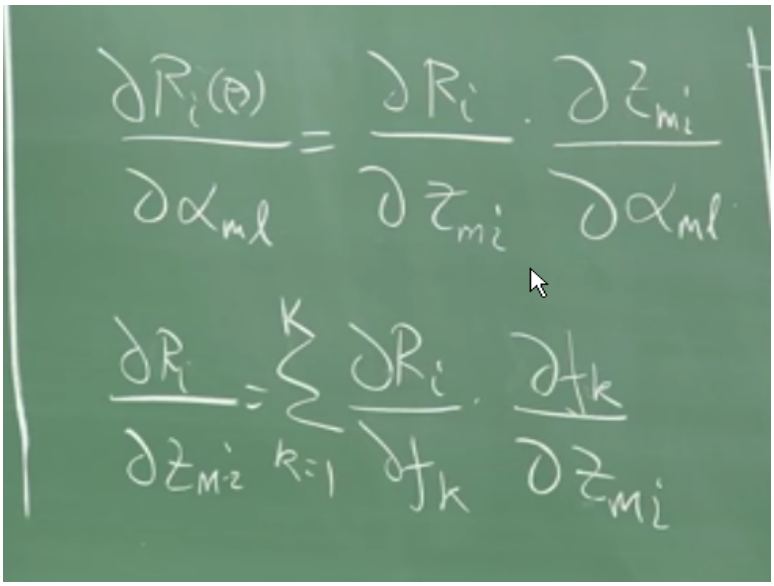


So I just need to chain through ZM okay is it clear so then let us do each one of these in turn so this is rather easy so is that you have that already so what is $\partial Z_M$ by $\alpha_M$ what if they did be more consistent okay that makes sense right the derivative of $\sum$ yeah can you zoom in so the derivative of $\sum$ times $x_{il}$ right so $\sum$ prime of $\alpha$ transpose Xi plus $\alpha^o$ into $X_{il}$ so that is essentially the, the derivative of Zma with respect to $\alpha_m$ it is straight forward differentiation if you are having trouble with it I do not know now is the tricky part so I am looking at $\partial$RA by $\partial$ZM right.

So what is ZM it is the output from here right but unfortunately this K goes to all the output neurons right so ZM can affect the output through all the output neurons okay so far there is been a single path that we have been considering but at this point we really have to consider all the paths of reaching the output from M okay.

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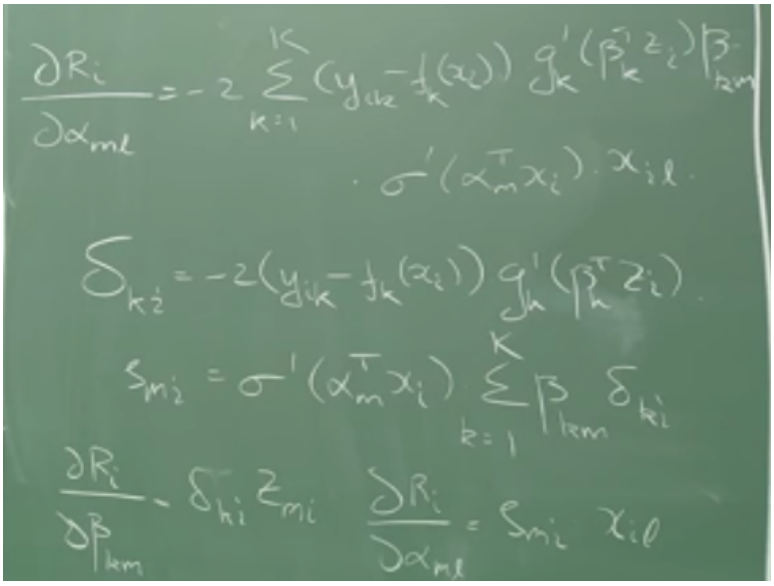


So what we really have to do is look at okay so, so ZM can affect RI through FK right so the derivative of FK with respect to Zm and RI anybody with respect to FK that is a chain rule again do this over all K because I can have multiple parts of reaching the output so what is ∂R/∂K okay ∂R/∂K it may which should be able to rattle it off just the derivative of GK so putting everything together.

I can write that is a big expression and I did nothing I just took this and wrote it here I took that and wrote it there okay I just took the product of the two terms so what we will do now is just to introduce certain simplifying notations let us think about it I have made my job a lot simpler so that is this term ΔK which ever define so ∂RI with ∂β is essentially ΔK into Zmi right ∂RI with $\partial\alpha_{ml}$ is essentially $S_{mi}$ into $X_i$ that is the Δ part right.

And there you have a β and then you have your ∑prime so this all put together gives me mass $S_{mi}$ so there nothing you just applied chain rule and done some manipulation to simplify this right if you go back and do it again okay you will find that it is very straight forward gradient computation but it took people a couple of decades to nearly a couple of decades to realize that they could do something as simple as this chain rule.

And apparently this technique which is very popularly known as back propagation so why is it called back propagation so when you take the input right and you compute the output that you are propagating the values forward through the network right but when you are updating the gradients so if you think about it so what you are doing is first you are computing the Δ's right and then you are propagating the Δ's back through that weights β's right.

So essentially what you are doing is Δtimes β it like when you are going forward you do x times Δ and Z times β right so here likewise you are doing something like Δ times β right so this is something like a back propagation of this Δterm through the weights so as to update the first layer weights right so that is why it is called back propagation okay so the forward thing is whatever you do this, this is the forward pass okay and the equivalent backward passes are given by that right so the actual equations are right.

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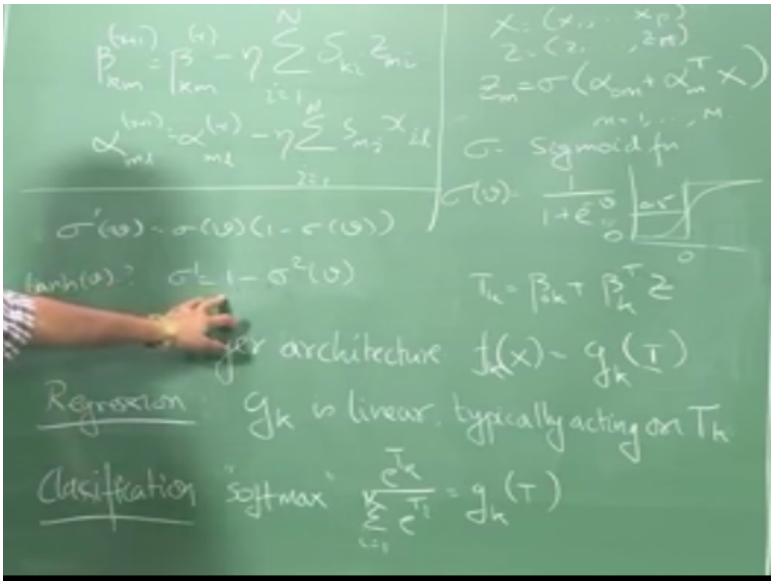


So we still left some things in there so I left a Gprime and a ∑ prime and so on and so forth so if G is your linear function great right what about ∑ prime ∑, ∑ is the sigmoid function then now you can take that derivative of the ∑ with respect to X and that is what you will get and if it is at tan H right instead of the sigmoid if I use the tan H function then my ∑ prime will be $1-\sum^2 V$ you can work it out but sadly easy differentiation always people get thrown off my back propagation but it is really nothing but differentiation and a lot of algebra right just manipulating things around it is nothing more than that everyone knows the chain rules right that is it that is it, it is just a chain rule.

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**Introduction to Machine Learning**

**Lecture 35**

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**Artificial Neural Networks IV –**
**Initialization, Training and Validation**

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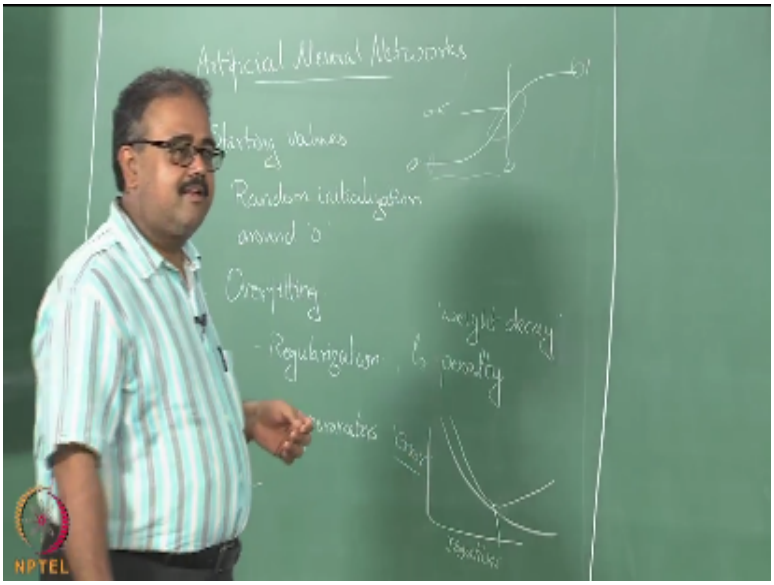


The first thing concerns the starting value of the weights right, so you have all this α and β you have this whole set of parameters in a neural network right, so we talked about gradient descent but gradient descent starts at some point in the weight space right, so you need to have an initial guess for what your α and your β should be right. So what should you do what are the good guess? set them all to1 is it yeah, so setting all of them to the same value whether it is one or

sometimes people say this 0 right setting all of them to the same value is usually not a good choice right.

More often than not you will end up in some weird part of your gradient space and you will find it hard to get out right it is very rare that you are going to come up with a actual solution where all the weights are the same right. So if you were going to start off with a solution where all the weights are same right it is going to be hard to specialize right, so typically what you do is you do use random initialization, but there is one more constraint let us be feeling really lonely one more constraint to the random initialization.

So what do you think that would be the constraint yeah of course you do not really want to have infinite weights yeah, so but then what should the min and max is, so you really want your weights to be rather small okay? So just think about what is the implication of having really small weights what is the implication of having really small weights, so you remember your sigmoid right, so if your weights are really small where do you expect the outputs to lie? You know your $\alpha$ transpose x or your $\beta$ transpose z will lie somewhere in this region because this is where 0 is right.

So this is 0.5 or 0 depending on whether you are using tan H or the sigmoid, so if you are using the sigmoid, so right around 0 you will have this 0.5, so if you look at this region this is almost a linear region correct. It is almost a linear region so I would really like to start off my network so that most of the outputs of the neurons are in the linear region. Why is that? Can you think of it and you have enough information to answer this question. The gradient will be larger right so the gradient will be larger if you are somewhere around here.

So even small changes in the input space or small changes in the weights will actually cause a large change in the output right, so if you end up going somewhere here or somewhere here right you can see that you are already saturated right, so you really have to drag yourself all the way here to see any change. So if you are somewhere in the middle you are more sensitive right to what the input right you are more sensitive to the weight changes right. So all of this helps you learn more rapidly, so this is one of the reasons you start off with random initializations around 0.

Of course you can make everything zero but then that will put you in a very weird part of the part of the search space right. So it is good to have some kind of randomness so that each wait can specialize to different things okay. It used to be the big bane of neural networks over fitting so why is that the case? You know all of you remember what over fitting is right, sorry yeah so I mean people remember what over fitting is yes, the training examples right there essentially you are fitting the parameters very closely to the training examples.

So you are not able to do any kind of generalization to unseen examples right and the reason why neural networks do this over fitting is exactly because they have too many parameters you have so many weights here. So if you remember we actually counted the number of parameters in a neural network right you know that M times 3 +1 + whatever K time's n / M + 1 well that is a huge number of parameters. So it is very easy for you to over fit so you have to be careful about it, so there are two ways of avoiding over fitting.

So can you think of what are the two ways of avoiding over fitting one we already know regularization right, so one way of avoiding over fitting is regularization, so what you do here is you essentially add a quadratic penalty for the weights right so you do a norm α squared +norm β squared and then you try to find the gradient with respect to that and then you try to minimize things, right it becomes a little bit more complex and if you add a squared error penalty right if you add a squared error penalty it is sometimes called weight decay right because it makes your weights go towards zero. So sometimes called weight decay, so can you add a $l_1$penalty

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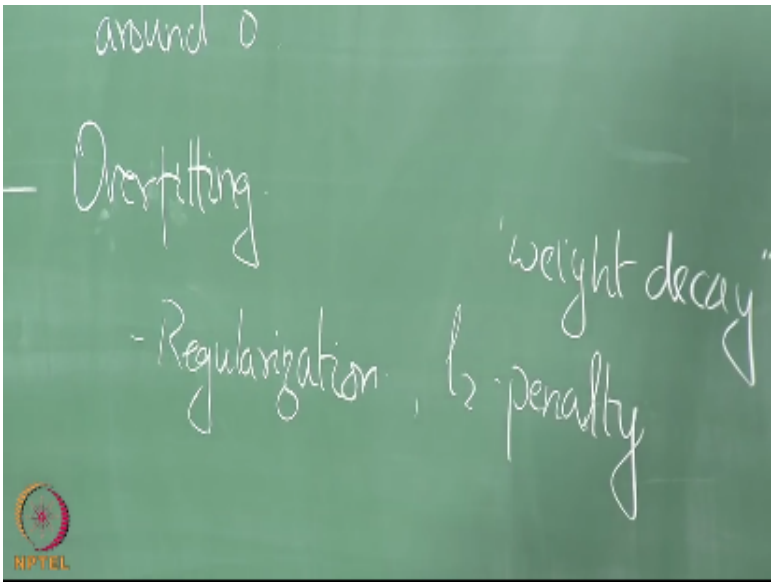


You could you could add anything I mean so it is it just makes it a little bit more complex but the question is does it induce sparsity right, they said another way of avoiding over fitting yeah you can tied parameters together to avoid over fitting that is good but in which wherever is new tied together? In fact one of the ways that deep learning actually has been made efficient despite doing this tied of parameters right but then the architectures that look at are very complex right so you could tie parameters together and try to reduce this okay.

So let me put that as a different kind of thing but it is actually a form of regularizing but it is yet another approach which is a purely empirical way of doing things which is to do what is called validation right I actually mentioned this in one of the earlier lectures right, so you train on a training set and then you have a validation set and then people remember I drew even a picture right, so as you are training so the error on your training set keeps going down right but the error on the test set or the validation set will initially go down, at some point it will start going up right maybe not that dramatically but go up nevertheless.

Like that right so the point here is where your right solution is okay, so I am putting it in quotes right solution okay, so what is the x-axis and what is they-axis on this figure? Whatever is your mission of error it could be miss classification error or whatever right, so fiction you ideally want miss classification error if it is the regression it is a prediction error whatever and the x-axis is

usually iterations right but you could also think of having a figure like this for complexity right. So you could say that I am going to keep adding more and more neurons right or more say I can keep expanding my M right I can keep adding more neurons in the hidden layer right.

I cannot change the new neurons in the input layer I cannot change in the output layer usually right because that is their depend on the problem that I am solving right when I can keep increasing the neurons in the hidden layer right but then these are more or less you know standard techniques for doing, I mean for avoiding over fitting not necessarily tailor to neural networks right. So you should remember that there was about a decade and a half when people worked a lot with neural networks in between right, they came up with many techniques for avoiding this kind of over fitting.

And they have explored many variations on parameter tying right and also many different kinds of regularizing so there are some really interestingly named algorithms for avoiding word fitting. so one that I particularly like is to be called optimal brain damage, essentially the idea was to remove weights from the network right so you train the neural network and then try to find out the sensitivity of the output with respect to certain weights okay. If I am changing this weight how much do the outputs change you know how much does the error change right.

So weights that have low sensitivity or right or yeah weights that exhibit low sensitivity on the error right well then remove and then you just retrained we keep doing this. So that way you are removing the number of parameters reducing the number of parameter heavily but we are not affecting the output too much. So like that there are many variations on it but these are the three things right that we have to think about.

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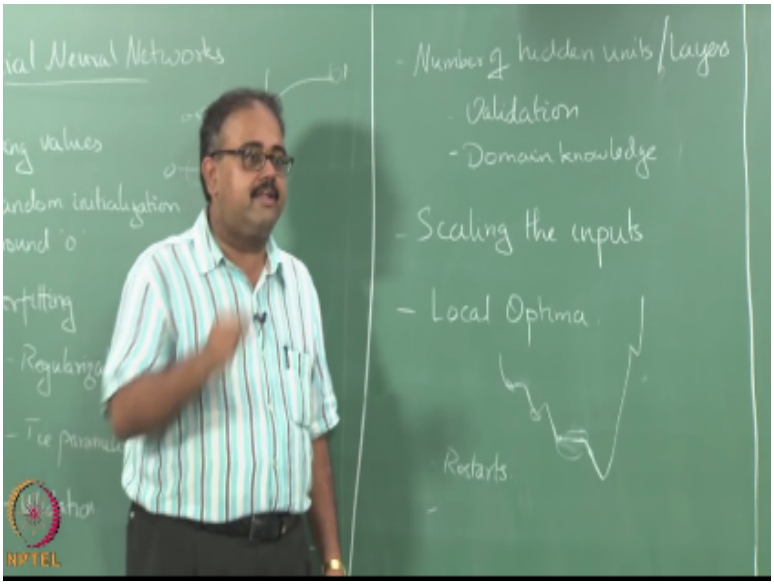


Related to the over fitting question you see how do you figure out the number of hidden units and layers. Like the very expensive way of doing it is to do a similar validation kind of a setup right keep increasing the number of layers and then our number of neurons and check that out it is incredibly hard right and so people came up with automatic pruning techniques right, people came up with ways of growing your neural network. So they start off by having one neuron so something very similar to your forward feature selection or what is the thing a stage by stage wise selection right, so people remember stage wise feature selection what did he do?

At stepwise stage wise or something different see you later right so you could do the same thing with neural networks right, you start by training a single neuron all right and then what you do is once that neuron actually starts making some predictions right then you train another neuron that actually nullifies the prediction error right. Now you know a third neuron that adds up both of these and gives you the output right, so you could do something like that right so you do not have to make a decision as to how many neurons you are going to put in from the beginning right as you go along you just keep training more and more.

But the problem is such a network was that it will not look like your layered architecture right I will start off with 1 neuron okay that will give an output then I will add the other neuron then they will go so all of this gets the input directly, then I add another neuron right so the layer

architecture is gone no wall right. So now how many layers this is our two or three you know so this neuron seems to be at the third layer right but it is connected directly to a neuron which is certainly first layer because it is taking inputs from here.

But then well you do not have to be very dogmatic about having the standard three layered architecture if you remember when introduced the standard three layer architecture I said there are a lot of different deviations from this that people have proposed right and will not be looking at most of those in detail. So these kinds of networks where you are actually trying to minimize the residual at every point they are called cascade correlation networks, so there are. So the most statistically sound ways to just do validation, what is the other way? To do it is kind of a cheat it is slightly better what can you do you can take an educated guess using domain knowledge.

I said you have some information about how complex the system is and then you can use ideas from that you can then try to see okay one layer two layers three layers right, so a lot of deep learning network that nowadays happen essentially it is more empirically driven you try one layer okay see what is the best you can do right see thoughts the best in performance, as you can get and then you try to add another layer and see if we can improve that another layer another layer until you are happy that you are performing well right. Of course you just cannot train the network into the ground you have to always make sure that you are not over fitting it but you can still this right okay.

So I did not explicitly mention this while talking about the numerical training but you can kind of imagine, wherever that I am going to be using the data as it as a real valued vector right I would have to worry about scale, so if I have one variable that has a very large range another variable which has a very small range, at the variable with a large rate it is obviously going to dominate my gradient computation. If you remember the gradient has a X the x ml component to it right that is the input variable is part of the gradient. So if the variable some of the variables can have a very large range and some of the variables might have a very small range and the large range variables will dominate the computation.

This numerically by being large they are going to dominate the computation right, so we do not want that to happen whether they are actually needed or not just by being numerically large right they will dominate the computation, so we essentially make sure that all the variables will have the same range right. So we talked about this in couple of other scenarios also but in this case again it is important so this is something which people typically forget. When they are using either neural networks or SVM's you try you take the raw data right and you just try to run it through a neural network or run it through an SVM and then produce a classifier right and quite often things do not look work that well right.

You might find some reported results that are much better than what you are getting by using is SVM, nine times out of ten okay the reason to fold with SVM is a two-fold with neural networks one thing you forgot to scale the input okay, the SVM's what happened? So you have those kernel functions we talked about right you forgot to tune the parameters of the kernel function you just took the kernel function as it is and you are trying to use it so that the performance will be bad, so you have to tune the parameters of the kernel function and you have to scale the inputs, if you do not scale the inputs sometimes the performance can be arbitrarily bad.

And this is a problem which SVM do not have and that is one of the reasons they became so much more popular than your networks in late 90s and early 2000s right, so the neural network error surface is fairly complex. So what do I mean by error surface? So what is it so error surface what will be the x axis the y axis the z axis whatever can you describe mathematically, what the error surface is? with respect to what aha on the what parameter is not the inputs okay, so the error surface is something which people have difficulty okay there is areas on I am making a fuss out of it.

The error surface is the function of the error with respect to the parameters okay, so as I change my $\alpha$ and $\beta$ how does the error change okay, so that is that is the error surface that we are talking about right and so how does the error surface how will it look like for the case of SVM and minimizing something quadratic they are right in terms of $\beta$ right, so it is actually very nice quadratic thing, so it always has a single Optima right when the optimal hyper plane formulation

the very nice thing about it is it has got one Optima and then if you run the optimizer on it you will always get that solution right.

The error surface for neural networks if you think about it it's got those stupid exponents in there right your sigmoid there is the derivative of, your sigmoid is in there so the error surface is going to look incredibly complicated right. So it is going to have lots of little valleys right the error surface is going to look something like this, ever even look something like this okay, so if I am doing gradient descent I might come here and get stuck, that looks like I mean whatever direction I tried to go there is increasing. So I might say okay this is the good place to be so I might just stop there right.

I could get stuck here what about here huh very slow or not at all because the gradient is 0 I mean if you are in the middle of a point the gradient is zero because it is flat it is flat, I declared that to be flat okay and so the gradient is zero at that point right and there you go okay and so you might not just move right you are essentially drifting around there you and whatever happens you are not able to make any progress. So the error surface can become really complicated like this right and this is just on one dimension right, they say this is a no one it is a single neuron with one input that is what we have drawn.

So imagine this generalized to a very large dimensional space right mp + mk dimensions right, so the surface can be really complex and again the plethora of solutions were getting out of local optima right, so we are not going to get into most of those and let us tell you one very practical way of doing it essentially do restarts right. So you start off with some random initialization close to 0 right you do gradient descent until you do not change weights very much. Remember those weights right remember those weights and remember the performance. No reinitialize the network again close to zero random weights close to zero.

You say different random see please all right and then rerun the experiments right and again you will go off to some other optima remember those and keep doing it. There are other techniques which people use right, so they can make the, you know there is something cleverer gradient descent techniques right which all of you to get over these local optima not all of them but at

least some of the shallow or local optima it allows you to get over easily. So for example this is a shallow local optima right with a little bit of effort I can actually get over and how do you provide their effort.

So think of it from a very dynamics perspective, so people have added something called momentum right, so if you have been moving in a particular direction I have been descending little bit a little bit a little bit okay do not stop just keep going in the direction for some more time right that is momentum right. So in this case these kinds of shallow things you can get over you know the gradient has become a slowed down significantly right this becomes 0 here but I will still be going in the same direction I went for a little while longer because I have momentum it is going to take me forward.

So that is going to get me out of this little valleys but if you are in a deep valley then still cannot get out right but so these kinds of tricks help right and then more recently with all the with deep learning, that one of the reasons that deep learning is become so popular most people do have very powerful gradient based techniques which allow you to navigate the error surface more efficiently a lot to avoid local optima, but allows you to navigate the surface more efficiently right good. So any questions so far, no why well I am going back to my zero yeah when I start restart a little again be small weights right.

I will be very far away from this is optima I have converged to after a lot of training, you maybe not see that, this is one thing which you should get your hands dirty then you will see what I mean. Even small changes in the starting weight configuration can lead you into very different Optima that so there is surface or so complex right and remember I am not just moving in one direction or the other I have a very large dimensional space in which I am moving right. So even though I am taking and I am constraining it to be around zero right, the volume that I can actually start in is very large because of the high dimensionality of the weight space right.

So and each random starting point can be very different because it is also possible that you start in the same location or start very close to the same location. I will end up with the same Optima but that is probability of that happening is very small especially with large networks. So if we do

the D start will actually end up with somewhere else. Any other questions so we have till this point we have the exam rate just checking yeah, so that is the question nobody asks how many times do you restart right.

There when you have a budget you just say that okay I am going to restart this many times right and yeah you might have actually re absolute minima but you may still be doing research that is one of the reasons I told you to remember the weights right, it could very well be that the best weight best solution, you got could have been oh the first one and then all the fuel further restarts that you do could actually be leading to worse collisions right. So I am not guaranteeing that we do a restart we will get a better solution the restart just allows you to explore different local optima and pick the one that is best.

It all depends on your budget right I mean if it is it is as expensive as to train the network the first time around right and it is really expensive to train the network if you are doing deep neural networks because the number of parameters are really large runs into several hundred thousand right and therefore doing are start is expensive, we do fewer of them and I also tried to come up with other gradient descent techniques, that allow you to avoid local optima. The whole idea behind simulated annealing is that you would want with some probability of ignoring the gradient right.

So what the whole algorithm here says at every point follow the opposite direction of the gradient right, you would descend the gradient direction you find which is the maximum ascent direction you descend it right. The whole idea behind simulated annealing is to say that no I allow you to ignore the gradient, you can move in another direction in fact you can move in the direction off the gradient also if you want right. So the gradient opposite direction of the gradient is a choice that is given to you can choose that or you need not choose them. As the number of iterations become larger and larger the probability of you choosing the direction of the grade up the gradient direction is going to become higher and higher.

This is essentially what we call the temperature parameter the temperature parameter is very high right if you can think of this particle that is going to be jumping all over the place right, so if the

temperature is very high you can move in whatever direction you want you are not necessarily constrained to following the gradient direction and that is the temperature becomes lower and lower you, then you are constrained to follow the direction of the gradient .So the reason is called temperature is because it is actually used in modeling physical systems right, so the so if you look at the Boltzmann distribution which people typically use in this context that is actually a parameter called temperature.

Which behaves very much like this right, so one another way to think of what the simulator annealing will do to your error surface it is like it will pull your error surface is flate if the temperature is very high it is like your inner surface is flat it essentially means I do not have any gradient information. I can move any which way I want right whatever direction I move it looks the same then I can this move randomly and then what happens I slowly start regaining the shape of the inner surface.

So what will happen is first the deepest dink will form okay the shallower ones will still be farther away and the deepest thing will the first tip that will appear in my error surface and likewise as I keep cooling it cooling it cooling it will completely go back to the original inner surface. That is one visual way of thinking about it I mean there are more formal ways of explaining why that visualization works but I do not want to get into simulated annealing today but that's another way of avoiding local optima okay done.

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**Introduction to Machine Learning**

**Lecture 36**

**Prof. Balaraman Ravindran**
**Computer Science and Engineering**
**Indian Institute of Technology Madras**

**Parameter Estimation I: The Maximum Likelihood Estimate**

More of a Bayesian approach to parameter estimation,

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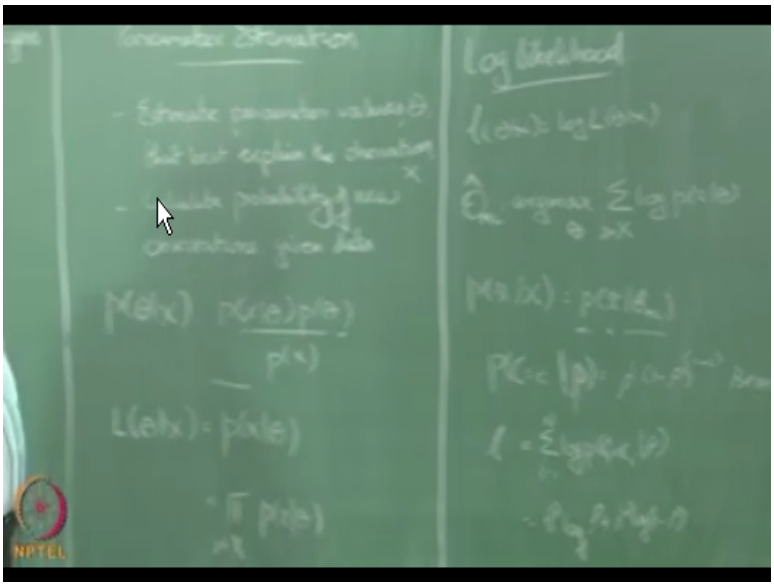


so I would say that there are two goals to us we want to estimate the parameter values right. That best explains some given data to us right. We already looked at this in the context of logistic regression right, so we assume some kind of a model that was generating the data and then we estimated the parameters of the model that somehow best explains the data right.

And we used the notion of a likelihood right we are just going to look at it again little in detail. And then go on and look at a couple of other ways of doing parameter estimation. The second thing we look second problem that we are interested, so essentially calculating the probability of new observations given the old training data right. So I am going to assume that the parameters are given by θ right.

The observation is given by X, so what I am interested in is per probability of θ given X right. That is all the familiar Bays rule; so what is that what is that prior, likelihood right. So I only looked at it right, so I am going to like write the likelihood does likelihood of θ given X right. So if I write it like that people get a little confused, so likelihood of the data right. I already mentioned this is likelihood of θ not data.

Its likelihood of the parameters given the data even though we write it as probability of x given θ. Is it clear? Why is it a function of $\theta_0$ of X? Because X is fixed in our context X is fixed right. I have given the observation X right I am interested in finding the parameter values θ. So the way I am going to set this up for a given X okay, for different θ. What is the probability of that X? Let us say I can consider five different θ for θ 1 what is the probability of x for θ towards the probability of X for θ 3what is the probability of X and so on so for.

So that gives me the likelihood of θ, sometimes people say the likelihood of X with respect to θ. In fact it is so widely used I do not know if it is even right to say it is incorrect anymore. Like pre pone a meeting, but so why are we interested in the likelihood why are we interested in the likelihood? Yeah! So, what I am really interested in is to find that θ that best describes the data right.

So essentially what I am interested in is finding the θ that has the maximum probability here. That given the X which θ has the maximum probability right, so the X is fixed right so this does not matter and if I really do not have any information about what θ is the best θ to start off with right. This is also irrelevant because it will be the same for all the θ correct, so if I want to maximize P(θ) given X all I need to do is maximize P(x) given θ.

Because this is constant that will also be constant across all θ all right, so it is enough if I look at likelihood right. So if I make an assumption that we make the assumption that my all my data samples are generated independently right as well right my likelihood as the product of the individual probabilities. Then we do not want product or probability so we typically end up using, people agree with that?

So suppose a new data point the X tilde comes what is the probability of X tilde with respect to X. Sorry, I mean given X sorry given that I have already been given some training data X okay. I am asking you what the probability of this new point X tilde is. What will be the probability right. In fact is exactly what we did in the logistic regression case, we found the maximum likelihood parameters for β maximum.

Likelihood estimates for the parameters β and then we plug them back in and say okay. This is how you estimate the parameters right. So let us look at a simple example, it is considered a simple coin tossing experiment right, so there is a random variable C okay. Which has some outcome lowercase C right, so lowercase C if it is one in his heads, if it is 0 it is tails okay. What is the parameter that I have did not coin tossing experiments probability of coming up its okay. Let us change the symbol still looks like P but it is a ρ so the probability of whether you come up with heads or tails right.

Given the parameter ρ okay, so what is the probability that should also look familiar? We already saw that right in the context of class label being 1 and class payable being 0 right. And the probability of coming up class 1 versus probability of coming up class 0 right. It is like heads and tails. Now the probability of coming up heads, well ρ power C is 3 well. If its 1 then it is ρ if it is 0which is tails since 1 -ρ okay.

So this is the expression in simplified form, I always have written I would have to write it as ρ if C=1-ρ if C=0 right. In stuff that I can write this using a selection function so what is this probability density called? Bernoulli yeah! Okay. The Bernoulli is so what is the likelihood when look like for each of the $I^{th}$ toss you would have an outcome lowercase $C_i$. So what is the probability that the random variable can be lowercase $C_i$ given the parameter ρ right.

And some this over all the right, so there are N one times head says occurred. Let us say and $N_0$ times tales has occurred right. Then I can simplify the summation as $N_1$ times log $\rho$ +$N_0$ times log 1-$\rho$ right simple enough. Now take the derivative of the likelihood equated to 0, and tell me what $\rho$ is right. So our common sense way of estimating probabilities from experiments is what toss coin N times find out number of times it turned up heads okay. Divided by N that gives you the probability of it coming up heads. And a turn out that is the maximum likelihood estimate assuming. That your coin is obeying a Bernoulli distribution.

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**Lecture 37**

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# Parameters Estimation II: Prior and the MAP estimate

Ok so far we assume that the whole motivation for doing maximum likelihood was hey I did not know anything about the parameters .Before I started the experiment right before I gather the data I did not know anything about the parameters suppose I did know something about the parameters. So what could you know about the parameters just stick with the coin tossing experiment give me something from the coin toss case yeah well yeah I know the high probability it is fair.

If you know it is fair or not that is not a prior information that is insider trading okay so with the very high probability and think it is it is fan right so he hands mean coin we look at his face I mean obviously he is not going to cheat me right. So I will assume it is a fair coin to begin with right so what I can do is I can have a prior probability of it being far being very high right now I can in fact think of having a Gaussian with the with a peak at 0.5 for ρ right. I can think of having probability of 0.5 for 0.5, for ρ being 0.5 there are two probabilities here all right do not get confused as a probability of the coin coming up heads and there is a probability of that being 0.5 right. So that is the prior probability we are talking about here and I am saying that I can think of it as Gaussian and our Gaussian is a good idea why not great probability is not only problem it can be even greater than one also mean either side is a problem right.

So what is a good distribution to do is useful you already seen that in your tutorials probability tutorials. Distribution that is β distributions limited between 0 and 1 in fact it seems to have been

invented for putting priors on probabilities. Right in fact it was so you can think of that as your prior right so I have some information about right I want to use that in my optimization right.

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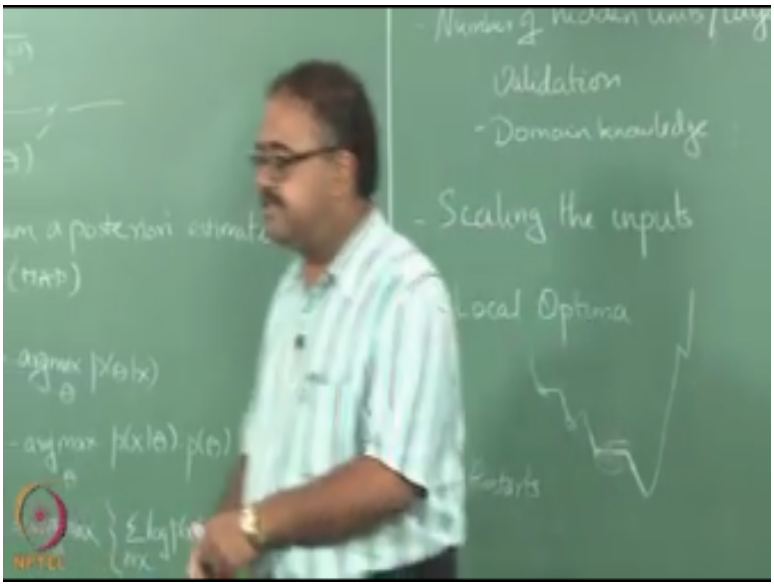


So these are called alright so we looked at the maximum likelihood or ml here now we are going to look at maximum a posteriori or map right so this is a priori information right is this prior information about as θ is the posterior information about θ right. But are we actually computing the posterior here we do not know it right so we will have to see that anyway.

So what we are interested in is finding out θ that gives me the maximum posterior right so if you think about it I do not have to actually compute the posterior to find out θ that gives me the maximum posterior why because X is common I can ignore that I can only down arc max on the numerator I do not have to do the arc max on the denominator right so I do not actually have to compute the posterior.

So as long as I have a convenient form of just dealing with the numerator I am happy right so i can just do the max over the numerator and of course I can take the logarithm because this is a

nasty term because it has a product in it so I take the logarithm converted the summation and basically do the max of this.

Right so there are a couple of things which I want to point out here so the one is yeah if you have some prior information you can use that right. So if you believe in the honesty of the person you can use point five right but there are other cases where I do not really have a prior information about how the true solution will be or what the true solution is right but I have some prior over what I want the true solution to be you have cases like that.

We could try to do this when we did rigid aggression or when we did lasso right we wanted the parameters to be small the way I achieve that was by putting a quadratic penalty right instead of that what I can do is I can say that hey I have this prior which is very low probability to high values of the parameter right I can I can make this prior you know is all of you know but β distribution I can have all kinds of weird shapes to the β distribution.

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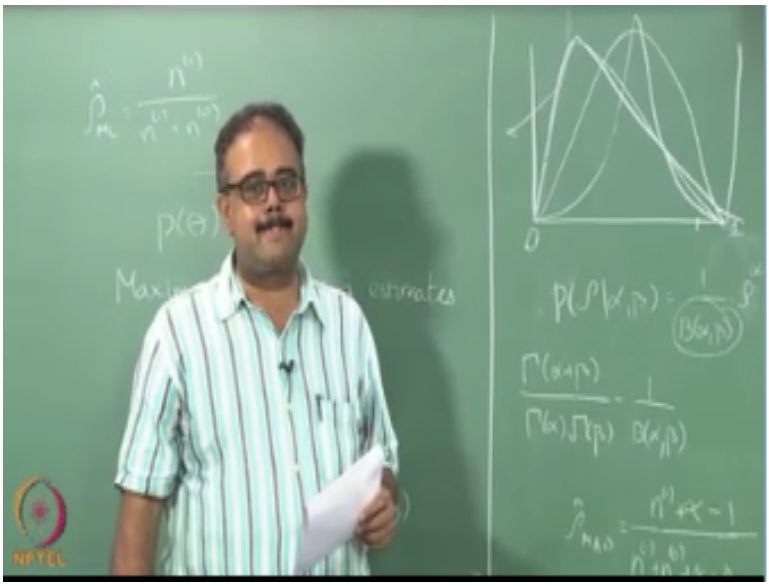

Right let us see nice prior right the honest prior right so is a really honest prior and that is something like the l1 prior l2 prior sorry wait so the probability of the higher value is β being high is small a row probability of ρ being high is small probability of ρ being small is high right

so this is one way of thinking about enforcing regularizer right does it make sense so I can use the priors for enforcing my regular the second say no do not give me things

That have very high parameter value of interested in smaller parameter values and if you this is just about single parameters if you want to talk about multi dimensional case you can also say that hey I will give you a low probability half having a solution which has more than thirty percent of the parameters nonzero so what will that enforce for me sparsity right that will enforce sparsity so then the probability.

So if I need to have a hence equation suppose I really need to have my ρ here my row is actually there right but I start off with a prior that looks like this will I will I reach my current estimate for ρ somebody said it depends so I'm happy depends on the amount of data I have right so it depends on the amount of data I have so if you have an infirmity prior grade the amount of data that you actually need is actually low.

If you have prior is correct right if you put the maximum probability on the right solution the amount of data unit is low but if you put these the prior the maximum weight on the prior on the wrong solutions the amount of data you need is going to go up significant amount of data is mean is going to go up significantly so I said to I made two points about priors right so this remember what are the two points price can be used for regularization okay so wrong priors need more data to corrector and a completely bullheaded prior can never been corrected.

So what is this it is actually the β distribution where I have written the normalize in a slightly different format you are used to seeing the normalizer as okay so what is this call that is the β function okay this whole thing is the β distribution so you actually have three β is here so the β distribution okay and the β parameter lowercase β as a parameter and then you have a β function thing okay.

So the thing to note here is that you are and your β parameters right almost act like as if you have seen heads and tails right. So your α is just increasing the count of your head s right and the β is increasing the count of your tails the make sense right α is just increasing the count of heads. So

if I had done the actually done the experiment I would have seen n1 heads right, but I am assuming I in addition.

I saw α-1 heads also right so if I am going to have a prior like this right can you imagine what would be the values of α and β as I will be less β will be more because α adds to the heads right so ρ would be higher if α is larger row would be higher so if I am going to skew it like this so they should have a β larger than α you can start reasoning about all of these things just if you understand what is happening so these things are sometimes called pseudo counts.

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**Introduction to Machine Learning**

**Lecture 38**

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**Parameter Estimation III**

So there is one thing that we are doing here see if you remember what was our two stated goals which I have erased of the board, what are the two stated goals that we had? One was to find some parameters that best explain the data, what is the second goal? Exactly right I am doing the best I am actually finding the best parameter like one single setting for the parameter that best explains the data that was given to me that is what we have been doing so far. But in terms of finding the best prediction for a new data point am I doing the right thing so far? is it the right way to do it right, so if you think about it right.

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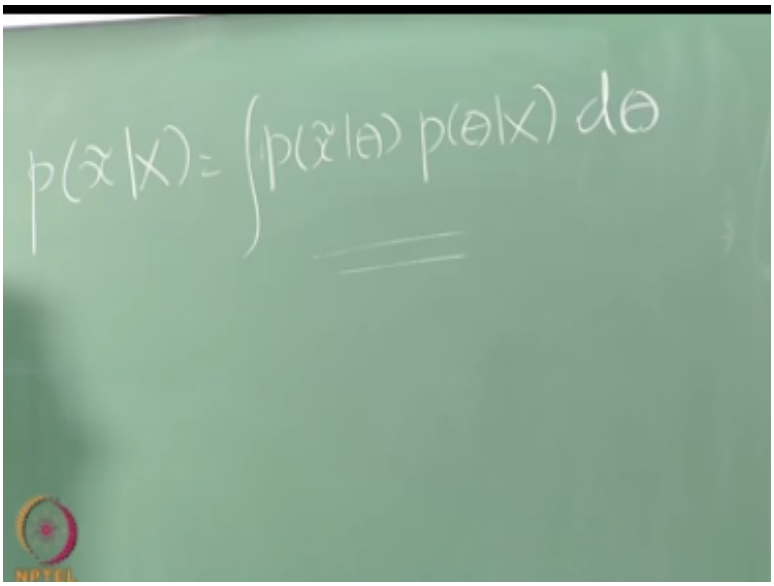


So probability of (~X / X), it is a probability of ~X given θ times the probability of θ given X summed over all θ if θ was a I mean first a discrete probability distribution but since we have been considering Bernoulli and other things it is going to be a integral over all θ righ.t I am not talking about the outcome I am talking about the parameterization so the parameterization is a continuous parameter right, θ is a continuous parameter.

So I cannot sum over θ it is not like I am only considering θ1 θ2 θ3 I am considering θ in the interval 0 to 1, right? So is integral over θ, right. So this is this is the actual outcome right but think about what happens in the case of the prior or any one of this case right I am only picking one θ it could very well be that there is another θ which also has a high probability of being correct.

But since I am picking only one θ I am sticking with that, there could be two different θ which I could have used them right in fact I should ideally be using all the θ because for a certain parameter setting some X ~ might have a high value right, so even if that probability of that happening is very small and I should still be accounting for that in my prediction, that makes sense why this is a much better predictor than using ML or MAP. But why do not people use this then computationally hard, why?

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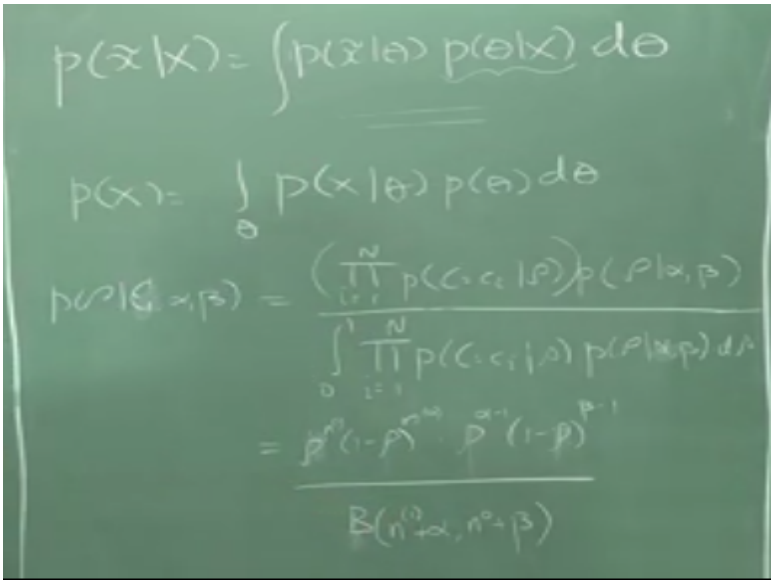


So far I was actually trying to avoid computing probability of θ given X right here I did that by assuming everything else was constant right and I just had to do the likelihood right here I said okay I am just doing a point estimate so I can ignore the denominator I can only do the

numerator right but when I go here boom, I have to do the full computation right this essentially means I need to know p(x) right.

And that becomes hard but computing that is actually harder to actually multiplied over all the data points that you have right so it becomes a little tricky right and what is P(X) by the way, no yeah but what is P(X), p(x) is a probability of seeing the data right what does the probability of seeing that I do not know that I have only given you the data right I do not I do not know the distribution from which the data was drawn that is exactly what you are trying to do so the θ gives you the distribution over which the data was drawn right, so what would be P(X) right so that is P(X), how do you compute that? Good point, so whenever we talk about parameter estimation right so you need to have some parameterize form of a function for you to do the estimation of the parameters right.

So if you remember in the logistic regression it was not Bernoulli it was the logic function that we were trying to estimate the parameters for and I also told you in when we looked at LDA I told you we could make a lot of different assumptions about the parameters in the LDA we made an assumption what did we assume it was a Gaussian right, anything else? Covariance was the same right this is for LDA right.

And I at that point I told you could use mixture distributions as well and you could use whole bunch of other things I also said you could use nonparametric techniques right but I told you it is a misnomer it is a misleading name because nonparametric really means that you just keep adding parameters and things like this so there is a very flexible very powerful modeling paradigm so you could do parameter estimation for nonparametric methods also right.

Where you have to actually figure out how many parameters you need as well so then the distributions you consider become more and more complex now we are looking at very simple forms right but the distributions become more complex and it is the parameter estimation consequently becomes harder right, so infact most of machine learning research nowadays is

essentially on parameter estimation for all kinds of different things like non parametric models how do you do the parameter estimation things like that lot of research is going into that. And a lot of powerful models have come out.

Let us go back to our Bernoulli case for a minute right I have Bernoulli and my prior is a ß distribution right now I can try to do this so this is this will be what, this P(x) given θ right that is P(θ),p(x) given θ is p(θ) divided by; make that makes sense right this just the is p(x) right this is P(ρ/X) right so X is your C, C is the set of experiments that we were talking about that right so P(ρ) given X is equal to P(x) given ρ x p(ρ) / p(x).

Next p(x) is given by integral over the entire row space which is 0 to 1 okay p(x) given ρ x p(ρ) d ρ and it is just the base rule I have it now, right. one thing you notice here what is gone here or nice convenient logarithms are gone right but it does not matter too much why? We are not doing any maximization here we are not have to take the derivative or anything now right and this actually interested in computing this whole functional form again and I am not interested in taking derivatives and trying to maximize.

So it is okay if we look like if I do not have logarithms but it just makes the whole thing more of a nasty right, when it turns out this is pretty easy to compute well so I skipped a few steps in between but you can figure that out so I wrote out probability of ρ given α ß which is essentially this right and this product I can write like this as we did earlier that we have done both of this before so what I have left out here is a normalizing function.

There should be a 1/ß, ß function of α ß right and then I have this integral also and it turns out that this whole thing including that normalizing factor is actually equivalent to the ß function of n0 + α I mean n1 + α and N0 + ß okay. These are P's these are ρ' s yeah if you remember the ß function right it is ρ power $\alpha - 1$, $(1- \rho)^{ß-1}$ and the normalizing factor is a ß function of α ß right, so it is $\rho^{n1 +( \alpha -1)}$, $(1 - \rho)^{n0 + ß - 1}$ and then there is the ß function of n1 + α, n0 + α this actually itself a ß distribution right.

So it is exactly the same ß distribution so we started off with the ß distribution as the prior over the ρ right and then we did this computation and the posterior turned out to be a ß distribution as well is very convenient right? Such distribution which allowed us to do this are known as conjugate pairs that are conjugate distributions so what are the two distributors are talking about here ß and ß and Bernoulli right.

So the data distribution was Bernoulli prior distribution was ß right if that is the case then the posterior will also be ß right people know the difference between Bernoulli and binomial, what is the difference between Bernoulli and binomial? Single trial is Bernoulli repeated trials is binomial right it turns out that ß is also conjugate prior for binomial, right.

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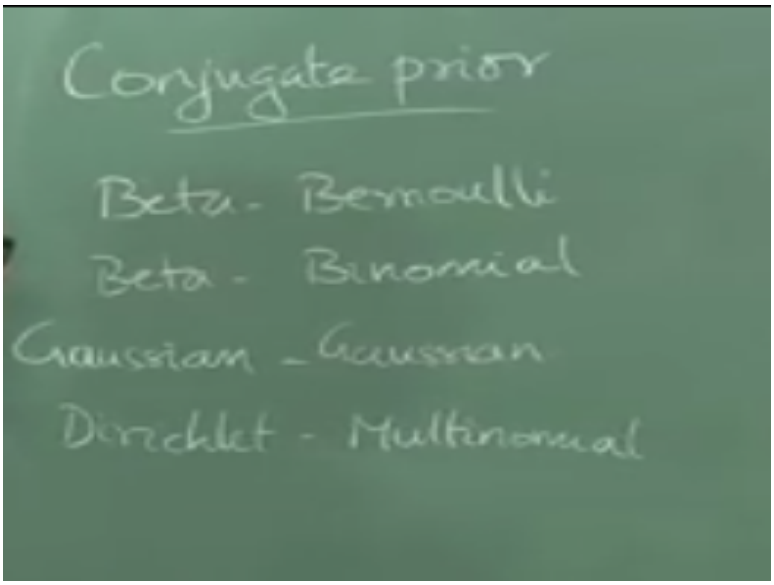

Any the famous conjugate pairs that you guys know? So both the data and the prior can because it so remember what we mean by the prior distribution right so the prior distribution is the distribution over the parameters of the data distribution when I say Gaussian-Gaussian that means that okay I am assuming my data is coming from a Gaussian and I am going to assume that the mean of the Gaussian is coming from another Gaussian right.

The probability of the mean is going to be given by another Gaussian so that is what I mean by a Gaussian-Gaussian prime like the like in the ß Bernoulli prior I am assuming that the probability of heads is ρ and the prior distribution ρ is a ß distribution, so when I say Gaussian-Gaussian I am assuming that the data is coming from a Gaussian distribution and the mean of the Gaussian is coming from another Gaussian distribution that is what Gaussian – Gaussian. There is also another very famous and so Derichlet so people about multinomial is what is multinomial?

It is a distribution that will describe multiple roles of a dye for example, binomial is when you have two outcomes multinomial is when you have multiple outcomes right, so the single experiment single trial version of multinomial is called not too many people know it and multinomial, binomial is called Bernoulli the single trial of a multinomial is called know the unimaginative name is called the discrete distribution okay.

So but so multiple trials is called the multinomial distribution and the prior the conjugate prior for it is an original a distribution which is nothing but the multi-level extension of the ß distribution so it is like the ß distribution when it is a multi-dimensional extension of ß distribution okay and there are a bunch of others okay so but there are several that are known and so typically what you do is you look at your data right look at the data and figure out what distribution is a good distribution for the data right.

So for example coin tossing experiments we figured out that Bernoulli is good right, so die rolls right we will figure out that multinomial is a good distribution what about text, people typically use multinomial distribution so you can think of having a very large dimensional die on one

word written on each side of it right so what is the next word you use roll the die that will tell you or the next word to use right so that is that is do not laugh I mean that is the seriously the model that people use for modeling text you know they use multinomial distribution so they assume that each word is a generated independent of the previous word sometimes when okay fine let another yeah.

So that each word is generated independent of the previous word and so you can model that as a multinomial distribution right, so they have actually have different names for it, it is sometimes called the Unigram model right also it is called the roughly a bag of words model right where the sequence do not matter and each word is generated in differently so many ways of describing the same idea right but at the end of it is nothing but using a multinomial as comical as it sounded that is what it means.

Having this huge die and rolling it every time I want to add a word to the document okay so that is multinomial right so once you have decided what is the distribution that you think is appropriate for modeling the data then you go and decide on what your prior should be right so sometimes the choice of the distribution for modeling the data is driven by not whether there is a conjugate prior is available for the distribution or not right so maybe there is a different distribution that is perfect for modeling data.

But because there is a very convenient conjugate prior for multinomial right people want to use multi no means so like that there are other instances where even though Gaussian is inappropriate for example people want to model discrete value data right the Gaussian cannot do discrete values right but then coming up with the distributions which have allowed discrete values and have nice conjugate priors is hard right so people just go with Gaussian sometimes you end up operate they use Gaussian.

Because it has a nice conjugate price so y and y are conjugate priors important, because an easy to do things in iterative fashion because once I run some data through the ß Bernoulli pair, okay I

am going to end up with a ß distribution over the parameters again so if I get more data and I can just happily just go ahead and do it and if I every time I run through it I keep getting a different probability distribution there is no functional form for me to stick these things into and then it becomes very hard for me to do this in any tractable fashion then I cannot come up with some parameter update equations or anything like that so it becomes very hard so, so the conjugacy is very important.

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Lecture 33 NOC

So we are going to shift gears and we are going to look at a very very popular supervised learning algorithm or supervised learning model I should say because there are many different algorithms for estimating this model based on decision trees. Can people at the back hear me? fine okay so decision trees have a very I mean very special place in all this machine learning stuff in that they are very widely used right and very poorly understood No in terms of I mean we talked about all these bias variance trade-off classification error we can show convergence we can show approximations and whole bunch of other things for all the linear classifiers linear regressors right we know lot about all the linear stuff right and some of the nonlinear stuff like SVMs and so on so forth again we have very strong theory we know about convergence and things like that. With decision trees, I can tell you what the problem is and I can tell you what is the best known heuristic for solving the problem, but I cannot even tell you how good the heuristic is. because there are isolated results under very special conditions people have some results on how good the heuristic is you know there is this best possible tree that you can learn right and how close would it get you to the best possible tree right so there are some very isolated results right but there is nothing that is really something out there that we understand well right and it is incredible because it is such a simple idea it is a very simple classifier right so it is more or less along the lines of how humans do their decision making right. So when you are trying to decide whether something belongs to some category X or some category Y right how would you think you go about doing it right you do not create a hyper plane in your head right. so typically what I end up doing is okay this is red okay it's not red okay is it round oh yeah it looks round and round and okay it's blue maybe it's that right essentially what you are doing is querying some properties of the object right or some properties of the entity at hand right do I want I mean do I think he's a studious boy or not right then then I can ask all kinds of queries okay does he show up for all the classes As you sit in the first row all the time, you see smiling after quiz 1.

So I can ask, I can build this thing, so I can ask the series of queries and then I can essentially I am building some kind of a characterization of the object and then I say okay great so this is this person is class 1 okay this person is class 2 right so on so forth that's the whole idea behind decision trees right you are essentially trying to if you think about what you are doing So trying to partition the input space into certain regions. The feature 1 has value x, feature 2 has value y, feature 3 has value z, and that gives me some region

in state space. So what do I mean by that? Let's take a.

.. So the first question, let's say it's a two-dimensional data set. Now let's forget about all the.

.. all the relevance to real day real life and things like this i have two variables x1 and x2 now i am going to ask the question first question here is a new data point is the x1 of this data point greater than 0.6 or lesser than 0.6 okay that is the first question i ask so what am i doing in in some sense i am right so i am splitting this into two parts it is greater than 0.6 it will be here if it is lesser than 0.6 it will be here right next question I can ask is okay suppose x1 is greater than 0.

6 okay then is x2 greater than 0.2 or not right then what do I do right so this region is now x1 lesser than 0.6 x2 greater  I mean x1 greater than 0.6, x2 greater than 0.2 and likewise this is x2 lesser than 0.

2 and likewise here I can ask the question is, what kind of question can I ask here? Come on, just say something random. Do not think too much about it. Make any conditions on x1 and then cascade it with x2, x1. You can do anything. So, typically they alternate, but you can also ask a question, is x1, given that x1 is less than 0.

6, is it less than 0.3 or not? Or you could ask, is x2 greater than, do not give me some root 2 or something, but 0.5. So as soon as I write that thing that it becomes 0.

5. Can I come to that little later? I am just trying to mimic the way we try to think of things. All of you were agreeing with me when I said you will think about attributes one at a time. Is the price okay? Is the TV screen of the right size? Then I am going to buy or not. I am just mimicking that process here. Then we will come to other things a little later.

It will be really truly amazing if your true class labels are going to lie like this. what is the problem what do you think is a likelihood that the class labels will actually be this kind of rectangular regions. it could be high I mean depends on what process was used to generate the class labels if the problem labels are generated by doing this kind of region splitting obviously they will we have to find the right regions but we are making some kind of an assumption right earlier when we made assumptions about linear right we are making some kind of an assumption about what the boundaries would be right likewise we are making assumptions here that will be rectangles. So if you do not want to make the assumption that there will be rectangles, then it is going to be little harder. So not only are these rectangles, there is something more special about these rectangles.

I mean they are all recursively generated right it is not like I can I can just take some arbitrary set of rectangles and tile the space right they are recursively generated by first splitting it into two and then splitting each section into two and so on so forth it turns out So this kind of recursive splitting is what is most attractable to handle and for most of our decision tree discussion we will stick with this. I will come back and address that issue little later about having more complex boundaries but almost all decision tree algorithms, all the approximations that we do for decision trees use this kind of recursive splitting of regions. like that how will you describe this region it becomes harder right see now each of these regions I can describe very easily right but if I start doing nested rectangles right it becomes a little tricky to describe the outer region. I can do something like this provided I am willing to accept that, so it is no longer a nested really no longer nested rectangle because I had to actually fragment the outer region. That will give you the inner rectangle, but the outer rectangle you have to actually exclude it, I have to specify the outer rectangle and then say remove the inner rectangle, so it becomes little harder to specify.

Yeah it becomes harder right I wanted to be easy I wanted to I want to represent this as a tree but then the way you are going it make it harder and harder so the biggest advantage of decision trees is the interpretability so for example this tree that I have this this region segmentation that I have drawn I can represent it as I have been talking to you about trees right where is the tree right the segmentation i have right i can represent it as a tree so what i will do is i will first ask the question is x1 less than 0.6 right then if you say yes i will go left then i will ask the question is  So I can very compactly represent this rectangle segmentation as a tree, you can see what is here, I am asking the question is x1 less than 0.6, if it is true I go to the right and then again I ask the question is x2 less than 0.5, it is true I go to the right and I say that this R1, so I am essentially here, otherwise I am here, then going to the other branch which is x1 is greater than or equal to 0.6 and this side and if it is less than 0.

2 i mean r3 if it is greater than 0.2 i mean r4 right so i can very compactly describe this segmentation as a  So what is nice about this tree, it is easily understandable. You show somebody, so you are building your this, you go out, you become a data scientist or a data analyst or whatever, some manager who makes like 10 times what you do but who has never heard of a hyperplane in his life, comes and asks you for a  to build a classifier here is the data build a classifier and then you tell him okay I built this classifier and this new customer you should label him as a buyer then he'll ask you why right so at that point you just talk to him about optimal separating hyper planes and and show him something okay well then the next day probably the manager is going to be earning infinitely more than you right so what you should okay so what you should be doing is showing him a

decision tree right because that people can understand right people even with an MBA can understand right so you can see what's my recommendation to you guys after you finish your B.Tech or anything right don't do a MBA but Anyway so that is easily interpretable right you say oh that is oh yeah you should you should classify him as a buyer because well this this on this parameter he is so much on that parameter is less blah blah and then there you go right the biggest advantage of decision trees is the interpretability it is always easy to explain the decision tree to people in fact so much so that at one point when neural networks were at their peak you know you know what is the biggest problem with neural networks no the opposite of distant trees incomprehensibility right this is interpretable and neural networks are incomprehensible essentially you say okay here is a black box I do not even know what hyper plane it is learning right if I get if you think optimal separating hyper planes are hard I cannot even visualize what the neural network is learning right so here is a black box you throw in all your data at one end something will come out at the other end you just take it on faith right So welcome to the church of neural networks. So that's essentially how neural networks are working.

So when the neural networks were at their peak, there was this whole line of research where people took a neural network that was trained on your training data, et cetera, et cetera, and then tried to construct a decision tree that will give the same decisions as the neural network. It will give the same class labels as the neural network so that you can actually understand what is happening. It sounds weird but remember that now I am no longer using whatever other heuristic I had for constructing the decision tree. I am using a decision tree which mimics the neural network. So if the neural network learned some complex function of the data, I am trying to build a decision tree that mimics the complex function.

So it is a different decision tree that I would come up with than the one I would have constructed if I had used any of my decision tree learning heuristics on the data. from the beginning. So that is value to doing this. So people do see that there is value to doing this because neural networks do something I cannot understand but they seem to work. They give me a wonderful answer and I do not know what that answer means.

So can I use something for which I know what the meaning is and try to understand it in terms of that. So that is how useful decision trees are. Even if you have a more complex learning mechanism at hand sometimes for interpretability sake so you can use decision trees. How expressive are decision trees? right if you remember we had the discussion about neural networks I said if you have two layers of weights there is three three layers of neurons and you can basically represent any Boolean function right I just the branching factor might be very high but then you can represent any Boolean function so neural networks are universal approximators in that sense right so what about right you

can it can be an universal approximator as well right I can just keep dividing dividing and subdividing this space okay it is just that my tree might become very very large right as long as there is some kind of guarantee on the function yeah does it mean that you can repeat some variable yeah I can repeat some variables yeah that is fine yeah yeah I could do that yeah I mean in fact yeah he was pointing out in the beginning I could have as well drawn another line here yeah The discussion was when we were discussing about this line versus that line. Sure, I can keep doing this if that is your question.

Now this becomes a much more complex tree. So now I am splitting here once more. I will have another branch here. I will have another branch here and another branch somewhere there. So it keeps becoming more and more complex.

But the point is conceptually you can represent anything. So it is powerful in the sense that it is a universal approximator. And it is just that the number of parameters can grow unbounded. And that is another thing nice thing about it, it is non-parametric remember we talked about what non-parametric means right, so distant trees are actually non-parametric it can just keep growing right we can keep adding parameters as you go along.

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**Regression Trees**

So as with the linear methods we will first start by looking at regression right, so we will see how to use decision trees for doing regression. So far I have just told you how to do the partitioning right.

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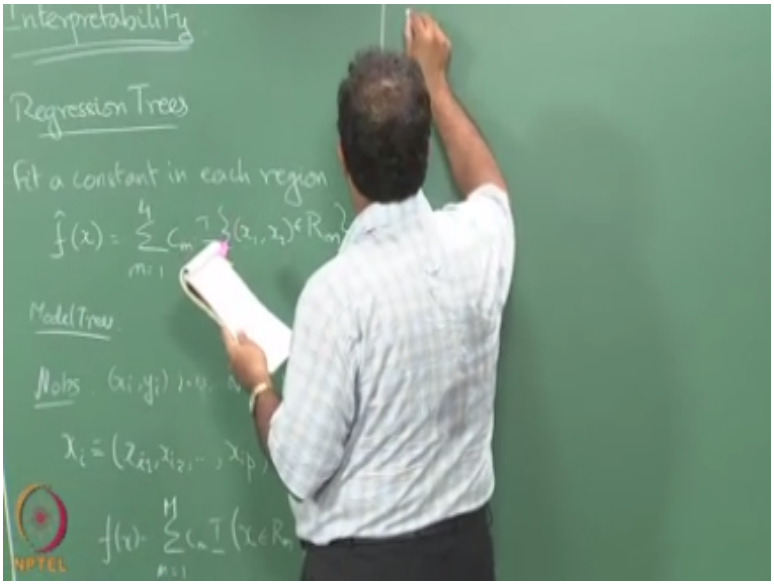


Let us look at regression tress, so I split the region into four right, so I will first see if the data point that comes to me right, whether it lies in region 1 or region 2 or region 3 or region 4and for each of those regions I am going to have a some constant that I will output right, so if you think about it the function that I will output from here right, so we will have one value in this region

right one value in this region right, another value in this region another while in this region so it will be like a piece wise constant thing right.

So people understand that right, you are not going to test my 3D drawing skills right, so you can see that there will be one output for any point in this region one output for any point in this region, one output for any point in this region, one output for any point in this sense and in some sense it is similar to KNNs, because you are assuming that there is a piece wise constant assumption about the function that we are trying to model, right.

By the second parametric or nonparametric, parametric okay, what are the parameters, so is it parametric resonance is depends on N becomes larger what happens, so we can apply one of those ideas here instead of fitting a piecewise constant per region you can fit a linear function on that region right, I have done the splitting right, so I am going to have some training data right, so some of the training data points will be here, some will fall here some will fall here and some will fall here I can take all the points in R1 and fit a plane to that.

And I can take all the points in R2 and fit a plane to that likewise for R3 and R 4will that be better or worse than fitting a constant? Better. no actually it does not depend actually it is always better right, it is always better in the worst case you will fit a constant I mean if constant is going to be really better you will fit a constant because you are minimizing squared error and anyway end up doing that right.

So what is the problem with that little bit more work here that do more work and there is variance the significant variance we will come to the variance bit yeah, but the significant variance but such things are called model tree sometimes, model trees in fact you can do more complex stuff itself it does not have to be linear, a linear is easy right, I can do any kind of regression I want on this right, I can fit I can use a neural network if I want and to learn a curve only on the data points that lie in R1 right.

Not a good idea usually because I have already divided my entire training data into at least 1/4 if not smaller right, I mean some of the regions could have much smaller data some could have

more right, but still I am cutting down on my training set that is available so it is going to be harder to train okay, so I am going to erase this stuff and try and generalize this or let me do it again. So I have n observations as you all know right, so this is our usual setting so where the input comes from Rp and the output comes from okay.

So the function that we are trying to learn is Cm and Rm right so sorry, the set of parameters that we have to estimate our Cm and Rm correct, so we need to know where are we splitting, how are we splitting the region and having found the regions right, what is the constant that we will fit within that region okay.

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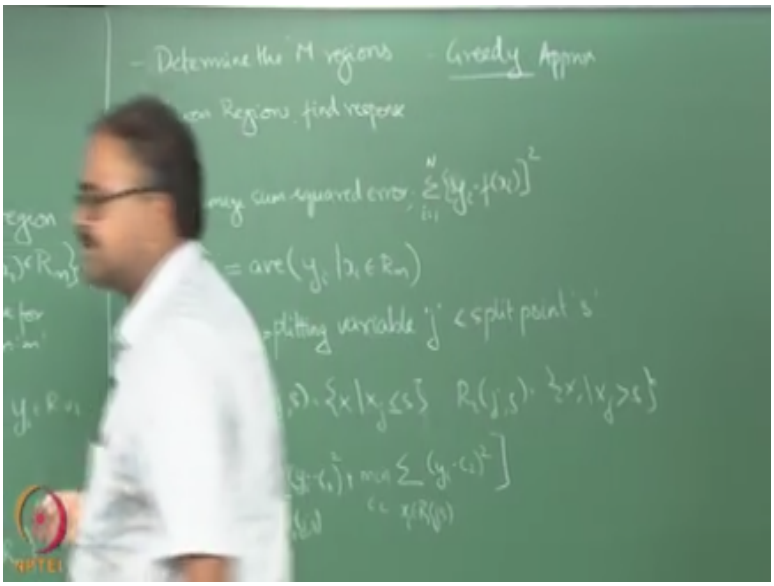


So there are the two questions determine the M regions and given the regions find the response right, but one thing to note is unlike KNN or anything the M is not given to you right, the M is something that you discover from the data and that is why I said it is nonparametric right, you can actually have more regions if the data requires it you can have lesser regions M is not given to a priori right, but sometimes as a regularizer you can decide to fix M as well you can say I do not want a tree that is more than four levels d as a complex d measure but again that is derived from the problem definition the model itself is not parametric.

So let us look at the second question first, because it is easier we actually have a proper answer to the second question right. Right, so this is essentially what we are trying to minimize right, and so we can try to do this region wise right, because the output that I produce for one region does not depend on the output I produce for another region so I can do this minimization region wise right, so I can pick yeah, every point would have it is own box kind of that. Okay, yes. Could likely yeah, so we will come to that will again address this question later, right.

Yes, so assuming that there is some amount of so you will not okay, let us step back into this you are going to get training data at best what you would do is you would have regions set within a region only 1.6 right, so that is not saying that I am going to fit it tightly around that point right, so all of our for that might be only one point but still there will be some kind of regional segmentation that is happening on the input data right.

Second I have I am going to introduce some kind of regularization that prevents me from doing that okay, so that will not be recap that is exactly what he was asking so you could end up with that that is what I am saying and we have to find some way of regularizing it so that you do not do that right. Right so if you going to minimize this region wise but anyway right now I am talking about given a region right so that is easy so we will not be over fitting things so given the region right.

So I am going to find out what should be the output of the mth region right, Cm is the output of the mth region right, that is what we assumed, what should it be give me a simpler I am fitting a constant I am not fitting a straight line here, average of all the points okay, like average of all the points which lie in the region and take the yi is corresponding to the points lying in the region and take the average okay, that is the best response that is that is easy okay that is done.

What is the harder part, finding the regions right, in fact it can be shown that finding the best possible Rm set of RM right is actually NP-complete right, and is NP-complete now in the again NP exactly NP-complete right, so you can show that finding the best possible Rm is very, very hard right, so we have to come up with some kind of approximation so essentially we use a

greedy approximation. No, they just tell you what XII I told you I told you what the training data is right, yeah this is all the training data is you get xi, yi your job is to find the regions and find the region I find the response.

Yeah, yeah find the best region you can given it a region you can tell me what the performance is right, but then finding what the best such segmentation is actually hard you have to search through the combinatorial really many segmentations. So the way we do it is following right, yeah, okay, now for a given M so I want to find the smallest M such that I get that performance smallest region, smallest region yeah, the smallest region said for which I get the performance that is really ideally what I am looking for right, smallest M sorry, given an M finding the region yeah, in general that is also hard but I want to find it for the smallest M as well.

Ideally you want to find it for the smallest, ideally like there is some data if you are right, like you would have to either specify the M and then you find the best or you see that when the best and find the smallest M for which you can. Ideally it should be find the best and then find the smallest M for which you can do the best right, but we end up doing compromise on that as well so what will what we will do is you just making me go do this all out of order by asking leading questions.

But what we are going to end up doing is we are going to say okay, here is a greedy algorithm right, greedy algorithm find the best that the greedy algorithm can do okay. Now find a smaller tree okay, that will achieve close to the greedy algorithms best performance again I will have to make a compromiser  I cannot say that give me the smallest tree that will give me the same performance as the greedy algorithm.

Because if that is exists one in fact greedy algorithm would have found it right, along the way as it was growing right, and therefore we have to say that okay give me a smaller tree right, that is close in performance to what I get with the greedy algorithm right. See remember we already made an approximation by assuming we are doing recursive partitioning right, so you cannot get the best possible performance okay, that we have given up right by choosing a tree representation right.

So this is a lot of approximation that is why I said right, I mean there is no good understanding of how decision trees eventually work if you ask me two specific questions like okay, how good an approximation will this greedy algorithm converge to right, suppose my performance the best possible performance on this the Bayesian optimal error on this dataset is say 93% performance I can see 7% is optimal error okay, how close to optimal error will a decision tree algorithm get to no answer right.

While you can answer some of those questions for things like logistic regression and consider some splitting variable j so what I mean by splitting variable the splitting variable is the question that I asked here, okay this variable here in the question so x1 right or x2, so in this case of splitting variable is x1 here the splitting variable is x2 okay, and what is the split point it is the number on the other side right so 0.6, so in this case the splitting variable was x1 and the split point was 0.6 okay.

Consider some splitting variable j and a split point s, okay so I am splitting my input data into two parts one where the jth variable is less than s, less than or equal to s the second part where the jth variable is greater than s okay, let us still to get to two parts so what we really want our j and s such that okay, I am seeking j and s is that if I fit the best value for the points that lie in R1and if I fit the best value for the points that lie in R2 that is what the inner minimum minimization is right the sum of this is minimized, so sum of the squared errors over the two regions is minimized right ,so the j and s actually influence which data point goes to R1which data point goes to R2 right.

So once I decide which points go to R1 and R2 I have a fixed optimization problem that I solve right, which one we already solved there. Yeah, I am just talking with the full data set I am at the root right then we can worry about the recursive splitting part right. So make sense for people so far yes, I want to find j and s such that this happens right, how do you solve this minimization problem.

So we can do this. No, this is not classification right, this is actually a regression I am solving right, so all these data points in our one I am going to output one value for all the data points in R2 I am going to output another value so you can think of saying there came grouping all R1 into one I am going to output one value and grouping R2 into one I am going to output one value find the right grouping such that I can output a value such that overall error is minimized okay.

So the first thing know it can be slightly better than that right, or worse I do not know I will tell you when depends a little better or not okay, the first thing you have to note here is I am going to do this for each and every j that I have okay, find the s and then I am going to pick the best j right I'm going to do this in turn for j=1, j=2, j=3 from 1 to P I am going to do this, and then find the best s and that will give me a value for the objective function, right.

And I can use this to compare which j is better that I do not have to do this jointly okay, so that is the first thing you have to notice okay, so given a j right how will you find the right s, so once you have fixed a j you can think of it as just a just a line right, I have to find that s at some point so that I can split everything to one side to R1 everything to other side R2, exactly so what are the steps I should choose for s, so he was talking about recursive doubling that is one way of doing it if you have no other clue right.

And then you have to come back and then you have to search through thee so people know about recursive doubling I start off by looking at 2 then 4 then 8 and I keep doing that at some point some sign will change right so I mean I will be fitting it one way then I will actually my error will start increasing again, so I stop and then now I will have a window of some power of 2 right, so I have to look between 8 and 16 and then I will do a search through that that is one way of doing it.

But there is a slightly better way we can do it any guesses something better than that imagine you are trying to do this not imagine so remember that you are trying to do this from data, I give you a training data set. Exactly, so order the training data along ascending order in that coordinate right, and then just keep hopping on that. Suppose I have data here, here, here and here somewhere there right.

So now if you think about x2, let us say x2 is my, or x1 is my splitting criterion so there are only five different values of x1that actually occur in my training data right, so that is x1 equal to this, x1 equal to this, x1 equal to this, this and this right.

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It does not matter if I consider any other values for x1 because that is one of these five values will give me the same split, right suppose I consider this a splitting point does not matter I could have as well consider that as my splitting point you see that right. So I do not have to consider it have to go smoothly along x1 I can just use any one of the data points that has come to me already right, so that is the easy way of searching for a splitting point in x1essentially what we will do.

Right, we just start it one of the reasons I already used either less than or equal to here it is not easy, so how much work do we have to do for finding one splitting point nxn logn or because the sorting part is it okay, you do not have to sort here yeah, you do not want to sort here that is what yeah okay, you can get away you can just go through whatever already good yeah, it does not have to sort here. No, if you sort I mean the computation becomes a lot easier but for computing

the complexity you do not have to sort, right you can leave it as it is you can it is NP N for each feature and you have P features right, so the amount of work you have to do is NP.

But if you sort then life is a little easier but you do not have to when you are writing the code you will know what I mean, but yeah an NP is the amount of work you have to do for one feature one level right, great so now what we do I have found the optimal j and optimal s have a the optimal and all our assumptions right, because I am actually doing an exhaustive search over j and s right, I am not doing any approximation here I am doing an exhaustive search so given the assumption that we are going to do something greedy and we are going to split on one variable at a time so we are finding the best possible variables great.

Now what do I do, I actually create the two sets R1 and R2 at having found the best possible splitting point and the splitting variable and the splitting value I find the two regions R1 and R2 and then I go into R1 I do the whole thing again, assuming that R1 is my entire data set likewise I go into R2 and do the whole thing again assuming R2 is my entire data set right. Thus, it makes sense to consider the j again the j that you split on say some j* right does it make sense to consider j* again yes, what does not make sense is to consider j* along with the same s.

In fact does not make sense to consider j* along with any s greater or lesser depending on which side you are right, so you can progressively you keep pruning your search based on, it but you do not have to worry about it because it is automatically taken care of because you are only looking at the values that are present in your data point, I am not written those things down right, you want me to write down the whole process all of you remember it right, j* is the one that gives me the minimum in this that I am considering each feature in turn right.

So j* is the feature that I finally choose to split on and s* is the value that I finally choose to split that j* on okay. there is another question somewhere okay, good okay great, so how far do we go so this is a question that we all had in our mind from the beginning right, if I do not put any restrictions on it I will keep going until I have one data point per region right, great yeah, so that is a really other people actually notice it could very well be that the number of features ends right.

How can you, yeah j can be repeated, but if j cannot be repeated can end up with something like this there is only one data point per leaf per region, where that is no more than one data point per region. Now you could do that but answer my question, we are allowing features to be repeated right, can you still end up with a point where you cannot grow the tree anymore but you have more than one data point per region.

If you keep getting your betting point as your border if your region you can traverse any further. Not, they are the same data point we can repeat it I never said your exercise have to be unique right, now it is not so it sounds like a trivial thing but no it is actually important I mean you should think about it right, so this is very important in cases where the yi's are different the xi's are same and the yi's are different there is no way you will get 100% correct maybe if you are assuming that it is a deterministic process truly underlying process is deterministic and it is corrupted by noises, but what if it is a stochastic process truly a stochastic process is generating all of these things for you right. Yeah, sure you can you should there is no question of you allowing it is life happens to you.

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**Introduction to Machine Learning**

**Lecture 41**

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**Computer Science and Engineering**
**Indian Institute of Technology Madras**

**Stopping Criteria and pruning**

Right anyway so the question is when do we stop.

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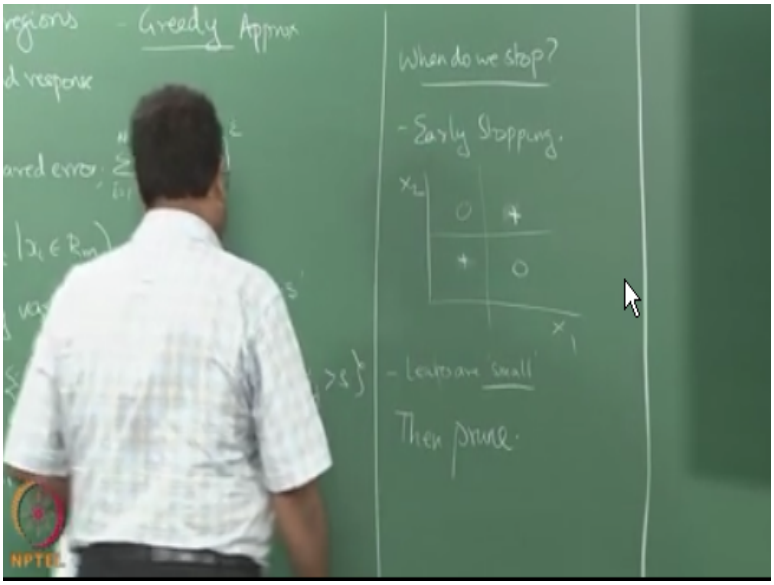


So that is one technique called early stopping okay, where you say that. Hey, I am considering all of these regions right all of the split points but the amount of improvement I get in my error is very small and therefore I stop. I come to some point so I let us say I have like several regions here now so I consider $R_1$ okay where I consider all possible ways where I can pick an $X_1$ all possible ways where I can pick gets to and try to split and the error does not change by much.

I can stop I do not have to keep going into smaller and smaller regions even if there are many data points here I can stop right is it a good idea so let us go back to XOR right so this is not a classification problem we are talking about regression so the excess have an output of + 5 this 0 is output of -5 right and you are trying to fit this. I can try to look at splitting it on $x_1$ so I have $x_1$ let me speak I have $x_1$ and $x_2$ right so if I were to try to split on $x_1$ right what will I do right.

And the only thing I can do is this right so if i split anywhere else I will be just keeping a all the data in one side right and all the other mean other one will be empty right this the only meaningful breaking point right and what will be the prediction error here whatever is the half of it i will be predicting the average of this right so it is 0 and 5 then I will be fighting 2.5 so it is essentially $2.5^2$x 2 right.

What would have been the prediction error if I had kept the entire region as one. Same thing the average--average sum square will be the same right so if I split on $x_1$ I am not getting any improvement so let me not split on $X_1$ okay what about $X_2$ same we split on $X_2$ I will not get any improvement let me not split to $X_2$ that essentially means that I will just give you the average output for all the four data points.

But if I split on X well and then split on $X_2$ okay now I can do really well right so early stopping is usually a bad idea because we will miss out such these kinds of interaction effects if we stop to early. Good point yeah! So in this case I mean one of them is a trivial split right this case is easy but in general yeah so you will have to take a call yeah! Case like your institutions day careful of them at one time but food is easy see yeah.

So just think about think about this optimization problem right if I m going to split on two at the same time this optimization problem becomes harder that instead of minimizing over J and S I allowed to minimize over $j_{1,}$ $j_2$ $S_{1,}$ $S_2$ it becomes harder and harder sure you could think of other ways of optimizing it right so but the most common way of doing it this main reason people do not do two variables at a time is the interaction effect I made.

So if I start looking at two variable Saturday then I can start thinking a whole bunch of other things so why do I look at j greater and alone right I could think of other combinations hey $J_1$/x xj/xk right X JXK I mean so then it just starts exploding so they said okay fine we will just do it this way and make sure that we grow a large tree very large tree so that we actually capture the interaction effects.

In fact you stop when the when the leaves are small I think of a tree right so the leaf of a tree is a region right so the region small I put it in quotes is not the extent of the region it is the number of data points in the region right so--so small would be like two or three or five or something of some really small number depending on how large your data set was you keep growing your tree. So if you view some of the standard tools right.

There will be an inbuilt parameter right which says how small the leaf should be right and you might have to go and prettily with it if you are going to use a decision tree function from either VICAR or MATLAB or something right they have an inbuilt parameter that says how small is the leaf and they stop right so for VICAR it is 2 now you might want to change it to five or something I am not sure what is the limit in MATLAB but you might want to change that.

So that is that is a parameter that you have to fix right and it matters, it actually matters. like he pointed out so if you if you set it too small then you might miss things like this right and then what you do? You build a very, very big tree right this is what I was telling you. You try to use your greedy algorithm and get the best possible tree that you can write so a tree with very very small leaves this kind of the best tree that you can build right right.

Once you get there now you are going to ask the question okay what is the smaller tree that I can get that performs almost as well as the big tree that I have right so there are two ways of doing it the first one she called reduced error pruning case is rather simple so I each leaf then the smallest the largest leaf is smaller than the threshold effect okay so every leaf should be less than that size so basically I mean I can stop each branch independently.

So whenever a leaf reaches size 2 I do not split it anymore so I keep doing but other branches can continue growing so the tree does not have to be of uniform hydrate at some part some sub tree might be shorten some sub tree might be longer right if you remember that picture here so I kept drawing lines in only one region so that means that path alone would have been a much deeper sub tree and others would have been much shallower that is fine.

so reduced pruning is something very simple and so I have a training day training set I built the tree fully on the training set and then I have a validation set may we talked about validation set long time back I have a validation set now what I do is I start greedily or not greedily it is very safely pruning away my internal nodes right so what I can do when I erase the only tree I had onboard here right.

So what I do is I have this prediction that I am making right I will replace right an internal node with a leaf it does sorry exactly I am just joining the region is together now I see the performance of this with respect to the validation set is I had the original performance on the whole tree right now I look at the performance with respect to the validation set right it could go down right it could go up depending on how the validation set is right.

When it because the tree was constructed only on the training set when you do the pruning the error might actually go up I mean the error might go down sorry right for on the validation set if the error improves our state is the same I will keep this right but if the error mug becomes much worse right I will put it back try otherwise so I as I use the Yi's for making a prediction right and then I keep doing this in turn.

Yeah that could cost could cause more variation agree provided mean usually when you have a large enough validation set so you can actually trust it right and then you try this again right and then if does not work keep going yeah so it is like to have this region but in stuff that I just treat this as one reason now once I you collapse the region I again do average on this whole region and you start at the output right.

See once I collapse this question is each one of this could have been outputting a different value right what will you do with the combined node right so I will take all the data points in the combined node take the average output and we use that as the new output for this right I could take the average of these two but why is that not a good idea the number of data points could be different right so it is it not be truly the average of the outputs right.

So if there are having the same number of data points then I can take the average of these outputs and use it otherwise they should okay so I keep doing this suppose I was able to prune right and now I have pruned this and I have ruined this as well then I can go back and try to prune that also right I can replace this whole thing with this and see how the performance is on the validation set right.

No reduced pruning is only on one thing you see the problem we do cross validation is I will end up with five different trees after the pruning now the question is how do I combine the 5trees right yeah so exactly so see that is this a very same pruning works it is only one validation say does not use cross-validation right so in that for that reason it is not that popular anymore I am just introducing reduce air pruning because is easy way to think about pruning right.

But like issue is pointing out first of all the variance will be very high depending on what you pick for the validation set right you will end up with a very different tree right so just like when already decision suffer from very high variance and the reduced pruning will actually make the variance worse but this is conceptually easy way of thinking about pruning and if I introduce a more complex pruning method right.

Then a little harder right yeah sorry as long as you are improving sure I will come to that I have a whole class planned on all those model selection methods right since he knew about cross validation he asked me the question I answered but I will come back to that right a whole --whole lecture planned on the model selection okay so cross validation is something guys should never forget.

Once you learn the other kind of pruning which we are all familiar with is called cost complex tree pruning right where you have your error function right and you also have your share in the name also have a cost for the complexity okay like you had here $\beta^2$ in your ridge regression and things like that right and norm β. so you already know about this kind of cost complexity measures right so we looked at that in ridge regression we looked at that in lasso and things like that.

And here what we essentially do is we grow the full tree right and then what we do is for every possible non terminal node that you can collapse right you collapse that non terminal node so it should mean that the entire sub tree underneath it you consider as a single region and replace it with the average prediction for the single region like that you can collapse each of them on terminals and create many many different trees right.

So each of this is a sub tree of the original tree right so what do you do is once you created such a collapse tree you look at the average prediction error of the tree it essentially look at the prediction error for each data point divided by the number of data points you get the average prediction error and add a complexity term right the prediction error plus some size of the tree right

So what is the complexity that we are really if so T is at three and α as a parameter will come to that so what is the complexity measure you think is good for a tree number of leaves right number of leaves number of regions you are split into so that is a measure that we use so when I say size of a tree it is the number of regions that the tree splits the input space into so α is a parameter that controls how small a tree I want.

Large α means small trees small α means large trees so now I essentially find my T okay which is a sub tree of the original tree right so it is not any arbitrary T tree okay I have original T tree that I have grew that I grew with this procedure right with this procedure I grow a tree and then I stop when the leaves are small and then what I do is I try and collapse each of the internal nodes of the tree and you can do this in a slightly better fashion right.

You can try to collapse from the lowest level on up and then stop at some point and things like that but you should remember that it could very well be that may be collapsing one sub tree alone might not give you much of an improvement right but if I collapse everything above it right it might give me an improvement why so maybe collapsing this alone does not give me an improvement but collapsing here when might give me an improvement.

No see the point is the error reduction might be small right but then I might not have gotten rid of enough nodes and if I get rid of this whole thing that I get rid of a lot of regions the complexity of my tree comes down significantly so even if I am making a slightly higher prediction error I might be willing to accept that because I have reduced the size by such a significant amount right so that is one of the reasons you consider all possible things.

May be pruning lower down might not simplify the tree enough for you to accept error reduction but if you go higher up the tree you might actually get the same error reduction right I mean whatever by reduction error worsening right but you must have you met a reduced entry by a much larger amount therefore you are willing to accept that right so it is actually no a great idea to just go bottom up all right so --so the small things to remember right.

So that's something that you pick since the magic word has been introduced so you pick α who ask the α question okay since the magic word has been introduced you pick α by cross-validation I will tell you what cross-validation everybody okay so all of you understand what validation is right so cross-validation essentially is kind of a multiple rounds of validation and instead of just using a single validation set.

You in fact try to use all parts of your data as validation in a very systematic fashion okay we will talk about this more detail later but just to give you a rough idea and so this is clear so what we are doing here yeah no, look at all possible collapsing right so basically what i mean by collapsing remove an internal node the entire sub tree structure underneath it whatever regions it was covering you consider that as a single region and you replace it with that.

So I can choose any internal node to collapse full sub tree and yeah concept is an expensive process so i said if it is expensive you can come up with other mechanisms of ordering it right but the best way to do it sorry nope nope decision trees there is nothing that is optimal I mean right I mean everything is hard right so any questions on any other questions on cost complextive pruning.

Likely yes but we do not know until you actually fit it you wouldn't know for example in the XOR case what we call it over fitting or not so you would not know right until until you fit the data you do not know whether you are or fitting or not so you have to grow the whole tree and if you are over fitting then when you prune you will actually end up removing it I mean the error will obviously on the training data the error will obviously be lower when you over fit right.

And that is why you need the complexity criteria right so when a prune if I am do not lose too much in terms of accuracy then I am happy to so any other question so so essentially what you do with the α as you pick a good choice of α right and then try to do the pruning on five different validation sets then pick another choice of α right on the same five validation sets you pick another different α right on the same five validation sets.

And then pick an α that gives you the best it depends on how you are normalizing the prediction error and as well as what is the expected size of the tree you are going to see right if the prediction error lies between 0 and 1and the tree sizes are order of, order of 10,000 right you would really want your α range to be small right where the tree is also of the order of say5 or 10 nodes then the α is could be larger.

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**Introduction to Machine Learning**

**Lecture 42**

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**Decision Trees for Classification-**
**Loss Functions**

So look at the probability of a data point in M region, belonging to class K which is $P_{MK}$ right not talking politics but so you estimate that by there is no counting the number of data points of class k and region m and dividing it by the total number of data points is fairly safe further this is how I do the prediction right so what about how do I grow a tree to do classification it is exactly the same as this except that I do not use square error right can I use square error why not exactly.

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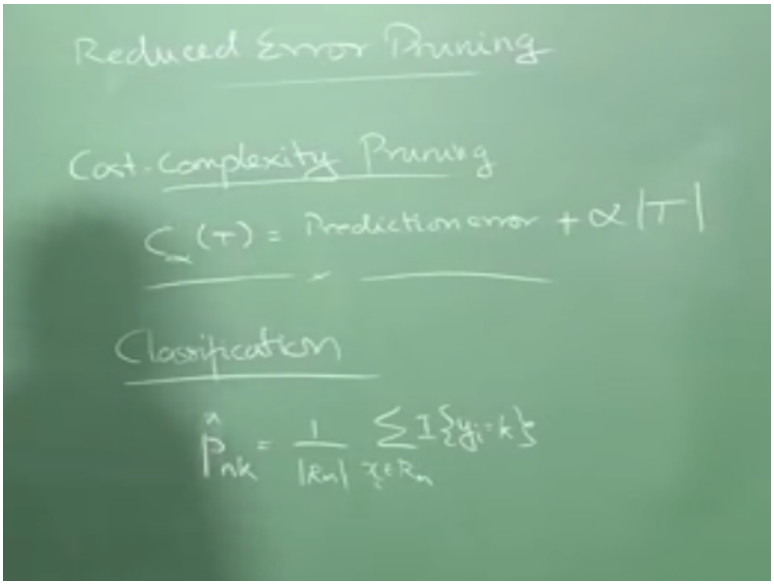

So it depends on how I encode it right I mean earlier in the linear regression you are kind of faking it by encoding it as indicator variables or whatever right so here I am going to have actual outputs I can still look at the distance of the prediction vector to the indicator variable vector right and try to do that right I can still do that but there are better ways of doing it right so the first thing I can use is the miss classification error.

So denote by K(m) the class label that I am going to assign to the entire region M just like we did the $\hat{C}(m)$ as the, the response that I am going to assign for the entire region M so k of M is the rest for the class label I am going to assign for the entire region M okay that just say arg max of this okay. so now the Miss classification error is I am going to count all the data points in RM which do not have K(m) as their label right that is a Miss classification right is all those data points the label I will be outputting K(m) right for all the data points in RM.

I will be outputting K(m) has the label so all the data points in RM which do not really have k of M has their label or misclassified right and divided by the total number of data points that gives me the average miss classification error is there some way to simplify this 1- $P_{m\ K(m)}$ okay so because the fraction of data points that will be correctly classified or $P_{m\ K(m)}$ right.

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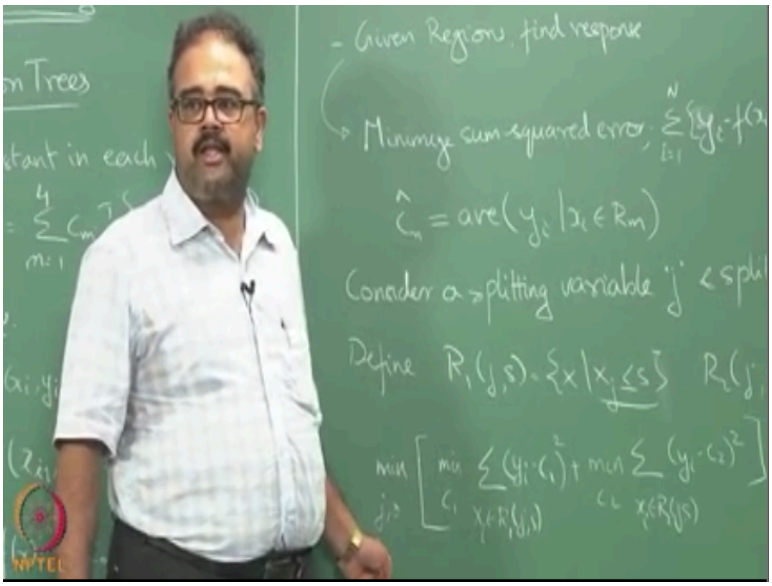

Because the two label is K(m) I am outputting K(m) there will be correctly classified right so the fraction that you be misclassified is 1- $P_{m\,K(m)}$ okay so that is a Miss classification error so how do I use this essentially I plug it in here right find the split point and splitting variables such that the Miss classification error in each of the regions is minimized at the sum of the Miss classification error in each of the regions many ways.

So remember that this has a very specific solution for minimizing so you do not really have to do this minimization is a fixed process right as soon as you find the region you just take the most abundant class in that region and set that as the class label for the entire region okay right is it clear I will do the how you used to miss classification error rate.

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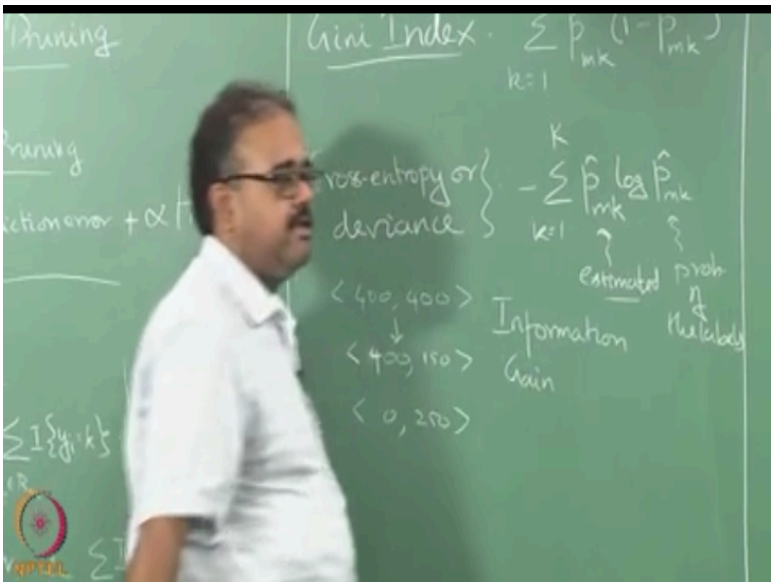


So the next thing we would like to look at this so one of the downsides of not being able to say anything very theoretically formal about decision trees is that it leads to dogmas so there are two camps of people who are very sure that this is the right way to do decision trees right they, they just keep fighting each other right and there are two very, very popular measures for doing classification using decision trees okay.

So the first one is called the Gini index ok so the Gini index was actually originally proposed by economists to look at disparity of wealth right so let us look at the wealth distribution in a population okay so are there more rich people than poor people or their lot more poor people than rich people I mean how does a disparity of the distribution of wealth okay that is essentially introduced that so in that in some sense you can roughly see that right.

So are there more class one data points than class two data points or anything else suppose I have K classes okay in this particular region are there more lot more class one data points than 22 k if I am able to split my regions like that then I am doing something good right because I can output the class level as one and I will have less error correct so if I am able to split region says that the class distribution is actually skewed within that region.

Then I am doing something good and if the class distribution is uniform within that region then I am doing something bad that because that is not a good region because whatever class table I output I am going to have a lot of error but if the class distribution is skewed in favor of one class over the other then I can output that class in fact the ideal leaf would be so skewed.

There is only one class present click so the skewness measure is what I have to look for and the more skewed the data is the better so the Gini index is actually more popularly given by this form so I do this for each region so this is for a single region I do this for all regions so the other popular measure is cross entropy or deviance but it is more popularly known by the name I will give it to you in a minute right.

And this is given by this expression this looks familiar to you guys Shannon's entropy kind of thing races cross entropy where is the cross part you have P hat $_{mk}$ and P hat $_{mk.}$ there so why do they call it cross entropy okay it turns out that that they see the true output label distribution that you have right from the data that is given to you right and this is what you do for estimating this is the estimated label distribution.

And since you are using an unbiased estimator for the probabilities you end up actually estimating the true probabilities so that is why it is called cross, cross entropy this, this is

supposed to be the that is the estimated okay and since you are anyway just counting the number of labels of each class and then dividing it and doing this so it is essentially end up with the same thing okay.

So the first one is the output label distribution this is the estimated one and so if you end up with the same thing right so another way of thinking about it is if you look at the prevalence of the labels in the data and I give you100 data points right essentially if I am going to randomly pick a data point and look at the label right so this is the probability of seeing label k correct so going back to your ideas of Shannon's entropy so if I have if I have a sequence of 100 things I have k possible symbols that can occur right.

And this gives me the number of bits I need to encode these k symbols given the relative frequency of those symbols right if I had not done the splitting right if I had not split into M regions right if I had kept the data as a whole I would have required some number of bits to encode the output level that make sense suppose let us look at it this way so I have my data so there are 400 data points of each class.

I will require some amount of bits to encode this right half, half the entropy is I mean the probability is half and half I will need some amount of thing to encode this suppose I split it up so that I get I get two regions one gives me 400,150 other gives me 0 and 250 that is how many bits do I need to encode the output variable here none right always the big improvement.

I do not need any bits for encoding the variable here and here I will need some but that certainly be less than this because we know half of the worst case right so in, in terms of the number of bits that I need for specifying the label I have some improvement when I do the split when I go from 400, 400 when you go from there and I get these two splits the number of bits I need has come down right.

So I have gained some information by doing this split right so how much information have gained? sorry right so the original entropy minus this quantity gives me the amount of information I have gained right so sometimes this is also known as the information gain criteria

because of that right so either you, you minimize the cross center of PR you maximize the information gain let us information gain is essentially some constant minus this so that is information again.

Therefore you maximize the information gain or minimize entropy so again the process is very simple you for every feature J you try to find that split point S such that this or this is optimized right one of these three things but the most popular or actually the Gini index and the cross entropy so one thing I want to point out so when you are splitting this into two things right and then I have to find out the overall cross-entropy are devious right.

So what I need to do is so the entropy of this will be weighted by 250 the entropy of this will be weighted by 550/800 right both of these cases so I will have to have some kind of weighted combination of the code or the Gini index whatever it is I have to have a weighted combination of the Gini index of the individual partitions or the deviance of the individual partitions so I have to be careful about that just do not add the M up okay.

You have to use the weighted combination so for this it is fine because it is per region yeah so again you have to be we have to make sure you are combining it appropriately right yeah there is only one output will come right only one symbol is present there is only one symbol present you do not need any bits to encode it because that is only symbol this present class one will not happen so, so 400, 400 means class one there are 400 data points class 2 there are 400 data points 0 to50 means class 1 there are zero data points class 2 there are 250 data points.

The symbols I am talking about are the classes right here there will be no occurrence of class 0 and only class 2will occur okay so one again one other caveat we are using this for classification you are doing cause complexity pruning right almost always you are supposed to use the Miss classification error because that is eventually what you are trying to optimize so you grow that tree with whatever error measure you want but when you prove the tree use the Miss classification error.

Because at the end of the day I am going to evaluate you based on the Miss classification error not on the Gini index or information gain or anything and these are in some sense they are relative measures we are good for comparing one feature against the other right but the final performance measure is only miss classification error right so use that when you are doing the protein ok so I will stop here.

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**Lecture 43**

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## Decision Tree – Categorical Attributes

We will continue looking at decision trees so like I said there is a little bit more about trees that I wanted to look at and then we will actually do a, an example today of how to construct tree, so starting from data set I will actually constructed decision tree right okay, so we already looked at a couple of issues with regard to addition trees what are the what are the things we looked at a well how will you we need to talk about that yet.

So when we look at how will you pick a cell a splitting attribute right what is the splitting value in the splitting attribute, so how large you should go a tree right and how do you prune it okay these are the issues that we looked at right whether several other questions that we could ask right, so when we talk about splitting attributes and split points inherently we are assuming that our attributes are continuously right. So that we can talk about a split point right so what happens if I have categorical attributes.

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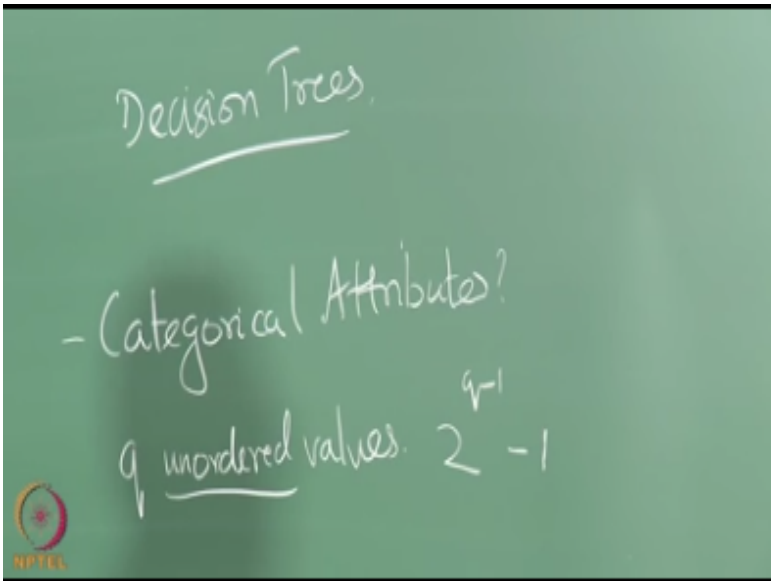


So no categorical attributes things that take some discrete values right, so it could be things like color red, blue and green, right or it could be things what you normally would believe our continuous variables like age right but for a variety of reasons they have been recorded as discrete values young, middle-aged old right, so most surveys and things like to if you look at it when you answer these things you know they do not ask you for an exact age they ask you are you lesser than 25 or in between 25 and 34 or greater than 35 or things like that right.

So somehow it is discretized either by for reasons of anonymization or for convenience or whatever you end up discretizing the values right, so you quite a lot of circumstances in fact especially if you are involving with the involved with medical domain right or with as kind of marketing kind of a domain right you will end up having discrete attributes right, so in which case what is the meaning of a split point, yeah I put this binary yes right suppose it is color right, so we think about it if I have q right so that is a key part here think of q unordered values right age itself has some kind of ordering in it.

So I can still fake it you know I might have only 3 different age entries but I can think of it young or middle-aged and old right young and middle-aged or old right it does typically usually

does not make sense to look at young and old and middle-aged you know because it is I mean I can say you can fake it you can think of it as a ordered attribute and then you can do this but suppose it is unordered right.

So essentially what you would really have to do is think of splitting the values into two subsets, so like I said let us say take color as an example right I might have to put okay red, blue and yellow into one group and green and I do not know give me a few more color names that is about how I how much I know magenta, purple okay so all of this into another group right and so like that right I have to figure out some way of splitting it into two okay, so how many possible splits are there like that.

I have q values to 2 power? good yeah, right so that many possible combinations is not really not going to be feasible for me to go over all of them in order to pick the split point right, so that is exactly what we were doing right if you remember algorithm from the last class you are actually going over all possible split points and we said there are only finitely many such possible split points because we only have to look at n of them right but now you have to look at even though I have only n values I mean n data points for training and I potentially have to look at $2^{q-1}$ split points right.

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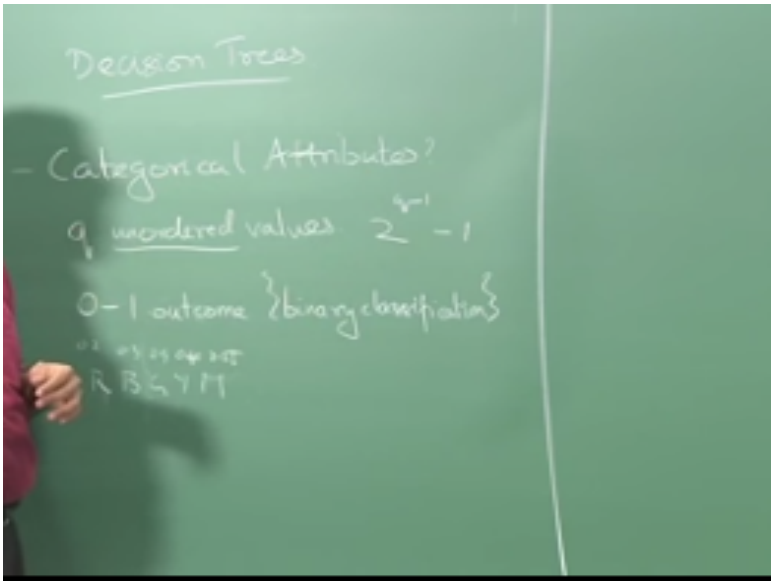


So that is not going to be feasible so there are 2 ways of handling this actually there are 3 ways of handling this the first one is if you have a we have a 0 - 1 outcome basically that is what people would call a binary classification problem if you have a binary classification problem you can do one clever trick what is it that you can do any ideas no I do not want to explore the number of attributes right I am making it some something very restricted here right I am looking at binary classification problems.

So says something that you can think of that you can do here, so what exactly are you looking at when you are trying to find a split point what you are trying to do is trying to make sure that your prediction right when I do it on one half versus other half is more accurate and the prediction I did on the data as a whole before the split right so that is exactly what you are looking at from the slip point trying to find a split point such that it is more accurate than the other right.

So what you can do essentially here is you can pick one of the classes let us say you pick class one right let us say I have 5 predictors or I am sorry not yeah I have 5 predictors, so predictors or this the unordered values right, so let us say I have 5 values for a particular thing let us say colors right red, blue, green, yellow, magenta right as I say I have 5 colors, now what I will do is I will

take red okay I look at all the data points that have color red okay I will see what fraction of them or class one right.

Then I will take all data points that have color blue I will see what fraction of them class one likewise for the other 3 colors it make sense so far I look at each color figure out what fraction of that color that data points having that color or of class one now I will arrange them in some order ascending order let us say of this probability when I say fraction it means what fraction that is a probability that a data point having color red will be class one right from the training data I shall ascend arrange it in ascending order of this probability.

Then I will just treat it like any other ordered variable and then I will split right does that make sense, so why does this help us think about it a little bit right suppose I suppose I have put it in some order right let us say that. So red has 0 .2 % of the data having class 1 right let us say suppose something like this no need not be why should they be I am just looking at each fraction of the data points with color red that were class 1 more than one color no this one attribute that says color of the data point okay whatever is it the one attribute that says color of the data point right.

And that way that attribute can take 5 values red, blue, green, yellow or magenta right suppose it is taken color red I look at what fraction of those data points that have color red or of class one right suppose I find that there are 10 data points that I have color red and two of them are of class 1 then it is 0.2, so obviously so these numbers do not have to sum to 1, right because they are only for that right, now we can tell me what is a good place to split this okay before somebody asked me a question about what if it is exactly 0. 5 okay.

There you go oh 0.2, 0.3, 0.4, 0.45 and 0.55 see how we go about doing the lot of one thing good point yeah so yeah, so you know how to do this come on pick up pick a thing and tell me what you know the Gini index and you know the you know Gini index or information gain or something like the right all of you know that all miss classification error let us use miss classification error as the splitting criterion, so for me to find an optimal split I do not really have to consider R and Y going to one part right B, G and M going to the other side right.

So it will either be here or here or here or here or here I mean that is a really bad attribute to pick if it is here right, but it will only be left to right all right, so these are the only subsets I need to consider I do not have to consider all of the other subsets right it does not make sense you can intuitively see this here right, so since this fraction of the class one keeps going up right, so either you break here or here or here or here, so that is a heuristic for this right in fact with a little bit of thing you can show that for two classes right.

You will get the same optimal split right by using this method as you would get by exhaustively searching through all the splits I am seeing too many puzzle looks we did this decision trees day before yesterday if remember decision trees okay, so split points if I have categorical attribute split points are going to be like subsets of the values the attributes can take right, so a split points would be okay do I consider our red and green to one side blue, yellow, magenta to the other side that is good potentially a combination right.

So in this case I am saying you do not have to worry about all possible subsets all you need to do is after you have done an arrangement like this okay wherever you choose to split depending on the criterion you are using so wherever you choose to split right so everything to one side will form one subset ever thing the other side will form another subset and these are the only subsets that you need to consider while you are trying to find the optimal space you do not have to consider all the $2^{q-1}$ subsets okay fine.

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**Decision Trees - Multiway Splits**

So what about multi class problems many, many classes not just 0 and 1 and I have like five classes for echo design labels from right what about multi class problem for multi-class problems this kind of simplification is actually not possible right so we have to end up using some heuristic or the other right so typically what people do is they end up doing some kind of very rough clustering on the values that attribute can take right and then try to define split points based on that and yeah so I am not going to go into the details of the heuristics I mean if, if at all you are going to use this and you will probably be using a packet but will be good to read up some of these.

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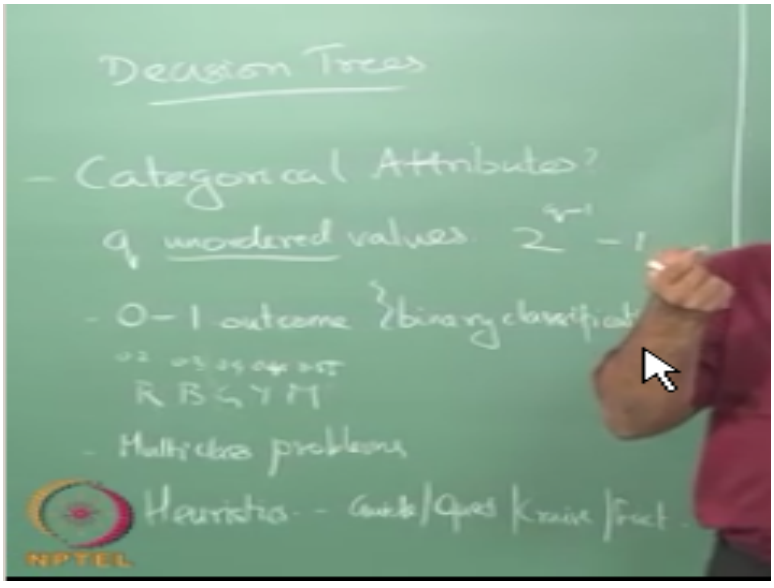

If you are interested as you can imagine as soon as you enter heuristic territory right everyone can have their own favorite heuristic so there are many that have been proposed in the literature is guide quest Clues fact the one says annoying things in many of this machine learning data mining literature as people sometimes go out of their way to come up with pronounceable acronyms.

So people have clustering algorithms called chameleon imagine how much work they must have gone to produce chameleon as an acronym and so in fact I think in fact what you essentially end up doing is you use some kind of indicator variables right for each of these right this is something I think you suggested that right this is an indicator variable for each of these dimensions and then they try to do some kind of dimensionality reduction on that I try to pick a discriminating direction right.

And then project on to that and then use that dimension for splitting suppose I want to spit on color right I will not do it on color so I will create 5 variables okay which is essentially one variable or color is red one variable for color is blue one variable for color is yellow and one will for color is magenta but I will not use those as Boolean variables right and I will try to find some kind of a projection from this 5 five dimensional space on to a single dimension and then flick that single dimension as a continuous dimension and try to do my projection on it essentially ends up doing some kind of clustering instead bring some kind of clustering on that one dimension.

You talk about clustering little later but you but you know what Clustering is I already told you what the problem is in the very first class okay the other approach to doing this is to is to do multi very multi-way splits, so what do I mean by that if okay if I decide to split on color or I have to evaluate color in so splitting it into two groups I will split it into 5 groups in our case in our example because they are 5 values color can take I will split it into five groups right so in my decision tree instead of always looking like this will suddenly start looking like that.

So what are the problem with multi way split so why do not we use multi-way splits all the time too much computation in what way not each of the class right why are we determining the split point for each of the class talking about an attribute that describes the data right this is that some confusion people are having here when it when I talk about categorical attributes I am talking about attributes of the data other than the class label the class label will always be categorical right if it is continuous then it becomes a regression problem right.

But then the values that are describing the data itself you normally assume that X comes from RP right I was telling you that that need not be the case right if suppose I am filling out a survey form you in stop filling in a rage or something will going to say less than 25 or something right so in such cases how will you test on that variable right how will I split on that variable that is a question we are asking so in now instead of saying that you see less than 25 and between 25and 35 will go left and greater than 35 and greater than 45 will go right.

Instead of saying that and say okay this will be less than 25 this will be between 25 and 35 this will be between 35 and 45 will be greater than 45 or something then splitting it all the ways in one go really does not it is a little bit more computation because when you are computing the score of each attribute that you have to do some additional work but it is not too much okay what is bad is it no yeah but moving you always remove the whole sub tree right yeah so interpretability becomes a casualty right.

So because if you are going to have multi-way splits becomes harder to interpret so the tree becomes very sprawling all right so remember as one of the biggest advantage of decision trees is that they are easily interpretable now if I am going to say okay there is a ten way split and then you have to go down the 10 way split and go down further then it becomes harder to interpret right.

So like if she was saying you might lose insights right that is essentially saying that some amount of interpret interpretability is lost but there is another problem with having sprawling trees yeah, so variance is more but the related problem to that right is the fact that if you do this multi-way splits the amount of data that is available might come down drastically right, so each path might

have suppose let us say Magenta is an Rare color right so here that has nothing to do at this 0.55 okay magenta might be a rare color right it might be just only 10 people in my million customer database ever have magenta color shirts okay.

Right but then 55% of them might be positive I do not know see that that has nothing to do with it right so how predictive it is of the positive class is nothing to do with the size of the population there right but the problem is I will only have 10 people here on which to make further decisions right so if I am going to do this multi-way splits I run into data scarcity problems very quickly does it make sense I mean I know I really cannot ask you questions and exams or things like that with all of these things more like practical guidelines for you to when you actually start using these algorithms.

What are the things you should be watching out for right it should we are using decision trees you should make sure that you are not running out of data points very quickly if you run the some branch in your tree becomes sparse quickly right then it becomes harder for you to trust the trick okay and this is related to the variance question because we are making decisions based on very small number of data points then naturally the variance is going to be high here decision branch that is what I say.

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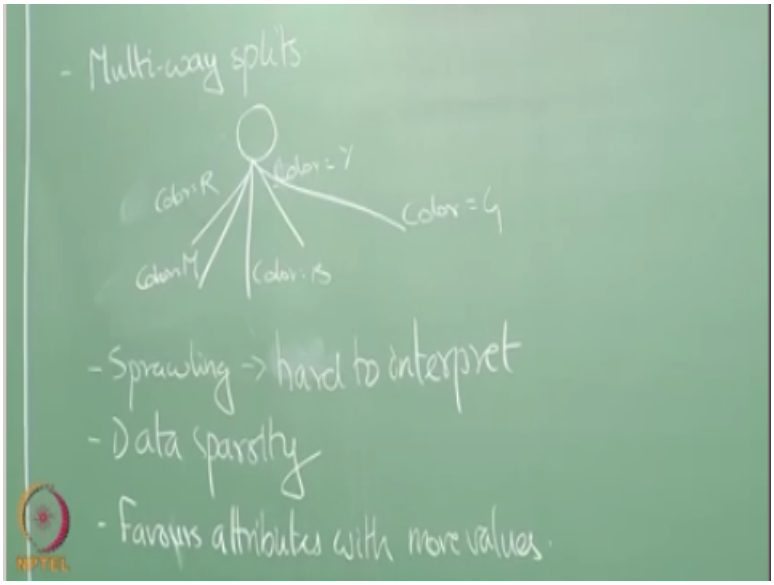


So if you want me to actually fill in some things here there are no two choices when you pick and that is the whole point I am eliminating the whole question of splitting again picking a split point right so at the color attribute I will say a color not is sorry color equal to R you go that way like that, so this is essentially how your free will look up, so this how it is going to look like yeah exactly so this is though so see you remember you compute the quote-unquote the utility of splitting on a particular variable right.

So you pick a split variable and then you find optimal split point in that split variable and then you look at what is the least quality whatever you can achieve rate we look at squared error we looked at entropy in whole bunch of other things, so you essentially look at that so here instead of looking at the best possible split point once you pick an attribute you split on all the values attribute can take and then compute the measure whether it is squared error or entropy or whatever it is you can compute the measure.

So all the measures we talked about where in contingent on the split being binary right essentially you just looked at $R_1$, $R_2$ for simplicity sake but I could have had $R_1$, $R_2$, $R_3$ , $R_4$, $R_5$ and I was computing the expression if you do not and if you cannot relate to what I am saying

this flip back if you actually were taking notes you would know what I mean right, so we have any way looking at the squared error measure right I wrote down as $R_1$ and $R_2$ where $R_1$ was it is lesser than J and where $R_2$ is greater than J or something or S whatever those split point.

But here we do not there is no choice of what is an optimal split point here once you pick an attribute you split on all the values it can take so it becomes prowling so that leads to so if we are doing this multi-way splits it is green natural please favor attributes with more values, so let us say I have color and then I have something else like a tan color has 5 values and ages like 15 different bins I have split age into right so when I split on color I will split into 5waybranch when the split on age I will split into 15 way branch right of course there can be exceptions but I would more likely to find pure leaves when I split into 15 then when I split into 5 right.

I split it into 15 ways and more likely to find leaves that are pure and if I split into 5 ways I am less likely to find leave set of pure right so just pure in the sense they have the same class right so this kind of multivariate tends to favor attributes with more values right, so that is not necessarily the best way of doing the splits because you might not be generalizing properly later right, so for this people use all kinds of tricks, so they are very popular decision tree algorithm called C 4.5 which uses something called gain ratio.

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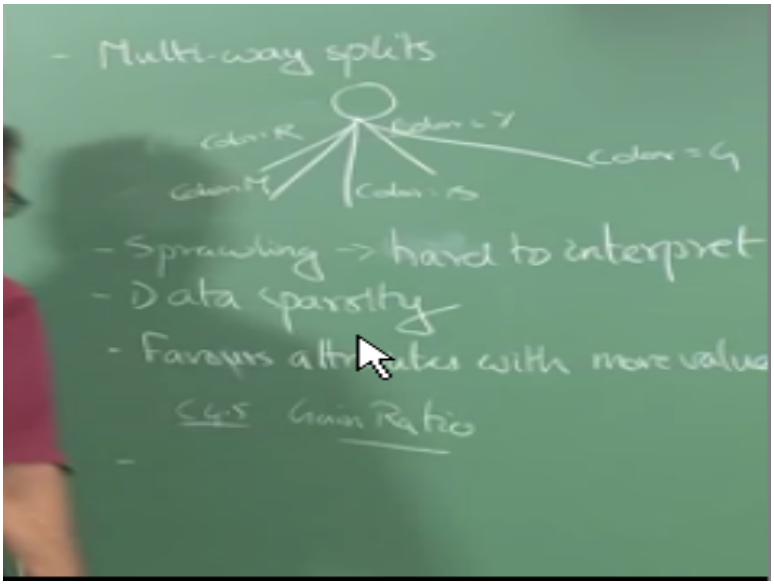


So people recall information gain as you spoke about in the last class it is related to entropy right the information gain thing so information gain tells you how much less information you need right by splitting on a particular attribute for encoding the class labels right, so what again ratio says is hey forget about the fact that I have this way this variable suppose I split the data into 10 ways randomly how much information would again vs. splitting it into 10 ways based on this attribute you see the defense I take the data split it into just randomly split it into 10 groups right or I take the data and split it into 10 groups based on this attribute okay.

So that ratio is what I will use so if I can just figure it out arbitrarily split the data into 10 groups under out of still gain the same information as spitting on this attribute then I do not want to split on this attribute it is no better than random right and heaven forbid ration is less than 1 I really do not want this right, so the ratio should be higher than,  so that is what I will be looking for so I can instead of using information gain I will use gain ratio in likewise you can order this for any of the attribute as any of the measures that you use that you can always adjust it for random splits. So that is essentially what we end up doing right.

So you gain anything special about expressive power for the tree we are doing multi-way splits as a tree become more expressive in the sense that can it represent more functions than you could with binary splits know whatever I do you do it is multi-way spits I can do a recursive binary splits and I can achieve that not adding to the explicitly it just avoids the question of picking a split point right that is not a trivial thing okay.

You have to come up with all kinds of heuristic to split bit points no pick split points but still if you can okay the recommendation is to avoid multi-way splits and stick with binary splits but in some cases just easier to do this especially if the number of ways in which you will split is small enough if you know if you are not going to split it into 20 different things or 50 different things right you can still do multi way splits like 5 or 6 should be fine.

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**Introduction to Machine Learning**

**Lecture 45**

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**Decisions trees – Missing Values, Imputation, Surrogate Splits**

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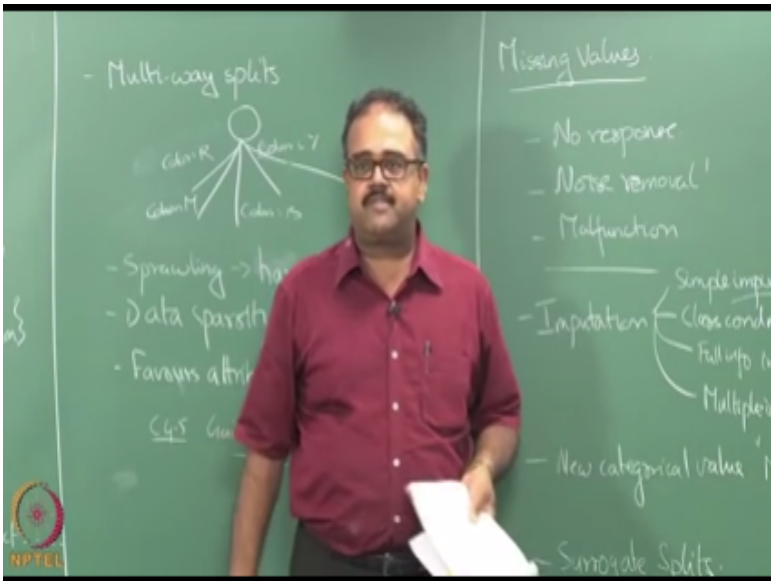


Missing values right, so what I mean by missing values? Some classes that are a different thing we come we will talk about class will much later. Suppose I have again let us go back take a real scenario right, you are filling in some survey questionnaire and then I am going to take all the data from you and then I am going to build a decision tree that allows me to predict whether you will buy a machine in my or buy a new computer from my shop or a new TV from a shop or something right and then the next somebody comes in and I am supposed just look at you and say okay he says he is going to buy a computer he is not right.

So I should be able to classify people like that but then when I fill in the survey I would just not answer some questions. So i will have a data point right so i am assuming that x is down from RP right or whatever the space that I am drawing $x_i$ from but for each $x_i$ assume I already know the values $x_i$ to $x_{ip}$ and so far we have never talked about the case where some of these might be unknown, some of this might be missing they might be missing for a variety of reasons right. So one could be right there could be a no response in a surveyor something right the other could be could be due to noise removal.

What do I mean by that? I look at my patient record data I find that somebody has a temperature of 223, so obviously some noise there right do I make it 22 or 23 when or 22.3 right nothing seemed straight right, I will tell you what scale it is in but still the still does not seem right so what do you do just remove it okay. So let us just assume that the nurse did not record the temperature of this patient right, so you remove noise from your data you might lose some attributes hi everything else about the patient I just do not know whether he was running a fever or not when he came into my clinic right.

So likewise you could just not have recorded it right that is equivalent to no response, so that guy messed up might have come with alike a bleeding right hand with it is just hanging off a wrist or something and you are not going to say hey first get this temperature and put it in there I do not want any missing values right, so this thing says it might not just get recorded you know so those kinds of things are there is an equal until no response, all right so anything else yeah exactly that is what is a malfunction right.

So it just that you might be recording sense of data from somewhere and the sensor just turns off for a while it may be it over heated or something went wrong and just for just a while you do not see any of this data being recorded, so the variety of reasons why you could have missing values in your data in fact if you work with real data right more often than not you will have significant missing values. In fact when I work with some data I have had cases where people have given me data where some attributes were missing in more than 80% of the data point's right.

So what you do in that cases remove attribute itself okay, you mean I shall not worry about the attribute because you are not going to be able to use it in any practical setting right so we just removed a trigger itself, but in other cases if it is missing only in like 5% or 10% of the data points you do not want to throw away that data point like throwing away 20% of the data point is still a big thing right and yes and you do not want to remove the attribute also because it is available in 80% of the data points and you do not want to throw the attribute away you do not want to throw the data points away right.

There are two things you can throw the column of a can throw the row way right if it is missing in more than 80% you can directly throw the column away, but it is somewhere in the middle right summer small numbers then you do not know what to do throw the column or throw the attribute throw the rope, do not both exactly. So there are lots of different ways of handling this missing, very the statisticians have studied these new ones right so they have come up with many techniques for handling missing values and why am i bringing it up while we are talking about decision trees and not other classifiers.

Because there are some techniques which is peculiar tradition trees which are not available for other classifier. I am going to talk about all of these right so that I mean in general also you could use some of these techniques the first one all right so we will give it a fancy name called imputation. That imputation is essentially filling in a value for the missing attribute right, so how do you fill in the value for the missing attribute, you the mean the simplest thing is to do the mean you could do a regression on the attribute right.

So you could regress on the attribute in fact what is the best way of doing regression on the attribute? You should do it in a class conditioned fashion, use the class also because you are talking about the training data here right, and so use the class also as in part of your regression or part of your averaging. So what you can essentially do is okay to take all the data points that are of class 1 and use those to predict the value of the missing attribute for in that set of data points right suppose I have like a hundred thousand data points and let us say thousand or of class 1and of which 4 of them are missing some attribute 3.

Let us take those thousand data points and I can i will do a regression and predict for those 4 data points i will use the remaining 9996 data points as my training data and fit the curve and now I can predict what that one missing point is, so why is this kind of conditioning on the class useful? You use that feature class, so if there is any kind of variation right the correlation between that feature in the class right this will help me preserve it right. If I am going to do this across the entire 100,000 data sets I lose the correlation I will lose the effect.

At least for these attributes it will get polluted right but this way I will be able to retain it, do not lose anything you do not do anything by doing it this way if that is correlation you actually preserve it if there is no correlation you do not lose, anything sorry? Exactly was asking just saying what if there is no correlation do not lose anything by doing it this way right. So this is imputation the different ways of doing imputation you can use the mean you can use the class condition mean right you can use regression for doing the imputation and there is something anymore complicated technique.

That something called multiple imputation on using the regression for doing the imputation is also called full information imputation, I told you only the statisticians have been at it for awhile right, so they have all kinds of the full information imputation is it because you are using all the known attributes, for predicting the unknown attribute right we are doing the mean you are only using that attribute in the same attribute in other data points and multiple imputation is a little weird thing. So what you do is you use all the data that you have right and setup a probability distribution over the missing attribute values right.

Like I said I have 996 data points in which that attribute is not missing right I will use that and figure out okay for if for red what is the probability for blue what is a probability for green what is a probability for discrete see there for continuous values i have to pick some distribution, let us say I pick precaution and else okay I will find what is a mean and the variance of the Gaussian that will predict the missing attribute value okay. Now what I do is I draw samples from this distribution and use those samples to fill in the missing values.

I will get one data set again other set of samples and fill in the missing values i will get another data so that is called multiple imputation. So i can create multiple copies of the data point by repeatedly sampling from this distribution and in some cases this has much better variance with much lower variance than using some of the other method. So even though this entails significantly more computation okay, so imputation is one that is another handle this I just introduced a new value for the variable right and I will call it missing.

Why would this be useful? exactly so there might be some kind of systematic reason for which the data goes missing and if I instead of trying to somehow guess what the value should be if I actually pay attention to the fact that it went missing right that would be useful, so I did not see who said that okay yeah, so infact it is actually a very practical practically useful thing it because quite often the reason it goes missing is that is a specific reason for it, and you can in fact the fact that it is missing might be predictive of and you know.

So how likely is my patient to recover the temperature reading is missing, so those kinds of things, so use something called surrogate splits, so what is a surrogate split? okay so surrogate splits actually a slightly different function, it works for imputation and this can be used during training itself right but the circuit splits thing we typically use during testing you can also use it during training is suppose. The basic idea is this for every attribute right that I have I will try to pick another attribute okay, that tends to split the data in the same way right.

Suppose let us say again let us take the same example I have 100,000 data points I split on attribute say 3 can I get two groups right this has says 70,000 data point this as another 30,000 data points. I split on some another attribute let us say 4 okay again I get two groups one has 68,000 data points other has 32,000 data points and not only that it turns out that the intersection of the 70,000 and 68,000 is something like 65000, on the intersection of the other two is something like 25,000.

So essentially three and four give me more or less the same splits, we are finding correlation we are not really reducing it here, so we are finding correlation what we do is if we have selected attribute 32 split on our tree and then we suddenly find that attribute 3 is missing in the data

point we just split on an attribute 4 and behave as if we split on attribute 3 and go on. So this is what it means a surrogate right it is like putting proxy right, so I have attribute for can put proxy for attribute 3 and then he just continued working with your tree right.

So that is essentially what circuit splits up and it does is it exactly finds, that right it actually looks at correlation between the attributes and tries to exploit that okay, so as you can see that imputation right and adding this new categorical values could work with any kind of classifier that you are working with right. As long as you have a way of handling categorical attributes it is just one more value that you are handling, while the surrogate split something very specific to trees right likewise we are going to look at fragment which is also something very specific to trees.

So this is a little subtle so what I am going to do is the following right, so I come to a point I am going to make some query, I am going to make this x3 <5 that is a query that I am going to make it is a variable x3 <5, that is a query i have to make at this point in the tree right and what do I find my data point does not have x3. That is for categorical drive talking about categorical attributes, so it will be like okay I am going to RM V YG missing like that that or if i am going to do it in two subsets I will put missing into one of the subsets right.

But suppose this is there x3 <5 so what do i do x3 is missing right what I do is I look at that all the data points for which x3 was not missing okay, I will see what fraction went down this way let us say oh 0.6 went here 0.4 went here, so what is 0.6 all the data points that did not have x3 missing 60% of those we are < 0.5% of those had $x_3 > 0.5$, so now what I do is I am looking at one data point right one data point that came here I am going to split it into two right, so it is going two point six of the data point is going to travel down the left and 0.4 of the data point is going to travel down the right.

So what I do is it is essentially I am actually letting the data point travel all the way reach a leaf right and the leaf is going to make some prediction, so some probability it is class one some probability risk class to some probability is class 3 right. So the 0.6 part will make one prediction the 0.4 paths will make one prediction, I make a weighted combination of the two predictions

and I output that as my final is it is seems like quantum mechanics name. So let us say I go down I finally reach here and I say its class 1 with probability I do not know 0.6 and class 2 with probability 0.4 and this one winds down somewhere.

And I say this is class 1 with probability 0.2 and class 2 with probability 0.8 right, so overall think that i will report is the probability of class 1 is okay, that makes sense no this is maybe this is a bad choice, do that make sense I am using it only once right. So what is the meaning of saying 0.6 of the data point goes down this side is essentially I will go all the way down and I will say that finally I will use the 0.6. I am not using the 0.6 anywhere here and these telling orders the semantics of saying 0.6 goes down this way.

So the reason we are carrying this weight along is at some point further down the line if I have another missing attribute and I decide to split it I have not been splitting one I will be splitting only 0.6 right, so this can get weird right, so I can have a data point which has multiple missing attributes traveling down more than two paths it will reach multiple leaves and then I eventually combine all the leaves so this is called when I am calling this fragmenting method right.

Again this is pretty unique to trees so if you think about it this is somewhat similar to doing multiple imputations. Of course the whole idea is to use training data to make a prediction on this data point right the whole the subject is predicated on using the behavior of other data points to predict output.

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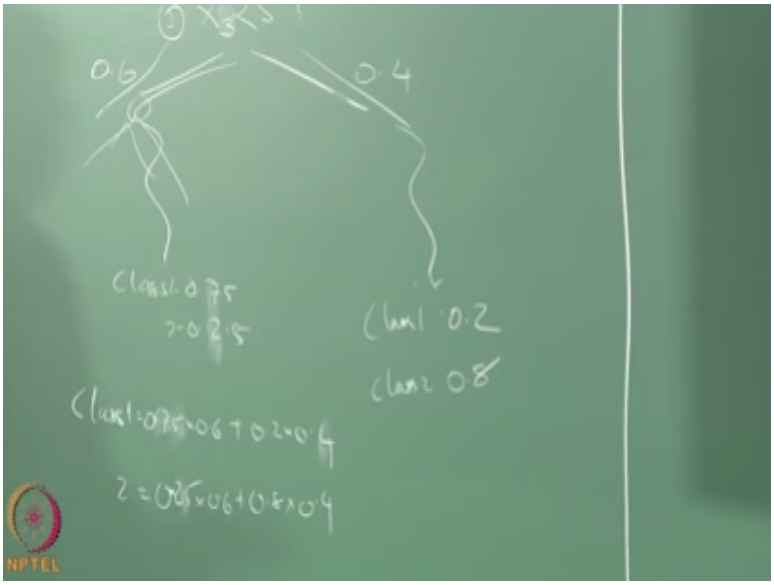


Of new data right, so it should not matter right.

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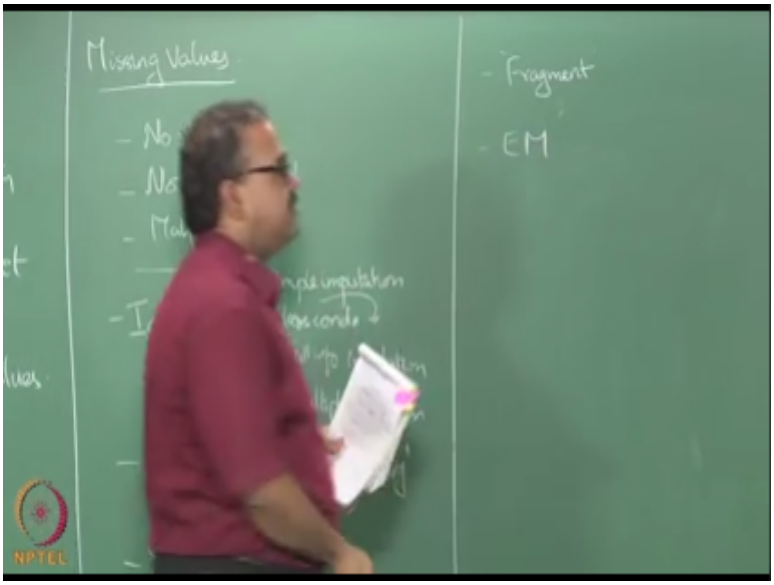


So the last way of handling missing values is something called am expectation maximization right and it is going to keep cropping up all over the place as we go along but we will do it we will actually formally do with deal with expectation maximization much later. So just be aware that when we look at EM this is one of the applications of EM okay and link to missing values I am not going to get into this is a pretty involved thing and in fact if you think you have been having difficulty with any of the concepts we have covered so far in the class you will not seen anything yet right so am is the one thing which everybody struggles with when they look at it first time so we will come to that later.

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**Introduction to Machine Learning**

**Lecture 46**

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**Decisions Trees – Instability,**
**Smoothness, Repeated Subtrees**

So the next thing would not talk about is instability.

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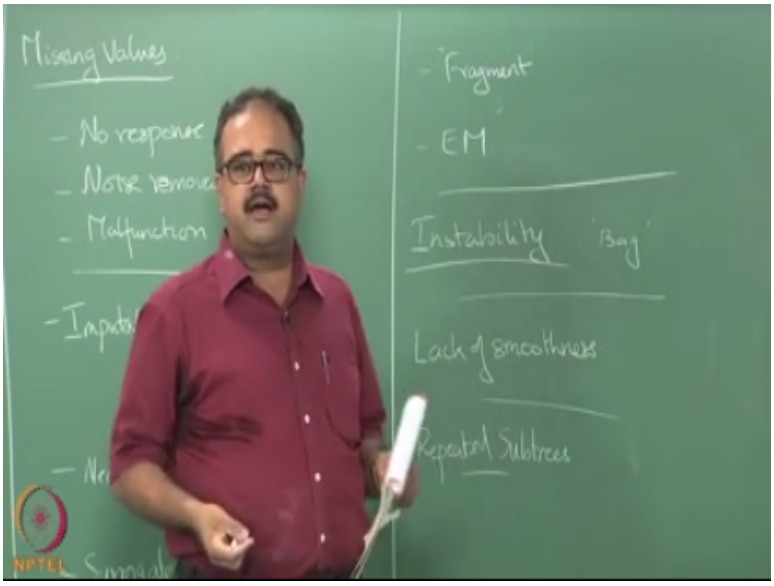


So Decision trees are pretty unstable so what we mean by not stable all of you know what we may not, stable small changes in the training data will could cause potentially large changes in the decision tree, so what could happen things that you split at the root might go somewhere down because of some changes in the data right and if the data you start out with small where is the sample size is small then the variation is going to be very high right. So I said any way of

getting around it some of it some of some regularization helps to some extent the solve the pruning and stuff helps to some extent.

But still not a lot right because the variance is really high these things is really unstable but still trees are very useful, so what we will do is we look at a very specific technique so that that is what we are going to do one of the ways of doing it yeah so there is a very specific technique called bagging right, so that not the next not the next four classes from now okay I will do bagging in more detail right but the basic idea is that to minimize variance is not just with trees you could do it with any unstable classifier.

So what you do is instead of training it on the data that is given to you train on slightly different versions of the data right maybe you can just take say 70% of the data randomly choose 70% of the data and then train a tree randomly choose another seventy percent train another tree keep doing this and then somehow combine the class labels predicted by all the copies of the trees.

That you have trained so this allows you to have slightly more stable classifier what is the problem with this yes anything else yes anyone else and I think I heard somebody say that you lose the biggest advantage of decision trees, which is simple comprehensibility so instead of having 13 or 15 trees or 100 trees now again you have the problem of keeping your job and your manager ask to explain what happened right so now I have 100 trees and somehow they make this magical prediction and you do not know.

So that is a problem so smoothness in your prediction is something that you are looking for a condition trees are not going to give, you that right there will always be this jagged jumping around especially regression trees right, so when you are talking about making some predictions if you are using a piecewise constant fit right for every region you are going to have some amount of jumping around so if you are looking for a smooth function for doing your prediction this is not going to work so you have to do some kind of post-processing after you build the tree in order to smooth the predictions right.

And there is nothing you can do this nature of the beast so the trees are so much convenient in other ways but smoothness is a problem, so if you are looking for a smooth fit right for your prediction that is not going to happen and problem of having repeated sub trees, so what do I mean by that multi way splits it does not have to be multi waste receiver in binary splits you can get into this problem think of XOR right how will XOR look like right I will split on $x_1$ if $x_1$ is 0 I will go down the left branch $x_1$ is 1 I will go down the right branch and then what do I do in the left branch.

I will text on the test on $x_2$ if $x_2$ is 0 I will go down one branch $x_2$ is one I will go down the other branch and likewise I will test on $x_2$ on the other side with 0 I will go down one branch I will go is 1 I will go to the otherwise so we can think of it these two sub trees are kind of similar right, so it could very well be that I split on one attribute but everything underneath it could be similar the tree structure is very similar but I cannot collapse it because I end up with different conclusions right.

So if x, $x_1$ was 0 and $x_2$ are 0 I would be outputting 0 right but $x_2$ us 1 and $x_1$ was 0 I would be outputting one so the outcomes are different so I cannot really club the two trees but then the test that I do are exactly the same right, so this shin trees are prone to having this kind of repeated sub trees right so you could have the same test set of tests that are actually implemented in many different points in the tree so it just makes it three more complex.

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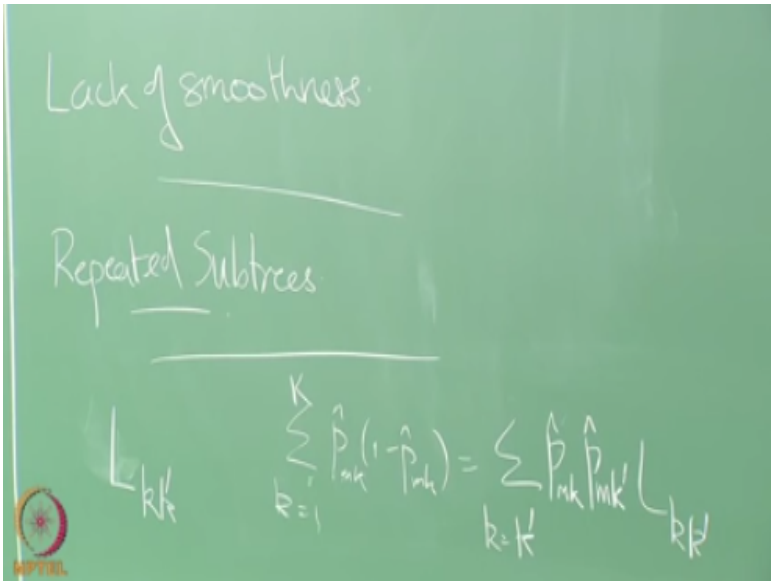


But there might be other ways of reordering things, so that you get with the simply the XOR is a bad case right so if you reorder x are you still get with this we still end up with the same kind of repeated structure but there might be other cases where you might have just done the splitting in the normal way but end up with too much repeated structures but if you had we flip the ordering of some variables even though it is not the best variable to pick at some point but you might end up with a more compact tree but finding that is finding that is very hard finding that ordering is very hard as I so you have to just live with it just pointing out some of the caveats.

So far we have assumed that we are dealing with the 0/1 loss function so what is the 0/1 loss function for classification right yeah as good as a mile I mean I do not care there is no ordering in my class labels if I miss if I do not predict it correctly I penalize you with one if I predicted correctly it 0 but there might be cases where some miss classifications are more acceptable for you than others right so what do you do in such cases they so you are going to have some kind of a loss value right.

So I am going to have some kind of some LKK` which is essentially the prowl loss that I will suffer by classifying the data into k` when it is actually class k correct so I am going to have this

so how do I accommodate that in the decision tree setup, how do I account how to accommodate that in the SVM setup optimal hyperplanes by the way if I am missing one thing we never actually talked about how you use SVM for multiple classes I will ask me the question what is the margin maximum margin.

What is a margin mean and you have multiple classes that is a topic for another day I will come back to that but yeah, so as she is not immediately clear right so how we do that so suppose you have neural networks like we all know about neural networks right, all of you are familiar with back prop by now I suppose right so how will you accommodate this kind of thing in back prop I think of ways of doing it, so it turns out there is no easy way of doing any of these but what you can do is at least in decision trees.

Try to incorporate this when you are computing your GINI index or your what information gain whatever right, so whenever you are looking when you are doing that so you can figure out okay what is the probability that I will miss classify this so what is the GINI index expression that we have right, so this is what we had right so this is essentially this was the probability that a data point in region m will be in class K right times the probability that data point in region M will not be in class K right.

So as another way I can write this which is essentially right so probability that the point is k + k and probability that is K` so essentially from here to here what I need to do is take out all the terms where with $P_{mk}$ hat and some out some of our the remaining so which will be 1- hat $P_{mk}$ right, so that is essentially what I did so for each k, I take $P_{mk}$ hat out from here and sum over the remaining things and I will get this expression okay so this is some way of saying let okay the original probability is k okay.

And the estimated probability is K` this is the someway of looking at it so here what I can do is I can add my $L_{KK`}$, so you have to actually work this out for all of the measures that you are going to work with so if you are going to have a neural network mean squared error criterion you are minimizing or cross the deviance at your cross entropy or minimizing whatever is error function

you are minimizing lot to figure out what is the appropriate way to use this class information this class specific loss information okay.

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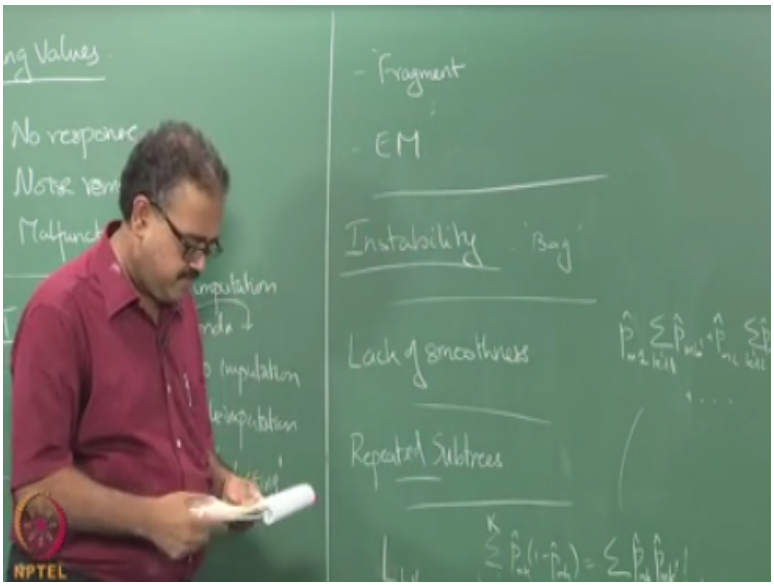


There was a not equal to the right yeah, okay without this yes they are equal without this year equal so essentially what is what I am doing here is so for every k right I am writing one term like this so he this I can simplify like this right, so that is $P_{m1}$ hat $\sum k` != 1 P_{mk`}$ hat + P hat m2 $\sum k` != P_{mk`}$ hat right like that I can do that so I will get case terms and this summation is essentially $1 - P_{m1}$ hat and this summation is $1\text{-}P_m$ hat to write like that so that is essentially what I get here right so like that you have to work it out for everything so if we have a different loss function okay.

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**Introduction to Machine Learning**
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Lecture 41 NOC

Thank you. Thank you. Thank you. Thank you. So this is a sample dataset and we'll try and build some classification trees with this. So here age is young, middle age and senior, income high, medium and low, student yes or no, Boolean variable, credit rating fair or excellent.

This is a target variable which we want to predict whether the person will buy a computer or not. So the first way we'll work on this problem is using entropy measure and a multi-way split. So what is the first step in building a decision tree? So we have to select which among the, so this is our target. So we don't consider this.

We have these four attributes. Which one should we pick? In a multi-way split, as we just saw, there is no issue in picking split points. So we just need to fix on which attribute to select. So how do we do this? So we'll use the entropy measure. Right, so if we start with the attribute age, so first of all we see how many data points have the value youth, 1, 2, 3, 4, 5 and what are the corresponding class labels.

So 2 have no and the remaining 3. Just a minute. So we have three no's and two yes. So we just apply this formula. So we are calculating cross entropy for the variable h for this data set.

So we have  Is this clear? Yes. Somebody had a question. It's fair and excellent. It really doesn't matter. I mean the data set is pretty generic.

It's just a toy example. So this is applying the cross entropy formula. But we have three different values. So when we are trying to calculate the entropy, this particular formula is applicable for when we consider one region. Remember M is the region.

But when you consider the decision tree, it will be split into three different regions. And we want to calculate the overall entropy. So this is for the youth. Then we will have to add to that what we will get for middle age and what we will get for senior. So if you can just add that, do that.

So if you look at middle age, you have 1 is yes, 2 is yes, 3 and 4, both are yes. So what will be the entropy? Do we need to calculate this? Right? for the senior. And while we are doing this, we also have to weight it according to the number of data points that go down that branch of the tree. So in this case, overall, there are 14 points, of which youth, there were five. So we'll have 5 by 14.

And for middle age, it doesn't matter, but there are four points. And for the remaining, it's 5 by 14. So how many are there for senior, if you can calculate that? We have one, two, yes. same 3 and 2 split. So we will have, is this ok? Can somebody calculate this and tell me the value? Is there any questions with this, please? How am I getting? So you look at, let's say, youth.

So that's one particular value. And how many of them are labeled no and how many of them are labeled yes? 2 are labeled yes and 3 are labeled no. So you are applying this formula. So what does this mean? In class region, what is the probability that this attribute will have a particular class? So in this case, with 2 by 5 probability, data point labeled with youth will be? 5 youth, there are 2 attributes. Yeah, that's right.

Has somebody calculated this value? Can somebody tell me? What is the cross part here? So, essentially this should be the actual distribution of the data, the true distribution of the data. So, if you knew the process that was generating the data points then you could then this is where you would use that like if the age is youth then what is the true probability of that data point being classified as yes or no. Here we have an estimate which we can use. The estimate is going to be off from the true distribution. So that is why we are calling it cross entropy.

However we do not know the true distribution. So in its place we are using an unbiased estimator which is the sample average. That is why even though we are using, it seems more like an entropy formula than a cross entropy formula. The notion is actually cross entropy. There is supposed to be a true distribution which we are supposed to use.

Anybody with this value? right so next so this is the entropy will get when we use the attribute as the root right so we can make a decision tree have age as the root attribute which we test on and this is the entropy would be good so next what we do is we do the same process and we try and find the entropy when we use the attribute income student and credit rating is that clear so can Maybe one row can go ahead and calculate it for income and the other rows can calculate it for the other two, student and credit rating. Same calculation, follow the same pattern. I'll give you about a few minutes. Can you go ahead and do the calculation? Important point is that you will get some values for entropy and which one will you select? The one which has the highest or the lowest? The lowest.

The lowest, because it's a measure of impurity.

And we want to eliminate that. So in this particular case, you'll find out when you do this calculation that this is the best attribute to use. So in your decision tree, you'll have at the root, age. And you'll split in three directions, whether it is youth, middle age, or senior. Now what happens here? So at each of these, once you've done this, It's a recursive process.

Now you eliminate from your data set everything which doesn't have age is equal to youth. So you consider only that portion where age is equal to youth and you repeat the process. We've selected the attribute at the root to be age. So now we have this partial tree. So at this point what do we do? Already we have used up the attribute age so we are remaining with three more attributes but in this branch we consider only those data points where age is equal to youth and we apply the same process to select the next variable.

So this whole process is continued and it goes on up to Either you can build the entire tree where each leaf is pure, or you can stop at one point. And then what do you do if it's not pure? You give it that label, which is majority. Is that clear? So the same process you can do with the Gini index. It's just a different formula that you have to apply. But the entire process is exactly the same.

Instead of using this formula, you'll use the formula for the Gini index. Now the next thing we'll look at So this is the case where we actually wanted multi-way splits. But what happens when we want to restrict ourselves to binary trees? We looked at some advantages of why we would require binary trees. So in this case what we do is, let us say, the same can be done with the Gini index. I will just continue with the cross entropy formula.

So again if we want to build a tree, a binary tree this time we still have to figure out which variable to choose but in addition we have to figure out split points. So just now we saw how to avoid the problem of right. So we order the variables, if it is a, for example a student is a yes or no, it is a binary variable, so there is only one split point. But in the other case where there are more than two possible values for an attribute, you have to figure out the split point. So let us do it for this age variable.

So you have you have 5 data points which have age is equals to youth and out of them only 2 have the positive class level, so that is for youth. So for middle age we just saw that all data points which have age equal to middle age are classified as yes. So here the probability is 1 and for senior it is 3 by 5. So our ordering will be right so essentially either we have a split point where we consider youth in one set and senior and middle age

in another or we do this now note that this particular attribute age has a notion of order. You consider youth, middle age and senior, there is some order.

But for this exercise we have treated it as an unordered variable. If you wanted to use the inherent notion you would just use youth, middle age and senior. So now what we do is the same process but now we consider all data points lying in youth as for example to start off with let us evaluate this so we will do cross entropy. Right? So here, we're trying to split the data in terms of a binary split, where on one branch we have age equal to youth, and the other branch is age equal to senior or middle age. And we'll try to find the cross entropy.

So in the previous case, because we didn't have to find split points, we could evaluate the cross entropy once for each attribute but here we will have to do that multiple times depending upon how many subsets we have to evaluate. So you will have one, in this case there are only two possible split points, so you will have a cross entropy value which will put youth on one side and the rest other two on the other and similarly you will have an evaluation for this part. So that is just for cross entropy variable. Then you do the same for income, student and credit rating. And finally among all of those, whichever is the lowest, you pick that as your attribute and split point. Is that clear?

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**Introduction to Machine Learning**
**Lecture 48**
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**Evaluation and Evaluation Measures I**

Okay, so that leads us into some things to talk about.

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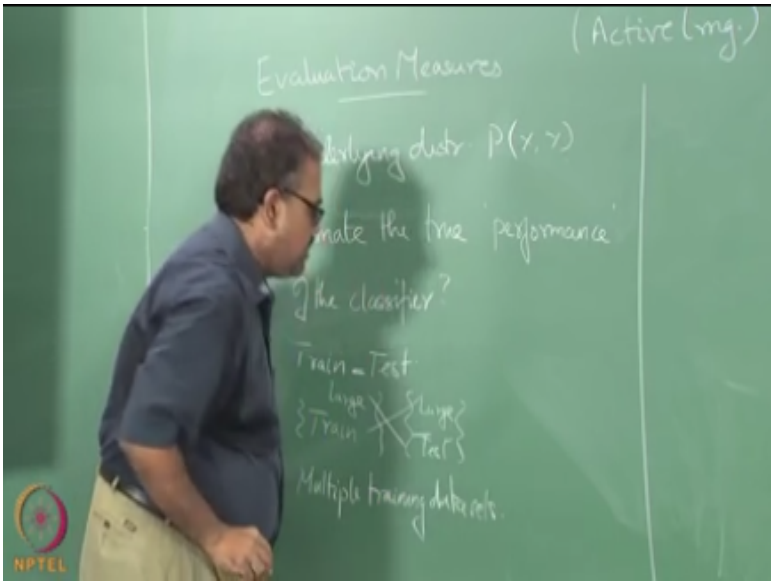


I am going to talk about evaluation measures for a bit today so there is maybe like a little bit of hodgepodge class okay, I am going to talk about a few things which do not really fit into a bigger theme right, so for as far as evaluation is concerned we have spoken about some very standard think so far right, so in classification we talk about come on give me something yeah ,what would be the evaluation measure in classification, miss classification error then cross entropy then Gini index okay.

Gini index is not a evaluation measure right, Gini index is more of a parameter selection mechanism right, I can after and finish the classification I can actually compute some kind of a deviance right, and I can say okay this method giving me better deviance then that method and things like that or the same thing for 0 1 error I can use squared error right squared error right, squared error of your thing to a target variable and anything else can I use, penalties in what sense, okay.

So there are a couple of things that we had to be careful about here, so there is something which I optimize right, to get to what I want right and there is something which I use for evaluating what I finally get if you limit ourselves to the classify supervised learning scenario right we are not even started unsupervised learning limit yourself to supervised learning scenario more often than not what we really want to evaluate ourselves with is the performance on the entire data distribution right, so I have some distribution of data right, I do not know the distribution apriori it derive go back to the very first class we started talking about something serious I mean not the one with the pictures right, the one with the Greek in it, right.

So in that class we talked about that being an underlying data distribution right, and that we did not know about this data distribution the only way we know anything about this data distribution is through training data points, right is through the samples that are given to us right, so there is an underlying so we had this distribution so what I am really interested in is finding out when you give me a classifier how well it is performing with respect to this distribution, right.

So in that sense how well it is performing I am not really interested in figuring out the square the ridge regression loss or anything like that right so I am using that to come up with a single classifier, but at the end of the day when I am looking at how well this classifier is performing with respect to the underlying distribution I have certain measures, so one of them is the 01 loss, so I do not care how we arrived with the classifier right I just want to look at the 01 loss and then I can do that right, that 01 loss gives me the miss classification error, right.

So ideally that is the evaluation measure that you should be using okay, so sometimes what people do because they are optimizing a different objective function they choose slightly

different evaluation measures that may can make their method look better right, so squared error could be one evaluation measure, if you are doing classification 01 losses the measure that you should be looking at if you are doing regression while little tricky but square error is the most widely accepted measure for looking at regression.

But then you can look at other things also like deviance and other things you can use it for classification right, but having said that how do I estimate it is easier for me to write classifier or I will pick one I am going to pick classifier but some of what I talk about now works for regression as well right, and how do I estimate the true performance of the classifier, so what do I mean by that, I give you a sample data right I give you some sample drawn from P(x,y) right.

So based on the training that I that is all the information I have right, and I can use some of the data for training. I see find the parameters so how do I find out how good these parameters are or how good is my there are two questions to ask right, so the first question is how good are my parameters that I have found okay, the second question to ask is how good is the method that I use for finding the parameters is if you give me a slightly different data will I perform better or worse right, how will I perform right.

So I need to know something about the technique right, suppose I am proposing a new technique and I want you to go use it on your data later on right, but I should convince you that you can use it on whatever data you have right, so that means I have to convince you that my technique is good for finding the parameters this is the two things here for a given set of parameters I have to figure out how good they are right, and I also need to tell you how good my overall mechanism for finding these parameters are, right.

So for a given set of parameters how do you find out how good they are, on the training data I said good enough on the testing data, cross-validated okay, so if you okay, so we will get to this thing right, so one thing if we spoke about earlier was to split the data into train and test right, so if I estimate parameters on the training data again write it on the test data that will give you some performance, okay is that a good estimate of the true performance of the classifier.

Why not, Test data might not be independent of the training data okay, the training data may be biased I met a wharf the model then you are doomed right, no, that is actually a very, very valid point in fact that is something which you will face in real life right, but the assumption mostly we make in theory is that the training data that is given to you is a sufficiently representative data of the true distribution okay, that is not the case then you are doomed anyway, right.

So you assume that it is the, so in real life that happens what do you do, in real life that happens okay mean you cannot avoid it, you cannot just say okay, I am assuming it is a properly representative sample of the underlying distribution then what do you do in that case, come on he is telling that you are not sample the entire range of P(x,y) so what is exactly so figure out where you are deficient right, so sometimes the most obvious thing is what you have to do, we have to sample more right.

But then do not blindly sample I mean of course you blindly sample you may actually return the same samples from whatever region you already have right, so what you should do is you should be more careful in how we do the sampling so you can use this is where you try to understand what the data is all about right, so you try to understand how the data is distributed the data that is given to you is distributed and figure out if there are parts of the input space which you believe we are important but are not covered in the data right, go and try to sample from that region, right.

So there are the different names for this okay, so one popular thing that people call this why this call, is called active learning, because I actively I am asking you for samples so I am not passively learning from the samples that were given to me okay, the learning algorithm comes back and says hey I want to know more about this part of the state space give me some samples from there right, I want to know more about this part of the input space give me so this is called active learning methods.

So your question this valid point of this discussion is yeah, one train one test is usually not a good idea right, so what do you do there are two things which you can do we can try to get multiple training sets from their data right, you can try to get multiple training sets from the data

and try to so nobody is asking the obvious why are you getting multiple training sets from the data why not one large training set, okay why not pull everything together and then create one large training set.

Yeah, close yeah, so I will have to spend a whole week on weak classifiers right so we will come to that right, so it is a see that is one very, very amazing property that will look at later which probably the next class and later means pretty soon, on how you can take a lot of not so good classifiers right, which are just better than random of course it has to be better than random right, it cannot be worse than random so classifies that are just better than random and give you an accuracy of 51% okay.

So in the two class problem that is just better than random okay, I can take classifies at giving accuracy 51% and they can produce arbitrarily powerful classifiers okay, it is an amazing, amazing insight that came about a couple of may decade and a half ago now maybe do not I am old more than two decades ago right, and it is one the girdle price and things like that is it an amazingly wonderful inside and we will talk about that right.

So that completely revolutionized machine learning once right, people then started saying so I really did not, do not have to build this super optimized classifier I can build a lot of this almost moronic classifiers but the operational word is almost right, I have a lot of them right, I have a lot of them and I will be able to do really well.

In fact in many, many applications that we have worked on right in real life where I have worked on with real data, I find it very hard to beat these kinds of classifiers you can think of whatever optimal classifier you want to come up with right, but beating these kind of groups of weak classifiers is actually very hard in practice alright so we will talk about that but no, that is not what I meant here I still have a point to make, right.

So yeah, so even if you do one large training set, to do one large one train one large test set right, you can get away with it provided largest large enough right, provided large is something that is dense in your input space right, if the large is so large that you essentially plaster your entire

input space right, so any point in the input space if a pick there will be one point very close by in the training set that is what we mean by dense I mean there is an actual more mathematical characterization of dense.

But if my training data is really dense in the input space then it is fine right, then I can get away with just doing one sample like one very large sample, but usually what is going to happen is you are not going to get such a large sample, so you are going to get a much smaller sample than that and therefore if you just use one sample and try to make an estimate okay, then the variance people remember what is the variance of the estimate, we talked about this again no, no that is unstable I am talking about variance yeah sorry, on data of similar the size we train a lot of models on data of similar size the parameter estimation I am going to make will be varying a lot right.

So it turns out that instead of doing one sample and then trying to train this if I take many, many samples right and then find the parameters on these samples individually right, and then take an average of those okay, turns out I can show that the variance will be lower in that case then what is the variance you are talking about the variance in the parameter that we are estimating, okay and what is the parameter we are talking about estimating here so here is the point where I am going to confuse all of you.

But the parameter I am talking about estimating here is the error is the miss classification error right, so I have the classifier right, what I am trying to measure is a miss classification error and that is what we have the whole discussion was all about I am trying to estimate the miss classification error right, so what I do is I start off with many samples of data right and on each sample I train a classifier separately again then I look at how the performance is on the test data and then take an average of all these performances and then I can tell you okay, if you give me a new data new set of data I expect to make this much error on the test data.

I am trying to figure out what the performance of the algorithm would be on a unseen train later I remember I was telling you I want to know how good my algorithm is right, so this way I can estimate the performance of the algorithm on the unseen data right. so there are many ways in

which you can generate this, the many ways in as you can melt this multiple training data sets right, so many of these or I have strong roots in statistics and were typically designed in errors where the amount of data available to you was small, right. The amount of data available was small and we are trying to see how you can fake multiple data sets with a small amount of data okay, so the first technique is known as bootstrap okay.

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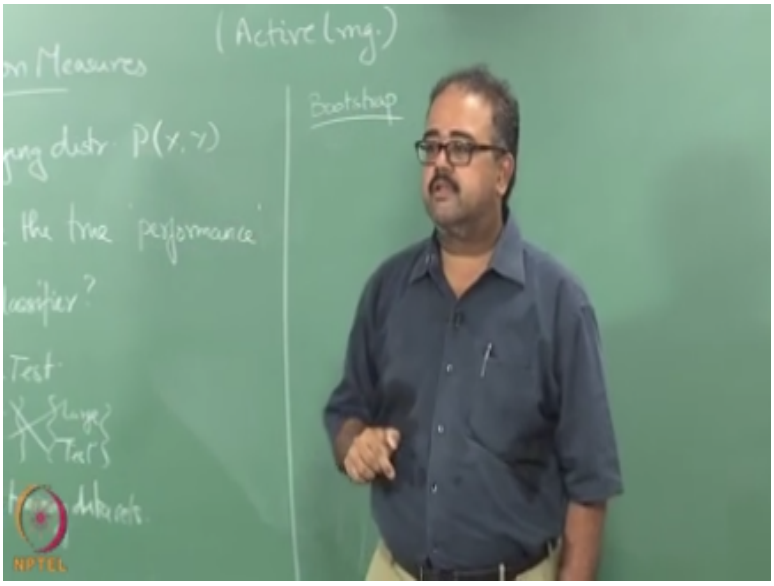


So bootstrap is actually a very powerful statistical technique it is used in a variety of different places we will come back to another use of bootstrap a little later.

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**Introduction to Machine Learning**

**Lecture 49**

**Prof. Balaraman Ravindran**
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**Evaluation and Evaluation Measures II:**
**Bootstrapping and Cross Validation**

Right anyway so I will just talk about one news of bootstrap here so the idea behind bootstrap is very simple right so I have a large sample right I have a large sample of data so what I am going to do is I am going to sample from this data with replacement.

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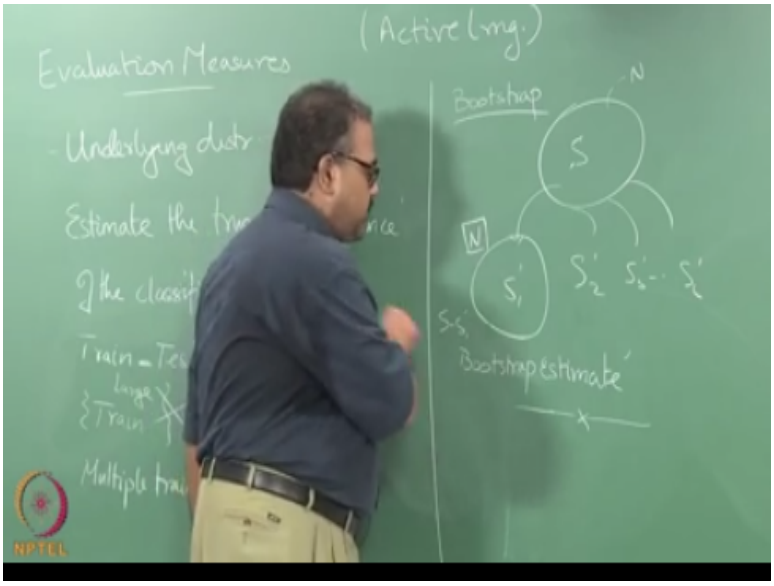


Let us assume that the set S has N elements in it so what I will do is I will sample from it another set S prime that also has N elements then sound like a big thing right. I mean if I wrote sample with the if a sample without replacement I will essentially be duplicating it but I am sampling

with replacement so what is the idea behind doing this no so if you remember I said that the assumption that we are making is that exactly the assumption that we are making is that the data is truly representative of the underlying distribution right.

In which case given the data right the best approximation I can construct to the underlying distribution is the discrete distribution right defined on this data in one sense if I do not make any other assumptions all I can do is I can construct a discrete distribution on the data. It is a probability of sampling this point $x_1$ is equal to the number of times $x_1$ appeared in my state set S divided by the size of s right.

So how do I simulate this distribution sample form S with replacement is it make sense to people right I am going to assume that yes in some sense the set s is representative of the underlying data distribution and I am going to simulate the underlying gator distribution by using the discrete distribution form by S. So what do I mean with the discrete, discrete distribution I do not know may be the underlying p is actually Gaussian or whatever.

But my s has only N elements so only these end points will have some nonzero probability of occurring so that is what I mean by the discrete distribution so I can just construct the discrete distribution from is and I will sample from that that will give me a S` right so I will call its S` like that I can do that multiple times to get right I can go up to SL prime right I can do that you can create many many many such samples okay.

So now what I do to get a bootstrap estimate of the classification error so this kind of a sampling to produce this L sub subsets I have done is called bootstrap sample oaky. So wonderful once I would derive such samples I can find out the bootstrap estimate of the quantity that I want right so in this case error so what will I do I will try on s1 prime right and what will I test on I will try non S1 prime and I will test on s - s1 prime right.

Because I am sampling with replacement so some of the data points will get left out right so whatever gets left out I will sample I will test on that so likewise I will try another classifier on s2 prime right using the same method if I am using back drop for training I will use back prop

and train on s1 prime okay and test on s-s1 prime likewise I will try on s2 prime and test on s-s2 prime.

Like way so I will get how many estimates for the error L estimates I will take an average of that that gives me the bootstrap estimate for their right and you can show that the bootstrap estimate will have a lower variance then just the error estimate on just using s and just randomly splitting into test and train it will have a lower variance right so what I go again--again I want to be clear what do I mean by lower variance here.

If I give you another training data point of size N right and then you do the same thing you do you do to estimates one just train on the original set that is given to you and test on the test set once or blue this bootstrap estimate likewise I give you another data point net the data setoff size n another data set of size and so on so forth so now you have two estimates for each of these okay the second estimate will be more consistent than the first estimate.

That is what I mean by lower variance okay so that is a bootstrap estimate okay so this is sometimes I forget the exact number so its estimate of the error then but it is some parameter that i am estimating about the whole process that's why I said right so this is one way of estimating the performance right or thing right I mean you can estimate anything you want on s1 prime s 2 primes 3 prime s 4 prime you can do whatever right.

You can estimate the variance of the data on s1 prime just the variance office one prime not all here for anything that you can do this for each of those yell things and then you can have a bootstrap estimate of the variance right so you can do whatever--whatever statistics you want you can measure on these L sets right and then you can call that the bootstrap estimate okay so good stuff is a very general technique it is not just for error measurement.

Yeah so if you have been in--in a proper statistics course by now should have gotten into the variance reduction properties of bootstrap and we could have shown some interesting results right but I'm just going to tell you that I variance Goes Down and you can see intuitively it goes

down and I will leave it at that okay so roughly about sixty--sixty nine percent of the data okay on an average rate sixty-nine percent of the data will be in s1 Prime.

And the remainder or sixty-three percent of the later sorry 63 will be in s1Prime and remainder will be in the test data right. So this is also sometimes called that ever get 0.632 bootstrap or something like that because sampling with replacement leaves a certain fraction of the data in the in the sample and leaves another fraction of the data in the test set okay. So it is sometimes that fraction also denotes what bootstrap estimated this right.

Remember okay so this is one way of doing it and this works fine provided I had a large enough sample to begin with right so it had large enough sample to begin with so suppose your sample is smaller suppose your sample was smaller you do something called cross-validation K fold cross-validation.

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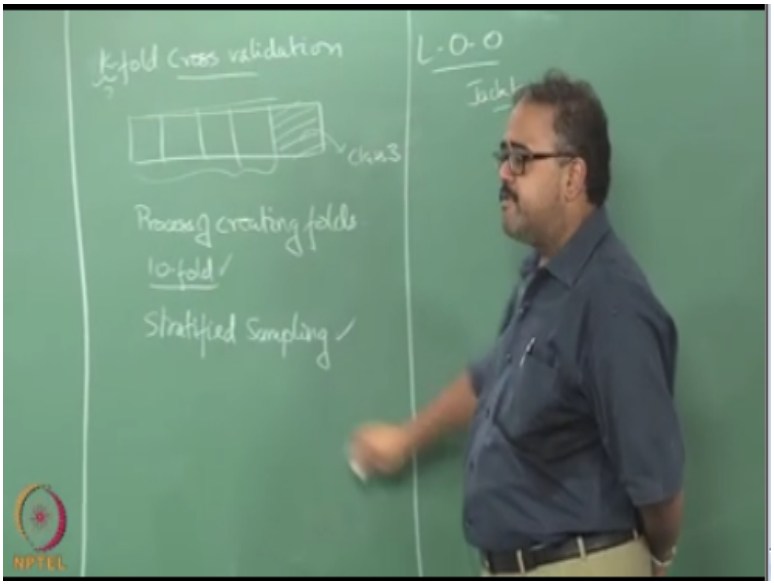


So what you do is you take your sample okay you divided into multiple bins and it divided into K bins what I do I train on some K-1 bins and I test on the last bin right the clips I use the first k-1 bins as my training data and one bin has the test data next what I do and servicing the last bin

I use the second last bin as the test data and everything else has the training data right suppose I break this into k bins I will have k different estimates right.

I will take an average of those so which will give you a better estimate bootstrap or cross validation depends yeah okay that is not that is not the end of it we are done we want it depends on the size of the data I mean if you have a sufficiently large sample that your bootstrap assumption is true right so you might get a better estimate with bootstrap right one of the nice things about cross validation is that every data point is in the test set at least once.

Is it correct exactly once every data point is in the test set exactly once so in some sense whatever number that you are reporting is essentially the average over here performance on the entire sample that has been given to you right entire sample that has been given to you at some point you are using this so there are a few caveats that you have to worry about the first one is what is K right and the second one is the actual process of creating folds so what should be k 5 to 10 yeah okay.

Yeah so those are the numbers typically used five or 10 okay those are the numbers typically used and so depending on the number of folds you have okay you have stronger variance reduction properties the more the number of folds the more the variance reduction property provided the folds are large enough that if you have K data points there is no point in creating k folds but people do okay.
That is a very special kind of cross-validation I will come to that later and I'm just going to leave you with the short form of it okay so we will come to that so which is exactly creating k folds if you have k data points it is called leave-one-out leave -one-out cross-validation okay leave one out that is what L over--over stands for right that essentially means that you will train on N- or K-1 data points in tests on one data point so in some cases.

No--no there are actually in there are in cases where this still gives you good estimates okay and earlier version of this was the one of the first do not ask me why it is called jackknife but one of the earliest is this kind of variance reduction technique used for a parameter estimation it was

called jackknife a jackknife is very similar to leave one out okay. So going back here so I would recommend that do not split it into so many faults.

That you have very little data point left in each fold right and so the typical number do not do not go more than 10 folds right if you manage to get 10 folds out of your data right then you should just be happy right that gives you good enough variance reduction so typically people in a report empirical results do not expect you to do more than 10 folds right. But then if your data size is small right people end up doing only five folds right.

There are extraordinary cases even when your data size is not small when you have to do fewer folds right. So let us think for a minute suppose I have am solving classification problem right so I have created these folds right and this is entirely of class3 okay and there is no class3 in this data right. Since you think this is odd this can happen quite frequently if you are dumb about how we split to your data so the data has come to you sorted by class level.

I will give you the class well sorted by class level and you do the fivefold splitting by serial number right so what will happen is you will have all your class go into one fold right the other 4 folds will not have any data point of class3 now if you try to test on this what will happen yeah right because you do not even know that exists class3 okay a training you didn't even know there was class3 so we are going to get 100 percent error right.

How do I avoid that shuffle the data in fact I do better than that I I do what I call what I call stratified sampling so I am not going to make much progress today so we will have to stop with the cross-correlation I have so much more that you need to talk about will do the next class so stratified sampling so stratified sampling essentially says that when you create the folds try to make sure that the class distribution that you had in the original data is maintained right.

Suppose I have five five data points of class 1 and 10 data points of class to right and I am splitting it into five folds right I should make sure that there are two data points of class two and one data point of class1 in every fold right so the class distribution the 2: 1 ratio is maintained in

every fold. So this is called stratified sampling so this is something that you have to do so the recommendation is due 10 fold then you do stratified sampling.

And now can you answer my question why even though you have a lot of data you might be forced to have smaller number of folds class imbalance right so i might have very few data points of one class what if i have only 10 data points of one class right if I do tenfold sampling then forced at the cross-validation right I will essentially put one data point of that class into each fold that might not be sufficient for me to get a good good enough estimator.

And I might want to do a smaller number of force five may be 3 of course the other things which I could do but this is some case where you might want to you work with a fewer number of folds then what your data would suppose right so if you want to have a more formal description of cross-validation, bootstrap and leave one out and all of that you are encouraged to read has t write the elements of statistical learning book has very very nice discussion on all of these things right so right now I will stop here right.

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**Introduction to Machine Learning**

**Lecture 50**

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## 2 Class Evaluation Measures

Okay, good. So today we will look at some measures that are typically used in classification. And so, I will primarily focus on two class problems okay. The main reason is that more often than not the classification problems will encounter is for two classes, you know want to identify it as belonging to a class or not belonging to a class right so multiclass classification is little rarer to combine right.

And frankly two class classification has a lot more richer set of measures that people have proposed and usually the ones that we use for multiclass classification or extensions of these measures to multiclass okay.

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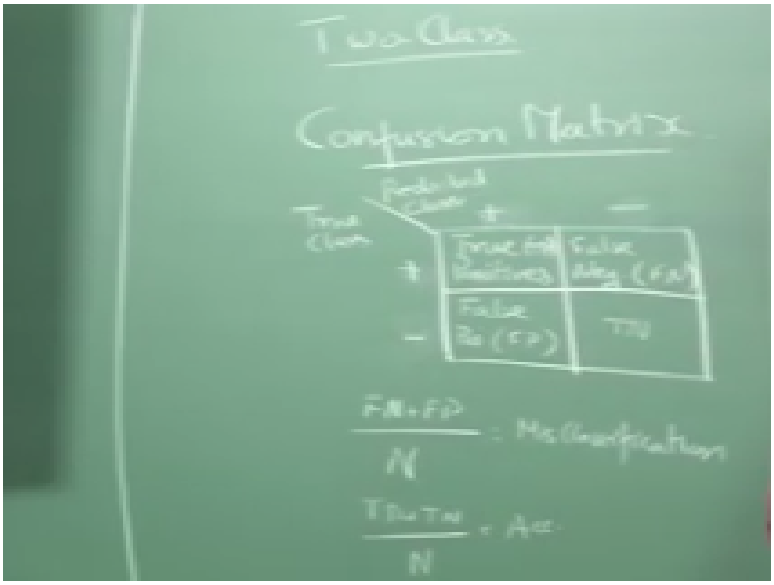


So the first thing I want to introduce you to is something called the confusion matrix okay. So is nothing to do with the understanding of the course material or anything so far okay, it is something completely different right. So I say it is a 2 class problem, let us say the classes are 0 and 1 right. So I am going to say that, so I am going to form this matrix so the true classes are on the rows, the predicted classes are on the columns right.

I mean it does not matter if you do the transpose of this as long as you remember the meaning of the numbers that going here okay. Let us, I am going to change this slightly, so I am going to call class instead of calling them 0 and 1 as we have done so far okay. So I am going to call them positive and negative. So it makes it little easier for me, so what is the positive class, typically the class of interest to us right.

So what we will denote as positive class is the class of interest to us, what we denote as negative class is the class that we do not want right. So when I say that in some problem that the positive class is that person is suffering from dengue, it does not mean that dengue is something positive okay. It just means it is a class of interest to us, so just remember this okay. So when I say positive class it is a class of interest okay.

And more often than not your positive class in the population will be small right, hopefully right, I mean not too many of you have dengue right. So the positive class will be small and the negative class will be large right. So and we have to worry about getting the, what does it mean, of course yes. I mean that is my interest, so I am a doctor, I need patients to pay me right. So obviously the people were sick or more interesting to me then the people who are healthy right.

So that is the class of interest and the negative class is the other class, I am not so much interested in right. So things that go in here right, so with the true class is positive right and I am also predicting it to be positive okay, these things are called true positive okay, otherwise known as we will denote them as TP right, there are true positives okay. And what about these guys, they are true negatives right.

They are actually true class is negative and the predicted class is negative there are true negatives right. So that is why I said, you just had to remember which the true positive, which is the true negative and other things here. So whether you write it this way or whether you write the transpose it does not matter right, as long as you flip these three things, these two things around so what about these guys, false right.

So what about this, some of you know about true positive false for everything right, everyone has been telling me all of this, so where have we encountered this before, nowhere so glaringly obvious okay good, great.

So now what is the most common classification thing that we know about this classification error right so accuracy right so -1 the miss classification error is known as accuracy right what will be the miss classification error here where n is the total number of data points, right this is what miss classification so what about accuracy.

Yeah so n is the total number of radar points okay it is not the total number of negative points in case somebody is worried about that but I need a symbol for total number of negative points also I will introduce something later right this is known as accuracy right as you can see is 1- miss classification error right and what are the other things that we know of that are popular right so

you can take a lot of different ratios of these numbers right and come up with different evaluation measures right.

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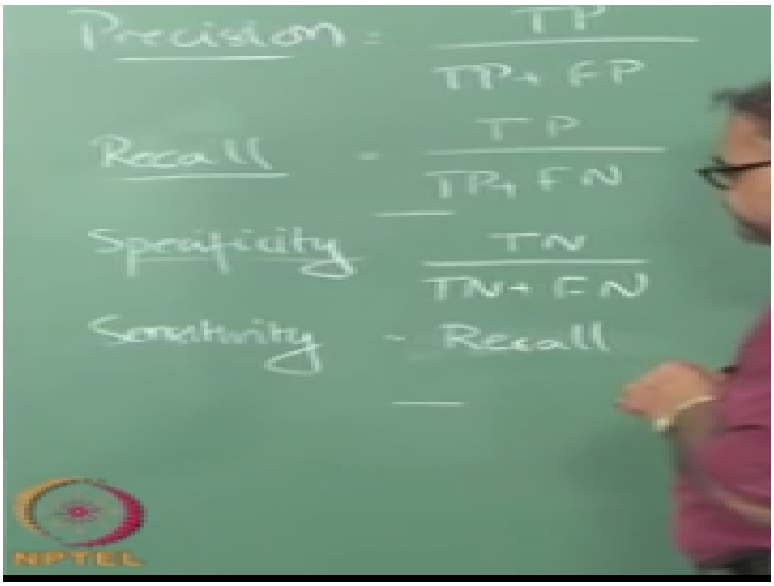


So I will talk about a few popular once so anyone know what prison is true positive so what does it mean so the classifier is going to tell you so many data points are of the positive class right the classifier is telling you so many data points are of the positive class how many of them are really positive right so the denominator is also those points which the classifier is telling are positive, right so true positive + the false positive are all the points that the classifier tells you are positive and the true positive are the once that are truly positive right this ratio give me the what is called the precision right.

So precision can be defined per class if you will if you want to rights so here I am doing this only for a two class thing that is why we are talking about positives but suppose I had k class problem I can treat it like a 2 class problem and I can give you the precision for any class, so suppose class I am interested in some class k okay I just keep that as the positive class and everything else as a negative class right now I can create a true positive true negatives, so I can actually talk about precision for class k precision class I and so and so forth.

It can for a multi class problem I can talk about precision right but typically this defined for 2 class problem, so there is a complementory measure for precision know as recall right so what is recall all is negative, so what is recall essentially there are so many positive points in the data right there are positive points in the data of these positive points what fraction is the classifier telling you are positive.

Right we get that all the positive points in the data which is true positive + false negative right how what fraction of it is the classifier telling you are positive so that is recall right so one way of thinking about it is so precision recall actually originate from information retrieval they are not originally for the classification domain they are originally proposed as a measures for evaluating information retrieval, so what do you mean by information retrieval so there is some repository of documents right.

Then you type in a search query right and then I give you back results corresponding to that search query right suppose there are 10 results that I give to you right of this 10 how many truly relevant to the query, what is that precisions suppose there are 50 documents in repository that are true relevant to the query how many of them appear in that 10 that is recall right so that is why it is called recall so how many of them to by actually recall from the repository so I have this huge repository of documents, how many of the truly relevant documents to by recall from the repository so that is why is call recall and procession you can see this is actually miss normal for people who are used to measurements right, what is precision in measurements sorry, how there is no not enough closeness in precision right.

No, elec guys come on how many elect guys are still in the class, 2,3,5,6,7,8 okay yeah, so what is precision sorry, how we elects valid precision if we go to measurements right, precession is essentially how many digits you are going to actually measure it to okay, that is precision you did not be accurate okay, the accuracy tells you how correct you are right, we can be less precise and very accurate and it can be very precise and very inaccurate I mean I can give you absolutely random number to 10 decimal places so I will be very precised, right.

But I cannot make any guarantees about the accuracy, but precession here is nothing to do with that okay, so precession here is essentially all to do with correctness, right so if I giving you 10 answers how many of them are correct. So why is this is a good measure, when this is a good measure already I told you an example, information retrieval right, so what characterizes information retrieval if you think about it, why cannot I use accuracy in information retrieval as a measure, why do I need something new, right.

Why cannot I use the following measure okay, I type in a query I give you back 10 results okay that means of the remaining documents okay, so I have rejected all of them right, so these 10 documents I have classified as being relevant to your query the remaining documents I have classified as not relevant to your query, so among those I have classified as a relevant how many are truly relevant, among those I classified as not relevant, how many are truly not relevant and I can do an accuracy, right.

I can do miss classification error, so is that not a good measure for information retrieval. Yeah, so there will be like I said 50 documents at extra relevant to your query but you might have a 10 million document copy though if you Google you have several 100 billion documents as your copyrights right, and of which 10 or 15 might be relevant to your query right. So if I just use accuracy as a evaluation measure right, I need to be really, really precise in the measurement sense to make out any difference between two algorithms, because they mostly they will be correct.

And because of large fraction of the data I am going to say is irrelevant and I will be correct right, suppose I have two million documents and I am returning 10 to you right, and there are only 50 things that are relevant so basically I am right mostly right, a few things I miss here and there but I am correctly most of the time, because I have said large fraction of the irrelevant documents are truly irrelevant, right so that way I am good right, so that is not a good measure.

So when there is extreme class imbalance right, accuracy is not a good measure right, only 1% of your data is of positive class then if I say everything is negative class I am 99% accurate, but my precession will be what, you can define it to be 0 but in mathematically it will be undefined

because I said everything is negative class I have no true positive, no false positive okay, so but you can define it to be 0 if you want and recall will also be 0 in that case, okay.

But quite often so what you will find is that if you try to increase you precision right, your recall will fall, when you try to increase your recall your precession will fall right, why is that so if you want to pull in more of the positive class right, if you try to pull in more of the positive if you want to predict suppose there are 40 documents that are relevant and I want to predict all 40 of them as a being relevant so the easiest way to ensure that is predict everything as being relevant, right so may recall will be very high.

Right which will be 100% right but the position will be too low depends on what is my universe of documents precision will be too low so and if you want to have very high precision what we have to do predict only sure documents select no documents that go back and define zero base zero as one instead of defining it has zero right.

So we will define zero and zero minute ago and define it has one right because if undefined you can do whatever you want with it right so select no documents is saying as 100% precise obviously because you cannot point out the mistake I have made in giving documents back to you so we will recall and we will suffer because recall will be zero right.

So there is always precision and recall right so we have to figure out where you want to pitch your algorithm right so typically people draw what are known has PR curves right precision and recall curves and how do you think this PR curves look like, like this, like this yeah like that here it really want to be here right.

You do not have high precision and high recall but then you can compare algorithm so you compares again this PR curves right for example again let us go on I will tell you little bit more as we get long right so there is another measure which is especially popular in medical literature call specificity so what is specificity something different from what we are seeing so far it is what does it mean yeah so what does it mean what does its schematic of this.

So schematics of this if I say that something is if I have a high sensitivity sorry if I have high specificity it means that if I say something is negative then it is for truly it is really negative right so why it is good thing to have exactly right so this is very useful in medical test so I run the test and I said that you do not have malaria right or well must be topical you do not have dengue right.

Then you really should not have dengue it should not be say okay he said he does not have dengue but then really has as a 50% of chance he actually has dengue even though the test says that you do not have dengue okay that is the bad thing to have right so in such cases specificity is very important so if you are building a classifier right that predicts whether a person is suffering from a particular disease or not.

Then it should have a high specificity okay the flips side of specificity is right is a terminology that comes from medical domain right so sensitivity is true positive by true positive plus false positive and specificity is too negative by true negative by false negative okay it looks like at the other things that you leave out here okay this is specificity this is sensitivity.

The two measures that you just like a precision and recall and information retrieval in medical literature you have sensitivity and specificity okay sensitivity is just like precision well specificity is the opposite of that right so sensitivity says okay how likely you have to sensitivity is recall I think sensitivity is recall sorry yeah sensitivity is recall so it just says how likely are you to diagnose the disease right.

If there are so many patients with this particular disease how likely I am to find a patient with the disease right so what fraction of the patient is fully discover if there are the disease so that is the sensitivity and specificity is essentially if at all you do not have disease how likely is that you do not have disease okay right.

So for regression we already looked at a classify measure so more or less is the same thing right so basically you look at squared error right so all the interesting things are with the classification right for regression we look at a squared error or you can do an absolute error also if you want to

evaluate the how good your regression fit is you can do absolute error we can do squared error whatever if we can use so one more thing which I want to talk to you about.

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**Introduction to Machine Learning**

**Lecture 51**

**Prof. Balaraman Ravindran**
**Computer Science and Engineering**
**Indian Institute of Technology Madras**

## The ROC Curve

So when I build the classifier typically right, so what is going to happen if I'm whether I am building this hyper plane business or whether I am using a discriminant function right, there is an additional thing which I can use to tune how am I going to finally assign the labels right, so let us say I have a two class problem so then you can say that I learn a discriminant $\Delta$ and typically what we do if $\Delta$ is less than 0 we assign it to one class, $\Delta$ is greater than 0 we assign it to another class, correct.

Yes ,so what if I tell you that no, no if $\Delta$ is less than some $\theta$ you assign it to one class if it's greater than $\theta$ assign it to another class, right so what will this let us do let me move things around a little bit right, so essentially what will happen is I can by moving by changing this $\theta$ I can take some points which are originally classified as positive and make them negative right, so one way or the other rights i just keep sliding $\theta$ around, right, so essentially.

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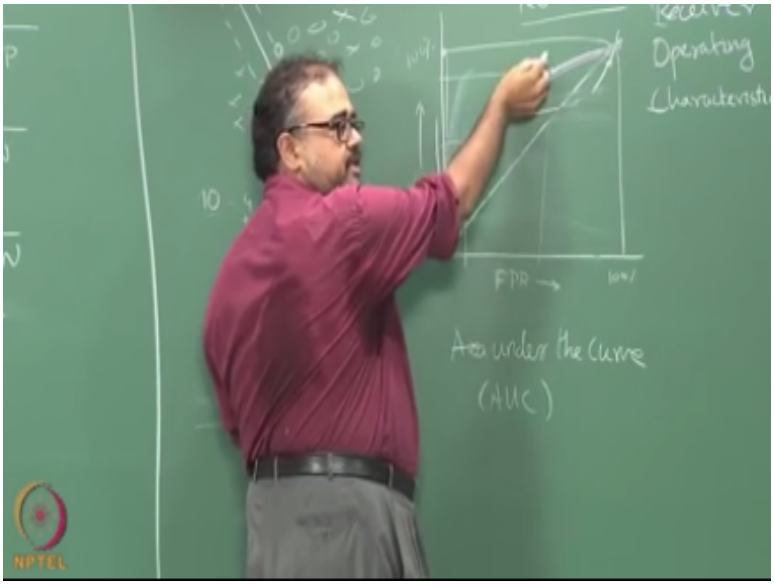


Right, so that is a line that marks where they are equal to 0 right, when I say that is greater than some θ so that essentially means I could either have a line this side right or I could have a line that side right, and as you keep moving this what happens when you move here so this will become 0 that will become class O right this is already class 4 now this will additionally become class O at likewise when I move it that side now this will become class x right, so I can actually change where I pitch my line right and I can get different performance right.

So what is important here is okay, I have figured out that right but then I can move this that way or that way is little bit I can change me what is it whatever I want to do right, so I can increase the precision of my classifier by just saying okay I have learnt the classifier right I have learnt whatever it is i have learnt the hyper plane but instead of looking at 0 right, I will just move it a little bit that same right when looking at the point where the probability is 0 I will move it a little bit that side or that this side so that I can change my classification that I give right.

So given that I can do something like this right, so how do I know i have got a good classification right, how to put it I put it in another way right, so I am asking you to give me a classifier right, but I am not going to tell you right what is the precision I need or what is a recall

I need from this classifier right, so you give me a classifier and then i am going to figure out what is the θ I need to set so that i get a good precision or a good recall or a good the classification error whatever it is.

Next I want to be able to tune this and figure out where I am going to settle down right, so one way of summarizing all of this is something called the ROC curve right, so the x-axis is a true positive rate the y-axis is the false positive rate so what I mean by this so at any point I am going to look at how many true positives could I possibly get right, now that is my denominator how many of them have actually obtained that is my numerator, okay let us take a simple example.

Suppose I have, I say I have 10 data points okay I am going to make it very simple let us say I have 10 data points right and 4 are positive 6 are negative okay, 4 are positive 6 are negative right and I have a classifier okay, so of these 4 points it gives me 3 here and 1 here right, yeah, so I got it right okay, fine so it manages to tell me that 4 of the negative points are actually negative right, and 3 of the positive points are actually positive right and one of the positive points it classifies as negative and 2 of the negative points it classifies as positive right.

So the true positive rate for this is essentially 3/4 right, the false positive rate for this is false positive 2/6 there are totally 6 things which I can say as false positives right of which only 2 I get as false positive so I am not 2 bad right so that is the thing, so typically what you what you would want is maybe I flip this thing around right, is that TPR on this FPR and always get confused with this right, so when I make when I say something right I need to go up yeah, so that is TPR and this is FPR right, yeah so I ideally want a curve that goes like this okay.

So I should get a 100% true positive rate before I start saying anything is false positive right, does it make sense so here is 100% and here is 100% so at this point I am classifying everything as positive at this point I am classing classifying everything is positive right, so i will have a 100% true positive rate because i got everything is positive and I 100% false positive rate so everything that I could say is false, falsely positive I am saying is falsely positive.

So at this point I am classifying everything as positive right, but then what I would really like to see is as I when should my false positive rate start going up after I have achieved 100% in the true positive axis right, after I have classified every positive point as positive then if I still ask you to move your classifier more that side then I should start getting the negative points as positive right, so you should really go up all the way here before you start moving this way right.

And what about that guy this is essentially random behavior right, I mean so for every true positive guy I get a equal fraction of false guys as positive so essentially I am this flipping coins and telling you whether it is positive or negative right, so I am just flipping coins and telling you that this essentially gives me that line right, so the further up you are if the probability of you saying positive is higher.

I am tossing a coin like you give me a data set it will give me a data point I will toss a coin and will tell you whether it is positive or negative okay, if the probability of it coming a positive is slow then it will be somewhere here the probability of it coming a positive is high then I will be somewhere here right, makes sense right so that is this line this is bad right, you want to be above that line you never want to be below that line right this is essentially random that you want to be above that line.

So typical curves you find will be something like this okay, so obviously you will not get that right ideally would like to get some curves like this right, make sense questions so far so the steeper this rise is the better it is for you right, so sometimes what happens you get curves that rise very steeply and then do that then a good or a bad ROC curve. Yeah, nobody said depends yeah, of course we all know the right answer is depends.

Depends on how yeah, what type of performance I am looking at right, so when will this be good right, so when I want to achieve a middling true positive rate right, so this is about middling rates about 50% true positive see that means of all the people with dangue I want to capture at least 50% of them okay, without putting too many people on quarantine right, so this is a good classifier.

But then if I see if you really want to get 90% right then this becomes unacceptably high false positive rates, well this might not be that bad right want to get 90% so this might not be that bad a false positive rate right, but then at this point this classifier is better so if i want for 50% false positive rate this classifier is better for 90% false positive rate this classifier is better, so there is no sure fire way of saying this is better that is better without knowing what you want from it just because I drew a curve that went below the random line really does not mean that this classifier is uniformly bad, right.

So if you want to show true dominance you have to show that one curve is above the other throughout right, this white line truly dominates the pink line that then now in such cases I can say white is better than pink right, but not in these cases in this case this curve is actually better for some points some operating points it is not better for some other operating points, so nobody asked me what ROC stood for.

Yeah, receiver operating characteristics I think I wrote curve next to it ROC curve okay, so this essentially was used in olden days when people are talking about radars and things like that so false detection true detection versus false detection right, and then you choose your operating point right based on how much, how sensitive you wanted it to be do you want it to capture everything that came your way in which case it shows a different operating point for your detector right, so that is why these curves came about and you can use it for the same purpose in your machine learning evaluation right, you can use it to figure out which point in the space of parameters that you want your classified operatives okay, makes sense everyone get what ROC curves are all about okay.

So the thing with the ROC curves is unfortunately and people do not really use it this way when they have when they run experiments right, so what do they end up doing they do not want to look at the curve right they do not want to look at this curve and try to sit down and do an analysis and write papers because they want to run 100 of experiments they want to generate 100 such curves and they wanted an automatic way of comparing the curves,

So what they did was they use this measure call, so area under the curve the uses measure called area under the curve or a AUC right, when they typically mean the area under the curve they do not mention it but they mean the ROC curve right, so the area under the curve is essentially this, so the assumption is the larger the area under the curve the better your classifier is because the ideal classifier is the one that goes like that right, so it gets all the positives before it gets a negative so the area under the curve is one for ideal classifier right.

And the area under the curve is 0.5 for random so if you are somewhere between 1 and 0.5 you are better than random so higher the value the better it is okay, all of this is fine provided all your curves are of similar shapes right, but you could get funky curves like this so what do you think went wrong here so the data is something like this right, so there are lots of 0 here and soon so forth and then are a few more x is that and then a few more 0 there right, so if I want to get those x as correct I have to get lot of the 0 as x's before I get those x is correct so that is essentially what is happening here.

I am getting more and more of the 0 both as x is here and then suddenly i get those last two pieces last two x's and then I get them right, so that is essentially what this means so that your negative class okay, is lying between two bunches of positive data points so you are getting all the initial set of positive data points and then before you get the remaining positive data points you are getting a very, very large fraction of the negative data points and then you are getting the positive data points right.

So this could be an indication that your encoding feature encoding is wrong right or you need to go to a different dimension so you need to do an expansion of the dimension so that you can get all the positives before you get the negative ones, but people unfortunately do not actually look at the ROC in fact there is all this code bases for generating AUC directly you just done your experiments feed the data feed the classifier to your AUC generating package and then out comes the string of numbers nobody even plots the ROC and looks at it anymore right, so you can actually get insights about what is happening by looking at ROC okay.

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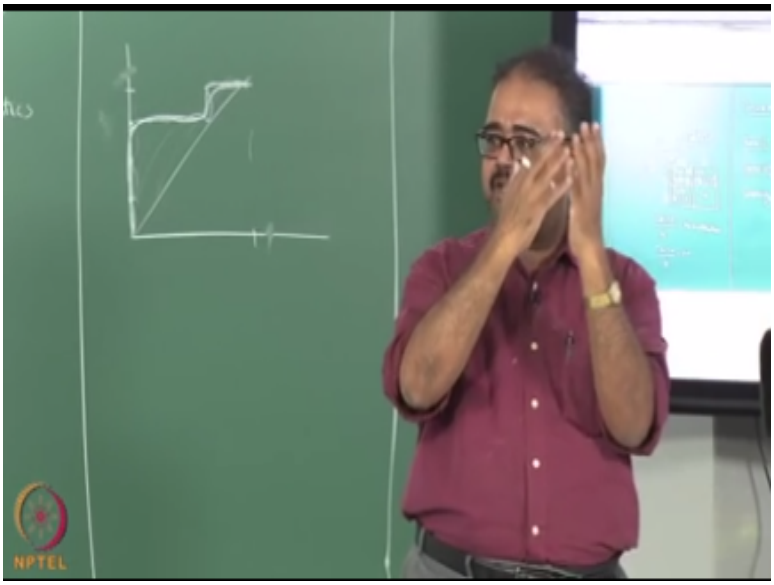

So how would you actually go about plotting the ROC okay, so here is a very simple way of thinking about it right ,so I am going to take all these data points right and arrange them in descending order of their likelihood of being positive it could be anyway I choose to do the right so if I am going to slide this thing around okay, the farther I am from the hyper plane right, so the less likely I will be further I am on that side from the hyper plane the less likely I'm going to be positive right so I will start off from here and then I will slowly increases or if I am doing a neural network I can look at the probability of the classification right.

I can look at what is the probability this data point is going to be positive right, so like that so whatever is it I am going to arrange it in descending order let us say that I do not need to do the particular class so let us say that I am re arranging it like this right, so this is the true classes I have arranged it in this order okay, now I choose a threshold above which I will classify this as positive right, so let us say choose a threshold here right, so what do I get no, $1/4^{th}$ right, so I will just go up by $1/4^{th}$ here right.

Next what do i do next I move it down one more data point then what we get up I move up by another one for if I see another plus I move up with another $1/4^{th}$ likewise next down I will move

up to another $1/4^{th}$ okay, I am assuming all of this is $1/4^{th}$ so that means that will be 1 okay, now what do I see say negative point what do I do, I do not go down I go right I go right by 1/6, and I go right then another negative point I go right another negative point I go right, right and then then right, right ,yeah.

So my ROC curve is actually like that because I have only ten points it does not look like a curve anymore right this is all the estimates I can do but this is my curve right, that make sense right and this is my random in fact I should cross the random curve at some point because 3/4th what is that 2/3 okay that is fine I will still be about random okay that is fine right, so that is my, that is how I draw the ROC curve right, fairly simple right.

So you arrange it in descending order of it being positive right, so whenever you see a positive data point as you go down that list you keep going up whenever you see a negative data point you keep going right the step you go up is one by the number of positive data points step you go right this one by the number of negative data points. Once you do this now you can compute this in the end of this curve fairly easily. So the probability with which my classifier thinks this point is positive right, or whatever measure weight whatever measure I am using so it could be the distance from the hyper plane whatever is the measure the closer it is to the hyper plane the less likely it is to be positive right.

I mean the further it is from a hyper plane the more likely it is positive, so whatever is the thing I arrange it according to the criteria that I am using right or if using a discriminant function the larger the value for the positive things then the higher up the ranking it will be right, so like that and I just do this in descending order of what is the probability i think it will be positive or what is the likelihood i think it will be positive right.

So this is roc curve, any questions on ROC yeah that is why i said you start of from the leftmost end and then you keep ranking them according to whatever you start of from the one end of it and then you keep ranking them according to that when there are other ways of actually once you train a classifier when you do the hyper plane thingy there other ways are figuring out what the probability should be right, so and from that you can you do not have to actually shift the hyper

plane around so you figure out what the probability of the classification will be and then arrange them in the descending order of that.

Because you can always say that okay, get my SVM to run give me the output give me the distance from the hyper plane for all the data points I will convert that to a probability and then I will use the probability for making the prediction if it is at least point three probable that it should be positive and classified as a positive class and in decision trees again it is easy to do probabilities so how will you do probability in decision trees, look at the leaves the data point lies in and look at the fraction of data points belonging to one class right, then you can do probability in decision trees.

So likewise SVM is the only thing that is tricky but you can there is a way of converting the distance from hyper planes into probabilities, anything else I want to say here so I am not sure if I am going to get to this later so let me actually make the note now there is a another of supervised learning problem right we talked about regression, we talked about classification that is a another problem called ranking problem learning to rank right, so it is inspired by information retrieval right but it is used in a variety of other settings right so I am not interested only knowing it is positive or negative okay, that is not the problem I want to know a ranking okay, so where is this appropriate.

So suppose I am yeah, so this is something which people do when they are trying to match protein structures so I will have one structure for a protein right, so and I want to figure out all proteins that look similar to this right which will probably have the same functions that as the original protein that I'm looking at so that I am not interested in you telling me if there is similar or not similar I actually want you to rank it right, or there is another question.

So I want you to build a recommender system for me right, I want you to predict whether I like a movie or not like a movie right, but I do not want you to end up giving me like or not like I want you to give me a ranked list of movies so okay, this guy is coming in okay these are all his history last movies that he has seen okay, and he is an old guy so these are the set of movies I am

going to recommend to him right, but in an order so this is what learning to rank means I am not just interested in yes or no answers but within the s i am interested in ranking order right.

It turns out that one of the ways of improving the performance in the ranking right, is not to look at this precision and recall and Felicity and sincerely in thing it is trying to directly optimize the AUC, trying to improve the things essentially what it means is the more relevant items you are trying to push up in the ranking right, so improving the AUC means what you are essentially making the curve go steeper and steeper right, improving the AUC means you have to make this curve rise steeper and steeper that is why this one will get a higher area under the curve right.

So that they essentially would correspond to making more and more of the positive points come higher up the ordering right, because this is a rank order right so you are trying to push these higher up their ordering so that essentially gives you the ranking effect.

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**Introduction t Machine Learning**

**Lecture 52**

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**Minimum description length &**
**Exploratory Analysis**

So in minimum description length principle the idea behind is very simple right among equally we already looked at it in some form or the other okay, so lasso is one way of thinking of it as minimum description right among all equally good classifiers I would like to pick the one that requires the least amount of bits to describe it right. So the description length should be as small as possible given that it has some acceptable levels of performance right.

So if you think about it so what does this tell us well if the classifier is very complex then I am going to need a lot of bits to describe it right if a neural network with a lot of weights the support vector machine with a lot of support vectors right or a decision tree with a lot of branches, so the more information you have the more detail the classifier is the more number of bits I will need to describe it right.

But then the better that it gets in performance right ideally why would you want to make it more complex only because it is making fewer errors right then you have to come up with some way of trading of this the description of the classifier versus the error it makes right ,you have to have to specify what the classifier is you mean you have to decide on how you want to specified suppose it is number of support vectors right so I will have to tell you what are the individual support vectors you remember support vectors or α times the Xi's right xi yi right.

So I need to tell you what the xi are I need to tell you what the α are for me to specify a SVM completely to you all right. So how many bits do I need for specifying those α and the xi, so the

xi yi I can take it as a product and I do not have to describe the way I separately but I need something to describe that and maybe to describe the α also and or maybe I can describe α xi to you if you can use that somehow to produce the inner product but if I have it there the kernel version of it then I cannot do that right.

So I account pre multiply the α into that right if it is a linear thing I can do this I can give you α xi so there are things to think about it so how do you encode this right, so you want to write a program to implement SVM in end of the day right what is the point in me doing a learning algorithm and then not letting you use it right. So for me to communicate to you how we implement it I need to give you the description right.

And the second part is the errors are there right, so I make some mistakes right and I have to tell you on a training set let us say there is a fixed training set and on that training set I also want to tell you what are the errors I made right the smaller the amount of errors I make the lesser the number of bits n heat to tell you how many errors I made, makes sense. So for me to make small errors I met need a complex classifier that will need more bits for me to describe the classifier right.

So if you want to reduce the number of bits I want to describe the error I might increase the number of bits I want for describing the classifier and vice versa, I am sorry just to set up a trade-off on equal footing right I am just talking about information on both sides now right. So that is I can make our classified arbitrarily complex right and we keep giving me better and better error right or I can make a classified very simple than it can give me a lot of error so how do I do the trade off ?

Between discrete the size of the classifier and the amount of error I make, so to put these things on an equal footing so that we can compare let people talk about the amount of information required in both cases you need some amount of information here and some amount of information here. So the more information you need here the lesser you need here, so that is the trade-off right.

So that is the idea behind minimum description length right and this huge theory behind it right it is actually a proper Bayesian approach and we talked about Bayesian learning at some point we also talked about ml and map estimates and so on so forth you can show that MDL is actually a proper Bayesian approach and people have derived a lot of complexity measures performance measures based on MDL right.

So I never talked about it earlier that but I think you guys are all ready to now read up on your own about MDL right so the brief introduction I gave should be sufficient right.

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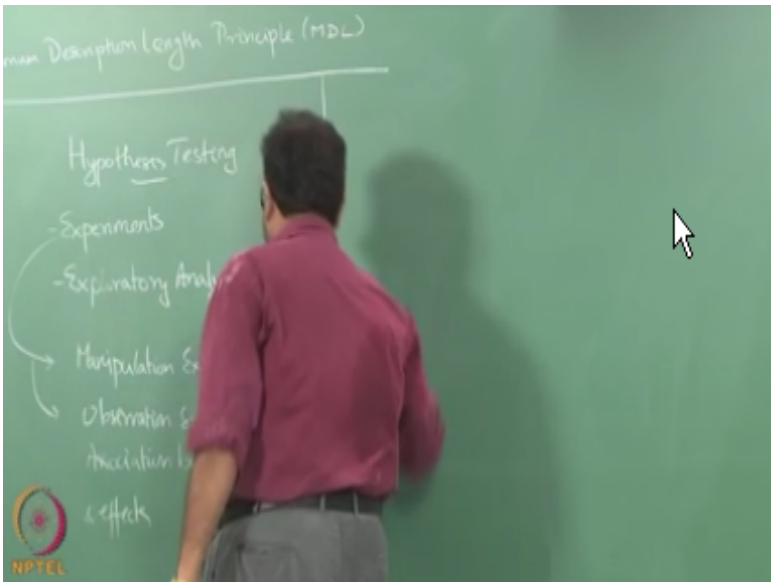


So if you think just one minutes stop and think about how people actually use machine learning right it is very heavily empirical right when doing a lot of math and other things or pseudo math in the class so far right but really at the end of the day when you start using it right it becomes heavily empirical it is actually a very applied subject believe it or not I mean of course you guys are all finding it out now with all the programming assignments but it is actually a very, very applied thing.

So whenever we have these kinds of empirical work right so you have to do our experiments right you really have to do experiments there is nothing like you know an analytical solution to your machine learning problem right. So when they give you a data set right you really have to experiment with the data to figure out what is it that you are going to do so all the theory and everything that we study now is all fine but when you actually get down to doing something you have to run experiments you have to do all kinds of things.

So you have to do experiments you also have to do some kind of some kind of exploratory analysis. So in fact we have not really talked about exploratory analysis at all in this in this course I have to do a lot of different things, so I actually do that whenever I teach my flavor of data mining I do that so what do you do with external analysis right so there are many things that you have to do first you have to figure out.

So how distributed variables are you know so we have to figure out so what is the range of the variables I give you data right I do not do you do not really know what the data is all about I just give you an the simple form is I give you an excel file at the complex form is I give you like few terabytes of data on a disk right but then let us say I give you a file and then you have to figure out what are the different variables are there right and what kind of values do they take right and what is the variance of these values right or their outliers on these is that some values that I can ignore right.

So the whole bunch of things that you have to core and do some kind of exploration right or that way variables that are important to my prediction maybe know about then we talked about some variable feature selection and things like that but all of this you have to do essentially you have to understand the data before you even think of what is the machine learning algorithm I am going to use.

So if I give you some data you do not just straight away plug it into a decision tree algorithm or straightaway plug it into an SVM right so you have to go around try to understand what the data is all about right. So that is part of it is through exploratory analysis and say a little bit more later

but as far as experiments are concerned typically they fall under typically they fall under two kinds of experiments.

So there is the manipulation experiments and observations experiments to who do you think this mean so in observation experiment I basically try to figure out correlations I try to figure out associations between variables right I would make a lot of observations and then I try to say okay if this variable whenever this variable was at this level then the output was at that level right so maybe I can observe the whether, so it is essentially trying to find associations between right associations between factors and effects.

So what do I do in manipulation experiments in manipulation experiments is essentially these are things where I have some control over some of the variables that constitute the experiment right, so typically I set up these kinds of manipulation experiments whenever I want to test a theory a causal hypothesis right whenever we want to say that A causes B right. So I cannot stop yelled you know from happening right.

So I mean those kinds of that does not make any sense but I can say something like okay, so by learning algorithm A is better than my learning algorithm B right whenever the load on my system is high right. So I can actually make a hypothesis like this that A is better than B whenever the load on the system is high right, no I have made my models I have model A I have model B I want to make a make a statement that saying that yeah is better than B forget about under heavy load.

I want to make a statement that okay you have a learning algorithm A he has learning algorithm B I want to make a statement that learning algorithm A is better than learning algorithm B. So when will I make such a statement when can he make such a statement haha right, so that is the whole thing that we are going to worry about here I am going to test what I mean what I call.

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**Introduction to Machine Learning**
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Lecture 53 NOC

So we will move on to another topic which is essentially ensemble methods. So what do people do in ensemble methods is that instead of using a single classifier or regressor, you tend to use a set of them in order to make the same prediction. these end up improving some performance or the other of the classifier. Statistically speaking more often than not they end up reducing the variance of your classifier, but that also ends up giving you better empirical performance at the end of the day. So we are going to talk about several approaches for ensemble methods. I will start off with one that is familiar to familiar to all of us called bagging.

So, what is bagging? Why did I say it is familiar to all of us? So bagging stands for bootstrap aggregation. Do not ask me how they derived bagging out of bootstrap aggregation. But the idea is very simple. All of you know about bootstrap sampling.

All of you know about bootstrap sampling. So what I am essentially going to do is, I am going to create, give me one training set of size n. I am going to create multiple training sets. of size n by sampling by replacement, and then I am going to train a classifier on each of those sets. I am going to train a classifier on each of those sets, and how will I combine the outputs of the classifier? Sorry.

I can do a majority vote or average what? Sorry. Well, that would end up doing majority vote. So, average what? Average the, if you can, if your classifier is going to produce probabilities for the class labels, I could do some kind of a weighted average of the probabilities. If the classifier is just going to give me 1 or 0, I end up essentially doing majority vote. Does it make sense? So, the idea is very simple.

Bagging idea is very simple. I am going to produce lot of classifiers. So, I am going to call them fi, so it could be regression as well, so it does not have to be classification. This regression I just take an average of the outputs of all the classifiers, so each fi is trained on one bag which I have produced from the from the original sample like this is another back derivation from the word bagging right so it is bootstrap aggregation but then each of those bootstrap sample you produce is sometimes called a bag right. So, if I produce B

bags, then I eventually divide by B to get me the prediction.

And if I am doing it for classification, I can produce majority vote or average the probabilities. There are few things to note. So, bagging reduces variance. So, in effect it ends up giving you better classifiers normally than what you would get by training on a single sample of the data or producing a single classifier. It is particularly useful when you are dealing with unstable classifiers.

It can take an unstable classifier and produce something that is more stable. That is just a fallout of reducing variance. It can take an unstable classifier and produce something more stable. So, one thing that you have to be careful about when you are bagging is that if you bag bad classifiers, the performance can become arbitrarily worse. uh something that has a classification accuracy less than 0.

5 less than or equal to 0.5 two classes sorry it's when you change the uh data on which you train the classifier right you are going to end up with a different classifier oh yeah but they are actually the same method but you are training in different on different sets of data yeah are you initializing You could as well if you want to, but that is a good point. If you initialize two different variables, different values for the parameters, you introduce an additional source of variance there, but you could. There is nothing stopping you from doing that. It is just that you have to be careful about how you do the analysis.

If at all you are doing a variance analysis, you have to be careful about how you do the variance analysis. So, by that he brings up a good point. In some of the ensemble methods that we talk about, the ensembles will typically be of the same kind of classifier. The only way we are distinguishing one classifier from the other is by training it on a different data, except for one approach, which I will tell you later, where typically we use, we end up using different kinds of classifiers. So, data would be the same, but the classifiers would be different.

What parameters are different? This is more about regression, but anyway go on. I am not using different variables. It is like suppose you are using a neural network, right. So, you need to have an initial starting point for the weights, right. So, if I use a different random starting point, so that was his question.

Should I use the same random starting point or should I use a different starting point, right. And even then going back to your question, right. So, you think about it this way. So, we are talking about f of x. Instead of that, think of it as… So, this h i will give me whatever features I want from x.

Even if I want to run each classifier on a different subset of the features, it will just be that that will get enrolled up into the classifier I can still do the averaging if I want right so that's not an issue but that is not the question he was asking anything else on this okay so If you throw a bad classifier into the mix, your performance can become arbitrarily bad. So that's something that you have to guard against. So bagging is a very, very intuitive, very simple thing. And a couple of more practical things about bagging is that It is what they call embarrassingly parallel. You can run how many ever instances of training on bagging you want.

Some of the other ensemble methods we talked about are going to be inherently serial in nature. So you will have to run one after the other. Suppose you are looking to run this on many large data sets. So doing bagging is kind of easier because you can run it. one bag like classified trained on one bag does not depend on a classified trained on the other bag in any way right so they can be trained independently and trained in parallel okay so that's the first thing Next thing we talk about something called committee machines.

This is nothing big. All it says is you train a lot of different classifiers. Sir, by doing bagging, let us say we have some 10 classifiers or something. Should all the individual classifiers perform well on test data or do they just have to learn the training data? Do they have to generalize well? so each classifier you typically train it using whatever is your normal training procedure right so if you normally you would expect it to generalize well on the test data right so you would want to produce classifiers that generalize well on the test data right but I mean, that's a call to be made. I mean, if you don't want to test each and every classifier, sometimes people just test what they call the bagged classifier. The combined prediction alone is something that they test.

They just train each classifier on the data that is given. And then they test the combined classifier. There are multiple reasons for wanting to do that. So one is that, typically, the classifiers that you use in bagging are not very powerful classifiers. So, the chances of them over fitting are low.

So, you do not really try to do a validation on the test set to make sure that it is not over fit and things like that, because the classifier itself is not very powerful classifier and then you just go ahead and test it on the test the combined classifier on the data. So, why you would want to test the combined classifier on the data, you will want to know whether you should produce more bags and things like that. So, the nice thing about bagging is that. because you are using at any point of time you are only using a weak classifier to fit the data right and not a weak classified but not necessarily a you know very strong classifier to fit the data the chances of you over fitting is very small even if you increase the number of classifiers in in the bag even if increase the number of bags I can do this

for 10,000 samples 10,000 such bags I still would not over fit. The danger of over fitting is no more than training it once.

So, that is a nice thing about bagging. I can keep making the classifier. I can reduce the variance in my estimate more and more, but I am not getting into any danger of over fitting. So, that is a nice thing. So, the other thing is committee machines.

You don't have to really think about anything. You don't even have to think about how, my God, how do I parallelize this thing? I mean it's a term that people use in architecture when they are trying to think of parallel computing so things like this is embarrassingly parallel so I can do at whatever levels of parallelism I want and things like that maybe I am misusing the term but yeah but it is really easy to parallelize right you can just run it on different samples separately. Yeah, so what is embarrassing about it? I mean why do you even have this whole parallel computing field to study something that can be parallelized so easily? So I'm really embarrassed to be working in parallel computing. I'm just making it up.

Okay, commuting machines. Can I move on to commuting machines? Any other questions? Okay, so commuting machines. It's a very simple idea. So I'm going to train, I have given a data set and I'm going to train a lot of different classifiers on. on the given data and then I am just going to combine their outputs based on some kind of weighting mechanism. So what could be the weighting mechanism? I'll train a neural network, I'll train a decision tree, I'll train an SVM, whatever it is.

I train many, many different classifiers, and then I have this set of classifiers that have already been trained, and I have to combine their output. So how do I go about doing this? There are many ways in which you can combine their output. So I am just taking this classification from the textbook, elements of statistical learning, not that I completely agree with it. So in committee machines, suppose I have m classifiers, the weight assigned to each classifier is 1 by m. So I treat all classifiers as being equal.

So that is called a committee machine. I have many different classifiers. I have the outputs of all the classifiers. I am going to give each one of them an equal weightage or equal vote.

So, I call that a committee. Then, we go on to something more interesting called stacking. no bagging is at the same classifier same algorithm but trained on different samples of the data right in committee machine it's the same data but trained on different algorithms right so I have SVM I could have neural networks I could have anything right all right it could be says it could be neural networks with different number of neurons I

mean I'm not saying that it has to be a completely different algorithm okay it's just a different classifier it could for different settings of the parameters and so on so forth right and so stacking the stacking is like committee machine so i have many many different classifiers right but what i am going to do is instead of arbitrarily assigning a weight to each of the classifier what will i do what can i do i could do that but with stacking what do i do i learn the weights So that is a natural thing to try and do. So I have the prediction that is made by each of the classifiers. I go ahead and I learn the weights. So another way of thinking about stacking is the following.

So I use each of these classifiers to generate a feature for me. So this way it is called stacking. So I have a set of classifiers. They all output some, it could be a probability vector or it could be just a class label or whatever. Classifier 1 comes and tells me, I think this data point is class 1.

Classifier 2 comes and tells me, I think this data point is class 2. Classifier 3 comes and tells me, I think the data point is class 1. Now what will happen is, my input to my next machine learning stage will be class 1, class 2, class 1. and now again it is a machine learning algorithm now I can run whatever machine learning thing I want it could be linear regression because I am interested in finding weights right so doing some kind of regression seems to make sense right but then you know problems with regression classification all of you know that right so you might want to use it for classification or you might want to use some other method for class you might want to use logistic regression for classification whatever it is right but then the inputs to this stage or the or these outputs of the first stage of classifiers and the target is the same target as the first stage, the same class label, right. So one way of thinking about it is like you are stacking these classifiers one upon the other.

So I first have some set of classifiers, they produce features for me, right. The features are essentially what is the class, what they think are the class labels, right. And then I learn to combine the features. I learn a predictor based on these sets of features.

So that is another way of thinking about it. Make sense? didn't make sense right let's take a classifier f i mean there are i there are some people are actually saying they didn't make sense i'm trying to make it a little more explicit let's take a classifier some f i right so it basically it operates on x1 to xp that is going to give me something. It is going to give me a real number or some class label. So that is basically the function. It is either classification or regression or whatever. So now what I am saying is I am going to try another H.

That is going to take as input It is going to take this f1 to fm as input, so fi is the first

level classifier, I have m of them and then h is going to take, h of x is going to take f1 x to fm x as input and it will produce a whatever is the thing I am looking for, real number or thing. So if you go and look at the structure of h to make it explicit, let us say I want h to be a linear function. So, that will essentially mean that h will look something like this. So, this is essentially saying that I am taking the outputs of all these classifiers, I am combining them in some kind of a weighted fashion.

The same way I train any of the FIs. Yeah, the same training data. The same training data that you had used for FIs, you will use the same training data for H. Maybe, maybe not. Depends on the kind of classifier that you are using. I mean, H is a completely different training algorithm.

I can see your confusion, so my initial training data is going to look like this, I do not know initial training data is going to look like this, this is my x and that is my plus 1. So, corresponding to this there will be a training data for h which will be f 1 of this So, now I have only two elements here, but this is f 1 of this and this is f 2 of this and the same plus 1 comes in here. So, I can do this. So, the dimensions do not have to be the same.

So, this is stacking. So, stacking is very powerful method and in fact, you can do all kinds of weird things with stacking. In fact, these weights that I am learning, I can make them functions of x as well. What does it mean? what does it mean if my weights are functions of x? Not type, depending on where in the input space the data is coming from, I might want to trust one output more than the other. Suppose it is in say the top left quadrant of my input space, then I trust F1 and F2 maybe a little bit, but then if it is in the top right quadrant, then I trust both F2 more and F3 less or something like that, I can actually do that also.

So, stacking this function can be arbitrarily complex. That is why I did not want to write the linear thing first because it will bias you into thinking about simple linear weighted functions, but this H can be arbitrarily complex. So, if you think about it, in fact we are doing something like this in the in neural networks. So, the first layer right gives you features your complex features some some some hyper plane is being learnt in the first layer itself and it produces a complex feature and the second layer takes all these complex features I have produced and it learns to produce a final output right the only difference is the first layer is not trained in this way right the first layer is not trained directly using the training data it is trained using the back propagation error or whatever is the training algorithm you use it is not directly trained using this data so that is the difference right but we already looked at things like this right these are all some kind of general additive models they are called additive models fine so any questions on this So are you meaning that training like this just simplifies your the way you are doing like

basically all your classifiers can be linear but a combination by training a combination of linear classifiers you will be able to train much more complex classifiers using static. The basic idea is that when my classifiers need not necessarily be linear classifiers see the thing is  So any of the classification algorithm that we are looking at comes with its own biases in terms of what are the class of functions it can fit and so on so forth. And it could very well be that across the entire input space the function is so complex, the final function I want to learn is so complex that no individual classifier can actually fit it or if I try to fit it with a single classifier I might end up with something that has too many parameters.

So when I do this kind of this layer wise training, so the  I can get by with whatever I know as simple classifiers in the first stage. I could use decision trees. It need not necessarily be linear. So decision trees are simple enough.

I do not have to grow it all the way out. I can stop at some point. I can use decision trees. I can use neural networks, whatever I want as my FIs. And then later on, try and combine the outputs from here.

You could. Given how much success people have had with deep learning, I would suspect that if you work on this carefully, yes you could have multiple levels of stacking. People do that. I mean it is not that, it is not that. The reason it is called stacking is because people actually did multiple levels.

So you could do multiple levels of this. But then the question comes how do you group these things and so on so forth. Do I run it on every, do I give all my first level classifiers as input to all my second level classifiers or should I group them somehow to form a hierarchy, blah, blah. So those kinds of issues would arise. As long as you can address them sensibly, then you can go ahead and do multi-level stacking. One thing which I wanted to mention, sometimes when people want to run competitions and so on and so forth, they do something very clever, they do not expose the actual data to the  They give you the outputs of the first level classifiers and then the actual class label.

They do not tell you what the features were that they measured. They give you the outputs of the first level classifiers and then they give you the class labels and then now all you need to do is train your second level classifier. Take the first level classifiers output as inputs and train the second level classifier and see what you can do with it. In fact, it ran for a couple of years.

I do not know. I am not sure they are still running. There is an ensemble learning competition which essentially does that. this also allows you to have some amount of data

privacy. So, I do not have to release x, but I am releasing some amount of simple functions computed on x, and then you build whatever classifier you want on it. So, it is hard for me to reverse engineer, because I do not tell you what f is even. I only tell you the output of f, I do not tell you what f is, I do not tell you what x is.

So, it is very hard for you to recover  You can't essentially compute F inverse. So that's another nice thing that you can. So the reason that there are so many approaches for doing this is because there is no one clear winner under all circumstances. So it depends. So in fact, like I said, so stacking is something that you can use under a variety of circumstances.

You can even use it under cases where you cannot do bagging. How so? mean I can use stacking when I do not want to have you want to give you access to data right. So that is one case another case is the data set is small enough that bagging does not make much of a sense on that right. So it is not really truly representative of the underlying sample and I am not really sure where you want to do that in which case I can use the different biases given by my multiple classifiers as my  as my variation right to praise my ensemble. So, there are different ways in which you can do this.

Next thing I want to talk about which is more interesting thing yeah. Say I have one single stack of classifiers and I run like I train them on a single data set and then I get the percentage of points misclassified and then I normalize them through all the classifiers. What do you mean by normalize? Like say 3 percent, 5 percent like I normalize them to. For each of the classifier you know what is the percentage of data points they got wrong fine. I normalize them to 100 percent and then I assign them to beta 1 to beta n how is that different from learning this when betas are not a function of x.

That is one way of finding the betas yeah there is nothing wrong with that. Like when betas are not a function of x, so how else would we redistribute the betas? That is one way of estimating beta. I mean, what do you mean by saying how else would you redistribute beta? I can think of many ways of doing that, right? Take the classifier that has the smallest error and give beta 1 to that and make everything else 0. Why would I even want to give weights to classifiers which give me higher error than the lowest error? Give me an answer, why? They are totally neglecting the possibility of them having a better chance.

No. They could be making errors in different things. So I might have a 1% error. Another guy might have a 3% error. But he might actually be capturing that 1% correctly, the one that I make the error on. The other classifier might get it correctly.

So I don't want to completely neglect other classifiers also. So that is the reason why you have to go about trying to be more clever about the weight assignment. So I can do this proportional to the errors. In fact there is another more Bayesian approach for doing this.

I can look at the likelihood of the classifier. given the data. And I can assign the weights proportional to the likelihood. The higher the likelihood, the higher the weight. And I can do some kind of normalization on that, so that my beta sum to 1. And I could do that as well, instead of just looking at the error.

The error would be a frequentist way of looking at it. A Bayesian way of looking at it would be to look at the likelihood of the data and do this. I mean, there are many ways in which you can derive the data. And so stacking just takes it to the extreme. Say, OK, fine, I'm not going to worry about a specific algorithm.

I mean, I'm just going to let it learn from the data directly somehow. So yeah there are pros and cons and whole bunch of things. So yeah empirically if you want to validate it finally do cross variation and then test against all of these ways of generating the joint classifier right. But then analytically also you can try to analyze some of the variance of the combined classifier and you will end up hitting a wall at some point that says that hey it depends on the characteristics of the variance of the data. So that's basically, I mean we are entering territory where you can come up with whole bunch of things. Ensemble methods are about like a thousand papers out there, research papers out there, which have proposed lots and lots of variations, right? So think of something, and before you try to publish it, do a literature survey, you'll probably find it there. Okay, so that's how crowded this space is.

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**Introduction to Machine Learning**

**Lecture-60**
**Boosting**

**Prof. Balaraman Ravindran**
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Okay, so with one of the most popular and most some sense mind-blowing thing with the non-thermal method space right, so boosting the original boosting work original analysis of the boosting work comes from theoretical computer science community not necessarily from an empirical machine learning community right, so, therefore, they were that they looked at having some oracle that had a probability slightly greater than 0.5 of being correct right. Then they try to see how you can better predict somebody who is just above 0.5 okay.

By combining many, many search Oracle's right I can keep improving my accuracy of prediction arbitrary close to 1 right, so that is the amazing part right I start with each predictor as the accuracy of 0.5 plus some epsilon just better than random that I can combine a lot of them and produce something that as accuracy close to 1 right, so this was a very big result that came out earlier and so we are going to look at some kind of simplified version of it, so they remember the goal that distinguishes I mean the main thing that distinguishes boosting from the other methods is that boosting is inherently serial okay.

So boosting is going to build this ensemble classifier in an incremental fashion right, where at each stage I am going to try and explicitly reduce the error produced by the previous stage, so this is something that you have to keep in mind you cannot write it just cannot come up with some ensemble method and call it a boosting method and I have seen that happen in many papers that I have reviewed people just write something that has multiple classifiers in it, so it is boosting, because they have read somewhere that boosting is a very hot area and people papers

in boosting get accepted, so they come up with any classifier and an ensemble method and call it boosting, boosting has this very specific property that at every stage right, you add one more classifier to the existing ensemble right, and this is done in such a fashion as to reduce the error produced by the classifier up till that point okay, makes sense.

Sorry, you get the choice as to what to add next right, so that is that you choose it such a way that you minimize the error that they have not at least you reduce the error okay, so not necessarily minimize. However, you reduce the error of whatever has happened the prediction till that point, okay, does it make sense, so that is essentially what boosting is sometimes you can think of it as error boosting, sometimes they call it error boosting and so on so forth.

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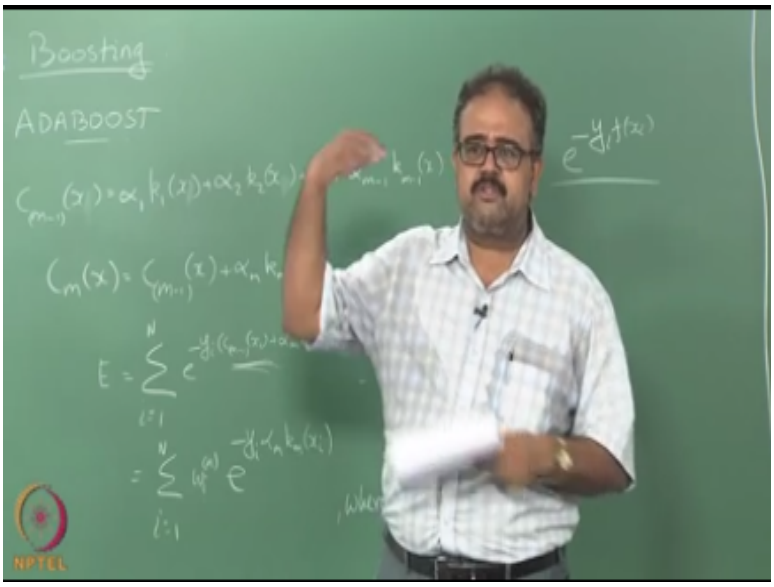


The one very popular and one of the original boosting algorithms is called ADABOOST okay, so let us. It is going to I am going to put up a tutorial for you guys to refer to okay I will use the notation from the tutorial so I will not translate it to the notation in the textbook, okay, so when you read the textbook you have to do the translation yourself, so that is one of the main problems when you have too many different disciplines contributing to the same field right, machine

learning has people from computer vision, people from statistics, people from AI and all other disciplines contributing to it and each one of them brings their notation to the mix, right.

So it becomes harder to keep track of everything, but currently, the I output is necessarily complicating my dress, okay, so I am going to denote by C(m-1)(x)th stage classifier okay, that is obtained by basically adding the outputs of all this individual classifies so α1, α2 to αm-1 right, so I am going to add up these are the weights and k1 is a classifier that I added in the first stage right, k2 is the classifier added in the second stage and so on so forth and km-1 is a classifier added in the m-1 stage okay.

And then basically I want to produce that okay, yeah, the rest of the class, so there are a couple of things which I should point out here one of the most obvious ways of doing this forget about Erebus one of the most obvious ways of doing this is to say that okay I am going to take this guy right, look at the residual error you know I can think of this as a prediction problem right, and look at the residual error of the predictor right, and then train a classifier km to minimize the residual error, right.

So what will be $\alpha_m$? Essentially how to make sure that this whole thing is along the direction of the residual, so we talked about this earlier right, when where did we talk about this, forwards stages we ask stage-wise or stepwise, stage-wise, so when we talk about stage-wise feature selection we talked about something similar right, so you could think of something along the same lines here instead of thinking of selecting features right, I am just selecting classifiers right.

So I can just take the residual error of $c_{m-1}$ and then use that to train $k_m(x)$ and then add it here right, in fact this can be one, it can be one does not matter because the $k_m(x)$ will actually align itself in the direction of the residual so I can just add it here so it is fine right, so that is a simplest way to do this thing and it is actually a good way to do it if you are doing regression let that make sense and I can take this as they can take the residual error and then train my $k_m$ to actually go in the direction of the residual.

So I can actually do this, so you can get a boosting like algorithm for regression just by training it along the direction of a residual right, but when I am doing classification that is not necessarily the right thing to do so people come up with different kinds of loss functions and then they try to improve the classification, so the loss function we look at is the exponential loss, so people remember the exponential loss, I talked about it when we are doing SVM's is exponential loss okay, $e^{-yif(xi)}$.

So we looked at the exponential loss earlier so we will essentially continue with that, so I will sum over all the training points right, so that is the exponential loss for the $m^{th}$ stage classifier right, people agree with me on that, so that is essentially what I wanted to write right, now expanded the cm. I have written it as this expression in the bracket here. Yet, can people see me at the back I see not me, but I am kind of hard to miss right, that makes sense okay great.

So this thing we already know right, so there is no control we have over that that thing we already know that is given to us all we need to find is $\alpha_m$ and $k_m$ right, so I am going to rewrite this as what is that, that is the last function we are going to be using like that is exponential loss function so for classification we looked at, if you remember we looked at the different loss functions and when we looked at hinge loss right, and I said exponential loss is one of the loss functions and this is how we defined it so essentially I am using that exponential loss function here.

And I mean this was not the way ADABOOST was originally derived okay, ADABOOST was derived in a completely different way and later on about five years after they publish ADABOOST they kind of discovered the connection between this kind of stage ways modeling right, forward are additive stage wise additive modeling and exponential loss they said okay, I can do forward stage wise modeling with an exponential loss function I end up with ADABOOST that connection was discovered five years later.

But now almost always people except in the theory community, in the machine learning community is always introduced like this okay. So where $w_i^m$ is sorry, m the same thing I wrote here, so the weight of the $i^{th}$ data point at the $m^{th}$ stage right, the weight of the $i^{th}$ data point at the

$m^{th}$ stage is essentially $e^{-yicm1(xi)}$ right, so what does this mean what is exactly this expression if you think about it, it is a loss I have incurred on that point x(i) up till the m-1 stage right, that is essentially the right just the loss that I have incurred on the ith data point up till the stage m-1 right.

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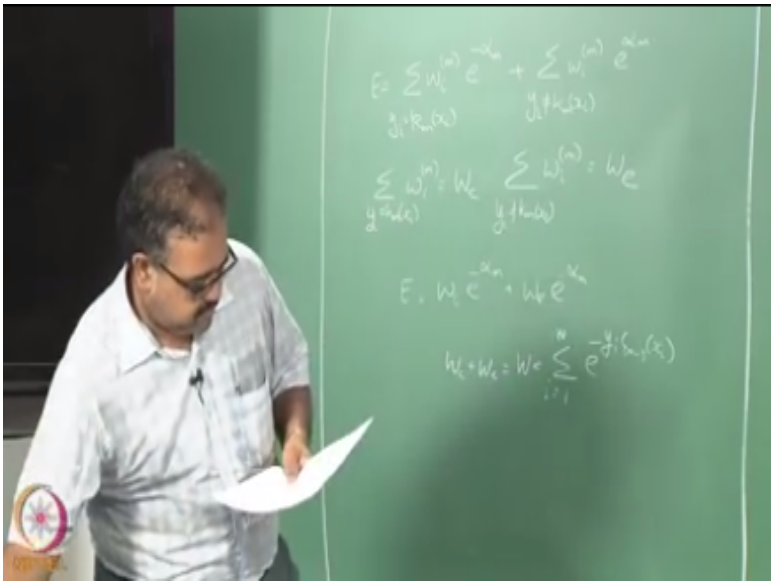

Okay, now I am going to break that sum up into two components, so do you think of these two components, no, no, no this is varies say $e\alpha^m$ so when will I get $e^{-\alpha m}$ when I am correctly classified it, when I will get $e^{\alpha m}$ when I am misclassified it so these are all the data points such that right you are the correctly classified data point is all the miss classified data points right, is intuitively you can see where we are going with this, so what is the best classifier that I can find at the $m^{th}$ stage.

Well, the best classifier can find the one for which this $\sum$ is empty the right, and get everything correct classified correctly, so that is the best classifier. But now increase the cracks right, remember our classifiers are all weak classifiers, and exactly that is a basic assumption we are starting with right the classifiers are all weak classifiers I can do only slightly better than random, so I have to get nearly half the data points incorrect, right.

So which half should go here which half should come here, and then we can move one data point from to here that here to make it better than half, so which half should go here which half should come here intuitively you tell me, which will incur less penalty, what is small half, it is half man what is small half that will be what clear me, can be more clear as to what is small means, no that is a valid way of interpreting small half tell me, $w^m$ 's right, so all the w's that have a large value should come here because they get $e^{-\alpha}$.

The w's ensemble small value should go there because they get multiplied by $e^{\alpha m}$, so what are w's is a small values the ones that I have correctly classified up till the previous point w is the large value are the ones that I have incorrectly classified up to the previous point, so at the $m^{th}$ stage what I should be looking at is try to get the data points which I misclassified from the previous stage, try to get them correctly as many as possible right.

So that is essentially the intuition behind ADABOOST, so at every stage what you do is you try to look at the previous stage see which are the data points you misclassified it tries to get them correctly in this stage, right. It is okay if you make mistakes on data points that you have correctly classified till the previous stage why is it, okay. And because those classifiers can adjust for it okay, then we will look at how we will do this again right.

So I am going to call, so it is all the weights of all the data points I got correct at $m^{th}$ stage right, likewise weight of all the data points I made a mistake on at the $m^{th}$ stage right, so then I can write my e's simply as okay, so if you think about it the value of $\alpha$ really does not matter in my choice of $k_m$ right, regardless of the value of $\alpha$ right, regardless of value of $\alpha_m$ whatever argument I gave you this no holes right, the idea is to see how much of the weight, the weight you can push to Wc right, and how less of the weight you keep in We.

I mean there is total of weight right, there is some total W right, W is a constant okay, Wc + We is a constant, the goal is now to see how much weight you can push into Wc as posted, km will be the classifier that rise as to my WC, so how we do this well you can use you can classifier they images that can assign based data point, we discussed very briefly in session the case right,

we can assign ways to data points and you can essentially multiply the error that you make on a data point by the corresponding weight, right.

So the error that you make you multiplied by the corresponding weight so that you can use weighted minimize other ways of doing this, so one way people see what I am saying over km right, you see what you are supposed to do to get your k, the $k_m$ is such that maximum weight goes into Wc they are splitting your W into two parts and depending on what data points are making mistakes on right, the data points you do not make mistakes on contributed Wc the data points you make mistakes on contribute of e.

If you want to see how much larger you can Wc. We basic that is the classifier you have to find, before that you use some kind of a payment method, so one way of achieving this is do the following you are saying weights to all the data points now what you do, if you go and sample some of these data points according to their weights, create a new training set by sampling from this data points are given things according to the weights, so what does this mean points for which the weight is higher you get to sample more often into this data sets, points for which the weights are very low I do not even appear in the data set, right.

So the points appears multiple times in the data set then when you are trying to minimize the training error you are likely to get a point correct, so instead of using a directly using a weighted training algorithm people simulate that by sampling from the data weights okay, so what has happened unfortunately because of this I change of tends to compact by bagging and boosting in the minds of people and if you look at some of the data mining text books especially some of the earlier data mining textbooks exciting and boosting we needed will described in a very similar fashion right, what do you do in bagging whenever you add a new classified.

So in the older textbooks how they describe is that what you do in bagging is every time you generate a new sample you sample uniformly right, with a replacement right you with sample replacement and boosting the differences every time we generate a new sample you use the prediction error from the previous thing there is the only difference between bagging and boosting right.

But operationally if you think about it, so there is the only difference between bagging and boosting, but then boosting is inherently serial and then there is this error minimization property right, but that never comes across and people just tend to think of boosting as bagging with the different sampling distribution right, whether it is incorrect at the fundamental principles of the two things are very different okay.

So we have found $k_m$ now right, so we all know how to find km you do some kind of weighted error minimization you find $k_m$, so what is next, what is next we need to find $\alpha_m$ right, see regardless of what value of $\alpha_m$ you choose the minimizer is the for km is the one that gives you maximum weight into Wc, correct. But then having chosen a $k_m$ I now have to choose an αm that gives me the error detection, so how do you go about doing that.

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In fact we can do our, so set is equal to 0, so $\alpha_m$ is essentially 1/2 ln 1- the error rate it is error rate is essentially the weight of the data points on which you are making a mistake divided by the total weight right, so this is for the km classifier alone right, We is the data points on which the $m^{th}$ classifier alone makes the error not Cm but km correct, so that is what we divided these

things into right, so this is the thing where km makes error, so essentially that so the data points on.

So essentially, it tells you how good the classifier if the classifier is really good right not just on the data points that you are interested in but on the entire data set. If the classifier is very good then the weight will be high that is the classifier has an error of 0 what will happen rate will be infinity because the only classifier you will need right you have a header of 0 on all the data points why do you need other classify just that one is enough right.

But then suppose it has a very high error, error close to 1 where it will be 0 okay, so depending on how good the classifier is this way it will vary okay, and then anything else that you have to do I have found km, I have found αm what do I have to do, I have to change my W's now for the next stage right, so what is my Wi is $e^{-yiCm-1(xi)}$ right, so now it has to become $e^{-yi.cmxi}$ so what is the right best way to do that, this multiply the existing W by $e^{-yi.kmxi}$ right, does it make sense after you have done that you come here okay, I do not erase that part right. So because you need the αm here for your update, so once you find the αm you come back here and change the weights of all the data points by this amount okay, as it makes sense so that is a plain simple version of ADABOOST okay.

So in fact we can show that the exponential loss function is closely related to the deviance right. An equally popular version of boosting called logic boost exists, where we use the deviance the logistic function right, the log odds function that we used for logistic regression you can use the same error function and then derive all the update rules that we just did for the exponential loss function you can do the same thing for the logit function the log odds function also. You can come up with similar update rules okay.

So the recent ADABOOST is so popular is because it deals such very simple updates right, if you think about it all the computation you do is okay, you find a classifier that minimizes this weighted the error right then you come back and compute this $\alpha_m$ and then you go back and change the weights and then repeat until you are happy with the performance of the total

classifier right, and both with basting I mean bagging and boosting the commend both here start basting things anyway if you do both decision trees are very popular classifiers for this, okay.

In bagging it seems to make sense right, why you want to bag decision trees they are notoriously unstable, so if you want to, if you bag decision trees you get more stable estimates, why would you want to bag, why do you want to boost decision trees, are they weak classifiers. Exactly, so what do you do with decision trees you can do the most extreme thing you can just have one node, just have the root node right, one node what can you do with one node decision tree.

Yeah, that is somewhat like linear right yeah, so somewhat likely linear I agree. However, people call it decision trees right, so one node decision tree because of the way I choose which feature I pick right, I will use information gain or Gini index or one of those things okay, I will at least take 50% classification otherwise I would not even split right on that 50% will I will be better than 50%.

I will be better than random even if I split on one node right, so I will split on one node and or maybe if the performance is too weak I can perform I can do a two level tree okay, these are called decision stumps, I do not build a full tree, but it is like chopped off at a very close to the root right, so one not two levels of the trees. However, they are v-classifiers and they take very little time to estimate and I can do many, many, many of these very quickly essentially what I do is, I boost these decision terms okay. In fact, there is one result in the book if you look at it right.

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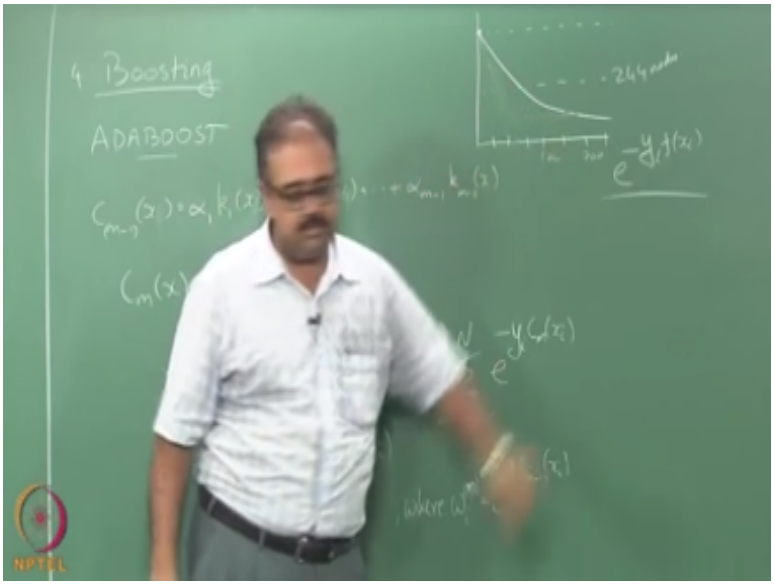


So I do not remember the exact scale on the y-axis. However, the x-axis is the number of levels of boosting that they do right, and so on so forth, the number of levels of boosting that they do so 100, 200, 300 and so on so forth, so a single stump it gives some performance level at that height okay, just one strum the best single stump gives you a performance there. They trained it on the full data, and they get a performance here, and this is like a 244 node tree, 244 notes is a fairly complex thee they built and that is the performance that they get right.

And then they did boosting the start here obviously with a single node right, and then they do boosting, and then they find that the tree the performance just keeps improving as I do ADABOOST. Remember, and these are all single node trees okay, they are all single node trees and so essentially by the when they reach 100that means they have only 100 nodes basically. They are way better than the 244 nodes that you get with a singletree right, and they reach 244 notes they are like more than twice as good as the single tree they built with 244 notes.

Because the objective function you are minimizing is something very, very different right at every stage, you are changing the function and you are focusing your efforts on actually getting to the harder parts of the space right so that essentially it is little magical. This is more dramatic is it something like this but look at the book crippled for the exact figure right, so this is amazing

posting is very powerful. I talked briefly about the random forest in the next class which is you do not even have to do any decision making right, you do things randomly but then do a lot of them right that is also very powerful.

So random forest is not a boosting technique by the way the random forest is a bagging technique right, but then that is also very powerful.

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**Introduction to Machine Learning**

**Lecture-61**
**Gradient Boosting**

**Prof. Balaraman Ravindran**
**Computer Science and Engineering**
**Indian Institute of Technology Madras**

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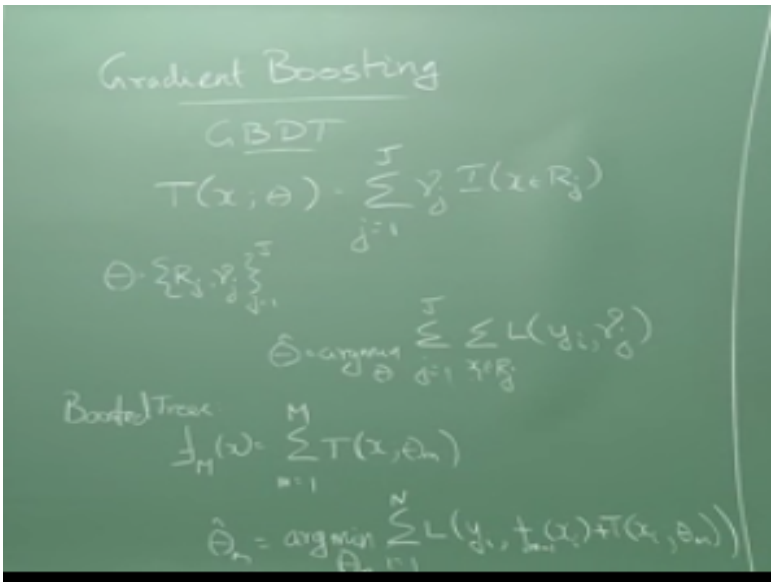


Right, so I want to talk to you about the interesting idea known as gradient boosting. So all of you remember what boosting is about right. What is boosting? What is bagging? Boosting yeah, so boosting is specifically a stage-wise process where at every stage you try to boost the classifier from the previous state says that the error is minimized right, an error is reduced not necessarily minimize, but the error is reduced right.

So that is a characteristic of boosting, so at every stage, you have to look at the errors from the previous stage, and you are trying to reduce that right. So we looked at AdaBoost right, one of

the most popular boosting algorithms. I then told you that AdaBoost uses exponential loss and that it is related to the logistic loss. So you can use the logistic loss function and derive a boosting algorithm called logit boost right.

But it is very similar properties to AdaBoost. However, AdaBoost is more popular, especially from an analysis point of view and things like this because it is not nice properties, okay. There is yet another approach to boosting that is gaining a lot of currency recently. It is called a gradient boosting, at not well recently would mean in the last decade or so, compared to AdaBoost, which is several decades old right.

So sometimes they even call it gradient boosted decision trees right, gradient boosted decision trees because you use this specifically in conjunction with trees right. And in fact, in many applications now, gradient boosted trees are getting hard to beat right. And so, we just reintroduced some notations that you might have forgotten right. So I is an identity function, which is 1 if X belongs to Rj is 0 otherwise right.

And so, this is summing over all regions, so R1 to 2 RJ and γj is essentially the output I am going to produce if x lies in RG right; this is the regression tree thinks. So what is my θ here? It is all the RJ is the specification of the RJ, and the γJ is for each of those regions. So that is my θ. And typically, we pick some loss function if it is regression; it is going to be squared loss and then right.

So I look at the loss incurred when the actual output is YI. The output I am giving you is γJ, so for all data points X that belongs to a region RJ the output will be γJ, so this is essentially the loss there and sum this over all regions, and this is the rectum just recapping the decision trees for you right. And then we looked at greedy methods for finding RJ right, and given an RJ; we knew how to fit γJ right, so given that we looked at some greedy search methods right.

So you can do, you can do boosting with trees also just like you did boost with other classifiers you can do boosting with trees right. So I have M trees so essentially it is taken some of the output of all the M trees that gives me my boosted tree right, remember that I mean this is not a

single tree okay it is now a forest, then I have a correction of trees a collection of trees is a forest right.

So it is a forest, so I do that, and the difference here is, can people at the back see this right. So this is essentially when I find the parameters for the mth tree right, so I am going to look at the classifier or the predictor that is formed by the first M-1 trees right. And then I am going to find that tree okay, whose output I will add to this predictor right, and you search for computing the loss right.

So for every data point in my training data right I look at the way I look at the output produced by the m-1stage 3, I look at the value that is added by the mth tree, so this is the output produced by the mth tree I look at the value added by the mth tree to that right. And then I will compute the loss function okay make sense yeah.

So the basic area is that every point this is just forward stage wise addition that we like whatever we did before introducing boosting right that is exactly that. So now this becomes boosting because I am explicitly trying to figure out what the residual error is from here okay, and trying to adjust for the residual error using my tree the new tree I am learning right. So when will it be the residual error, when it is a regression task, and squared error is my metric right.

So and the loss function is squared error right, and I am so trying to solve the regression problem right, then essentially what I will have to do here is take the residual error. So whatever the tree does not explain them-1 stage tree whatever that does not explain, so that error I will have to explain using this right. So if you think about what we are doing here, so you will first build one tree to predict your output as best as possible right.

The predictor function as best as possible will build a tree, then what you will do is okay you will take the residual of that, build another tree that predicts the residual as well as possible and add the output to this okay. And then take the combined thing find the residual of that build the third tree which will predict the residual and add it back to this and so on so forth, you just keep doing this right.

So that is essentially what boosting increase means, so you still not come to the gradient boosting part okay. Finally, it will look very similar to what I am telling you now, but we have still not come to that part yet right. So as with regular decision tree learning given the RJ's right finding the $\gamma J$'s is easy given the RJ's finding, the $\gamma J$'s is easy. But the problem is finding the RJ's, in general, it becomes a little tricky finding the region's becomes a little tricky because I have to take into account the other tree's output also right, in general, right.

But I am talking about the squared error. It is very easy because things nicely decouple right when I am talking about a squared error. I do not have to worry about FM-1 after I compute the residual right. The residual could have been generated by any classifier any regress righty; it does not have I do not even have to worry about the fact what generated the residual was a tree right, I do not have to worry about it, all I need is just the residual.

The residual then becomes any function right, so with squared error boosting becomes just like learning a series of decision trees, nothing special about it. But if you have other kinds of loss functions, then we will have to worry about how to accommodate it, but at least in this case of squared error loss okay.

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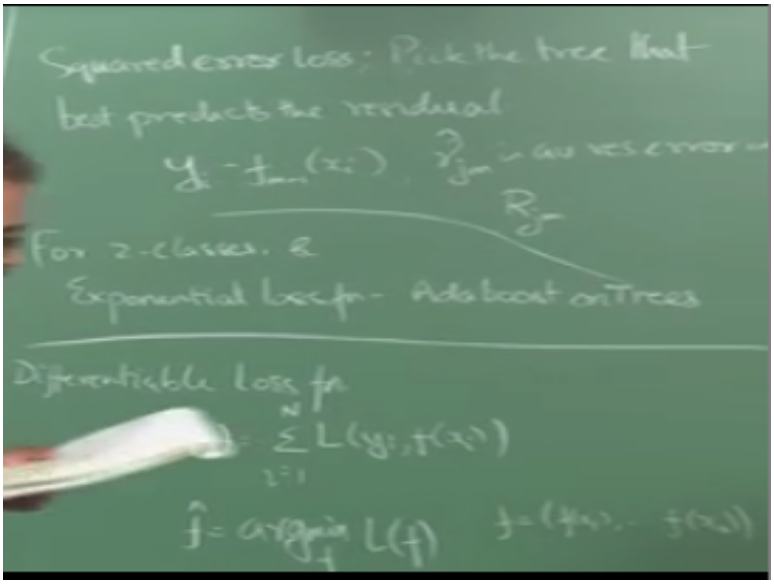


So that is essentially your target function right, and what will be the γ hat that you will need just be the average residual error in the Jth region right. Any questions, so in fact, there is another case where it becomes simple, which is essentially, so for two-class problems and exponential loss functions, what you think we get, it becomes the same as doing ADABOOST with trees okay, the two-class problems.

But, so it turns out, there are tricky things here right, so if there is if it is a multi-class problem okay, then things do not decouple as nicely. So if you have two-class problems and your loss function is exponential loss okay you can show that this is essentially the same the computation that we are trying to do here right that minimization everything that we are trying to do here essentially reduces to the same solution that you get if you did the ADABOOST derivation on decision trees okay.

But these are the two cases where this thing simplifies that, for example, if you are trying to use deviance as a loss function, then things do not decouple this easily okay. So these are things that you have to keep in mind, so this is one part this is essentially telling you how to do boosting with trees the regular way okay. So let us look at something else now, so I suppose I have some differentiable loss function some loss function which is which I can take the derivative of..

So if I want to do, so if you want to take a numerical approach to optimize this kind of a loss function typically what will I end up doing I will start with some guests for a solution right take the gradient of the loss function for the parameters at that solution point we will number gradient descent right. Then I will compute the gradient, and then I will move in the opposite direction of the gradient then I will move a small step in the opposite direction of the gradient go to a new place and compute the gradient again and then move again and so on so forth until I converge to the right answer right.

So if you think about what this is doing this is something like okay take the initial solution, okay, then I add another solution to it which is essentially the gradient times something then I add something more to it, so I will add something so essentially the solution I am computing a sequence of additions that they have done on the basic solution I started with right.

So even though one way of thinking about it is at every point, I give you a parameter vector, but the parameter vector itself is composed of a sequence of additions. So I can think of it as first starting with initial guess for my parameters, then adding something more to it, then adding something more to it, then adding something more to it, and adding something more to it, till I come to the final answer right.

So that is one way of thinking about it, so let us try and write this down a little formally. So now I am going to see what I can do about this f is for the time being ignore the tree constraints I will come back to the trees later, time being let us ignore the tree constraints right. So what I need is, so just when I am trying to do this numerically, I am just operating with a single data set right.

So when I say f what I have looked at is a point in RN, so what does that mean let f means okay what is the value of F at X1, what is the value of F at X2, X3, X4 up till Xn so when I impose constraints on F, then I will be restricting the kind of vectors I will see. But in general, when I am talking about F in this context, I just mean like an n-dimensional vector right. So you can think of it as a point in n-dimensional space right.

So typically, what you do is you start with some solution right F0 you start with some solution let us call it H0 right. So you can think of it like this I start somewhere here that is my F0 right. And then, I compute the gradient and move in the opposite direction right, so I take a small step in this direction, so I come here that gives me a new set of parameters right. So this is 1 θ, this is another set of θ, and this will give me another F.

But instead of saying that this will give me another F, so I am going to say that okay this is one F. I add something to it right, so that gives me the second F. So what I am computing in every step is the amount that I add to the previous solution to derive my new solution okay. So I am calculating θ and F here, so what I have here is θ corresponding to every θ I have here there will be an F corresponding to every parameter setting I will have that will be output vector F when I change θ this values will change right.

So when I am here I have one solution right, so when I want to go here that essentially means that whatever F vector I have here I will have to change each of those coordinates by some value so that I will end up here right, those values I change the coordinates by it is my H vector right, is it clear what we are doing here.

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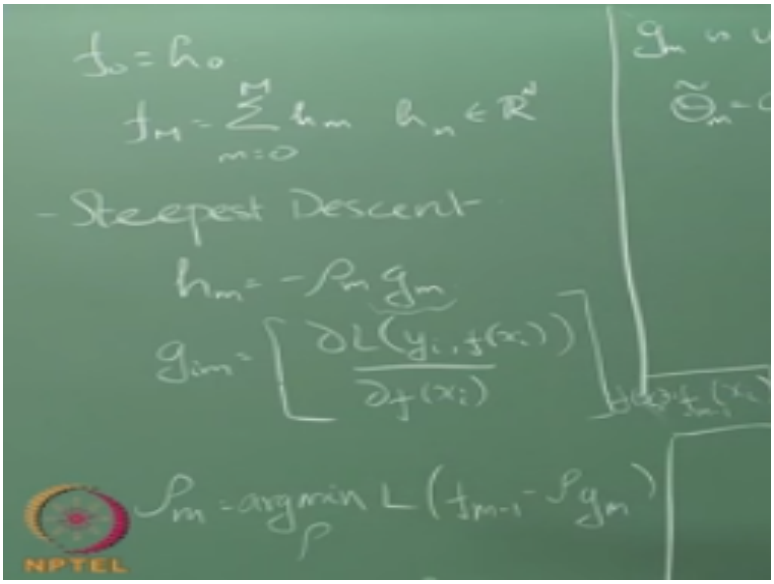


So, what is the normal mechanism by which we will do this right So, I have been using the same example so far right that steepest descent will pop up right even the other way of doing this optimization right. But steepest descent is the one that we are all familiar with the one that I have been using as an example here so far right. Since I have not chosen any arbitrary parameterizations to form a θ right for the F, I have not chosen any parameterization θ or anything right.

So the parameters of F are the output set each one of the input points to see the way I characterize my function F is looking at okay, what will be the value of F at X1, what will be the value of F at X2 and so on so forthright, I do not have any other parameterization for it. So instead of finding your δL/δθ you find that I am writing it as δL/δF okay. So F(xi) is essentially the output of F at Xi and what is F here, it is Fm-1 because I am determining the M stage I am looking at the m-1 guess for my function right.

So the steepest descent direction would be saying that okay, so Gm is the direction in which you have to move because that gives me the direction in which the or rather - Gm is the direction I have to move. After all, that gives the direction of steepest descent. And ρm gives me the step size I have to take in that direction how larger steps I can take in the direction. So how is the flow I am determined you should look very familiar right?

This is exactly how we did the ADABOOST derivation. Now people are thinking about it ADABOOST derivation like this yes, we did go back and think about it okay. So very similar not exactly the same thing, but very, very similar the exact the same steps that we did right we first found out which way we have to change it right. Then we found out what the step size should be, and the way we did it was okay, I have already had a classified till m-1 stage okay, what should I do at the M stage.

To minimize the error, so this is exactly what we did, the idea behind each of the steps are the same. The mechanics might have been slightly different right. So once I get this, then I do okay, right is it clear people are doing so far right. So whatever we are trying to do right, is nothing, so there are two different parts here okay, if you people are getting confused. Hence, the first part here talked about boosting trees okay, the second part here talks about taking some differentiable loss function and trying to do some kind of a stage-wise process on it okay.

I just took your normal gradient descent procedure and told you that you could think of it as a stage-wise process, I guess as we did with boosting, we can think of it this additive model right. So whatever you will learn here right, so now the thing is how we connect up the two will somehow account for that later. I do not want to erase anything from the board because what we are doing right now is connecting the two parts right. So I do not want a erase anything from the board, so you can see both that and this well we are looking at this right.

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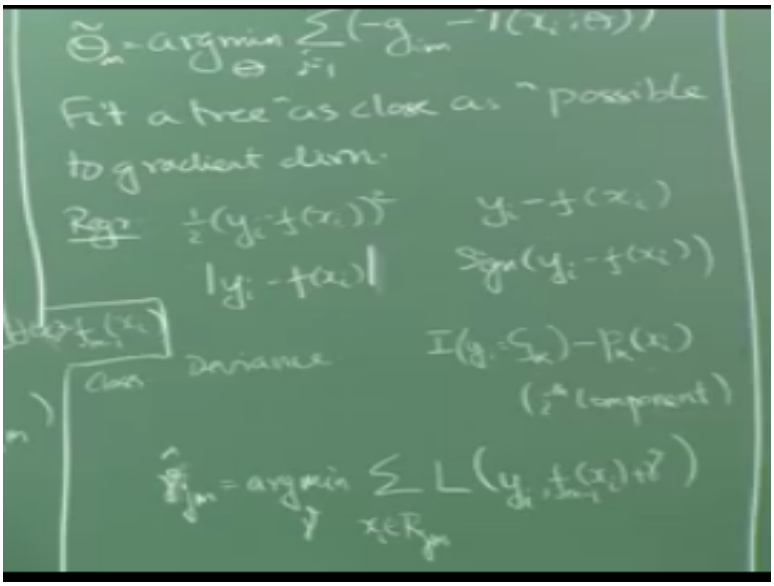


So far, so Gm, Gm is some kind of an unconstrained maximal descent direction I do not have any constraints on f for anything else right what is the maximal descent direction and essentially get Gm. So now what we are going to do is to say that hey, all of this is nice, but I would like some parametric forms for what I am doing right; otherwise, things become too complicated. So what I am going to do is I am going to fit a tree.

So what I want to know is Gm right, so if you think about it, so I need to compute Gm, and instead of doing this in this arbitrary unconstrained form, I am going to build a tree that approximates Gm as closely as possible okay. So you should note something here what is it you should not, so for all this while I have been very carefully writing L for the loss function well, I am trying to keep this as generic as possible right.

But here I wrote a squared error loss function because it does not matter what the problem that I am trying to solve is, it does not matter what this loss function is okay. After all, what I am trying to solve now is trying to approximate a vector I trying to approximate a direction right by a tree essentially I am always solving a regression problem here right. So if you think about it, Gim is going to be some kind of a vector right. Gim will be some kind of a vector. All I am trying to do is predict the value of that vector component right.

So I am just doing regression regardless of whether my original problem was a classification problem or a regression problem or whatnot, I could use any loss function here, and I could use any loss function here this is the crucial difference you should appreciate right, for the actual or solving the problem that I could be using a different loss function here okay. But when I am building the Mth stage decision tree, all I will be doing is regression, because all I need to predict is what is that particular gradient descent direction for that input value Xi right.

So this is what I am going to do, how am I going to go about doing this. Well, it depends on what this loss function is okay, so is this loss function, so if the problem I am solving is a regression, this is what I am solving, this is the squared error loss function okay, this is not this okay, this is that. So if this L is squared error okay, then what do I get here is essentially what is this, give this is Gm essentially the gradient of the loss with respect F(xi) right, the gradient of this for F(xi) is essentially the residual Yi-F(xi) right.

So this is basically – Gm if you would think correct. So now, what happens if I am doing regression with squared error loss function, and I am trying to do this gradient boosting right. So I am trying to build a new tree that predicts the direction of the gradient what do I end up doing, I end up predicting this is the residual right. So I end up predicting the residual and here what we said if you are making the squared error loss pick the tree that best predicts the residual.

So that was derived from just the basics of boosting regular boosting. Well, here I am talking about a technique that can do boosting on trees regardless of what is the underlying loss function right, but it does boosting using trees okay. So that is a cool thing about gradient boosting right, so you always are solving a regression problem as far as the tree is concerned, and solving regulation problems using trees is very easy right. Your solving regression problems using a tree is always regardless of this loss function.

If you remember when we are deriving this boosting update, I said squared error, and for two-class exponential loss, the boosting form is easy right. But now if you busy doing the gradient formulation of it okay regardless of what you are doing with the loss function, you can

still do boosting with trees. So that is why gradient boosting decision trees have become very popular now because you can do all kinds of cool stuff with it.

So what about this right, and suppose you are doing classification let us take deviance as the loss function right, we will remember deviance, we looked at deviance multiple times right, and turns out that. So, if the ith class is Gk, then it will be 1 minus that the data point Xi is in class K, the probability of data point Xk belonging to class K. So it will be 1 minus that, and if the actual class is not K right, then it will be a minus probability of Xi belonging to class K okay.

So this is like the ith component of it, so ith component of my Gi okay. So again, what I have to do, I have to take this expression to plug it in here and do regression again, I will take this expression to plug it in here and do regression right. So all you need to do is figure out what is the derivative of your loss function with respect here F(xi) right, and then once you find out the derivative, you just do the regression for that for each stage in your decision tree okay. So what will be my $\gamma j$ this is $\gamma j$, such $\gamma$, $\gamma jm$, what will be $\gamma jm$? So earlier, we said it would be just the average residual error in Rjm right.

So, in this case, it is going to be, so once I have found out what the actual regions are right, once I have found out what the region. So this will give me the regions right, so once I find out what the actual regions are, I will do the following. So what have we done here, okay? So earlier when we are finding out the, when we are building decision trees right, so the way we found out the regions right.

If you remember the way we found out the regions were we postulated a split point right, and then for that split point we figure out what is the best $\gamma$ in both halves of the tree right. Then we took value for that, and then we kept looking at all the possible split points right, the splitting variables and split points for all the possible combinations and for each one of them we evaluated what the resulting residual error would be right.

And based on that, we pick the split point, so here we are doing something different when we are splitting picking the split point right, we are going to pick the split point such that the residual

error in predicting the gradient is minimized right. I am only predicting the gradient here; the residual error in predicting the gradient is minimized. But when I finally decide what the output that I am going to give in each of the final regions right is.

So, in the regular decision tree building, I would have already solved the optimization problem right by the time I reached that point. So I know what the solution that I have to give is, and it is the one same thing which I use for splitting criteria. But in this case, when I finally give the outputs, I am going to look at the loss function right of data points that fall in that region look at what already exists, which is Fm-1 of Xi right.

And look at what I need to add to bring up the output so that the loss is minimized. Whatever is the loss function, so when I am doing the loss function here I will no longer be using the squared loss I will be using one of these, I mean this is quite a loss I could be using the absolute gloss or I could be using deviance or whatever is the measurement that I want right, I will use this loss function the loss function I use there, I will use that here in order to figure out the outputs okay.

So you let this sink in a little bit, so it is a pretty cool idea right. So I have somehow come up with a mechanism where I can use decision trees in a very powerful way, because finding regions. At the same time, I do regression is very easy with decision trees because we know how to use squared loss and there are lots of tricks and optimizations that you can do when you are searching through with square loss right.

Now we looked at some of them, but there are many other things that you could do so what I am going to do is for all my tree growing part right I am just going to use the regression trick right. I finally have to give an output at that point I will use whatever is the true loss function I want right, that is why it is called gradient boosting. So I use the gradient at every point to boost my performance right.

So the way I fit my, so if we think about it, this might not necessarily be an ideal way of doing things, why is that right. So if I want to predict γjmhat that truly minimizes forget about the Rj's right, that truly minimizes the loss function, I might want to split the space differently right. But I

am amusing the gradient information to split the space, then whatever splits I get whatever regions I get by using the gradient information, I am using the same regions in order to reduce the error also right.

It is fine as long as I am unconstrained right, as soon as you put in the constraints of trees it is not entirely clear that the tree that you want for getting the representing the best γhat is the tree that you want for representing the best GM it is a very, very subtle point you have to think about it a little bit. But more often than not, it turns out to be fine, okay, but there is no guarantee that the best three for predicting GM is the best three for representing your γ's okay.

That is two slightly different things, but still, it turns out to be fine right. So all of this discussion collapses if you move to L being the residual, I mean the squared error right, L is squared error everything looks the same you already solve it here is an easy thing to do. What I want to point out is that it is essentially the same square error mechanism you can build boosted trees for any loss function that you want, it can be regression, it can be classification.

So whatever it is, you can build a decision tree. So one thing that you have to be careful about is that you do not overfit the things any time right. So you have to be careful about not overfitting the data because you are only working with the n data points always right, you can build a very complex tree that will try to overfit the gradient for just the training data right.

So that is not a good idea, so try to keep your tree down the complexity of your tree down. Quite often people choose the size of the tree a priori. And then, you might end up adding more trees than necessary because he chose too small a tree, but at least you will avoid overfitting right. And so that is it for gradient boosting.

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**NPTEL ONLINE CERTIFICATION COURSE**

**Introduction to Machine Learning**

**Lecture-62**
**Random Forests I**

**Prof. Balaraman Ravindran**
**Computer Science and Engineering**
**Indian Institute of Technology Madras**

So now we know that trees are great candidates for boosting as well if you are using gradient boosting right. However, trees are great candidates for bagging as well as right. What is the important property in bagging that we talked about? What does bagging help us reduce variance, right? So bagging allows us to reduce variance. You can show that the reduction in variance is highest if the classifiers that you are building okay, or not correlated right.

So I am building many, many classifiers, and the classifiers are predicting the same output right. So if the classifier parameter set I am estimating or somehow if we can make them uncorrelated right, then the reduction in variance is maximum it kind of intuitive right. If the classifiers are very correlated, there is no point, they are not different classifiers right, and they will give me the same output, so the variance will be high.

So if you can somehow make the classifiers uncorrelated, then the reduction in variance is high right. So there is a particular relationship between the amount of correlation between the classifiers and how much you pay in terms of the reduction variance right. So I am not going to derive that, I just point it out to you, so if you want, you can look it up; it is there in ESF right. And if you think about what we are doing with bagging right, we are taking one data set right, and you are sampling with replacement from that right.
So the probability of the trees that you are generating being correlated express rather high, the probability of the trees that you are generating being correlated is rather high. So it can become up with some way to reduce the correlation between the trees you are constructing right.

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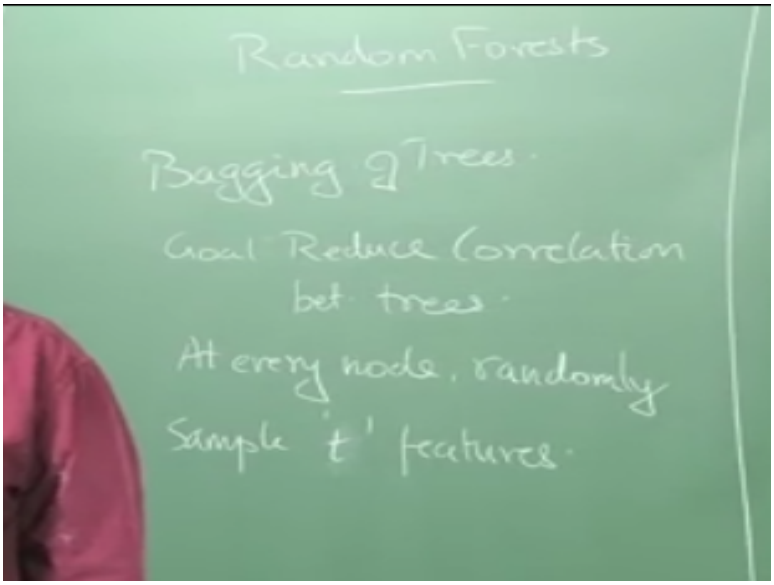


I am going to be doing bagging right, but the goal is to reduce the correlation between trees. The people who came up with the random forest had a very simple idea for doing this right, and you start doing bagging as you would normally do okay. So you have your data set, then you create a bag by sampling with replacement from that data. Now, when you start building the tree on this data set, so what do you do at every node right, sample some P features from your feature set, and use P for the regular feature description right..

So let me use a different, so we have a total of P features right your data points come from some RP space, you use randomly sample some T features from that P features okay. Find out which is the best split point, which is the best split variable among these P features alone, split the data go down to each of the subsets repeat the same process, sample, and other T variables right not necessarily this joint, and just sample okay, sample again T more variables okay.

And then, try to find out which is the best split point among these T variables and keep doing this. So what does this get us to see if you had worked with the same data set right, which I mean if I am just done bagging at the root level it is highly likely that each one of the bagged trees

would have picked the same attribute right, just because you have sampled it again right? So it does not mean that the very predictive attributes will get discarded.

So at the higher levels of the trees, it will look very similar right. But now you are getting rid of that I said okay. I had chosen randomly. I have chosen T variables, and only from them, I am choosing the best variable; therefore, I am reducing the chances of the trees looking similar. We can show that this leads to a significant reduction in the variance in the bagged estimate, and random forests are performing very well.

Random forests are competitive with gradient boosted trees in some applications and vice versa, right. So boosted trees are better than random forests in some applications, and random forests are better than boosted trees in some applications. And because sometimes, till sometime back there are very efficient random forest libraries and people use random forests a lot right. But now there are also very nice libraries available for gradient boosted decision trees. And therefore, try everything and see which works right.

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**Introduction to Machine Learning**

**Lecture-63**
**Naive Bayes**

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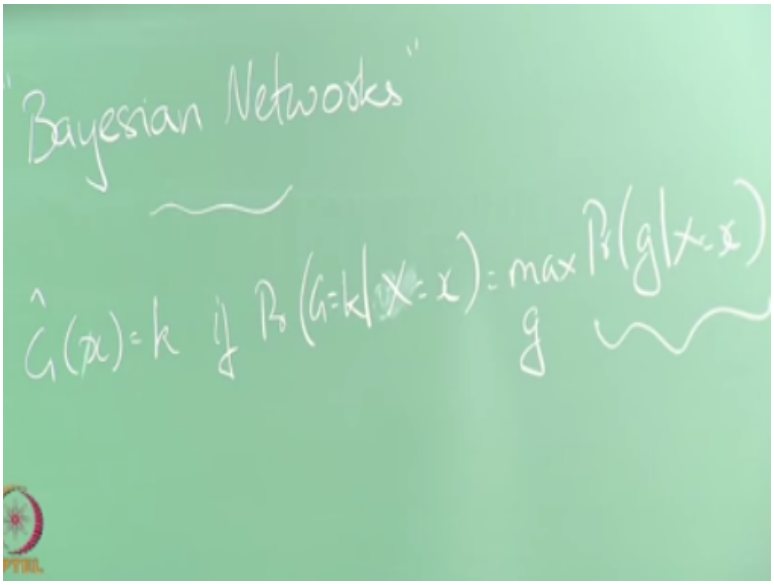


We mentioned the Bayesian classifier or the Bayes optimal classifier in the very first class. We talked about the nearest neighbour methods we probably have one 'X' and how do you know what its class is. The probability of the class given a data point because you have only one data point in your training and therefore we took an average over a region and so on so forth. We will use that estimate over a region for finding the probability. So, this is how we motivated what KNN right the K nearest neighbour classifier right.

$$\hat{G}(x) = k \quad if\ P(G = k \mid X = x) = max_g P(g|X = x)$$

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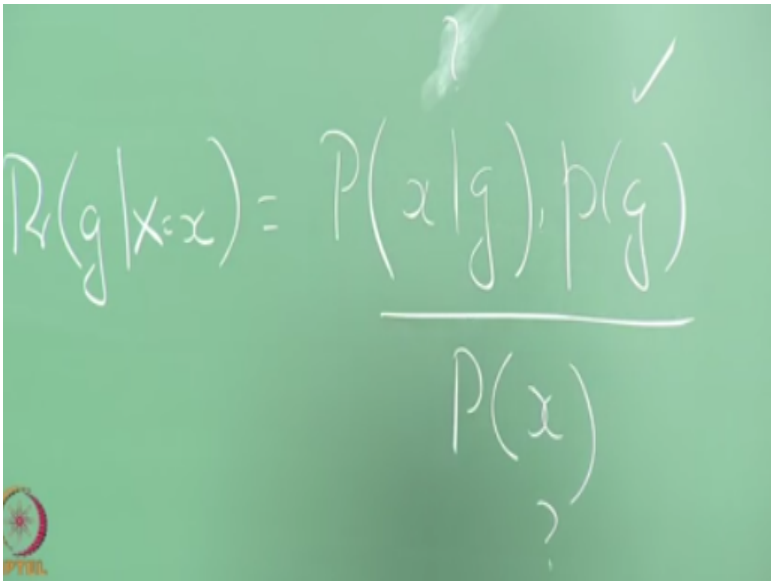


So now I am going to take a slightly different tack right, so what we want we want the probability of right 'g' given 'x', we want the probability of 'g' given 'x' right, So we have our friend the Reverend how many of you know that Thomas Bayes was an ordained priest okay, so we have Reverend base to help us outright, so is essential. Right so then we have a lot of quantities that we can estimate this, you can estimate this from data. How will you estimate that from data?

**Bayes Theorem**

It describes the probability of an event, based on prior knowledge of conditions that might be related to the event. It is expanded as,

$$P(X = x) = \frac{P(g)P(g)}{P(x)}$$

$P(g)$ – Class Conditional

$P(g)$ – Class Prior

$P(X = x)$ – Posterior Probability

$P(x)$ - Data Prior

Okay, we will come to that, Can you estimate this from data? It is the fraction of a particular class you can do that, or you can make assumptions about the class densities, and you can estimate the parameters of those from the data and so on so forth. That is fairly straightforward to estimate right, so what about this guy? But in general, if you wanted to do the max, yes if you don't want to do the max then this becomes a question is how do I go about estimating this right.

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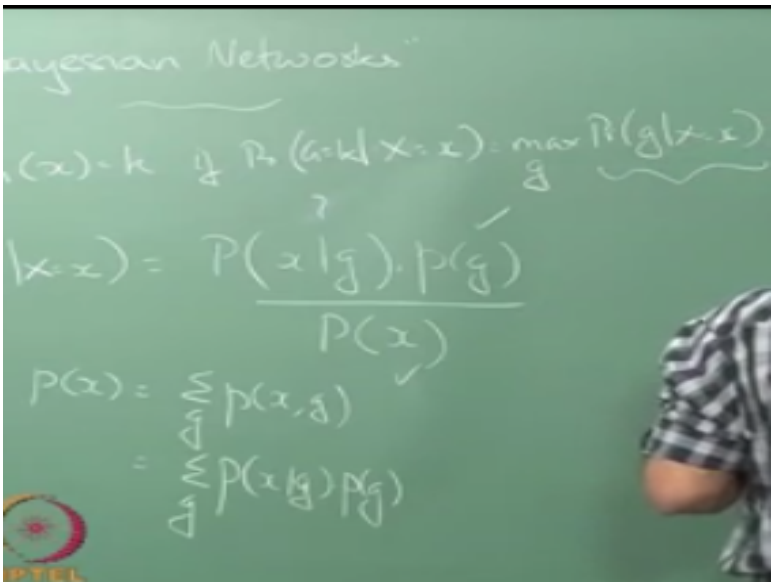


So there is one way of doing this how do you do that, so you can assume that you have a is what is the probability of 'x'. That essentially some of the numerator for all possible 'g' so I get my denominator so this I can do so all we need to know is how to estimate that right, so can you estimate it?. Yes, it is the distribution of data points given the class. But the main problem we will face is the sparsity issue.

The $P(x)$ can be estimated as,

$$P(x) = \sum_{g} P(x, g)$$

$$= \sum_{g} P(g)P(g)$$

Often enough we might get one sample here one sample there and so forth. This will not cover the entire data distribution to get a good estimate of the probability distribution. This is especially happens in really high dimensional spaces. Suppose we are assuming X comes from some R power p. If p is a hundred-dimensional thing p is 100, so if X is a data point in R power

100 right, so what is going to happen data points are sparse in this vast space right. This makes the estimation hard, so we have to make assumptions about the distribution.

Theis is called as class condition distribution because their probability of x conditioned on the class right. So sometimes you also call them what likelihood yeah I thought somebody would say likelihood before anything else. Still, people are just keeping quiet, so this also called the likelihood but is the class conditional distribution right so what is the difference between likelihood and class condition distribution, so likelihood is a function of 'g' right I remember I kept repeating that multiple times when we looked at likelihood when I am conditioning it on some parameters $\theta$ okay. Still, it is a function of $\theta$, so 'x' is the same.

But when I am talking about class conditional distribution I am talking about the probability of X okay it is an a function of x conditioned on g okay good, we will just go back so the most the simplest assumption that we make.

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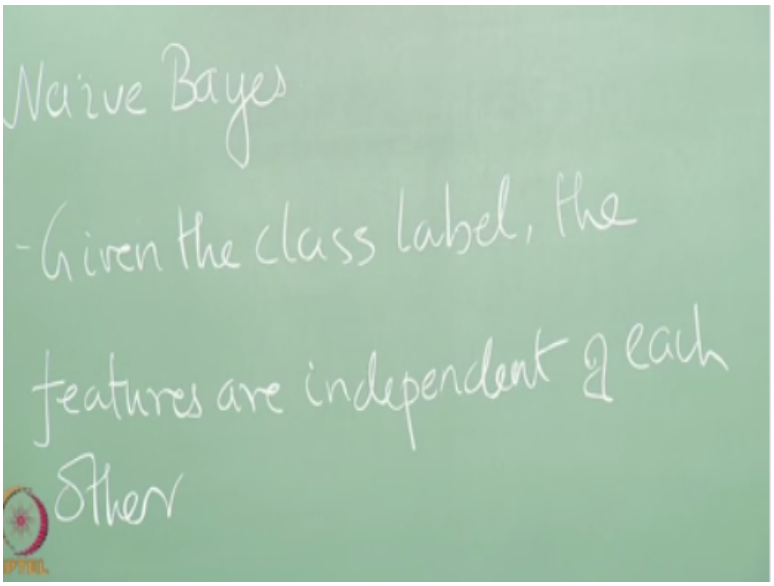

To get this to be tractable is called the Naive Bayes assumption right, so what does the Naive Bayes assumption tell you it says that given the class label the features are independent of each other right.

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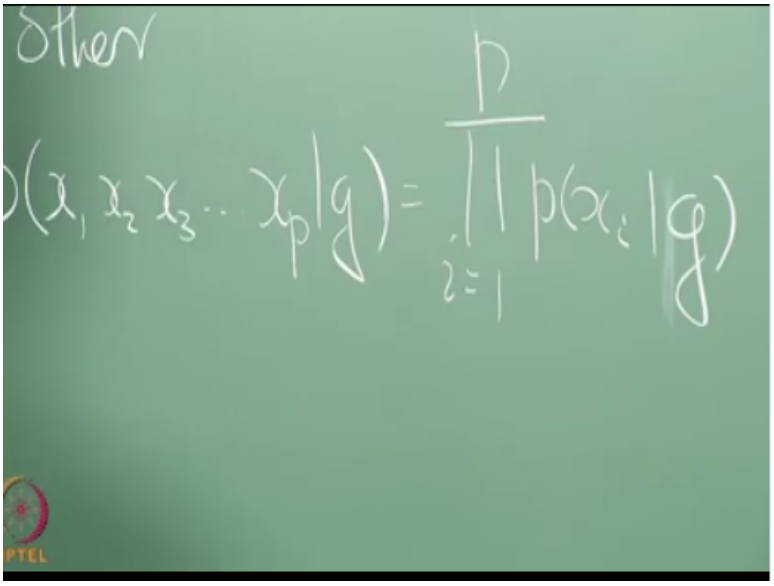


So what does this tell us it says that if I have the probability of that, I can write this as this is called as Naive Bayes assumption right. Once I do this now it becomes very easy to for me to estimate the parameters. In how many data points has "x_i" taken a particular value of that particular class. First segregate all the data points by class and then in class one in how many data points did "x_i" take the value of zero in class one how many data points did "x_i" take the value of one and class two how many in the how many data points did "x_i" take the value of zero and "x_i" I take the value of one so on so forth.

Do you want me to say "x_i" takes the value of 0 "x_i" takes a value of 1, "x_i" takes a value of 2, "x_i" takes the value 3. I know that just takes a lot of time so just making it binary so that it is easier for me to speak right, so it could be anything right, so "x_i" could be real value this, for example, our always our setting is our R power p right in that case what do you do some binning you could look good that is one very valid option even though many textbooks do not

recommend that but not that they do not there is not that actively not recommend it just they do not even talk about it okay.

But that is actually a valid option, and I will tell you why in a minute but the usually recommended option is to have some kind of parametric form for these marginal distributions these are conditional marginal. If you think about it these are marginal if so this was the Joint Distribution right then this is just the marginal that is the conditional Joint Distribution, so this is a conditional marginal so for the conditional marginal they ask you to assume some kind of parametric form and the usual one that they suggest is a Gaussian right.

Based on the Naïve Bayes assumption the conditional joint distribution could be written as,

$$P(g) = \prod_{i=1}^{p} P(x_i|g)$$

$P(g)$ – Conditional Joint Distribution

$P(g)$ – Conditional Marginal of the "i" th feature

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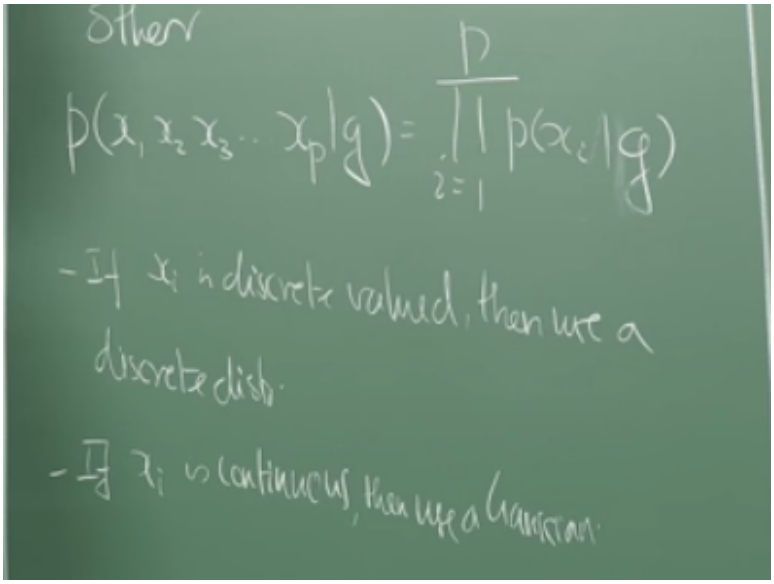


Usually, this is some kind of a Gaussian form for this conditional marginal right can you read it at the back can okay you can read it okay fine now because the thumb didn't stay up long enough, so I was not able to make sure which direction it was you just did something like that and right but then what is the problem in using a Gaussian all of you know the problem just think for a minute or lesser okay not a big issue there is a much bigger issue. Too much inductive bias but the wrong kind of inductive bias why is it wrong what do you know about the Gaussian.

Sorry, close so what does it mean the Gaussian is unimodal? Suppose there are two different values of "x_i" which are separated right which is very probable right for this particular class if you use the Gaussian what will you estimate the most probable point as, is there mean okay, so if we say 3 is very popular very likely to occur and 5 is very likely to occur your Gaussian will say 4 is the most probable value for x given the class, so you do not want that to happen.

So that is the reason I said binning makes sense, but then the problem is you have to find the right kinds of bins because when you are using the discrete distribution, there is no I mean it can be multimodal right I mean it is no notion of like you unimodality there right so I could have one output having very high probability another output having very high probability I could have like 10 different outputs having high probability is everything else having low probability it could be anything right.

So there that I do not have to worry about these, but then the problem here is hard to find the right binning, so that is a whole set of lectures by themselves, right so how do you bin your input variables there are many ways in which you can bin input variables in there you have to keep coming up with clever tricks depending on the application you have and so on so for this is actually not trivial right and you could do that.

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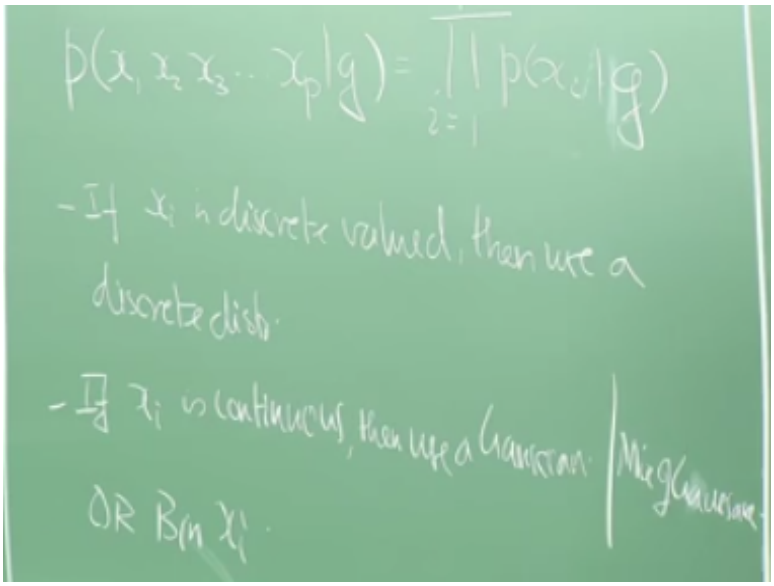

But of course, the Gaussian is just a simple example if you know that the data is going to be multimodal right what should you be using? The mixture of Gaussian and not a single Gaussian. Naive Bayes to work seems to be a very simplistic assumption is even called Naive Bayes even though I have never heard used in any papers or any literature so T. Friedman says that it is also known as the Idiot Bayes algorithm.

So Naive Bases sounds a little better more sophisticated right yeah do you think Naïve Bayes will work well. It did work pretty really well right weren't you amazed you didn't know how simple it was at the time compare it against SVM's did not we compared against the SVM's should so it turns out that the let us say for examples like text classification where you are the data dimensions are inherently very high right it is incredibly hard to beat Navie Bayes you know it looks like it is something so simplistic right.

I mean look at the assumption you are making I have a lot of texts I want to classify them as politics or sports okay that is one very simple problem that is out there is a standard problem that people use for text classification is a standard data asset I want to classify a news article as being sports or politics. And what I was saying here is given that it sports the probability that I will see cricket is independent of the probability that I will see football am.

Not only that the probability I will say cricket is independent of the probability i will see say Dhoni in the document sounds like nonsense right well if you are talking about Indian media yes right, but in general it seems very surprising, so that is because we are trying to assign all kinds of semantics to what is happening and but the algorithms that we are trying to use whether it be SVM or anything else or really not into the semantics of these things right, so we are only worried about it because we have all these others the knowledge base superstructures that we have built.

And that we are trying to look at the data through that right at the end of the day if you look at it is more a question of things occurring together co-occurring and in a very large document space, so the probability of any words co-occurring right is kind of diminishing right, so if I do not

know it is a document about cricket right and if i see the word Dhoni then I will say okay now maybe the probability of me seeing something with actually a sports-related term goes up because I did not know whether it was a sports document or a politics document before I looked at it right.

But given that I know it is a sports document and the probability of me seeing this anyway be higher it is not going to change appreciably because the word Dhoni appeared you see the reasoning if I had not known anything about the document because the word Dhoni appeared in it right the likelihood that it is a sports document goes up right well given the nature of Indian cricket the probability that is politics has not gone to zero yet right so but if I had known that it is already at the sports document knowing that okay it is Dhoni the words Dhoni appeared in it does not appreciably change the other probabilities okay.

So that is essentially the idea here because the number of words is very very large if there are only like ten possible words or ten possible values these features can take then there will be appreciable change even if I know that value for one feature but each of those words can take something like 10,000 different outcomes. Hence, it ends up becoming a pretty good approximation right when you are doing it in practice if you remember I told you something like KNN would be bad if you are in 100-dimensional space or a thousand-dimensional space.

And imagine text is a very large dimensional problem right, so what will be the typical dimension for text classification yeah so naively modelling it you can get 24,000 right you can do some kind of feature reduction and so on so forth try to reduce it to smaller state space. Still, it is pretty large right 10's of 1000 dimensions so if you try to do KNN in that things are not going to work that well. However, people still do that right so all these cosine similarity-based retrieval systems and other things that people used to do in the past but all nearest neighbour kind of techniques.

And not that they do not work but something like Naive Bayes when you want to do classification works tremendously well in this you can see this I'm challenging you to go try it out against SVM right you will find that maybe there is like a couple of percentage difference

right and the math is so much simpler there are a few things which I should point out about Naive Bayes one of the biggest advantages of Naïve Bayes is that I can have mixed-mode data. I can have some of the attributes being discrete-valued some of the attributes being continuous-valued and everything.

And it is very it is fine for Naive Bayes so what other classifiers can you say that? Trees right anything that is based on trees are also pretty robust when it has kind of mixed and you do not even have to do any kind of normalization of those attributes also. I can just keep each attribute one thing can run from one to a million other one can run from 0 to 0.1 is all fine right when I am looking at other more numeric classifiers in the sense of the kind of computations that they do they do distance-based computations and things like that.

There I have to normalize things a neural networks if something goes from one to a million. Another one is going from 0 to 0.1 the feature that runs from one to a million will overpower the feature the runs from 0 to 0.1 right I should Ideally be normalizing everything to 0 to 1 right so those kinds of things I do not have to do in Naïve Bayes it is all clean. Nothing the second thing I do not have to do is any kind of feature encoding normally i do not have to do any feature encoding I do not have to convert this into some kind of code.
So I do not have to take red and convert it into some code, code words so that I can feed it into my neural network wait how will I feed right into my neural network you have to encode it somehow right whether you use the RGB value for the colour or whether you use some other encoding for the colour we will have to do something about it right I do not have to worry about all of that thing this the same thing in addition trees as well right.

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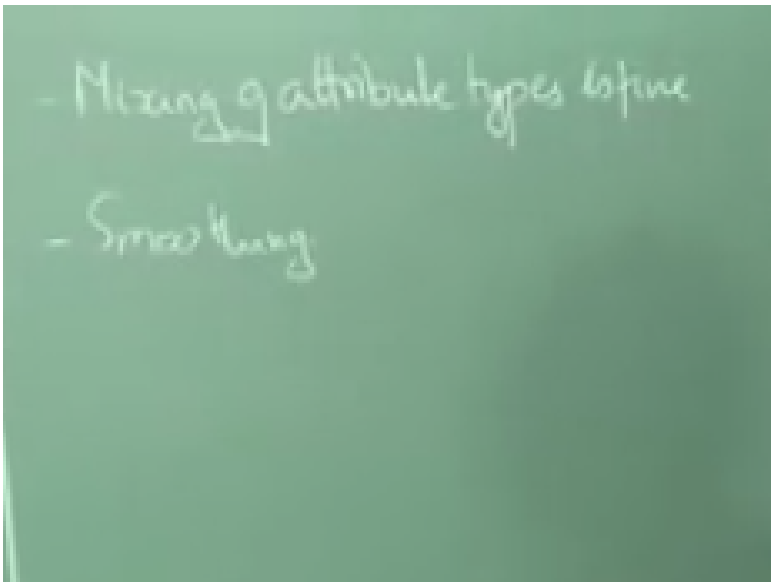


So that is one thing the second thing I want to talk about is how do you handle missing values right if 'x'is continuous if 'x_i' is continuous we are all fine right we are anyway using a Gaussian the Gaussian has an infinite supports, or something else which you have never seen in the training data comes will always assign some probability to it but what about discrete value things right, so I have trained you I have trained my classifier by looking at data right that has red, blue and green.

In that thing and in the test time somebody comes along with yellow, I can assign it a probability of 0. It says different so what if I do not have all the 'x_i' one more example I mean things like neural networks would cover that right I mean something I have not seen beforehand as long as have some encoding for it will just take care of it I do not need to have seen it in the training phase we do not know "x_i" is going to come or not say we have to somehow account for all the unseen "x_i".

So there are multiple ways in handling this in Naïve Bayes one thing which you can do is you can just ignore that attribute do not multiply it in and make it zero just ignore it look at all the attributes you do know the probabilities for okay and then multiply that is a problem with it what is the problem sorry you are assuming the probability is 1 right, so you will be overestimating the probability, so you will have to come up with some mechanism by which you will normalize that

right if you are using lesser features that they come up with some mechanism by which you are normalizing them right.

So that is something which will have to think about the other one is called smoothing right smoothing is essentially similar to what [Student] was saying earlier is that you assume that everything is everything that you could see right has occurred at least once in the training that means that you will give it some probability at least you will not make it zero and will also take care of this overestimating problem you will not make it one it will make it like 1/10,000 or something that will be a very small probability that you are signed to all the unseen values in the training data.

So when it test time at least you will not assign zero value to that data point right if everything else is very probable except that the colour is yellow right so you will not make the probability 0 the probability will get depressed significantly. Still, it at least not go to 0 okay, so this is one thing the problem is smoothing is the following if there are lot of values that you do not see I suppose that there is a color I see only four colours in training. Still, my actual colour spaces 256 right so I mean right or my actual colours space is 64,000 okay, and I see only four colours in training what will happen.

I am just talking about practical issues here if I do smoothly in such a situation what happens. It means you are training data is messed up, so you have to think about it. Still, if you use smoothing blindly in this situation what will happen exactly you will smooth the heck out of your probability distribution right you are taking this probability of one you are dividing it among 64,000 guys, so everybody should get a count of one at least right suppose I have 10,000 training points right the best account I can hope for is if all the 10,000 of the same colour it will get 10000 by 64,000.

No it will get10000 by 74000 each one of the 64,000 I will count at least once. It 10,000 has happened so 74,000 it will be 10,000 by 74,000 will be the probability for the colour which occurred in all your training data points okay that is a really small value for the colour so smoothing you have to be very careful when you apply to smooth right so if there are too many

unobserved values and smooth it blindly like so you will essentially lose all the information in the training data you will have to come up with other mechanisms like [Student] was pointing out there is something wrong with your training data first right.

So you have to go back and try to fix that see if we can generate more representative sample so that is those are the things that you should look at, we know how to do P(g) right, so we know how to do P(x) once we know how to do this and this we know how to do by doing this so parameter estimation is taken care of and this all the ancillary things.

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**NPTEL ONLINE CERTIFICATION COURSE**

**Introduction to Machine learning**

**Lecture-64**
**Bayesian Networks**

**Prof. Balaraman Ravindran**
**Computer Science and Engineering**
**Indian Institute of Technology Madras**

One of the things that you could do along the lines of this independence assumption is trying to be more nuanced about your independence, so what do I mean? So just do not make this assumption that everything is independent of one another given the class right so think of something like this I want to look at this joint distribution $X_1$ to $X_P$.

Consider the Joint Distribution

$$P(x_1 \ldots\ldots x_p)$$

So I am going to say things like okay, I am going to write something work I am not stopping. So what I am saying here is given $X_2$, $X_1$ is independent of everything else if you know what the value of $X_2$ is $X_1$ is independent of everything else.

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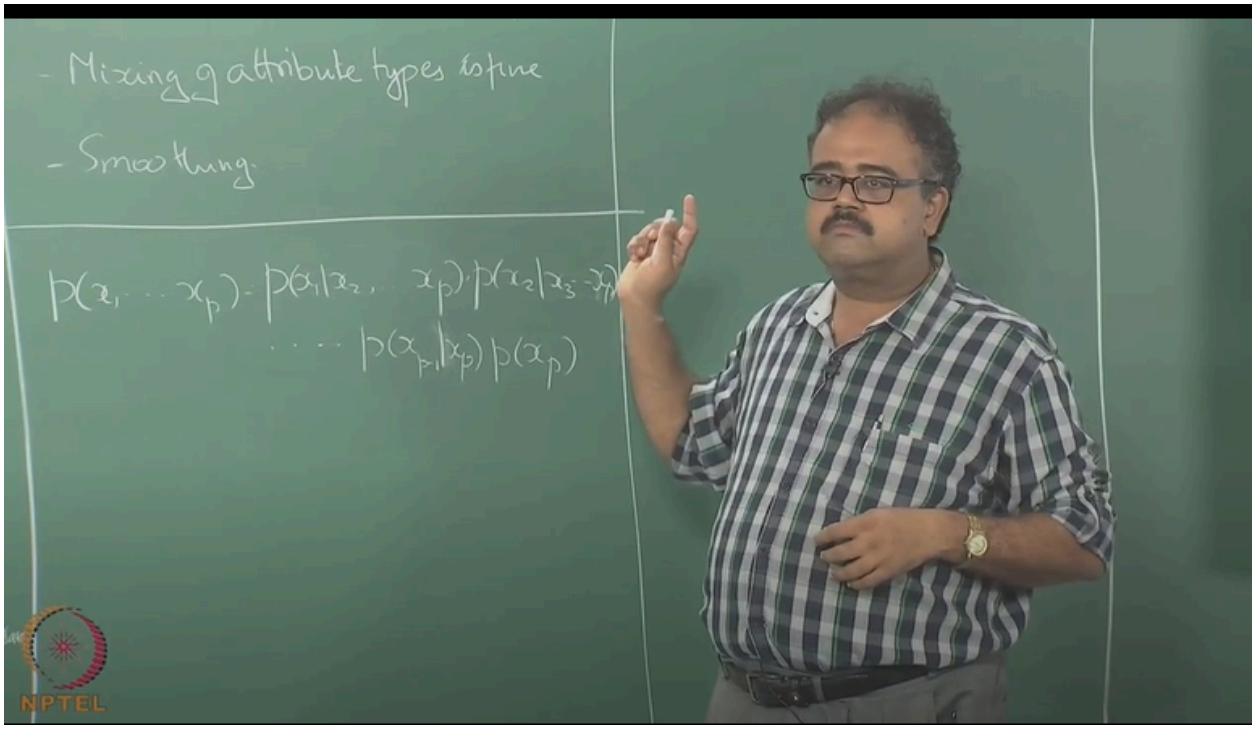


So I can write the probability of $X_1$ to $X_P$ as probability of all conditionals. To find out what is the probability of $X_1$ given $X_2$, $X_3$, $X_4$ up till $X_P$. Is like trying to find the probability of $X_2$ given $X_3$ I mean either some arbitrary ordering of choice and $X_1$ to $X_P$ right, it could be any other ordering now the probability of $X_2$ given $X_3$, $X_P$ and so on so forth.

The Joint distribution can be factorized as

$$P(x_1 \ldots\ldots x_p) = P(x_2, \ldots x_p)\, P(x_3 \ldots x_p) \ldots\ldots\ldots\ldots P(x_p) P(x_p)$$

Now I am going to tell you that, okay this is you can always add a conditioning 'g' if you want right this makes my life easier if you do not put everything conditioned on 'g', right. If you want, you can do that as well okay. How likely that variable like $X_1$ depend on value taken by other system variables. Especially, if I am going to have 30 and 40 variables how likely is that $X_1$ is

going to depend directly on all the other 30,40variables in the system right, is it not going to happen right.

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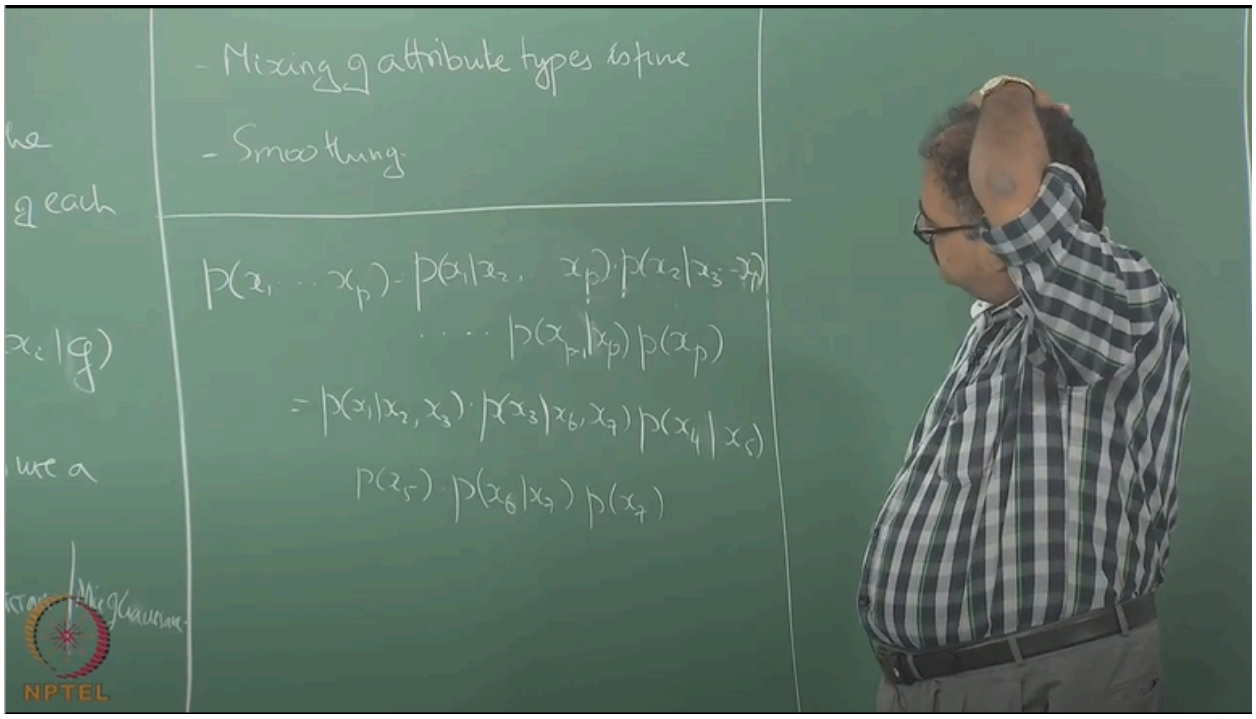


In reality, the dependency might be simpler than above,

$$P(x_1 \ldots\ldots x_p) = P(x_2, x_3)\, P(x_6, x_7)\, P(x_5) P(x_5) P(x_7) P(x_7)$$

So what will happen is? Let us say that this is equivalent to say it is something like the probability of $X_1$ given $X_2$, $X_3$, probability of $X_3$ given $X_6$, $X_7$ the probability of blah, blah, blah right up to get another example. Oh, well just okay right, so maybe my system is like this, so what does it mean? So, $X_1$ is dependent only on $X_2$ and $X_3$ given $X_2$ and $X_3$, $X_1$ is independent of all the other variables in the system. Likewise given $X_6$ and $X_7$, $X_3$ is independent of all the other variables in the system, right. Given $X_4$ and X, well-given $X_5$, $X_4$ is independent of everything else.

And $X_5$ is independent of everything else just by itself right it is independent of everything else and $X_6$ depends only on $X_7$ and $X_7$ is independent of everything else. It is just one way of writing it right, whenever I say $X_6$ is dependent on $X_7$ I can always flip it around and say $X_7$ is dependent on $X_6$ I am not talking about causal directions here I am not saying that $X_7$ causes $X_6$ right.
The probability distribution can be factored in the form of $X_6$ given $X_7$ into $X_7$. Otherwise I can also do it as the probability of $X_7$ given $X_6$ the probability of $X_6$. Just keep in mind there is nothing sacrosanct about this way just a convenient way of representing.

Does it make sense? Like I said if you are worried about the classification scenario, you can add conditioning on "g" everywhere. This kind of factoring things is more powerful than just using it for classification. And you can use it for learning about any probability distribution okay does not have to necessarily be about classification you can use it for representing any probability distribution okay.

So one way of I mean this looks a little hard to track right, one way of specifying these kinds of conditional independence relations is to use a graph, so what will I do in this case I will have a graph that has seven nodes. So one node corresponding to each feature, right. What are the features here? More generally the features here are random variable say $X_1$ is a random variable that will take the value in whatever range $X_1$ can take and so on so forth these are all random variables.

So I am going to have it. Is there something that is missing here? So I have connected the graph. So I have $X_1$ okay, so $X_1$ depends on $X_2$ and $X_3$ so I will put arrows. $X_3$ depends on $X_6$ and $X_7$. And $X_2$ depends on $X_4$, $X_4$ depends on $X_5$, and $X_6$ depends on $X_7$. So, this graph structure right gives me the dependency or independence/conditional independence relation I wrote in that expression that right makes sense.

So if you remember, when I was talking to you about the interpretation of conditional independence that I said if you do not know what the class is then the Dhoni and cricket might become dependent. But if you know what the class is the Dhoni and cricket are independent, right. I mean occurrence of the words right I was telling you about that at likewise right if I know

what $X_2$ is right $X_4$ and $X_1$ are independent right is it clear if I know what $X_2$ is then $X_4$ and $X_1$ are independent.

But if you do not know what $X_2$ is then $X_4$ and $X_1$ become dependent, so what do I mean by that if I know something about $X_4$ then I can tell you something about $X_1$. So if that is a little confusing we will try to make this concrete let us say this can take values 0 and 1 and this can take values 0 and 1 not that is not confined to binary things right, Boolean things what makes it easy for me to write.

So let us say that the probability of something like this. So $X_2$ copies $X_4$ with a high probability right and likewise, so that is $X_2$ $X_3$ so I will have to write a table like this for $X_1$ right and yeah. So basically it says that when $X_2$ is zero, probability of $X_1$ being zero is low and the probability of $X_1$ being one is high. Likewise $X_2$ is one the probability of $X_1$ being zero is high the probability of $X_1$ being zero is low that is what I am saying that okay, so now if I know what $X_2$ is right. Let us say that I tell you that $X_2 = 0$.

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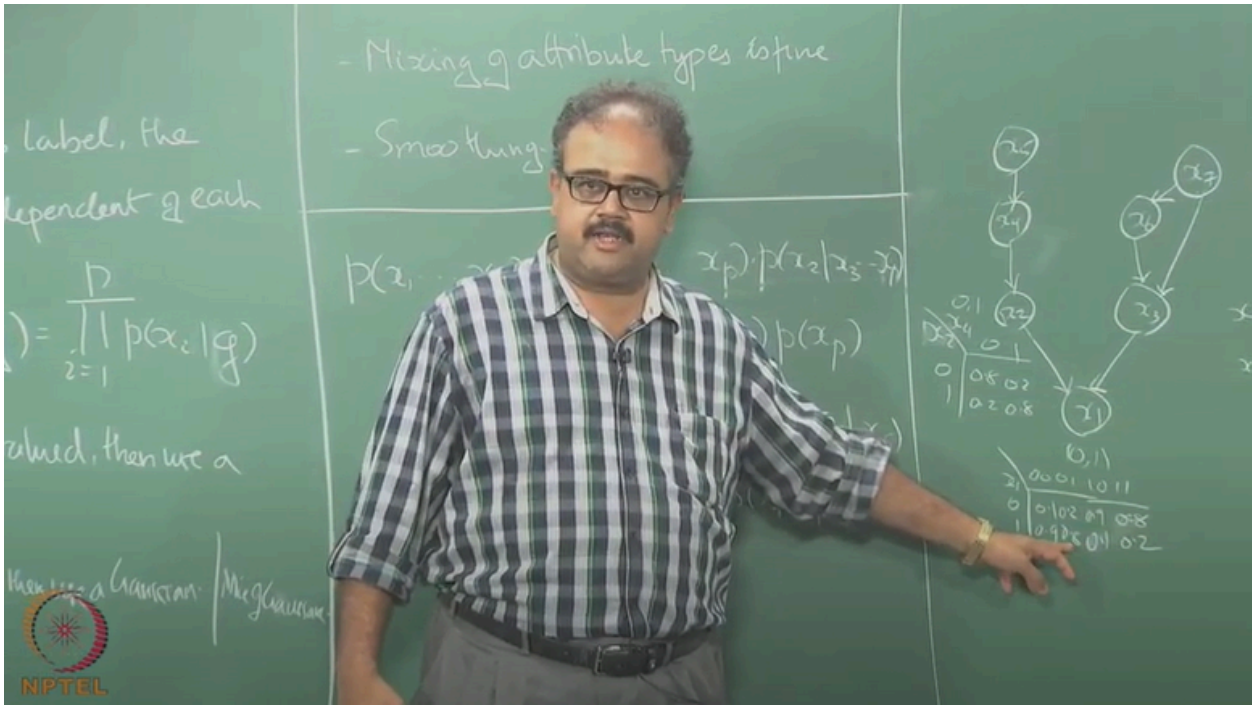


Now the fact that $X_4 = 1$ and if I say $X_2$ is 0 then you know that the probability of $X_1$ being one will be high, right. Regardless of the value of $X_3$ because that is the way I have written this thing down. But if I know the value of $X_3$ also then I will know that okay whether it is 0.9 or 0.8 right. Now if I tell you that $X_4$ is 1, it does not matter because the only way $X_4$ will give me any information about $X_1$ is through $X_2$, but I know $X_2$ is already 0. But suppose I do not know that $X_2$ is 0 I, but I tell you that $X_4$ is 1 right immediately what do you know that the probability of $X_2$ being one is higher right. Therefore, the probability of $X_1$ being zero is higher, right if I had not told you the value of $X_2$.

But if you are told you the value of $X_2$, I say that okay $X_4$ is 1 right. Still, there is a small chance that $X_2$ can be 0 right so $X_2$ can become 0 in which case the conclusions you can draw about $X_1$ completely changes is a very dramatic example, but this is not always so dramatic. Still, the point

I am making is because of the way I have drawn these arrows right if $X_2$ is not known to know $X_4$ will tell me something about $X_1$ if $X_2$ is known the knowing $X_4$ will not tell me anything more about $X_1$.

So everything that I can know about $X_1$ by knowing $X_4$ I already extracted by knowing $X_2$ is it clear so this is this whole idea of conditional independence and why this kind of graphical representation helps us right. So knowing $X_4$, $X_3$ right not $X_2$ I know only $X_3$ but not $X_2$ still does not disconnect me from $X_4$ right because the paths are very different so $X_2$, $X_4$ can still leak influence $X_1$ if I do not know $X_2$ but know $X_3$ is it clear okay, I will come to that right.

So this is the initial setup right, so what these kinds of graphical models do or rather this kind of these are called, Bayesian network right sometimes call sometimes called Belief network and then sometimes called a Bayesian belief network okay. In literature, we will find all the terminology you will find a Bayesian network, Belief networks, and Bayesian Belief Networks. And so in the Bayesian network is a DAG it has to be an acyclic graph and because if it has cycle in it, you are basically messed up, right because the semantics of the thing right so we will talk about a graph representation which does not have any arrows even right which is an undirected graph.

So there are undirected graphs you can start talking about cycles we will come about come to that in the next class. But when you are talking about a directed graph representation right it has to have no cycles because it has cycles then $X_1$ depends on $X_2$ $X_2$ depends on $X_3$ and $X_3$ will, in turn, depend on $X_1$ therefore what will happen? This thing will get completely messed up. So you cannot write out a factorization like that right.

So one way of thinking about it as a set of conditional probability distributions and each node is going to have a set of the conditional probability distribution. $X_1$ is going to have a distribution which gives you $X_1$ given $X_2$, $X_3$ likewise $X_2$ will have a distribution associated with it which will give you the probability of $X_2$ given $X_4$. So if I take the product of all these conditional probability distributions, I will recover the Joint Distribution of those variables okay.

So that is the semantics associated with it right take the product of all this conditional probability distribution I should recover the Joint Distribution of all the variables so if you are going to have cycles then that property will no longer be satisfied. So, we do not want cycles in this case right. And what is this here? So it is a DAG where each node is a random variable, okay and each edge represents a conditional dependence great.

So because of the nature of the graph, we are drawing right, so the graph encodes a lot of separation rules so what we mean by separation rule it tells me that $X_1$ is independent of $X_4$ given $X_2$ right. So I would say that $X_4$ and $X_1$ are separated by $X_2$ right, so likewise can you say something about $X_6$ and $X_1$ separated by $X_3$ what about $X_7$ and $X_1$, $X_3$ right what about $X_6$ the would $X_6$ separate $X_7$ and $X_1$, $X_6$ will not separate $X_7$ what about $X_6$ will it separate $X_7$ and $X_3$ no, right.

If there is a directed path from $X_j$ to $X_i$, any node along the path will separate $X_j$ and $X_i$ provided that is the only path. If there are multiple directed paths it has to appear on all of those directed paths. Else you have to have a set of nodes, these two nodes together will separate $X_j$ and Xi. So you will have to select one representative from each of those directed paths then it will be separated. Because we will have to consider directed edges here this is not called separation it is directed separation or d-separation come on, obvious right, so d-separation.

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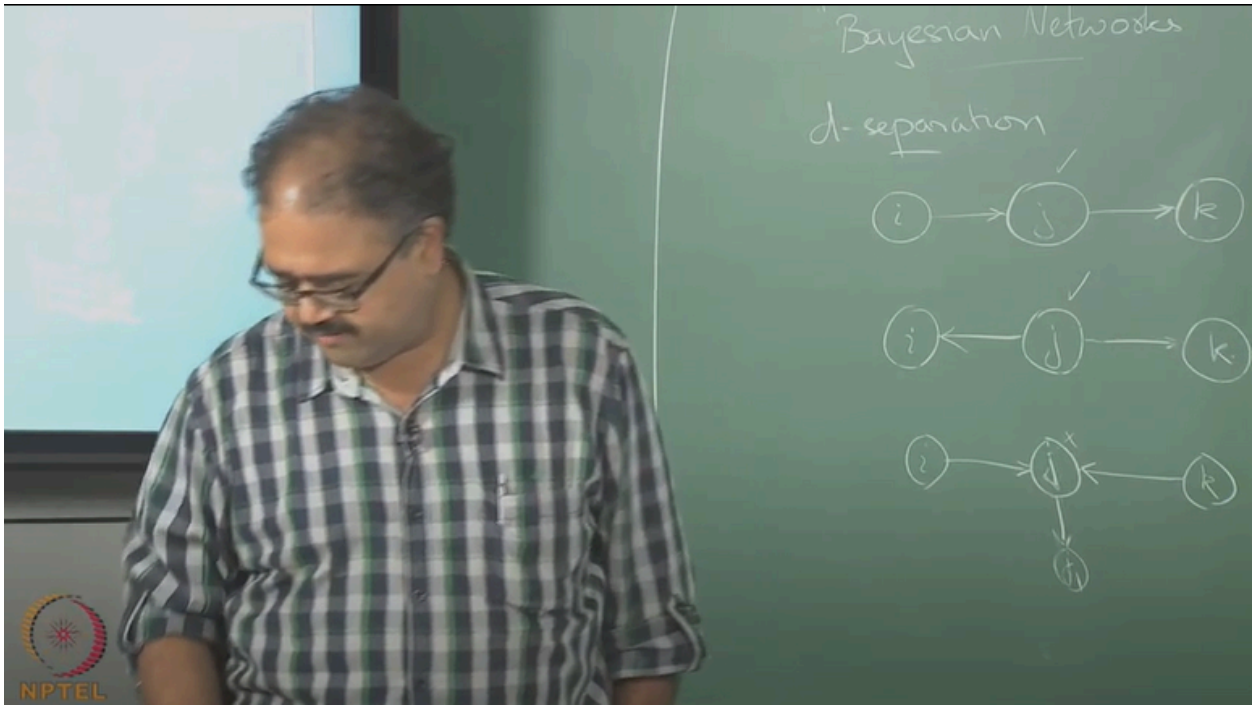


So there are 3 d-separation rules very simple d separation rules okay, j d-separates i and k that is a rule we already saw okay. So, what do you think about this hmm if I know j, i and k are independent if I know j, i and k are independent, right they are separated if I do not know j they are connected, so knowing j okay separates them. Likewise knowing j will it separate i and k here? Yes again think about this right suppose I did not know j right, but I know i let us say something like this right.

So I know that $X_1$ is has a value of one I know $X_1$ has a value of one right and I know that $X_1$ will be 1 right, only when or rather I will know that i is one with a high probability if j is 0 right, so if

I know that i is one then I know that the probability of j being 0 is higher as soon as you know the probability of j being 0 is higher then I will know something about k right because there is a direct influence from j to k.

But if I knew j I know that j is 0, I do not care what i is, i can be anything but knowing i will not tell me anything more about k than I get by knowing j right. So, in this case, knowing j separates i and k, knowing j separates i and k is there anything else that we need to worry about any other combination. Sorry, i,j,k all the way yeah anything else is substantially different convergent say.

So what do you think? Knowing j, actually something else, it does, not knowing j separates i and k knowing J connects i and k think about it knowing J connects i and k because let us go back here right. So I know that Xi is 0 I know that $X_2$ is one, no, I know that $X_2$ is also 0 right I know that $X_i$ is 0 and I know that $X_2$ is 0 then what about $X_3$ both are 0 the probability of $X_3$ being one goes slightly higher right. Because this is the case where both zeros occur with a higher probability. So if I know $X_2$ is 0 I do not know anything about $X_1$ I cannot say anything about $X_3$ does not matter in fact, $X_1$ and $X_3$ are independent, and $X_2$ and $X_3$ are independent if I do not know $X_1$ right. If I know $X_1$ then $X_2$ and$X_3$ become connected let that make sense is right. In both these cases knowing j separates in this case not knowing j separates. It is slightly stronger, I can look at any descendant of $j_1$, knowing any of the descendants of $j_1$ will end up connecting i and k. As soon as you know the value of $j_1$ I can make an inference about j and now that will help connecting i and k.

So these are the 3 d-separation rules so it is great to see the d-separation rules it did not talk about the values of the probabilities. It is just a representational thing so I can plug in whatever values I want all I am saying is just from the structure of the network I can tell you something about the separation properties right. So the actual probability values could come in later the values I use there was only for illustration purposes did not necessarily be that right this is the structure of the network itself tells you that what are the separation properties okay. Any questions on this? This is clear. I can give you a very large graph right and ask you okay Are A and B separated if I know C, D, and E okay, what should you do then? Sorry, you have to find out all the paths directed and undirected between A and B because. So,I have to look all paths

between A and B and figure out the connection. We have C, D and E are the variables that are known and all the other variables are unknown. Then find out whether knowing those variables disconnects the path between A and B,. So you have to apply the third rule for the unknown variables. If all the paths get disconnected between A and B then you say that A and B are d-separated by C, D and E so this is a kind of analysis that you can do to make sure that you understand your system properly right.

So one of the original motivations for proposing these kinds of belief networks it is kind of DAG representation for the variables was to study causality was to figure out causal relationships. These kinds of networks are also called as causal networks. Typically when you talk about causal networks you do not associate conditional probabilities. It is just you that A causes B kind of relations. Don't wirry about the probabilities in that in this setting.

So the same representation can be used for representing causality also let A causes B right so that kind of relationships can be represented using the same representation but in general, when you are using this as a Bayesian network you do not imply any causality, is something which you have to keep in mind when you are using it in practice right. So you are not implying any kind of causality right when you are using this direction this does not mean that I believe that $X_2$ causes $X_1$ right.

When you are using it as a causal Network model, yes, okay when you are putting in an arrow here that means you have thought about it and you believe from the physics of the system or whatever it is that $X_2$ actually causes $X_1$ and it turns out that when you are trying to do this learn this graphical structure by just observing the system looking at the data and trying to infer this kind of graphical structure between the variables the most compact structure that you can derive will turn out to be the one that corresponds to the actual causal nature of the system right.

If you do it incorrectly, you will end up adding a lot more spurious dependencies between variables. That if you are doing it in the correct causal ordering, then you will end up having a much more compact graph than you would if you are doing it at Willy-Nilly way right. What is the use of doing all of this?

So essentially we are interested in answering queries about variables right, so no class, no lecture on graphical models or Bayesian networks is complete without you looking at the earthquake network at least once okay the very, very popular network for historical reasons. So you have a burglar alarm in your house okay, and the alarm rings okay the alarm could ring because of two things right it could ring because there is a burglary it could also ring because as, okay.

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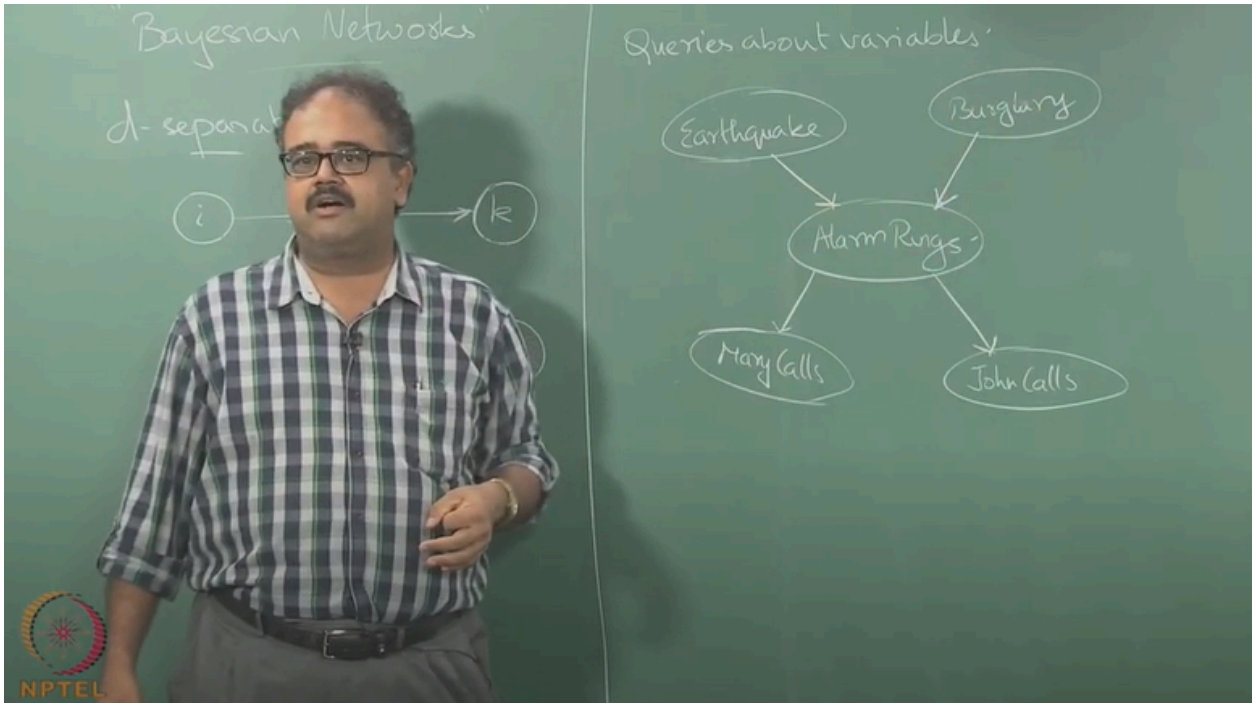


So this network was originally made up by Judea Pearl okay is one of the early pioneers in the study of causal networks and belief networks and so on so forth. Judea Pearl lived in LA in California he was not thinking of wild animals he was thinking of something else remember what I call this network, okay. So probably the most two common occurrences in California earthquake and burglaries, fire alarm know it is a burglar alarm I am not interested in the fire alarm I am interested in burglar alarms okay.

And it turns out that Pearl had two very nice neighbours right who would call him at his office and tell him, hey your burglar alarm rang okay with some probability, so this is so if the alarm

rings Mary or John will call Pearl in his office and tell them that hey your alarm rang. So, you can think about the causal directions here right so the alarm will be caused by the earthquake or burglary right and then Mary will call, and John will call if the alarm rings came both of them might call, or none of them might call because they are all probabilistic things right.

So now I can ask questions like this. Mary called me and said that there is she thought she heard the alarm okay. So, what is the probability that the alarm rang? Both Mary and John called me and said both of them thought they heard an alarm, what is the probability that the alarm rang? I know there was an earthquake in my place, but both Mary and John did not call me what is the probability that alarm rang? They are dead, is it? Good point.

Few Queries to Earthquake Network,

$P(Alarm\ Rang|Mary\ Call)$

$P(Alarm\ Rang|Mary\ Call,\ John\ Call)$

$P(Alarm\ Rang|Earthquake)$

No, but if that is going to happen, I should have had an arrow like this. Since they do not have the arrow I am going to assume that it is not likely that is what I am saying right if the earthquake is going to directly influence John and Mary's behaviour whether they are the question of their mortality or not or other things right, so I would need to put an arrow directly between earthquake and Mary just assume that Pearl and Mary live in different earthquake zones right there is this one small fault line which will only shake Judea Pearl's house and go away go on guys this is illustrative example do not take in too, much too hard.

So the point here is I can ask all kinds of queries on variables on this right I can ask even other things like that hey Mary called, what is the probability that a burglary happened? Right, Mary and John call, what is the probability that a burglary happened? Things change or not, change or not you know everything to answer that question. Yeah, it changes, it will change because if both

Mary and John call then my belief that alarm rang goes up right if it believes the alarm rang goes up then whatever belief I had about burglary happening will also automatically go.

Now we know why these are called belief networks right so when I say Mary call then my belief on whether the alarm rang or not changes right base I have some belief on alarm ringing. Hence, I think okay if nobody calls the law must not run if Mary call cell I will flip this backward right here this will be the only probability of Mary given the alarm, right. So this is the probability I will have here but now given that Mary was one what is the probability of A being one and given that Mary and John or one what is the probability of A being one and given the probability of A being one what is the probability of burglary happening right.

So all of these things I can do all kinds of reasoning about, about the system based on just this whole model that I am learning right, so I can ask all kinds of questions I can ask questions about joint distributions so what so Mary call okay, what is the probability that the alarm rang and that John will call? it kind of redundant probability but still you can try to ask those questions like this right or I know that the alarm rang somehow I know that the alarm rang what's the probability that Mary and John will call me so I can ask all kinds of questions I can also conditional questions I cannot join probability questions they can ask conditional probability questions right.

And if you think of this as classification problem I can ask questions about okay I know these five variables what is the probability that this is class one, you got around our Naïve Bayes problem right I said what if you observe some variables whose values I never see before how will you estimate the probability I can still do that I can just assume that the variable is unobserved. I can estimate the probabilities that will give me give me valid answers that given that I know A, B, C what is the probability that class is one. I might have another D, E, F, G, H, I, J, K, which I have not observed that is okay.

So I can ask all these kinds of questions that given partial data now I can ask questions about classification right, Or given class labels I can ask questions about conditional class densities right the given that it is a document on cricket right how often will I do not know let me not pick

on any cricketers. Given that it is a document about football how often will the word Ronaldo and goal occur together in my document right yeah if you leave it to my son he'll see you say the probability should be 0. But I know this is almost religion right the camps, anyway right.

So that is the whole idea right so these kinds of questions these kinds of queries. I asked about these variables, so we call this problem as the problem of inference on the graphical model is the problem of inference on the graphical model essentially is to figure out all these conditional probabilities or the marginal probabilities that we are interested in the right. So we looked at Naïve Bayes right.

So can you think of drawing the Naive Bayes assumption as a graphical model every node is independent, every node is independent, is it so that it will be like this is that my graphical model. No, what is the Naive Bayes assumption given that class they are independent, so where should the class be? The top or the bottom and of course I can draw it wherever question is the direction of the arrows if you let me draw it at the top.

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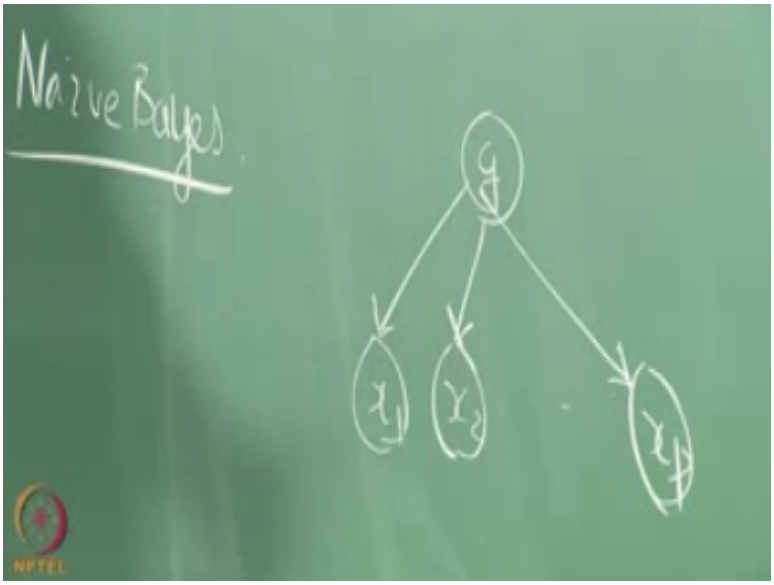


So people tell me how the arrow should go down, right. You will be surprised how many times people draw the arrow up? The reasoning is the variable values are the one that causes the class to happen right. So if $X_1$ is this $X_2$ is this $X_3$ is that then they should all be influencing the class

variable that for the arrow should go up well that is a fairly valid argument it is just that it does not capture the Naive Bayes assumption that is a different kind of assumption that you are modelling there right.

So each variable is somehow affecting the class; this is essentially the opposite of Naive Bayes. Okay instead, is essentially a complete model right, so since all the variables are influencing the class that if in fact if you think about it like given that class what happens in that case? All the variables get connected; it will be this case right. So if all the arrows were going up it will be this case so if j was known all the variables get connected right in this case its opposite of Naive Bayes that given the class all the variables are dependent on one another that is the assumption if you draw the arrows upwards the arrow should go down and up and down its relative rate arrow should go away from the class node.

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**NPTEL ONLINE CERTIFICATION COURSE**

**Introduction to Machine Learning**

**Lecture-86**
**Multi-class Classification**

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**Indian Institute of Technology Madras**

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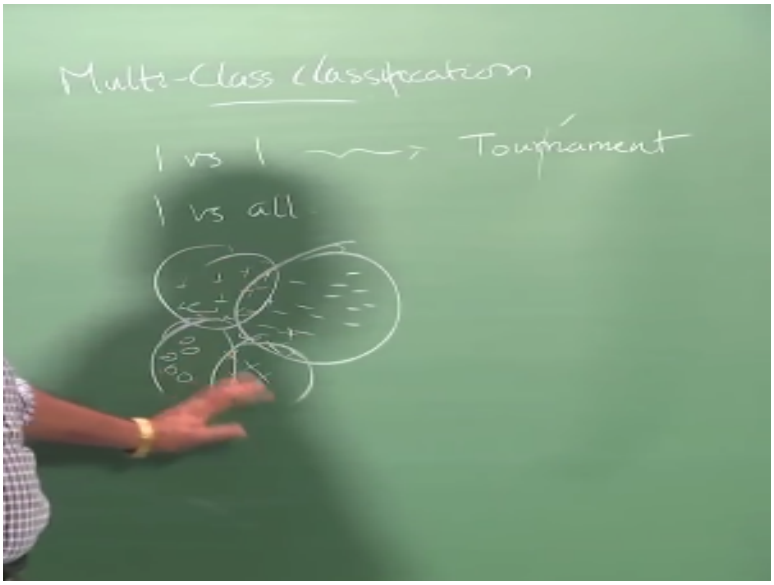


There are some classifiers that we looked at which are naturally multi class classifiers , which are they you know neural networks yeah with a little bit of work here they are multi class something which is more immediately multi-class this decision trees is immediately multi-class no need to worry any fiddling around with anything else Naïve bayes and all the Bayesian classification that we looked at read all those are immediately multi class classifiers and then there are things which we looked at which are inherently two class classifiers .

SVM's is one of those every popular of those and any other two class classifiers that you know logistic regression is inherently a two class classifier but we have multi class variants of it I did the two class classifier in detail in class but there are multi class variants of logistic regression but again the two class one is the one that is best understood and any kind of discriminate function based classification that we looked at are inherently to class classifiers .

You mean you can think of ways of converting them into multi-class classification but inherently to class classifiers so suppose I give you a very powerful mechanism for constructing a binary classifier, can you solve the multi-class classification problem using that let us make it even more concrete I give you an SVM I will give you this packaged code for an SVM I am telling you this is the best possible SVM implementation but it does only binary classification can you use that and convert it into a multi-class classification think how do you do that? what is the advantage of one versus one? potentially be balanced . hopefully I mean depends so you still have an unbalanced class classification problem I might have 30 classes in which each class has 10 data points and one class has 10,000 .

So then one versus one will be a problem then but one versus all will always be a problem even if you have equal number of data points in every class one versus all will be a imbalance problem but the disadvantage of one versus one? how many classifiers you need in one versus one? n choose 2 , ? so that is a large number of classifiers I give you a hand 100 class classification problem so how many classifies will you need within 1 versus 1 large number .

So you are in the time and we actually want to the classified we need not run them all like at the time interval timer feel to the creation which would probably be a one-time process yeah we would need all okay and when you run when you actually want to classify how many do you need how many would be actually if I whenever you guys while you get rid of but one person any person do you how come because you for example if you have class A B C and D huh, I will run the classifier for a versus b a versus c a versus b then you run it for a versus B you through one of them out then suppose it was yeah b versus c and one come over here.

You could do that but then for that you have to be little bit careful because the guarantees that you would have or slightly weaker so this is really not called 1vs1 okay so that is one version of running 1 vs 1which is called a tournament is essentially what you are suggesting is run at tournament . So you train lot of 1vs1 classifiers and then you run a tournament so you need to train that many classifies but we are deploying it you will have fewer numbers to use .

But the problem with that is suppose you have A versus B classifier was weak then you would throw out A incorrectly and suppose we are running A versus B and then B won but if you have done here A versus C and A versus D also A might win against all of those and then B might lose to C and D then it becomes an issue so a would have gotten two votes well be would have got only one vote and then what do you do so classifies are good then tournaments or be great if classifiers are weak and you have problem in tournament that you might eliminate things a little early .

And then another problem in the tournament is you can identify only the most likely class but if I wanted to give ranking of class labels okay that cannot be done with tournaments okay but if you have hundreds of class labels , so you have some you have to give up on something so essentially give up on the correctness and they essentially try to run a tournament if you have a lot of class labels try running a tournament on this .

So the scikit- learn implemented tournament automatically you know in this one versus one yeah that is that is fine what about SVMs? It supports multi class SVMs but there is nothing called multi class SVM here we have to do one of these I will take it back that is a multi classifier we will take it back but there is not work scikit-learn yeah so but this might be something you might want to employ okay now going back.

So I told you that it is possible that we have severe class imbalance in a 100class classification problem even , so you have one class that has like a million data points and each of the other classes have like a thousand data points so what would you do in that case so we spoke about some ways of fixing the problem the class imbalance problem so weighing some one class more

than the other under sampling over sampling did we talk about this at some point I vaguely remember discussing class imbalance in the class, no we discuss class imbalance in the class .

So there are different ways of fixing that you could try that alternatively you could try some kind of a hierarchical classification , so what you do in hierarchical classification is you essentially try to split the classes into two groups okay. And then say that okay first level I will see whether it goes into group one or whether it goes to group to the next level or in within group one I will try to assign it to a specific class or I can split it in to groups three and four and then within that group three I will assign it to a specific class , you could do some kind of error K classification .

So what is the challenge here? So sorry choosing the hierarchies yes using the groups so unless the groups come to you somehow from the domain itself sometimes you could have like people classify web pages and then you can go and look into some open directory project or something like that and there are nicely classified web pages for you . So you have a hierarchy of web pages there you can then look at the classification down the hierarchy.

So you will start off with saying okay entertainment versus news and then within news you could have say politics sports and then within entertainment you will have movies and I do not know and sports comes into the news or entertainment. So wherever so you could I will have this kind of hierarchies and then you can use this hierarchy to give you your hierarchical classification in the absence of that how would you want to do this?

If you want to induce the hierarchy I give you a flat set of 100 labels if you want to induce a hierarchy on this hundred labels how would you do this? Hmm, based on the number of data points, clustering it is clustering so what do you do with clustering how do you do clustering in this case just cluster of the data points blindly that person you are essentially solving we do not know which way to solve it so how will you cluster it here people are throwing up all kinds of terms now .

So the point is you have, so the intuition is the following I have this class conditional densities I know okay this is class one what are the data upon this is class to what are the data points , so I

would like to group them in such a way that the class condition densities belonging to one group are very different from the class conditional densities belonging to the other group does that make sense .

Suppose my data is like this so I have all my class one here class two here class four here and class five data points are here four okay, so which what is what is a grouping that suggest to you itself suggests itself to you one and two should be one thing and three and four should be the other . So if you think about it so this is the class condition if we are assuming these are drawn from Gaussians and I will have a mean here and some variance over this and the mean will be somewhere here and some covariance oh these look closer then the means of these distributions.

So that is a basic idea one way of achieving this is to say that okay I will do clustering and then I look at the class labels that fall together which class labels fall together more often this is very nicely done . So what if my classes are actually like this the classes look like this now it is harder to harder to separate them out , so what you can possibly end up doing something like this and I find some clusters some groups of data points like that.

Now predominantly in this I have class one predominantly in this I have class two predominantly in this I have class three in predominantly this whatever prominently in this class three class four now I start looking at which is which of the clusters are similar and then I can do some kind of predict which values the training data is given to you . So you have the training data you are just being clustering on the training data.

So the training data will tell you what the class labels are , this is there is this really a formal way of doing it I am just giving you tips practical tips forgetting addressing some very large problems, sorry so I have done these clusters now I have clusters and I can I can figure out which clusters are similar to which cluster which clusters are close that I have some description like suppose I am using some kind of a Gaussian model for describing my clusters I will of some description of the clusters .

So I can I can now figure out which cluster is close to which cluster I will talk about hierarchical clustering depending on today or Friday, Friday stands for next class okay. So today or Friday I will talk about the hierarchical clustering then you can see that okay there are four clusters and then these two clusters get merged first and then these two clusters get merged so at that point I can say hey okay now I am going to say all the classes which are more prevalent in these two clusters should go together the classes that are more prevalent in these two clusters should go together.

The data points it belong to those clusters you go and build a classifier first classifier you build on all the data points that separates these two clusters from these two clusters that spacing so I do not want to do a distinction at the very beginning I do not want to do a distinction between this and that this and this and so on so forth . Then how does this help us in class imbalance?

Yeah what if you originally you had class imbalance what if originally 1 class had a million points and all the other classes had a thousand points each, yeah if that is supported if the data supports that million and this is extreme but say ten thousand to one class about the we do get real data like that. I did not get it, we were making clusters on the glass labels so it does not matter what the size of the cluster is so all points belonging to same level would fall into in the same cluster.

How it goes is class imbalance, no it will not cause imbalance I am asking how will it relieve you from class imbalance we are getting is less than has only see I am not I am not using the clustering itself to do the classification I am only doing the clustering turn that I can group the class labels and then I go back and try to solve the classification problem after that yeah. So I suppose all of you are going to try out different things.

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**Introduction to Machine Learning**

**Lecture-60**
**Boosting**

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Okay, so with one of the most popular and most some sense mind-blowing thing with the non-thermal method space right, so boosting the original boosting work original analysis of the boosting work comes from theoretical computer science community not necessarily from an empirical machine learning community right, so, therefore, they were that they looked at having some oracle that had a probability slightly greater than 0.5 of being correct right. Then they try to see how you can better predict somebody who is just above 0.5 okay.

By combining many, many search Oracle's right I can keep improving my accuracy of prediction arbitrary close to 1 right, so that is the amazing part right I start with each predictor as the accuracy of 0.5 plus some epsilon just better than random that I can combine a lot of them and produce something that as accuracy close to 1 right, so this was a very big result that came out earlier and so we are going to look at some kind of simplified version of it, so they remember the goal that distinguishes I mean the main thing that distinguishes boosting from the other methods is that boosting is inherently serial okay.

So boosting is going to build this ensemble classifier in an incremental fashion right, where at each stage I am going to try and explicitly reduce the error produced by the previous stage, so this is something that you have to keep in mind you cannot write it just cannot come up with some ensemble method and call it a boosting method and I have seen that happen in many papers that I have reviewed people just write something that has multiple classifiers in it, so it is boosting, because they have read somewhere that boosting is a very hot area and people papers in boosting get accepted, so they come up with any classifier and an ensemble method and call it boosting, boosting has this very specific property that at every stage right, you add one more

classifier to the existing ensemble right, and this is done in such a fashion as to reduce the error produced by the classifier up till that point okay, makes sense.

Sorry, you get the choice as to what to add next right, so that is that you choose it such a way that you minimize the error that they have not at least you reduce the error okay, so not necessarily minimize. However, you reduce the error of whatever has happened the prediction till that point, okay, does it make sense, so that is essentially what boosting is sometimes you can think of it as error boosting, sometimes they call it error boosting and so on so forth.

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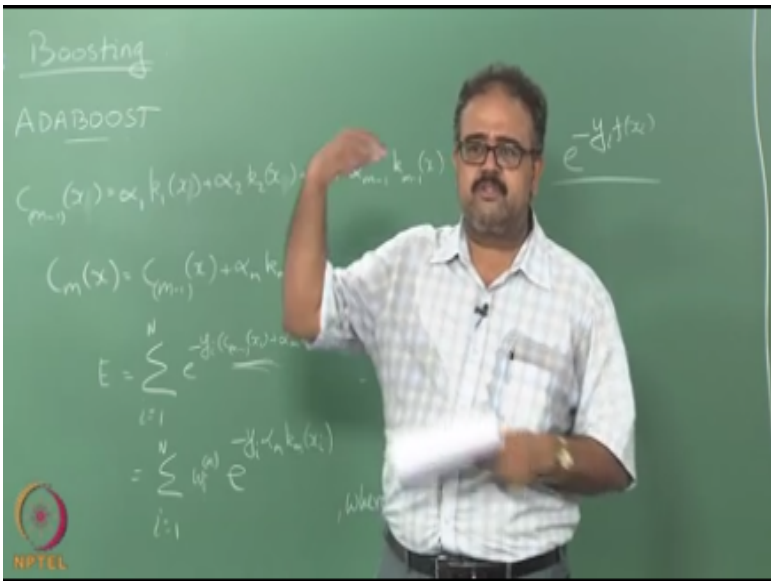


The one very popular and one of the original boosting algorithms is called ADABOOST okay, so let us. It is going to I am going to put up a tutorial for you guys to refer to okay I will use the notation from the tutorial so I will not translate it to the notation in the textbook, okay, so when you read the textbook you have to do the translation yourself, so that is one of the main problems when you have too many different disciplines contributing to the same field right, machine learning has people from computer vision, people from statistics, people from AI and all other disciplines contributing to it and each one of them brings their notation to the mix, right.

So it becomes harder to keep track of everything, but currently, the I output is necessarily complicating my dress, okay, so I am going to denote by C(m-1)(x)th stage classifier okay, that is obtained by basically adding the outputs of all this individual classifies so α1, α2 to αm-1 right, so I am going to add up these are the weights and k1 is a classifier that I added in the first stage

right, k2 is the classifier added in the second stage and so on so forth and km-1 is a classifier added in the m-1 stage okay.

And then basically I want to produce that okay, yeah, the rest of the class, so there are a couple of things which I should point out here one of the most obvious ways of doing this forget about Erebus one of the most obvious ways of doing this is to say that okay I am going to take this guy right, look at the residual error you know I can think of this as a prediction problem right, and look at the residual error of the predictor right, and then train a classifier km to minimize the residual error, right.

So what will be $\alpha_m$? Essentially how to make sure that this whole thing is along the direction of the residual, so we talked about this earlier right, when where did we talk about this, forwards stages we ask stage-wise or stepwise, stage-wise, so when we talk about stage-wise feature selection we talked about something similar right, so you could think of something along the same lines here instead of thinking of selecting features right, I am just selecting classifiers right.

So I can just take the residual error of $c_{m-1}$ and then use that to train $k_m(x)$ and then add it here right, in fact this can be one, it can be one does not matter because the $k_m(x)$ will actually align itself in the direction of the residual so I can just add it here so it is fine right, so that is a simplest way to do this thing and it is actually a good way to do it if you are doing regression let that make sense and I can take this as they can take the residual error and then train my $k_m$ to actually go in the direction of the residual.

So I can actually do this, so you can get a boosting like algorithm for regression just by training it along the direction of a residual right, but when I am doing classification that is not necessarily the right thing to do so people come up with different kinds of loss functions and then they try to improve the classification, so the loss function we look at is the exponential loss, so people remember the exponential loss, I talked about it when we are doing SVM's is exponential loss okay, $e^{-yif(xi)}$.

So we looked at the exponential loss earlier so we will essentially continue with that, so I will sum over all the training points right, so that is the exponential loss for the $m^{th}$ stage classifier right, people agree with me on that, so that is essentially what I wanted to write right, now

expanded the cm. I have written it as this expression in the bracket here. Yet, can people see me at the back I see not me, but I am kind of hard to miss right, that makes sense okay great.

So this thing we already know right, so there is no control we have over that that thing we already know that is given to us all we need to find is $\alpha_m$ and $k_m$ right, so I am going to rewrite this as what is that, that is the last function we are going to be using like that is exponential loss function so for classification we looked at, if you remember we looked at the different loss functions and when we looked at hinge loss right, and I said exponential loss is one of the loss functions and this is how we defined it so essentially I am using that exponential loss function here.

And I mean this was not the way ADABOOST was originally derived okay, ADABOOST was derived in a completely different way and later on about five years after they publish ADABOOST they kind of discovered the connection between this kind of stage ways modeling right, forward are additive stage wise additive modeling and exponential loss they said okay, I can do forward stage wise modeling with an exponential loss function I end up with ADABOOST that connection was discovered five years later.

But now almost always people except in the theory community, in the machine learning community is always introduced like this okay. So where $w_i^m$ is sorry, m the same thing I wrote here, so the weight of the $i^{th}$ data point at the $m^{th}$ stage right, the weight of the $i^{th}$ data point at the $m^{th}$ stage is essentially $e^{-yicm1(xi)}$ right, so what does this mean what is exactly this expression if you think about it, it is a loss I have incurred on that point x(i) up till the m-1 stage right, that is essentially the right just the loss that I have incurred on the ith data point up till the stage m-1 right.

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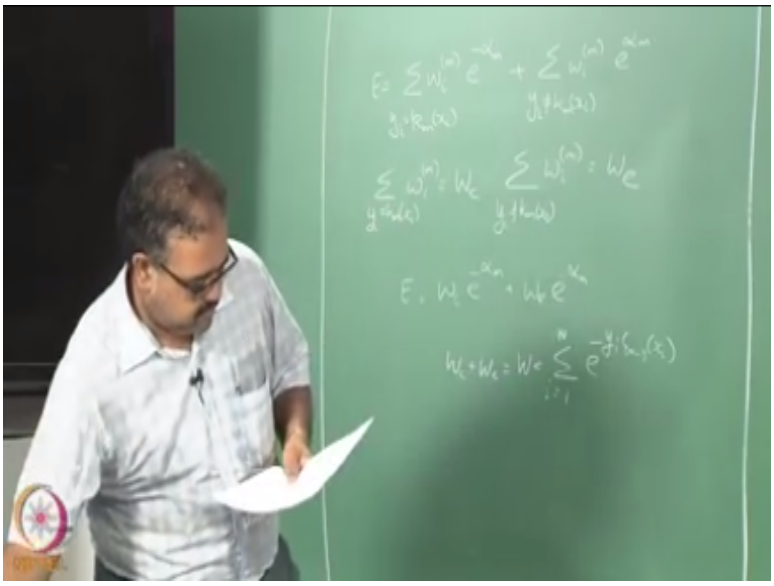

Okay, now I am going to break that sum up into two components, so do you think of these two components, no, no, no this is varies say $e\alpha^m$ so when will I get $e^{-\alpha m}$ when I am correctly classified it, when I will get $e^{\alpha m}$ when I am misclassified it so these are all the data points such that right you are the correctly classified data point is all the miss classified data points right, is intuitively you can see where we are going with this, so what is the best classifier that I can find at the $m^{th}$ stage.

Well, the best classifier can find the one for which this $\sum$ is empty the right, and get everything correct classified correctly, so that is the best classifier. But now increase the cracks right, remember our classifiers are all weak classifiers, and exactly that is a basic assumption we are starting with right the classifiers are all weak classifiers I can do only slightly better than random, so I have to get nearly half the data points incorrect, right.
So which half should go here which half should come here, and then we can move one data point from to here that here to make it better than half, so which half should go here which half should come here intuitively you tell me, which will incur less penalty, what is small half, it is half man what is small half that will be what clear me, can be more clear as to what is small means, no that is a valid way of interpreting small half tell me, $w^m$ 's right, so all the w's that have a large value should come here because they get $e^{-\alpha}$.

The w's ensemble small value should go there because they get multiplied by $e^{\alpha m}$, so what are w's is a small values the ones that I have correctly classified up till the previous point w is the large value are the ones that I have incorrectly classified up to the previous point, so at the $m^{th}$

stage what I should be looking at is try to get the data points which I misclassified from the previous stage, try to get them correctly as many as possible right.

So that is essentially the intuition behind ADABOOST, so at every stage what you do is you try to look at the previous stage see which are the data points you misclassified it tries to get them correctly in this stage, right. It is okay if you make mistakes on data points that you have correctly classified till the previous stage why is it, okay. And because those classifiers can adjust for it okay, then we will look at how we will do this again right.

So I am going to call, so it is all the weights of all the data points I got correct at $m^{th}$ stage right, likewise weight of all the data points I made a mistake on at the $m^{th}$ stage right, so then I can write my e's simply as okay, so if you think about it the value of $\alpha$ really does not matter in my choice of $k_m$ right, regardless of the value of $\alpha$ right, regardless of value of $\alpha_m$ whatever argument I gave you this no holes right, the idea is to see how much of the weight, the weight you can push to Wc right, and how less of the weight you keep in We.

I mean there is total of weight right, there is some total W right, W is a constant okay, Wc + We is a constant, the goal is now to see how much weight you can push into Wc as posted, km will be the classifier that rise as to my WC, so how we do this well you can use you can classifier they images that can assign based data point, we discussed very briefly in session the case right, we can assign ways to data points and you can essentially multiply the error that you make on a data point by the corresponding weight, right.

So the error that you make you multiplied by the corresponding weight so that you can use weighted minimize other ways of doing this, so one way people see what I am saying over km right, you see what you are supposed to do to get your k, the $k_m$ is such that maximum weight goes into Wc they are splitting your W into two parts and depending on what data points are making mistakes on right, the data points you do not make mistakes on contributed Wc the data points you make mistakes on contribute of e.

If you want to see how much larger you can Wc. We basic that is the classifier you have to find, before that you use some kind of a payment method, so one way of achieving this is do the following you are saying weights to all the data points now what you do, if you go and sample some of these data points according to their weights, create a new training set by sampling from

this data points are given things according to the weights, so what does this mean points for which the weight is higher you get to sample more often into this data sets, points for which the weights are very low I do not even appear in the data set, right.

So the points appears multiple times in the data set then when you are trying to minimize the training error you are likely to get a point correct, so instead of using a directly using a weighted training algorithm people simulate that by sampling from the data weights okay, so what has happened unfortunately because of this I change of tends to compact by bagging and boosting in the minds of people and if you look at some of the data mining text books especially some of the earlier data mining textbooks exciting and boosting we needed will described in a very similar fashion right, what do you do in bagging whenever you add a new classified.

So in the older textbooks how they describe is that what you do in bagging is every time you generate a new sample you sample uniformly right, with a replacement right you with sample replacement and boosting the differences every time we generate a new sample you use the prediction error from the previous thing there is the only difference between bagging and boosting right.

But operationally if you think about it, so there is the only difference between bagging and boosting, but then boosting is inherently serial and then there is this error minimization property right, but that never comes across and people just tend to think of boosting as bagging with the different sampling distribution right, whether it is incorrect at the fundamental principles of the two things are very different okay.

So we have found $k_m$ now right, so we all know how to find km you do some kind of weighted error minimization you find $k_m$, so what is next, what is next we need to find $\alpha_m$ right, see regardless of what value of $\alpha_m$ you choose the minimizer is the for km is the one that gives you maximum weight into Wc, correct. But then having chosen a $k_m$ I now have to choose an αm that gives me the error detection, so how do you go about doing that.

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In fact we can do our, so set is equal to 0, so $\alpha_m$ is essentially 1/2 ln 1- the error rate it is error rate is essentially the weight of the data points on which you are making a mistake divided by the total weight right, so this is for the km classifier alone right, We is the data points on which the $m^{th}$ classifier alone makes the error not Cm but km correct, so that is what we divided these things into right, so this is the thing where km makes error, so essentially that so the data points on.

So essentially, it tells you how good the classifier if the classifier is really good right not just on the data points that you are interested in but on the entire data set. If the classifier is very good then the weight will be high that is the classifier has an error of 0 what will happen rate will be infinity because the only classifier you will need right you have a header of 0 on all the data points why do you need other classify just that one is enough right.

But then suppose it has a very high error, error close to 1 where it will be 0 okay, so depending on how good the classifier is this way it will vary okay, and then anything else that you have to do I have found km, I have found αm what do I have to do, I have to change my W's now for the next stage right, so what is my Wi is $e^{-yiCm-1(xi)}$ right, so now it has to become $e^{-yi.cmxi}$ so what is the right best way to do that, this multiply the existing W by $e^{-yi.kmxi}$ right, does it make sense after you have done that you come here okay, I do not erase that part right. So because you need the αm here for your update, so once you find the αm you come back here and change the weights of

all the data points by this amount okay, as it makes sense so that is a plain simple version of ADABOOST okay.

So in fact we can show that the exponential loss function is closely related to the deviance right. An equally popular version of boosting called logic boost exists, where we use the deviance the logistic function right, the log odds function that we used for logistic regression you can use the same error function and then derive all the update rules that we just did for the exponential loss function you can do the same thing for the logit function the log odds function also. You can come up with similar update rules okay.

So the recent ADABOOST is so popular is because it deals such very simple updates right, if you think about it all the computation you do is okay, you find a classifier that minimizes this weighted the error right then you come back and compute this $\alpha_m$ and then you go back and change the weights and then repeat until you are happy with the performance of the total classifier right, and both with basting I mean bagging and boosting the commend both here start basting things anyway if you do both decision trees are very popular classifiers for this, okay.

In bagging it seems to make sense right, why you want to bag decision trees they are notoriously unstable, so if you want to, if you bag decision trees you get more stable estimates, why would you want to bag, why do you want to boost decision trees, are they weak classifiers. Exactly, so what do you do with decision trees you can do the most extreme thing you can just have one node, just have the root node right, one node what can you do with one node decision tree.

Yeah, that is somewhat like linear right yeah, so somewhat likely linear I agree. However, people call it decision trees right, so one node decision tree because of the way I choose which feature I pick right, I will use information gain or Gini index or one of those things okay, I will at least take 50% classification otherwise I would not even split right on that 50% will I will be better than 50%.

I will be better than random even if I split on one node right, so I will split on one node and or maybe if the performance is too weak I can perform I can do a two level tree okay, these are called decision stumps, I do not build a full tree, but it is like chopped off at a very close to the root right, so one not two levels of the trees. However, they are v-classifiers and they take very

little time to estimate and I can do many, many, many of these very quickly essentially what I do is, I boost these decision terms okay. In fact, there is one result in the book if you look at it right.

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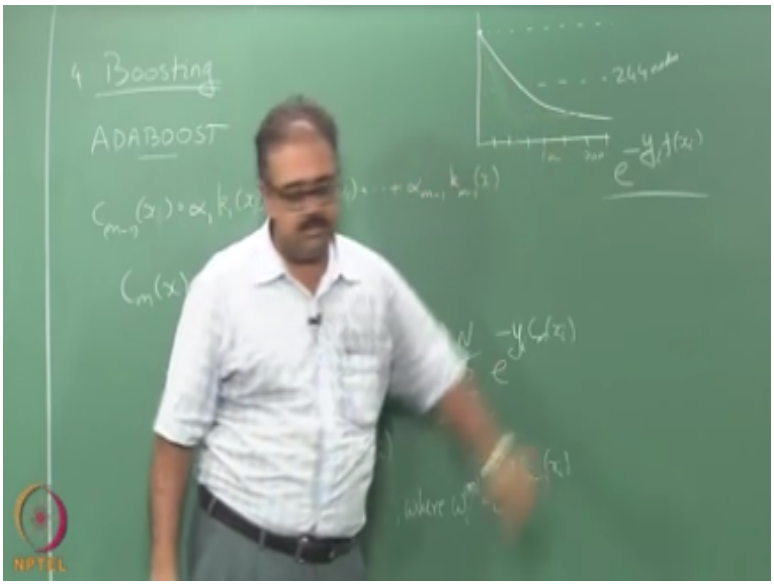


So I do not remember the exact scale on the y-axis. However, the x-axis is the number of levels of boosting that they do right, and so on so forth, the number of levels of boosting that they do so 100, 200, 300 and so on so forth, so a single stump it gives some performance level at that height okay, just one strum the best single stump gives you a performance there. They trained it on the full data, and they get a performance here, and this is like a 244 node tree, 244 notes is a fairly complex thee they built and that is the performance that they get right.

And then they did boosting the start here obviously with a single node right, and then they do boosting, and then they find that the tree the performance just keeps improving as I do ADABOOST. Remember, and these are all single node trees okay, they are all single node trees and so essentially by the when they reach 100that means they have only 100 nodes basically.

They are way better than the 244 nodes that you get with a singletree right, and they reach 244 notes they are like more than twice as good as the single tree they built with 244 notes.

Because the objective function you are minimizing is something very, very different right at every stage, you are changing the function and you are focusing your efforts on actually getting to the harder parts of the space right so that essentially it is little magical. This is more dramatic is it something like this but look at the book crippled for the exact figure right, so this is amazing posting is very powerful. I talked briefly about the random forest in the next class which is you do not even have to do any decision making right, you do things randomly but then do a lot of them right that is also very powerful.

So random forest is not a boosting technique by the way the random forest is a bagging technique right, but then that is also very powerful.

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**Introduction to Machine Learning**

**Lecture-66**
**Undirected Graphical Models - Potential Functions**

**Prof. Balaraman Ravindran**
**Computer Science and Engineering**
**Indian Institute of Technology Madras**

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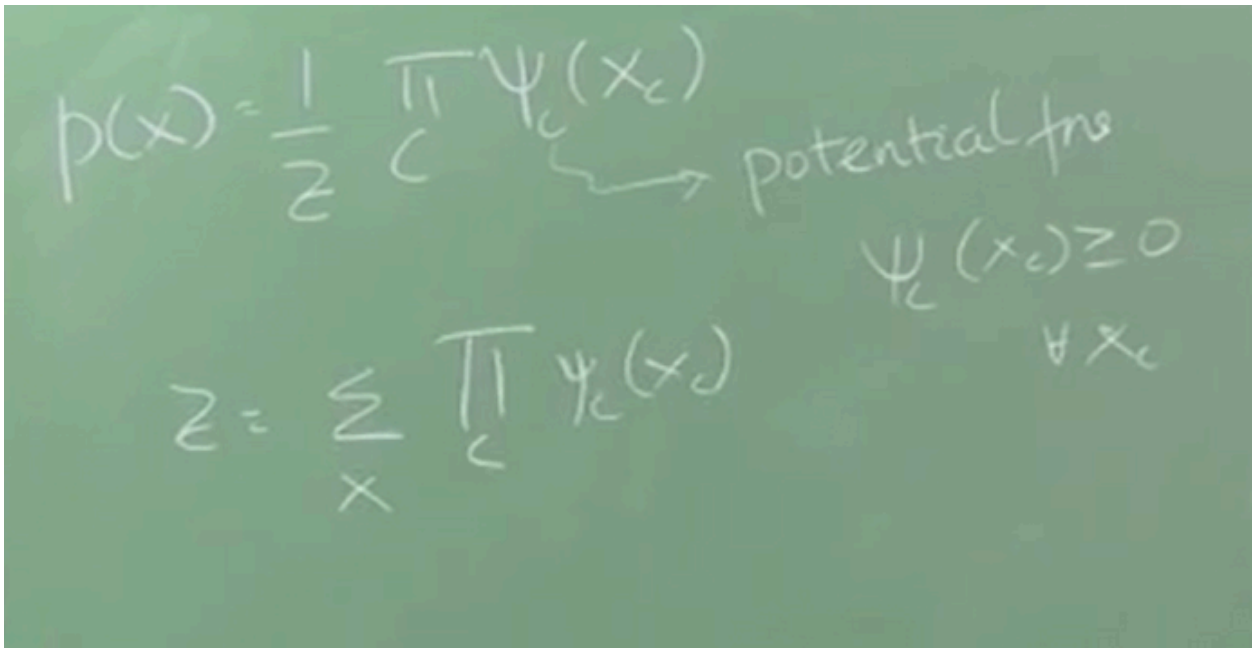


$P(X) = \frac{1}{Z}\pi_c \Psi_c(X_c)$

$\Psi_c(X_c) \geq 0 \qquad \forall X_c$

$\Psi_c$ – Potential Function.

$Z = \sum_X \prod_C \Psi_c(X_c)$

Z- Partition Function (Normalizing Constant)

We have ‘Z’ because as opposed to directed case potential function could be anything and not restricted to probability.

So we sometimes call these. Sometimes call this Ψ as potential functions right, so Ψ(C) is a potential function associated with the clique C right and Xc is the set of variables that are participating in clique C.

Right, so this is a product over all of these cliques okay there is a small problem here right, so we have to make sure that whatever we are writing is a. Probability function right is how do you make sure of that. Right, so you basically and have some kind of a normalization factor where Z will essentially be whether the integral or the sum or whatever if you're looking at.

Looking at discrete values we have been talking about binary so far right so if you are looking at binary value at the variables essentially this will be sum over all values that X can take right. Suppose you have n binary variables the sum will run over 2 power n entries right sounds like a really bad idea right anything where ever you're summing over 2 power n elements seems to be a bad idea.

And it is so the most the biggest difficulty in using undirected graphical models. so why did not you have this in the director graphical models. We chose the factors cleverly, right the factors were chosen to be conditional distribution, so when it took the product it was it is guaranteed to be a distribution but here we have no such restrictions on the Ψ right this is what makes it even more confusing I have no restrictions on the Ψ can be anything right so I can go run up till three million I do not care right Ψ can be any function right Ψ can become negative oops can it. No, right the only condition I have is Ψ has to be non-negative yeah okay that is a good question this Ψ have to be positive or non-negative right.

So it is okay for me to have zero probability for some configuration right. So Ψ can be so Ψ non-negative then that is all the condition I need, so the only condition I need is that so for all values that Xc can take so Ψ has to be non-negative right.

So Z this also sometimes called the. Partition function which is the terminology that comes from physics right, so I am not going to get into the explanation of it but sometimes also not a separation so if you reading up something and somebody mentions partition function okay essentially the Z that they are talking about. So that makes it a little tricky right, so we are saying that your. This thing is not restricted in the interpretation. Right here, so I see is not restricted interpretation it can be anything and as long as you can do this normalization will get a probability okay, so I am going to write down a very powerful theorem.

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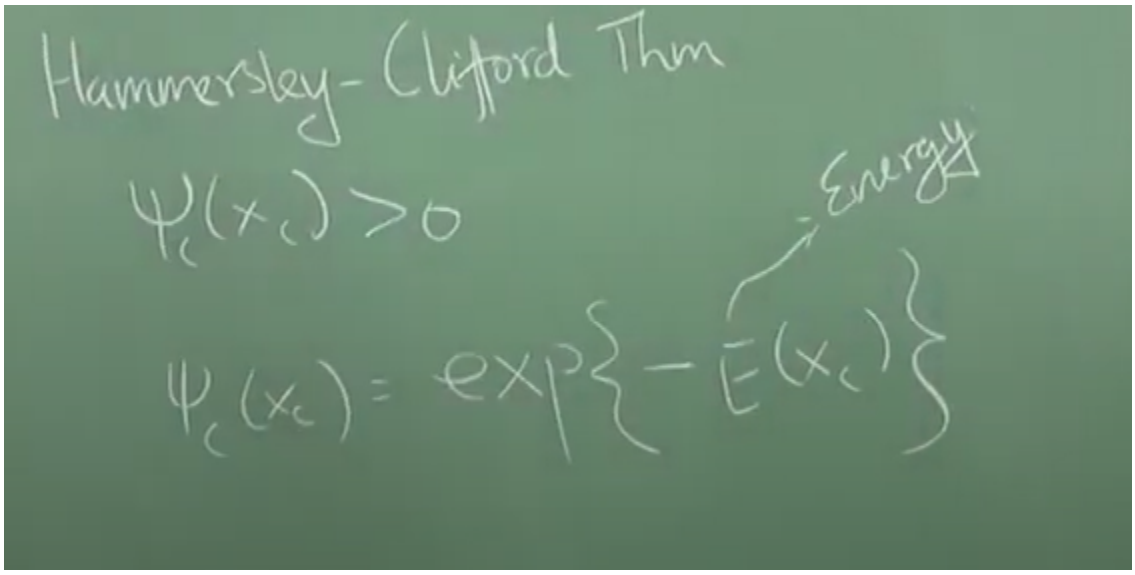


Hammersley-Clifford Theorem

$If\ ,\ \Psi_c(X_c) > 0$

$\Psi_c(X_c) = exp\{-\ E(X_c)\}$

$E\ -\ Energy\ Function$

Right, so the Hamersley Clifford theorem or the Clifford Hamersley theorem it says that for any probability distribution right for any probability distribution that is consistent with this kind of a factorization over a graph right. So any probability distribution that is consistent with this kind of a factorization over graph right the condition here is a little stricter, so the condition says that.

It cannot be 0 right so the condition says it has to be positive not non-negative so what the Hamersley Clifford for theorem says is if a probability distribution okay that is consistent with this kind of a factorization that is means of such a factorization exists okay then that probability distribution can also be expressed by using factors. of this form.

So that makes our life easier, so then now my energy function or what they the E function is called the energy function all of this comes from physics right, so you will see all this energy and other things here energies and potentials you will see that so the energy function right can be anything, now so as no restrictions can be negative it can be positive right well as long as it is real I suppose not complex right, so if the energy function can be anything so this is known as the.

So essentially what the Hamersley Clifford theorem tells us is that so if you if you write your probability as a product of Exponentials right offset these kinds of factors then there exists a graph representation where this kind of a factorization can be obtained in like ways if you are able to write a factorization like this on a graph then you can have it is expressed as a product of Exponentials right, so each factor is an exponential so essentially my probability will be the product of exponentials right, so this is actually a very powerful result because it allows to simplify a whole bunch of computations right I do not have to consider any arbitrary form for my $\Psi$.

Yes immediately you see this right I do not have to consider an arbitrary form for $\Psi$ it is just an exponential okay, now I have to consider an arbitrary form for my energy but now that we started talking about it as energy, so we can start applying our intuitions from physical systems.

Right, so what should be a state with a high with a high probability. Exactly we know that right the state with the high probability should have low energy so when you start looking at the data right I find that configuration Xc which is most popular most prevalent in the data right and I am going to assign that the least energy and I do this for every clique.

The power of nomenclature, so I call it energy and now everybody understands what the graphical model is doing right, So if the energy is very high so it is going to be e power minus the energy so the probability is going to be low or rather the this factor will be low right and this Ψ C is going to multiply herein your numerator right therefore the probability that you assign will be low right if the energy is very low the e power that will be high relatively right and therefore the probability you will assign will be higher.

So that is essentially what we are going to do right, so as far as undirected graphical models is concerned so how do you decide what these energy function should be it depends on your The prevalence of that particular configuration in the data right, so what will be the energy of X1 is 1 and X2 is 0 and X3 is 1 okay how often did the combination occur in the input right and what should I do with that?

Nothing I can just use the count as the energy right because energy is unrestricted do not have to worry about normalizing it or anything right., I just use the count as the energy for that counting the data I can use that as energy, so the higher the count the higher the energy and therefore more prevalent the more probable that configuration will be sorry higher the count okay then one by the count is energy then use one by count as energy right, so the higher the count the smaller the energy and then the more likely the configuration will be right sorry yeah,

So that is the easy enough to do all right  so I wanted to look at before I go on to do some inference thing I want to look at one or actually two popular graph (inaudible) yeah. So this can be zero so that can be at most one yeah. Anyway we are interested only doing probabilities here right,

So that is not that is not that much of a much of a problem right. Then again that is a beauty of the Hamersley Clifford theorem that is essentially it right it tells yeah it is fine you still can represent any probability distribution you want by looking at as this product of exponentials right. So that is the thing now since all of us now understand the undirected graphical models I start with the simple undirected graphical model right, and.

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Right so this kind of a simple lattice like structure right so undirected graphical models are also sometimes called Markov random fields just like directed graphical models are also called Bayesian networks right undirected graphical models are also called Markov random fields right so people if you have heard of the term Markov random field somewhere right so one of the most often new structure in the with Markov random fields is like it is kind of a lattice structure okay.

So what this is really tell you it tells you that this variable okay is independent of everything else in the network given the four neighbors right then this variable is independent of everything else in the network given only these two neighbors I mean these lattices can run for like you know, 32 cross 32 or sometimes 256 cross 256 people typically use these kinds of lattices for modeling images.

Okay, so it is random you agree with the right, so all of these are random variables right so I have this collection of random variables okay it is called Markov because in this particular case right given the immediate neighbors right I am independent of everything else right so if you think of what the Markov assumption is in a probabilistic models right so it is a stochastic model with Markov assumption says that given the immediate predecessor your independent of the past right, so that is a normal Markov assumption right so given the immediate predecessor you are independent of the entire in the past so here what I am saying is instead of the predecessor because there is no notion of direction here so I am since there is no predecessor here, so instead of that I am saying given the immediate neighbors I am independent of everything else that is why it is called Markov right.

So let us take Xi right so Xi is independent of everything else given all right so now what people do is they try to use this for by making all kinds of predictions. Right, so I am interested in labeling every pixel in an image, so I have a big image right I want to label every pixel in an image give me an image labeling task.

I want to label every pixel either foreground or background and this is the guy standing there or is it the tree behind him right so I want to label it as foreground or background right, so now it is a two label task right so each of these random variables will take one of those values what are the values it will take it will take whether it is a foreground or is a background okay.

Now here is the additional assumption I am going to make I am going to make the assumption that the value of the pixel I am going to see. The value of the pixel I am going to see depends on whether it is a foreground pixel or a background pixel right nothing else.

The value of the pixel I am going to see it depends on whether is a foreground pixel or a background pixel and nothing else right now essentially what I will do is. I am going to assign more random variables right. Each one of them stands for the individual pixel.

Each one of the stands for an individual pixel so now what will I do is I will observe these pixels right I initially will observe the pixels, so what will be my potentials here what are the Ψ how many how many Ψ do I need one for each edge right. So that is the maximal clique here I cannot do anything better than that so for every edge in this graph I will need a Ψ but for every edge in the graph I will need a Ψ right.
So what I will do is I will observe these pixel values okay, so some something some values I will observe okay then what I can do is I can figure out what should be this level of label of this pixel right just with this knowledge alone right because of there is this potential right I can kind of convert that into a potential on the node alone.

You see that because I have observed the values for the pixels right I can essentially take that entry from that thing right, so there will be one column associated with that pixel value right so

People giving me blank stares we are talking about a function of two variables right so I call this Xi will call this Yi the pixel value Xi Yi so I will tell you what Yi is right then what will have we left with. A function on Xi right if I tell you what Yi is I will be left with a function on Xi correct so I can convert given an observation right I can convert these potentials right into potentials on Xi alone.

Yi is a part of the graph but the way use it will always be that I am given the Yi so here is an image give me the labels which is the foreground which is the background, so I will always know this Yi right so given the Yi I can convert this edge potentials into, node potentials.

Right, so this essentially from now I from the function of Xi, Yi it will become a function of Xi alone okay so if you look at many such I mean graphical model applications right, so you will actually find that they will always reduce this into node potentials and edge potentials.

So it looked like they are defining something called a node potential okay which is like a potential function on single variables and then they will be defining edge potentials which are potential functions on pairs of variables okay in reality something like this would be happening for you to assign node potentials okay the node potential are essentially marginal some kind of

information you have about the marginal's okay, so in this case I am telling you how the marginal information comes.

It is not complete it is not the complete marginal okay given the pixel you can make some guess of work what the Xi should be so that is my node potential so I already can reduce all of these edge potentials between a node and a pixel okay between a label and a pixel I can reduce it to a single potential on the label right. So now having done that right what can I do so that is there will be some potential for a label here okay there will be some potential for label here.

Right and there will be some potential for these two labels happening together. Right, so essentially this is telling okay this is a background okay what is the pro likelihood this is also a background if this is a background what is it likelihood this is also a background okay if this is the background what is the likelihood that is a foreground.

So like that right so for each of these edges so I have where the edge can change where the label can change that information I have, so finally when I assign the final labels to this so what will I do I will find that configuration of labels okay that gives me the lowest energy. And essentially that would mean that this potential should be. Low right if I suppose this I say is background and this I say is foreground then when I say label is B here and label is F here that entry in the potential function should be low.

Right people get that, so that entry in the potential function should be small so like that I need to do this for all the pairs here so it is a very hard problem so because it severely constrains so I have consider all possible pairs and I have to figure out where we low entry occurs across this pairs, so for example so if I say this is background and this is foreground and this is B and F and this gives me a low value right but I say B and B for this and this gives me a high value right. I you can possibly turn this into an F right but then I the whole thing might come around and then this might get changed back into an F B right so then I might go around or not have to figure out what I see right potential to pitch this and that is the inference problem, so inference problem is really hard in the undirected case right so in so much so that when you have loops like this right when you have loops in the graph like this exact inference is impossible to actually give you the

right answer is impossible like quite often we end up giving some kind of an approximation right. So where can you give exact answers when there are no loops right so undirected with no loops is a.

Tree right so on trees you can give exact computation but as soon as you have any loops in that then you have to do some kind of approximation there are some very special cases but we will not go that right you just want you to get an intuition of what the individual factors would mean right. So how will I determine what these factors are okay so let us look at this right I want to look at. I want to look at the that right so what am I going to do.

What is it really it is? Oh I want to assign a low energy to the configuration that have occurs most often in my data right so what I will do is look at the label data so I look at each pixel's label right the label data will tell me which pixel is foreground which pixel is background right then I look at for these two pixels in the image let us say I have only a three by three pixel image right but for each of these two pixels I look at them right, I will figure out okay how often was this foreground and this also foreground how often was this foreground and this background how often was this background and this foreground how often this background and this background, so all these four things I look at just count that from the data and then I will set the energy to be some inverse of that count so the one with the largest count will get the smallest energy right it pretty simple right, so like that I can do this for each and everything right. So that is just one thing so the second thing I have to do is then start looking at this thing.

Exactly that is what I think that is what is particle sorry you are looking at my back you can see from this I said there should be a potential function for Xi and Yi as well right so what that what I will do I look at the data again so look at okay when the pixel value was this okay what was the label right so I will do that I will do the co-occurrence information okay now things should start looking a little fishy to you guys I mean the pixel can have a lot of values depending on how I am encoding my color or brightness or something like that right.

So that I could end up having a very large distribution there itself right so even if you assume that my pixel is going to have 256 levels of brightness right so for every value so it is to 512 probability 512 counts that I have to make right.

512 counts looks suspicious right will I have data to actually make accurate estimates of 512 individual counts well I could if I have very large volumes of data but typically what you do is you don't for Yi you do not do the explicit counts like this right you try to learn the factor Yi by some kind of a parameterized function right.

So you could use logistic regression right so figuring out what the what is the probability of Xi given Yi right that logistically regression can tell you right, so you encode your pixel using whatever things you want right so you can look at you can look and even more funny thing so you can do funky things you can make this a function of all of these pixels right so because you are moving away from your Markovness, but once you start thinking of doing a logistic regression you can come up with very powerful classifiers right.

So typically the Yi, Xi probabilities right or other the Yi Xi factors are learnt in a different manner they just do not do the maximum likelihood estimate you do some other thing that typically the most popular choices using logistic regression right you can do other things but then the Xi Xi potentials you can learnt using maximum likelihood estimate simple maximum likelihood provided they are small enough okay so that is basically how you train this Markov random field and it turns out that they are pretty powerful in terms of working with images and in a very wide variety of setting right and people use MRFs a lot right. And there are variants of it which is called conditional random fields so people use those also tremendously so.

Very popular and powerful classifier and training it can be a pain right, so I just give you a simple example right value just did the counting and stuff like that when you have a very large model right very large graph right training it can be a pain but then people because the data sparsity right, so I had to look at all possible combinations of all variables right and so it becomes a little tricky and so you have to come up with the cleaver ways of training the models.

Okay they say it is particularly this part of it is painful right not Ψ not completely (inaudible) just that the inference processes is hard like you are selling your head you might you may keep going in circles it may take a long time for you to actually converge to a probability and so on and so forth yeah.

So there exists a proper assignment to this so that yeah so but the finding it is hard yeah Clifford Hamersley let tells me that no I am not worried about loops right as soon as there is a clique there is a loop.

Clique loose but there is only one potential right, so it is fine I will be doing the inference on that potential alone I would not get into this loop business so the loop business started off saying that okay I will make one inference one assignment here one assignment here one assignment but if this had actually been a clique right then I will know what is the combination of assignments to these four variables that has the lowest potential, I would have just done that right so I would not have to runaround chasing the things so that is the right so.

But I still have to do the chasing run because this has other things it is involved in right so the only way I can ensure that I do not do the chasing around this if I have one connected graph a complete graph right one if a complete graph then of course there is no chasing around and then what so what is the difficulty.

There is no factorization I am back in the system as well not have worried about the graph right, so if I have a complete graph there is no factorization. So but that is only way I can ensure that I will not get into.

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**Introduction to Machine Learning**

**Lecture-67**
**Hidden Markov Models**

**Prof: Balaraman Ravindran**
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**Indian Institute of Technology Madras**

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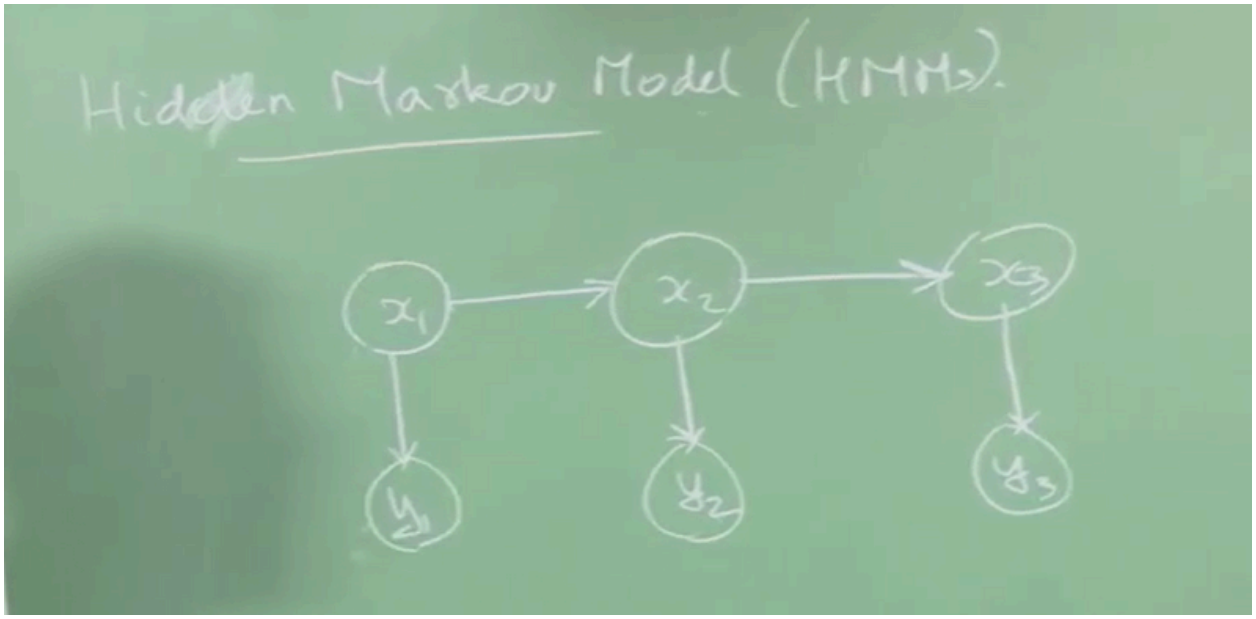


Right so people might have come across HMMs in other contexts right, and you might be wondering what does he mean by saying it is a graphical model, but it is a graphical model right so here I will stick with a concrete example, I am going to have a sequence.

Right, so I am going to have a sequence of random variables right I am going to assume that it is Markov in the traditional sense, so what would that mean.

It is left to right, and hence that is the earliest point in time. Okay, so there is no time it is a sequence right. This is the first element in the sequence okay the second element in the sequence the third element. Because it is Markov so this is this will be dependent only on the previous one. So knowing this value makes this independent of anything that came before right, so this is essentially the graphical model version of Markov right just have a chain directed graph, so it is sometimes called a left-to-right model. So this is the Markov part right. I am also going to assume that. I have a set of observations that I make.

Right, this could be like the pixels we are talking about right, so I have labels for the images right labels for each pixel on the actual pixel value right. So likewise in the hidden Markov model so I will have random variables which could be labeled right, so these are labels. Right and these are the entities that are being labelled. So give an example of such a situation they would find.

Media is a little tricky because you also have the spatial dimension to it right, so the text is why I asked how many of you are in NLP other day. So text right I might want to do say part-of-speech tagging. And I want to take each word again I want to assign a part of speech to the world right all I want to assign each word I want to take each word. I want to say that whether the word is part of a phrase or not part of a phrase. For example, I would need to figure out the United States of America is a phrase .

Is that so is that some way of doing this automatically right, so people have come up with ways of doing there are many things there are tasks in the NLP called chunking, so people have come across chunking, no, okay? So chunking essentially says that I am going to take a piece of text right and break it up into some meaningful chunks it could be noun phrases verb phrases whatever right, so but some chunks right, so that is called chunking right. There is another task called shallow parsing right so shallow parsing essentially breaks it up into phrases right but does not look at the structure of the phrases I just want to give the phrasal structure of the sentence. Hence, there are many tasks which people do this right, so people use hidden Markov models a lot in text right and the other place where people use hidden Markov models a lot is in speech right because speech is inherently a. Speech is inherently a forward process, so people use that a lot in speech. You can use it in videos except that you are individual random variable will now

become something that covers the entire spatial dimensions. This becomes a little more complex hidden Markov model right.

So great, so I am going to call these after that as X's. Well, these are words right. This could be words, and these could be whatever label you want to send to them you can say that okay so is it part of a chunk or not part of the chunk what is it this is part of a name of not part of a name so I could do all kinds of labels.

That is the assumption we are making to make the model easy. Okay, so yeah we can relax this assumption there are models that the more complex model that does this specifically the conditional random field that I mentioned when I was talking about Markov random process. It relaxes the assumption of this dependence.

Right, so this Markov assumption is still there right. Still, this dependence is relaxed, so that is typically what you are looking for I can look at any word in the sentence and then I can label the seventh word so that is that assumption is this is like right now I have to look at only the seventh word and the sixth label I cannot look at the sixth word I had to look at the sixth label and the seventh word right, and that is all the information I have.

So one thing to note here is a hidden Markov model essentially says that your Xi give rise to the Yi right does not go the other way. This gives also gives ways to other problems later on. So the X's are essentially you have labels, so you do not know the labels you don't see the labels but whatever you do not see right that part of it is Markov okay.

And whatever you see is or the labels okay and they are just given influence only by the I am sorry. Whatever you see are the words then they are influenced only by the labels that are the assumption you are making so very strong independence assumptions right. Still, it turns out that it works like Naive Bayes works right so this also works in many situations right, but then, of course, it also does not work in a lot of situations, so people have come up with other models I just wanted to introduce you to HMMs.

I do not get to see X1, X2, X3 and I only get to see Y1, Y2, Y3 and X1, X2, X3 I will have to guess. I mean you can also think of this as a hidden Markov Random field if you want, but they do not use the terminology. Still, hidden Markov models are something that is used quite often, and there are inference techniques which people have specially honed for HMMs right. Still, it turns out that the same things work on many of the graphical models that you will see you will see in practice right, so they have all kinds of things they have an algorithm called the Viterbi algorithm which essentially tells you what the probability of the labels given the text, no it does not give you the probability of the labels given the text it gives you the. MAP estimate and it gives you the most probable label sequence given the text.

But then I can use it on any kind of this kind of an inference process right, so I can use a Viterbi algorithm for any kind of a MAP inference process is instead of looking at the probability distribution over X. If I want to answer the query which is the most probable configuration over X this is the MAP estimated we looked at that right maximum likelihood MAP and then the full Bayesian inference which cases me the full distribution right so far we have been talking about knowing the full distribution. Still, you also want to do the MAP query right. So that is essentially what Viterbi will give you right, so we are talking about estimating a potential between Xi and Yi right, so we are talking about estimating a potential between Xi and Yi. Hence, the question is $\Psi$ (Xi, Yi) independent of i with regardless of where in the pixel the label occurs right sorry we're in the picture the label occurs right is the relationship between the pixel the label the same right.

When I say that the $\Psi$ is independent of I that is what it means right that need not be the case right suppose on the edge if a particular pixel value occurs it might have a higher probability of being a background right I mean it is very edge, you typically not going to frame a picture so that the foreground goes to the very edge rate. Still, then that might happen, but the problem is supposed I have a 256 x 256 image then I have to learn a classifier for every one of those positions or to learn a classifier that gives me the probability of Xi given Yi for every one of those positions.

And that is a very large problem right; it is a very hard problem. Hence, what we typically assume that Ψ is an independent of Ψ (Xi, Yi) is independent of i so I am going to estimate the same model across the entire image right so of course, you can try to be more clever I mean if you know some things about the about the task we can be trying to be more clever and say that okay I am going to break it up into four classes of Ψ okay for Xi and Yi.

I will break it into four classes I will use one class here rather than the bulk of the image and so on so forth you can do things like that but yeah but typically without any prior knowledge you just assume that it is the same all along all right. This has to be different right, and if that's also the same all through then, you are we can use one pixel and predict the whole image the label for the whole image Yeah, do we end up doing that you could do that also yeah Ψ (Xi) Ψ (Xi +1) right whatever that could also be taken as being independent of i.

You lose more modelling freedom that way, but you could do it that way as well yeah depends on how much effort you want to spend in doing the building the model yeah.

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**Introduction to Machine Learning**

**Lecture-68**
**Variable Elimination**

**Prof. Balaraman Ravindran**
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**Indian Institute of Technology Madras**

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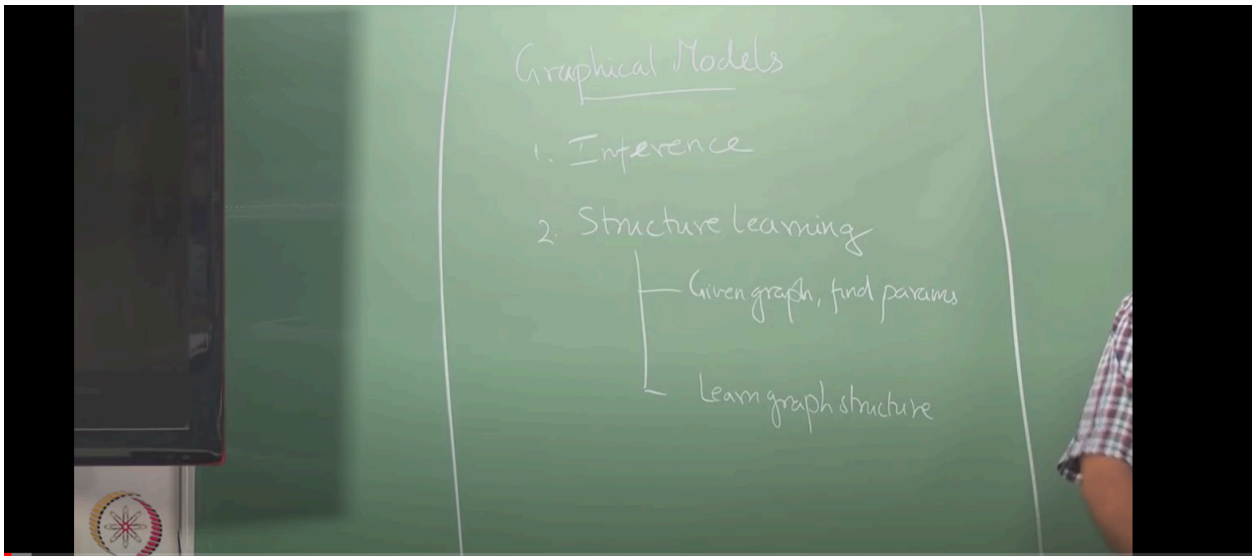


Right, so we are looking at graphical models we looked at both directed and undirected models right. And I said the thing of interest was, so two things are of interest. The first thing is.

Given a model right, how do you do inference using the model right? So what is the inference question, inference question is trying to answer queries on marginal's right. So I give you a very complex joint probability distribution, I want to know what is a probability that there is an earthquake, yeah that is not a very complex system.

So what is the probability that there is an earthquake right, I can also ask for conditional marginal's given that John called what the probability that is a nice case is. So these are things we looked at right, so it turns out that this itself is a hard problem and for large graphs, you will have to come up with ways of approximating given this right. So I will kind of motivate why there is a hard problem in a minute.

And the second problem that we are interested in is, What was the first problem, sorry, exactly. So what could be the second problem? No, find the model then, how do you derive the model right, but you are close right, so how do you find the model right, from the raw data may I will give you training data, I will give you a lot of data how do you find the model right. So the Bayesian network structure learning itself is a hard thing.

So the simple problem is even in the structure learning they split it into two things right. So I should probably put this down, the first problem is inference.

Right, so here there are two components with, so given the graph right. So I will give you the graph find the parameters and in the directed case that would be finding the conditional probability distribution. So once I give you the graph I know what the conditional probability distributions I need are, I can just go to the data count and find it out.

And in the undirected case what it would find the potentials, so given the graph find the potentials right. As soon as I give you the graph, you know what the potentials that you need to estimate right are. So you will have all these edge potentials, you will have node potentials, and you have clique potentials, so you will know what the potential that you are estimating right is, and you just go and estimate the potentials right. So this is essentially the learning problem given.

And the second problem would be to find a graph right. So one of the things you should look at finding in trying to find a graph essentially you would need to find that graph structure right, that supports them in conditional independence that is present in the data, it is directed graphs or undirected graphs whatever graph structure you are learning. So you have to infer what is the conditional independence that is present in the data, and you have to find a graph that will

support that. So essentially you will have to, there are many ways of doing it people start off with a completely connected graph, and then they start knocking off edges right. And then you can do some kind of cost complexity pruning as you do in decision trees right. So you could have a much more complex graph that then you can try to prune things down so that you can do a tradeoff between the number of edges you have. So the variety of algorithms that people are proposed for graph structure learning. So this part is the easy right given graph find parameters are easy, how will you do that just like conditional probability distribution estimates right. So you can very easily do that for directed graphs just counting.

Look at the data, see how many times Mary called when there was an earthquake right, or when the alarm rang how many times Mary called and then you can essentially fill in this conditional probability right. So those things we can do a straight forward right. But learning the graph structure is a little bit involved to get into that because it is a lot of, you know a lot of structure that we have to build in before you can. So I am going to now go back to inference, so the inference is the interesting part right.

So let me start off with an example right. So I am taking this example from. So for a long time, we did not have a really good book on graphical models, and then Koller and Friedman wrote this over complete book on graphical models I mean it is like it has everything that you would need to know about graphical models and more right.

So it is like this huge stone right, but it is a fantastic book, it really is a good place to start right. So why I am saying it is a good place to start this, this is still a very active area of research right, probabilistic graphical models and every year newer techniques, newer breakthroughs keep coming up. So it is like, it is not like you can write a book and say okay everything you need about graphical models is captured in the book right so because it is still evolving field. Right, I am going to draw a really large graphic here.

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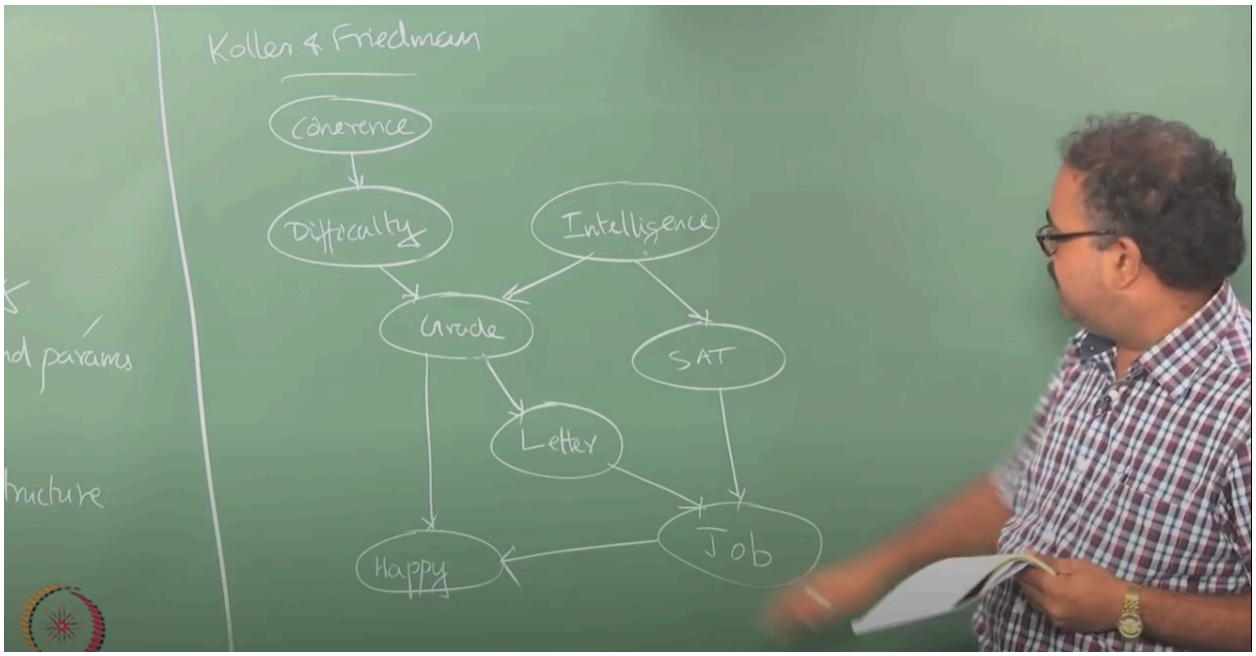


Okay, it is a small thing which Daphne Koller came up with to capture some fraction of her interaction with students right. So depending on the difficulty level and the intelligence of the student okay, the student will get some grade in the course right. And the difficulty level of the course depends on how coherent the teacher is right.

So the coherence influences the difficulty level, okay, and then the difficulty level intelligence influence the grade right. And so depending on whether the student got a good grade or not in the course, the teacher might give him or her a letter right letter of recommendation if the grade is bad, then the probability of getting a letter is very small, as the grade is good the probability of getting a letter is very high right, even there that happens.

And whether they get a letter of recommendation from the teacher or not right, it influences whether you get a job, and whether you get a job and whatever grade you did influences whether you are happy or not right, this is like, so sometimes you might be very happy for having done very well in the course even if you even though you do not find a job.

I am just giving you the structure here because this is sufficient for us to talk about some of the difficulties in the inference process right. When you are actually solving problems in this, you would need the probabilities, but we are not going there right. So they just give you the structure. Suppose I am interested in answering a.

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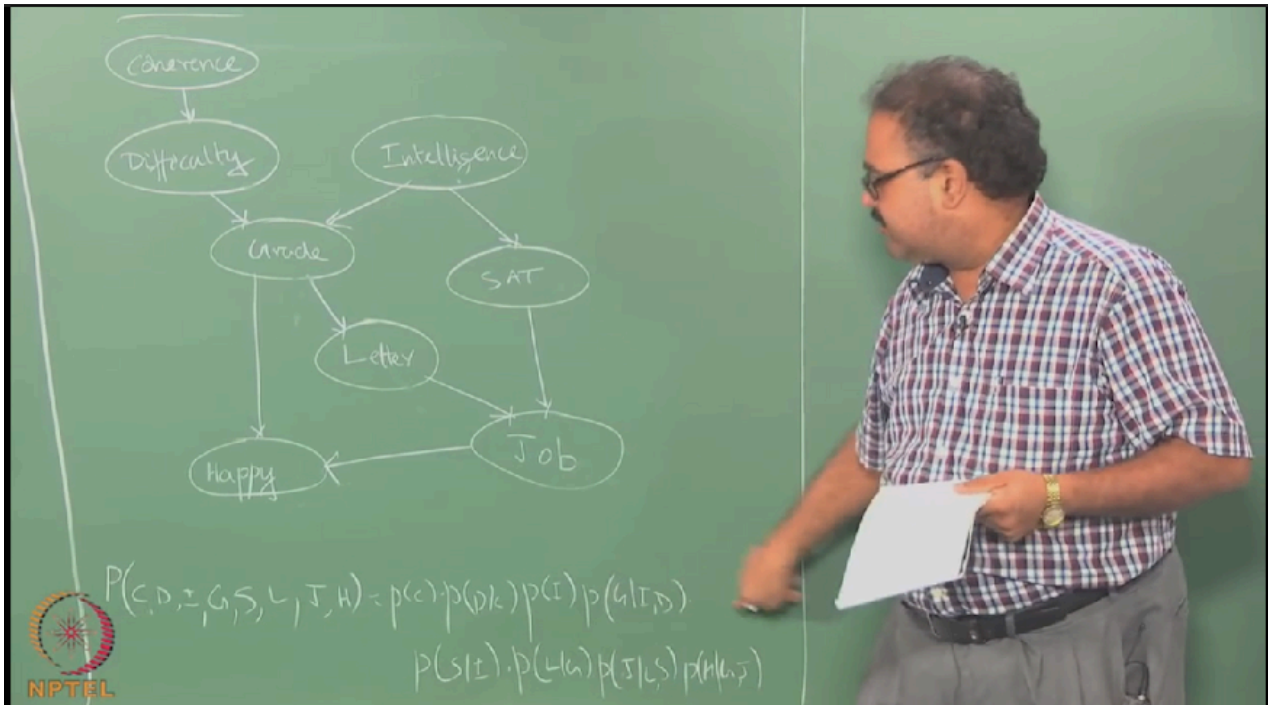
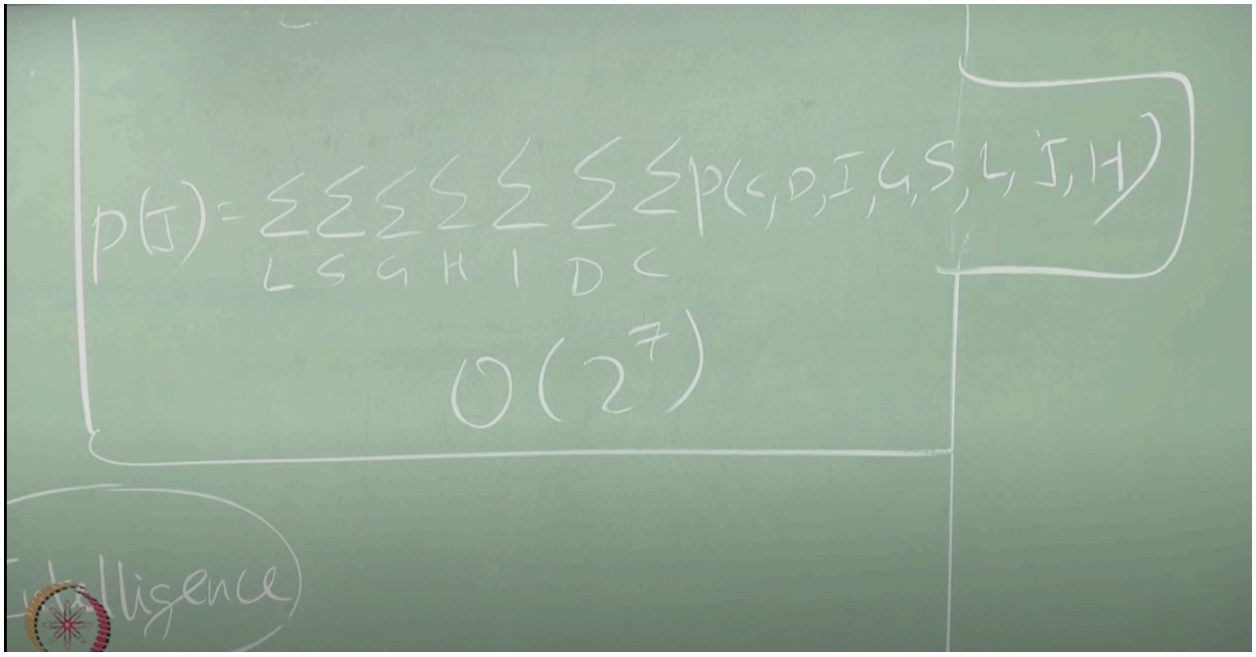



Let me write this out now, probability C, D, I, G, S, N okay.

So you people see what I have written if you cannot, you can write it from the graph directly, so you do not really need me to write this out. So right, so this is a probability of coherence times, the probability of difficulty given coherence times, the probability of intelligence than the probability of great given intelligence and difficulty, so on so forth, I have just written out the joint distribution you can just look at the graph, and you can write out that yourself easily right.

The Joint Probability distribution of the variables can be factorized based on the above graph,
$P(C,D,I,G,S,N) = P(C)P(D|C)P(I)P(G|I,D)P(S|I)P(L|G)P(J|L,S)P(H|G,J)$




What is the probability that a student in this universe will get a job; in this universe, I mean the universe is captured by this way. So what is the probability that person will get a job right, so what will you, how will you go about doing this essentially this will be. Okay, right, so if you think about it is essentially the order of $2^7$ computation if everything is Boolean right.

$$P(J) = \sum_{L\ S\ G\ H\ I\ D\ C} P(C, D, I, G, S, L, J, H)$$

Assuming everything Boolean, we will have $O(2^7)$ computations.

So it looks odd right I mean so it means running this over the entire table running the summation over the entire table is not correct. So the whole idea of us making inference was doing this factorization was to make this computation simpler right. If I did not have the factorization right, I essentially would have had to do this computation. So yeah, so this is some set of running over this very large table.

So now what we are going to try and do is try to make the summation simpler by pushing in some of the sums right, pushing it into the maximum extent possible so that what I sum over okay, as a smaller table as possible right. Right now and all my seven sums are running over the

entire joint distribution right, I want to rearrange this in such a fashion that each sum runs over as smaller setup as possible right. So how will I do that? Just for the same question here

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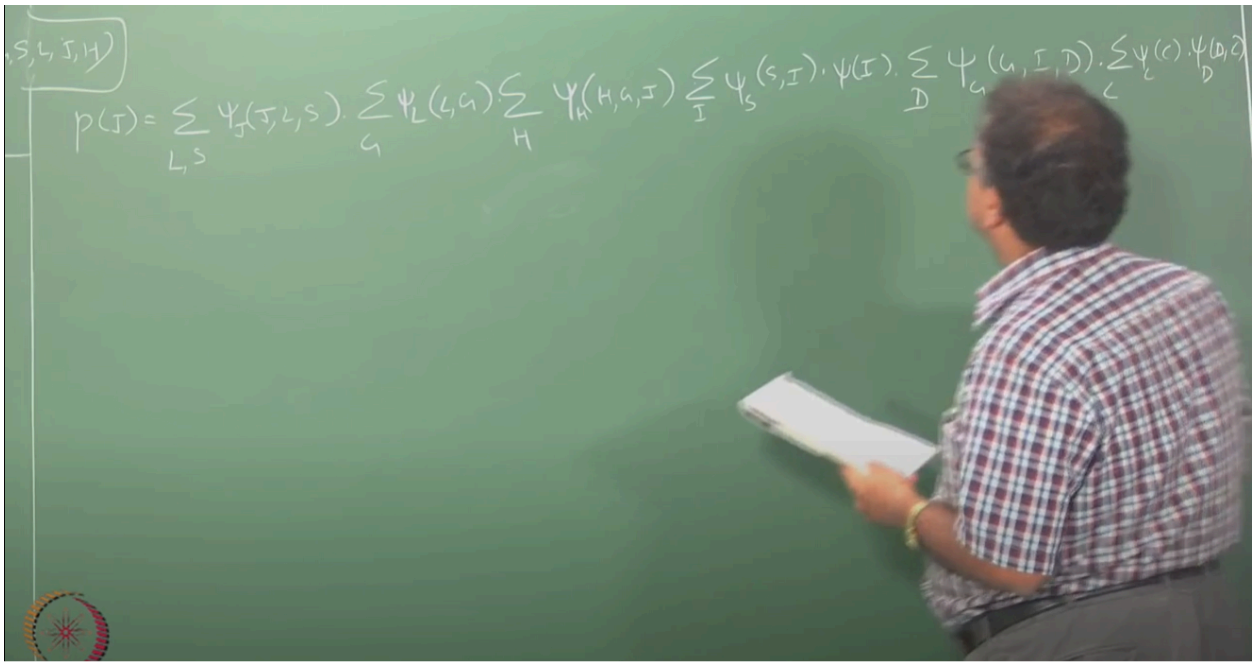

We can simplify the above summation form of P(J) as,

$$P(J) = \sum_{L,S} \Psi_J(J,L,S) \sum_{G} \Psi_L(L,G) \sum_{H} \Psi_G(H,G,I) \sum_{I} \Psi_S(S,I)\,\psi(I) \sum_{D} \Psi_G(G,I,D) \sum_{c} \psi(c)\,\Psi_D(D,C)$$

Ψ – Factors of the graph.

In directed graphs, Ψ's will be conditional probabilities.

So I have moved from the conditional distribution to the potential formulation right, but you know what this means this is essentially the conditional distribution here so you can actually think of that having been represented as an undirected graph. Also, we can use the same technique that I am doing here even with undirected graphs right.

So that is the point I am to make that point I just switched over from this notation to this notation. So in this particular case, these factors happen to be conditional distributions, but they could be factors that you get from here. So in which case you probably have to have some kind

of normalization going here right. So if you are going to use this as an undirected model then you have to have some normalization to take care of, so is it correct, so what I have written is correct right.

So the notation I am doing here is essentially this it takes in J L S as arguments again returns a distribution over J. So that is what the J here stands for, so it takes J L S as arguments and returns the distribution over J right, or some function over J, this takes L&G has arguments okay and return something over L. So that is what this is right, so this is essentially the probability of J given L, S or something like that the equivalent to that in my potential notation. So that is the thing I am marking here okay, is it clear?

So now you can think about it, so the C runs once over only those two tables they are small tables, so C has just one in two entries in it right Ψ(C) will have only two entries in it right. Whether the teacher is coherent or the teacher is not coherent right. And Ψ (D, C) will have how many interests in it. Four entries in it right, how many independent entries in it.

Yeah, two okay it will have only two independent entries, not three right. Because given the course is not, given the teacher is not coherent, what is the probability it is difficult. So automatically 1 minus that gives me the probability it is not difficult right, even the teacher is not coherent what is the probability is difficult and 1 minus that gives me the probability that it is not difficult right. So I only have two parameters, so you can see that I am reducing the parameters tremendously.

So here what would I have had, I would have had $2^{8-1}$ parameters right, the full joint distribution right if I specify $2^{8-1}$ parameters and 1 minus the sum of that will give me the last one. But here look I have tremendously cut down, so this has one parameter, this has only two parameters right. So likewise this is going to have.

Four parameters that are for every combination of I, D you are going to have one possible outcome for the other is 1 minus that right. So for every combination of I, D you need to have one parameter so you will have four parameters, so likewise here you will have one parameter

again, here will have two parameters so like that, so you are reducing, if you take the product is much, much smaller than the 2 $^{8-1}$ that we had right. So that is the power of doing the factorization, so the number of parameters you need for specifying the joint distribution comes down significantly.

You can start pushing the sums in so that this sum runs over only if the small number of elements right. Likewise, this sum runs over a small number of elements and so on so forth, and then I can complete the entire joint distribution right. So this kind of an approach right where you push the sums in is known as—variable elimination.

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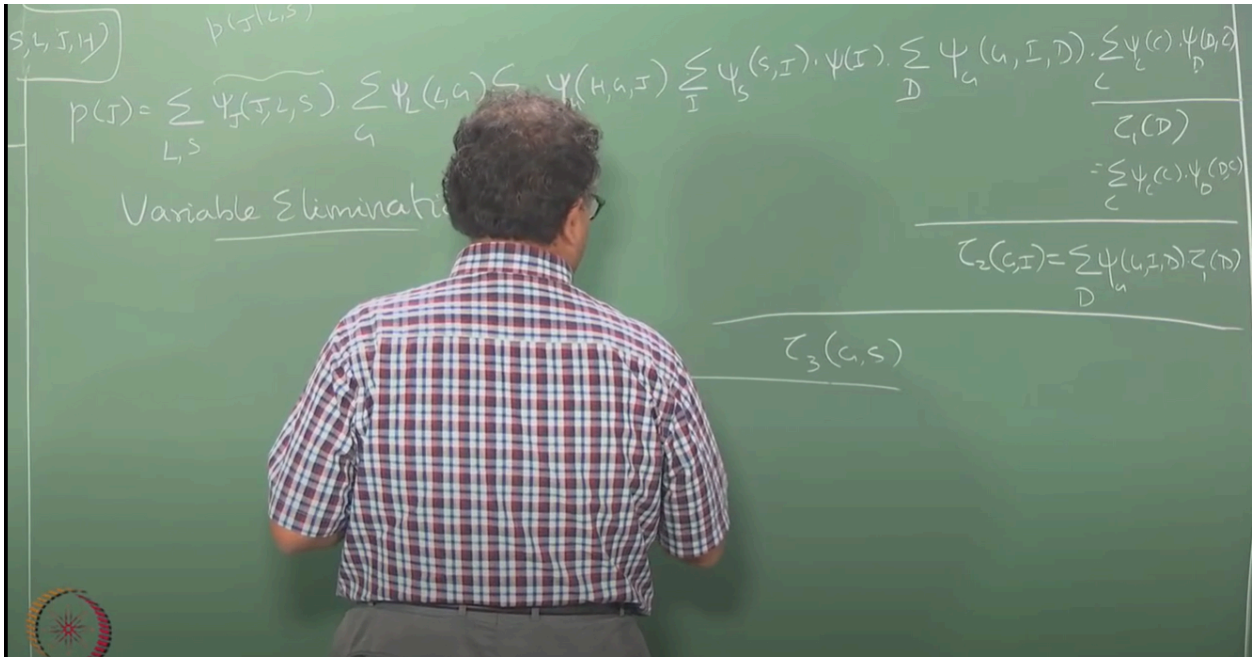


$$P(J) = \sum_{L,S} \Psi_J(J,L,S) \sum_{G} \Psi_L(L,G) \sum_{H} \Psi_G(H,G,I) \sum_{I} \Psi_S(S,I)\,\psi(I) \sum_{D} \Psi_G(G,I,D) \sum_{c} \psi(c)\,\Psi_D(D,C)$$

Variable Elimination: A exact method of inference,

Using this method, we can reduce P(J) further,

$$\tau_1(D) = \sum_{C} \psi_c(C)\psi_D(D,C)$$

$$\tau_2(G,I) = \sum_{D} \psi_G(G,I,D)\tau_1(D)$$

We can continue performing this creation of $\tau_i$ factors with estimated parameters finally to estimate P(J).

Right, so for small graphical models okay, this is a good way to do inference right. It is not an approximate way of making an inference; it is an exact way of making inference right it gives you the same result as you would have gotten if you had summed over the entire distribution okay. So it is called variable elimination, and so the advantage is like I said they have an amount of computation that you are doing you will be minimizing right. So how much computation would you be doing, what will be the maximum, what will be the largest table that you are summing over?

Exactly it depends on how much you are able to compress the things and how much is actually able to eliminate the variable. So the more variables are supposing variable eliminates the faster will be your computation right. So think about what you are doing here, the first step is marginalizing over C right.

So I am going to say that you marginalize over C right, and you end up with a factor over D right I am going to call it some $\tau_1(D)$ right so what will $\tau_1(D)$ look like.

That is $\tau_1(D)$ . Right, next what do I do what I am I marginalizing over D right, so this guy this whole thing I am marginalizing over right I am going to call that factor $\tau_2$ and what will be a function of G and I?

So I keep doing this next I'm eliminating I, so what we will end up with the factor over G & S right yeah. Then what we will end up with I will have this guy as it is right am eliminating H right, so there is no H here so $\tau_3(G,S)$ will continue propagating beyond this point right but I will also introduce a new factor called $\tau_4$ which will have you can see that right at this point I

just trying to try for you to get an appreciation of what the computation is happening right at this point you will have $\tau_3$ you will also have $\tau_4$ right.

So when you compute it till $\tau_3$you have eliminated you eliminate a $\tau_1$ you eliminated $\tau_2$ because you have rolled up everything into $\tau_3$ but we do $\tau_4$ you are not able to eliminate that right so $\tau_3$ is still carries on to the next level. Now we eliminate G, so what you get at $\tau_5$ so eliminate G, so we will have J left-right we will have S left, and L will now get added here finally you will get depending on what order you do this thing in. Here I will first sum over S I get this then sum over L, I get my P(J), okay so this is essentially the or how you will be doing the elimination.

So as and when you are doing the elimination, and you are creating this new factors so what we should be thinking of is it is as if you are adding a new potential it is as if you are changing the graph right. So when it did this right well, I did not let us certainly really add anything new D is already there right what about this. So now I create an edge between G and I and not a big deal G&I already existed right but what about this now I create a potential between G and S right. So when I come to this point, so it is like I am adding a. Another connection between G and S right so likewise anything else is happening. Anything else. J & L is already there JSL, JSL.

Right, I need to have a clique for me to have a potential JSL I need to have a clique, so I am essentially like adding an edge between S and L right. So you can think of the way we are doing this is essentially like we are making this larger some of these potentials are making larger and larger right.

So, in this case, it turns out that luckily none of the intermediate steps that we are creating makes a large table right none is nothing is larger than any of the existing tables right so we could choose a bad elimination ordering I can choose a different order.

So here the order we chose was C, D, I, H, G, S, L okay, so that is the order in which we eliminated the variables started off with the right-hander at C, D, I, H, G, L okay, suppose I did this.

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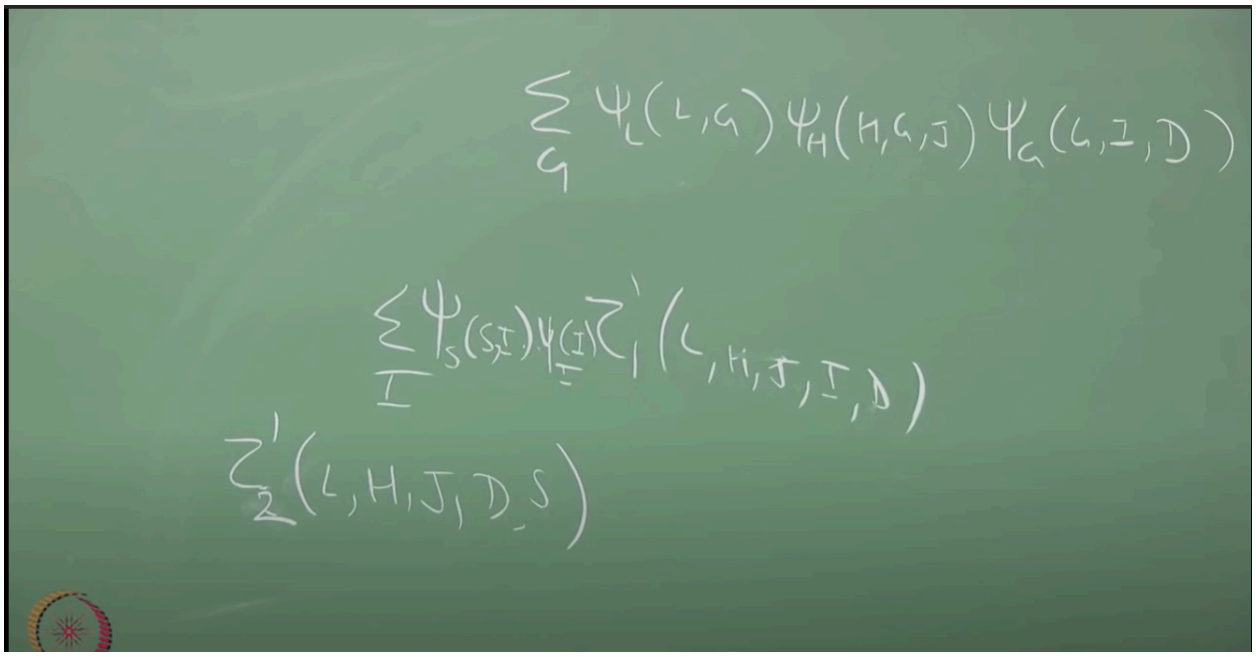


Order of Elimination:

P(J) =

$$\sum_{L,S} \Psi_J(J,L,S) \sum_G \Psi_L(L,G) \sum_H \Psi_G(H,G,I) \sum_I \Psi_S(S,I)\,\psi(I) \sum_D \Psi_G(G,I,D) \sum_c \psi(c)\,\Psi_D(D,C)$$

Consider a new order of elimination for the same graph,

G,I,S,L,H,C,D

To eliminate G, we have to all factors that have G,

$$\tau_1^{'}(L,H,I,J,D) = \sum_G \Psi_L(L,G)\,\Psi_H(H,G,I)\;\Psi_G(G,I,D)$$

We have a 5-way table if we eliminate G first.

Now we eliminate I,

$$\tau_2^{'}(L,H,J,D,S) = \sum_G \Psi_S(S,I)\,\Psi_I(I)\;\tau_1^{'}(L,H,I,J,D)$$

Likewise, we can eliminate other factors. This shows how the order is important in variable elimination.

They start off eliminating G right. So I can sum over G, and I have to put in all the factors that have G in it right, so what were the factors that had G in it I will sum over G I will do right from this side right so I will have Ψ(L) L G Ψ(H) when I summed over G over all these factors. So now I am going to create my new $\tau_1$ right so I will call it $\tau_1^{'}$ so $\tau_1^{'}$ will be a function of everything in that that is not eliminated right. So G has been eliminated so what it will be so L H J I D ouch. Now I created a 5 a table there. By choosing to eliminate G first right, I have created a 5 a table so that is a large table and now I am going to sum over this.

So now will be summing over a table which has $2^5$ entries right so that is a bad thing right so next one what I have eliminated next try to eliminate I next so what I will do that so I will have $\tau_1^{'}(L,H,I,J,D)$ is there any other factor that has I Ψ(I) this doing right and what will this do it eliminate I right but it will add S to the factor so my $\tau_2^{'}$ will be a function of L, H, J, D, S now I have another five-factor table, in fact, this is the worst possible elimination order okay to give you the really bad picture right that is the worst possible elimination auditory then I eliminate S.

So what do I do in that case well I add JLS also to the mix right a large JLS also to the mix I will eliminate S that but J and L are already there in the factor so, in fact, this will come down so my T 3 will have only L H J d because I eliminated s right then I will eliminate L right nothing a

new gets added that the only thing that is left out to is C right yeah so by the time I come to C, yeah so everything else will get eliminated.

So finally I will be left with a factor that contains only D&J and then finally eliminate D so what will happen when I eliminate S we are done to yes okay what happens on eliminating L I will end up with a factor that has HJDS then what happens if we eliminate H I will end up with a factor that has JD, JD what L right no L is already gone eliminate H will just end up with the factor that has JD I have a factor that as JD I will also have the this is the C's the last two factors will still be there the Ψ(C) and Ψ(DC) that those two factors will still be there right everything else will get eliminated, and then what I eliminate C that means those everything all those factors will get eliminated I will be left with a factor that has only D&G.

And finally, eliminate D, okay but what I had done along the way is that I have created a big clique herewith five variables in is right, so if you notice as we went along so even though this looks like a clique of four variables okay it was never created as a clique of four variable said that at best I only did a clique of three variable just two different cliques of three variables it looks like a clique of four variables but we never generated the clique right but in this case we actually generate a clique of five variables so it can become very large right. So it turns out that the complexity can be related to this the complexity of running inference on this graph can be related to the size of the largest clique you generate along the way right,

So, these kinds of edges that we generate like this right are called fill-in edges yeah. This one G&S this one yeah also eliminating I right. So when you add this thing I mean well I did not want to erase everything but When you add this fill-in edge that essentially when you remove that so this not really a clique.

So this is not really a clique this is only a this the edge is the maximal clique, in this case, so your question is I do not have a potential that says I GI and S right, so when we did the original ordering we never did a G, I, S potential that is because what you pointed out. So I was eliminated, and therefore we only have GI was already existing right we have a potential corresponding to G, I, S in the beginning.

No why should we have one corresponding to G, I, S no we do not need one corresponding G, I, S, so we do not need one corresponding to G, I, S, so you do not need one at all in the inference also when this fill-in edge is added that those things are not there right. So we only have to worry about those fill-in edges which actually leave you with a clique is what I am writing the size of the largest clique. In the elimination, ordering is called the induced width of that ordering.

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**Introduction to Machine Learning**

**Lecture-69**
**Belief Propagation**

**Prof. Balaraman Ravindran**
**Computer Science and Engineering**
**Indian Institute of Technology Madras**

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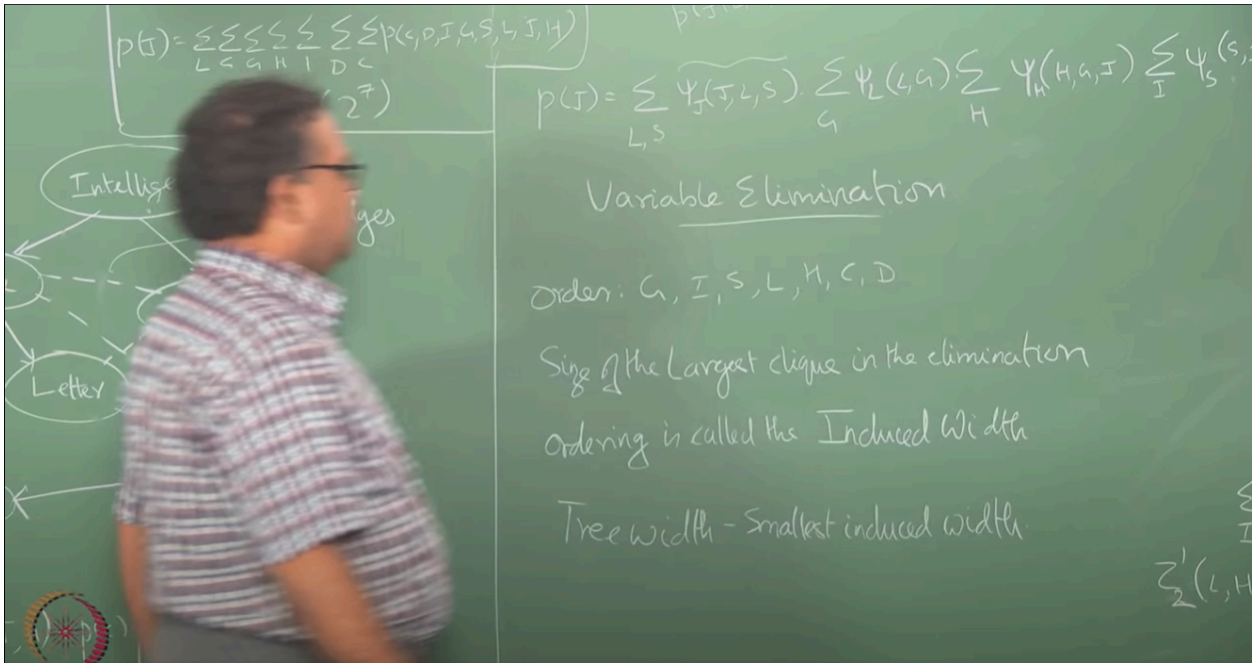


And we have a concept called the treewidth of a graph, treewidth of a graph which is the minimal induced width, the treewidth of a graph is the minimal induced width so what do I mean by that so across all possible variable elimination orderings that you have right, so you can find out what is the induced width for every elimination ordering right, and there will be some order that gives you the smallest induced width that is your treewidth.

The complexity of Variable elimination is

$O(K^{\text{Tree Width - 1}})$

Tree-Width – Smallest Induced Width

Where k is the number of values for each random variable, so, in this case, we assume k is 2, so it will be the order of $2^{\text{Treewidth}}$ right. So what would be the treewidth for a tree? One or two depending on how you count tree width some people count treewidth that is the size of the clique -1 okay, in which case it will be 1 right if you count as the tree with as the size of the clique it will be 2.

Because I can eliminate it from the leaf to the root right, at no point we will add a larger factor so every time I will be just collapsing one edge at a time right, no point will be adding a larger factor so if I eliminate it from the root to the leaf then that might be problems right, is some arbitrary ordering, but the smallest thing is to eliminate from the leaf all the way back to the root so every time you will be just removing one edge, right.

So if you think about it the smallest elimination ordering for us here right, also started off with the kind of like the single node hanging of here. Right, so you have to eliminate C first if you eliminate C at the end then you ended up adding some other nodes along the way right, so getting rid of here somehow coming from the outside inward right, or going from inside out was a bad idea, so that is essentially well. So for trees, variable elimination is great right, because it does things kind of in the best possible way you can expect it right.

But still, there is a problem, what is the problem?

I asked you to find p(H) right if I had asked you to find p(H). Right, you basically have to redo this computation all over again right, so many of these tables that you computed internally could actually be reused that if I want you to find p(H) in fact.

Up till this point, everything can be reused; in fact, some of this computation also could be reused right, inappropriately modified form right.

So but then you end up doing everything all over again right if I asked you to do p(H) right, so there are more efficient techniques where you can keep caching these things away right, the most popular of this is called.

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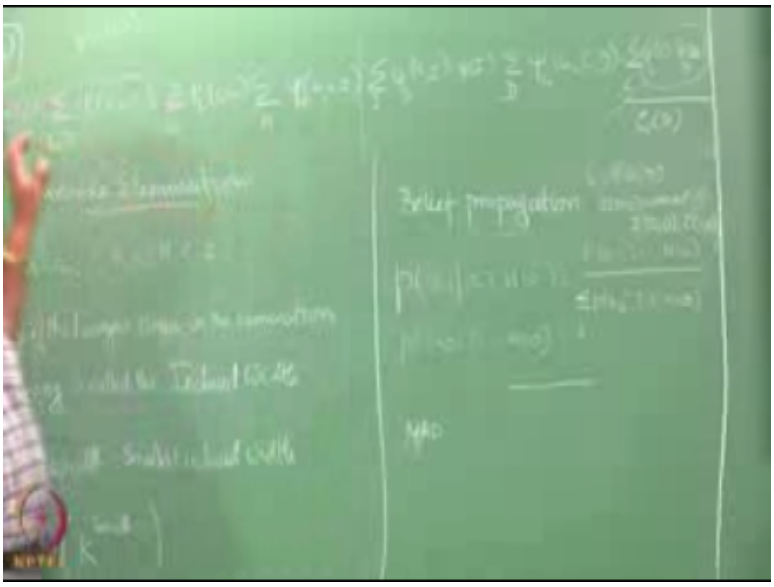


Most popular of this is called belief propagation right, wherein you have some kind of an incremental way of computing these $\tau$ factors right, that you have by passing what is known as messages between the nodes.

Okay, and the nice thing about belief propagation is it allows you to reuse a lot of the computation that you have already done for answering different marginal queries, okay any question so far.

So I can answer any marginal queries you can see that right, so I basically have to sum out all the other variables and have to find out now appropriate variable elimination ordering right, and then do it.

So the trick here is finding the right ordering, so I gave you the right ordering right, but in arbitrary graph finding the ordering is actually NP-hard right, finding the right order is NP-hard so you just have to do the best that you can and tree it is easy you can immediately see in trees you can go from leaves to the root but in an arbitrarily structured graph right it is hard to find out what is the right ordering, okay.

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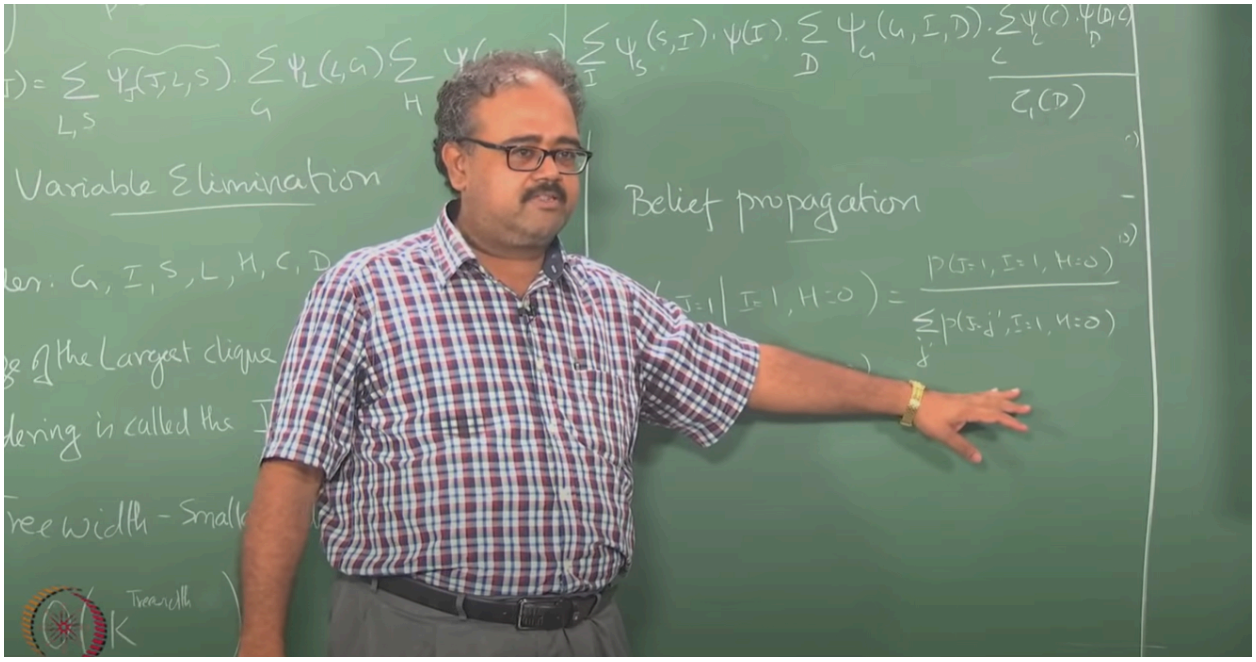


Other Queries,

P(J=1 | I=1,H=0)

P(J=0 | I=1,H=0)

We can compute it as,

$$P(J=1 \mid I=1,H=0) = \frac{P(J=1, J=1,H=0)}{\sum_j P(J=j,\ I=1,\ H=0)}$$

Great, so what about queries like, what is it probably that I probability that I let us say get a job, given that I am intelligent, but I am not happy. Right, and also what is the probability that I do not get a job. So this is essentially a conditional marginal line right, so I condition on some variables, and I want you, I want to know the marginal right. So we know that everybody here is intelligent so once we figure this out if J=1 is lower than J=0.

What should you do, be happy somebody said that yeah, so be happy yeah that is it, that will actually put you in we do not know we have to evaluate the marginal for that right, that does not guarantee your job but being happy at least well leaves you happy right.

No, that is one of the things for getting a job right, you want to be happy, so if you choose to be happy already it becomes irrelevant, anyway so what do you do for this.
Right, so this is essentially I can just compute this as.

Right, it is essentially ratio of two marginal's, but this is one marginal, so this is another marginal what is this a marginal over, it is marginal over I=1, H= the marginal probability of I=1, H=0 right, so this is a marginal probability of J=1, I=1, H=0.

So essentially I will have to eliminate all the other variables I will be left with one table, and from that table, I can read this value right, so if I eliminate all the other variables I will be left with one table that has J, I and H as the entries in it right, and I can just read off the entry corresponding to J=1, I=1, H=0 right, and this one again is another marginal so once I know how to compute marginal's I can also answer questions about conditionals.

So last thing then we will stop there will be the end of graphical models for now right. Suppose I am not interested in marginal distributions right, so this is another inference query I mentioned

this in the last class right, the second kind of inference queries we would be interested in our MAP queries right, so why would we be interested in MAP queries?

In fact, many of the classification and other things we will be talking about we are only interested in MAP queries quite often right, so I will give you some image that I want you to label the image, and I am interested only in the MAP estimate of the label right, I want you to give me a label I do not want the distribution over the labels I want you to give me a single label so in which case I need the MAP estimate.

So which label is the most probable according to the posterior right, so that is essentially what I need so for finding the MAP estimates so what do we do in this case. So what I am going to do is once I decide this kind of an ordering right, I am going to replace the sum here with a max.

I am going to replace the sum with a max. Right, so I am going to say look this is the max over C right, of something right so these probabilities I will compute then what you do is, when I do this max all over and I finish the computation right, I will get some probability right, the probability is the MAP probability, the probability of the most probable point.

How do I recover the most probable point at this computation is only for a probability right, so when I say do a max over C what does it mean for every value of D you will be entering one value right, you have a $\tau_1(D)$ right, so $\tau_1(D) = 0$ will be that probability right, for which is maximal you look at. Essentially you will have when you finish this computation right, so you will have some factor that is called a $\tau'(C,D)$ when you finish taking the product will have some $\tau'(C,D)$ right.

So $\tau_1(D)$ will essentially be something like this $\tau_1(D) = 0$ will be max of $\tau'$ of, max($\tau'(0,0)$ $\tau'($ 1,0)) right, this is what I mean by taking the max, so my $\tau_1(D)$ will be the max of okay when D=0 and C=0, D=0 C=1 now these two entries whichever is the largest I will put that as $\tau_1(D)=0$ likewise for $\tau_1(D)$ =1 I will take $\tau'$(0,1 and 1,1) whichever is larger I will put that there, okay.

Now I will be eliminating D here right, so for each value of $\tau_2$(G, I) right, I will figure out which is maximum across D=0, D=1 right, I will put that in my $\tau_2$entry, so likewise I keep going until I finally get a product right, and now how will I recover the actual point now I have found out which is the most, what is the probability of the most probable configuration, how do I find out what is the most probable configuration.

I keep track of which one gave me the max right, so I keep track okay here for $\tau_1$(D) = 0 did I get the max from C=0 or did I get the max from C=1 right when I had D=1, did I get the maximum C=0 or did I get the maximum C=1, so I keep track of in every stage I keep track of which entry gave me the max right, and then once i finish the computation I just go back and read out the max entry so that essentially gives me the MAP the probability of the most probable point and also this most probable point, right?

So if you have a tie, then you can choose 1, so it will give me at least one of the most probable points okay, so that is essentially how you do MAP estimates and yeah, so it is rather bad because it is exponential in the largest factor, but it is useful for small graphical models, because most of the other methods have significant overhead in setting up the entire process, right. Suppose you want to do belief propagation you will have to set up the data structures corresponding to the messages and it is a little bit of overhead terms of computing, right.

If you have a very small graph like the earthquake graph okay, the earthquake graph, you can graph you can make an inference by. Inspection right, just look at the graph and make inference you do not have to even do any computation right, so some slightly larger graphs like this right, so where you have to actually do some computation you can do variable elimination it is fine. But when you start talking about running it on images right, so we told, I told you right we have like a lattice like structure one node for each pixel in the image, and then I want to run this on a 256x256 pixel image, right.

Then you really need some help right, and then such cases variable elimination is not the right thing to do right, because the first case the treewidth can be large right, for that and so there are more efficient ways of doing it and even belief propagation right, in this kind of directed acyclic graphs right belief propagation is actually an exact algorithm right, even though it is pretty efficient it is an exact algorithm so can still be time-consuming. So people look at answering queries in an approximate fashion, so I will not be able to tell you what the exact probability is, I will not be able to compute the exact that same MAP probability right, but I might be able to give you the MAP is the actual point that has the highest probability.

But if you ask me what the highest probability I might not be able to give you that accurately is, so those kinds of approximations we are willing to take so that we can make inference efficiently.

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**Introduction to Machine Learning**

**Lecture-70**
**Partitional Clustering**

**Prof. Balaraman Ravindran**
**Computer Science and Engineering**
**Indian Institute of Technology Madras**

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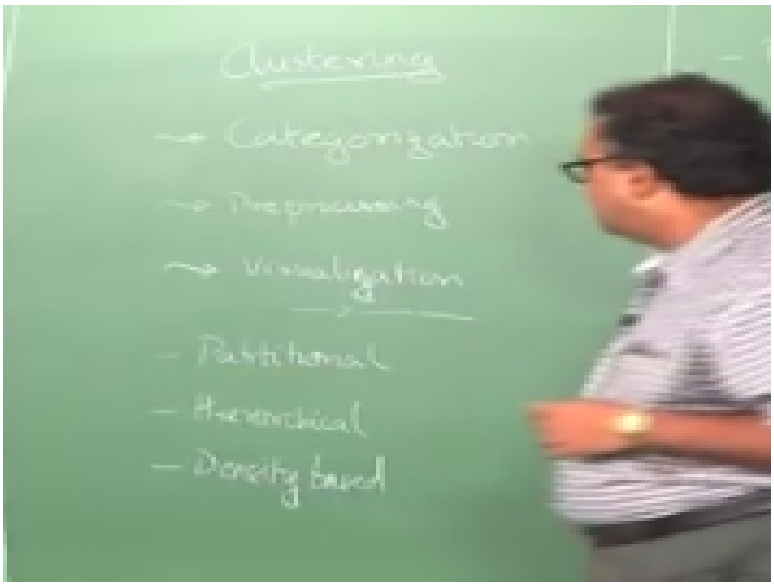


Clustering so far you know what clustering is right, so we did in the very first class we looked at clustering, so all you know what clustering is and essentially is the idea is to group together data points that are similar right, so what you are essentially trying to do is find a partition the simplest clustering problem is stated as follows you want to find a partition of the data points such that the similarity between the points are belong to the same cluster is maximized right and the similarity between points that belong to different cluster is minimized right.

So that essentially the thing and so if I give you a set of n data points right and I ask you to partition into k clusters, right how many clusters are how many different clustering are possible so when I say a clustering it is a set of k cluster okay so how many clustering's are possible if I give you n data points and k clusters and before condition that none of the clusters should be huge empty number right, so nice question to ask in exams but it is a huge number, huge number of clusters right.

So it is just impossible for you to exhaustively search through all the possible clustering's and then come up with the one that is best right, so inherently clustering is all clustering algorithms we will look at are all some form of an approximation or the other right to, to the actual base solution, right in fact some statistician so consider clustering a ill-defined problem right and do not dive into solving it you know, okay that is a ill post problem I will not solve it right so thing is it is a real problem.

People do not so all kinds of a place where we look at clustering so there are two main things that we do it clustering right, so the first one is as a machine learning as a task as a data mining task in itself right, I am interested in producing clusters right so I am interested in producing clusters right so this is some kind of a what we will call some kind of categorization I want to take this data point data is set of data given to me and then want to categorize them into different groups right in it of itself.

I am interested in doing clustering right clustering is also very valuable as a pre-processing tool right, so why would I want to do clustering as a pre-processing to so I can take a very, very large data set if I have a cheap way of clustering it right I can cluster it reduce it to a few data points right I can take a say 10m no dataset.

Like 10m for 10m items and then I can say I am going to cluster it into 10,000 clusters right, pretty large let we take a long time to do it but then if it has only say 10, 000 representatives of this 10m data points so I want that to sample 10, 000 data points from this 10m but I wanted to it in such a way that they are as representative of the data is possible so what I do is I use clustering

right and so 10, 000 is a large number of clusters right so each of those clusters is going to have few 1000 points right.

Not very large right so then I can just go and pick out one the representative for each one of these as suppose to sampling directly from a 10m node space, okay so that gives me some kind of leave here right so pre-processing, the other place where we want to use clustering right and any other things we can think of our clustering is useful, exactly right so for visualization again cluster is something there is very useful right instead of just looking at a large table of data something like that if I looked at it pictorially and I show you that okay here are one set of data points at belonging to one group here is another set of data points and stuff like that then it makes it lot easier for you to understand the structure of the data that you are looking at right.

So clustering allows you to visualize the data and understand any kind of special structure or structure in the feature space right when you say special structure does not mean that it is actually 2D space right I mean structure in the feature space. So is there any kind of structure in the feature space that you are able to understand that right. So it is a very valuable visualization tool and also very valuable for you to understand something about your data right.

So in fact when I talk about categorization right so in and of itself it can be the problem that you are solving or it could be something that you do as a people are seeing step before you go and actually solve the problem, so I have been interested in finding out how many classes are there in my data so I do not know I am not be given class labels a priory okay I do not give class labels a priory but can I tease out class labels right can I say that the data contains seems to be coming from three different distributions right and then I am going to say each of those is a separate class condition distribution, and I can assign a class label to all of those distribution rights.

So it is like I am going to look at the data try to understand that and then see okay these are people who are likely to finish a course right these are people who listen to all the lectures but not write the exam I do not know what to be so but we not look at the data at here okay. So this kinds of things which you could do that are what I am saying, clustering is just not a one short okay give me the data here, or the clusters in you are done with it in fact quite often it is not even

called clustering it is called cluster analysis because you stop with the clustering alone you actually have to go and figure out what the clustering is telling you.

Classification in some sense is much easier in some, so you are given something you are basically returning the labels and more often than not you are done right but clustering it is usually a step on the way to something else okay right. So there are many ways in which you can do clustering so the most popular of this okay are called partitional approaches right so partitional clustering and I tell you what is the others are I get to them on okay.

So the first thing is partitional clustering then hierarchical and then density-based, and they are not really disjoint classifications right, but typically methods get the index to the end of this three and then there are many, many other smaller things raise, so somebody wants to get a paper return the over I propose something that is neither partitional large hierarchical and then they will propose a new algorithm and thing like that, so they are to sell their papers and not necessarily individual classification right. However, these are the three main classifications right and so look at each in turn right.

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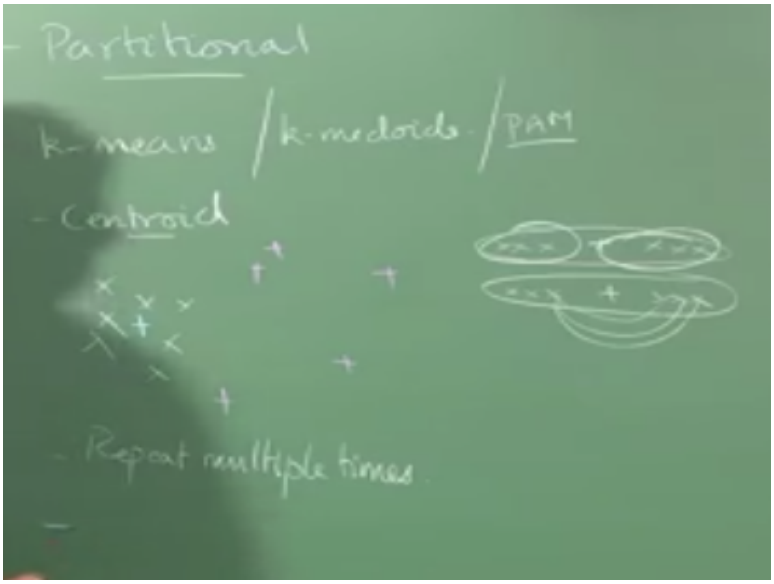


Partitional clustering things K means, everyone knows what K means as of right, so why do we call them partitional clustering? Clustering is partitioning the data right so why I am calling this as partitional clustering. So I am essentially these are methods which search through the partitions directly right the final partitions that I want right they search through the partitions directly, so that is why they are partition clustering methods okay.

So suppose to hierarchical clustering methods right which does not says through the space of partitions on the entire data set right they first try to do it into two groups right they do not search through all the partition suppose I am interested in K things right they do not do all the K they do not do the K clusters right they can start off with two clusters and then split the two clusters into four and so on and so forth okay.

They can go down into there, they do search directly in the space of k partitions right while k means exactly does that right this when what did you do in k means I am sorry so what you are doing in k means you start off with k guesses for the centres of the clusters right so there are few things that we need to know so the first concept here is the centroid so what is the centroid well this et center of cluster right so essentially you take the average along each coordinates right to suppose I have 10 points it belongs to the cluster it is in a five-dimensional space.

For each dimension, I take all the ten coordinates take the average right so. Finally I will end up with the single data point, so that is the centroid right suppose I have some set of data points like this right so the centroid could be somewhere there now that I am exactly computing the centroid, but it will be somewhere there so right sure so what is a how do you find the perfect solution, yeah so there are many ways of cleverly initializing this right we talk about vanilla k means right so the many ways in which you can look at a clever starting point right.

So the other things which people do they what they do is they start off with 10% of sample of data and then they repeatedly run clustering on that and figure out which are the good centroids for the 10% of data and then that use has an initialization for clustering the whole data right. So the idea there being rather than repetitions of very small when you are doing the 10% of sample of whole data so this is the one way you do it the other way of doing it is to do this kind of an initialization where you try to move to the further corners.

But then that has it one problem right so essentially it will make it more sensitive to outliers right because you are trying to put your centroid the beginning centroid at the edges of your space right so it will make it a more sensitive outline, so those there are issues. Yeah that is k means another's so I do not think he meant that I think he meant something else we will come to that right so might there are I mean many heuristics again so I think the current came in variation champ is I think k – means ++ of some thigh right.

So that gives you a very good initialization, and then your drawing a clustering from there and works well right but k – means is an approximate thing so when you look at EM right so you actually see a more well-founded derivation for k – means so right now I am just going to introduce it to you as a heuristic right but later on when you look at EM right the canonical EM problem that you solve right first canonical EM problem that you look at will be K – means.

The variant of k usually means we call Gaussian mixture modules but it is essentially like k means okay so what you do with k – means is that you first pick centroids for all your clusters right a randomly right of course I have k clusters then I will just choice k centroids at random right that could even be like that right choice k centroids at random and then I will have my data

points so what I do is I assign each data point to the centroid that is closes to it right I say in each data point the centroids that is closes to it okay now I forget the old centroids.

Now I have k groups of data point's right and for each of those groups I try to find the new centroid okay this is the actual centroids the first once I started over of centroids where really not centroids but we keep we still call the centroids anyway so in the just to keep the terminology uniform right now I recomputed the centroids then I go back and assign each data point to the centroid that is closes to it.

Yeah whatever distance measure you are using right so you are in some RP space right remember our data points come from some p dimensional space so in that space whatever is nearest you will you could use Euclidean you could use whatever distance measure you want it choose the appropriate distance measure right so k – means or any of these distance-based computations that you do right works best okay any of the distance based computation you do works best if everything is real value right not even integral right.

So everything is real value that works best if you are going to categorical attributes you will have to think of a different distance measure that you will have to define and I do not know if may I can talk about 1 or 2 things that people use for categorical things but the most popular by far is using some kind of Jaccard similarity, right so I have lots of categorical values that the thing can take and then I say okay how many of those it actually is similar on right so how many dimension I have may actually agreeing with the other one on so if I have the same value in that categorical dimension then I will say one if I do not and the I will say 0 and the I will try to find how many I agree on right.

And that could be one measure but then defining a centroid there is little tricky right so categorical variable may centroid might come with a value of you know red 0.5 where I mean right 0.5 times red + 0.5 times green right so what would that mean no do not add up the colors please I mean I said brown or something I do not know what red and green would up with but do not know that does make sense right so we have to we, very vary about using k – means when you have categorical attributes.

Yeah so what I mean by probability vector like one of n kind of encoding right 1 hot encoding right yeah you could do that but then really what would be the mean centroid value for that 1 hot encoding, yeah that is a interpretation issue so you could still try k – means I mean you do not really have to interpret what the centroid means unless your returning the centroid as a representative point right if I am returning the centroid as a respective point then you get into problems.

So just going to do clustering on it you can go ahead and do crusting on it you do not have to really interpret the value of the centroid right so there are ways of handing this when you have categorical attributes right so k – means is the simplest way then there is something called k medoids right which kind of gets around this whole issue of having to generate an artificial centroid right so instead of centroid it essentially uses a the equivalent of median right so the mean cannot be not actually be a data point but the median is always a data point right.

So likewise medoid I always a data point so it gets around this interpretation issue so computes the centroid and takes the data point closest to the centroid as the representative so the medoid right and the you also have other kinds of things which is called partitioning around medoids right so we do partitioning around medoids you work with data points as the representative of the clusters right and you do not ever generate an artificial point you always choose a another data point as a centroid I will describe that more right.

So those in such cases you do not have to worry about this you know meaningless attribute values being generated but you still need a distance measure so you will have to come up with some kind of distance measure which categorical attribute so if you use 1 out n or 1 hot encoding or anything else you still use some kind of a Jaccard similarity right so then do that okay.

Right so going back let us finish up the simple k means algorithm right so what you do with k means is so once you have done the assignment to the centroids if you forget the centroids estimate new centroids and keep repeating this until you no longer make any changes right sorry done?

No, no labels is necessary, there are no labels, there are unstable is learning from right. So if you have labels and other things, I mean depending on what your application is you might want to look at labels right, that is what the scameans is concerned, the question is the following. And you start with initial guess for the centroids right, I assign data points to the closest centroid right, recompute the centroids, then reassign the data points the closest centroids, I keep doing this until the centroids no longer change.

The question is have you done? Consider a, yeah, okay. Al right, how many clusters you want me to put them into? Three clusters I am done, you are getting close. So the probability of recycling is really small yeah, yeah whether other things which I can show is slightly more dramatic right, okay. These are my two sets of data points. Obviously, they are in two clusters, arent they?

These are the two initial centroids I start off with okay. So what are the two clusters I will get? Okay, and I compute the centroids for those, where will I end up with, and I reassign the data points to the centroids, I will end up with the same thing right so because of my bad initial choice for centroids, I do end up at a point where the centroids do not change anymore, but it is a really, really bad clusters.

Yeah, randomly chose those two points right, I randomly chose things. Exactly, so my question to you was if the clustering does not change or you are done, the answer is no okay. In this case surprisingly one of the few cases where the answer is not it depends, the answer is no okay. Because if you just do it once right, you are not done, so that is the thing with K-means in, you will have to repeat it multiple times right.

And please every time you said different random seed for choosing your initial starting point right. If you just the same random seat then you will get the same thing right. Yeah, so we will come to that right. And right, so the people understand why you have to do this multiple times right, because you will get stuck in some kind of local optima right, and you have to start over again with a different random set of K centroids right.

And then keep doing this yeah. Good point, so there are different measures right, I never told you about cluster evaluation measures so far right. So there are different ways in which you can evaluate clusters right. So more of the more popular is this some kind of a dispersion measure right, so I took a diameter, so the way the diameter is different there are two definitions unfortunately for diameter the literature.

So the first one is the average pairwise distance between the data points that belong to each cluster right so if you took that right, so I will take the pairs, pairs of data points right, I will measure the distance between the pairs. So but, then I will also have to consider pairs like that right. So I will look at the distance between every possible pair of points that belong to the same cluster right, and I will take the average okay, that will be the diameter of this cluster.

And the average of the diameters for all the clusters will be the quality of the overall clustering okay some kind of a dispersion measure, right so how much spread out my data point this right alternatively if you have if we are completing centroids right we can also compute the average distance of the data point to the centroid right and take the average and cross all the clusters.

And use that as your quality measure side so the either one is referred to as diameter in the literature so some call the and the most popular one is the average pair ways distance okay, is it fine right and I take it back so the average distance to the centriod is called the radius of these average is the centriod is called the radius of the cluster right.

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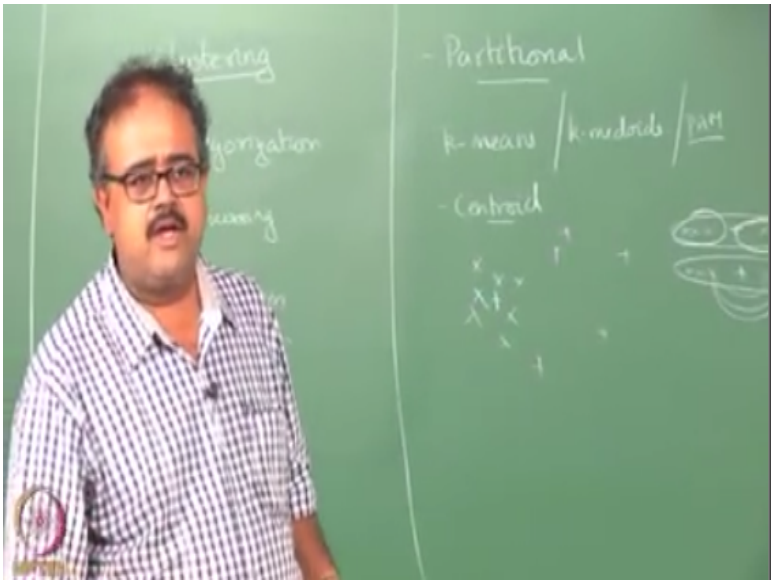


So the diameter the second definition of diameter is the max of the pairwise distances right either to the average of the pairwise distances right or the max of the pairwise distances both could be call the diameter but the radius is essentially the average of the distances to the centroid, okay so this is one measure right so there is an another measure which I could think of which is essentially is called purity which did you guys look at purity already, will never gave it your in the first then assignments of array.

So purity essentially tells you for each cluster that you have what fraction of the data points belong to a the same belong to one class what is the largest fraction of data points and belong to one class, okay so purity can be used only when you have data sets that have classed label associated with that right so suppose I have 10 data points I am going to one cluster six of them are in class one two or into class two or in class three, so the purity of the cluster is 0.6 right because out of 10 or in class one two out of 10 are in class 2 so 6 out of 10 is largest thing.

So 0.6 is the purity of the cluster so like this so whatever right so which are the level of the hierarchical or evaluating it at we will have some set of clusters, right we just evaluate the purity if you purity is a measure that it choose right so there is matter so purity mixing right, so something related to purity again if you using label data sets we can use something like entropy.

Right and what would be the entropy when the class distributed right look At the class distribution in the cluster so the P 1 fraction in the class 1 and the P 2 fraction in class 2 so that we can do the -P1log(P1) so that is entropy so that their classes are more are less evenly distributed then the entropy gives you better measure than the purity so the purity measure is the 0 .5 it is not clear to be weather the other 0.5 belong to the other class or the belong to the many other classes.

So that is the thing so these are the measure that reduces and then we having the zillion this is the satiation will be the clustering is the ill posed problem because there are many different in the clustering and then trying to list them in the popular one okay and the other one is called as the rand index which is typically used when you have the reference clustering and we have the reference clustering and I am trying to achieve the reference clustering so what is the rand index is the for the good point I don't have class labels.

So have the reference clustering but I do not have really but I do not have the reference clustering so the new comes in to the I do not have the class label so in the data points to the classes so the new set of the data and the new data point would not come I will learn new set of the data and I should run this algorithm in the in the that set of the data and then produce the clustering right.

So that the point here is I am not really looking to the reproduce exactly that cluster right what I want to do in that is the following I give you data point and run a cluster algorithm so I would like in the cluster on the training data it look like this so give me a new set of the data points to learn the clustering algorithm on it I will get the set of the cluster and then I little bit more comfortable with the set of the cluster and then the whole data point.

And then manage the reproduce the reference cluster a new set of the data points I get like actually get map to the whole data points itself so itself it slightly different problem so I am just evaluating the cluster algorithm and the clustering algorithm not the simple algorithm is produce

so then we have the ram index and the one instants could be that we have nicely separated classes but many of them.

So that is the other case I am not talking about it so thermal index is typically used to every paie points in our data set so the F I and the F J belong to the same cluster in the reference clustering and then belong to the same clustering the cluster that they produce give the score of the one if the F I and the F j are the reference cluster and the difference cluster and the cluster that they produce and then give the score of the one and then how many pairs are there and Nc2 the pairs of them and then divide the some by Nc2.

And the what fraction of the pairs of the data points have you cluster correctly right what is the nice thing about rand index is suppose these are the more thing in the clustering that the original clustering is given to you right and then the end of the slitting points and then splitting in the two right the original clustering and the larger cluster and the original cluster and the original cluster.

And then this is the original cluster. and the rand index is not suffer a lot and the suffer greatly it is the only there cross and the cross cluster pairs that in the penalize within the cluster parts all fine so it gives you little bit in the terms of the lens in the terms of being in there stricter than the other than the other and the reference clustering okay these are the different measure and we can use whatever you want right because it is not the rand index and then we can use diameter or radius in the case.

Right if we have the label data points and the we can use the label purity or the entropy or the full measure of these okay so we can list about the hundred of these measure and the people are used in the literature so just pick the favorite one yeah good point so how do you know which cluster is what? right so I have to figure and then I have to the alignment on the cluster right see I have some K1 cluster I have to use that is my reference cluster and my reference has k1 cluster and I have to use K cluster so which how do I align this k to the k1, right. So rand index gets rid of alignment problem okay. Oh okay no clusters are- yeah. Yeah there are minimum phase in which you can do this optimization right.

There are literally 100's of paper out there explaining to this kind of optimization so the question is yeah what is overhead and implementing this optimization is that whether in your particular application whether the overhead is justified because you are getting a significant improvement over the usual we have doing things right. So what people do as standard implementation or things which kind of give you good results across a variety of domains?

So if there is something very specific for your domain then you have to welcome to try to this optimization right and that is what make this engineering discipline as supposed to I supposed to theory right, so yeah a many things if you can do so went back right so now I did this I do tis again I keep doing this multiple times and what do I do with all the clustering I produced, take the best yeah that is right.

So is not nothing so you just keep doing the clustering and what are my evaluation measure you have so you take the best according to that evaluation measure, okay. So we threw away the rest so you have repeat it a few times and then take the rest, great. Okay let me re-writing it okay you have to repeat k means multiple times do not please do not come to us after doing k means once and say that Oh it does not seem to work okay.

So I will guarantee you that one time it would not work okay and we do not know but just making that in fact emphatic. So let us go back to the question, how do you fix K? Okay that is one answer what is your answer, domain knowledge that is yeah exactly I mean depending on use of requirement domain knowledge right. So domain knowledge is one way of fixing K right is there are other more systematic way of fixing K.

Try all ks but then how do I know which k is good so I tell you one thing say suppose I am using diameter as my measure, right. Larger the case and better it is exactly so the right way of doing it would be to say that I am going to have some kind of a complexity performance trade off right but it is incredibly hard to implement in K means right incredibly hard to implement in k means.

So but you can do that I mean people have come up with that so especially if you are going to take a basin approach to cluster right if you take a bayesian approach to cluster right how will

you implement the penalty for the size so the number of k the prior what you will do with the prior who said prior what I will do with the prior, how will you make it penalize larger K exactly, so reduce the prior probability for large K, right.

But the you do the searching the main problem with this is most of this optimization thing is for finding cluster work well if I fix K but if K becomes a parameter and optimization it becomes incredibly hard that is why I say it is hard to implement right if K becomes a parameter and optimization it becomes incredibly hard to solve the problem right for the fixing a K right you can think of K mean as a approximation to solve the fixed K assignment problem right.

Even if for a fixed K if I have to do something like K means to solve it if I am going to make K a parameter and try to solve the larger optimization problem and it becomes little tricky so suppose to that.

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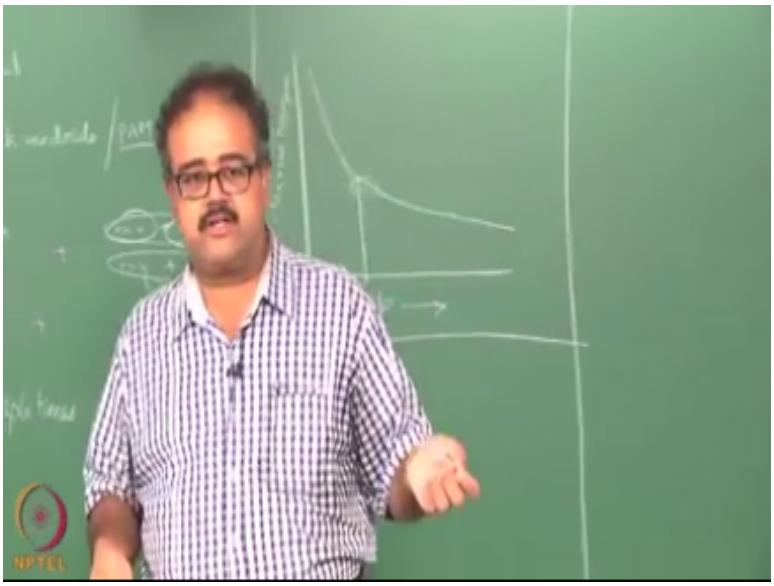

So what people do is practical thing is draw a curve between K and let us say draw a curve between K and diameter right, so what would you expect to see? As K becomes larger decrease we will see something like that we will keep going down and keep going down yeah you keep going down to 0 when depends on how many data points you have right so if you at K = n it will become 0 I do not have to do all of them I already told you what the right value of K is here I have shown you an example say criteria what is the thing it bring, so may k initially decreases you can see a rapid decrease in the diameter right, after some point what happens the thick slows down right, so if you think about it that is a significant change of slope somewhere around here, right.

So that essentially quantifies your good curve complexity verses performances trade of, so wherever there is this change of slope right, you pick that point and say that this is the right k . Usually it is you do this when you are trying identify a small number of k right, if you your is very large right, then you are probably using clustering for some kind of pre-processing rather than as the final means okay, so if you, we said I remember right, I told you that there are different ways in which we use clustering.

So if your clustering is very, very rare the case very, very large right, you are probably doing it using it for some kind of pre-processing right, we which case the exact choice of k does not matter that much right, when you are trying to use it as a algebraic visualization tool even visualization choice of care does not matter that much some of you using it as a actual end in itself right, you want to do clustering, you want categorization in those cases typically you have k smaller, right.

But it can run k up to a few 100 that side, right and then you can find the right value of k like this right, but if determining the right value of k is of some crucial importance to you and k is very large then I would not recommend partional methods at all, right so I would recommend other approaches for doing the clustering okay. So this is called the bend now what did you say it is that name for this it is very descript and they must call the Knee method, so it bends right.

So it is called the Knee method for define k okay, so in fact you can use this for in other cases also where you want to do this kind of complexity verses performance trade off and the optimization problem because very hard if it is know in the k into the complexity parameter also if you throw into the optimization, optimization problem because very hard so can use this kind of a empirical method for determining the actual value, okay, good and we move on right.

So what about k medoids , it is like k means so you find the centroid right, and represented and it uses the k is closes to the centroid right, so the every point of time you have a representative which is in actual data point, right and then when you do the assignment you assign it to the medoids that is closes to you, okay not a very interesting thing.

But the advantage of this is when you move from average to median, mean to median what is the advantage, in statistics it does not get effected by outliers same thing here, so when you move from centroids to medoids okay, it becomes little bit more robust to outliers right. Suppose of all this is the data points right, and I try to cluster it right, so I will end up getting a centroid somewhere there right, then but the medoid might be this a slightly better you know not that much but it is better than having a centroid that is out there, okay.

So one important thing I forgot it in an application of clustering something call outlier mining, so what is outlier remaining that find outlier right, I mean that is nothing how big deal about it, so why do you want to find outliers, delete term is one of it anything else sorry, fraud detection right, so I want to do any kind of anomaly detection so outliers would be anomalous data points, so who said fraud detection show me yeah okay, yeah.

Yeah so the outliers would be some kind of anomalous data points and therefore you would want to find them I am not interested in deleting them from the data set where I am probably interested in deleting them from the real world right, so that I want to catch these things and put them safe guard against them and things like that. It will be useful for understanding yeah; it is the one of the initial thing I told you right. so instead of randomly sampling around the entire state space right generating 10000 samples from the million sample data base to clustering with k = 10000 and sample from each cluster, that gives you more samplings.

I mentioned that in one of the uses in very beginning okay great. So we have done k- means, we did k mean right okay, so PAM is called the partition around medoids right. In fact it is incredibly expensive algorithm, nobody uses PAM any more. When it was proposed it was very big thing and but then people came up with faster ways of doing PAM, all of these work on very small data sets okay. Really large you want to do 10million data points things like that. PAM is nowhere near competition, any of medoids or not at all competitive.

On a very large data sets and any way I will just talk about it because it was very interesting algorithm right. So in partition around medoids, so what you do is okay. So I have some data points, I start by assuming some, say I am doing some clusters, I will start by assuming some two dada points as my initial medoids right. So let us say that this as 1 medoids right and unfortunately assume that this is the other medoids right. Now I look at the quality of the clustering, let us say I use radius as a measure for the quality.

So I will assign all the data points close to right, end of this is one cluster and I look at the average distance of the data points to the medoids count keep that as my point to medoids, I will keep that as my quality of clustering, right now what I will do is for every medoids right, I will consider swapping it with the non medoids right. So I will say, I will make this a normal data point right that we make that a medoids. Now if I make that a medoids what is the change in the quality of the clustering?

Likewise for this medoids I will consider each non medoids intern and consider swapping with it, or whichever gives me the best improvement in clustering, I will keep that as the my new non medoids and then I will go look at the other medoid. Now I will consider swapping this with each one of the data points intern right and then I will be swap it here. Sorry anywhere, so this how it works, it is very expensive as I told you right. So for every time you do the swapping you do the order n thing.

I just have to check the distance, so checking the distance rather order in computation right, so essentially I end up doing $n^2$ computation for very swap, that I have to make. People made all

kinds of interesting observations and they came up with the ways of cutting down on the number of the computation . So when I make a swap I do not have to go through each and every PAM, so only those change cluster membership. Only those data point change cluster membership I have to really look at it.

Data points belong to the current cluster right the medoid of that change; obviously I have to do re computation for that right. Among all the data points do not belong to the clusters, only those change clusters memberships, I have to evaluate this right. But then I have evaluated the cluster membership anyways okay. So I do not have to do new things but still I have evaluated the cluster membership. So but then you can again organized it little more efficiently so for.

So depending on how you organize this computation people come up with a variety of different things that is PAM I can remember, that is partition is on medoids right. People are interested I can give points to read out on more PAM things like that, like I said it is not very that widely used in the community so we will skip those things. So what is the problem with K- means PAM addresses? Initial random really does not matter anymore because I am any way considering everything; I am any way considering very possible pair gone.

Then well it is medoids it is no long that affected outliers right, what about the choice of K, we still have to choose k that is still there, it is not gone away right. Right what about the issue k-means if you have real value attributes, works well if you have attributes, if you gotten rid of that. I still need a distance measure right. So if I am going to have categorical attribute better have a distance measures that takes care of categorical attribute, still the close problem remain with PAM okay great.

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**Introduction to Machine Learning**

**Lecture-71**
**Hierarchical Clustering**

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**Hierarchical Clustering**

Hierarchical Clustering.

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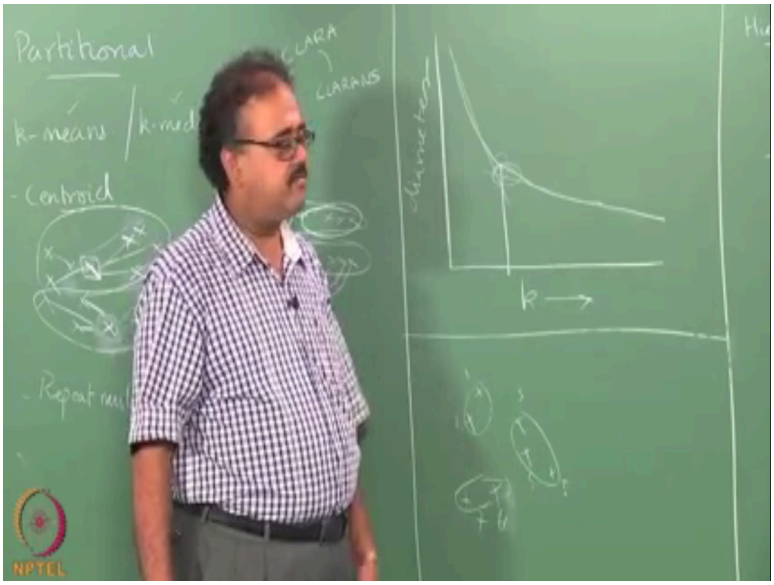



So in hierarchical clustering what I will do is I will start off with there are many ways in which to do this , one way to do this is to say that I will start off with each data point being a cluster off its own .

Each data point is a cluster of its own and then what do I do I try to merge them into larger clusters .

this is not a special distribution of data points okay just put these things here that is I mean individual clusters  data points are individual clusters then what I do is I compare distances between the data points and say maybe merge these merge these merge these merge these .

So when I draw a line like this it means these two have been merged then likewise these two are merged and then the third data point was merged with that and then these two got merged .

I should have one two three four five six seven one two three four five six .

So I initially start looking at this way okay one and two are close together let us merge them three and four are close together let us merge them five and six or six and seven are close together let us merge them after I have done these things okay next thing I look at okay five is close to three and four okay, let me merge them and so on so forth  and next what we would want to merge?

That should bring you to the question how do you measure the distance between clusters you know how to measure the distance between data points and how do you measure distances between clusters
even how do you know that five is close to three and four
you could do a variety of things  there are many different measures that you could use .

so centroid based sometimes how that another thing that people use is called single link distance you know what a single link distance is?
I look at two clusters. I look at the pairwise distance of taking one point from this cluster and another point from that cluster .  I look at all possible pairs
I look at the closest to such a pair,  look at the closest such pair and then I use that as my distance between the two clusters. What do you mean by not equal every possible pair?

there are five here to here I do ten pairs okay,

you could, what do you think that is called that is another decision we should call average link clustering okay the single link clustering single link clustering essentially takes the closest data points if we take the closest data points here which is closer I already merge this this is one cluster of now that is another cluster this is in the cluster which is closer these two are closer at least or closer the question now boils down to is one closer to three then four to six.

here is not very clear to me but I will take your word for it okay now that is basically done all the data points are merged at this point

I did four and six because there that a single link more I mean single link is by far the most popular hierarchical clustering distance measure and then there is another one called what do you think complete link would mean not summation, farthest max.

I look at pairwise distances and the max of these pairwise distances is assigned as the distance between the two clusters

in single link clustering distance between these two clusters was given by the distance between one and three and the distance between these two clusters was given by the distance between four and six that is single link clustering in complete link clustering the distance between these two will be given by two and five or one and five were okay.

Whatever two and five and the distance between these two will be given by seven and five , what you think is larger? Oh my god!

It is hard to make it how to make 2 and 5smaller is it?

I'll make it easier for you there, yeah two and five is moving two and five smaller than 7 and 5 if I am doing complete link clustering then I would have merged these first and then I would have merged .

I could do a centroid based distance equal to a single lane complete link I could do an average link anything else you can think of yeah, I could do radius based. What do I do? I will take the two clusters. I want to find the distance between these two clusters. I essentially merge the two clusters and find their radius. If I want to find the distance between these two clusters I will merge these two clusters and find their radius.

the smaller the radius the closer the two clusters are

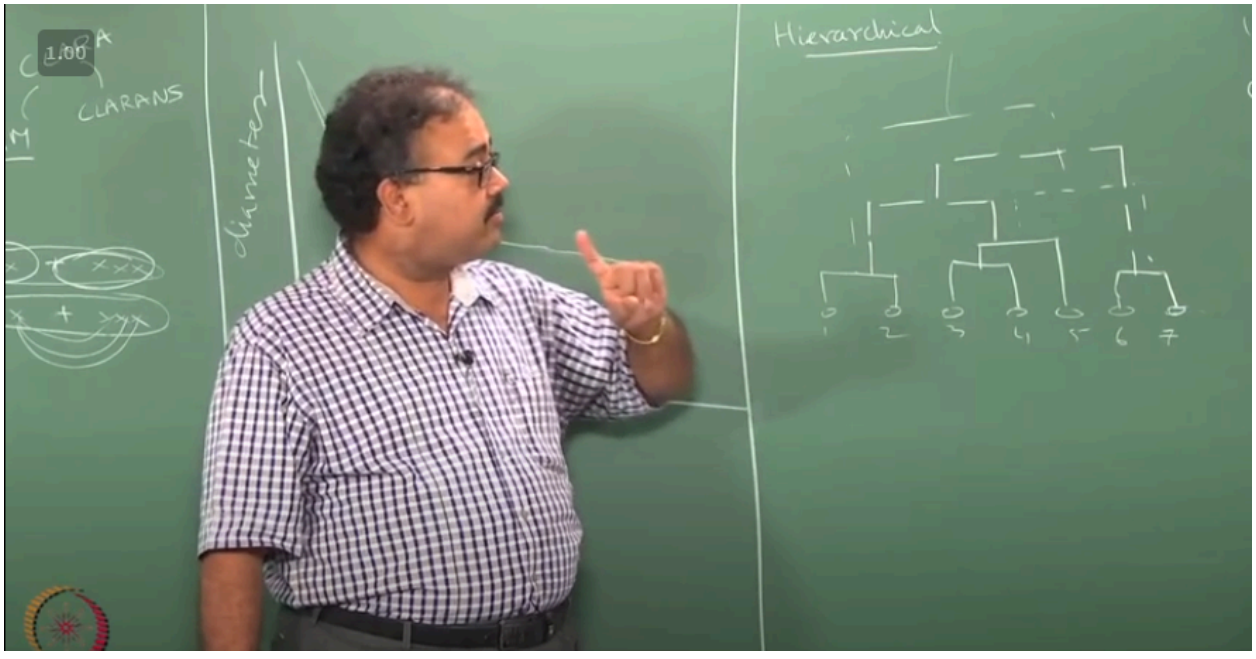
no centroid I am looking at the distance of the centroids of the two here, I am merging the two and then finding the centroid of the merged cluster.

it is different

I like and similarly I can do that you know diameter

I can merge the two clusters take the diameter, the smaller of the two is the better these are essentially more useful for comparison purposes I want to know whether cluster one is closer to cluster two or to cluster three then I can merge the two find the diameter and then make a decision it is not really a true distance measure in the sense that I cannot work by say what is a distance of cluster 1from cluster 2 okay diameter does not make sense .

But then if you want to say it is Cluster 1closer to cluster 2 then to cluster 3 then I can use the merge diameter and I can make those decisions okay, all of these are valid ways of doing hierarchical clustering yeah, you could I mean define whatever you want these are popular ones yeah and yeah they do use other distance measures as well for doing hierarchical clustering okay.




**MEASURES:**

1. **Single Link**
2. **Average Link**

3. **Complete Link**
4. **Radius**
5. **Diameter**

well now it is a nice thing about hierarchical clustering once I choose this distance measure , think one minute stop and think these are meta distance measures I still need to decide on point-wise distance measure .

that could be an euclidean measure when I say single-link I said that distance between the two closest points but what is that distance? That could be Euclidian that could be Jaccard similarity that could be whatever you want absolute deviation whatever then you can look at whatever distance measures you want.

that I still have another distance measure, that is dependent on the data type that distance measure typically depends on the data type, well this distance measure typically depends on the kind of clustering that you are looking at . Once I had a tree like this, what you did not have to choose here? K I did not have to choose K here, once I have a tree like this how do I recover clusters if you think about it when you completed my clustering all I was left with was one cluster ?

How do I recover the cluster? Traverse the tree, how would you traverse the tree? Now when they start here at a single cluster if I come here I have individual data points yeah, I basically have to figure out me point okay I will break here if I break here how many clusters do I end up with three clusters if I break here how many clusters I end up with two okay I could choose to break at me point in between and then say that I will take that many clusters I get .

How do you choose which point to break it at?

I can do the knee method again . I can do a K versus evaluation and I can get this thing, what is the advantage of doing hierarchical clustering is I get that entire graph generated to me in one go .

I am not actually having to rerun everything for different choices of K I get the entire K graph that knee graphs are generated for me in one shot you do not, you know well yeah you do not as many as you can get see the point is the goals that you do not get or the ones where it was very hard for you to find a breaking point .

essentially if you choose a threshold at which you are merging their points , all these things get merged there is no real reason for you to choose three over seven or something if you don't get anything for five six in between after three you move to seven, maybe we did not make any difference to look at four five six usually that is what we will end up
great.
depending on what kind of a measure you choose you end up with different kinds of clusters for example if we choose a single link?

What does a single link say? That closed the distance between two clusters is the closest data points so I could have a cluster that like that another cluster that is like that and another cluster that is like this okay

okay we tell the two you should merge now?

where well give me names leave me numbers one two three. So for you single link clustering I will merge one and two if you single link I will merge one and two I will basically get this humongous very long cluster .

So that is the problem with single link clustering you might end up with very long clusters essentially the points at one end of the cluster to the other end might not be very related at all that is the problem with single link clustering yeah on the other hand if I had used complete link clustering I would have merged two and three first and then I would have merged it with one but if I use complete link clustering is highly unlikely that I would actually produce say such elongated clusters as one and two in the first place , single link clustering tends to produce very tight small clusters .

And at some point you then merge a lot of clusters but then you will merge them at very high levels in the tree. It is the lower levels in the tree you'll be getting smaller clusters. Okay ah where is the tree here? that thing that figure I drew there is a tree okay but it is not called a tree in the hierarchical cluster in literature it is called something else
Dendrogram, you know what the dendrogram means?

It's a tree yeah dendrogram means tree just said they went to a different language and pulled out the one loaded three okay, dendrogram is a tree . So that is a dendrogram and what are these levels I am talking about that yeah no not really iterations they are the levels at distances at which I merged .

So when I merged one done to , I had some threshold I start off with one I say anything that is within point one distance unit of it I will merge into a cluster there's nothing it stays as one and then I say okay within point two point three point four now at point four great, now I have two at the level of point four I have merged one and two and also turns out that level of point four emerged seven and six and three and four .

That is why all of this should be at the same level but you can think of this being slightly different because three and forests are slightly farther apart then one and two all , these levels are essentially the distances at which you are merging there . So that is why five gets merged with three and four at the higher level because if I slightly farther away from four then three is that is the reason for the levels and then these two get merged at a higher level because we are farther away, the levels in the tree at which you merge or essentially the distances rate at which you are doing the merging okay.

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**Introduction to Machine Learning**

**Lecture-73**
**The BIRCH Algorithm**

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So the idea is I want to cluster large I want a cluster really large data sets right very large data sets and the way I am going to do it is the following right
so I am going to do a very rough clustering at the beginning right I am going to do a rough clustering at the beginning where I am going to produce tight clusters right, but many of them and i provide a lot of small, small clusters.

So essentially taking the suppose I have a million data points will reduce it to some 100,000 data points, but each one of them will be very, very small diameter clusters right I produce these things and then what do I take one representative point from each cluster, and then I try to do my hierarchical clustering on that
so initially I do something that little fast ends the fast and dirty right.

So I might get the correct clusters I might get a little bit off here and there but, but then I do a second pass through the data, and I will fix everything right so the way I do this as follows
I do one pass through the data produce some rough clusters right, and then I take the centroids of all these clusters right and then I do a proper clustering algorithm we only work with the centroids I do not worry about the million data points I reduced it to10,000 centroids 100,000 centroids.

Whatever I only work with these 10,000 data points, so I know I know small enough that I can fit it in memory right memory is so large nowadays you can fit a million data points in memory but

let us scale it appropriately right you have a billion data points and you are not able to fit all of that in-memory then reduce it to something small that you can fit in memory right so now what you do after you have done the clustering right.

You will have a new set of centroids for all your clusters right now what you do run through the data once more. I assign each data point to the closest centroid, so essentially you are making two passes through the data once to produce clusters which are very tight right and then you take the representative points and then you do whatever you are you can do an iterative clustering algorithm on top of that so once you are finished with that so what you do you go back and reassign the data points to the nearest centroid.

So the two passes through the data so that is one of the things that they should think about when you are working with very, very large data sets how do you minimize the number of passes through data right so if you can do it with one pass through the data great!
if you cannot we try to keep the number of passes minimal right if think of k-means rated very plain the k-means how many passes are you making through the data as many iterations.

As there are so every time you do an iteration, you need access to all the data points because I need to compute the distance of every data point to the centroid so I need to read the data all over again right so with a very, very large data just going through that might be plain if I can't store everything in memory right so if I want to read the data set all over again from the disk then it becomes a pain.

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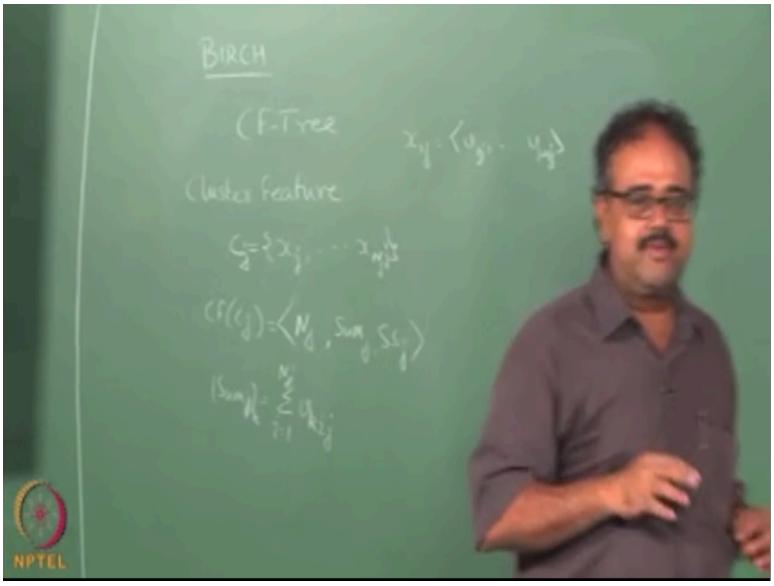


$$C_j = \{x_{1j}, x_{2j}, \cdots x_{nj}\}$$

$$X_{ij} =< v_{1j}, \cdots, v_{pj} >$$

$$CF(C_j) =< N_j, Sum_j, SS_j >$$

So we need to have something better than that right so what they came up with is a data structure called the CF tree came up with a data structure called these an of the tree where CF stands for

clustering feature it is CF stands for clustering feature or cluster feature right so what is the cluster feature it is a three tuple.

Suppose I have a cluster right that consists of some endpoints like right so I will denote the cluster by seeing if the cluster C consists of some endpoints I will denote it as x1 to xn right in fact I should do something more than this I should say probably cluster j consists of cluster j consists of Nj points which I will denote as x1j to xnjj so it is consists of NJ point so this is the crystal so the clustering feature corresponding to the cluster J will be the number of points in the cluster okay.

And the sum of the individual coordinates of the data points suppose my $x_{1j}$ is consists of some $b_1$ up till Vp right will be too many indices but so it is a point my $x_{1J}$ is a point in a P dimensional space right so I essentially each one of them is a point in P dimensional space right so this will be essentially so my sigma ∑ is call the sum j.

$$(Sum_j)_k = \sum_{i=1}^{n_j} V_{hij}$$

And the I will come to that in a minute so some j sum j again a vector wait some j is a vector

and okay so some Kth coordinate of some j essentially the sum of the Kth coordinate of all the data points some of the coordinates in individual coordinates Sum j is a vector sum of the vectors here that is my data points as vector concerns and sum of the vectors the, the reason I did not do that as sumof the vectors is sometimes people have been confused in the past.

When I read some of the vectors, they end up adding all the coordinates
I did not I do not want the summation ∑ also to run over k right so basically if you think of proper vector addition it is just the sum of all the data points right. Still, this notation is

convenient for us when I want to write they will take the square of the individual coordinates and add them up, so these are this is what I call the clustering feature.

And the claim that they make in the BIRCH paper is that this clustering feature is sufficient for you to do some form of hierarchical clustering without actually not knowing the identity of the data points and suppose I give you the clustering features of two different clusters right is there any way you can compute the distance between the two clusters I give you the clustering feature of two different clusters.

I give you CFJ and CFI right can you compute the distance between cluster I and cluster J.
I've given you the sums the squared sums as well as the number of data points so if I divide the sum by the number of data points I get the centroid right I can go ahead and compute the distance between the centroids that gives me one way of measuring the distance between clusters that I do not need to know the individual data points all I need to know is the sum of the vectors that is one thing it said anything else I can do I am sorry I have sum square also we can use that I can do the radius right.

So I am going to leave that tears of homework okay show me that you can compute the radius in terms of some and SS right so I can actually compute the radius of the merged cluster also right based on the sum, the sum of the squared sums of the coordinates right so these are the things that I can do I can even do the diameter ok so I will leave you towards that out so you can do this you can do the centroid distance you can do the radius.

And you can do the diameter distances you can't do a single link complete link ok, so that is gone right but remember what you are doing at this stage is trying to the initial stage with the CF tree is trying to find small clusters the Custer's of small diameter so that is what you are interested but I am not interested in doing single link or complete link at this point I what I am trying to do when I use the CF tree.

The CF tree I will tell you how to build the CF tree a minute so what I am interested at this change is only in data reduction I mainly interested in making data reduction I am not really interested in actually finding the final clusters right so it is fine so we can use either the centroid measure or the diameter or the radius and then keep some kind of a threshold right so what do I do now is I start off from the root ok so the root is going to have so one cluster right.

So you going to have a clustering feature that is going to be there so as, as and when the data points come in the right I start updating the clustering feature at the root right the first data point comes in the clustering feature will be1 comma the data point comma the squares of the coordinates of the data point that will be the first one okay then the next, next data point comes in then what do I do 2, the sums of the two vectors right plus the sums of the squares of the coordinates.

But I do this only if the two points are close together and if they are too far away if the diameter is very far-right very large then I do not merge them I make it into a different cluster, and I actually do the clustering feature there right, so I keep doing this right I keep going until I come to some point where I have too many clusters that I have thrown it at this point right.

So essentially I start breaking the cluster into two breaking this leaf this node into two so some clusters will go here right, and some clusters will go here right and what do I do in

This place I insert a new clustering feature that summarizes all these data points enter a new clustering feature that summarizes all these data points so why am I doing this so when a new data point comes in the right.

I look at these two clusters and figure out which of those two clusters is closer to this data point ok and then root it down the trees right so if I keep all the clusters at one level then every time a data point comes in I will be doing a very large linear scan to find out which cluster the data point should belong to does it make sense and suppose I suppose I start off with one cluster then I produce ten clusters.

Now a new data point comes in I have to check against all the ten clusters to see which is the best cluster to put the data pointed right now what I am trying to do is organize this in a more efficient manner so that instead of taking the order of 10 right it is going to take some logarithmic factor here when I have to make two comparisons will take order of two now depending on how many I put in here.

So I suppose I divided into two and then put five here and we will take order of six right, but then it is a tree becomes deeper then you can see the savings right the savings will be tremendous, so it is you have some kind of a logarithmic compression, so that is essentially what we do here yeah okay so initially what I do is I have 1 node okay.

So I start keeping track of all my clusters here right suppose I reached in clusters let us say 10 is the maximum it I reach 10 clusters a new data point comes in I want to create 11th cluster sorry various factors it depends on what is your memory size right so and in fact, you have to go even further down actually so it depends on your page size not just on in-memory size right want your every leaf to span multiple pages right because a single axis could actually lead to more page faults right.

So you want to see if axis one entry in a leaf I would really like the entire leave to come into the and stay there so there are all these kinds of consideration when you start talking about really large data right you have to start going into the system level of things right so you have to know how things are stored you have to know how things are accessed right, right so here let us say I pick 10 for somehow right I said 10 is the number I am going to store and what are these numbers.

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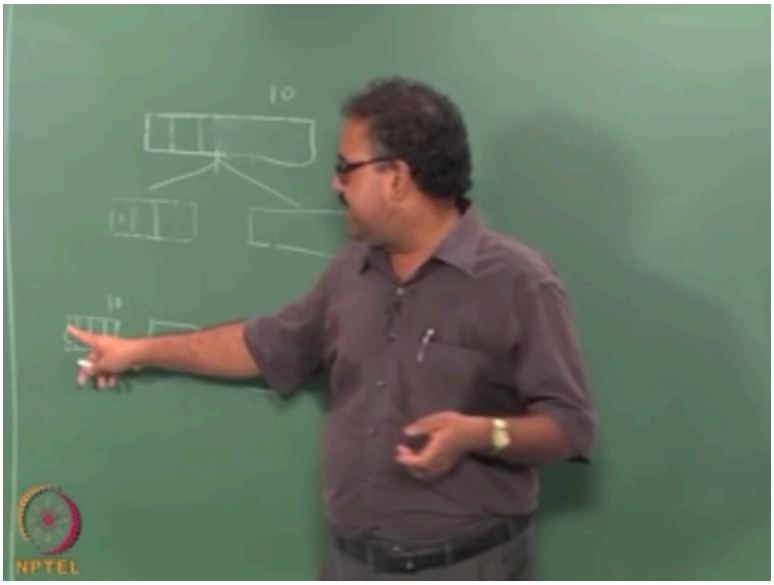


These are essentially three things like this right each one of these only three things like this at some point I say okay even if have a 4kb page even like, like 100 clusters I should run out of space right I will say no, no I want to break it right so then what I do is I split this and I put 10 here and I say another 10 here I am sorry put a 5 here put another 5 here right and then I will start off with two entries here right instead of having 10 entries it all go down I will get two entries here right.

And each one of them will be so the first entry will be summarizing the lab the first child the second entry will be summarizing the second child right now a new data points come in I can try to shove them into this right and then at some point it will come to a point where this thing gets filled up then I will split it into two right so I can do one of two things I can actually push it further down right or you can split it into two and add another entry.

There which is better not by this like 10 rate I mean I have 10 I have space to store 10 CF features there right so I could either make it broad branching at all I could make it deep which is better broad it better be deep is better draw this better so if you choose to go broad when do you go deep then what gets filled up when this gets filled up and then you only push it down from there is it, no, no yes right that is ready to do its work done.

But do not doubt yourself what you said was correct okay I just wanted to see how sure you are up to your answer so essentially you keep broadening it right, and this gets filled up then again you split that everything goes down one level okay so keep doing this and until you have gone through all your data points.
So that the small point here is to notice, I am not doing a numeric example and if people are interested in a numeric example of how BIRCH works.

You can refer to the BIRCH paper so we will put up the BIRCH paper online right we will write? Yes! We will put up the best paper online, okay. So what we what will happen is suppose a data point comes very, very beginning right, so it will get associated let us say with this cluster so, and for that cluster, I compute the clustering feature added to the clustering feature in that cluster right but then later on as I keep going down more and more data points will come here.

And some more data points will come here and so on so forth it is likely that if you if, I try to insert the data point again into this class three right it may go into some other cluster because things move so there is this significant order effect right so sometimes what people do with the BIRCH is after they grow on the tree to some point they stop okay, and then they try to do some kind of rebalancing okay that is just an audition optimization step.

You need not do the rebalancing step you could grow the CF tree in one shot right more often than not it will work so the end of the clustering what you will have is you will have a lot of right lot of leaves at the bottom and each one of them containing say ten clustering features right and these are the final clusters that you want reservation or these your final clusters know right.

You remember I told you the first phase when I am constructing the clustering features the sea of the tree all I am interested in finding on tight clusters that I can then use as representatives for doing a further clustering right so what do I do now is go into each of these clustering features compute the centroid I can do that right divide some by Nj so corresponding to each entry in each one of the leaves can get one data point which is the centroid right.

And then I start with these centroids and do a clustering all over again here I can do whatever I want I can do single link I can do a complete link whatever so because I reduce the number of data points is something small and manageable right I can do whatever clustering method I want on these data points right and then what I do is the final centroids I get after I do that clustering I go back and look at the data all over again right.

So I can the data once when I build the CF three right now is can the data again after I finish my clustering okay and then assign the data to the nearest centroids you need a new point in you can just as any recording video now I need to cluster the whole data right I need to know what cluster you belong to the right so I need to know that right so like I said I cannot trust the identity here.

So initially I said okay this clustering feature got the data point I now so the same clustering feature would have percolated down to the leaf but I cannot say that okay this clustering feature.

whatever cluster it goes to that is the cluster that the data point belongs to because that would have drifted away this is the centroid could have drifted due to the later data points.
So I cannot do that is not the, the mapping between the CF the clustering feature and the data point is not, not static it would have changed or drifted away. Therefore I have to do the second round of assignment people understand BIRCH say I really like BIRCH you know because it is a very simple algorithm and it also allows you to do handle very large volumes of data I usually ask people the implement BIRCH in one of the programming assignments okay

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**Introduction to Machine Learning**

**Lecture-74**
**The CURE Algorithm**

**Prof. Balaraman Ravindran**
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Cure sorry it is its abbreviation yeah the clustering using representative points

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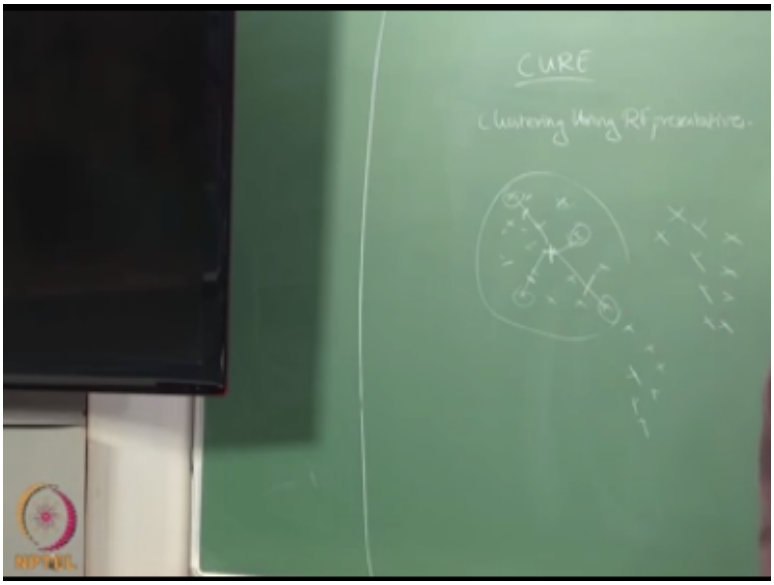


Okay fine so with CURE again was touted as a an algorithm for handling lately large datasets that CURE again this is an algorithm for handling large data sets, so the way you do this here unlike BIRCH where you actually go through all the data points so CURE what you do is you sample a large fraction of the data okay, as large a fraction as would fit in your memory comfortably sample a large fraction of the data and then you do some kind of clustering on that right.

So what you do in the clustering then you after you've done your clustering right, so you start off with some initial clustering of the data right, what you do is let us say I have something like this right some kind of weirdly shaped data like this side once I have done some clustering so I will take the centre okay, I will take the data point farthest away from the centre-right so this is let us say this is a cluster I take the data to point farthest away from the centre let us say that is it okay next I take the data point that is farthest away from this data point.

In that then I take a data point that is farthest away from roughly do not worry I do not measure it with a scale or anything right I will take the data point that is far the farthest away from both of them to put together right, so I compute the distance from both and then add it up and then, let us say that one and then I figure out one more if I need it one more right I will take that one right so now I will take these as may represent that one and take these as my representative points for the cluster.

Right, so a data point gets assigned to that cluster which has the closest representative point basically I am looking at the boundary points, right I am looking at the representatives or things that delineate the boundary of my cluster right I do not take too many points right I take some number of Representatives lately like 3 or 4 representative points at all or say ten representative points so that they do not completely trace out the boundary, but they give me some idea of what the boundaries okay.

So this is why representatives I reassign the clusters I reassign the data points right so I have some clusters, now I find the representative points I find the representative points for this cluster I find it for this cluster now I reassign the data points to clusters so a point goes to the cluster which valve whose representative point is closest earlier we used to reassign them to the close a Centroid now will be assigned them to the closest representative point, now for each cluster I will have 4 points right for each cluster I will have 4 points.

So what I will do now is I will forget the clustered memberships right this is just like how k means works right so k mean what will I do I will reassign a data point to the closest centroid now I will reassign the data point to the cluster that has the closest representative point initiate whatever the whole thing is done out of a set of samples, have a set of samples I drew from the data for points for each cluster Oh yes, among the sample post but this one thing I forgot about the representative points1 point I forgot about the representative points apologize.

So once you identify these guys right so you do not take them as your representative points because they become too susceptible to outliers okay, you do what is called shrinking right you move them some fraction alpha α towards the centroids, let us I find the points at the boundaries right, and then I shrink them a little bit towards the centre-right so that I do not get too influenced by the outliers, and the shrinking is done by a fraction of the distance of the centroid so that the farther away for you are from the centroid.

So the greater the shrinkage, okay, so this hope you find the representative points, and once you find the representative points, you go ahead and keep doing this on the same sample right. So the idea of taking the sample is the sample is small enough to fit in your memory right so that I do not have to go back to the disc to read the data for the second iteration they still a couple of stages to go this is the first stage of CURE at all sample I cluster using this representative points okay, so remember that the sample.

I hope there is representative of the whole data set, but it might not be right I have taken some as large a sample as if can from the data right into my memory, and then I do this clustering around representative points right is it the clustering the representation based clustering is clear right, just like k-means instead of assigning it to the data point with the closest centroid I send it to the data point with the closest representative. It is a parameter yeah cm some m you choose like, in this case, we chose it to be four right, yeah was it as I said you start off with an arbitrary clustering you start off with some arbitrary clustering okay, and then you start pick

representatives find the centroid of that clusters that you have right and then go to the corners shrink them right do not forget the shrinking step.

Shrink them and then you get a representative right, so just like the centroids are not real data points the representatives also need not be real data points okay because once you shrink them, they are no longer so when you find them they are real data points then you shrink them then they are gone right they are no longer real data points okay, now you get m such sorry I will be using all the data points of the point but only the sample, so we did not bring the clustering I am not worried about anything else.

That I have not sampled right now what I do is I keep doing this until I have converged to some clustering right once you have converged to some clustering, okay what I will do is I will remember those representative points suppose I have k clusters I will have M times k points it will remember those k representative points I will forget everything else right, now I will pick another sample from the data I will repeat this, in fact, the recommended way to do this is to partition the data initially randomly into some large number of bins like say k bins at partition the data randomly into k bins and then do run CURE in parallel on each one of those K bins.

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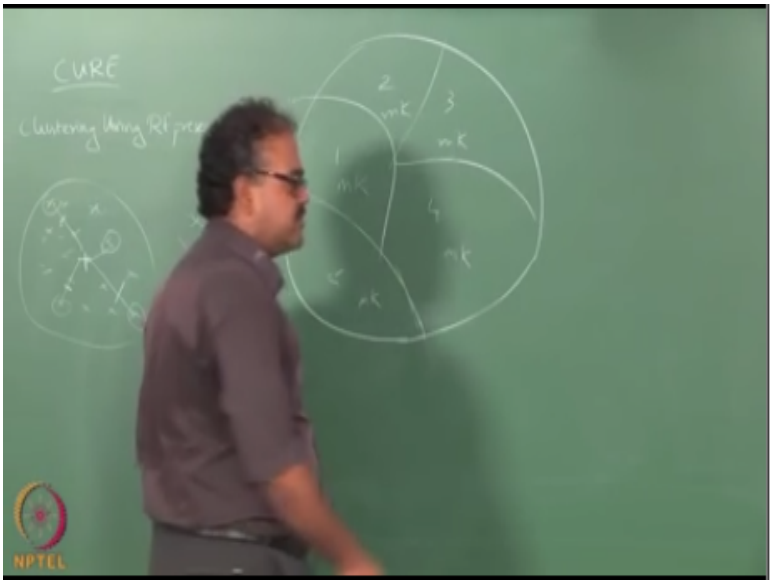


So if you have a lot of machines so what you can do is you can take your entire data set, okay this is not meant to be a geometric representation of the data right, so now when I say I am going to divide it into two parts, I am not saying take data that belong to one part of the input space and assign it is I do not know this right this is randomly split the data right and say that okay here there are five bins again in each bin I will run CURE independently right and what we will end up with some mk points for each of these bins.

Now, what will I do is I will throw in all the mk points into a single clustering problem and one CURE on that mk what about 5, 5 times mk points again, again you will get a set of representative points right for each cluster will get a set of representative points then I will go back and assign each data point to the cluster which has the closest representative point right and on each part I run CURE right I will end up with some clusters right and for each cluster I will have a set of m representative points.

Let us say I end up with k clusters in all of them I will apriori define k right I will end up with k clusters in all of them right and so there will be some mk points from here where m is a parameter that I have chosen already which is the number of representative points let us say 4

now what is happening is I have a small fraction right for each cluster I am going to have 4 points is a small number with my number of data points will be very large the number of clusters will be small right I will have 4 times.

The number of clusters a small number right I have now 5 such divisions so I will have 5 times m times k points case number of justice k clusters just like k means you define k, so but the point here is at every point at no point am I looking at a larger clustering problem right so I will split the data right so I am looking at some fraction of one-fifth of the data or one-tenth of the data, and that is the largest data set size I am going to look at this is why CURE is an algorithm for handling very large data.

So I can take a very, very large data I can split it up 10 ways the second spirit of 15 ways 20 ways and I will do each one of these clusterings. Hence, if have if I am looking for 100 clusters and for representative points, so that is basically 400 points returned from each one of these clustering's right and then what I do is 5 of these right, so I basically end up with 2000 points right, so I have100 clusters for representative points per cluster right and 5 such problems I have solved so end up with 2000 points.

Very, very small number right so I go and run CUREon that again right again I will get 100 clusters on this 2000 points right, bad idea 100 classes on 2000 points is a bad idea end up with 20 points per cluster, so what you should typically be doing is whatever final clustering you want to end up with this should be much larger than that, this K that you use here suppose I want to end up with 30 clusters then I can use 100 clusters suppose I want to end up with 100 clusters I better end up using a larger number 200 clusters or something okay.

So that I love enough data points to cluster finally and then I do the clustering suppose I end up with 30 clusters each will have four representative points I will pick those four representative points I will do another scan through the entire dataset right how many scans I have done through the entire dataset so far one I had done only one when did I do the scan? To do the

partitioning right, so I did the scan once to do the partitioning the second time I do the scan I will have m representative points from 30 clusters.

And I will go back go and assign the data point to the cluster that is the closest representative point right, so at every point, I do not solve the problem that is larger than what I put into this one partition this allows me to do these things very rapidly. If I have multiple processors that can run right, I can do this in parallel this stage can be done in parallel and whatever cluster representatives it returned and then I do in a second round. Hence, CURE is if you do the implementation correctly.

It is rather fast another nice thing about CURE is that because I am doing this clustering around representative points I am not really limited to looking at convex clusters like I have with k-means rights okay means if I have two centroids right so basically this will be the separating FF three or four centres I will basically be looking at some convex shape around the centroid right, but if we have since you have multiple representative points, I can actually have non-convex shapes also disadvantage of using non-convex shapes this is overhead right so if the data is small you really do not want to get into the CURE kind of setup right because the overhead is large.

And you have to maintain so many representative points right so every time finding a representative points involves you getting the centroid and doing this computation to go to the edges and then shrinking them right so it is additional overhead that k-means you just find the centroids and you just move on here you have to do an additional computation, so that is the overhead okay good, so any other any questions on this I said a question is no it usually defeats the purpose I am assigning it to the data point that has the closest representative point right so the centroid is will be anyway surrounded.

By the representative points, I am taking so it will anyway end up going to the same there might be a small change here and there, but people typically do not include the represent the centroid, in the representation, this is that we have now to find a set of two parameters of pair? Yes but

you get some advantages was it right it runs on large data sets which is stop and think about it you could do and claim it run some clusters right, so the and the second thing is you can get non-convex clusters.

So if I have fun funky shapes, funny shape clusters, then you get those in a sense? Representative points are not real data points they are fictional points like centroids so when it does the reassignment of points to clusters, and representative points never get reassign I mean they are not points at all remember I told you to go to the edge you find real points, and then you shrink it by a factor α towards the centre, so just another parameter we have to select so three parameters now it is drink it by some factor towards the centre so, therefore, they are not real points just like the centroid does not really get reassigned neither make the representative points.

So the whole point of doing the shrinkage was the reverse the outliers right, now you could think of using the mid idea more computation without replacement picture yeah without a replacement that is why I showed the partitioning typically sample without replacement and CURE you could I am not saying you cannot I mean see the point is for every, every variant that you propose you have to either convince me empirically that your variant is better or convince me theoretically that your variant is better.

No that is this let is go feat will generate papers right and let me stop and think what is it that it buys you know is there some kind of statistical parameter that will become better by sampling with replacement is a sampling without replacement for example will you get more stable cluster estimates if you sample with replacement, so regardless of what how many ever how do I speak the samples right so if you say that I will get the same cluster centroids maybe then that is a valid thing to do right if you cannot show anything like that.

Then not clear right, so the thing with which has more overhead sampling with replacement or sampling without replacement has more overhead really? With replacement has less overhead if you are only sampling, okay if I am interested in partitioning the data I can just look at some

random permutation of indices right and then just chop the data at some points right so that is like sampling with sampling without replacement, but I have exhausted the entire sample space.

So if I am going to do that for sure right then it can do it efficiently right then that will actually have lesser overhead than sampling with a replacement but why the sampling with replacement have lesser overhead than without replacement when I do it with replaced with the replacement I do not have to keep track of what I have already sampled from doing without replacement I actually have to remove it or allowed to have some bit that sits there.

Let us say this string has been visited so how now how do I change my sampling distribution so that I avoid those data points which I have already sampled yeah, what is it now you can you can pre-roll your sampling yeah so if you are going to just sample one at a time then it is a then it is a problem right if you can pre roll your sampling that is essentially what both of you are saying that you can what I am saying is you a priori generate all the random seeds you want right just to the permutation and then just keep chopping off the whatever is the ID at the topmost is a way of implementing that can be pretty efficient.

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**Introduction to Machine Learning**

**Lecture-75**
**Density Based Clustering**

**Prof: Balaraman Ravindran**
**Computer Science and Engineering**
**Indian Institute of Technology Madras**

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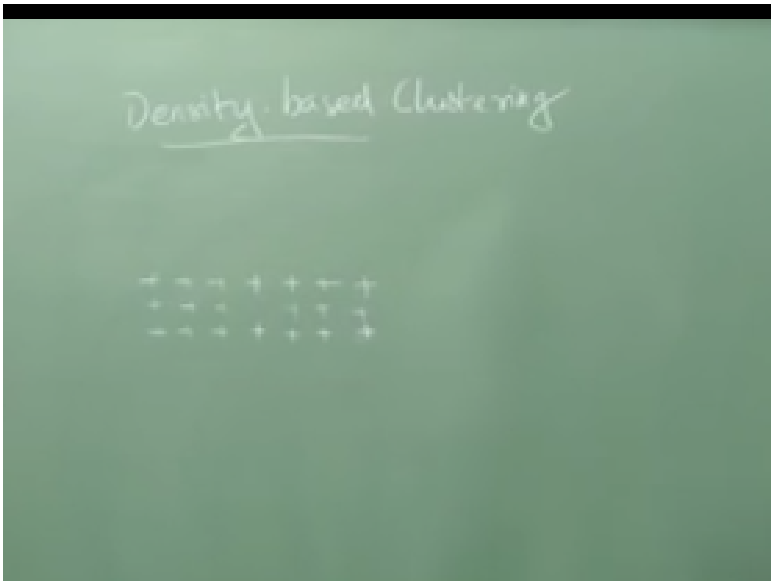


What are the two classes as a row of pluses there in the row of pluses at the bottom, so there are two clusters here right.

So we will not touch these data points right

we do the following,

now do you see two clusters side by side so what essentially defines clusters is not the distance between the data points that are more like the density of the data points.

Naturally, when we think of clusters, it is really that where data points are dense is one cluster. Then we tend to draw the boundary between clusters where the density is lower right so we did not change the data points in the initial set of data points you are very happy to say that okay the cluster or the top and at the bottom right so what cost you to change your clustering when added the additional data points.

The density went upright the density went up differently, therefore, you said oh okay the least dense point is no longer between the two but vertically right no longer horizontal but vertical was the low-density point right.

If you run k-means, you might get anything we do not know and depend on where you start with right.

So the question is if you have your intuitive notion of clustering has to do with density so why do not you try to come up with a clustering algorithm that captures this notion of density right and try to come up with a clustering algorithm that captures this notion of density.

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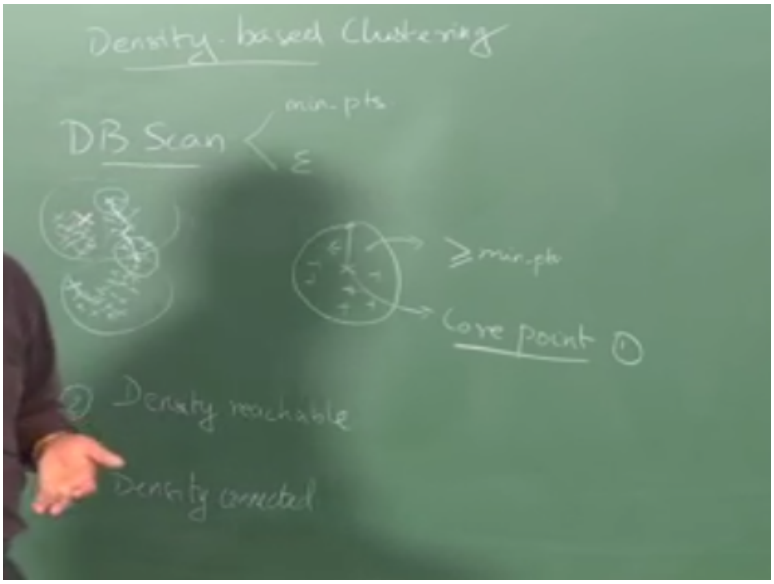


So there is a very popular clustering algorithm called DB scan,
It's a very popular algorithm called DB scan that does density-based clustering okay so Dbscan has a lot of terminologies that they define right once all the terminologies are described then the clustering algorithm itself becomes nearly trivial okay.
but the terminology takes a while to get through, so the basic idea is very simple right
suppose we have data points like this right.

You see two clusters here clearly two clusters right incredibly hard to get k-means return these two classes and if you run k-means what will happen is you will end up with the centroid somewhere here right another centroid somewhere here right and it will say these data points are one cluster and these data points another cluster is the biggest drawback with k-means right.
so what DB scan says is that two points belong to the same cluster okay.

If we can get from one point to another right by moving only through only through dense regions only through points that are close by right two points belong to the same cluster right suppose we take let us take let us say we take this point let me take this point right in fact they look pretty far away and if you look at the direct distance between them they are pretty far away in fact the points of other cluster which are closer to this than this point right but when you look at it you think that they this thing is one cluster this thing is another cluster.

So what is the intuition here is that can keep hopping at no point do we take a very big hop right we can keep hopping to things that are nearby and we can go from here to here right, so if you take these two points the blue and the brown one right so if we take these two points the no way we can hop from here to here right because there is a this nice gap here right there is no way we can get from here to here only by going through dense regions right is that clear.

So that is the intuition that we are trying to capture here so what is it that we should define now first what we mean by a dense region right so that is essentially what we have to define so we will start off by defining something called there are two parameters that DB scan uses one is called min points other one is ε (epsilon) there are two parameters so min points essentially gives you some kind of a threshold on how many points would you consider as being dense right.

And ε (epsilon) gives you the area over which you will perform the count is a min point says okay if you have five points okay then you are in a dense neighborhood but where do we count these five points okay in a radius of ε(epsilon) around me you count the five points okay count in the area ε (epsilon) around me if you find five points then you are in a dense region and if you make the ε (epsilon) very large then it might encompass my entire input space then everybody will be dense so it does not make sense.

The ε (epsilon) has to be small likewise if we make my min points 1 point 1 not 2 points that means everything will look dense unless make me ε (epsilon) very small so these are actually complimentary things we can control my min points and we can make them in points very small right and then make the density high right or we can make my min points large with the larger ε (epsilon) and that it then also we can make me density high the effects that you will see are different for both right so we let you think about it.

Which one we mean what is the effect of increasing min points versus decreasing ε (epsilon) okay so essentially what we am saying is okay take a data point right take a radius ε (epsilon) around it okay that that is that is a circle so take a radius ε (epsilon) around it and count the

number of points okay if this count is greater than min points okay then you call this a core point take a data point take a ready ball of radius ε (epsilon) around the data point right.

Count the number of points right if the number of points is greater than or equal to min points right number of points is greater than or equal to min points in that ball of ε (epsilon) then you call this a core point right a core point is a point that lies in a high density region that is the definition we have right.
So we say a point is you see that the point is density reachable okay.

A point is density reachable if there is a core point from which you can reach this point it went by traversing only through core points so this might not be a core point because is it the border rate we draw in the radius around it we get only one point here okay so this might not be a core point right but then if we start here let us say let us say this is a core point for sure a lot of points in the anything from here i can basically move two points within ε (epsilon) of itself.

Which are internal core points right so we can move to you can move to core points right and finally reach this that no point we should be making a jump greater than ε (epsilon) right because it is starting a core point we will have enough points in the neighborhood that we can actually jump to something within ε (epsilon) right so I make steps of size ε (epsilon) and I actually go through core points every time then we call them density reachable say point I is density reachable if there exists a core point from which I can reach here.

By jumping only from a core point to core point until the last step obviously every core point is density reachable because it Is reachable from itself right every core point is density reachable and then there will be these border points right which are density reachable from core points there might be other points which are not density reachable from core points which are essentially outliers okay right.

So these are the definitions we have so this is the first right these are the two quantities we need this is these are not definitions this is the first definition what is a core point second definition

third definition is density connected case I will say two points I and J are density connected if there exists a core point k from which both of them or density reachable.
That makes sense right so what is density reachable. I start from a core point and only move to core points right until I make the last hop to this case so no point I will be making a move greater than ε(epsilon).

And all the points I visit in the on the way will be core points ok that is density reachable density connected is if I and J that exist one core point from which both I and J are density reachable then the I and J are density connected okay
here is the next thing. So I and J are in the same cluster if and only if they are density connected. this is the definition of a cluster two points I and J belong to the same cluster if and only if they are density connected make sense?

Sorry! How do I implement it ? this so I start off with YS any point right I pick up a random point right I figure out whether t is a core point or not right then okay it was a core point great so I will keep that as my starting point for the cluster how do I determines the core point I pick up a point look in the neighborhood figure out if there are ε (epsilon) if there are min points within a neighborhood of Epsilon if that is the case then I will keep it right.

Then what I do is I look at all the neighbors of that point and look at all the neighbors of the point and each point in turn I will check whether it is a core point or not right, so each point in turn I will check if it is a core point or not so any additional points I encounter when I do this check I throw it into my Queue so I will keep going right if I reach a point which is not a core point okay so I will not insert the neighbors of that new neighbors of that point I will just stop there if we reach a point is not a core point I will just leave that exploration go back to my Queue to see if anything else is still there.

So I keep doing this until my Queue becomes empty so all the points I have examined from the time I started till my Queue became empty go into a single cluster sorry like it that first search moves like a depth first search you do that all these data points go to a single cluster now what I

do I go and start at a random point which has not been assigned a cluster so far and then do it if first search again till I find the whole clusters I do this and I am done.

So the nice thing about this is I am really doing only one scan through the data right so every data point I will actually look at it once right I will examine the neighborhood right and then I will go on but then the number of computation I will do will be still significant so DBscan is a slow algorithm even though I examine each data point only ones but the amount of computation I do an examiner data point is significant.

Because I am looking at the radius ε (epsilon) and then we have to find all the neighbors within the radius so unless I have a very efficient data structure that will return to me the nearest neighbors very quickly right so this can take a significant amount of time in running so there are some efficient implementations of DB Scan out there it is really cool in that it gives you all kinds of arbitrary clusters right.

And so these kinds of things today whatever I drew that that you would not be able to recover using k-means or even hierarchical clustering depending on the kind of cluster measures that you choose right cure might be able to give you or might not be able to give you this kind of clusters depends on again how your sampling that you do and what you start off with and so on so far so whole bunch of imponderables.

But again the same thing with DB scan depending on which is of min points in ε (epsilon) right you might get very weird results, right yeah so if you look at the data mining text book by Han and camber right Michelin Cameron book right so they have actually horrendous examples of DB scan in fact I think they did a significant optimization to find out which are the worst parameters possible form in points in ε (epsilon) and they give you results.

Because they wanted to write a paper that said hey we have an algorithm that does better than DB scan so they said oh DB scan can perform really badly if you give it bad parameters so let us give it back parameters and then we will beat it right. So I think this is assuming that they were

actually being more fair than that but the way they look at the results it looks like that I mean DBscan looks really horrendous and they are a method which is called chameleon.

It is an acronym yeah so I think C stands for clustering I am not sure good I sure about that either so yeah so yeah it is a nickname DB scan right I just start off with some arbitrary point and then I look at look at the exam examine the clusters and soon so forth, so what happens if I happen to start off at a boundary point instead of a core point I will examine it I will say okay it is not a code point I will throw it away right so it will never get assigned to any cluster.

So it will kind of be all by itself as an outlier even though it should belong to a particular cluster that is one thing the second thing is suppose I want to vary ε(epsilon) and I want to run the clustering again I basically have to do it all over from the beginning right so what optics does this gives you a clever way of ordering your data points such that right for different values of ε (epsilon) you can recover the clusters very quickly for the same value of min points min points is fixed but ε (epsilon) changes right.

So it gives you a way of ordering your scan through the data such that for different values of ε(epsilon) I can recover the clusters very quickly and is a really cool idea right.

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**Introduction to Machine Learning**

**Lecture-76**
**Gaussian Mixture Models**

**Prof. Balaraman Ravindran**
**Computer Science and Engineering**
**Indian Institute of Technology Madras**

I will be talking about Gaussian mixture models and the Expectation-Maximization algorithm.

(Refer Slide Time: 00:21)

The plan is to start with introducing Gaussian mixture models and then talk about mainly how we estimate parameters for a Gaussian mixture model and then through that introduce what Expectation-Maximization is because that's the iterative algorithmic framework that we will be using for parameter estimation. That's in general and then we will come back to Gaussian mixture models and see how EM can be used for Gaussian mixture models and then talk a little bit about theoretical properties of EM and why it's interesting.

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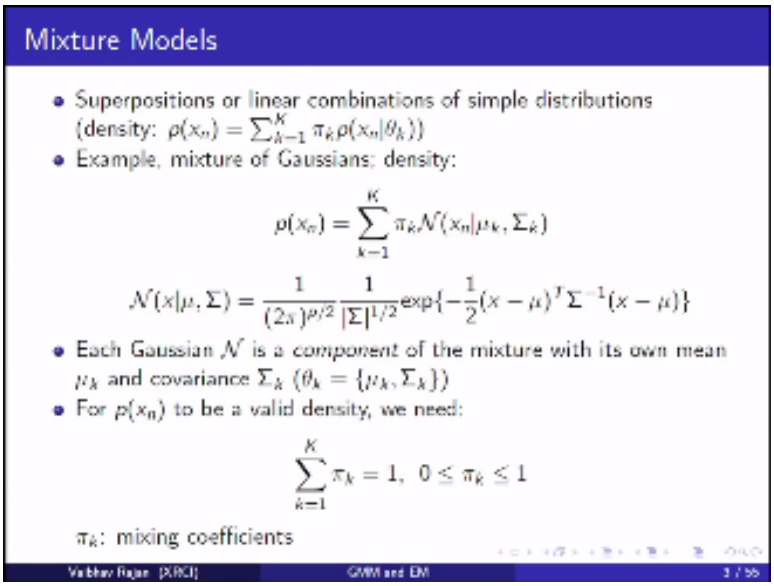


Mixture models, as the name suggests, are a mixture of models. Formally, they are linear combinations of distributions. So they typically have a form like this:

$$density: p(x_n) = \sum_{k=1}^{K} \pi_k \, p(x_n|\theta_k)$$

The density of a mixture model is a linear combination of other densities $p$ and different mixture models will have different forms for the probability distribution here.

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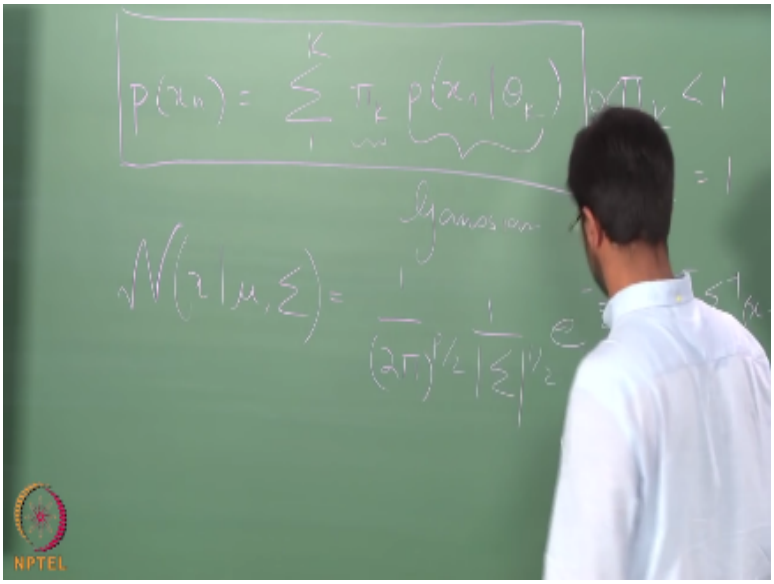

$$p(x_n) = \sum_{k=1}^{K} \pi_k \, p(x_n|\theta_k) \tag{1}$$

The density is given by a linear combination of different probability densities and here we have k components and each of these components have a mixture weight (or a mixing coefficient) denoted by $\pi_k$. So this probability here can assume different parametric forms. The most common is the Gaussian. And when it follows a Gaussian, this is called a Gaussian mixture model.

The Gaussian mixture model is one of the most commonly used mixture model. It can seen in a lot of different domains such as bioinformatics and speech processing. One of the reasons why it is used is because it is mathematically tractable but there are other nice properties too. I guess you all know what a Gaussian is but let me write it down anyway because we will use this very often in the coming few slides.

$$N(\mu, \Sigma) = \frac{1}{(2\pi)^{\frac{p}{2}}} \frac{1}{|\Sigma|^{\frac{1}{2}}} e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1} (x-\mu)}$$

So this is the form of a Gaussian. I am assuming you all know this but let's keep it.

So each component here is a Gaussian and each of these Gaussians has its own parameters - the mean parameter and the covariance parameter. And for equation (1) to be a valid density, we need the $\pi_k$s to be between 0 and 1 and also the sum of all the $\pi_k$s to be exactly equal to 1. We can show this mathematically:

$$\sum_{k=1}^{K} \pi_k = 1 \quad , \quad 0 \leq \pi_k \leq 1$$

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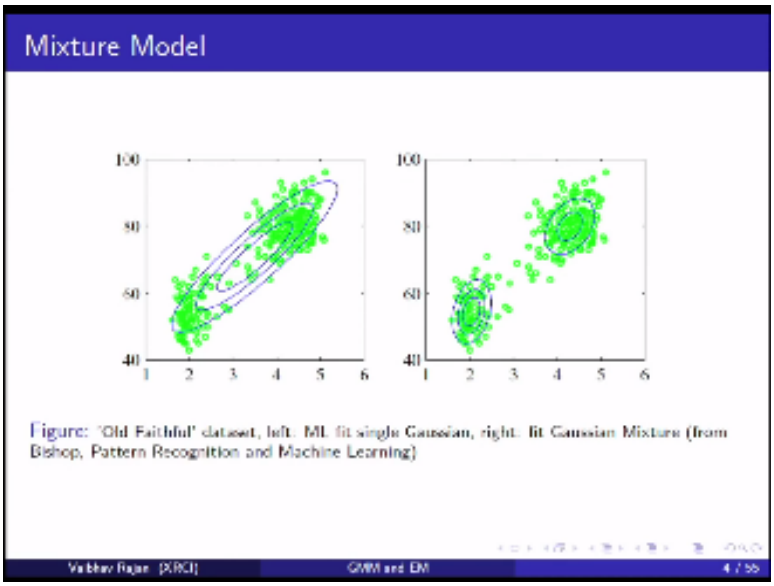


So why do we need these mixtures? Why do we need these superpositions of densities? Here is an example from the Bishop's book. It's a very well known data set – "Old Faithful" dataset. It is a two dimensional dataset and we have plotted the dataset in green points. On the left, we try to fit a Gaussian on the dataset. Now, visually it clearly looks not okay because the Gaussian is most dense around the mean but when you see the plot on the left it doesn't look like the data is most dense around the mean.

But instead of a single Gaussian, if we use two different Gaussians and try to fit a mixture of two Gaussians to this data as shown in the plot on the right it looks somewhat okay. The data is dense around the means of both these Gaussians and it looks like a mixture of two Gaussians would be a good fit for this data. So let me show you some more examples.

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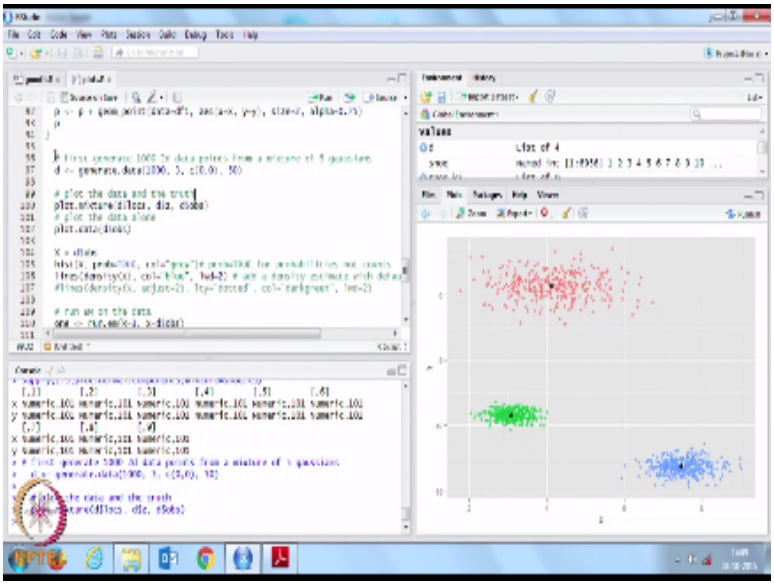

This is some R code to sample data from a Gaussian mixture. I am going to sample and then plot the data. Since I have set the number of components to 3 in this case, every time I sample data, it is sampling from three different Gaussians and you see a clustered kind of data which has three clusters.

If I change the number of components to 5 or 6, you will start seeing more clusters. The clusters need not be well seperated as shown here but can be overlapping as well. The first thing that comes to mind when you try to model data, with such a clustered kind of strucuture, is to try to use Gaussian mixtures because Gaussian mixtures can nicely fit such clustered data.

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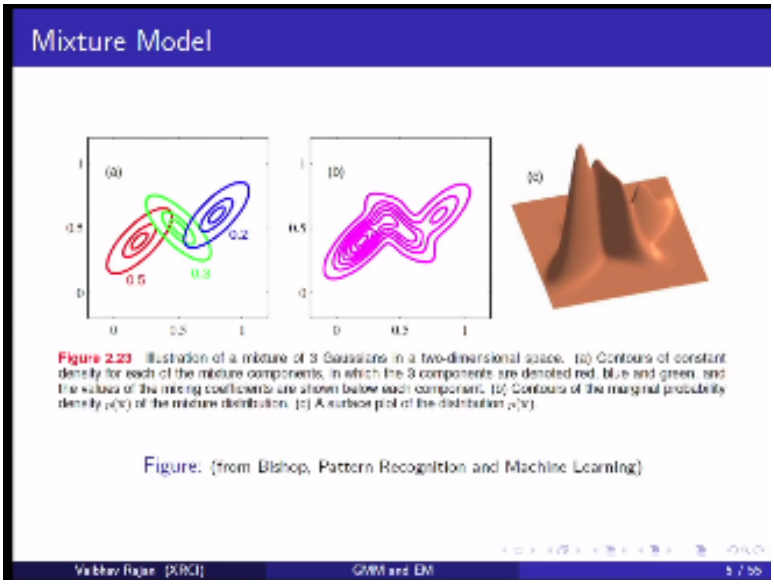


This is another figure from Bishop's book, which is an illustration of three different Gaussians. As mentioned earlier, these Gaussians need not always be well separated. In this case these Gaussians are overlapping because of the choice of mean and variance. From the subfigure on the left, you know that or you are being told that there are three different Gaussians. However, when you look at the data and try to plot the fitted density, it looks similar to the subfigure in the center. The subfigure on the right is the surface plot of the distribution.

Another thing that you should observe from the subfigure on the left is that it was generated from a three-component Gaussian mixture with weights 0.5, 0.3 and 0.2, which means that the red Gaussian is contributing most of the mass. It is again observed in the subfigure in the center when you plot the density. So the probability at mean of the red gaussian is the highest and then a bit lower for the green Gaussian and the lowest for the blue Gaussian. Let me show you some more examples of these densities.

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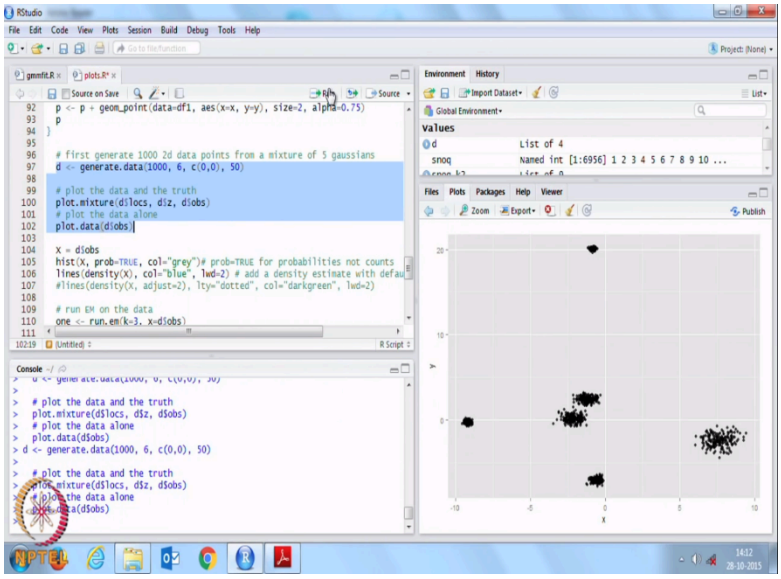


I generated the data shown here from six different components. Let me reduce the number of components to three.

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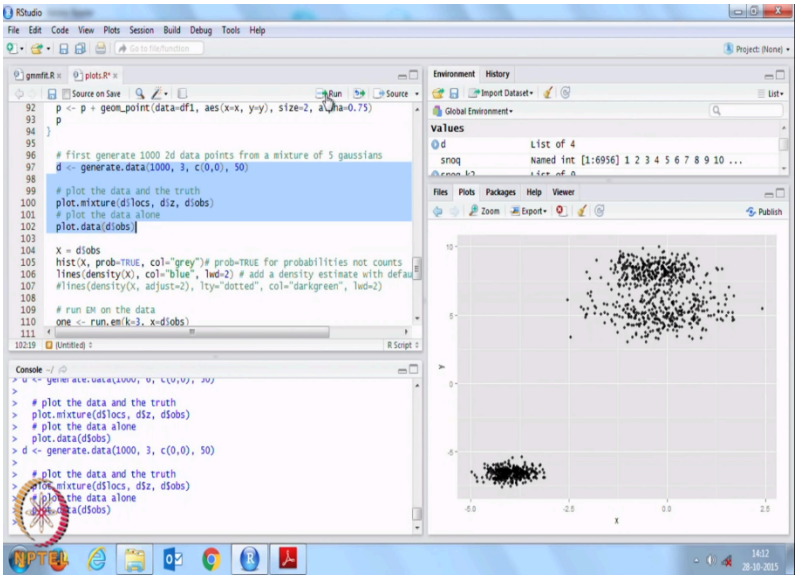


So here there are three components but two among them are highly overlapping.

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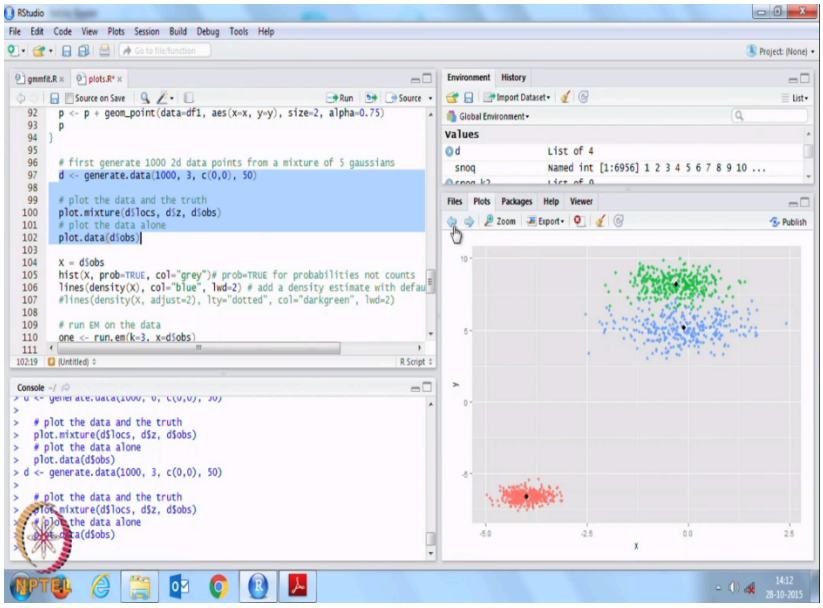


So if you see how the data shown previously was generated, it was generated by three Gaussians as shown here.

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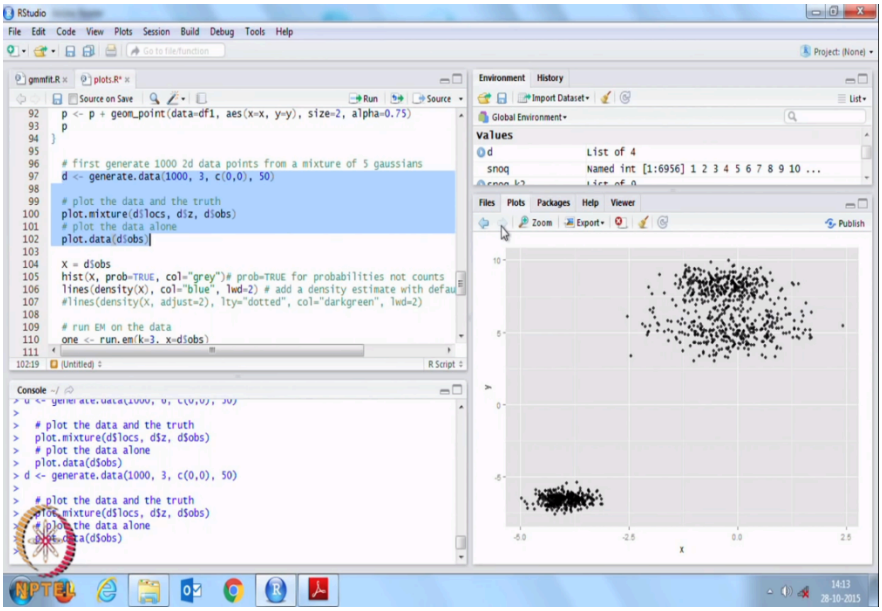

But when you look at the data, you don't know how it was generated. It looks like as shown here. So sometimes it may not be apparent that there are exactly three different components.

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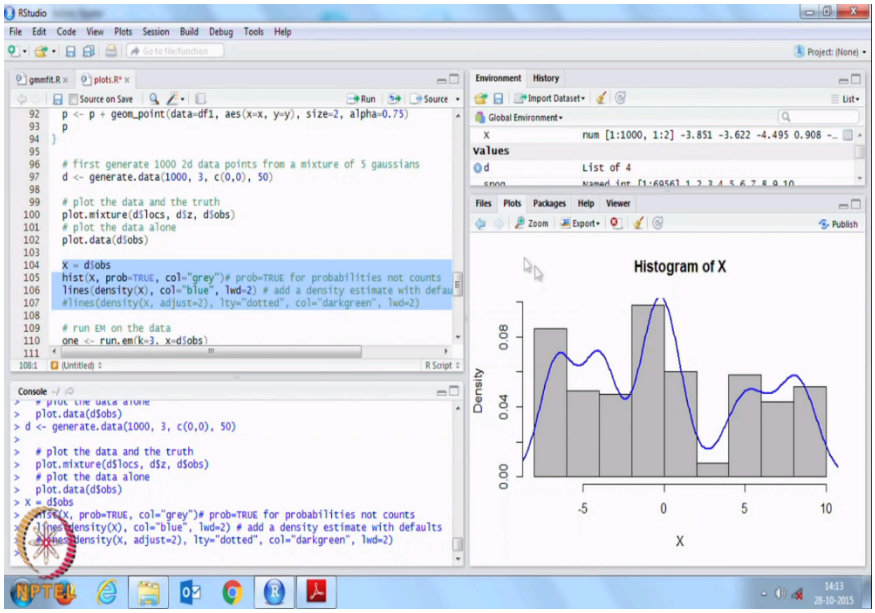


And when you plot the fitted density for the data shown previously, you typically see a density curve as shown here, which has multiple modes. Now this is a fitted density for a three component Gaussian. You see it does not necessarily have exactly three modes. So it depends on the samples that you have. So let us see a few more density plots.

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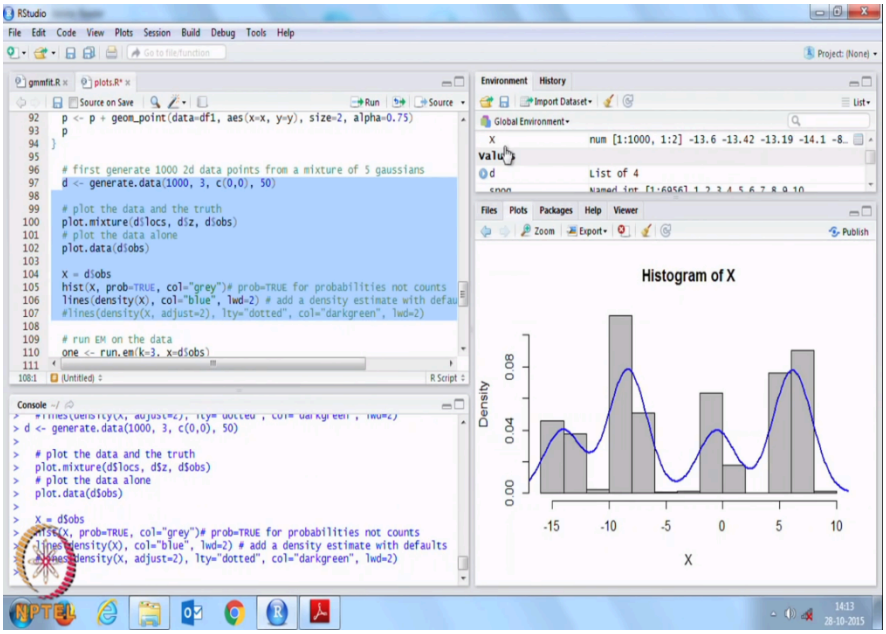


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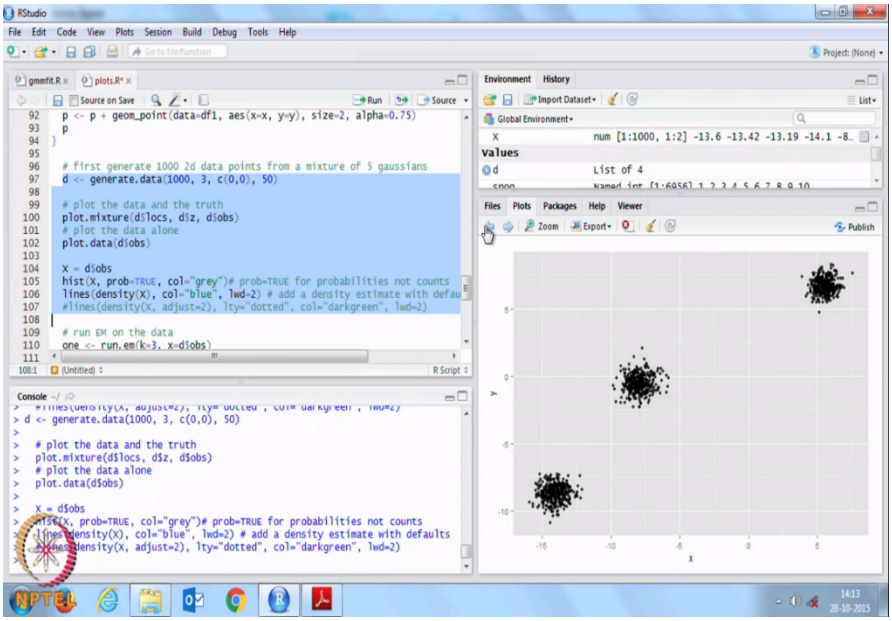


The data that I generated is shown here and the fitted density is shown in the previous figure. The fitted density has four modes.

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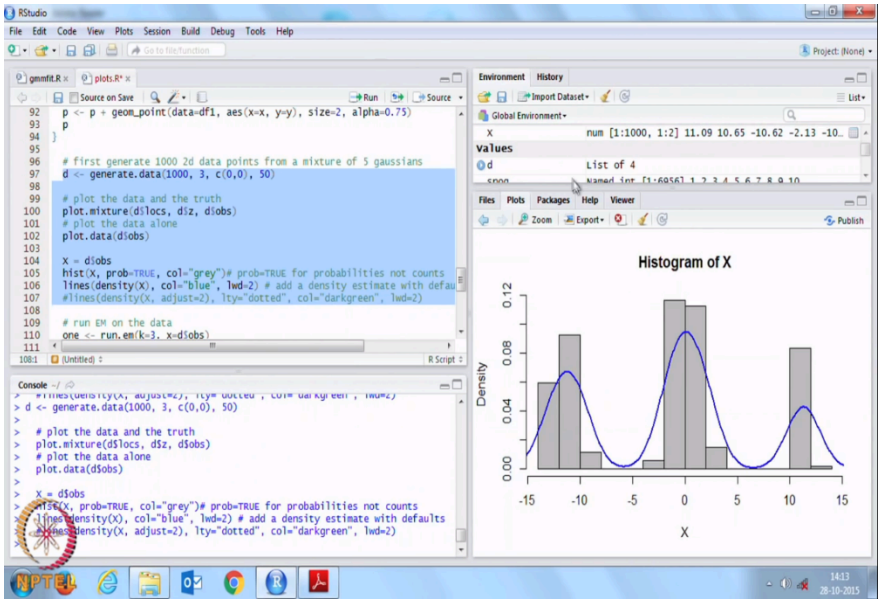


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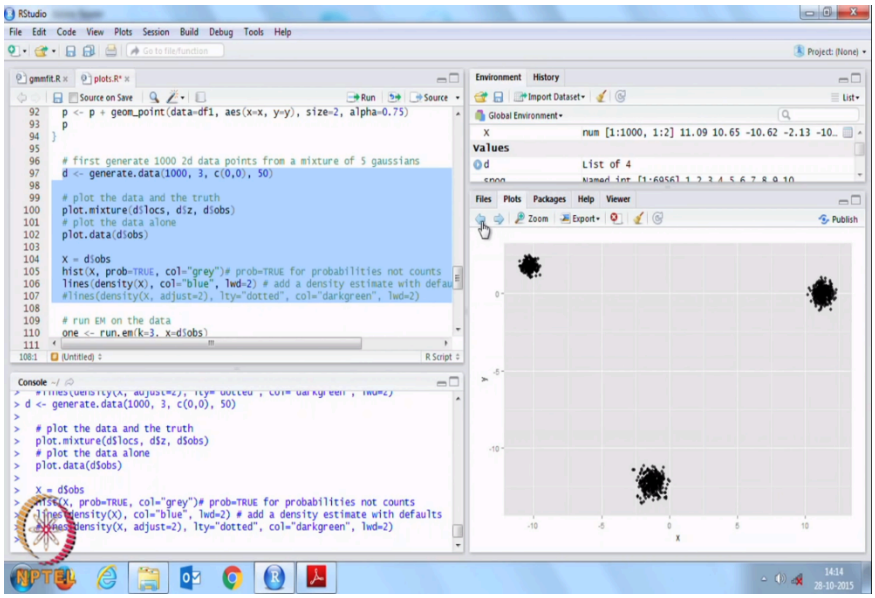


Upon running again, I generate the data shown here. It has exactly three modes as shown in the previous figure. The data is very well separated and forms three clusters.

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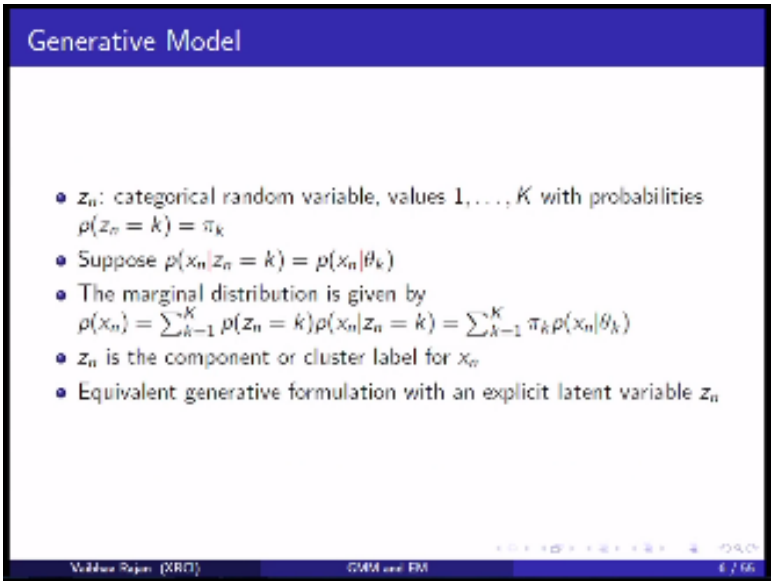


We can also formulate equation (1) as a generative model for selecting a component. Once you select a component, you have selected the Gaussian corresponding to that component. As you know the parameters of the selected Gaussian, you then use them to sample data from that Gaussian. So that would be a generative model for a Gaussian mixture.

To make it more formal, let us take $z_n$ to be a categorical random variable which takes values from 1 to $K$ with the probability of $z_n$ equal to $k$ being exactly equal to $\pi_k$.

$$p(z_n = k) = \pi_k$$

Suppose that the probability of the data $x_n$ given $z_n = k$ is the probability of $x_n$ given that you know the parameters for that particular component. I express it as:

$$p(z_n = k) = p(x_n|\theta_k),$$

where $\theta_k$ represents the parameters of the kth component.

So the marginal distribution can be expressed as the probability of $z_n = k$ (i.e., you select the component) times the probability of $x_n$ given $z_n = k$ (i.e., the probability of $x_n$ coming from exactly that component). By what we have assumed, the first probability is $\pi_k$ and the second probability is probability of $x_n$ given $\theta_k$.

$$p(x_n) = \sum_{k=1}^{k} p(z_n = k)\, p(z_n = k) = \sum_{k=1}^{K} \pi_k\, p(x_n|\theta_k) \qquad (2)$$

So this is an equivalent generative formula with an explicit latent variable $z_n$.

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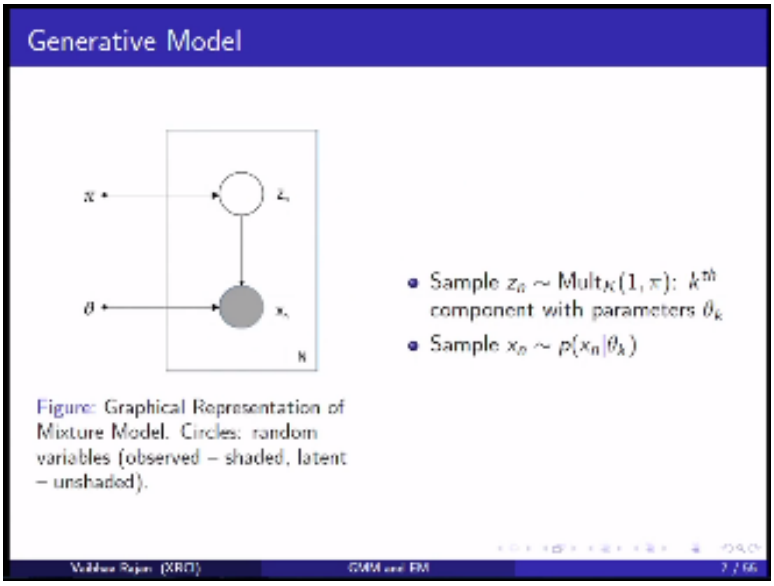


We can represent this graphically. The shaded circle shown here represents the observed data $x_n$. The unshaded circle represents the latent variable $z_n$. The latent variable is governed by the parameter π. Once you know the latent variable $z_n$, the observed data $x_n$ is generated by using the known parameters from θ corresponding to that latent variable $z_n$. So if you have to generate

data from such a model, you first sample $z_n$ from the categorical distribution, which just a special form of multinomial distribution with parameter 1.

And once you samples $z_n$, you get the $k^{th}$ component with parameters $\theta_k$ and then you sample $x_n$ from that probability distribution. So in our case the probability distribution is a Gaussian but it need not be Gaussian. It could be exponential or any complicated probability distribution.

We are assuming that there is a joint distribution and that the marginal distribution representing the probability of $x_n$ can be written as shown in equation (2). For each of the k components, we take the probability of $z_n = k$ and then the probability of $x_n$ being generated from that component. So this is exactly what the graphical model shown here represents. If we write down the probability distribution represented by this graphical model, it will be exactly as shown in equation (2), except that there is a summation term that takes care of all the different components.

But for a single component, it is the term inside the summation in equation (2). We are choosing a component and then sampling the data point given the distribution parameters for that component. So remember that the mixture model is just another distribution. One use of it as demonstrated earlier is to model clustered data but you can also model other kinds of data with it. As an example let's look at another data set from the web.

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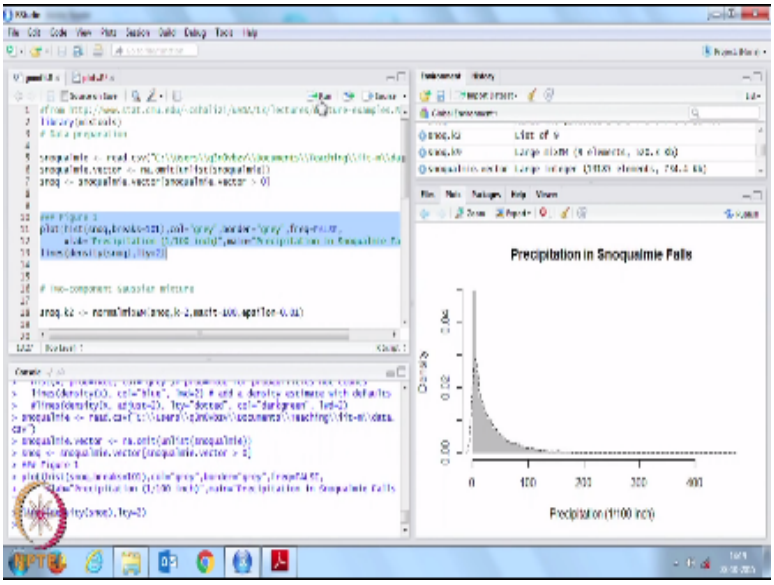


So shown here is the fitted density curve for a data set which records precipitation in the Snoqualmie falls. Suppose we want to model this density, we can model this with a Gaussian mixture. Let's see how it looks when we do it with a Gaussian mixture with two components.

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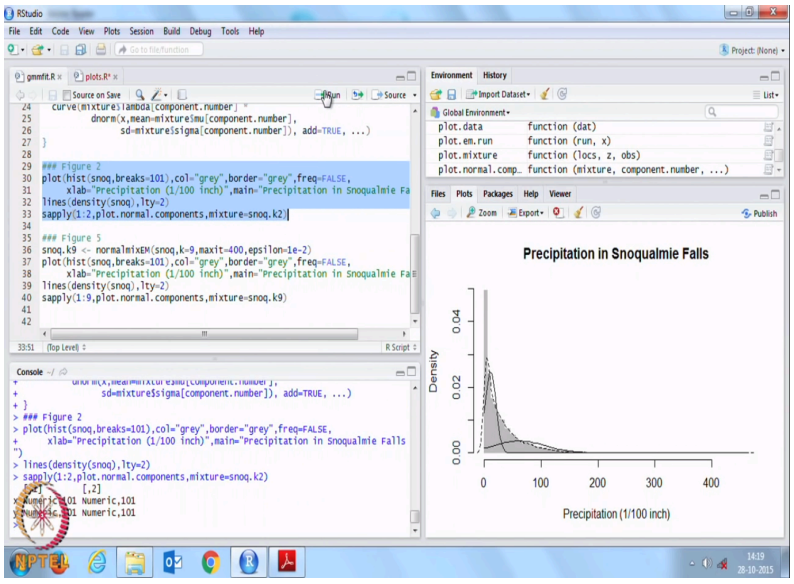

So we are trying to model the data with a Gaussian mixture. We use a Gaussian mixture with two Gaussians as shown in the figure. The Gaussian mixture with two components is unable to the model the data well. However, we can increase the number of components.

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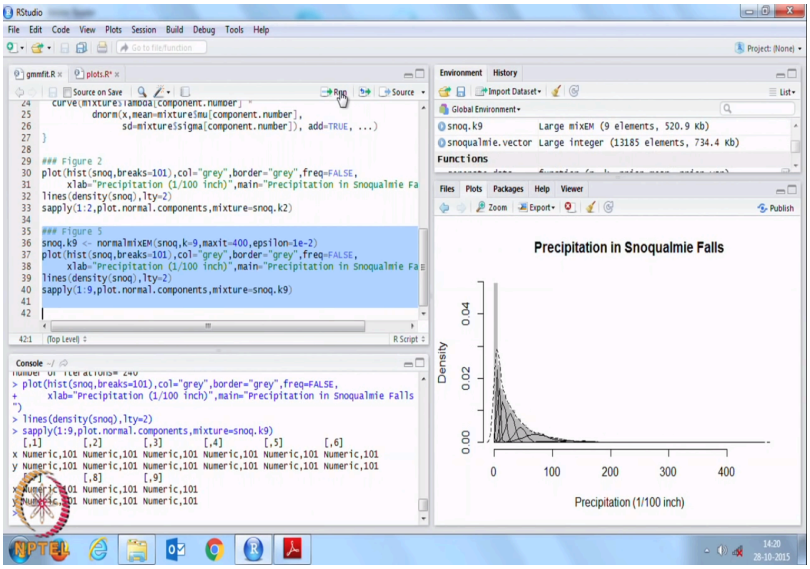


When use use nine components, the fitted distribution gets closer to the true distribution of the data as shown here. So a Gaussian mixture with nine components, in which each of the Gaussians have different mean and covariance parameters, models the data reasonably well.

So Gaussian mixture by that way is very versatile. It can model a lot of different distributions by just choosing the right number of components and choosing the parameters appropriately. So it models not just clustered structure.

When we are fitting the model, we want to estimate what the right number of components is and also estimate what the parameters corresponding these components are. So if you are just given the data, we don't know what those parameters are. The bulk of the lecture talks about the estimation of these parameters.

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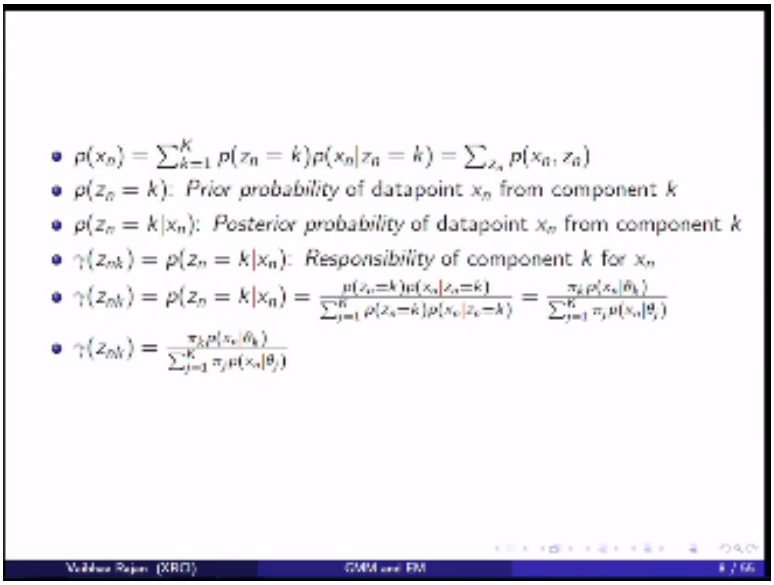


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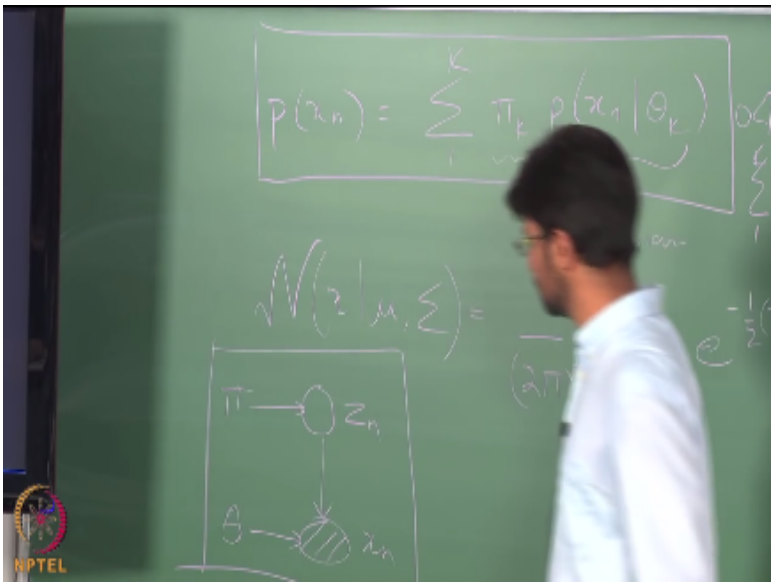


It would be good if you keep the graphical representation of the mixture model discussed above (and also shown here) in your mind as we go through the lecture because all the maths that we will see will sort of start making sense when you have this model in your mind. So when I talk

about some formula like the one in equation (2), the graphical representation can help us see that it is just the probability of $x_n$ being generated.

The generative model is usually more easy to think with. So the generative model would be that I choose the component $z_n$ and then once I choose the component, I choose the corresponding parameters from θ and generate the data point $x_n$.

If you are doing a clustering task, these components are the cluster labels. So suppose you have three clusters and you want the cluster labels, if you fit a Gaussian mixture model there, the component values 1,2 and 3 would just become the cluster labels in that case. So this would be a probabilistic way of doing clustering.

The probability of $z_n = k$ (i.e., the probability of $z_n$, the latent variable or the cluster label taking a particular value $k$) is the prior probability of the data point $x_n$ coming from the component $k$.

$$p(z_n = k): Prior\ probability\ of\ datapoint\ x_n\ from\ component\ k$$

Now suppose you are given some data set like the "Precipitation in Snoqualmie falls" data set, and you are asked to find what is the label $z_n$ for the corresponding data point $x_n$.

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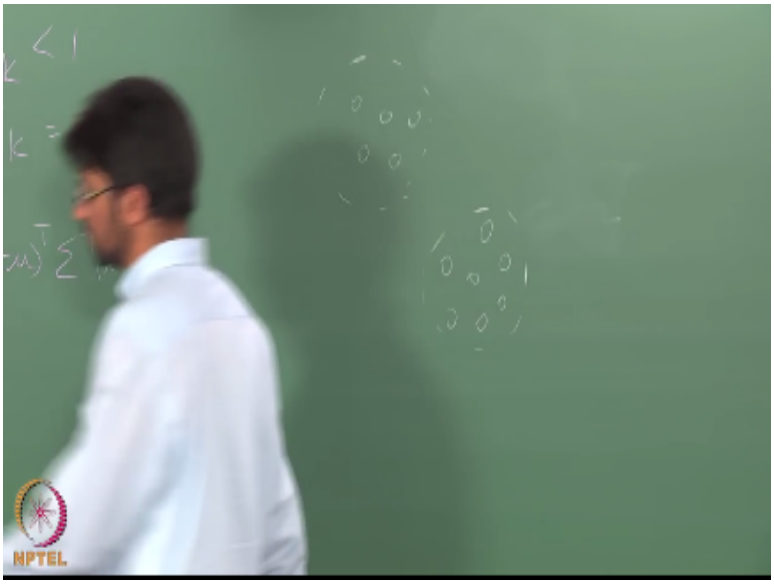

Suppose you have the data points as show here, which have two clusters. If you knew how the data points were generated, then we would able to assign the data points within the dotted circle above to the same cluster (say having cluster label 1) and assign the other points within the dotted circle below to the another cluster (say having cluster label 2). However, you do not know how the data was generated.

So once you are given this data, you have to infer what these parameters (π and θ) are and given that you are using the mixture model to fit the data, you have to infer the $z_n$ value for each of the data points.

So the $z_n$ values for all the data points in one component will have the same value (say $z_n = 1$) and the $z_n$ values for all the other data points in the other component will have the other value (say $z_n = 2$). Of course, for clustering the cluster labels can be interchanged among the components.

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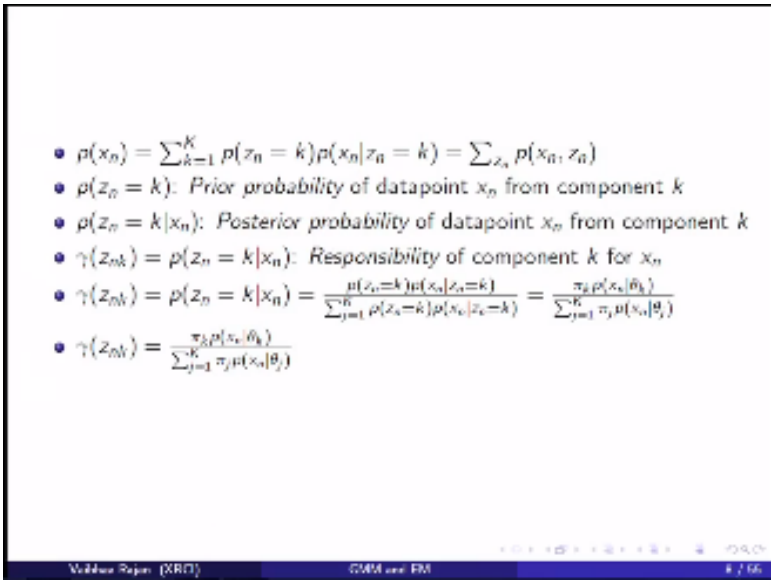


So this probability, the posterior probability of the data point x n coming from component k is so important that it is given a name of its own. It's called the responsibility. I am going to write that down as well because we will you reuse it again and again and again. Oh, he asked me not to use this.

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$$\gamma_{nk} = p(z_n = k|x_n)$$

The equation represents the posterior probability of $z_n = k$ given the data. It represents the responsibility of component $k$ for data point $x_n$.

So until now, I have described it in a way that, there are these different components and only one of these components is responsible for giving rise to a particular data point. However, that is from the generative point of view. But if you look at it probabilistically, each of the components is contributing something towards the probability of that data point and the weight of that contribution from component $k$ is given by $\pi_k$.

So your clustering need not always be a hard clustering (of this component versus that component), it can be soft clustering as well, where the cluster label can be probabilistic. For example, it can be cluster label 1 with probability 0.5, cluster label 2 with probability 0.3 and so on. So when you express it as a probability, i.e., $p(z_n = k|x_n)$, then you get the option of doing both hard clustering as well as soft clustering.

So now you can use Bayes rule to get derive the formula for responsibility.

$$\gamma(z_{nk}) = p(x_n) = \frac{p(z_n=k)\, p(x_n|z_n=k)}{\sum_{j=1}^{K} p(z_n=j)\, p(x_n|z_n=j)} = \frac{\pi_k p(x_n|\theta_k)}{\sum_{j=1}^{K} \pi_j\, p(x_n|\theta_j)}$$

So the equation is straightforward. When you substitute for the prior $p(z_n = k)$ in the LHS, you get $\pi_k$ in the RHS. Also, the probability of $x_n$ given that you know the component (i.e., $p(x_n|z_n = k)$ ) in the LHS can be substitued with $p(x_n|\theta_k)$, where $\theta_k$ are the parameters for $k_{th}$ component.

Observe that you don't know the responsibility values, $\gamma(z_{nk})$, until you know all the parameters. In addition to the value of $k$, you need to know all the $\pi_k$s and $\theta_k$s for all the k components, which in the case of Gaussian is all the $\mu_k$s and $\Sigma_k$s.

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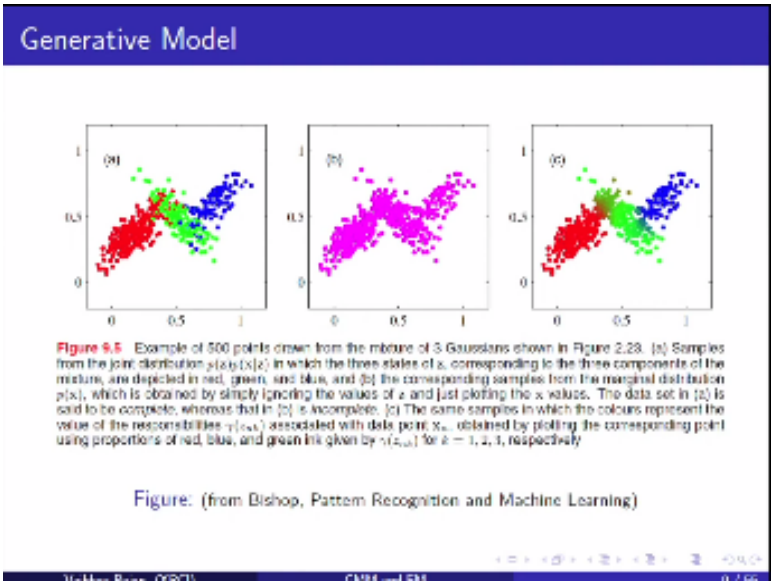


So here's another very interesting picture from Bishop's book. The data was generated from three different Gaussians - the red, green and blue Gaussians. However, you don't know how the data was generated and when you see the data, you see something similar to that shown in the subfigure in the center. Then you try to fit a mixture model with three Gaussians on to the data, and plot the repsonsibilities for each of data points as shown in the subfigure on the right.

So the data points plotted with pure red colour have been given the responsibility corresponding to component 1 (component red). The second component is responsible for all the data points plotted in pure blue colour, and component 3 (green component) is responsible for all the data points plotted in pure green color. However, in the border between these components, the data points are plotted with a mixture of green and blue (or red and green) colours depending on probabilities values (or reponsibility values) for the data point corresponding to each of the

components. These data points are not completely assigned to one single component (or cluster). This is an example of soft clustering.

You can see that there are mistakes in inference because if you do something like a maximum likelihood estimates, these will be most likely from a single Gaussian, and these blue points and these red points will not be identified correctly.

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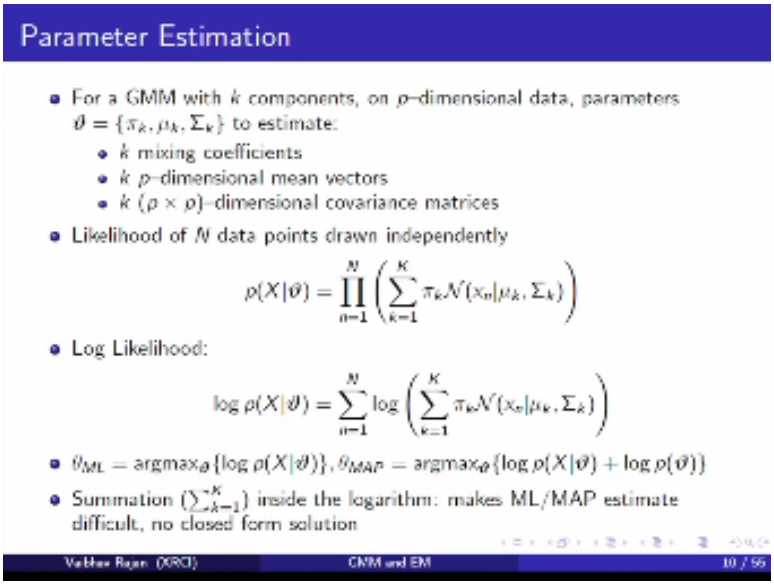


So suppose you are given data, and you want to fit a Gaussian mixture model to it, to either infer the clusters or to just fit the density. For either case you need to estimate the parameters of the model. If you have p-dimensional data and are using a Gaussian mixture with $k$ components to fit the data, we need to find out the k mixing coefficients ($\pi_k s$), the k mean parameter vectors ( $\mu_k s$) each of which is p-dimensional and the $k$ covariance matrices ($\Sigma_k$s), corresponding to each of the $k$ components. For now, we assume that we know the value of $k$ (i.e., the number of components to use) and will later come back to discuss how $k$ is estimated.

By observing the dimension of the covariance matrices, $\Sigma_k$s, we can infer that fitting the model is going to get difficult for high dimensional data because the dimension of the covariance matrices scale quadratically with the number of dimensions in the data. Dimension of the covariance matrix is $p \times p$, where $p$ is the dimensionality of the data.

To estimate the parameters of the model, we will take the standard route of maximum likelihood. The likelihood of N data points drawn independently, for the case of Gaussian mixture model with k components is given by:

$$p(V) = \prod_{n=1}^{N} \left( \sum_{k=1}^{K} \pi_k \, N(x_n | \mu_k, \, \Sigma_k) \right) \tag{3}$$

where, $\vartheta = (\pi, \ \mu, \ \Sigma)$ are the parameters of the model, and $N(x_n | \mu_k, \, \Sigma_k)$ is the probability for data point $x_n$ sampled from the Gaussian distribution corresponding to the $k^{th}$ component.

We apply logarithm to both sides of equation (3), which converts the outer product to a summation.

$$\log log \, p(\vartheta) \ = \sum_{n=1}^{N} \log log \left( \sum_{k=1}^{K} \pi_k \, N(x_n | \mu_k, \, \Sigma_k) \right) \tag{4}$$

The summation within the logarithmic term in the log likelihood computation shown in equation (4) causes lot of problems in the estimation of the parameters of a Gaussian mixture.

Since our objective is maximize the likelihood (and thereby also maximize the log likelihood) of the data $X$, we should be able to estimate the parameters $\vartheta$, by differentiating equation (4) w.r.t. to the each of the parameters in $\vartheta$, and equating them to 0.

$$\frac{\partial \log log\, p(\vartheta)}{\partial\, \mu_k} = 0, \quad \frac{\partial \log log\, p(v)}{\partial\, \Sigma_k}, \quad \frac{\partial \log log\, p(\vartheta)}{\partial\, \pi_k} = 0$$

for all the values of $k$.

However, the summation within the logarithm term in the expansion of $logp(X|\vartheta)$, shown in equation (4), poses problems in differentiating $\log log\, p(\vartheta)$ w.r.t. to each of the parameters in $\vartheta$ and we are not going to get a closed-form solution. This is one of the main problems for estimating the parameters of a Gaussian mixture.

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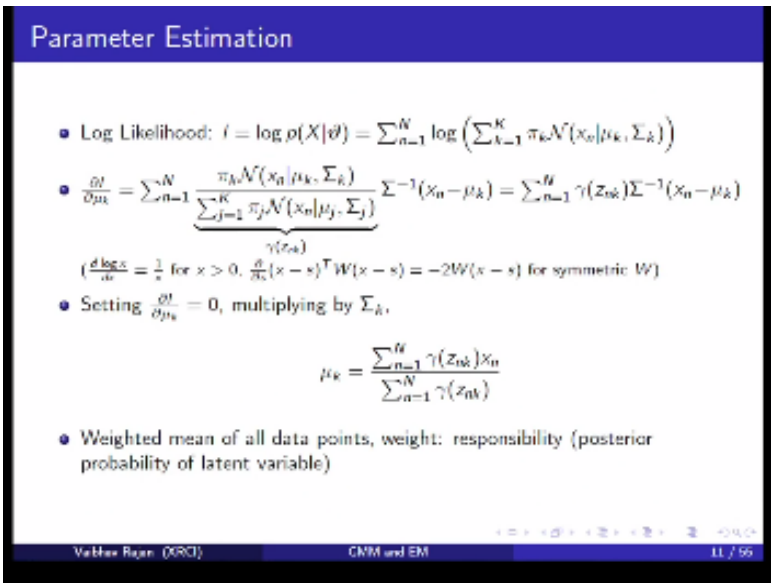


However, let's attempt to estimate the parameters of the model by differerntiating the log-likelihood with respect to each of the parameters of the model and equating them to 0. We will make one crucial assumption which is that we know the responsibility terms $\gamma_{nk}$, for each of

the data points $x_n$, for each of the $k$ components. So let us start with the parameter $\mu_k$ and see what we get by doing some algebra.

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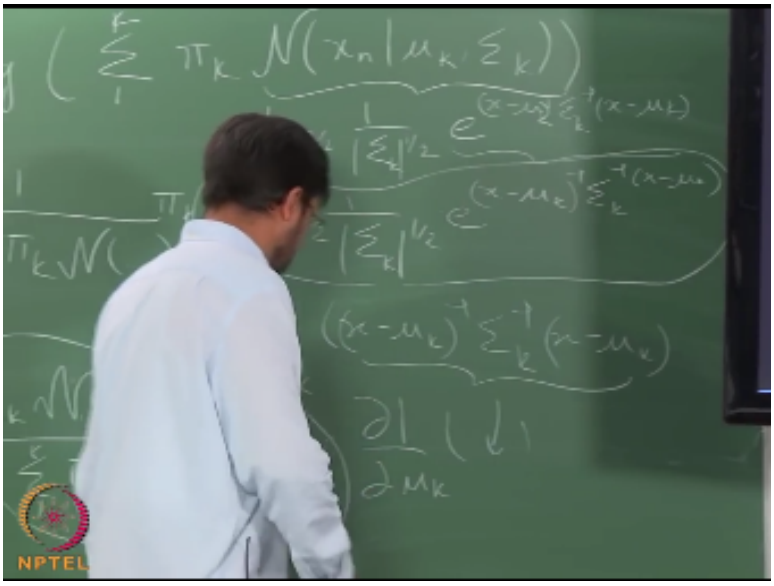

The log-likelihood is given by:

$$l = \sum_{n=1}^{N} \log log \left( \sum_{k=1}^{K} \pi_k N(x_n | \mu_k, \Sigma_k) \right) \tag{5}$$

where $\pi_k$ (not bolded) in the equation of log-likelihood refers to mixing coefficients of the mixture model.

Also, we know that the probability distribution of a Gaussian can be written as:

$$N(\mu_k, \Sigma_k) = \frac{1}{(2\pi)^{\frac{p}{2}}} \frac{1}{|\Sigma_k|^{\frac{1}{2}}} e^{-\frac{1}{2}(x-\mu_k)^T \Sigma_k^{-1} (x-\mu_k)} \tag{6}$$

Where, π (bolded) in the equation refers to the constant π (defined as the ratio of circle's circumference to its diameter).

Substituting equation (6) in equation (5), the log-likelihood can be rewritten as:

$$l = \sum_{n=1}^{N} \log log \left( \sum_{k=1}^{K} \pi_k \left( \frac{1}{(2\pi)^{\frac{p}{2}}} \frac{1}{|\Sigma_k|^{\frac{1}{2}}} e^{-\frac{1}{2}\left(x_n - \mu_k\right)^T \Sigma_k^{-1} \left(x_n - \mu_k\right)} \right) \right) \quad (7)$$

We now find the partial derivatives of the log-likelihood with respect to parameter $\mu_k$:

$$\frac{\partial l}{\partial \mu_k} = \sum_{n=1}^{N} \frac{1}{\sum_{j=1}^{K} \pi_j N\left(\mu_j, \Sigma_j\right)} \pi_k \left( \frac{1}{(2\pi)^{\frac{p}{2}}} \frac{1}{|\Sigma_k|^{\frac{1}{2}}} e^{-\frac{1}{2}\left(x_n - \mu_k\right)^T \Sigma_k^{-1} \left(x_n - \mu_k\right)} \right)$$

$$\cdot \frac{\partial(-\frac{1}{2}\left(x_n - \mu_k\right)^T \Sigma_k^{-1} \left(x_n - \mu_k\right))}{\partial \mu_k}$$

$$= \sum_{n=1}^{N} \left( \frac{\pi_k N\left(\mu_k, \Sigma_k\right)}{\sum_{j=1}^{K} \pi_j N\left(\mu_j, \Sigma_j\right)} \right) \frac{\partial(-\frac{1}{2}\left(x_n - \mu_k\right)^T \Sigma_k^{-1} \left(x_n - \mu_k\right))}{\partial \mu_k} \quad (8)$$

However, we know that the responsibility of the $k^{th}$ component for the data point $x_n$ can be written as:

$$\gamma_{nk} = \frac{\pi_k N\left(\mu_k, \Sigma_k\right)}{\sum_{j=1}^{K} \pi_j N\left(\mu_j, \Sigma_j\right)} \quad (9)$$

Substituting equation (9) in equation (8), we get:

$$\frac{\partial l}{\partial \mu_k} = \sum_{n=1}^{N} \gamma_{nk} \left( -\frac{1}{2} \frac{\partial \left( (x_n - \mu_k)^T \Sigma_k^{-1} (x_n - \mu_k) \right)}{\partial \mu_k} \right) \tag{10}$$

From the matrix cook book which contains complex matrix derivatives, we have:

$$\frac{\partial}{\partial s}(x - s)^T W(x - s) = -2W(x - s), \; for\, symmetric\, W \tag{11}$$

Using equation (11), we have that:

$$\frac{\partial}{\partial \mu_k}(x_n - \mu_k)^T \Sigma_k^{-1}(x_n - \mu_k) = -2\Sigma_k^{-1}(x_n - \mu_k) \tag{12}$$

(because covariance matrix Σ and thereby $\Sigma^{-1}$ are symmetric matrices)

Substituting equation (12) in equation (10), we get:

$$\frac{\partial l}{\partial \mu_k} = \sum_{n=1}^{N} \gamma_{nk} \left( -\frac{1}{2} \left( -2\Sigma_k^{-1}(x_n - \mu_k) \right) \right) = \sum_{1}^{n} \gamma_{nk} \Sigma_k^{-1}(x_n - \mu_k) \tag{13}$$

Equating $\frac{\partial l}{\partial \mu_k} = 0$ from equation (13) and multiplying by $\Sigma_k$, we get:

$$\Sigma_k \frac{\partial l}{\partial \mu_k} = \Sigma_k \left( \Sigma_k^{-1} \sum_{n=1}^{N} \gamma_{nk} (x_n - \mu_k) \right) = \Sigma_k 0 = 0$$

$$\mu_k = \frac{\Sigma_{n=1}^{N} \gamma_{nk} x_n}{\Sigma_{n=1}^{N} \gamma_{nk}} \tag{14}$$

It can be observed from equation (14) that the mean inferred for the $k^{th}$ component Gaussian, $\mu_k$, is the weighted mean of all the data points, where the weight is the responsibility of that cluster towards that data points.

It is important to note that we do not know the responsibilities $\gamma_{nk}$. We have assumed that we know the responsibilities, substituted it and did the math to arrive at a nice form for $\mu_k$, shown in equation (14). However, the responsibility $\gamma_{nk}$ has all the unknowns inside it, including mixing coefficient $\pi_k$, mean parameter $\mu_k$ and covariance parameter $\Sigma_k$, as shown in equation (9).

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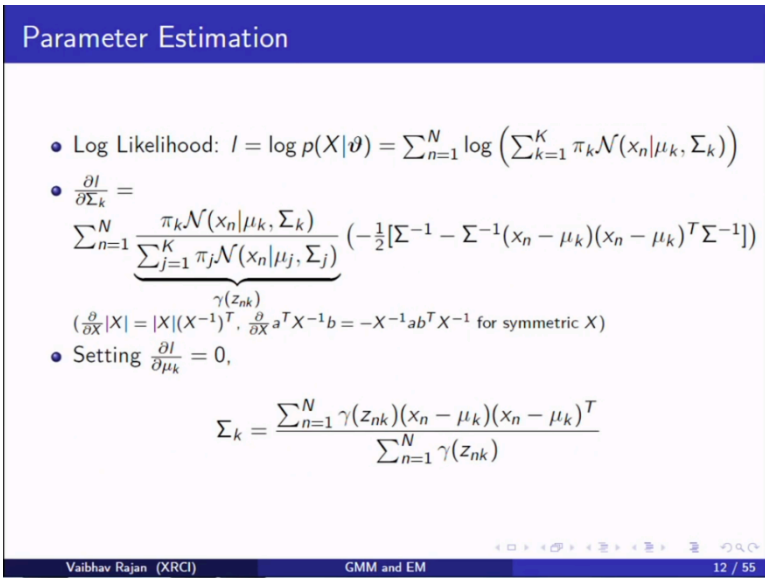


Similarly, we can arrive at a nice form for $\Sigma_k$ by derivating the log-likelihood of the data w.r.t. to $\Sigma_k$ and equating it to 0. Here also we assume that we know the responsibilities terms.

$$\frac{\partial l}{\partial \Sigma_k} = \sum_{n=1}^{N} \underbrace{\frac{\pi_k N(\mu_k, \Sigma_k)}{\sum_{j=1}^{K} \pi_j N(\mu_j, \Sigma_j)}}_{\gamma_{nk}} \left(-\frac{1}{2}\left[\Sigma^{-1} - \Sigma^{-1}(x_n - \mu_k)(x_n - \mu_k)^T \Sigma^{-1}\right]\right)$$

$$\left(\frac{\partial}{\partial X}|X| = |X|\left(X^{-1}\right)^T,\ \frac{\partial}{\partial X} a^T X^{-1} b = -X^{-1} a b^T X^{-1}\ for\ symmetric\ X\right)$$

Setting $\frac{\partial l}{\partial \Sigma_k} = 0$,

$$\Sigma_k = \frac{\sum_{n=1}^{N} \gamma_{nk}(x_n - \mu_k)(x_n - \mu_k)^T}{\sum_{n=1}^{N} \gamma_{nk}}$$

Responsibility is the posterior probability of the latent variable given that you have the data point. So essentially it's trying to capture which of the components generated that data point.

So right now we do not really have anything. We currently do not have the mean parameters or the covariance parameters. We just know that if we knew the responsibilities, which we don't know, we could get a nice form for mean parameters $\mu_k$s and covariance parameters $\Sigma_k$s. However without this assumption, the derivations for $\mu_k$s and $\Sigma_k$s does not hold.

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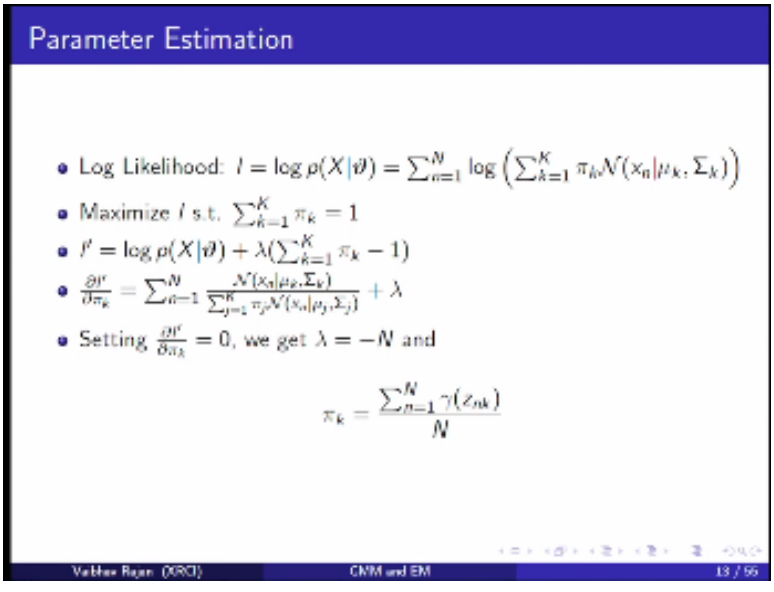

When we take the derivatives with respect to $\pi_k$, we have to be careful. We have to make sure that we use the constraint $\sum_{k=1}^{K} \pi_k = 1$. So we cannot directly differentiate the log-likelihood with respect to $\pi_k$. We have to use Lagrange multipliers.

So we just take a Lagrange multiplier here, take this constraint, and do the differentiation. The responsibility terms crops up in the derivative. If we set the derivative to 0, we can get langrange multiplier $\lambda = -N$ and we also get the value of $\pi_k$ as:

$$\pi_k = \frac{\sum_{n=1}^{N} \lambda_{nk}}{N}$$

So if you take the sum of all the responsibilities over all data points over all components, it is equal to $N$. So the $k^{th}$ mixture weight $\pi_k$ is nothing but the proportion of the responsibilities that are coming from the $k^{th}$ component towards all the data points, and you are taking the proportion that is given by those responsibilities with respect to all of it, i.e., the sum of it.

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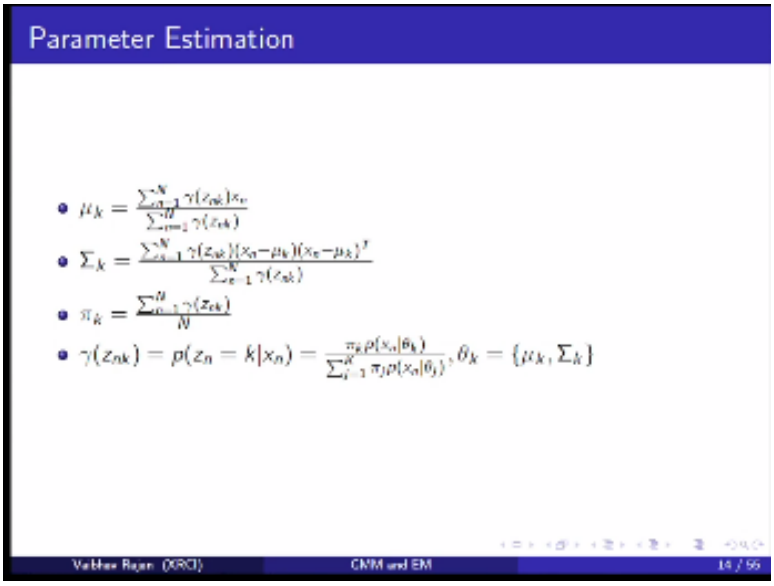

So let's summarize. What we found is that we have very nice forms for $\mu_k$, $\Sigma_k$ and $\pi_k$, given that we know the responsibilities, which is the posterior probability of the latent variable coming from the k component for the nth data point. Can you think of how you can use this to create an algorithm for estimating the Gaussian mixture parameters?

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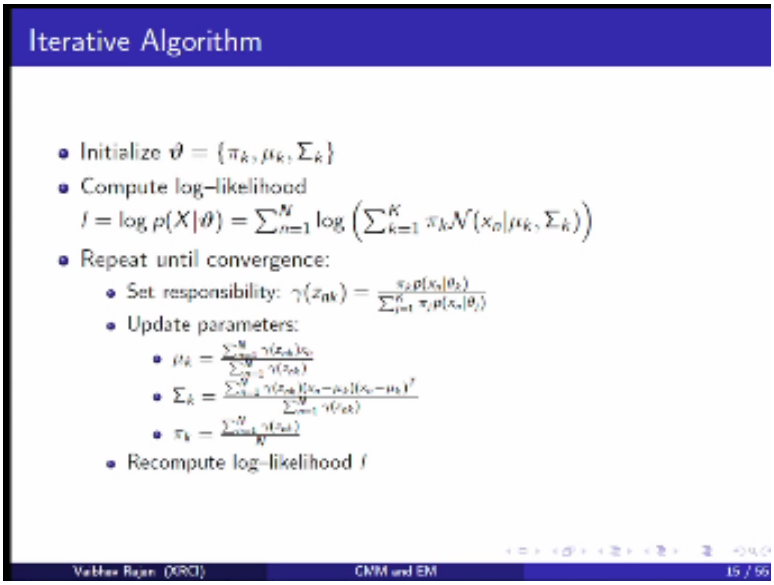


I will denote by $\vartheta$ all the parameters. We start with some guess of the parameters, and then we will compute the log-likelihood. We can compute it because we know or have guessed the parameters. Then, we will set the responsibilities because we know or have guessed all the parameters necessary for computing them. Since given the responsibility we can compute all the parameters again, we will use this to get a new guess, and that way we will iteratively keep refining our guess.

Now, this looks very ad hoc but is actually theoretically quite sound. The reason why this is a good thing to do is we will show that this is guaranteed to increase the likelihood at every iteration. We will see that when we understand how EM works. So this turns out to be actually an instance of the EM algorithm. It is quite intuitive.

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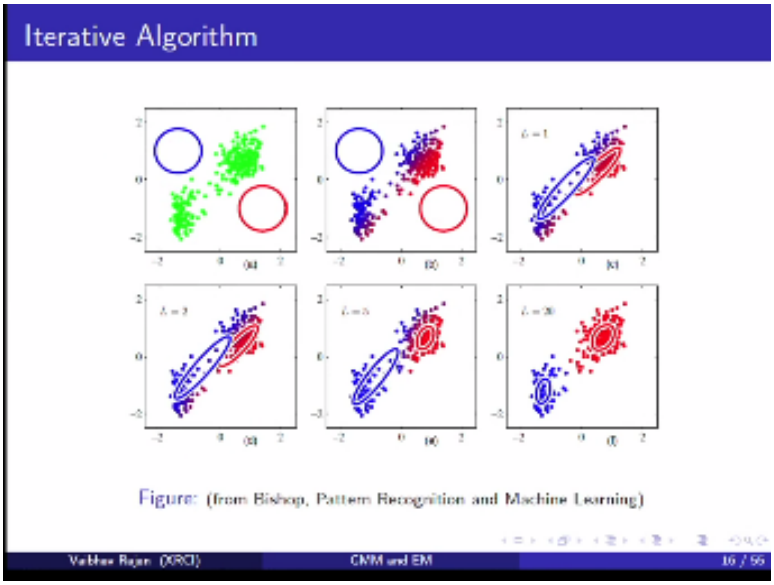


So this is an example of exactly that algorithm. We start with some data. We start with some guesses for the Gaussians, and iteratively we see that the Gaussians converge to nicely fit the data. The parameters that you get here will fit the data very well. The only problem is that it usually takes a long time to come to the right parameters. In this whole iterative algorithm, we are assuming that we know the value of $k$.

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**NPTEL**

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**Introduction to Machine Learning**

**Lecture-77**
**Expectation Maximization**

**Prof. Balaraman Ravindran**
**Computer Science and Engineering**
**Indian Institute of Technology Madras**

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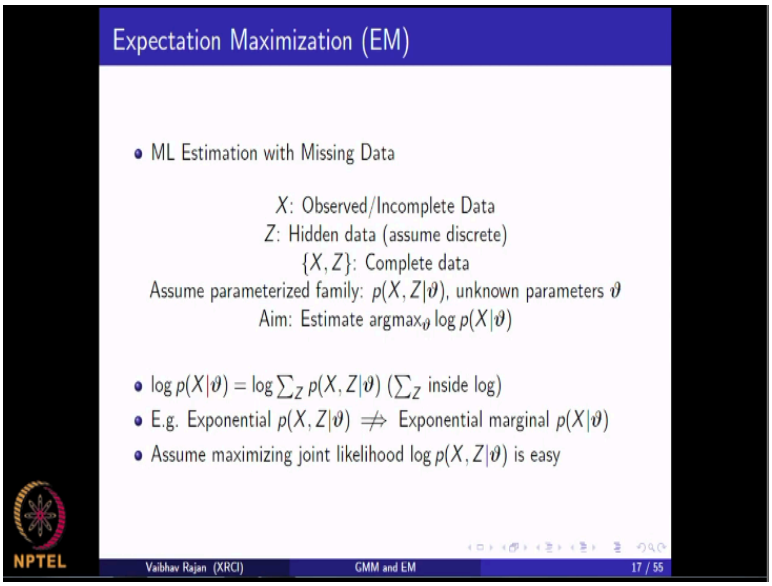


Expectation Maximization (EM) is a way to do maximum likelihood estimation. Initially, it was proposed as a way to do maximum likelihood estimation when you have missing data. Suppose you are given data $X$ which is what you observe and is known to be incomplete (i.e., there are some values that are missing), and you want to get the maximum likelihood estimates of the parameters which are unknown.

Here we are assuming two things. We are assuming that there is some parameterized family (i.e., you are doing some parameterized fitting) for which the joint likelihood is easy to compute. So we denote by $\{X, Z\}$ the complete data (the observed data plus the hidden data), and we assume some parameterized family from which this data is generated (like a Gaussian or an exponential, and we do not know the unknown parameters which we want to estimate.

So we start seeing connections with what we have seen in the case of Gaussian mixture because if you take the marginal probability here, you again see a summation coming inside the $log$, and this again poses problems. Also, the marginals need not be of the same family. For example, if the joint probability distribution is an exponential distribution, it does not mean that the corresponding marginal probability distributions also comes from the exponential family.

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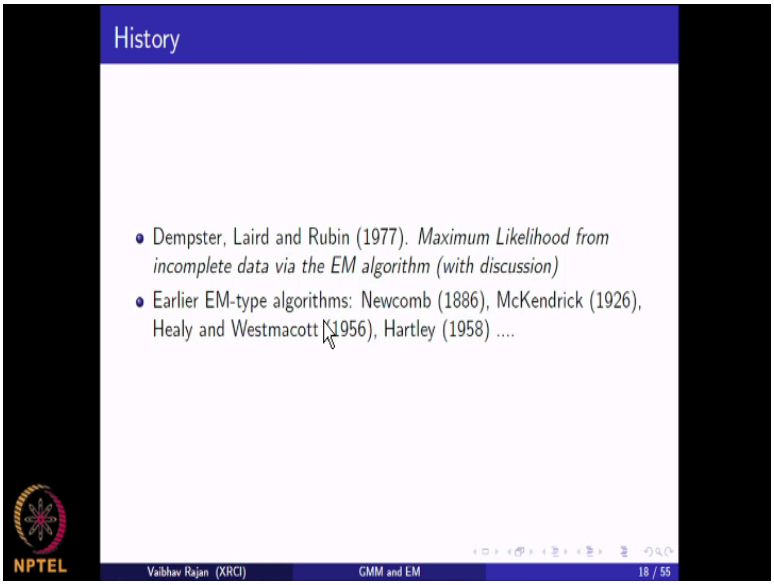


EM as it is most commonly used today was proposed by Dempster, Laird and Rubin in their 1977 seminal paper "Maximum Likelihood from incomplete data via the EM algorithm". Even before 1977, a lot of statisticians have developed EM like algorithms, but when we cite EM, we usually cite this paper.

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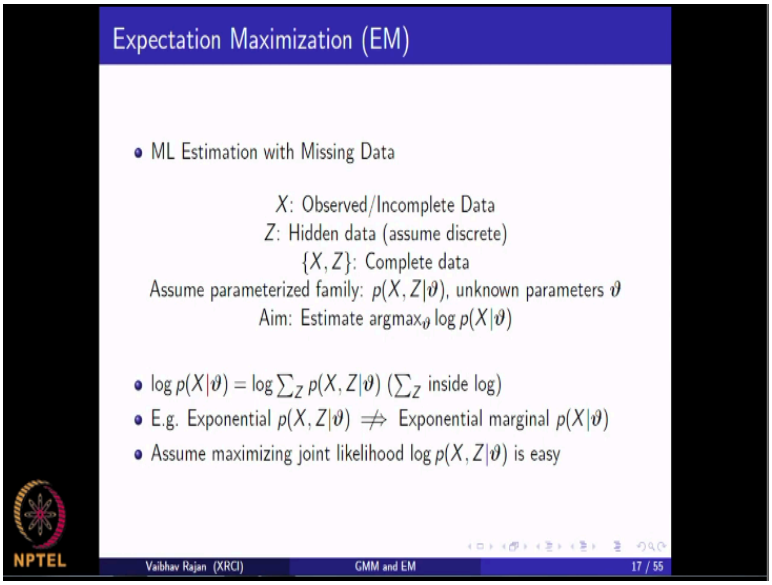


We have observed or incomplete data and hidden data. For the rest of the discussion, we will assume that this hidden data is discrete, but all the derivations will work if we assume the hidden data to be continuous as well. In which case, we just have to replace the summations with integrals.

The complete data is the combination of the incomplete data and hidden data. We assume some parameterized family for the complete data, and want to estimate the parameters. This is the problem that EM solves.

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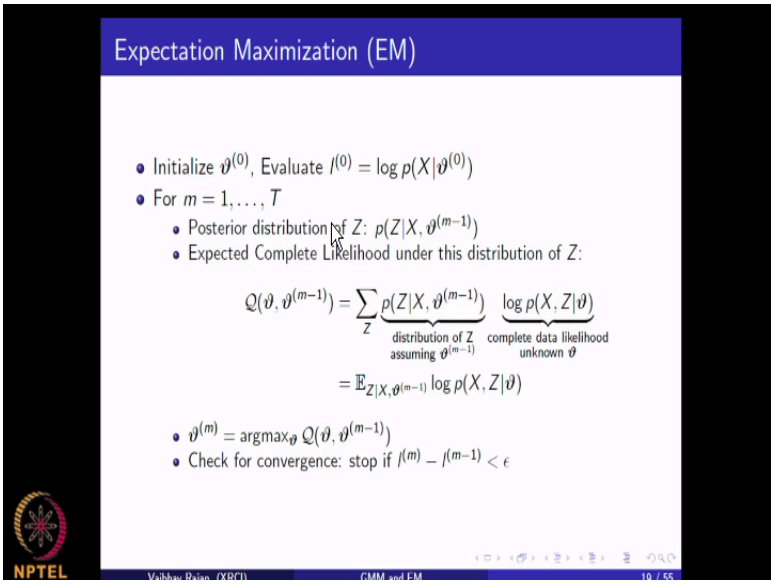


EM is an iterative algorithm, just like what we designed for Gaussian mixture. We again start with a guess of the parameters that we have. We evaluate our first guessed likelihood, and then we iteratively do two steps. First, we compute the posterior distribution of $Z$ given the current estimate of the parameters. After that, we compute the expected complete log likelihood under this distribution, which we'll call as the Q function. Now notice that this, this expectation takes the complete data likelihood. Here the parameters are unknown, and this expectation assumes the distribution of Z given the parameters that we have guessed in the previous round. And then we again get a new guess for the parameters by maximizing this Q function, and taking that argument $\vartheta$ which maximizes this Q function.

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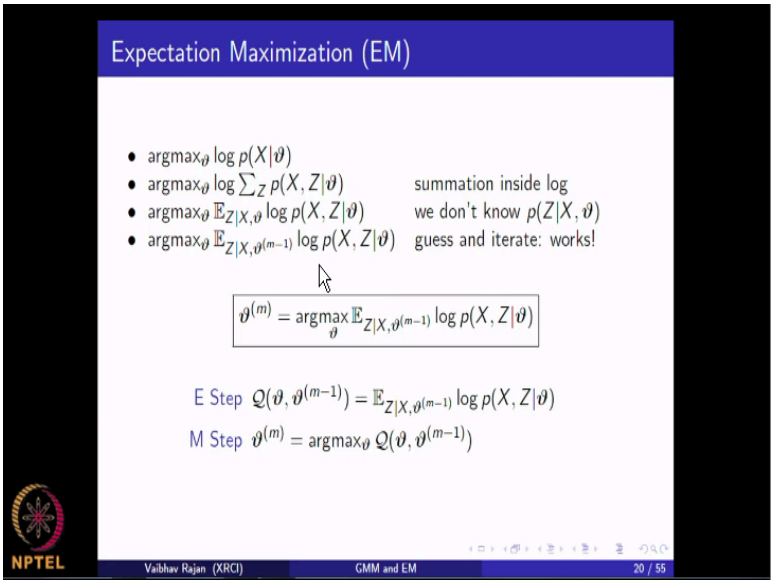


What we wanted was to get the maximum likelihood estimate of $\vartheta$. We see $X$ but $X$ is not complete. $X$ has some missing data $Z$. So we express the likelihood of observed data $X$ as a marginal distribution of the complete data, and we get into the same problem of the summation being inside the logarithm.

So we decide to not compute the maximum likelihood in this way, but instead compute the maximum taking the expectation of the log likelihood of the complete data under the distribution of Z. But we do not know the real distribution of Z because we do not know the parameters. So we take the guess that we had in the previous round. We compute the expectation of the complete data likelihood under the distribution of Z given the current guess of the parameters.

This works. We will see why it works. So the entire EM algorithm can actually be represented by just this one line. You start with some guess, and then for the next guess you calculate the expectation of the complete data log likelihood under the distribution of the missing data using the previous guess. This can actually be broken down into two steps, where in the E step you compute this expectation, and in the M step you maximize this Q and get the next set of parameters.

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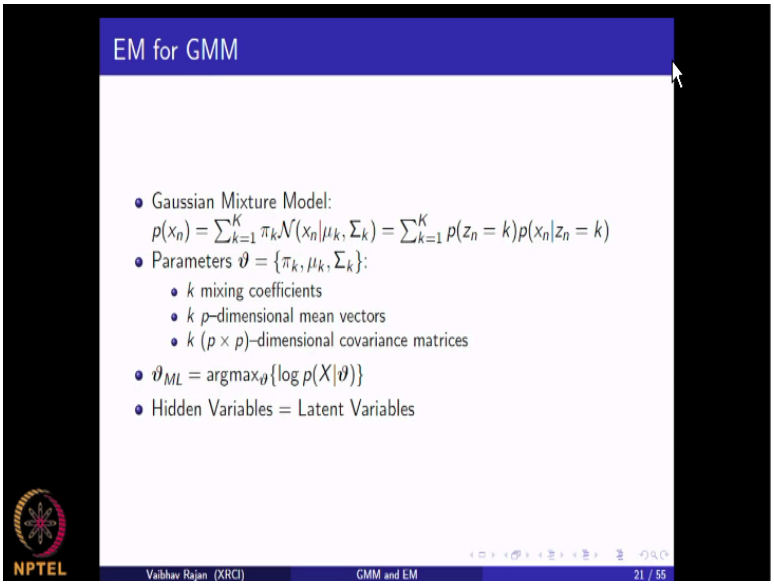


So let us see how we can get the EM algorithm for Gaussian mixtures. The key thing here is we did not say anything about hidden variables, but the trick here is to use these latent variables as hidden variables. This is how EM is used in a lot of different models, not just Gaussian mixture, but also in a lot of latent variable models. You assume that the latent variables are hidden, i.e., you do not know them, and run the whole EM machinery.

This slides recalls the Gaussian mixture model. We have the Gaussian mixture model, where we want to estimate all the parameters represented by $\vartheta$. We have $K$ components, and each of the component has parameters $\pi_k$, $\mu_k$ and $\Sigma_k$. What we want to find out is the maximum likelihood estimate.

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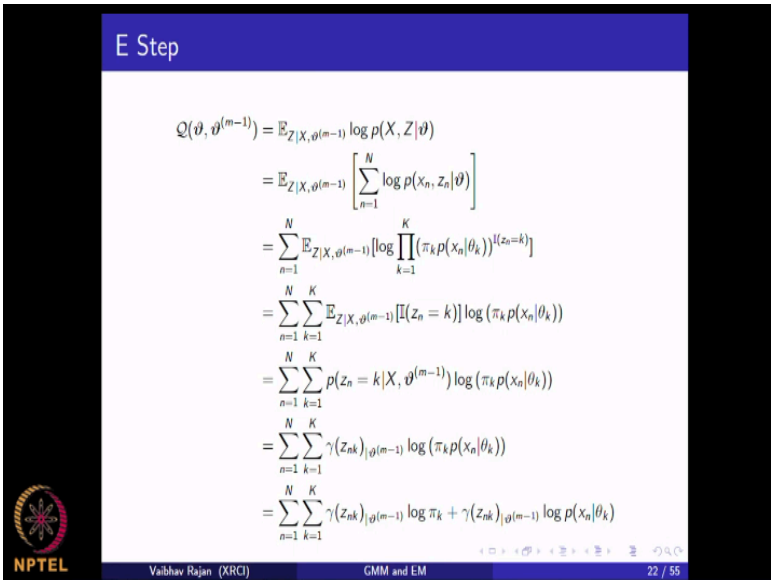


All right.

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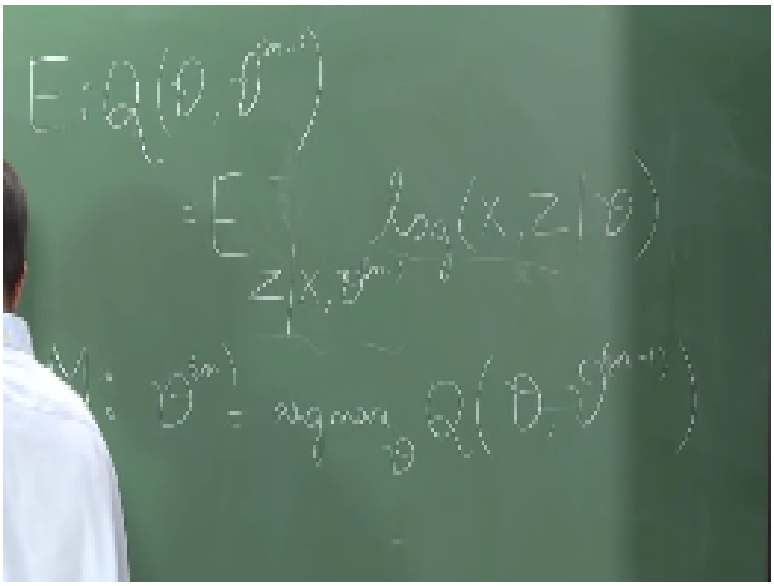

So let's write this down here.

$$E\text{: } Q\left(\vartheta,\ \vartheta^{(m-1)}\right) = E_{Z|X,\vartheta^{(m-1)}}\ log(X, Z|\vartheta)$$

The most important thing to remember is that this distribution is taken over the previous guess, where as the expectation is for the complete log-likelihood over the unknown parameters. The M step just gets the next guess.

$$M: \vartheta^{(m)} = argmax_{\vartheta} Q(\vartheta, \vartheta^{(m-1)}) \quad (1)$$

If we want to get the maximum likelihood parameters, we first need to compute this Q function. Then usually the M step is easier. It's just the expectation computation in E step that requires some work. Once you get Q, then the maximization step is just computing the derivatives.

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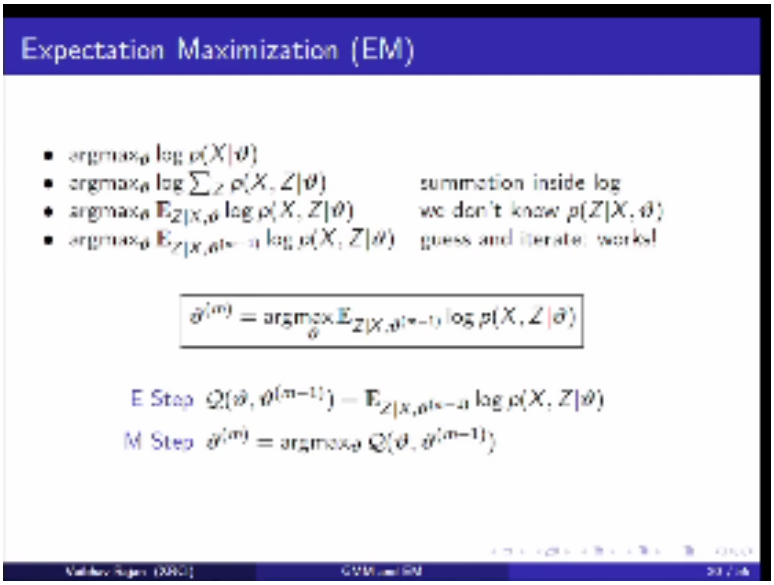


As you can see, this works only if the E step gives you something which you can easily maximize. In the case of Gaussian mixture, we will see that by using the complete data likelihood, we will be able to get something that we can easily maximize.

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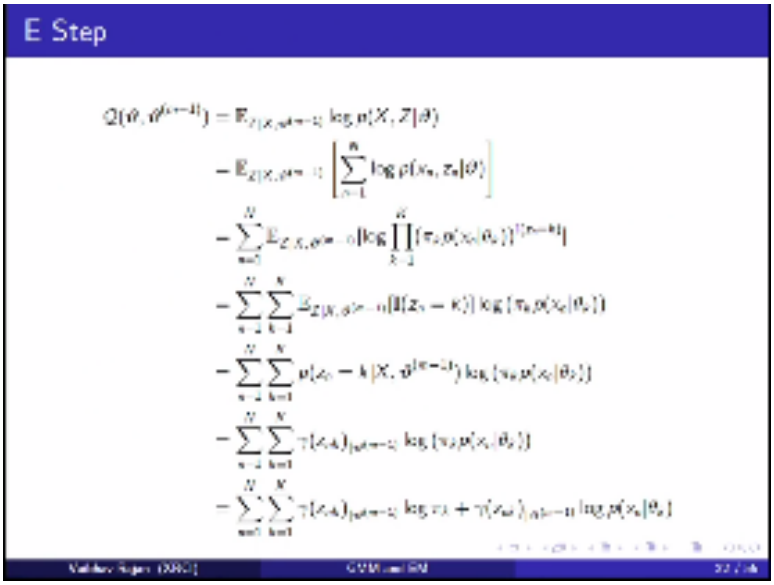


The Q function, by definition, is the expectation of the complete data likelihood under the distribution of Z given the previously guessed parameters.

$$Q\left(\vartheta,\ \vartheta^{(m-1)}\right) = E_{Z|X,\vartheta^{(m-1)}}\ logp(X, Z|\vartheta)$$

So the log likelihood here is the likelihood of $x_n, z_n$ given the unknown parameters, over all the data points.

$$Q\left(\vartheta,\ \vartheta^{(m-1)}\right) = E_{Z|X,\vartheta^{(m-1)}}\left[\sum_{n=1}^{N} \log log\ p(x_n, z_n|\vartheta)\right] \qquad (2)$$

We can take this summation outside by linearity of expectations. Rewriting the the complete log likelihood, the equation assumes the form:

$$Q\left(\vartheta,\ \vartheta^{(m-1)}\right) = \sum_{n=1}^{N} E_{Z|X,\vartheta^{(m-1)}}\left[\log log \prod_{k=1}^{K} \left(\pi_k\, p\left(\theta_k\right)\right)^{I(z_n=k)}\right]$$

(3)

So we can derive this formally, but intuitively it is very clear. $I(z_n = k)$ is just an indicator function which assumes a value of 1 when $z_n = k$, and assumes a value of 0 when $z_n \neq k$. So in this product, all the terms except one will get an exponent of 0 (and therefore, thos terms assume a value of 1). So only one term out of the K which will remain for each of the data points. The log-likelihood comes straight away from this formula after using the indicator function here, and this gives us the complete data likelihood.

So then the product becomes a sum when we take it outside the log, and the exponent term $I(z_n = k)$ comes down. Again the expectation can be brought inside by linearity, and so we get both the summations out.

$$= \sum_{n=1}^{N} \sum_{k=1}^{K} E_{Z|X,\vartheta^{(m-1)}} \left[ I(z_n = k) \right] \log log \left( \pi_k \, p(x_n|\theta_k) \right) \tag{4}$$

With respect to the distribution of $Z$ using the previously guessed parameters $\vartheta^{(m-1)}$, $\log log \left( \pi_k \, p(x_n|\theta_k) \right)$ is just a constant. So the expectation is only over the indicator function $I(z_n = k)$.

Expectation of an indicator function just gives us the probability, and we have the probability of $z_n = k$, again given $X$ and the previously guessed parameter values $\vartheta^{(m-1)}$. Log just remains.

$$= \sum_{n=1}^{N} \sum_{k=1}^{K} p\left(X, \vartheta^{(m-1)}\right) \log log \left( \pi_k \, p(x_n|\theta_k) \right)$$

So $p\left(X, \vartheta^{(m-1)}\right)$ is again the responsibility, which is the posterior probability of $z_n = k$. In this case, this responsibility is not with respect to the original parameters, but this posterior

probability is with respect to the guessed parameters. This has been indicated with the subscript. So it is the responsibility times the log that remains.

$$= \sum_{n=1}^{N} \sum_{k=1}^{K} \gamma\left(z_{nk}\right)_{|\vartheta^{m-1}} \log log\ (\pi_k\, p(x_n|\theta_k))$$

So, expressing the log of a product as a summation of individual log terms, we get:

$$Q(\vartheta,\ \vartheta^{(m-1)}) = \sum_{n=1}^{N} \sum_{k=1}^{K} \gamma\left(z_{nk}\right)_{|\vartheta^{m-1}} \log log\ \pi_k\ + \gamma\left(z_{nk}\right)_{|\vartheta^{m-1}} \log log\ p(x_n|\epsilon \qquad (5)$$

So what have we achieved? So we have got an expression for Q in terms of again the responsibility, but this time the responsibility is with respect to the guesssed parameters.

If you just look at the Q function in equation (5), you can see that it is easier to differentiate because the summations are all outside and the normal distribution term comes towards the end of the equation. The differentiation of the normal distribution term will be just like what you do in the case of fitting a single Gaussian. How did this nice mathematical form come about? It happened mainly because we were taking the complete data likelihood as shown in equation (2). The complete data likelihood gave us a nice mathematical form in equation (3), and due to the expectation, all the summations got pushed out in equation (4). So we got a nice form for $Q$ as shown in equation (5).

So is it only because the summation is inside the logarithm, we need to do all this? Otherwise, can we can skip the whole thing ?

Yes, but that comes in many contexts, not just in the case of Gaussian mixture. In a lot of those cases, EM is useful. If we could get the maximum likelihood easily, we wouldn't need to use EM for Gaussian mixture.

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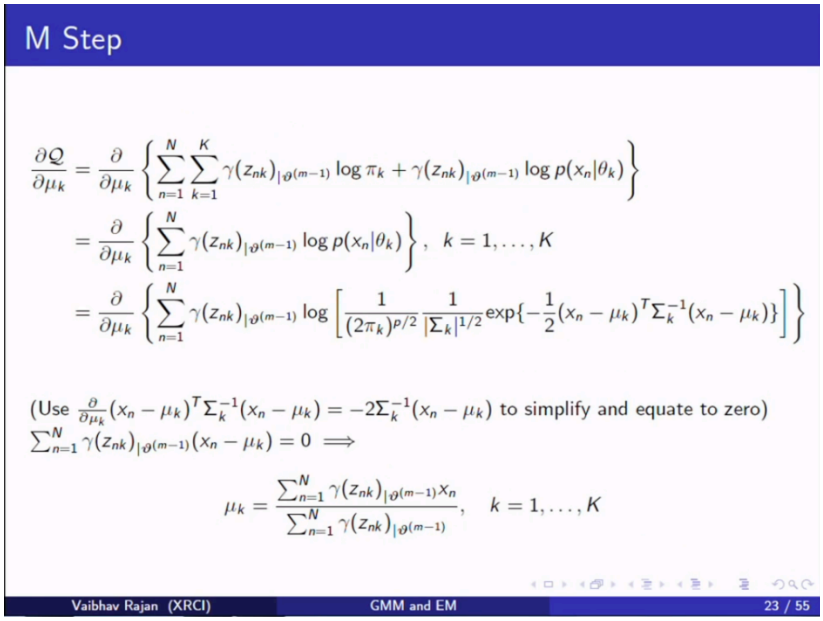


Now, the M step is just that which is shown in equation (1), which is we differentiate the $Q$ function with respect to each of the parameters. The $Q$ function here is the same $Q$ function which is expanded in equation (5). Now one thing to remember here is that we know the guessed parameter at the previous iteration ( $\vartheta^{(m-1)}$ ). So we know the responsibilities. So the responsibilities are just constants in this case, and so differentiating the $Q$ function with respect to the parameters becomes very easy.

So here we differentiate the $Q$ function from equation (5) with respect to $\mu_k$.

$$\frac{\partial Q}{\partial \mu_k} = \frac{\partial}{\partial \mu_k}\left\{\sum_{n=1}^{N}\sum_{k=1}^{K}\gamma\left(z_{nk}\right)_{|\vartheta^{(m-1)}}\log log\,\pi_k + \gamma\left(z_{nk}\right)_{|\vartheta^{(m-1)}}\log log\, p(x_n|\theta_k)\right\}$$

This whole first term inside the summations ( which is $\gamma\left(z_{nk}\right)_{|\vartheta^{(m-1)}}\log log\,\pi_k$ ) is not necessary.

We focus only on the second term inside the summations ( which is $\gamma\left(z_{nk}\right)_{|\vartheta^{(m-1)}}\log log\, p(x_n|\theta_k)$ ).

For each of the different components, we get the entire normal distribution within the equation here.

$$\frac{\partial Q}{\partial \mu_k} = \frac{\partial}{\partial \mu_k} \left\{ \sum_{n=1}^{N} \gamma\left(z_{nk}\right)_{|\vartheta^{(m-1)}} \log log\, p(x_n|\theta_k) \right\},\ k = 1,\ \ldots,\ K$$

$$\frac{\partial Q}{\partial \mu_k} = \frac{\partial}{\partial \mu_k} \left\{ \sum_{n=1}^{N} \gamma\left(z_{nk}\right)_{|\vartheta^{(m-1)}} \log log \left[ \frac{1}{(2\pi)^{\frac{p}{2}}} \frac{1}{|\Sigma_k|^{\frac{1}{2}}} \exp exp \left\{ -\frac{1}{2} \left(x_n - \mu_k\right)^T \Sigma_k^{-1} \left(x_n - \mu_k\right) \right\} \right] \right\}$$

There are no summations inside the logarithm. This is exactly the same derivation as for a single Gaussian. We use matrix derivatives to get very simple forms here.

Substituting $\frac{\partial Q}{\partial \mu_k} = 0$, we get:

$$\mu_k = \frac{\sum_{n=1}^{N} \gamma\left(z_{nk}\right)_{|\vartheta^{(m-1)}} x_n}{\sum_{n=1}^{N} \gamma\left(z_{nk}\right)_{|\vartheta^{(m-1)}}},\ k = 1, \ldots,\ K$$

We will see that we again get the same form for $\mu_k$ that we saw earlier for our adhoc iterative algorithm except that these responsibilities are with respect to the previously guessed parameters.

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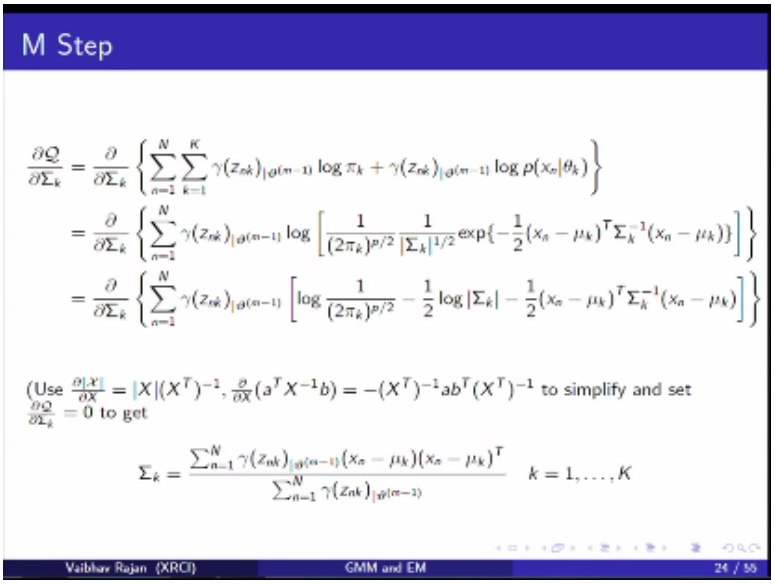


Here, similar to what we did for $\mu_k$, we find the derivative of $Q$ from equation (5) with respect to $\Sigma_k$ and equate it to zero.

$$\frac{\partial Q}{\partial \Sigma_k} = \frac{\partial}{\partial \Sigma_k}\left\{\sum_{n=1}^{N}\sum_{k=1}^{K}\gamma\left(z_{nk}\right)_{|\vartheta^{(m-1)}}\log log\,\pi_k + \gamma\left(z_{nk}\right)_{|\vartheta^{(m-1)}} logp(x_n|\theta_k)\right\}$$

Again, we do not need to worry about the first term within the summations. We only find the derivative for the second term within the summations.

$$\frac{\partial Q}{\partial \Sigma_k} = \frac{\partial}{\partial \Sigma_k}\left\{\gamma\left(z_{nk}\right)_{|\vartheta^{(m-1)}}\log log\left[\frac{1}{(2\pi)^{\frac{p}{2}}}\frac{1}{\left|\Sigma_k\right|^{\frac{1}{2}}}\exp exp\left\{-\frac{1}{2}\left(x_n-\mu_k\right)^T\Sigma_k^{-1}\left(x_n-\mu_k\right)\right\}\right]\right\}$$

You can simplify this further by first applying the logarithm here for each of these parts.

$$\frac{\partial Q}{\partial \Sigma_k} = \frac{\partial}{\partial \Sigma_k}\left\{\sum_{n=1}^{N}\gamma\left(z_{nk}\right)_{|\vartheta^{(m-1)}}\left[\log log\,\frac{1}{(2\pi)^{\frac{p}{2}}} - \frac{1}{2}\log log\left|\Sigma_k\right| - \frac{1}{2}\left(x_n-\mu_k\right)^T\Sigma_k^{-1}\left(x_n-\mu_k\right)\right]\right\}$$

Then, the derivative for the determinant is given by a simple formula. We can apply the derivative formula here, and by setting $\frac{\partial Q}{\partial \Sigma_k} = 0$, we get back the same form for $\Sigma_k$ which we found earlier.

$$\Sigma_k = \frac{\sum_{n=1}^{N} \gamma(z_{nk})_{|\vartheta^{(m-1)}} (x_n - \mu_k)(x_n - \mu_k)^T}{\sum_{n=1}^{N} \gamma(z_{nk})_{|\vartheta^{(m-1)}}}, \; k = 1, \ldots, K$$

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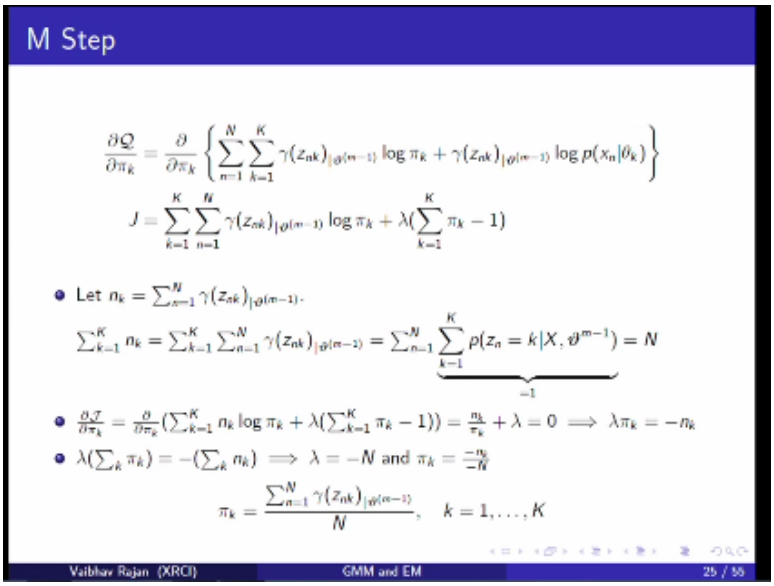


So similarly we perform the computations for the M step for $\pi_k$. However, this time in $\frac{\partial Q}{\partial \pi_k}$ the second term within the summations (which is $\gamma(z_{nk})_{|\vartheta^{(m-1)}} \log log\, p(x_n|\theta_k)$ ) is a constant with respect to $\pi_k$ and therefore goes away. We focus on the differentitation of the first term within the summations (which is $\gamma(z_{nk})_{|\vartheta^{(m-1)}} \log log\, \pi_k$). Since, the $\pi_k$s must satisfy the constraint $\sum_{k=1}^{K} \pi_k = 1$, we use Lagrange multipliers to get the formulation for $\pi_k$. The langrange function $J$ can be written as:

$$J = \sum_{k=1}^{K} \sum_{n=1}^{N} \gamma(z_{nk})_{|\vartheta^{(m-1)}} \log log\, \pi_k + \lambda \left( \sum_{k=1}^{K} \pi_k - 1 \right)$$

By differentiating the langrage function $J$ with respect to $\pi_k$, and setting the derivative to 0, we get:

$$\pi_k = \frac{\sum_{n=1}^{N} \gamma(z_{nk})_{|\vartheta^{(m-1)}}}{N}, \; k = 1, \ldots, K$$

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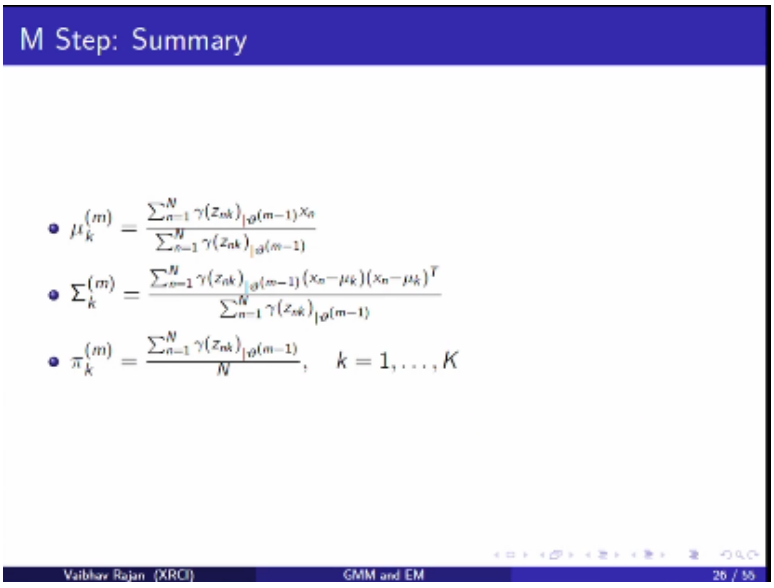


In the previous lecture, we first found sformulas for $\mu_k$, $\Sigma_k$ and $\pi_k$ by assuming that we know the responsibilities. Then, in this lecture, we used the E and M steps to find exactly the same formulas for $\mu_k$, $\Sigma_k$ and $\pi_k$ that we had found earlier in the previous lecture.

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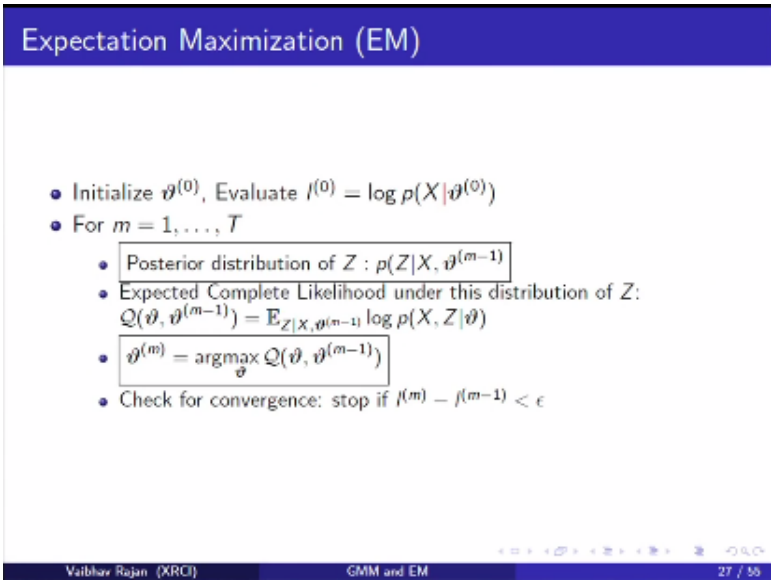


We plug the formulas for $\mu_k$, $\Sigma_k$ and $\pi_k$ into the EM framework. In the EM framework, we start with a guess for these parameters. We then iterate by first finding the posterior distribution of Z (which gives us the responsibilities and can be represented as $p(Z|X, \vartheta^{(m-1)})$), and then we find the expected complete likelihood under this distribution of Z (which can be represented as $Q\left(\vartheta, \vartheta^{(m-1)}\right) = E_{Z|X,\vartheta^{(m-1)}} \log log\, p(X, Z|\vartheta)$ ), and we finally maximize the Q function to get the new guesses for the parameters.

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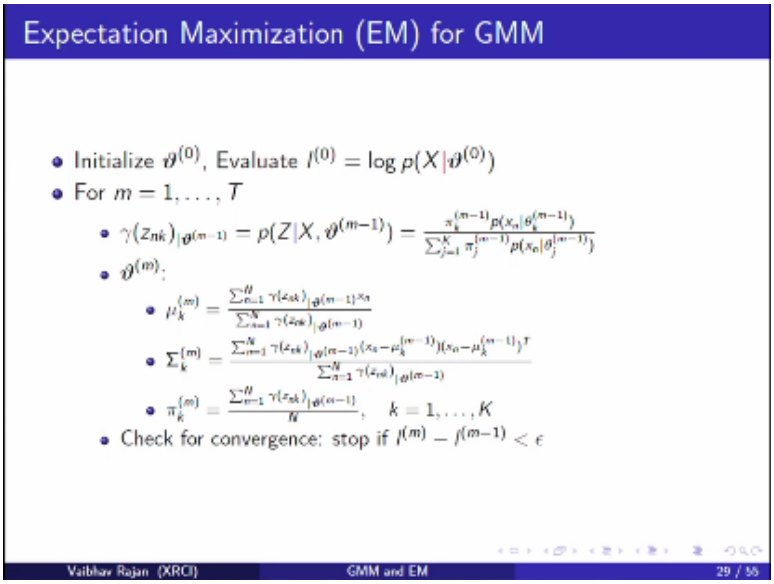


This gives us the next set of guesses, and this iteratively we can, we can check for convergence. When the likelihood does not change much we stop, and that is the EM algorithm for GMM. I have still not told you why this works, but we will see that, we will see the theoretical properties of why this is, why this works well. Yeah. I just wanted to show you that what we have got through, by doing all the math for EM is exactly the same as what we found during the iterative algorithm that we guessed.

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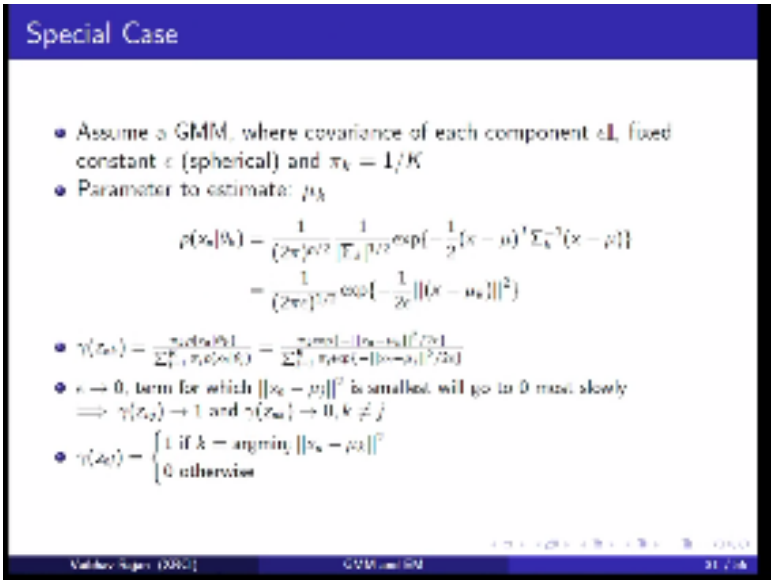


Let's look at a special case. Let's assume a Gaussian mixture model where the covariance of each component is fixed to be $\epsilon I$, where $\epsilon$ is a fixed constant and $I$ is the identity matrix. Covariance matrix of $\Sigma_k = \epsilon I$ will give us a spherical Gaussian. We also fixed each of the $\pi_k$ to be $\frac{1}{K}$. So each component gives exactly the same contribution towards the Gaussian mixture. Now the only parameter to estimate is the $\mu_k$s.

Since $\Sigma_k = \epsilon I$, the formula for the normal distribution simplifies to:

$$p(\theta_k) = \frac{1}{(2\pi)^{\frac{p}{2}}} \frac{1}{|\Sigma_k|^{\frac{1}{2}}} \exp\, exp\left\{-\frac{1}{2}\left(x - \mu_k\right)^T \Sigma_k^{-1} \left(x - \mu_k\right)\right\}$$

$$p(\theta_k) = \frac{1}{(2\pi\epsilon)^{\frac{p}{2}}} \exp\, exp\left\{-\frac{1}{2\epsilon}\left\|\left(x - \mu_k\right)\right\|^2\right\} \quad (6)$$

The formula for the responsibility also simplifies. We plug $p(x_n|\theta_k)$ from equation (6) and also substitute $\pi_k = \frac{1}{K}$ into the formula for responsibilities $\gamma(z_{nk})$, and we get:

$$\gamma(z_{nk}) = \frac{\pi_k p(\theta_k)}{\sum_{j=1}^{K} \pi_j p(\theta_j)} = \frac{\exp exp\left\{-\|x_n - \mu_k\|^2/2\epsilon\right\}}{\sum_{j=1}^{K} \exp exp\left\{-\|x_n - \mu_j\|^2/2\epsilon\right\}} \quad (7)$$

Now if we look at this expression in equation (7), and see what happens to the denominator as $\epsilon \to 0$, as $\epsilon \to 0$, the term for which the difference $\|x_n - \mu_j\|^2$ is smallest will go to 0 most slowly. So the responsibility for $x_n$ will tend to 1 for that particular component (which is the $j^{th}$ component in this case) because the numerator will be equal to the denominator in the limits, and for all other components (for which $k \neq j$), the responsibility will go to 0.

$$\gamma(z_{nj}) \to 1 \quad \text{and} \quad \gamma(z_{nk}) \to 0, \ \forall\, k \neq j$$

So this is the special case of hard clustering that was mentioned earlier. What it turns out to be is just setting the responsibility of $x_n$ to 1 for that component for which the data point is closest to the mean, and setting the responsibility of $x_n$ to zero for all other components.

$$\gamma(z_{nk}) = \{1 \ if \ k = argmin_j \|x_n - \mu_j\|^2 \ 0 \ otherwise$$

The responsibility is just the posterior probability of $z_n$ being equal to $k$. For the data point $x_n$, we want to know which component it has come from. For the data point $x_n$, the responsibility is one for that component whose mean the data point is closest to, and it is 0 for all others components. This means that if you look at the data and look at $x_n$, and want to know the posterior probability of which component it came from, it is that component whose mean that data point is closest to.

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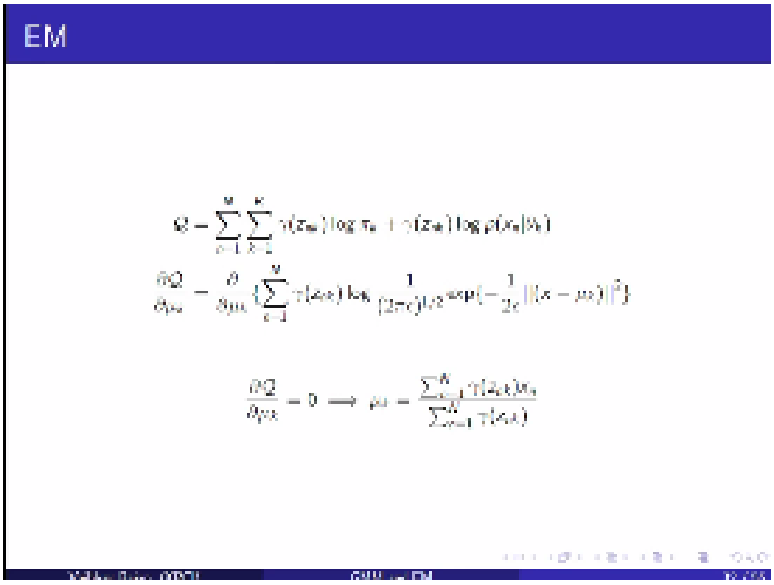


Let us do the EM for this special case of hard clustering in which for each of the components, we have set $\Sigma_k = \epsilon I$ and $\pi_k = 1/K$. The first step in EM is to calculate Q.

From equation (5), we have:

$$Q = \sum_{n=1}^{N}\sum_{k=1}^{K} \gamma(z_{nk}) \log log\, \pi_k + \gamma(z_{nk}) \log log\, p(x_n|\theta_k)$$

The formula for Q is the same as before because we are doing Gaussian mixture. It's just a special Gaussian mixture.

Substituting $\pi_k = \frac{1}{K}$, and replacing $\log log\, p(x_n|\theta_k)$ with the expansion from equation (6) for the this special case of Gaussian, we have:

$$Q = \sum_{n=1}^{N}\sum_{k=1}^{K} \gamma(z_{nk}) \log log\, \frac{1}{K} + \gamma(z_{nk}) \log log\, \frac{1}{(2\pi\epsilon)^{\frac{p}{2}}} \exp exp\left\{-\frac{1}{2\epsilon}\left\|(x_n - \mu_k)\right\|^2\right\}$$

We do the differentiation of the Q function which has this simplified normal distribution in it.

$$\frac{\partial Q}{\partial \mu_k} = \frac{\partial}{\partial \mu_k} \left\{ \sum_{n=1}^{N} \gamma(z_{nk}) \log log \frac{1}{(2\pi\epsilon)^{\frac{p}{2}}} \exp exp \left\{ -\frac{1}{2\epsilon} \|(x_n - \mu_k)\|^2 \right\} \right\}$$

$$\frac{\partial Q}{\partial \mu_k} = 0 \Rightarrow \mu_k = \frac{\sum_{n=1}^{N} \gamma(z_{nk}) x_n}{\sum_{n=1}^{N} \gamma(z_{nk})}$$

We again get the same formula for $\mu_k$, but the only difference is that this responsibility is defined in the way it was described earlier (for hard clustetring).

What is it basically saying? So for the k th component, only the responsibilities of the data points assigned to the $k^{th}$component will be 1, and responsibilities of the same data points corresponding to other components will be 0. So we take all the data points that are assigned to the $k^{th}$ component, take the mean of those data points, and update the mean parameter of $k^{th}$ component, $\mu_k$, with this computed mean value.

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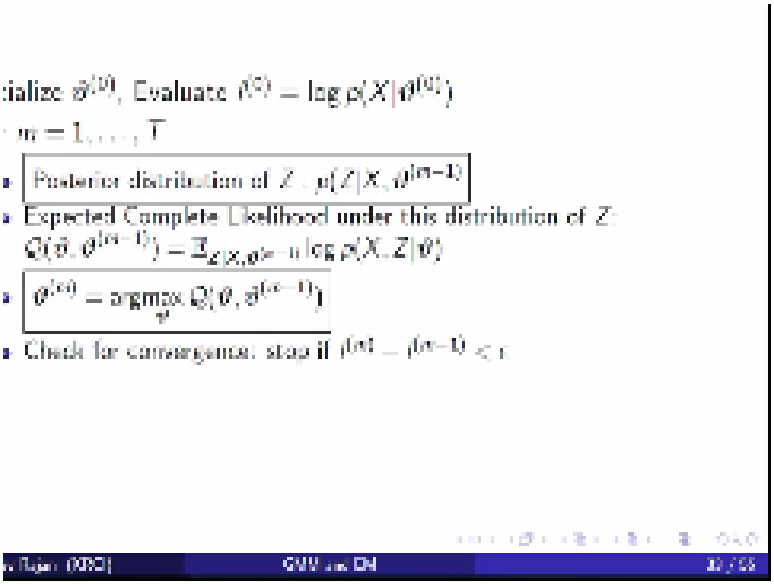

This is the general EM algorithm. We first calculate the posterior distribution of Z, which we saw is exactly given by:

$$\gamma(z_{nk}) = \{1 \; if \; k = argmin_j \|x_n - \mu_j\|^2 \; 0 \; otherwise$$

We assign the latent variable of $x_n$ to the closest mean, and then set the new mean as the mean of all data points with the same latent variable. This is exactly also the k-means algorithm.

So we are assigning $x_n$ to the closest cluster, with the cluster center $\mu_k$, and then reestimating the cluster centers as the mean of all data points that are assigned to that cluster. So k-means is just a special case of Gaussian mixture, where the covariance matrix is epsilon times the identity matrix. That is why we just have to compute the means, and do not have to worry about the covariance.

This entire procedure follows the same framework of EM that we saw.

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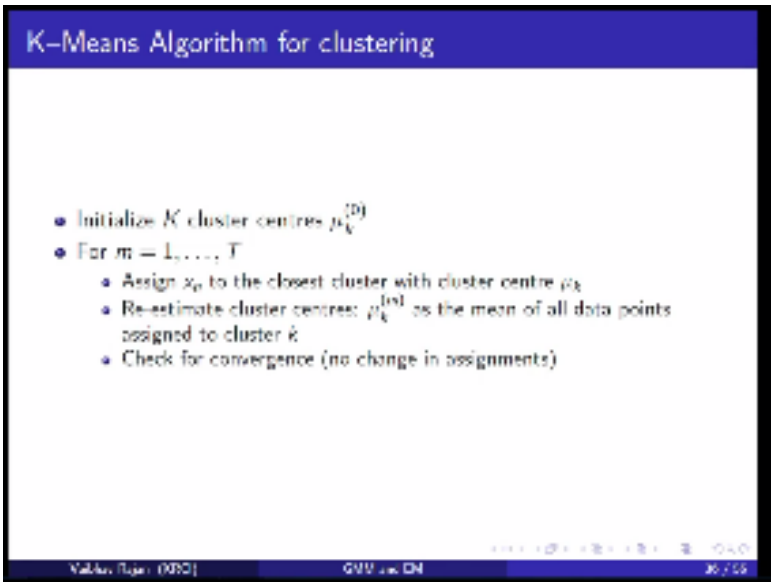

So I think we will stop here because after this we will talk about all the theoretical properties of EM. I wanted to show you one more thing.

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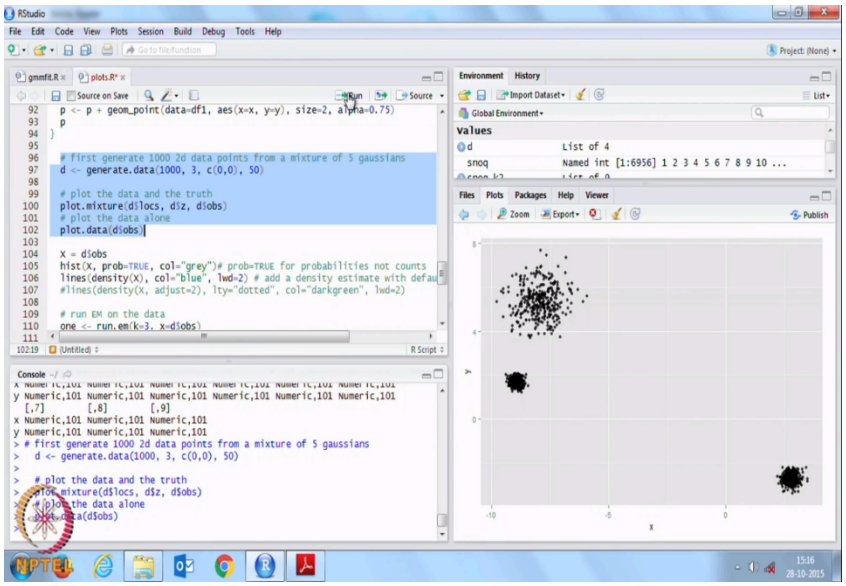


Let us consider this data.

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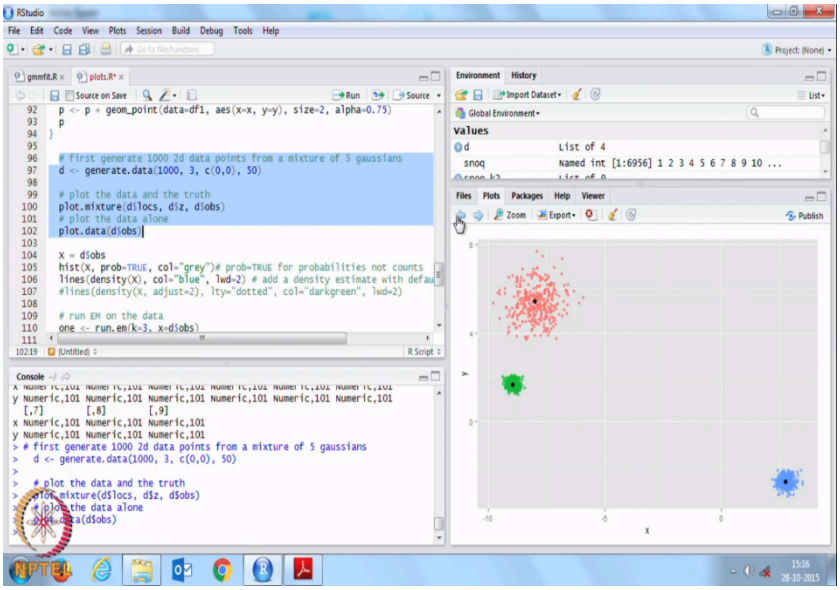

The data is generated from three Gaussians. The three Gaussians and their cluster centers are shown here. The previous figure shows how the data looks.

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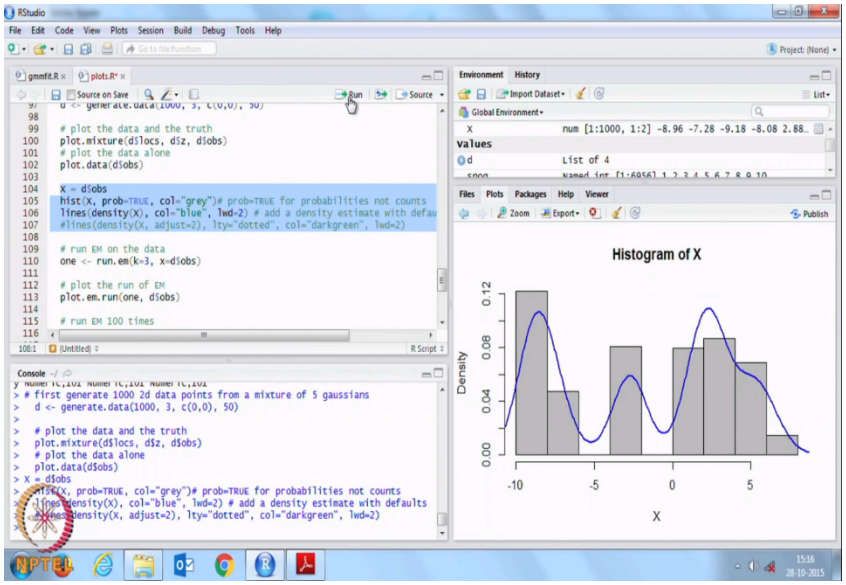


This is the density plot corresponding to the X coordinates of the data.

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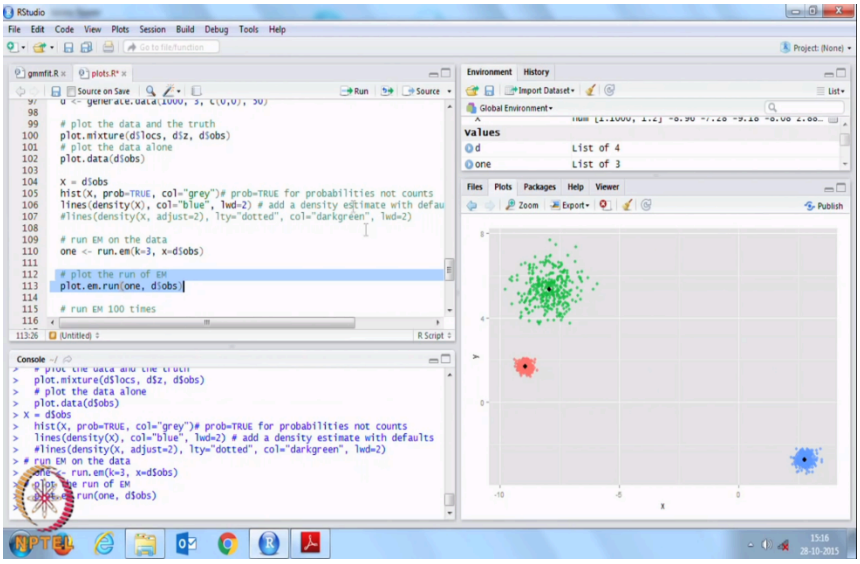


If we run EM once on it, and plot it, we see that it recovers the means and covariances exactly the way the data was generated.

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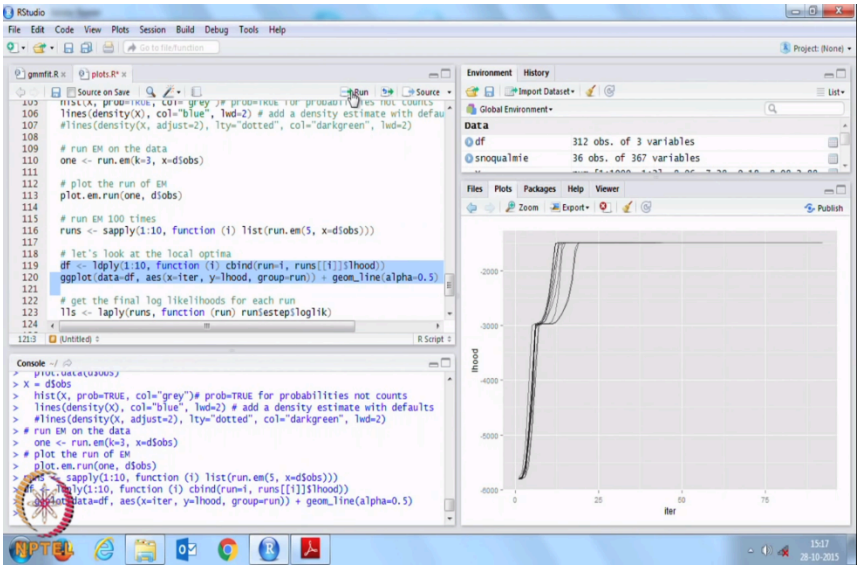

Now let's run the EM algorithm ten times, and plot it. In this plot, on the x axis we have iterations and on the y axis we have the likelihood. In each iteration, the likelihood keeps increasing until it reaches a point and remains steady there. So this is a property of the EM algorithm which we will prove in the coming lecture.

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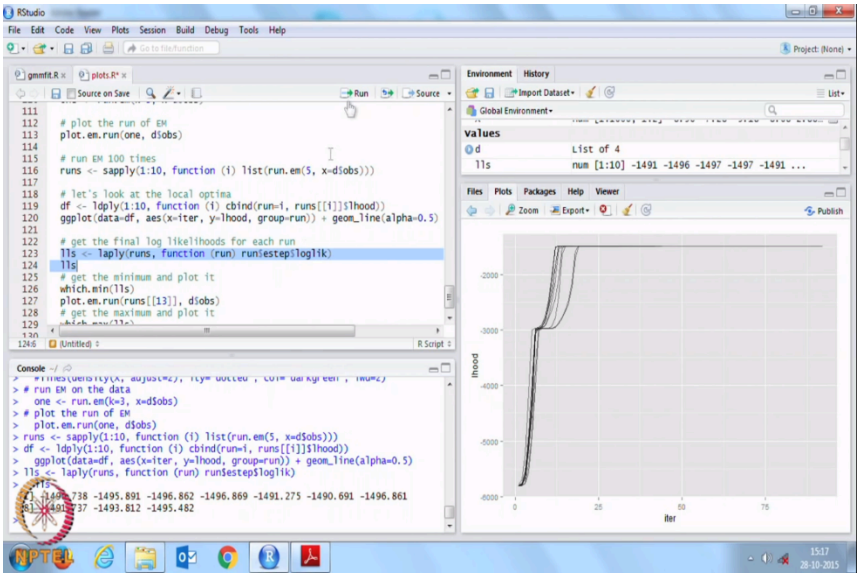


Let us look at these ten likelihood values. The 10 negative log-likelihood values corresponding to when we stopped the iteration for each of the 10 runs are:

-1490.738 -1495.891 -1496.862 -1496.869 -1491.275
-1490.691 -1496.861 -1491.737 -1493.812 -1495.482

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We always have a hard bound $T$ on the number of iterations because sometimes the likelihood may not converge. So we usually give an upper bound on the number of iterations as well and stop it there.

When we ran EM ten times on the data, the 10 final negative log-likelihood values were -1490.738, -1495.891, -1496.862, **-1496.869**, -1491.275, -1490.691, -1496.861, -1491.737, -1493.812 and -1495.482. The minimum negative log likelihood was achieved for the fourth run of the EM.

This is a very easy case.

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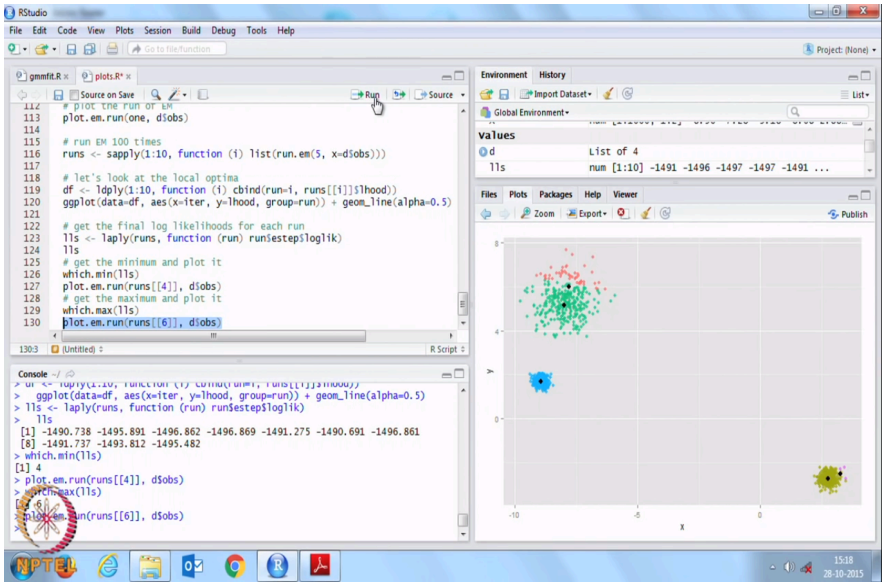


I gave $k = 5$, so it has estimated five different components. The means of the components are shown here. So it has tried to fit five Gaussians.

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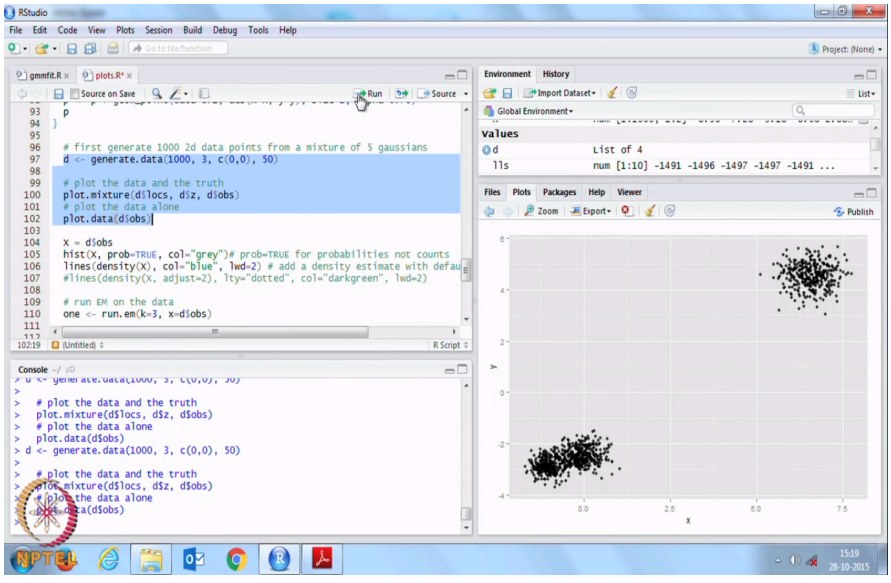


So this is a more difficult case because the three components are not very well separated. Now we run the EM algorithm ten times on this data. In this case, the data is as shown here.

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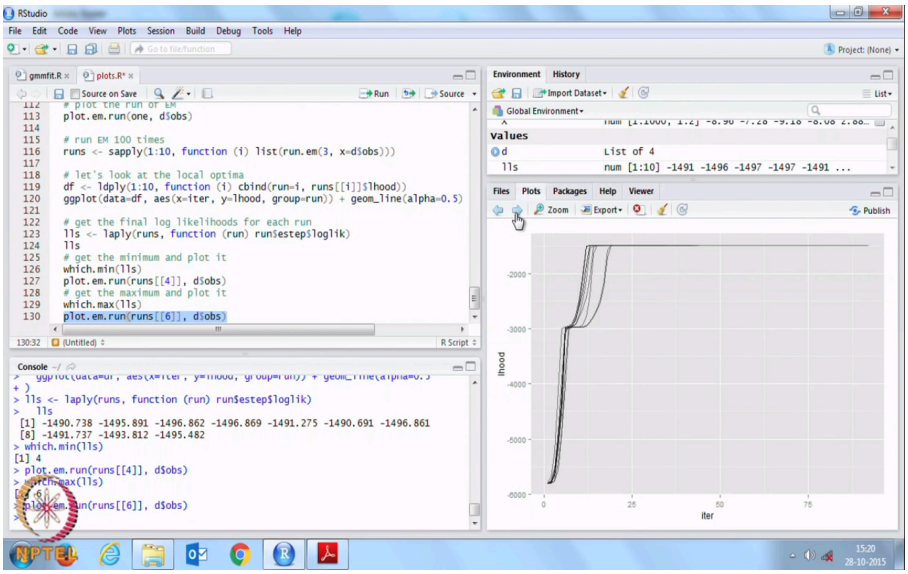


EM ran, the likelihood always increased.

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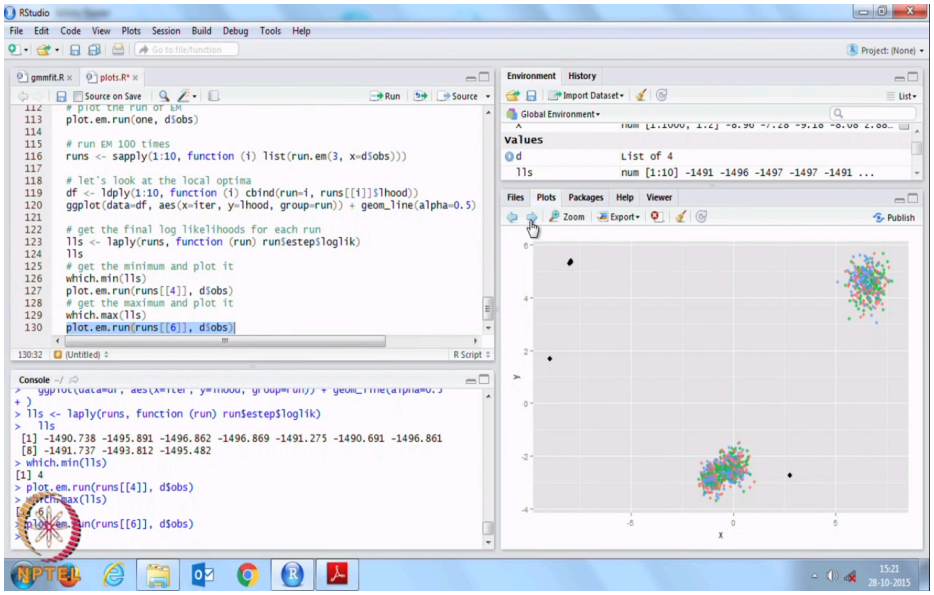

But what it estimated is shown here. So when the data is very well separated, like we saw earlier, EM will usually give very good results, but when the data is not very well separated like this, it starts giving very weird results. But no matter what it does, the likelihoods will always increase.

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**Introduction to Machine Learning**
**Prof. Balaraman Ravindran**
**Computer Science and Engineering**
**Indian Institute of Technology Madras**

Lecture 72 NOC

So we started with defining Gaussian mixture models, which are just a superposition of k different Gaussians. And in a general mixture model, instead of a Gaussian, you can use any other probability distribution. So the three important set of parameters are the mixture weights, the mean and the covariance matrices of each of the Gaussian's and there are k components, I have still not come back to someone had asked how do we estimate k, we will see that today. We saw some examples of how to fit Gaussian mixture models. We saw that Gaussian mixture models are good models when there is naturally good models when there is a clustered structure in the data so that each of those clusters can be nicely fitted with a Gaussian. Then we saw that the Gaussian mixture model can be very intuitively explained through the generative procedure where you assume that there is a latent variable that basically tells you which Gaussian to pick and then once you pick that you sample your or you generate your data from that particular Gaussian.

I think this is very important to remember as it makes sense of a lot of the math. Then we saw our posterior probabilities which are also called responsibilities. The posterior probability of the latent variable taking a value k given the data and we saw it coming up repeatedly in all our calculations. so the estimation assuming that we know k on p dimensional data the estimation problem is to estimate these pi k's mu k and sigma k for each of the Gaussians right and we saw that initially we first saw that if we assume the if you assume the we know the responsibilities then the math works out very nicely and we get very intuitive intuitive forms for the different parameters.

And then we designed an iterative algorithm that essentially guesses the parameter first and computes these responsibilities and then refines the guess in each iteration. And later we saw that this actually is the EM algorithm for Gaussian mixture models. So let's see this more carefully. In general, EM had been proposed for data which had some hidden data points not known when you get the data set. And so Z is, we denote that hidden data by Z.

And for the purpose of this discussion, we assumed it's discrete. And later, we saw that in the case of Gaussian mixture models, we can take the latent variables to be hidden.

That's a common trick used in many other models. And then we saw that EM is a good approach to take when the joint likelihood, the complete data likelihood can be easily parameterized. And if we make that assumption, then we see that we can get the marginal likelihood also.

So what is EM? The key idea is that, so this is the key idea. We take the expectation of the log likelihood of the complete data under the distribution of the latent variables assuming the guesses of the parameters that we have made. And instead of computing the maximum likelihood, we compute the parameters that maximizes this expectation. And this is the key idea of EM. Actually, if you remember this, this should be the main takeaway of this class, this formula.

So then we saw that if we use this formulation, then for Gaussian mixture models, we essentially get back the iterative algorithm that we had guessed. So this expectation is also called Q function in the literature. We get a very nice form for the Q function. The reason we get a nice form is one because we are using an expectation operator which pushes the summation to the outside and the second reason is that we get this logarithm of the Gaussian without any summation inside. The expectation pushes it outside.

That was the reason why the math worked out. and the derivatives became very easy to calculate for the case of Gaussian. We essentially got back the same formulas for mu k, sigma k and pi k that we had guessed earlier assuming that we know the responsibilities. So the general EM algorithm is this, guess the posterior distribution of the hidden data or the latent variables and then refine your guess by maximizing the Q function which is the expectation of the complete data likelihood under the distribution of Z with your current guess. And today we are going to see that this procedure is nice because it guarantees that the likelihood will increase in every iteration.

So whatever likelihood you start with, at every iteration the likelihood is going to increase. So that is what we are going to show today. This is the complete EM algorithm for estimating the parameters of a Gaussian mixture. and we also saw that if we assume that the only parameter to be determined is mu k which means we assume that all the gaussians are spherical with known covariance matrices and pi k is set to 1 over k then essentially what we get back is the k-means algorithm. So we can see this.

So this is the theoretical guarantee I was talking about. EM monotonically increases the observed data likelihood until it reaches some local maximum. It can also get stuck in some saddle point. So it doesn't give you the global maximum. It only takes you to the local maximum.

So let me show you that simulation that I had shown you last time. So, I generate some data. Three gaussians. This is what the data looks like. It was generated like this by taking those three means and covariance matrices.

This is what the fitted density looks like. If I run EM once, so this time EM did not do well. You can see what happened. The means that it inferred where two of them are here and the third one is here. Because these two clusters are very close together, it assumed that it is coming, it is being generated from the same Gaussian.

Let's now run this. So I ran this 10 times and what I see is that the likelihood for each of this run, the likelihood keeps increasing. Every time for each iteration the likelihood increases. Sometimes it gets stuck at a saddle point or a fixed point and then does not increase. So this is the typical behavior of EM.

In fact, this is a very good debugging tool if you're writing EM algorithms for your models. And if you see that the likelihood is not increasing, there's some bug in your program. Now let's see in these 10 runs, these are the likelihood values it got stuck at at the end, or it stopped at at the end, not necessarily stuck at. So if we see the minimum of these, this is the second one, and we see the fitted density when we use that run, you see the fit is not very good. If we take the maximum, the maximum likelihood among those ten runs, it is the ninth one, the fit is much better now, you see.

This time when the likelihood was in the 9th run, the likelihood was the highest among these 10 runs and the fit was also much better. So now let us prove that what we saw there is true in all cases, that it actually monotonically increases the likelihood in every iteration. So the main result we will need to prove this is Jensen's inequality. Are you all familiar with this? No? It is very simple. So if you have a convex function and you have a linear combination of these points x i, then the convex function applied to the linear combination is less than or equal to the summation is a linear combination of the of f where the function is applied to each of the x i's right and what we are interested in you might have guessed is the logarithm function because log is what appears in our summation and if we use the fact that minus log x is convex and put it here then we get this inequality It is the same thing.

The function is just the logarithm. So what we see is that log of a summation is always greater than equal to the summation of those lambda i times log x i. So let's start with So we have these latent variables or the hidden variables in some cases. Let us assume that q is some arbitrary distribution over the latent variables. We will not define what q is right now.

So because these are probability values, each of these q z n's for each latent variable is greater than 0 and the sum over all z n is equal to 1. So now let us take the likelihood of our data and we express it again as usual in terms of the joint likelihood with respect to the latent variables and then we just multiply and divide by Q of Z n. Now because of this condition Q of Z n is the same as lambda i is here, it follows the assumptions of Jensen's inequality. So we can apply Jensen's inequality here and get a lower bound on this expression. So basically take the summation outside and get the log inside and this lower bound just follows from Jensen's inequality.

All we have done is applied this inequality and lambda i's are the qz n's here because their probabilities the assumptions are true. okay so now this logarithm can be written as a difference of the numerate log of the numerator minus log of the denominator and what we get here now this expression should start looking familiar to you this is just an expectation is expectation of the complete data likelihood under the distribution Q right so this is something that we we have been working with in EM. And on this side, we have the entropy. So this entropy term is not going to play a big role here, but we are going to be interested in this.

So let's call this Q. Although it will be the same Q eventually, I have used a different Q here, because right now we don't know that it's the same Q. So what have we got? We have got a lower bound on the log likelihood. And we have proved this for any arbitrary distribution. We have not said that it is the distribution of the latent variables under the guesses of the parameters that we had. We didn't say anything about that.

So now the question is, which distribution Q should we choose? Any guesses? so what we have is a lower bound what kind of distribution would you like to choose no guesses think iteratively all right so we want since this is a lower bound we want the bound to be as tight as possible okay. So we will choose a queue such that we maximize the lower bound to reach the actual likelihood. That's a natural choice when you're dealing with bounds. So let's see how we can choose such a queue. To do that, let us just look at this expression again.

We will ignore the summation n because we will bring it back later but I have just not written it. So this is the original expression for the lower bound. The Q function is here. I have just written that again here. And now I am just expressing this joint likelihood.

I am factorizing it in this way. So, you have the probability of X n, Z n, the joint probability is just the probability of Z n and then the probability of Z n given theta. So, there should be an X n, Z n here. Yeah, it's fine.

No, no, it's fine. Yeah, it's fine. So this is just a factorization of this probability. And then I just separate it out in a different way this time. And what we get here is a term which is just the Kullback-Leibler distance between QZn and this probability, this distribution. It is the negative of the KL divergence between QZn and probability of Zn given Xn and theta. And this term is essentially summing over all Zn for this.

So this is independent of Q and we just get the likelihood back here. And here we have the negative KL divergence between these two distributions. If we want the lower bound to reach the actual likelihood, which we are getting here, we want this term to become 0. And that we can do by just putting q Z n equal to this probability, probability of Z n given x n theta. But again, we come back to the same problem that we don't know the actual theta, the curly theta.

But in an iteration of E m, we have guessed the value of theta, theta m. So we can use that value of theta m and use that probability distribution as q. So what we get if we use this value for q m which is the probability of Z n given X n and the guessed theta m values is nothing but this expectation which we saw coming up here. except that instead of q we are using qm which is based on the current guess value and we are getting the entropy term but this entropy term is independent of the thetas so when we maximize this in our m step of em this doesn't play any role and what we are eventually maximizing is this expectation of the likelihood under the distribution of z.

So let me again summarize what I did. I took the log likelihood. This is the likelihood that we are interested in for getting maximum likelihood estimates. Using Jensen's inequality, I got a lower bound. The lower bound was in the form of an expectation. which is the expectation we maximize in the E-step if we take Qm to be exactly this probability distribution.

And this probability distribution turns out to be exactly the probability distribution to take which will maximize the lower bound to reach the actual data log likelihood at that step. So what? What have we done? We have maximized. We have taken our current guesses and chosen a value of qm that will reach the actual likelihood with respect to the current guess. But that has not brought us closer to the real theta. We are still working with our guesses of the parameters.

Is that clear? So here comes the crucial part. So at the mth step, we took Qm, the distribution of Zn to be exactly this probability distribution, the posterior distribution of Zn given the data points and the current guesses. And we saw that this likelihood is exactly equal to the KL divergence plus the log likelihood. and because this KL

divergence becomes 0 at this point this Q function is exactly equal to the log likelihood which means the lower bound is tight after E step which is what we wanted and so maximizing Q after this is going to maximize the data log likelihood also to see that see this picture. So this is your current value, the guest value of theta. Now the E step and this red curve here is the actual data log likelihood with the original parameters that you don't know.

Now what the E step has ensured is that you get a lower bound using the Q function that we had. So that lower bound is L. So this is the this is the lower bound and which is exactly the expectation that we are trying to maximize. So you use this L function and you know that this is lower bound which means it is always lesser than the red curve. The important point is that at the E step this bound is tight which means this is touching the red curve.

And if you maximize this, you will get a new set of parameters which will increase the L value. But because this is touching it and because it is a lower bound, it will also increase the likelihood value with respect to the original theta. So it's a trick, because we cannot compute this likelihood, but we know the lower bound. We have computed the lower bound, and we are maximizing this. But the new values are guaranteed to increase the original likelihood also, because at this point the approximation is tight, and we are maximizing it.

So now again at the next step the E step will ensure that the lower bound that you calculate the green curve will be tight and once again you maximize it you will get a value somewhere here or anywhere here. and then the next value of theta is again going to increase the likelihood because every time you are at each step the e-step will ensure that you get a proper lower bound and you always get to the you always make sure that it is tight because of the choice of the distribution of q that we take at each step. Yeah because what if we get the saddle point, the likelihood curve need not always be like this right, so for example the likelihood value can be something like this. Well you can suppose Suppose it goes like this, then at this point it is not guaranteed to go up that way.

It will just be here in this region. So the usual problem with optimization. So we can say, now we can do this formally, at the m plus 1, m plus first round we have some parameters, that is the log likelihood of those parameters and we know that the q function is a lower bound, we proved it for any q, any choice of the distribution, small q and then this Q value was chosen by the previous iterations M step, so this equality follows, this is the maximum value of Q which maximizes the maximum overall parameters curly theta and then this by definition is greater than any Q here and because this E step bound is tight we get that this is equal to the logarithm, equal to the likelihood in the previous step

which is just the likelihood of the previous step. So any questions? Is this clear? Why it is increasing the likelihood at each point, at each iteration? So now that covers the basics of EM, now let us look at some strange cases. Sometimes what happens is when you are running EM, you tend to get very strange solutions.

And this could be one of the reasons. So I'm going to motivate this mathematically. So suppose you take your log likelihood that you want to maximize. And you set, now suppose you have two components. Well it does not matter, so take one of the components and set mu 1 the mean to be equal to x 1, one of the data points and set sigma 1 to be equal to some diagonal matrix of dimensionality p and take some pi 1. So this can be just split into two parts, we are looking at just one Gaussian here and when you plug in these values you essentially get this expression.

Now what happens if sigma 1 square the variance tends to 0, this likelihood essentially tends to infinity, I mean this total likelihood because this value goes to infinity. So this is a problem in general with maximum likelihood solutions. The likelihood will tend to infinity although the fit is really bad. So the pictorial representation is something like this. What you are doing is you are taking two Gaussians and you are fitting just one data point with one Gaussian and the other Gaussian is fitting the rest of the data points.

so this is in most real life cases this is not a good thing to do because yeah it is very very unlikely that the data has been generated by two Gaussian's like this, one data point from one Gaussian and the rest from the other Gaussian right. So any so when you try to do this with just a single Gaussian you think you will get this problem, why? Yeah but suppose you have, suppose I give you, take unidimensional case and I fit this 1 Gaussian here. This, it is, there will be a non-zero probability of a point coming from somewhere here, right? This is the mean. So intuitively, we would think that the blue Gaussian is what might have generated this data with so much variance. But there is a non-zero probability that the data has been generated from such a Gaussian.

So why will we not have this problem there? yeah so the maximum likelihood solution will never give you this maximum likelihood solution is most likely to give you something like this right when you work out the likelihood the likelihood for this the pink Gaussian is definitely going to be lesser than the likelihood for this and again this is just due to the mathematical form of the Gaussian mixture so this because of the summation this is really happening because it is possible that you can fit the data like that in a way that the likelihood goes to infinity. So how do you deal with this? The simplest way in a frequentist framework is to just keep when you are running EM you check whether it is happening or not and if you if it is happening then you just reinitialize the parameters you keep trying to detect such collapsing components and try to do it. And in general actually

it is better to restart EM several times because EM is, so as you know it can get stuck at a fixed point or a saddle point. So with different initialization parameters you can get much better solutions as we saw in the simulation as well.

The Bayesian solution is to take priors. You take priors on each of the parameters and it turns out you can work out the math and see that the E step remains the same and the only difference necessary is the additional term in the M step that we need to maximize and this usually solves the problem by choosing right priors. So now let us come to finding k. Till now we have assumed that we know the number of components and how do we find k? This is there is no really good solution to finding K. And what statisticians usually prefer and what works well in practice is to generate many candidate models. You look at the data and you assume that there can't be lesser than three components here, there can't be more than 12 components here.

So let's run EM for all these different values of K. And you choose that K which minimizes some criterion and there are different criteria that people have discussed. For example, it is something like the regularization that you do in other models. You basically penalize high values of k. So the AIC Akai-K information criterion is this.

This is just the log likelihood plus k. So minimizing this will give you the least number of components which can explain the data well. There is Bayesian information criterion which uses k log n, similar general idea. And then there are other approaches for finding k which are Bayesian non-parametric approaches where you assume some Dirichlet process priors and then the method itself automatically estimates k. So, the algorithm that we discussed in that form was given in 1977, so you can imagine that a lot of work has been done on EM since 1977. There are a lot of different kinds of EM algorithms, there are online versions that work on large streaming datasets, like I said EM is designed to find local maxima, so there are annealed versions that increases the chances of finding global maxima.

The simplest solution is random restarts but annealing does something more variation. So sometimes, so in the case of Gaussian we saw that the E step and the M steps they were computationally tractable. We could derive analytical formulas for these. But in a lot of cases, if there is time I can show you one, we will see that they are not computationally tractable and sometimes you need to do additional things. So there are variational versions of EM, there are stochastic versions of EM, Monte Carlo versions where you have intractable E steps, there is something called generalized EM which was one of the earliest algorithms where you have computationally intractable M steps.

Then when you have sequential parameters, dependent parameters, then there are other

versions of EM. And in general, EM is quite slow. So each step within EM, within the iteration is computationally not very expensive, but convergence is usually very slow. And it's especially slow when you have lots of missing data or lots of latent variables to infer. So, there are many approaches to deal with these Eitken acceleration techniques over relaxed EM and so on.

So to summarize like what I said the major advantage of EM is that it monotonically increases the likelihood, it guarantees that if you take any distribution, any mixture model or any likelihood computation where there are hidden variables or latent variables and you apply EM and if you follow the formulas carefully you will guarantee that the likelihood is increased except at fixed points. It is usually numerically very stable compared to other techniques like gradient descent. It is easily implemented. The interesting thing is that many problems can be modeled as incomplete data problems.

We saw that in the case of Gaussian mixture. There is no missing data in the beginning, but we assumed the latent variables to be missing. The disadvantages as I mentioned is slow convergence and there are no guarantees of finding global maximum and the steps may be analytically intractable. So the two standard references have very nice explanation for EM and there are very nice tutorials also available, matrix cookbook you should be familiar with to get all your matrix derivatives and this is the standard reference if you want to go really deep into EM, McLachlan and Christian's book on EM, the whole book is on EM algorithm. EM can always solve it but it may not be able to solve well.

If there are lots of missing data then usually it does not give good results. So there is some, there is a lot of work on, so these estimates that you are getting, you may need to, you may sometimes want to know how good those estimates are. So you want to get the standard errors on those estimates. So there is, so in fact that is one of the flaws of EM, it does not automatically give you that. But there are methods to deal with that. For example, there are some bootstrap methods that can give you estimates of the error, error estimates for the estimated parameters.

In that finding k, you had k max and k min, right? How do you get those two parameters? Yeah, they are also guessed. So that's something you have to guess based on the data that you have. So if you take the standard R packages like mclust or something like that, they usually have some default parameters, 2 and 12 or something like that. But then you can set them. So when m-cluster runs and gives you, like what I showed you in the simulation, when it runs and tries to find the parameters, it runs it for all those different values of k and takes the best one, best one with respect to the likelihood.
So it will be a good exercise I think like if you take if you take some different

distribution so take something like Bernoulli some very simple distribution and work out the math it will be quite nice to see how it works out and yeah even the other thing that I did not work out here this part is also quite simple to do. assume that there is a prior and see how it works out, but the general idea is clear right.

**NPTEL**

**NPTEL ONLINE CERTIFICATION COURSE**

**Introduction to Machine Learning**

**Lecture-80**
**Learning Theory**

**Prof: Balaraman Ravindran**
**Computer Science and Engineering**
**Indian Institute of Technology Madras**

Computer science folks, all of you are aware that theory in computer science means talking about hardness of a problems , so how hard is to solve and looking at space complexity, time complexity , and approximability . How approximate is the solution, those kinds of things . So we are going to try and see if we can give such a flavor of theory to machine learning as well .

So we are going to talk about how hard is the problem to approximate . So I have a perfect solution , but then I can get close enough to it , but how often can I get close enough to the solution. So that is the one kind of question that people want to ask. And then there is another question on how hard problems are to solve in general. So we are going to talk about hardness of problems. So different measures of the hardness of the problems .

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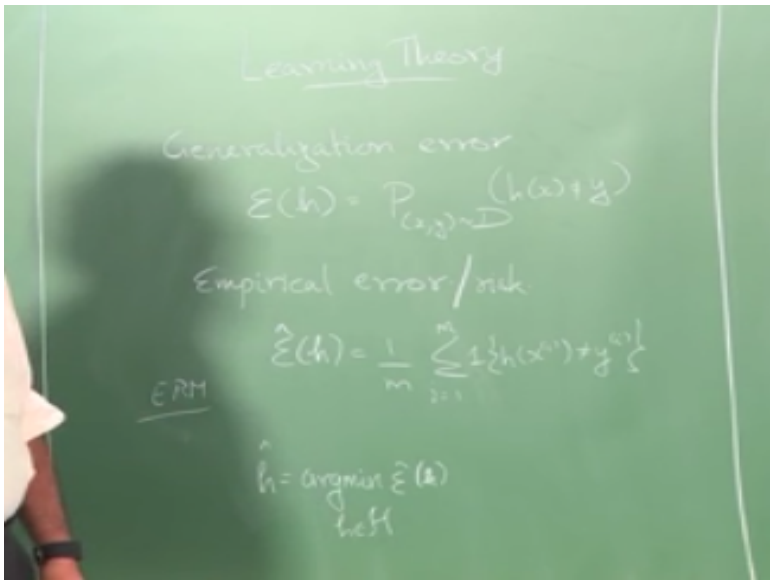


So typically we are interested in generalization error.

So what is generalization error?

So ε(h) is the generalization error of a hypothesis H, so I will denote my ε the error function, and of h means the error of hypothesis H .

$$\varepsilon(h) = P_{(x,y)\sim D}(h(x) \neq y)$$

The error of H this is the probability of the hypothesis H making a mistake on a data point X. When the data points X and Y, where X is the input and Y is the label, so where the data X and label Y are sampled from some underlying distribution D. If you are assuming that this distributing D is fixed, apriori and unknown we remember about all of this .

So we always talked about this, I talked about the$P_{(x, y)}$ earlier, but here we are talking about the distribution as D. So when the data is sampled according to this distribution D, so what is the probability that I will make a mistake ? So this is also the expected number of mistakes, because every mistake will count once. So this is essentially the generalization error.

But typically what is the error that we have access to ? We have access to something called an empirical error or sometimes denoted as empirical risk, which we will denote it by ε^.

$$\varepsilon\hat{}(h) = \frac{1}{m} \sum_{i=1}^{m} 1\{h(x^{(i)}) \neq y^{(i)}\}$$

So where xi and yi are the $i^{th}$ data point given to you in the training set , and 1 is the indicator function. This function will be 1 if this condition is true it will be 0 otherwise

So essentially when it will be 1, whenever I make an error, whenever h makes an error this will be 1, whenever h is correct this will be 0. So I will add it up for all the training data. I am assuming that I have m samples in my training set , I will divide by m.
So this gives me the probability of me making a mistake as estimated from the training data . So this is sometimes known as empirical error or empirical risk.

So typically I only have this quantity. So whatever is given to me as a training data. I can have many ways in which I can estimate this error, but this is all I have access to. So this is called the empirical risk. What I mainly interested in this is the generalization error.

So what we want to know is how good is this empirical risk estimate that I make in terms of measuring the generalization error, or how close is it to the generalization error. This is the question that we want to ask. See this has shades of hypothesis testing. So we are going to do a very different kind of analysis here. It has shades of hypothesis testing, but the kind of analysis we do here will be very different.

So the question that we are asking is, given that I can estimate empirical risk, what can I say about the generalization error. So before we go on to look at this in more detailed, I want to introduce a couple of results which would make easier for us to talk about. Before that let me talk about one thing. So most learning algorithms that we have, do what is known as empirical risk minimization .

So the answer that you will typically end up giving is,

$$\hat{h} = \text{argmin}_{h \in H} \hat{\varepsilon}(h)$$

so you will have some hypothesis class H, suppose you are looking at linear classifiers and your input dimension is say some P. So then you will be essentially looking at classifiers that are defined by $\theta_0$, $\theta_1$, $\theta_2$ or $\beta_0$, $\beta_1$, $\beta_2$ up to $\beta_p$ and then given by the inner product of that with the data point. And if it is greater than 0 you will classify it as one class, if it lesser than 0 you will classify it as another class. That is basically what linear classification does.

So when we look at the variety of linear classifiers, at the end of it you will have something like that. You will have some $\beta^T x$ and then you will have some function of $\beta^T x$ and then whether that is greater than 0 you put it with one class, if it is lesser than 0 you will put it in the other class. So that is essentially what the hypothesis class H would be. That would be in the case of linear classifiers.

In the case of neural networks, it will be all the classifiers that we can implement given the choice of number of layers and number of neurons per layer that we had. When I make some number of neurons, some number of layers choice, so for all those different values you can set for all those weights you will get a different classifier. That constitutes your hypothesis family subscript h. So what I typically would like to report is that h, has the minimum empirical error over all members of that family H. And that $\hat{h}$ is the classifier that we will report by doing empirical risk minimization. This is essentially called empirical risk minimization.

When this is ideal case, obviously we know that we do not get this. If you are using neural networks or training using back propagation or neural net and stuff like that, we actually do not find the argmin , you essentially have some approximation of it, then you just stay with it.

So likewise, depending on what classifier you are using you might not actually find the minimum. What you are trying to do is minimize the empirical risk anyway, because that is the only thing that we can measure. So we would really like to get a classifier that is good according to this, but we are not able to do that.

So we look at empirical risk, and then we find empirical risk minimization and we get this. So before I move on, I want to introduce a couple of results.

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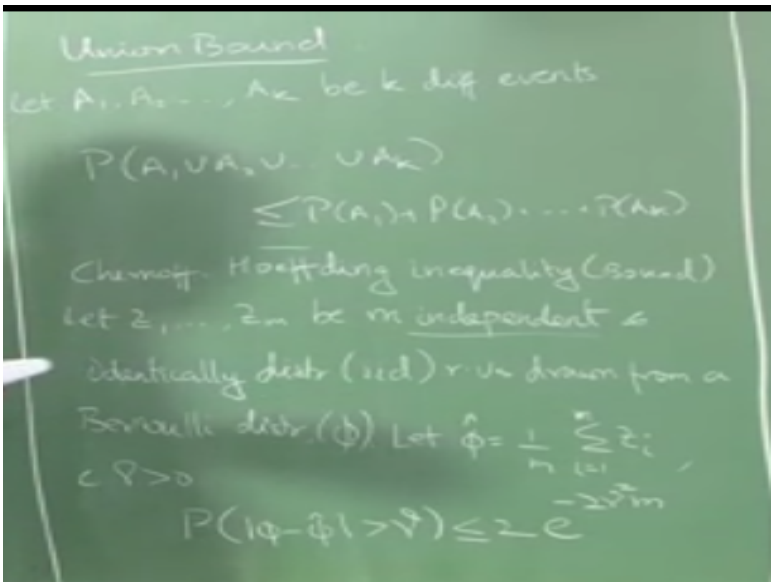


So the first one is called Union bound. Let A1, A2, ... to Ak be k different events, then ,

$$P(A1 \cup A2 \cup \ldots \cup Ak) \leq P(A1) + P(A2) + \ldots + P(Ak)$$

In most versions of probability theory this is taken to be axiomatic okay. So this is called the union bound.
So it is equal when, they are independent or they are disjoint. If we are thinking of them as sets they are disjoint.

So this is something which is variably called a Chernoff Hoeffding bound or the hoeffding inequality or the Chernoff inequality, when some subset of these two and some subset of that will be used for describing this result.

Let $Z_1$, $Z_m$ be m independent & identically distributed (iid) , drawn from a Bernoulli distribution.

So here I am stating this very specific to Bernoulli distribution but the Chernoff bound holds for in general.

There are other milder conditions, it need not necessarily be Bernoulli but as far as we are concern we are only interested in binary outcomes.
So can you guess what is outcome I am interested in ? Correctly classified or not correctly classified. So that is the outcome I am interested in so we are only considering Bernoulli in this case. In fact there is a version of it where you can also relax the independent assumption but gets more and more complicated.

In fact when you relax independent assumption you get some kind of the chromatic number or the interrelationship graph enters the picture. I still have not figured out how the chromatic number enters the picture, it gets really complicated lets so let us hope and pray that all the random variables we deal with out independent or you can think of this like that. But in some case that is not true you will have to worry about it.

So in this case essentially what I mean is, the probability that some Zi equal to one is say some $\phi$ .

The probability that $Z_i = 0$ is 1- $\phi$. So I will just keep it as some Bernoulli distribution parameterized based some $\phi$.
So what do we know about the Bernoulli distribution? $\phi$ is also the mean. We know the $\phi$ is also the mean. So typically the Chernoff Hoeffding bound is stated on the mean. But this version of the Chernoff Hoeffding bound is stated on the parameter $\phi$. But it is not in the role as the probability, but the $\phi$ is here is used in the role of mean.Because that is one thing I want you to remember like I do not want you to look at the Chernoff Bounds and then go actually flip through some other place and find that nowhere is the probability of outcome is mentioned, that will never be the case because Chernoff Hoeffding bound is stated on the mean.

Bernoulli distribution ($\phi$), Let $\hat{\phi} = \frac{1}{m} \sum_{i=1}^{m} Z_i$

So what is the $\hat{\phi}$? It is the mean estimated from these random variables I mean this all be familiar to you from the hypothesis testing case this $\phi$ is the true mean of the distribution and $\hat{\phi}$ is the mean that is estimated from these random variables the samples that I have drawn.

Right and $\gamma \geq$ zero, there is some fixed constant $\gamma \geq$ zero. So the probability that $\hat{\phi}$ is away from $\phi$ greater than $\gamma$. So what does this mean $\hat{\phi}$ is away from $\phi \geq \gamma$, is,

$$P( |\phi - \hat{\phi}| > \gamma) \leq 2e^{-\gamma 2m}$$

The 2 comes because this is a 2 sided inequality.
If you think about it, it is less than $\gamma$ or greater than $\gamma$. It is two sided inequality therefore the 2 comes so if you want to look only at 1 side then you can drop the 2 and the use $\mathbf{e^{-\gamma 2m}}$ .

So essentially what does it say if I have lot of samples? If m is very large , I have $e^{-\gamma 2m}$ , so it becomes smaller and smaller, as m becomes large. The probability that my $\hat{\phi}$ will be far away from $\phi$ becomes smaller and smaller. So this gives you a way of quantifying how many samples I need before my estimate that I am making. Before the mean that I have estimated is close enough to the true mean, with the high probability.

So $\gamma$ is something I fix apriori. $\gamma$ is something, I need you to be at least this accurate for me. Now go and tell me how many samples I need.
Alternative I can ask a question like, I have so many samples, how accurate I am likely to be ? Is it fine, is it enough to fix the number of samples ?
You have to be little bit more work.

So what is the probability that the error I am making is greater than $\gamma$ ? So I need supply $\gamma$. An error for me to find m. I need to supply m and an error for me to find $\gamma$.

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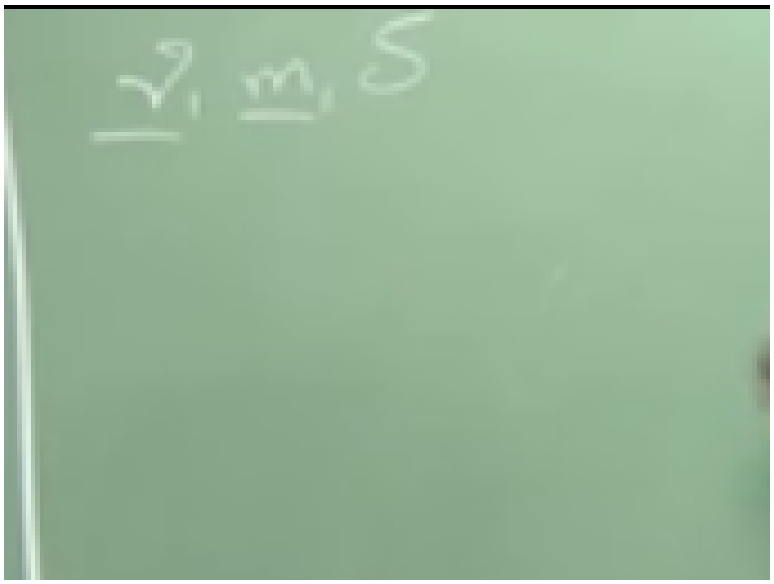


There are 3 quantities here. So there is γ, there is m and there is also the probability of the error.

So γ is the magnitude of the error. What is the probability that I am greater than γ. So that is a 3rd quantity. So I have number of samples, I have the magnitude of the error and then I have the probability of making an error of that magnitude. That is the left hand side. So I have 3 things here. So this equation has 3 things here.
So if I want to solve for γ so I need to specify the left hand side and I need to specify m, then I can solve for γ.

I say, I do not want to make a mistake more than 10% of the times. That means my probability should be 0.1 . I am only giving you 100 samples, then I will come back and tell you to be correct 90% of the times then you will have to say that even if I am so far away from the hand side I am correct. Okay only then I can give you the 90% guarantee. So that is essentially what I

means by saying there are 3 things. Here you have to specify any two then you can think about it in terms of the 3 one.

So now we have these 2 results for us. So usually that probability is denoted by sum $\partial$. okay so we have this 3 things.

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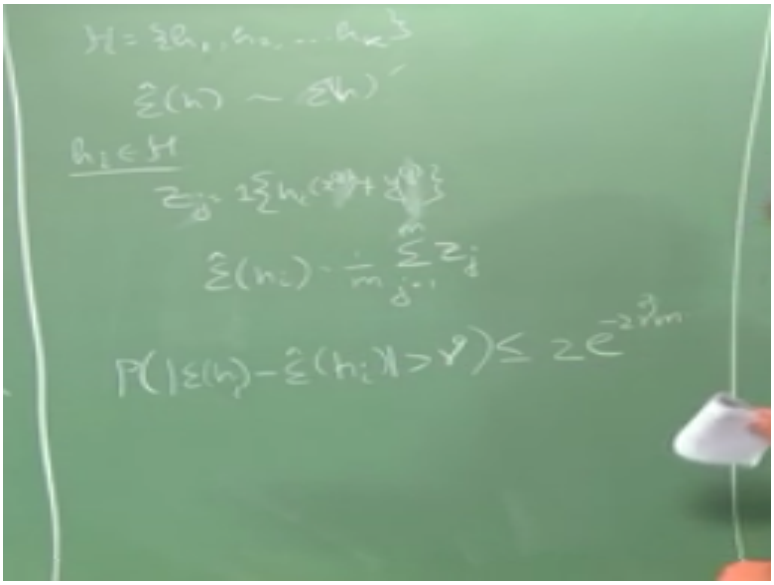

So we will start of a case where I have only k specific hypothesis in my hypotheses class. I am only searching thorough a space of k. k can be very larger, I am not telling how small or large k is but I am only saying, I have only k hypothesis in my class and I am going to search through this. It makes this slightly easier for us to developed some intution and the we can go on and talk about the infinite h okay.

So we want to look at how $\hat{\varepsilon}(h) \sim \varepsilon(h)$ . For some hypothesis h how does $\hat{\varepsilon}(h) \sim \varepsilon(h)$ .

I am going to fix some $hi \in H$ .

The Bernoulli random variables that wanted here as we mentioned earlier are going to be defined by so random variable z is if $h_i(x)$ is not equal to y. It will be 1, it will be 0 otherwise.

So whenever hi makes a error then this random variable will be 1 whenever hi does not make an error the random variable will be 0.

So we can go head and write, write zj for each xj as xi xj not equal to yj so if you remember we always make the assumption that the training data was given to us in an IID fashion, independent identically distributed fashion from very beginning we have been saying that training data is IID each sample was taken independently of one another and we used this fact even in the hypothesis testing case again we will use the fact here and that allows us to apply the Hoeffding bounds there allows us to apply the Hoeffding inequality.

So what is that we have so I have if I have m training points like we said we assume there are m training points, therefore I have this m independent identically distributed random variables okay where is the distribution that I am drawing this from, from the true distribution of the data so I am assuming that all the points are coming from the true distribution of the data, all the training data points have come from the true distribution of the data so that is something which is very important.

These are the two main assumptions that we are making here what is the first assumption, IID the second assumption the training data comes from the true distribution so that is the distribution according to which I want to evaluate the generalization error so these are the two assumptions you make , of course we can always relax this assumptions in fact quite of we need to relax this assumptions because in real life we will not be able to satisfy either of the assumptions .

But this is good enough to give a some kind of a intuition as to how things will work and then we can worry about relax in this assumptions later on.

So ε is already define there, so all I have done is take whatever expression there was there in the sum there and define that as zj and I get this. so already we know that so what is ε(h) that is the true mean. What is $\hat{\varepsilon}$ ? so it is estimated by taking m samples. So you can directly, go apply some Hoeffding bounds.

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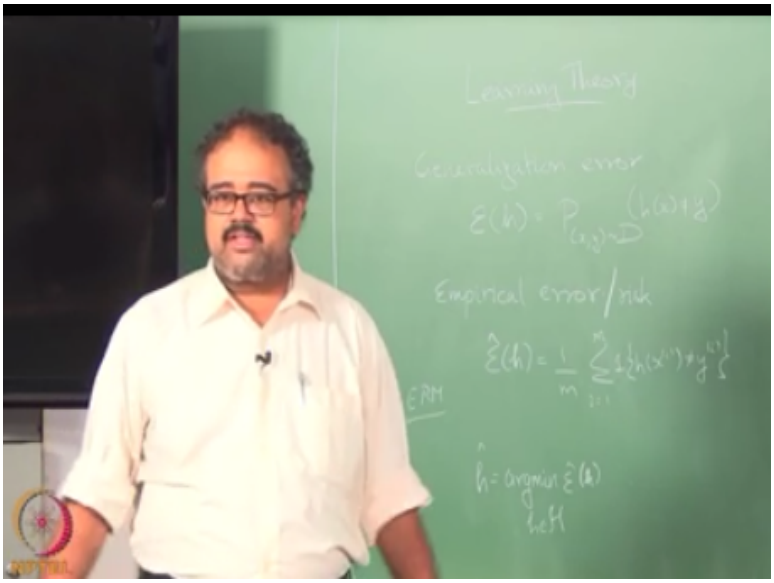


So if you have a very large class then you have to be very careful about minimizing the empirical risk, because you have you run the risk about fitting it , so yeah so essentially what you are trying to do in many of those cases is try to get a better estimate of the error on the better estimate of the you know generalization error actually, so when you are trying to do the validation .

So essentially trying to get a better estimate of the generalization error directly, so with all of these things or essentially to give you some notion of what is it, some notion of the complexity of the problem that you are trying to solve okay. This just gives you some notes of the

complexity of the problem that you are trying to solve and we will see that in the minute you will see that .

So in fact one way to avoid over fitting in neural networks is to have a very, very, very large training set if you have lots of ways and we need to have a very, very large training set that will kind of all out of this just give me a second to explain this.

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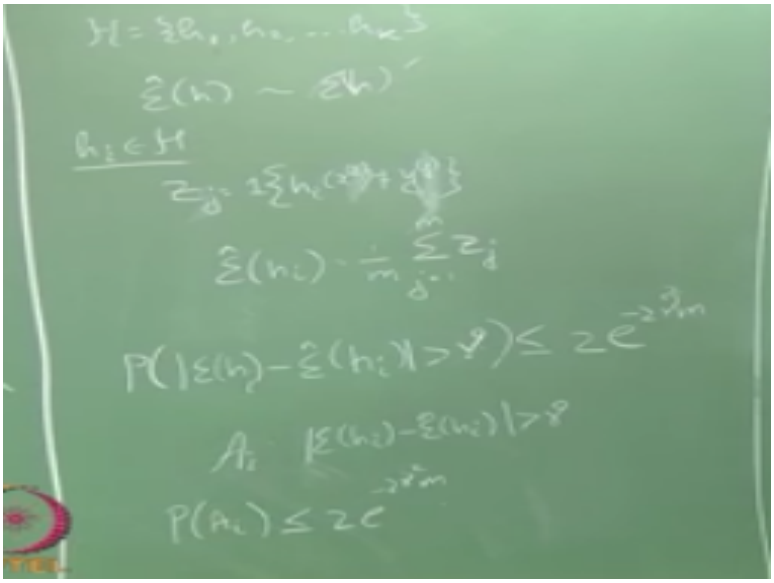


So what we are now shown here in the above image is that for a single hi from the hypothesis class if I have a large m then the error will be the probability of having a large error will be small, so or even say I will be pretty close even for small γ suppose I want to know what is a probability that this greater than ε greater than ε^ is the different between ε, ε^ greater than let say 0.01 so that is what I want to know so my ε^ should be within 0.01 of ε okay then what I will plug in here is 0.01.

Right so $e^{-0.01}$ so essentially getting to 1. Essentially getting to this since what I am saying is the probability of the error greater than 0.001 is less 1 then really tell me anything is just less than 1

at a I mean I know that, but then if my m is very, very large suppose I want this to be 0.01 this is $0.01^2$ then I will let say m is a million so then what will happen is I will have $e^{-20}$ or something, so that is a small number.

So if I say have a lot of samples then the probability of ε^ being 0.01 close to ε will be high . what this states is the probability that it will be 0.01 away from ε is low , so the converse is if it is the probability that it will be 0.01 close to ε, it will be high so that is essentially the result that we have what we have shown here using the hoeffding bound so I will show the proof of the hoeffding bounds it is not trivial so interested you can look it up this is not very hard okay it is just have to work it out that is all.

So but once you accept that on faith though you have the result but unfortunately this holds only for one particular hi that is not very interesting I am essentially what I have shown is you can give me like a 10, 000 different hypothesis and I can show you in for one of those hypothesis, if I have a lot of samples then it will be close.
What I really want to show is okay if you give me 10, 000 samples are give me a 1 million samples for every hypothesis in this hypothesis class, I will be close so every hypothesis is in hypothesis class I will be close so how where you going to do that ? use the union bound so what I will say is I will define my event Ai is I need those A1 to Ak , I will define the event Ai as so define this event $A_i$ as ε and ε^ being more than γ away so now this becomes probability of Ai.

$$\mathbf{A_i : | \varepsilon(h_i) - \hat{\varepsilon}(h_i)| > \gamma}$$

$$\mathbf{P(A_i) \leq 2e^{-2\gamma 2m}}$$

Right now this is essentially becomes probability of Ai.
So now I do union bound so what is the union of A1 to Ak what does it mean at least one of them giving a higher error , so essentially this reduces to.

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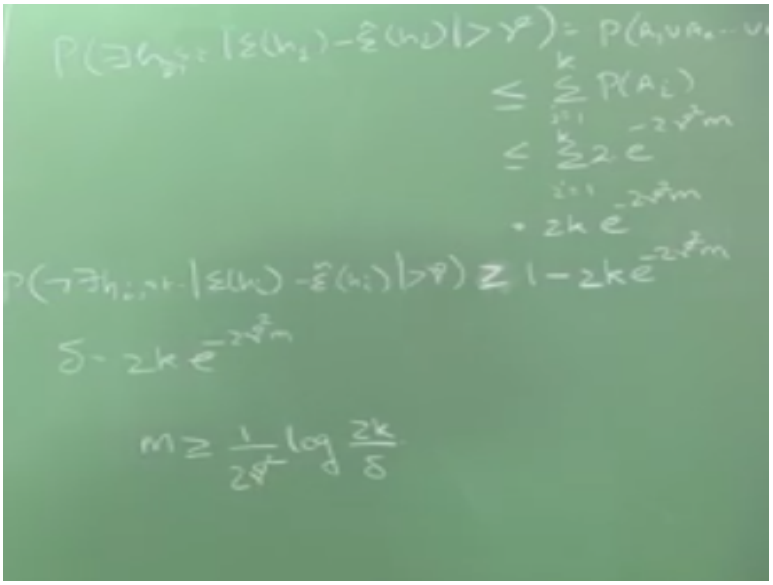


Where exist h such that it is that this is some hi okay at least one h so probability of at least one h for which there exist at least one h for which the error is large so this is essentially equal to the probability that there will be at least one error there may at least one hi which has large error is bounded by $\mathbf{2e^{-2\gamma 2m}}$ .

For one of them it is just $\mathbf{e^{-2\gamma 2m}}$ . For k of this it us just multiply by k this is essentially what we get from union bound.

So now what I need I need the probability that, that does not exist any classifier that makes a large error if I give it m samples okay that does not exist any classified that gives a large error so

what will that be 1 – this similarly 1 – that. This is simple algebra here so we got this and so this kind of a result which holds for all H in the complexity class where mean in the hypothesis class we are taken if because this now this results holds for all H and capital H these are called uniform convergence results, . So this is the uniform result.

Because it holds for everything in the this is more like a single result this is for a specific hypothesis well this result that we are giving that $(1 - 2k)e^{-2\gamma^2 m}$ .
That bound is for all hypothesis in this hypothesis class so this is called a uniform convergence result okay, when somebody says uniformly convergent that means set it works for all classifiers it works for all classifier, okay. So far note that I am not actually talked about finding the classifier, .

So what is ε, ε given a classifier what is the error in the classifier will be making in the overall population and ε^ is the error that it makes on the training data so I am just comparing the two I am not actually talked about finding the classifier so what we should not be looking at is, we should be looking at comparing okay I will come to that in a minute.

If you remember I said there are like three quantities in the beginning. So now I am interested in solving for m I want to know how much how many samples I have I want I can draw before I can give a certain guarantee on okay I can give a certain guarantee on the performance what I mean by performance here, whether my empirical error is closed to the generalization error I am not talking about the best generalization error when I am talking about performance here I am talking about whether my estimated error is closed to the true error, okay.

So how many samples should I draw before I can give some guarantees on the performance of the estimator okay. So I want to solve for m and I need to fix γ as well as the error probability as well as this guy so, k is fixed for me I give you the hypotheses class as soon as you give me the hypothesis class K is fixed for that so I am going to says that the probability should be utmost this quantity should be utmost $\delta$ like some $\delta$ I will give you the $\delta$ so this quantity I will fix so the probability should be 1 - $\delta$.

and I will give you the γ also solve for m . $\delta$ will be , now I can give you γ also so I have given some value I will give you some number for $\delta$.

I will say $\delta$ should be 10% so I will say $\delta$ should be 0.1 I will give you the $\delta$ I will say $\delta$ should be 0.1 .
I will give you the γ also okay, now solve for it. So what does $\delta$ equal to 0.1 mean that 90% of the time this event should be true okay so I will give you a γ I will give you a $\delta$ you find m for me. What this tells you is you chose a small hypothesis set to ensure uniform convergence.

So that it will tell you what the true error is it does not tell you anything about how good the true error is okay. So that is so that is what I kept re-iterating to say what performance means in this case, performance is not minimizing the error. performance is minimizing the error in predicting the error okay this is kind of a circular but this is trying to minimize error in predicting the error, okay that is all.Or in measuring the error.

$$m \leq \frac{1}{2\gamma 2} \log \frac{2k}{\square}$$

So this is called the sample complexity this tells you how many samples you have to draw so that what, so you are all your classifiers are within γ of the true classifier , the probability of that happening is at least 1- $\delta$, . So this kind of formulation are called PAC formulations, P A C so you know what is PAC is?

You know what PAC is ? probably approximately correct , so that probably part comes from that, . The approximately part comes from that so I am not telling you it is correct okay it is approximately part comes from that so I am not telling you it is correct okay it is approximately it is within γ of the answer but is it always within γ of the answer no, no it is with high probability it is within γ of the answer, .

So it is probably approximately correct but in many cases this is the best I can tell you because there is so much variation in the samples that you are drawing and the problem itself has inherently has noise in it say there is only so much I can do in terms of predicting it correctly.

So the another thing which I want to look at is I give you m and $\delta$ okay. Can you solve for $\gamma$? Right I fix m and $\delta$ solve for $\gamma$ what you get fairly straight forward ?

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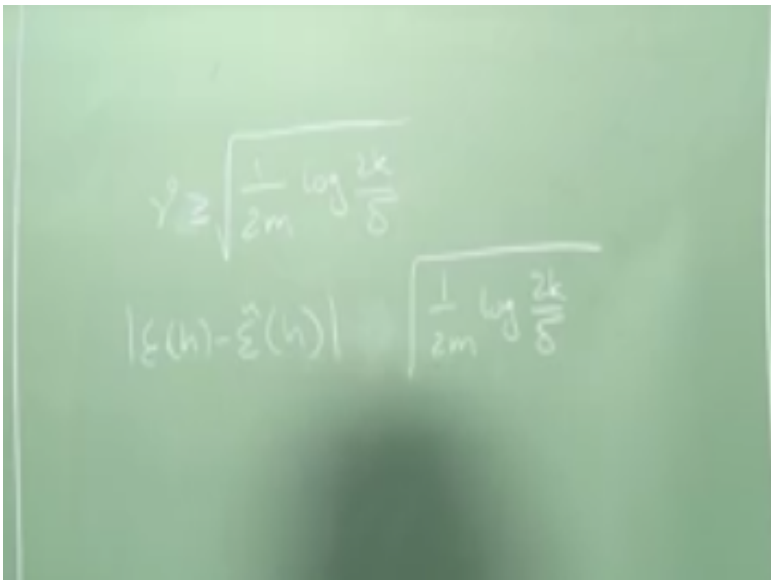


Okay likewise if you want to solve for $\delta$ you can try these things but this is fairly easy system so what is $\gamma$ really it is the error in the prediction of the error .s so it is $\varepsilon$ - $\varepsilon$^ is $\gamma$ .

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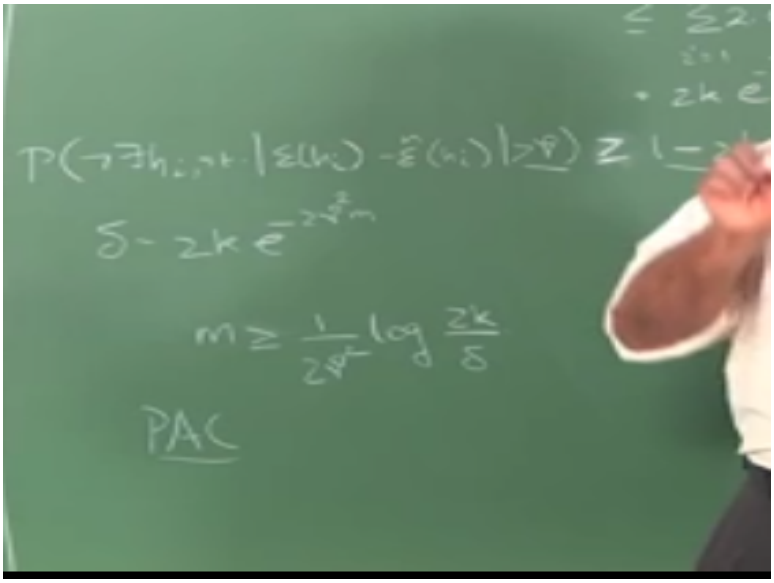


So my γ yeah so should be greater than or equal to here , γ should be greater than or equal to so my γ should be at least yeah, γ can be at most this small , γ can be at most this small but it can be greater so then I can give you the guarantee so that is essentially what we are looking at here , so this is not needed.

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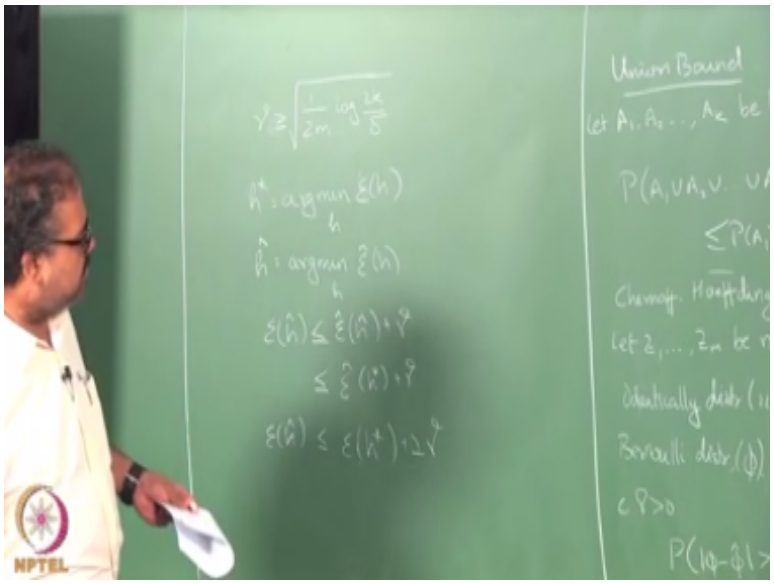


$$h^* = \text{argmin } \varepsilon(h)$$

$$\hat{h} = \text{argmin } \hat{\varepsilon}(h)$$

$$\varepsilon(\hat{h}) \leq \hat{\varepsilon}(\hat{h}) + \gamma$$

So we will define h* to be, h* is the true, the truly the best classifier that we have , in hypothesis class h , we will define h hat to be that classifier you will pick by doing empirical risk minimization , ε^ is your classifier you pick by doing empirical risk minimization okay, so knowing whatever we know can be write things , so $\hat{\varepsilon}$ , $\varepsilon(\hat{h})$ is less than or equal to $\hat{\varepsilon}(\hat{h}) + \gamma$. why is this true because of whatever we have shown all the value , so I am saying that , I have taken enough samples M so I can say with some probability this event will hold, .

Because of my uniform convergence with some probability that some probability 1-$\delta$ this event will hold , because this will be γ close to the error , the $\hat{\varepsilon}$ will be γ close to ε.

$$\varepsilon(\hat{h}) \leq \hat{\varepsilon}(h^*) + \gamma$$

$$\varepsilon(\hat{h}) \leq \hat{\varepsilon}(h^*) + 2\gamma$$

So that means that h* should either have a higher error than $\hat{h}$ or at least equal error. It cannot be a better error than $\hat{h}$ because we have better error than $\hat{h}$ then $\hat{h}$ would not have been chosen okay, does it make sense , this I am using by the visual of the fact that I did minimization here , and therefore ε^ h* should be worse than ε^ h hat.

Using uniform convergence again I can peel out one more γ from there , did it make sense what I have done here is it is exactly I went from here to here , from here to here how did I write this γ because of uniform convergence , similarly what I did was I took ε^ h* and I said that it will be within γ of ε h* so then I add up the γ here so I get 2 γ , so all of this will hold with some probability but I am assuming that I have taken enough samples and I have converged to whatever degree of accuracy I need. I have converged to some probability 1 - $\delta\delta$. I have converged here.

So once I have converged so this will hold so essentially what I have is that so what does is mean so the true error of the classifier I produce by doing empirical risk minimization is within 2 γ of the true error of the optimal classifier okay, does it make sense the true error of the classifier I produced by doing empirical risk minimization of which is $\hat{h}$ is within 2 γ of the true error of the classifier that is produced by minimizing the true error is essentially the optimal classifier, .

So essentially this gives you the guarantee that if I do empirical risk minimization , taking that many samples , then I will be within 2 γ of the true classifier anything else I need that, with probability 1 - $\delta$.

so if I take at least this many sample M , so the classifier that I find by doing empirical risk minimization will it be within 2 γ so our γ is say number you plug in here , within 2 γ of the optimal classifier at probability 1 - $\delta\delta$, so that is essentially the result that we can have, .

So let us put this together and write small theorem.

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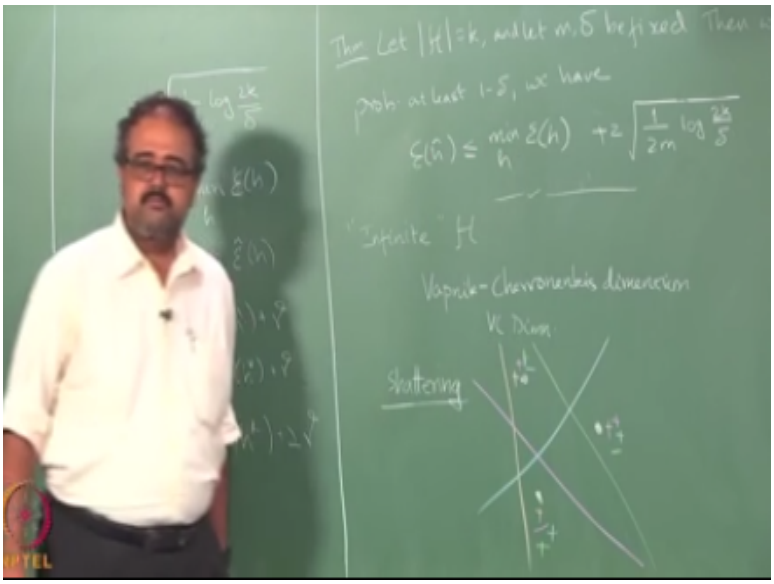


$$\varepsilon(\hat{h}) \leq \min_h \varepsilon(h) + 2\sqrt{\frac{1}{2m}\log\frac{2k}{\delta}}$$

Right, so essentially I have written our γ from wherever we solve for γ, I wrote the γ expression down there , and we get this okay, . But what is ε (h*) , so ε(h*) is actually minimum over h of ε of h.

so this is essentially the best possible classifier that you can build okay. So now we have this result we can think about, if my hypothesis class is small , essentially what I will be doing is I

will be searching over small number of things to find the lowest error so it is likely that my solution the best solution I can find in this class itself will be bad, if my class is small.

But if my class is small the second term is small , if the number of thing if the k is small the second term is small, but my k is large , then the likelihood of me finding a small error solution is high but the number of samples I will need will also become larger. So this is one form of bias variance trade off. So if the hypothesis class is small that means that regardless of how much data you give me I will always be making some error because I have only a few hypothesis that I can search through.

So that is like a bias you know it is like doing linear regression kind of thing. and this is variance because hypothesis class is very large then I will need a lot of samples for me to estimate the error properly, so that is the variance path so this is like one version of the bias variance trade off that comes in so that is the reason you need large hypothesis class so that you can be sure that you contain you have the solution in there.

But if you know exactly what is the solution we are looking for then you are better of using a much smaller hypothesis class okay, so that is essentially the to take away message here.

So we already know what is the sample complexity you need for this result hold for a specific $\gamma$ , so here when the $\gamma$ is given by this expression but if I give you a specific $\gamma$ then say something like okay, with some probability 0.1, I want to be at least 0.1, close to the true answer , the probability 0.1 I want to be at least 0.1 close to the true answer.

So what is the 0.1 close here it is 2 $\gamma$ , so it is 0.05 , $\gamma$ should be 0.05 , so I will plug in 0.05 and 0.1 here , and I know what k is because of the hypothesis class I have chosen I can always find out what the sample complexity m is, so given a $\delta$ and a $\gamma$, I can find the m, okay. So this kind of an analysis, this kind of sample complexity is sometimes called is usually called $\varepsilon$ $\delta$ where because people usually use $\varepsilon$ as a symbol for $\gamma$.

But in this case it will be $\gamma\delta\delta$ $\delta$ sample complexity or $\gamma\delta\delta$ $\delta$ PAC analysis.

Because I fix the γ I fix the $\delta$ and I ask you for the sample complexity , so this sometimes called γ $\delta\delta$ $\delta$ PAC, okay. So this is assuming you have a finite hypothesis class what do you do in the case of a infinite H ? any thoughts about how it extent this analysis an infinite H?

So in a practical setting I will be typically implementing all my machine learning algorithm in a digital computer this is kind of a cheating argument but it is fine so I will be implementing these in a digital computer and digital even though they are implementing in infinite class of things they are limited by their by the numerical procession .

So let us say I use 64 bits to the present floating point numbers then only finitely number of finitely many classifiers I can represent with 64 bits . So the problem is it is a large finite number but I can still go back and if you look at that number I have there if you look at the m that I need you can think of m is being actually ignore a whole bunch of things here but m is order of $(1/\gamma^2)\log(k/\delta)$.

So you can always say that okay regardless of how large my hypothesis class becomes this is going to be log of that order of log of that so that will be a significant reduction. Unfortunately the hypothesis class becomes exponential suppose I have d I have d numbers I need to specify one classifier and have 64 bits , so how many hypothesis do I have? So how many bits do I need for representing one classifier? 64 times d so how many do you have and then how many classifiers I could have 2 power 64 times d, that is K.

I plug in K here then it becomes D it becomes order of D. So if I have a 1000 parameters then I need about order of 1000 samples not bad. 1 by γ, log $\delta$ all I mean it depends on what you except out of it you chose a large γ large $\delta$ I am just joking , yeah of course you always have those things the γ, $\delta$ and other things actually play role there but if you think about it I mean I have d parameters I mean to need to get at least order of D samples to solve the problem okay, how do you think it can do it less than order of d if I am do it less than order of D? Then many of my parameters are all tie together they are redundant.

So this the power of the big O notation you can hide a lot of things under the big o umbrella because I hiding all your 1 by $\gamma$ square and log of 1 by $\delta$ you are hiding under the big O umbrella but it is not a bad thing okay so this is 1 way of thinking about it it is not the greatest way of thinking about it but this is one way of thinking about it in fact people use a rule of them suppose you are training a neural network which has a say 10000 weights they use the typical rule of them they uses you need at least 10x the number of x, so if you have 10,000 weights you need 1,00,000 data points.

At least in fact this is a very useful rule of thumb if you are only using feed forward neural networks so remember this if they count the number weight you have and ten times that how many data points you need at least for you to give anything useful. But then there is a better way of doing it . call the Vapnik - Chervonenkis dimension otherwise known as the VC dimension .

So given a hypothesis class we can define the VC dimension of then hypothesis class I will finish in a few more minutes okay. So before I define a VC dimension I need to introduce the notion of shattering , so give some set of points okay let us say some set S, I give some point x1 to xk let us say a hypothesis class H is set to shatter the set S if for every labeling you can give for that set S, there is some element in the hypothesis class which actually separates the classes a binary classes okay.

So every possible binary labeling that give on the set okay I have a hypothesis in my hypothesis class that separates it from one class from the other okay is it clear, so let me draw a picture that will make it clear let us say that, so that is my set s okay and my hypothesis classes all straight lines okay. So think of all possible labeling I can do for this . I can basically set on say everything to one side of the line is + everything to other side of a line is - great.

So now what we do everything to one side is + everything to other side is - so likewise I can keep going I keep going as long as there are different number of colors here and then guys can kind of intute . So give the set of three points you can just see that so if I flip the + and - it is exactly the same so you have to only worry about the unique things. so if the flip the + and - it is just the same.

So if I make this + and these two - the same thing will work . So is there anything else that we need to consider? So + + -, + + +, + + - that should be a - , that will work anything else so I have to consider 2 pluses and one - two - and one + is just flip of it okay and three pluses I have considered and three - is just the flip of three + okay, so anything that we can to consider? That is it ; I will leave anything but anyway even if I left out something you can make it up , so you can easily see that.

So straight lines so hypothesis class of straight lines shatters three points in space , what about four points? Yeah we talk about single straight lines, so what about four points? Is there any configuration of four points which can be shattered by single straight lines, no configuration of four points that can be shattered by single straight lines that then automatically applies a five six seven eight nothing .

So the VC dimension of a hypothesis class H okay is the size of the largest set that the hypothesis class shatters.

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Note that this is the size of the largest set high H shatters that does not means that H has to shatter all point of that size even in the three case co-linear points I cannot shatter if may three points like this and I label this + - + I know straight line that can separate this okay but there is some configuration of three points in fact lot of configuration of three points which I can shatter and therefore the VC dimension of straight lines is three .

What about VC dimension of pairs of straight lines parallel lines not arbitrary pairs. Let us say I will give you parallel lines what is VC dimensions of parallel lines, 2 parallel lines, they have to be parallel. However you want you can look at it, but it has to be 2 parallel lines, it can be 4 for sure. Can you do 4? You can always keep one of them as redundant straight line you can do as much as you can, and then you get you get straight lines. As everything else can do by 1 line, only the Xor case + + and - - you cannot scatter with the single line .

Remember perceptron, so it is exactly that, so for that you need parallel lines. Everything else you have 1 lines there and you keep next parallel line at infinity you are fine. What about 5 points? Now you can see why we can ask all kind of interesting questions . So I can give you this kind, next I can give say okay, instead of looking at parallel lines, just look at quadrilateral . So if it inside the quadrilateral it is class 1, class positive is outside the quadrilateral its - okay. Now what is the VC dimension, in fact there are uniform conversions VC dimension as well.

So I will have the TAs put up this write up online and there of course other material you can find out online. I am not going to do the proof because it is pretty complicated derivation , but then the nice thing about this is, at the end of the day just like we have staple complexity in the number of parameters . You can show even with the Vc dimension that it is polynomial in the Vc dimension . So if the data as the hypothesis finite Vc dimension the uniform conversion, require you to have order of the Vc dimension of samples.

And typically it turns out that for most kind of classifiers, most kind of reasonable classifiers that are out there, Vc dimension will be of the order of the number of parameters in the classifier. So if we think about it straight lines. How many parameters are there in this case? 2 or 3? 2 slope and intercept and slope that is all you have. 2 parameters and the order, in the Vc dimension is 3

close. This will be close to the parameters you have , so and that is why I said 10 time the f weights in the neural network rights all of these things and you can get that kind of rough intuition.

So I will stop here so if you have any questions feel free to fire away. So you could do pack line regression , but Dc dimension is different for classification. Yeah you can coarsen the regression problem little bit then try to do dimensional classification and PAC you can do for regression, essentially you are trying to look at. We defined a very specific variable random and did it, so you could define any random variable. Whatever the distribution is there, it is the amazing thing about the Chernoff Hoeffding bounds .

So the result holds on the expectation, the parameter μ whatever is the expectation of the distribution, so the empirically evaluated expectation will be the close to true expectation that is the result we have. With that you can change your random variable definition and you can get something appropriate for the regression as well. What does it mean? When you change the parameters it is the different classifiers that what I am concern, Do you mean same family of classifier? It depends on how you define hypothesis class .

So in at the end of the day what I am really interested in this is? What is the decision rule the hypothesis class is entailing for me. So I can say that I am going to define hypothesis class is the mix of decision tress and something it is up to you but typically you define hypothesis class as a single family which is differentiated with the parameterization. But we do not actually take a call on that, all I am telling you is that okay given k hypothesis , how are you are going to find out the.

So what I am finally, at the end of I am interested in the decision rule, okay given a data point what does it assign it to you? You derive the decision rule by the means of using a logistic regression or whether you derive the rule by means of using I don't know LDA it does not matter. For this kind of complexion, how you got to it that does not matter for me. Only in this context otherwise it matters, with the context of the complex analysis it does not matter.

Which is  is our finite procreation arithmetic is little disappointing? Because there I chose D parameters , if you have got it from some other I could re paradise thing and I can increase the number of parameters for representing the same decision surface. I want to define the straight line but I can actually increase the number of parameters that I am going to use to define the straight line, I will be saying that where ever I have ax square, I will have - ax square . so now I will have two more additional parameters for me to define one should a and other should be - a.

And it will take out the difference of $x^2$ so it will still be a straight line but I have increased the number of parameters and going back to our finite procedure, arithmetic argument I also have blown up my complexity, which is casually not correct, all I am interested is the decision rule, that is why the finite thing is little unsatisfied. But here we did not tell you how you could represent the line. So Vc dimension definition does not require you to know to represent the line. At the end of it you know that it is the hypothesis that I want to represent.

But the most compact way of representing the hypothesis parameters I would need. You do not have to worry about how we get to it. Anything else, I can chose any classifiers that I want from my set , foe shattering the data points, so any effect it will 1 corresponding to the, I mean what is the most powerful, so other one will shatter that would be Vc dimension. So Vc dimension if it can shatter of any arbitrary size VC dimension will be infinite. So if Vc dimension of the classifier is infinite none of the analysis will work, all the Vc dimension analysis works assuming that Vc dimension of the classifier is finite.

I should point out that most of the classifiers we looked at do empirical minimization except? Anyone know? SVM, is called structural risk minimization because they have an additional constrained that is there apart from the empirical they also try to minimize the solution size. They try to minimize the norm of the weight factor  so that actually gives rest to a different kind of minimization. So it does not do empirical, they called structural risk minimization. So you know who came up with SVM? So vapnik.

So he said came up with structural risk minimization this is the best you can do, so we need to have a different way of doing the minimization, then he motivated and then he said and came up

with empirical structural risk minimization you can do. We will go and do something else and he can do something else structural risk minimization and then he derived SVM with them, if you read original presentation of SVM  it will not look anything like how we presents SVM now days. If you remember we started off with perceptron and then we went from there.

He starts from the theoretical things, he said okay if you do empirical risk minimization it is the best you can do what else I can do and then I can improve on it and then he came up with structural risk minimization okay and then he derived all those systems okay.

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**Introduction to Machine Learning**

**Lecture 83**
**Introduction to Reinforcement Learning**

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**Indian Institute of Technology Madras**

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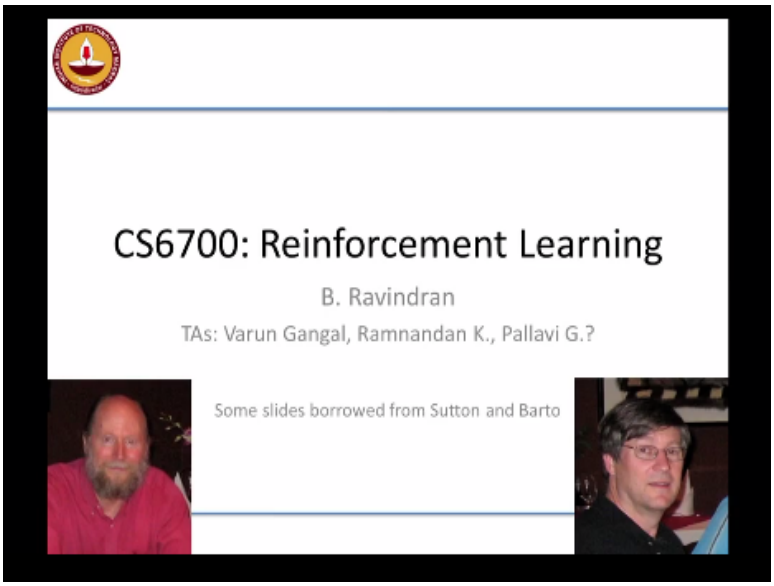


Reinforcement Learning is a very different kind of learning than what we looked at in ML. In Machine Learning, the idea was to learn from data, you know you are given lot of data as training instances for you, and then essentially you are trying to learn from those training instance as to what to do. And there were different kinds of problems that we are looking at, so one was supervised learning problem in which you were looking at classification, regression. So, in the machine learning class we looked at learning from data.

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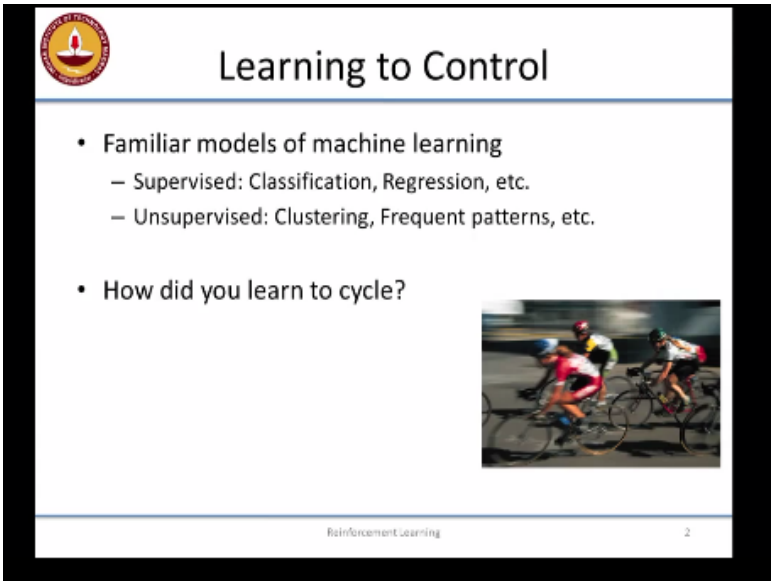


Primarily, so one of the models we looked at was supervised learning , where we learnt about classification and regression, and the goal there was to learn mapping from an input space to a output, which could be a categorical output in the case of classification, it could be a continuous output in this case is called regression . So if you have not been in the ML class do not worry about it.

Because this is just to tell you that RL is not whatever you learnt in the ML class. So if you have not learnt anything in the ML class then you do not have anything to unlearn, so do not worry. So the second kind of learning thing we looked at was unsupervised learning, when there was really no output that was expected of you. Since, therefore there was no supervision, the goal was to find patterns in the input data, I will give you lot of data points, you can find out if there are groupings of similar kinds of data points, can I divide them into segments.

So that kind of thing was called clustering or you are asked to figure out if there were frequently repeating patterns in the data. And so this is called frequent pattern mining or derived problems that was association rule mining and so on so forth.

How did you learn to cycle? Was that somebody tell you how to cycle and then you just followed their instruction. Fell down a couple of times and that automatically made you cycle; you have to actually figure out how to not fall down. So falling down alone is not enough, but you have to try different things. It is not supervised learning; it is really not supervised learning. How much time ever you think? Because now that I have given this talk multiple times, people are getting wise to it. Earlier when I used to ask these people, people say of course it is supervised learning, my uncle was there holding me, or my father was telling me what to do and so on so forth.

And best what did they tell you? Hey, look out, look out do not fall down. So that does not count as supervision . Or keep your body up and some kind of very vague instructions was what they were giving you. Supervised learning would mean that, so you will get on the cycle and somebody will tell you, now push down with your left foot with three pounds of pressure.

And move your center of gravity 3° to the right. So this is something, somebody has to give you exactly what is the control signals that we have to give to your body in order for you to cycle. Then that will be supervised learning , if somebody actually gives you supervision at that scale you probably have never learnt to cycle, if you think about it , because it is such a complex dynamical system and somebody gives you control at that level and gives you input at that level you will never learn to cycle.

And so immediately people flip and say that it was unsupervised learning because here of course nobody told me how to cycle therefore it is unsupervised learning, so if it is truly unsupervised learning what should have happened is you should have watched 100's of videos of people cycling figured out what is the pattern of cycling that they do and get on a cycle and reproduce it.

so that is essentially what unsupervised learning would be you just have lot of data and based on the data you figure out what the patterns are and then you try to execute those patterns, that does not work you can watch hours and hours of somebody playing flight simulator, and you cannot go fly a plane, so you have to get on the cycle yourself and you have try things yourself

so that is the crux here so what it is how do you learn to cycle is nether of the above it is neither supervised nor unsupervised it is a different paradigm.

So the reason I always start out my talks not just in the class but in general when I talk about reinforcement learning is because people always talk about reinforcement learning as unsupervised learning. It is just because you do not have a classification error or class label does not make it unsupervised learning. It is completely different form of learning and so reinforcement learning is essentially this mathematical formulation for this trial and error kind of learning.

So how do you learn from this case minimal feedback you know falling down hurts or somebody your mom or somebody stands there and claps when you finally manage to get on the cycle you know that is kind of positive reinforcement when you fall down you get hurt that is kind of a negative feedback how do you just use this kinds of minimal feedback and you learn to cycle so this is essentially the crux of what reinforcement learning is about trial and error. So the goal here is to learn about a system through interacting with the system it is not something that is done completely offline you have some notion of interaction with the system and you learn about the system through that interaction.

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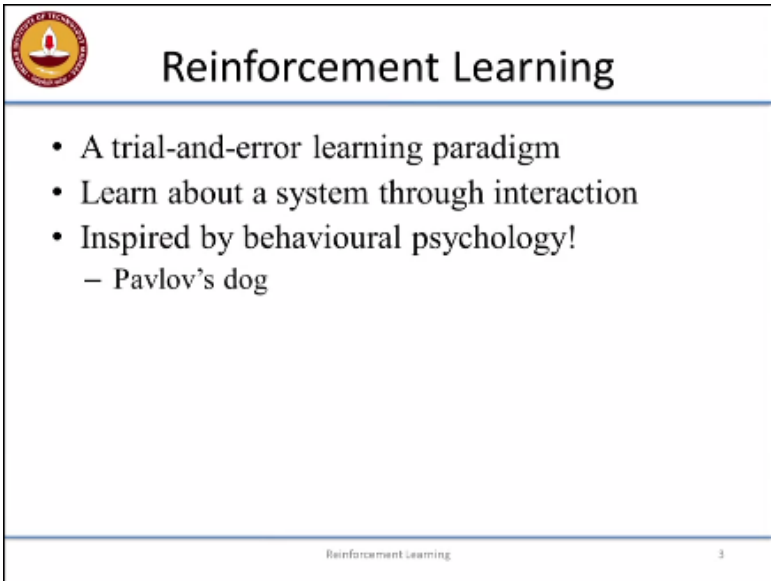

Reinforcement learning originally was inspired by behavioral psychology, so one of the earliest reinforcement systems that are studied with the Pavlov's dog how many of you know of the Pavlov's dog experiment what does the Pavlov's dog experiment. That is called a condition reflects so when the dog looks at the food and starts salivating it is a primary response because there is a reason for it to salivate on the site of food. Any idea why? Exactly. it is preparing to digest the food you know make sure the food is preparing to digest the food it starts salivating.

So, if then now if you think about it hearing the bell and it salivates what is it doing? preparing to digest the bell? No? so when you ring the bell and then serve the food the dog forms an association between the bell and food and later on when you just ring the bell without even serving the food the dog starts salivating in response to digesting the food that it expects to be delivered. So it essentially the food is the pay-off you know the food is like a reward for it and it is learned to form associations between signals in this case which was a bell like an input signal which was the bell and the reward that is going to get , so this is called the behavioral conditioning and so inspired by these kinds of experiments on then more complex behavioral experiments or animals now started to come up with the different theories to explain how learning proceeds .

In fact some of the earlier reinforcement learning papers appeared in the behavioral physiology journals. The earliest paper by Sutton and Barto appeared in brain and behavioral sciences journal. Just to go back I needed to say something about Sutton and Barto. This is a larger audience we can tell that about them. We are going to follow a text book written by Richard Sutton and Andy Barto but more importantly they are also kind of the cofounders of the modern field of the reinforcement learning so in 1983 they wrote a paper, “Adaptive neuron like element that learn the control behavior” or something to their effect . And that essentially kick started this whole modern field of reinforcement learning. So the concept of reinforcement learning like I said goes back to Pavlov and earlier, people have been talking about these kind of behavioral conditioning and learning and stuff but the whole modern computational techniques that people use in the reinforcement learning started by Sutton and Barto.

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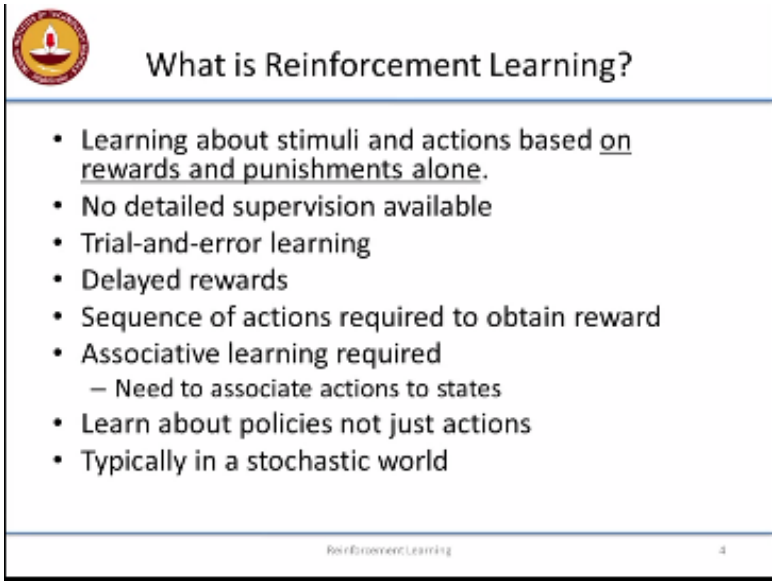


So what is reinforcement learning? It is learning about stimuli the inputs that are coming to you and the actions that you can take in response to it learning about the stimuli, only from rewards and punishments , so you are not going to get anything else food is a reward following down and scraping your hand is a punishment, so only from this kinds of rewards and punishments alone there is no detailed supervision available nobody tells you, what is the response that you should give to a specific input.

Suppose you are playing a game, there are multiple ways in which you can learn to play a game. So you can learn to play chess by looking at a board position and then looking at table that tells you for this board position this is the move you have to make, and then you go and make the move. so that is a kind of supervision that you could get you know that gives you a kind of a mapping from the input to the output and essentially you learn to generalize from that. So this is what we mean by detailed supervision. So another way of learning to play chess is just so you have an opponent and you sit in front of him and you just make sequence of moves at end of the move you win you get a reward somebody pays you ten rupees if you lose you have to play the opponent ten rupees, so that's all that happens so that's all the feedback you are going to get . Whether you are going to get the 10 rupees or going to lose the 10 rupees at the end of the game.

So nobody tells you given this position this is the move you should have made that is what we mean by saying learning from rewards and punishments in the absence of detailed supervision. is that clear. And a curial component to this, is trial and error learning because since I do not know what is the thing to do given an input, I need to try multiple things to see what the outcome will be, ?

I need to try different things to see if I am going to get the reward or not? If I do not try different things, I am not going to learn anything at all , so we will I can give you more formal mathematical reasons for why we need all of this as we go on but this is intuitively you can understand this as requiring exploration, so that you know what the outcome is? And there are bunch of things which are also characteristic of reinforcement learning problems one of those is that the outcomes ? the rewards and punishments based on which you are learning can be fairly delayed in time. They did not be temporally close to the thing that caused it. I mean while our playing a game let us say so you might you know now drop a batsman and then he goes on to score like 150 or something like that? So then you lose the match at the end of the day but the event that caused you to lose the match is the dropped catch that probably around the $12^{th}$ over? Or it could be much more convoluted causal effect? so and how many of you followed cricket, my god it is really losing its popularity yeah put your hands down. I am not going to give you a cricket example then.

So there a bunch of other things. So we talked about delayed rewards so rewards could come much later in time from the action that caused the reward to happen for example let us go back to our cycling case I might have done something stupid or I might have gone over a stone somewhere well I am cycling at a very high speed and might have been a small stone in the road and that will cause me to lose my balance . And though I will try my level best to get the balance back I might not and I finally fall down and get hurt. That does not mean that what cause the falling down is the last action I tried , I might have desperately tried to jump of the cycle or something like that but that is not what cause the punishment it what cause the punishment happened a few seconds ago when I ran over the stone, .

So there could be this kind of a temporal disconnect between what causes the reward or punishment from the actual reward and punishment so it becomes a little tricky how do you are going to learn those things learn the associations , so quite often you are going to need a sequence of actions to obtain a reward . it is not going to be like a one shot thing? It is going to be a sequence of action to get the reward.

So again going back to the chess example you are not going to get a reward every time you move a piece on the board ? you have to finish playing the game at the end of the game if you actually manage to win you get a reward. so it is a sequence of actions. And therefore you need to learn some kind of a association between the inputs that you are seeing in this case it will be board positions. Or how fast the cycle is moving and how unbalanced you feel and so on so forth. Two actions so inputs that you are getting sometimes which we will call state, and the actions that you take in response to this input that you are seeing , so this is essentially what you are going to be doing when you are solving a reinforcement learning problem, so this kind of associations are essentially known as policies, ?

So what you are essentially learning is a policy to behave in a world ? so learning a policy to play chess or you are learning a policy to cycle ? so this is essentially what you are learning you are not just learning about individual actions ? at all of this happens typically in a noisy stochastic world ? it does makes these things more challenging, so these are all the different characteristics of reinforcement learning problems. So reinforcement learning has been used fairly successfully in a wide variety of applications, ?

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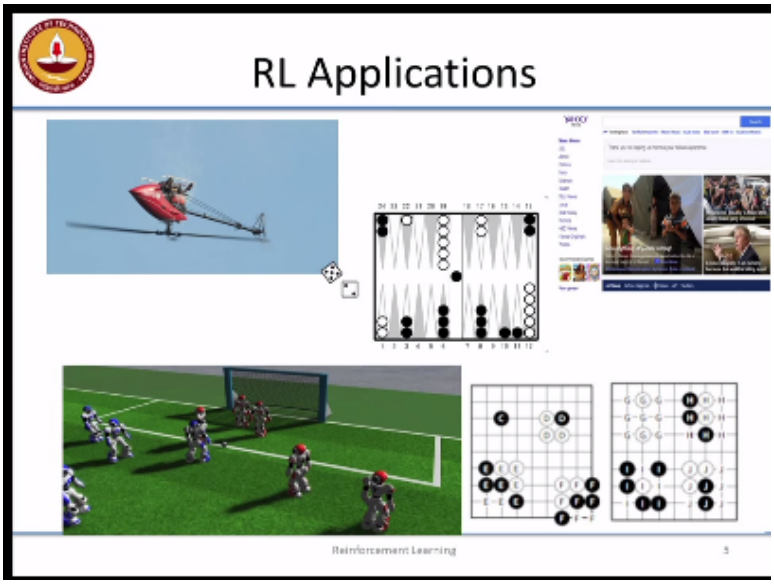


So you can see a helicopter there ? so that is not a cut and paste error here the helicopter is actually flying upside down ? so the group at Stanford and Berkley which have actually used reinforcement learning to train a helicopter to fly all kinds of things not just upside down and an RL agent can do all kinds of tricks on the helicopter so I will show you a video in a minute.

And it is amazing piece of work, I mean it was considered the show piece application for reinforcement learning. I mean getting such a complex control system to work and it actually could know things at a much finer levels of control than a human being could, well it is after all a machine so you would expect that. but the tricky part was how it learn to control this complex system from without any human intervention. And in the middle, so I have a couple of games there.

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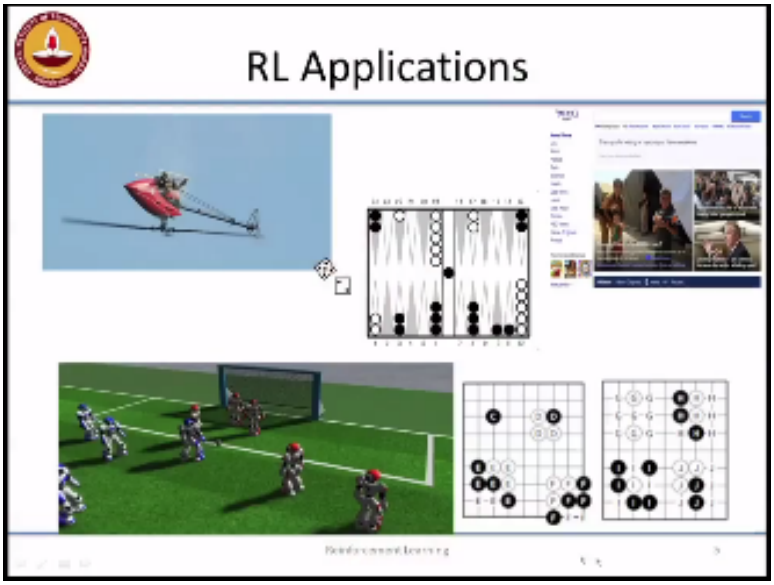


So that is can you see that? So backgammon is like a two player ludo , so you throw the dice you move piece around and you take them of the board , so it is a fairly easy game but then you have all kinds of strategies that you could do with it.

But it is also a hard game for computers to play because of the stochasticity and also because of a large branching factor that is there in the game so at each point there are many, many combinations in which you could move the board pieces around and then there is a die roll that adds in additional complexity, so people are not really getting great results and then there is a person Jerry Tesoro from IBM who came up with something called neurogammon think it was called neurogammon and that was trained using supervised learning under neural network.

And so if you have done it recently, it would have been called the deep learning version of neurogammon or something, because he did it back in the 90's early 90's so is just called neural network version of backgammon and it played really well for a computer program , so that is essentially the best computer program backgammon player at that point, and then Jerry heard about reinforcement learning he decided to train the reinforcement learning agent to play backgammon, .

So what he did was set up this reinforcement learning agent which played against another copy of itself, let them play 100s and 100s of games , rather 1000s and 1000s of games. So essentially what they did was so you trained one copy for like 100 move or 100 games something then you move it here , freeze it and then continue learning with this so essentially what is happening as you learn you are playing against better and better players gradually your opponent was also improving, .

And then this was called self play , so he trained backgammon using self play and it came to a point where the TD gammon as he is called it was even better than the human player of back campaign at that point in the world, so they actually had head to head challenge with the human champion there is a world championship of backgammon you know it is apparently very popular in the middle east and people actually have world championships as a world championship of backgammon and so he challenged the human champion which IBM seems to do a lot . I mean they challenge Kasparov two matches and things like this, so he also challenged yes, Tesaro worked for IBM he should realize , people who spend a lot of resources getting computers to play games well probably be working for IBM you know. So Jerry had this thing and it beat the world champion. So we have reinforcement learning agent that is the best backgammon player in the world, not no more best computer player or anything so we could actually make that claim and there is another game there which snap shot from the game of ‘Go’.

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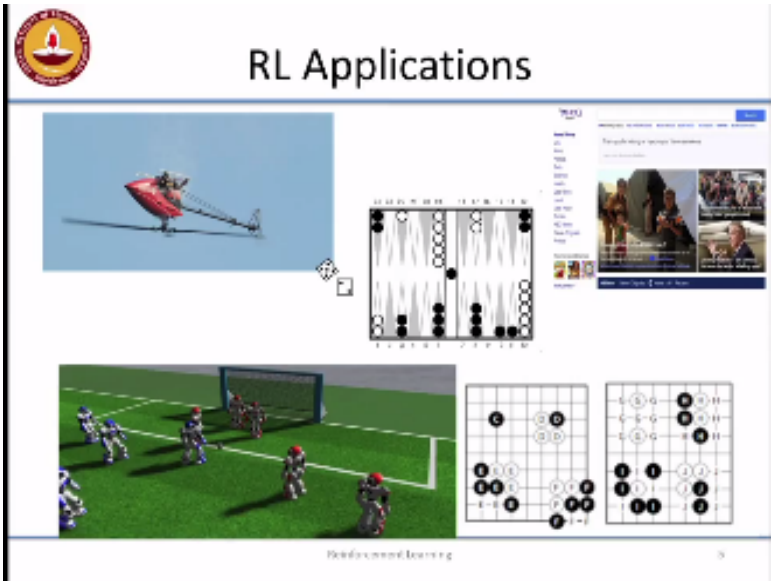


So people have played go? Oh come on, at least one or two people have played go? People have played Othello? . That's also a very few number. Isn't it one of those free games in Ubuntu? I thought everybody plays that in some point rather the well you rather play Othello than watch paint try you know?

But anyway so, Go is like more a complex version of Othello if you knew it. It is again a very hard game for computers to play because a branching factor is huge and it is actually a miracle that human even play this because the search trees and other things are really complex.

So this is one case which clearly illustrates that humans actually solve problems and fundamentally different way than we try to write down in our algorithm because they seem to be making all kinds of intuitively it is an order to able to be play go. So this person David Silver who currently works for Google Deepmind and but before that he spent some time with Jerry Tesaro at IBM and at some point along the way he came up with this reinforcement learning agent call the TD search that plays go at a decent level. It is still not like master human level performance but it performs plays at a very decent level.

So this is a, what I am pointing out here is things are typically hard for traditional computer algorithm or even traditional machine learning approaches to solve AI has a good success. And here is another example. There are some robots on the bottom left of the screen and so that is a snap shot from the UT Austin robots soccer team called Austin villa and they use reinforcement learning to get there robots to execute really complex strategies.

So this is really cool but the nice thing about the robots soccer application is that they do not use reinforcement learning alone but they actually use a mix of different learning strategies and also planning and so on and so forth which is going on the other studio . So they use a mix of different kinds of AI and machine learning technique in order to get a very, very completed agents it is very hard to beat and they are mean the champions ,I think two or three years running now in the humanoid league. And again hard control problems thinks like how do I take a spot kick you know those are the things for which they use reinforcement learning which it is really hard balancing problem so you have to basically balance the robot on one leg and then swing the other leg so then you take the kick. So, it going to be hard control problem, so they use RL to solve those.

And then up on the top is an application which will probably the one that actually makes money of all these three now all the others that is on essentially on using reinforcement learning to solve online learning. So online learning is a use-case where I do not have the feedback available to me apriori, so the feedbacks coming piece meal. So for example that is the case where we are having new stories that need to be shown to the people that come to my web page and when people come to the page I have some editors will pick like 20 stories for me and from those 20 stories I have to figure out which are the ones that I have put up prominently. And what is the feedback I am going to get?

Nobody tells me what stories that the user is going to like; I mean I cannot have a supervised learning algorithm here. So, from the feedback I am going to get is, if the user clicks on the story I am going to get a reward, if the user does not click on the story I am not going to get a reward, that is essentially the feedback that I am going to get. Nobody tells me anything before hand, so I have to try out things.

I have to show different stories to figure out which one he is going to click on and I have very few attempts to do this in, so how do I do this more effectively? People have done supervised approaches for solving this and it has worked fairly successfully, so but rei nforcement learning seems to be a much more natural way of modeling these problems. So not only in these kinds of news story collections, people use reinforcement idea in ads selection. How do I see some of those ads on the sides when you go to Google or some other web page ?

So how those are those ads selected, there might be some basic economic criterion for selecting for slate of ads. So here are the 10 ads which will probably give me the pay off and then you can figure out, which 3 of those ten am I going to put it out over here and things like that, you could use the reinforcement learning solution for selecting those. Ofcourse, this whole field called computation advertising, is a lot more complex than what I explained; But RL is a component in computational advertisement as well.

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**Introduction to Machine Learning**

**Lecture-84**
**RL Framework and TD Learning**

**Prof: Balaraman Ravindran**
**Computer Science and Engineering**
**Indian Institute of Technology Madras**

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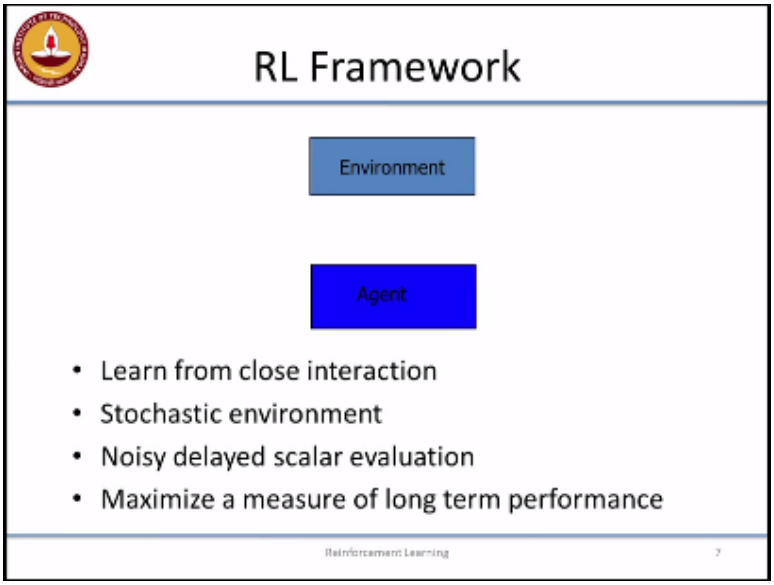


The crux in the reinforcement learning is that the agent is going to that is the learning agent it is going to learn in close interaction with an environment so the environment could be the helicopter it could be the cycle and or it could be your backgammon board and your opponent all of this could constitute environment a variety of different choices so you sense the state in which the environment is in you sense the state of the environment and you can figure out what is the action that you should take in response to the state .

So in apply the action back to the environment this causes a change in the state so now comes the tricky part so you should not just choose actions that are beneficial in the current state but it should choose actions in such a way that they will put you in a state which is beneficial for you in the future just capturing the queen of your opponent is not enough in the chess that might give you a higher reward but it might put you in a really bad position so you do not want that you really want to be looking at the entire sequence of decisions that you are going to have to make.

And then try to behave optimally with respect to that . So what we mean by behave optimally in this case we are going to assume that the environment is giving you some kind of an evaluation it is like falling down hurts when or capturing a piece maybe gives you a small plus point five or winning the game gives you like hundred so every time you win every time you make a move or every time you execute an action you did not get a reward or you did not get an evaluation from the environment .

So it could be just zero it could be it could be nothing so I should point out that this whole idea of having an evaluation come from the environment is just a mathematical convenience that we have here but in reality if you think about biological systems that are learning using reinforcement learning all they are getting is the usual sensory inputs so there is some fraction in the brain okay that sits there and interprets some of those sensory input as rewards or punishments .

So you fall down you get hurt I mean that is still a sensory input that is coming from your skin or somebody pats you on your back that is still a sensory input that comes from the skin and it is just another kind of an input so it could choose to interpret this as a reward or this as a collision with an obstacle something is brushing against my shoulder let me move it or you can just take it as somebody is patting my back so I did something good .

So it is a matter of interpretation so this is a this whole thing about having a state signal and having a separate evaluation coming from the environment is a friction there is created to have a clear cleaner mathematical model but in reality things are a lot messier you do not have such a clean separation and like I said so you have a stochastic environment.

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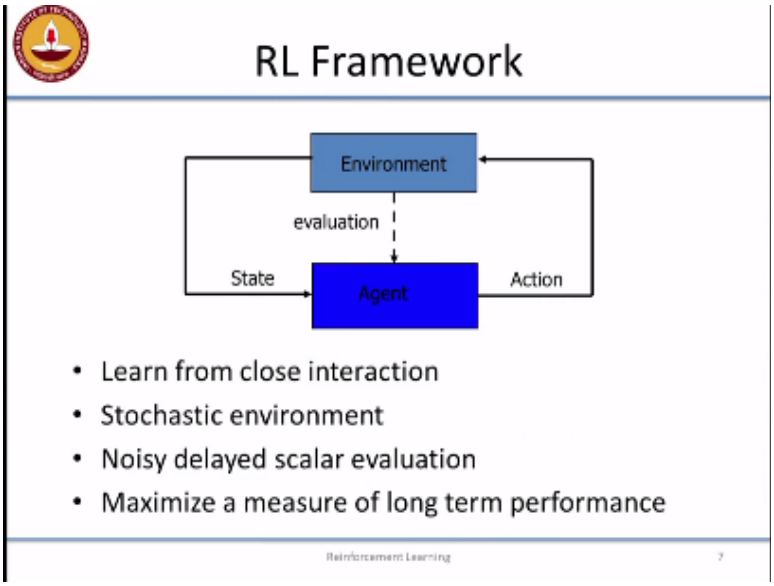


You have delayed evaluation noisy so the new term that we have added here is scalar the new term we have added is scalar so that is one of the things with the classical reinforcement learning approaches he said I am going to assume that my reward is a scalar signal so have we talked about getting hurt and having food and so on so forth what do all of this will happen mathematically is I will convert that into some kind of a number on a scale .

So getting hurt might be minus 100 getting food might be plus 5 winning the game might be plus 20 capturing a piece might be +0.5 or something like it so I am going to convert them to a scale and the goal is now know that I have a single number that represents the evaluation the goal is now to get as much as possible of that quantity over the long run okay, make sense . So if you have questions doubts stop me and ask.

So mathematically a scalar is easier to optimize not necessarily I am just talking about so it as like a cost function if you want to think about it in terms of in terms of control systems so this is like a cost and I am trying to optimize the cost all and so for the cost is going to be vector value and then I have to start trading off one direction of the vector against the other so which

component of the vector is more important so then it get into all kinds of super at optimality and of questions so it is not really clear what exactly is optimal in such cases so here again let me emphasize it is not supervised learning .

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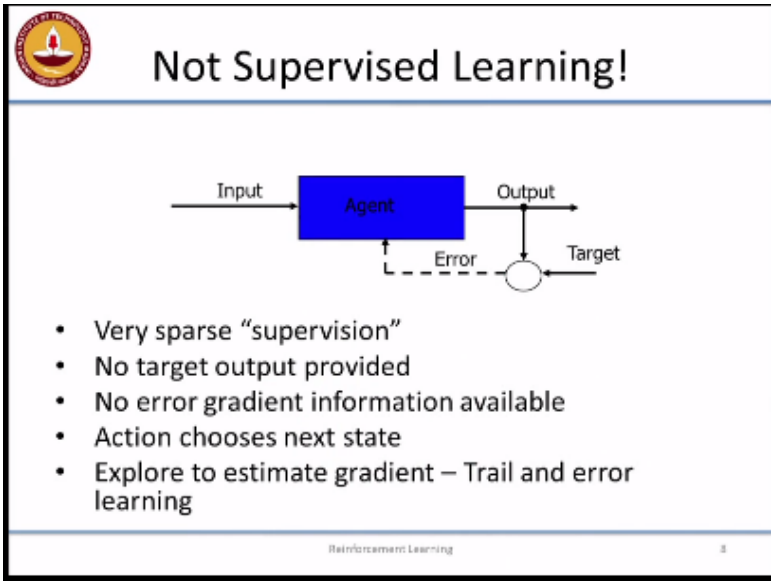


In supervised learning this is essentially what you are going to see there will be an input and there will be an output that you are producing and somebody will be giving you a target output okay so this is what you are supposed to produce and essentially compare the output you are producing to the target output and we can form some error signal and you can use that error in order to train your agent .

You can try to minimize the error you can do gradient descent on their work and do variety of things you can try to train the agent so here I do not have a target I do have to learn a mapping from the input to the output but I do not have a target and hence I cannot form an error and therefore my trial and error becomes very essential see if I have errors rate I can form gradients of the errors and I can go in the opposite direction of the gradient of the error and then that gives me some direction in which to change my parameters and that constitute the agent all major is going to be described in some way the error gives me a direction but now since I do

not know a direction so I just I do something I get one evaluations I do not know that the evaluation is good or bad so think of writing an exam I do not tell you the answer I just tell you three and so what do you do now do happy with answer should we change it should it change it in one direction or should he change it to the other direction.

See what makes it even more tricky is I do not you do not even know how much the exam is out of so when I say 3 it could be three out of three it could be three out of 100 , so it could be any of these things so you don't even know whether, three is a good number or a bad number so you have to explore to figure out A if they can get higher than three or three is the best the second thing is if I can get higher than three how should I change my parameters to get to become higher than three.

Let us I have to change my parameters a little bit that way okay I have to change the parameter rise a little bit this way so if I am cycling wait I have to push down a little harder on the pedal okay I will have to push down a little softer on the pedal to figure out whether I am staying balanced for a longer period of time or not I do not know that otherwise unless I try these things I would not know this is why the trial and error part.

So if I pushed on a little harder and I stay balanced maybe I should try pushing down even more harder next time so maybe that will make it better and then there might be some point where it too poor so I need to come back so this is how things which you have to try unless you try that you do not even know which direction you have to move in so this is much more than just the psychological aspects of trial and error there is also a mathematical reason if you want to adopt my parameters I need to know the gradient okay so that you need to yeah.

The reward is the one that you know that gives you the evaluation for the output so herein the supervised case the error is the evaluation for the output of the error is 0 then your output is perfect but then the way of gauging what the error is because you have a target which you can compare and from there you get the error so in the reinforcement learning case the evaluation is directly given to you as the evaluation of the output it is not necessarily comparing against a target value or anything you do not know how the evaluation was generated that you just get an

evaluation directly so you just get some number corresponding to the output and so maybe I should have done put an arrow from the top saying evaluation comes in from there.

But that is exactly where it is coming let us substitute for the error signal but it is just that you do not know what the evaluation is of course the way differs from the error is minor differences you typically tend to minimize error but you tend to maximize evaluation it is also not unsupervised learning so unsupervised learning has some kind of an input .

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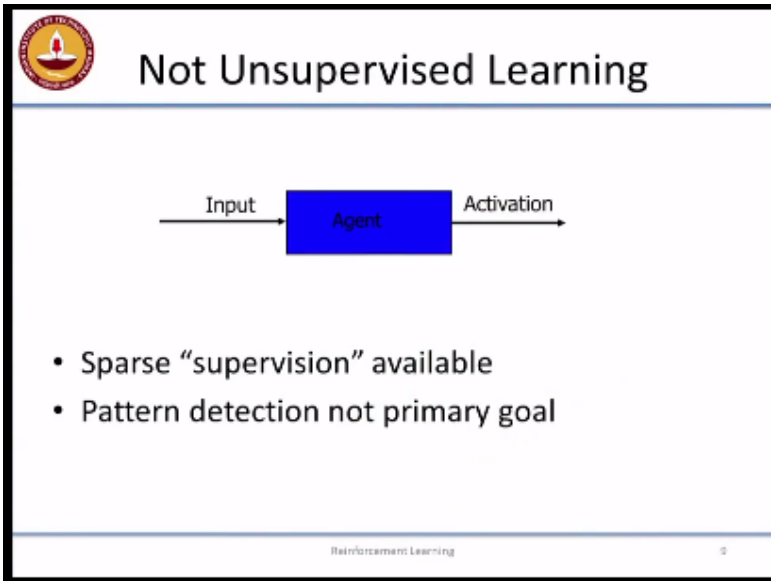


That goes to the agent and then it figures out what are the patterns for thee in the input here you have some kind of an evaluation and you are expected to produce an action in response to the input it is not simply pattern detection so you might want to detect patterns in the input so that you know what is the response to give but that is not the primary goal but in unsupervised learning the pattern deduction itself is the primary co so that is the difference .

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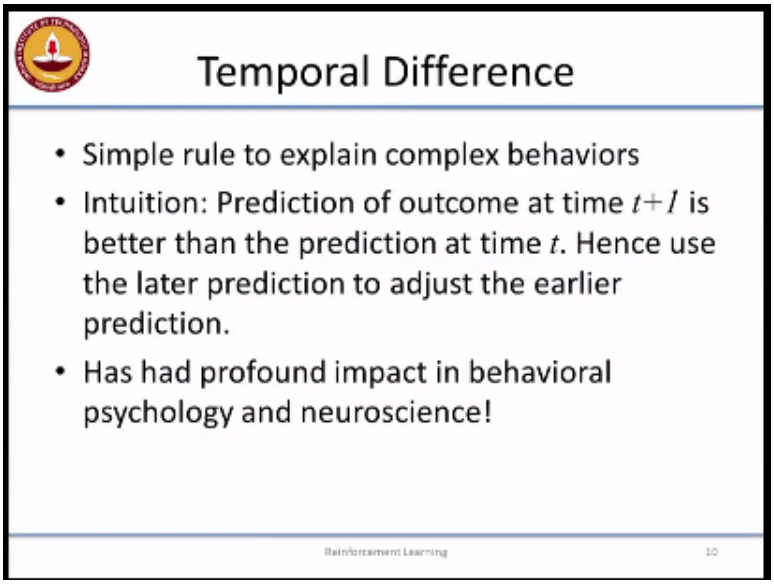


So here is one slide which I think is kind of the soul of reinforcement learning it is called temporal difference so I will explain a little more detail and in a couple of slides but the intuition here so if you remember the Pavlov's dog experiment what was the dog doing it was predicting the outcome of the bell you know if the bell rings there is an outcome that is going to happen it is predicting the outcome which is food is going to happen and then it was reacting appropriate to the outcome .

So most of reinforcement learning you are going to be predicting some kind of outcome that is going to happen since I am I going to get a reward if I do this or if I am I going to not get a reward I am I going to win this game if I make this move what am I not going to win this game all so I am always trying to predict the outcome. The outcome here is the amount of reward or punishment I am going to get this is essentially what I am trying to predict at every point .

So the intuition behind the what is called temporal difference learning is the following so the prediction that I make at time t+1 okay of what will be the eventual outcome let us say I am playing a game I am going to say I am going to win now I am very sure I am going to win down, . So I can say that with a greater confidence closer to the end of the game then I can at the beginning of the game so I have all the pieces set up and if I am going to sit there then and say I

am going to win the game it is most probably visual thinking but then you have played the game for like 30 minutes or something and there are like five pieces left on the board.

Now I am going to say I am going to win the game now I say I am going to win the game that is a much more confident prediction then what I did at the beginning so taking this to the extreme so the prediction I make at t + 1 is probably more accurate than the prediction I make at t , the prediction I make it t + 1 is more accurate than the prediction I make it t, so if I want to improve the prediction I make at t, what can I do?

I can look go forward in time then basically go to the next day let the clock tick over and see what is the prediction I will make a time t+1 with additional knowledge I am getting , I would have moved one step closer to the end of the game so I know I know its little bit better about the game I do not know how the game is proceeding I know I can may now make a prediction about whether i will not lose .

And use this go back and modify the prediction I make it time at time t, and t I think there is a possibility of say probability of 0.6 of me winning the game okay and then we make a move then I find out that I am going to lose the game with a very high probability then what will I do is I will go back and reduce the probability of winning that I made at time t so instead of 0.6 I will say okay maybe 0.55 or something .

So next time I come to the same state as I was at time t, I would not make the prediction of 0.6 I will say 0.55 that is essentially the idea behind component difference learning so it has a whole lot of advantages we will talk about it a couple of slides down but one thing is that the significant impact in behavioral psychology and in neuroscience so it is widely accepted that animals actually use some form of temporal difference learning.

And in fact there are specific models that that have been proposed for temporal difference learning which seemed to explain some of the neuron transmitter behaviors in the brain yeah, no see at this point I will be making a prediction about what is the probability of winning and it

could be for each of the moves if I make this move what is the probability opening if I make this move what is the probability of winning.

Let us say I make move two okay and then I go I see a new position the opponent response to it and then I decide oh my god this is a much worse move than I thought earlier, so what I do is I will change the prediction I make for move two in the previous state you see that the other moves will not be affected because the only move I took was 2 only about that move I have additional information therefore I can go back and change the prediction I make for move 2 alone.

So you can still have the 10 moves so they are not changing any of that yeah, seeing the prediction is not like I mean if an ideal world you should be able to take back a bad move , except if it is a parent playing with the kid I do not think those things are allowed , in fact I when I play with my son we have sometimes had to rewind back all the way to the beginning it will probably be me asking to do the rewinding not him, because he will be dropping me in someone of those games but yeah, otherwise you cannot you just make the change the prediction.

So next time you play the game it will be better at it, it not for that game well basically you have messed up or you did well I mean so whatever it is yeah, okay I hope I was not too boring in somebody fell over , that is known to happen yeah, so people sleep and I actually had a person sleep and they fall off the chair once.

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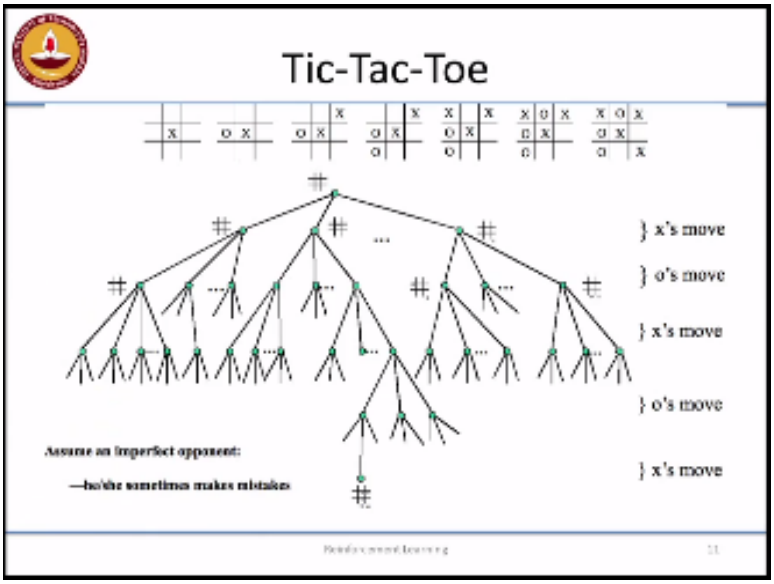


Yeah, I still cannot get over this okay, there is one time was going to teach a class , and I said was entering the class one person was leaving the class I said hey, what are you doing I suppose me in my class he said no, no, I feel very sleepy I cannot and I do not care if you are going to sleep this get back to the class , and he looked at me for a minute shortly I said okay, and they walked into the class went to the last place actually lie down on the bench on and I going to sleep okay, and he recently sent me a friend request okay, coming back to looking at the RL .

So listen looked at tic-tac-toe , how many of you have played tic-tac-toe good, even you put your hand up okay, good so in tic-tac-toe so you have these board positions , and so you make different moves, so in the first this is what I have drawn here is called a game tree , so I start off with the initial board which is empty , and there are how many possible branches there for people making moves nine possible branches , for excess move there are nine possible moves I have nine possible branches and then for each of these I will have like eight possible I am not sure this is the  TV and for each of this i have eight possible branches and they keep going, .

So what we are going to be doing is essentially trying to this formulate this as a reinforcement learning problem so how will you do this as a reinforcement problem , so I have all these board positions , let us say x is the reinforcement learning agent and o is the opponent , so initially given a blank board I will have to choose one among nine actions , so there state that I am going

to see is this the X's and O's on the board , and the moves I will be making are the actions . So in the initial position I have nine actions I make that do I get any reward not really there is no natural reward signal that you can give.

Essentially the reward that I am going to get in this case is if at the end of the moves if I win I will get a 1 if I do not win I get a 0, and if I win I get a 1 if I do not win I get a 0 , so what is going to happen is I am going to keep playing these games multiple times , and at each point yeah, okay, so there is a note here so what is it note say. You have to assume it is an imperfect opponent , otherwise there is no point in trying to learn tic-tac-toe why.

We will always draw and the way we have set up the game you are indifferent between drawing and losing so you learn nothing I mean basically, so you will not know even learn to draw okay, you will just learn to nothing, basically we learn nothing because you can never win , so you are never going to get a reward of 1 so you will just be playing randomly, so it is this is a bad, bad idea so let us assume that you have an agent that is imperfect , that makes mistakes so that you can actually learn to figure out where the agent makes mistakes, where the opponent makes mistakes and learn to exploit those things, okay, .

So your states are going to be this board positions as you can see we give you see a game that has been played out on the top of the slide , and the actions you take or in response to those board positions and finally at the end of the game and if you win you get a 1 if you do not win you get a 0 , Sir, in case like I mean does it have to be a binary sort of a reward system I mean could you have a scale whether there are three parameters you lose a 0 if you draw 1 by privilege 1.

Sure, you could even know other things like if you win it is 1 if you lose its -1. Yeah, you possibly could but you probably have to pay a lot of games because the perfect opponent it is almost impossible for you to start getting any feedback in the beginning , you will always be losing so it is going to be hard for it learn but you will eventually learn something yeah, it will take a lot of mozes level to learned something go back, so if I say that at every point, so we are

learning like at a particular stage the probability of winning and like it is what I am going to US state so you are storing information for each and every state that you have entered .

So how will it be different from exploring the proper next state space every time because after you have done let us say a thousand games or million games you would rather explored a lot of states I will have to store for each state the probability of you winning at that point. Yeah. And all that so I will that be different from exploring it again.

I know the probability of winning program why would I have to close, this still I am not even totally how you are going to solve it okay, let me explain that and then you can come back and ask me these questions, okay if you still have that okay, quit. So what the way we are going to try and solve this game is as follows , for every board position I am going to try and estimate the reward I will get if I start from there and play the game to the end , every board position I am going to look at the word I will get if I start from there and play till the end.

Now if you think about it what will this reward connected , so if I win from there I will get a 1 if I lose from that or if I do not win from that I will get a 0 , when I say what is the reward I expect to get starting from this board position , it is essentially this average over multiple games, it some games I will win some games I will lose, or I will not win like some games I win some games I will not win so what will this expected reward represent after having played many, many, many games.

The probability of winning , the reward is going to represent the probability of winning in this particular case , if the reward had not been 1 , if it had been something else if it had been +5 that you would have been some function of the probability of winning , half it has been +1 for winning -1 for loosing and 0 for draw well it is something more complex is no longer the probability of winning , it is the gain I expect to get , how what fraction of games I expect to win over the fraction of games I expect to lose or something like that, so it becomes a little bit more complex.

So there could be some interpretation for the value function but in general it is just the expected reward that I am going to get starting from a particular board position okay, so that is what I am trying to estimate , that is assume that I have such a expectation well defined for me , as you say I have such an expectation well defined, . Now I come to a specific position let us say I come to this position here , let us say I come to this position.

How will I decide what is the next move I have to make sorry, whichever next state has the highest probability of winning so I just look ahead to see okay where if I put if the x here , if I put the X here what is the probability of winning, if I put the X here what is the probability of winning, if you put takes here what is the probability of winning , I do this for each one of these , and then I figure out whichever has the highest probability of ending and I will put the x there, .

So that is how I am going to use this function does it make sense, yes, it is very important so this is this is something which issued understand this is the crux of all reinforcement learning algorithms , I am going to learn this function that tells me if you are in this state , if we play things out to the end what will be the expected payoff that you will get , whether the rewards or punishment or cost whatever you want to call it what is the expected value you are going to get and I want to behave according to a this learnt function.

So when I come to a state I look ahead figure out which of the next states has the highest expectation and then go to the state okay, great how do I learn this expectation. What is the simplest way to learn the expectations, this is especially keep track of what happens, essentially keep track of the trajectory through the game tree , you play a game you go all the way to the end .

So you keep track of the trajectory and if you win , you go back along the trajectory and update every state that you saw on the trajectory you update the probability of winning , it just increase it a little bit or you come to the end of the game and you found that you have not 1 , you go back along the trajectory decrease the probability of winning a little bit, .

Alternatively you can keep the history of all the games you are played so far , after every game has been completed you can go back and compute the average probability of winning across the entire history of all the games in which you saw that particular position , make sense thus easiest way of estimating this probability, .

But the problem with this is a you have to wait till the game ends , or you have to store the history of all the games you have played try to means all of these could be potential drawbacks okay, you can get around the history part by coming up with an incremental rule but the main difficulty here is you will have to wait all the way to the end of the game , before you can change anything along the way so tic-tac-toe was easy is like how many moves can you make in tic-tac-toe at best 4 , the fifth one is determined for you, .

So it is basically four choices that you can make , so and that is easy enough to remember , you can always wait till the end of the game and then you can always make the updates , what if it is a much more complex situation , what if you are playing chess, maybe you can wait till the end, so what if you are cycling maybe you can wait till the end exactly we do not know , this it depends on where you are cycling if you are cycling learning to cycling 90 meters it is fine, when you are learning to cycle somewhere on the Sardar Patel road you do not want even think about what end is there , so this is there are some tasks for which you really like to learn along the way, .

So this is where TD learning comes into comes to help , I do not think I have it slide anyway and I am not using the fancy thing where I can draw on the projection, so let us see if I can do it here . Suppose I have come here , and from here I have played at this point I know the probability of winning is say point 4 , so I came here by making a move from this position, so I said here late and we made a we know that the probability of winning from here is a point 3 .

But I made the move from here to come here , but here I had thought my probability of winning was let us say point 6 then I thought my probability of winning was of point 6 , but then I looked at my next states and I found that the best one was point 3 somehow , so I went there . But now since the best I can do from here this point 3 me saying point 6 here there is something wrong ,

so I should probably decrease the probability of winning from here . So why could it be, why could it have happened that I thought that was point 6, but the best among the next was point 3, the thing is.

So that whenever I came to came through this part maybe I won before it so happened that when I went through like this initially I would have gone through like this and played the game and i am the examples I drew I might have actually won some of those games . So I would have change this 26 but it is possible for me to get here by playing a different sequence of moves also .

So for example to come here I could have put the X first here and then here or I could have put the X like I did here I put the X first here and then here, that either way I would have reached this position , so there are many combinations in which I could have reached the same positions it just to be nice to these guys . To reach here there are different orders in which I could have put the O and the X here we have showing a cell specific order the O first 1st put here then put here that the x was first put here. Then put in it could very well be I put the X first here in the war first here and then I put the X here in the O here .

The multiple ways in this thing goes reach so sometimes when they play those games I lost, all sometimes and it played these games I won therefore it turns out that for due to some random fluctuations , so sometimes i win when I go through this place specific point and that is what I have a higher evaluation of winning but when I went through the other paths I had a lower evaluation of winning . But we know that really does not matter what path you went through in tic-tac-toe . Once you reach that point what is going to happen further is determined only by that point.

So what i can do now is take this 0.3 you should update that 0.6 down, so I am very confident here I think I will win with the probability of 0.6 but the best probability I have from the next stage is 0.3, therefore here I should not be so confident good point, that depends on the how stochastic your game is . So if you are game has a lot of variability then you do not want to make a complete you know commitment to a 0.3 so you might want to say ok now let me move it little

bit towards 0.3 . But if it is a more or less a deterministic game then you can say okay 0.3 yes sure let me go to all the way to, in the difference on the yeah it is misleading it is called game tree actually.

But it is a game graph in this case yeah, so as I said kid when this is this is an instance of temporal difference learning, so how while I use the thing to update this is called temporal differential learning okay. So there is one other thing which I should mention here if I always take the move that which I think is the best move now, let us talk about it I start tab I have never played tic-tac-toe before , so I play the game I play it once I get I get to the end I win. So now what I do I go back whether I am using temporal difference learning or waiting till the end up dating whatever it is I change the value of all the moves I have made in this particular game .

So the next time I come to a board position what am I going to do? I look at all possible outcomes everything except the one that I have played will have a 0 and the one that I have played will have something slightly higher than 0, I am going to take that, then in fact it will be like how many of you watch the movie Groundhog Day? It will be like Groundhog Day I will be playing I will be playing the same game again and again because that is what happened to give me a win in the first time around . That but that might not be the best way to play this .

So I need to explore , so I need to explore multiple options so I should not be always playing the best move I should always be paying the best move I need to do some amount of exploration, so that I can figure out if there are better moves than what I think is currently the best move . So tic-tac-toe there is inherently some kind of noise if your opponent is random but if an opponent is not random and if operand is also playing a fixed rule and if you are playing greedy, then you will be just plain a very small fraction of the game tree and you would not have explored the rest of the outcomes .

So you have to do some amount of things at random so that you learn more about the game . So here is a question for you, when I am estimating this probabilities of winning , let us say I have reached here I look down and the action that gives me the highest probability of winning say gives me a probability of say 0.8 what I want to explore so I take an action that gives me a

probability say 0.4 okay. So I will go from here to another action that has a probability 0.4 that another board positions that has a probability of 0.4 of winning. So should I use this 0.4 to update this probability or not.

No why? that you are questioning the whole TD idea and you are exploring you should probably wait for the or not just ignore it okay, any of any other answer because you are good or a bad move will be found out I have to update the value of that move I agree. Do I update the value of winning from the previous board position was the question so that 0.4 I will have to change but do i change the 0.8, that was the question the 0.8 was a probability of winning from here I look or whatever. So probably say I had a probability of winning of 0.6 from here I look at the bottom and the best probability of winning says 0.8.

But then I take because I am exploring I take an action that has a probability of winning of 0.4 all the question is do I go back and change the 0.6 towards 0.4 or do I leave the 0.6 as it is? Sorry that one where I am exploring , I mean this is we will be necessarily be less than 0. 8 this will be 0. 4 will be 0.6. So the question here is 1 way of arguing about this is to say that, a if I am playing to win I will play the best action from here and then the best action says 0. 8 therefore I should not penalize it for the bad action which is 0. 4.

Which I did to learn more about the system and that is one way of thinking about it another way of arguing is to say that hey, no this is how I am actually behaving now . So I should give you the probability of winning about I about the current behavior policy , this should not be some other ideal policy should be about to what I am behaving currently and therefore I should update it . So which one is correct first or the second questions? But this is something these are this is like I said ask you to think about the whole tic-tac-toe thing and many of these answers have relevance later on.

In fact there are 2 different algorithms one does option one does option two , so there is no answer or wrong answer answer is depends yeah, so yeah so this is a different things that you can think about in this but I told you about 2 different ways of learning with tic-tac-toe one wait till the end and figure out what the probabilities will be, the other one is keep adapting this as

you go along and both cases you not explore that is it to keep out here in both cases you have to explore otherwise will not learn about the entire game.

So this is where the Explore exploit thingy comes in okay yeah. Great question different algorithms deal with indifferent way that is one of the crucial questions that you have to answer in RL. So it is called the explore exploit dilemma , so you have to explore to find out which is the best action and you have to exploit.

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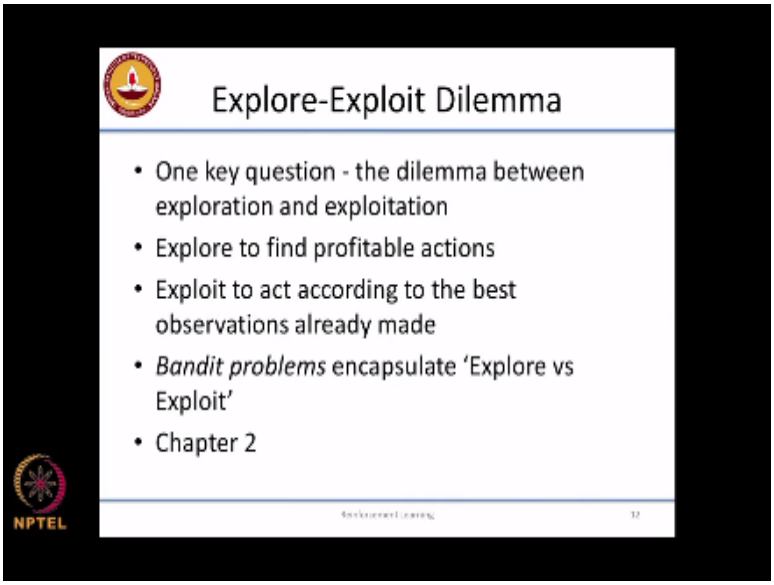


Whatever knowledge you have gathered and you have to act according to the best observations already made , so this is called exploitation . So the key one of the key questions is when do you know you have explored enough should I explore now or should I exploit now, this is called the explore exploit dilemma and a slightly simpler version of reinforcement learning called the Bandit problems okay. Some carefully called bandit problems they of course he is an expert on bandit problems here you can the Bandit problems encapsulate.

This explore exploit dilemma lot of people are turning and looking at a noticeable but, so this will ignore a whole bunch of other things like the delayed rewards you know the sequential

decisions and other things. Even in the absence of all of these other complications that even if I say that you are all your problem is you have to take an action and you will get a reward okay your goal is to pick the action that gives the highest reward. I will give you 10 actions you have to pick the action that gives you the highest reward , but the problem is you do not know what is the probability distribution from which these rewards are coming .

So you will have to do some exploration I have to actually do every action, at least once okay to know what will be the reward even if they are deterministic . So I cannot say which is the best action before I try every action at least once, if it is deterministic it is fine I can just try every action once and I know what is the payoff . But if it is too stochastic I have to try every action multiple times how many times you have to try it depends on the variability of the distribution.

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**Introduction to Machine Learning**

**Lecture-85**
**Solution Methods & Applications**

**Prof. Balaraman Ravindran**
**Computer Science and Engineering**
**Indian Institute of Technology Madras**

There are two classes of algorithms that we will be talking about. So one of them is based on what is called dynamic programming.

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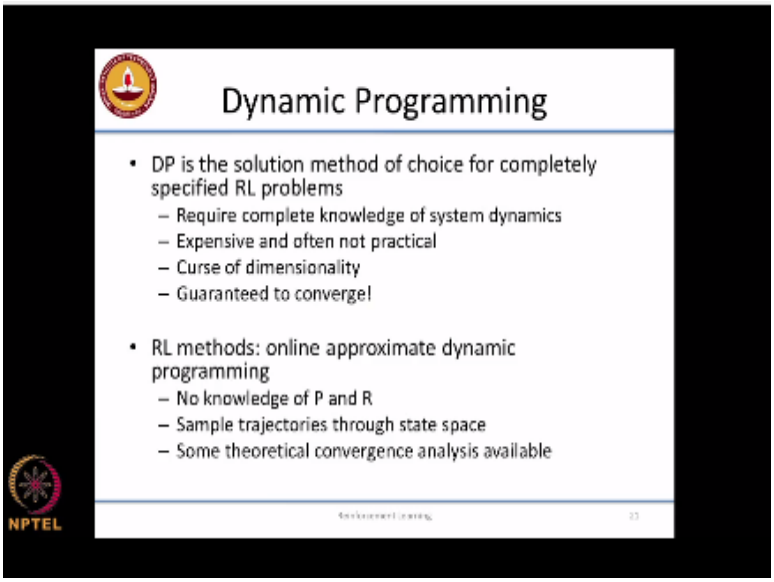


Now the essential idea behind dynamic programming is you will be using some kind of repeated structure in the problem . So I have to solve the problem more efficiently , suppose I have a solution that I can give for a problem of say size n . Then I will try to see if I can use that for defining the solution for the problem of size n+ 1, so some kind of a repeated sub structure in the problems . The very rough way of describing what dynamic programming is . So for example

one way of thinking about dynamic programming is I have this game tree , so I look at the values of winning or the expectations of winning from all of these steps . I will use these in order to compute the value of winning or the probability of winning from the state .

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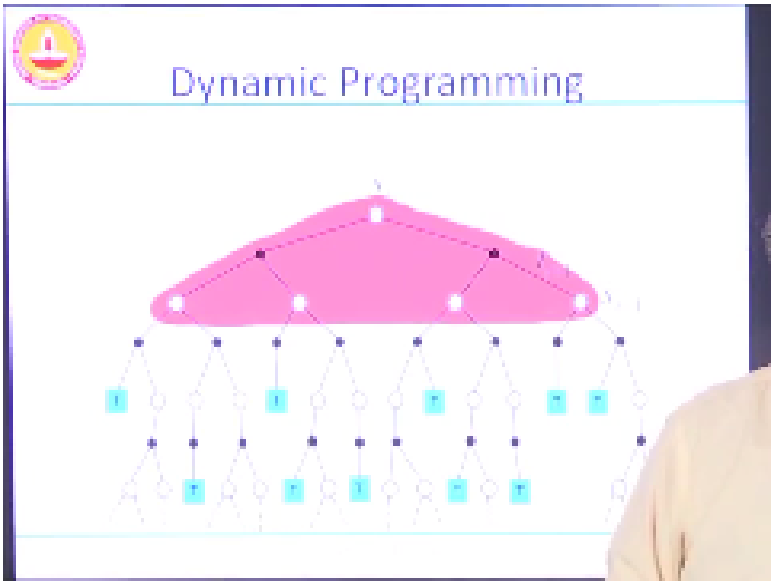


So if you think about it if from here if I am going to take say n steps, from here how many steps will be expect me to take? N minus 1 steps , so I look at the probability of winning when I when I have only n - 1 steps left , I will estimate that first I will use that solution for estimating the probability of winning when I have n steps only n -1 steps left I will estimate that first, I will use that solution for estimating the probability of winning when I have n steps left . So that is essentially the idea behind dynamic programming and so all you do now is instead of having the entire outcome.

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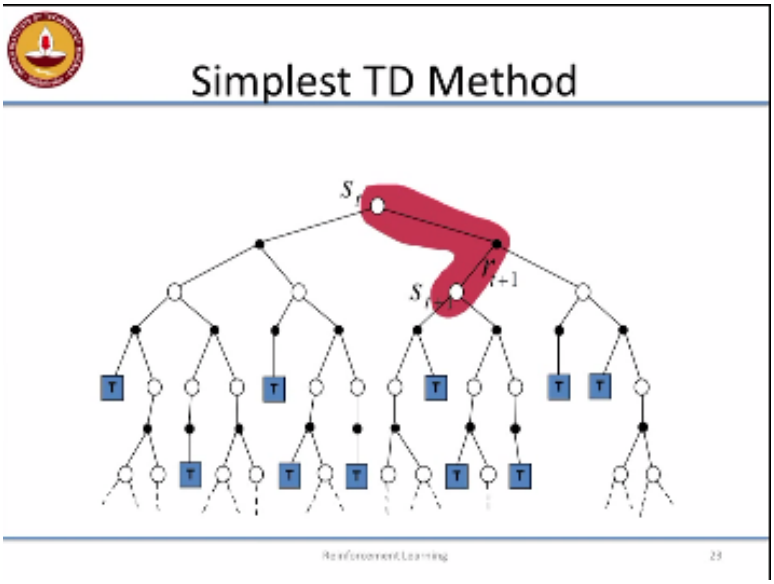


And using that for estimating the probability of winning here I am going to just use one step that I take through the tree, I use this what happens in this one step I will use that in order to update the probability of winning here. all s instead of using the entire outcome as a dynamic programming in reinforcement learning methods we will be using samples that you are getting through the state space okay this is the TD method.

In the other method I explained to in for tic-tac-toe what would you do your sample will run all the way to the end and you use that to update. So this is the two different ways of using samples here, so f T will be determined by the value of as p +1 so the value of s t + 1 should be computed for all and so depends on the steepest to again that should be computed first so would not this be the same thing as exploring the whole all the way down and then computing this if you are doing it for the first time the first time down the tree there will be no updates actually because st +1 is also be 0 st will also be 0 the first time we go down.

There will be no updates but once you reach an end then from there you start updating the previous day, so the next time you go down the tree then it will keep going further up whether in what the game have lost the game I actually I am taking the exact outcome that happened in that particular trial , at that particular game and I used that to change my probabilities but here I am not just taking the outcome of that particular game I am looking at the expected value of

winning from the next board position . So if I wait there all the way here and I say I won and it take this in update st then I will be only updating it with the fact whether I won or not okay but if I am updating it from st + 1 see I could have reached st multiple ways before .

When I am doing the updating from my st + 1 is essentially the average accumulated over all the previous trends that I will be using , so if I play all the way to the end and update it will be with a 1 or a 0 but st + 1 could be anywhere between 1 and 0 depending on what is the probability of finding so I will be using that value for updation that is a crucial difference I so there are many different solution methods so there are all this which are called temporal difference methods so these are all different algorithms.

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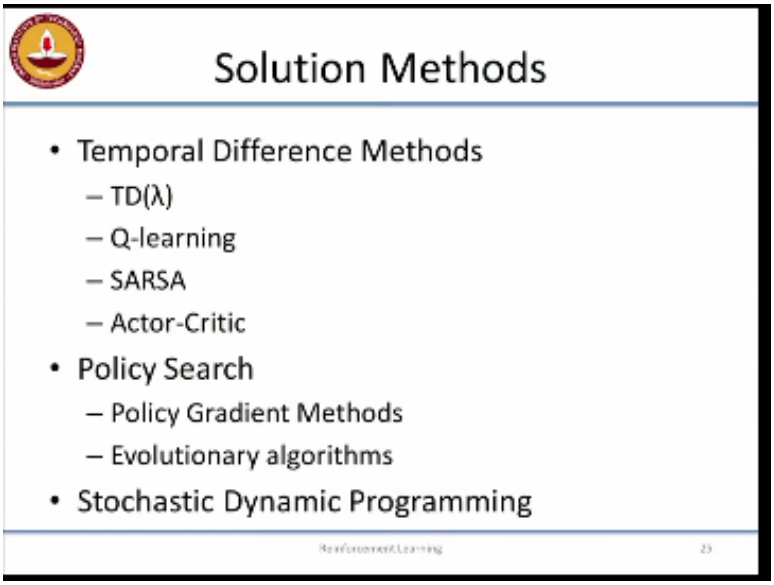


TD-λ, Q-learning, SARSA, actor critic so on so forth and then there is a soul search of algorithms which called policy search algorithms and then there is dynamic programming and their whole bunch of other applications for RL it so we could they are all over the place as you could see.

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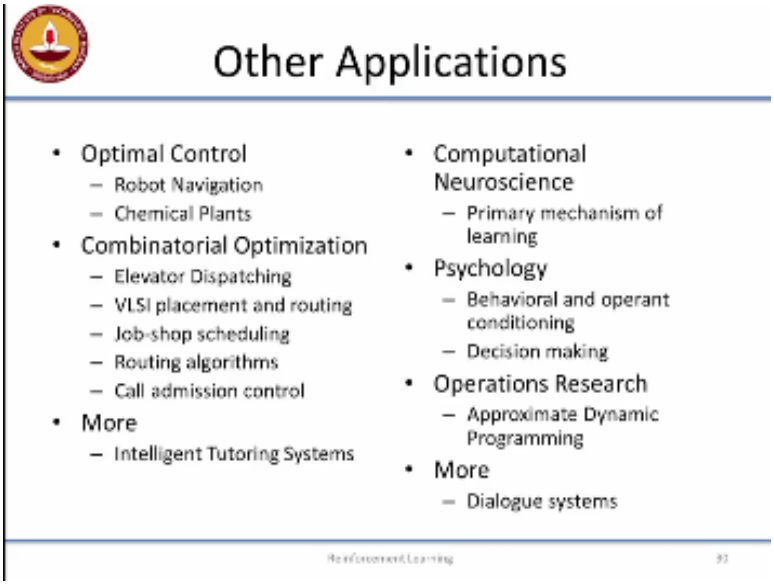


Optimal control optimization company too common a total optimization for psychology neuroscience so that is not a theory since I was asking is there anyone from biotech because biotech people do use reinforcement learning a lot and usually there are one or two people in the RL class so this is a surprise or maybe that is there was a bear trick with this according to the economic website maybe this is gave up in CS 36 and went back I do not know so here is the most recent.

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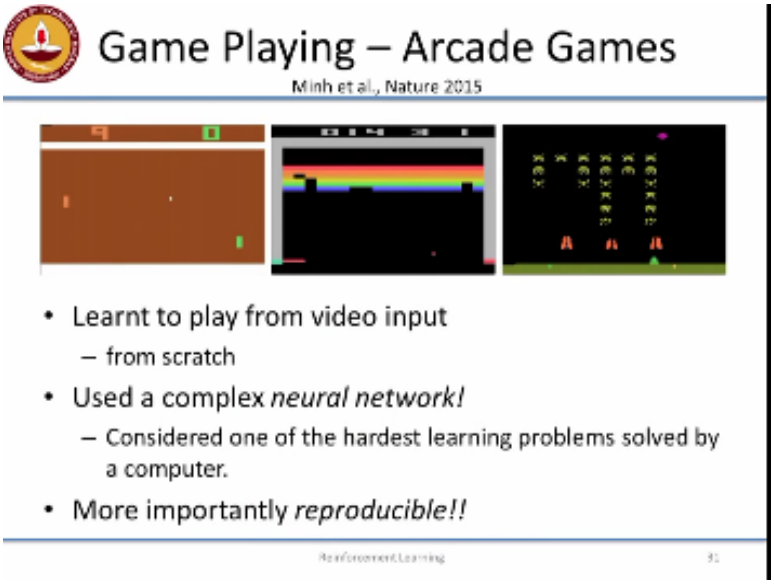


The hot thing that comes from came from RL more game playing and for a change is not from IBM and it is from Google but the company that actually built the first this arcade game playing engine was called deep mind and as soon as deep mind built a successful engine Google bought them and so now it is Google deep mind but it is a separate entity it is not part of Google deep mind operate out of London and they bring all kinds of interesting stuff many of the hot advances very recent advances in the last year or so. In reinforcement learning seem to be coming out of deep mind so what they did was how many if you know about this Atari games , everyone knows about Atari games people are getting tired really no one has played pac-man yeah, how about how about the pong, break outs, space invaders come on yeah anyway so what happened was this is team in University of Alberta okay, which put out this their what they call the Atari, the arcade learning about the Atari learning environment on arcade learning environment which essentially they allowed computers to play these games .

These artery games and what the deep mine fellows came up with is a reinforcement learning agent that learn to play this game from scratch I just by looking at the screen okay that is all the input it was getting just the pixels from the screen they are all pixels on the screen but given as inputs to it used to very complex neural network so it is a deep learning deep network and it is considered of the hardest learning problem solved by a computer and I think I believe it is a one the only computational reinforcement learning paper ever appear in nature , so usually is very

hard for non natural science people to publish in nature and kind of obviously it is usually Hard for computer science to publish in nature but this was totaled out as a next step in trying to understand.

How humans process in post blah essence of all kinds of marketing jargon but more importantly than anything else about this it is reproducing them so I told you about I think that is a warning sign for me to stop so I told you about the helicopter so there is basically Stanford and Berkeley or the two people who get the helicopter to fly I told me about the backgammon player that is like Jerry Tessaro is to are is the one person who gets the backgammon player to work .

Partly because all the input features he uses in there or proprietary but partly because has a very hard problem to solve what is the amazing thing about this Atari game engine is that this case our release of code you can if you have enough powerful GPUs you can set it up here and get it to play and get a reasonably working engine that plays those places Atari games that is the amazing thing about it that is reproducible as opposed to many of the other things other success stories other success stories you had in the past so I do believe I had just one more slide after this so let us see if this will work.

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Okay.

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Oh so if you are really doubts the green one is the learning agent. No this just sped up for you to see you mean it is not like the game is progressing at the same rate but you can see the score slow Mind you it was not given a reward okay it is just given the screen never got to reward for winning ideas and understand that the pixels on the top or rewarding and if we give it rewards it becomes cheating  yeah which is what they did they did they did add a game over which is misty the a heurists considered as cheating but they did not a game over sign so the longer you keep it going the better.

It is basically nothing this is getting boring  so this is learnt here so this is a seaquest you have to swim and then sink down get some things and come up so seaquest is a game that it never learned to do greatly on. Seaquest is not something which did learn to solve well so there are a few games like this, so they initially published the nature paper I think they had liked like 45 or 46 out of the 50 atari games they were able to play well and I think in 43 of them it had better than human performance and I think the current state is they have like one game that does not play well  and have better than human performance in like 48 of those.

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